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Xingxing Zhang
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Future Urban Energy System for Buildings

The Pathway Towards Flexibility, Resilience and Optimization

Springer

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Preface

Urban energy systems for building are undergoing an accelerated transition to achieve the goals of sustainability, security and resilience. The main drivers for such transition are the emergence of renewable-energy-source solutions, energy-efficient buildings, digitalization, automation, smart mobility, carbon emission reduction and increased social awareness. However, the climate change, pandemic crisis and socio-economic uncertainties aggravate the challenges of transition.

This book investigates three main characteristics of future urban energy system for buildings, including flexibility, resilience and optimization. It explores the energy flexibility by considering renewable energy integration with buildings, sector coupling and energy trading in the local energy market. Energy resilience is addressed from aspects of future climate change, pandemic crisis and operational uncertainties. Approaches for system design, dynamic pricing and advanced control are discussed for the optimization of urban energy system. Knowledge from this book contributes to the effective means in future urban energy paradigm to closely integrate multiple energy systems (i.e. distribution, mobility, production and storage) with different energy carriers (i.e. heat and electricity) in an optimal manner for energy use. It would facilitate the envision of next-generation urban energy systems, towards sustainability, resilience and prosperity.

This book targets at a broad readership with specific experience and knowledge in energy system, transport, built environment and urban planning. As such, it will appeal to researchers, graduate students, engineers, consultants, urban scientists, investors and policymakers, with interests in energy flexibility, building/city resilience and climate neutrality.

Stockholm, Sweden
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Xingxing Zhang
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The Importance of Urban Energy System for Buildings

1

Xingxing Zhang, Pei Huang, and Yongjun Sun

Abstract

Urban energy systems for building are undergoing an accelerated transition to achieve the goals of sustainability, security, and resilience. The main drivers for such transition are the emergence of renewable-energy-source solutions, energy-efficient buildings, digitalization, automation, smart mobility, carbon emission reduction, and increased social awareness. However, the climate change, pandemic crisis, and socioeconomic uncertainties aggravate the challenges of transition. Flexibility, resilience, and optimization are three aspects that can facilitate the envision of next-generation urban energy systems, towards sustainability, and prosperity. Such transition of urban energy systems could be a solution to the achievement of Sustainable Development Goals (SDGs). This chapter introduces the background, aim, objectives, motivation, and structure for the whole book.

Keywords

Urban energy systems · Climate change · Flexibility · Resilience · Optimization

1.1 Introduction

1.1.1 Background

According to the United Nations (UN), there are nearly 55% of the population who now lives in cities, and the number is expected to increase significantly by 2050 (Division, 2018). Cities account for almost 60–80% of global energy use and 70% of global CO_2 emissions, in which buildings contribute 20–40% of the energy use and 1/3 of the CO_2 emissions (The Strategic Plan 2020–2023 | UN-Habitat). As a result, cities and their energy systems need stronger commitments and ambitions to reach climate neutrality goals.

By considering the UN's Sustainable Development Goals (SDGs) (THE 17 GOALS | Sustainable Development), this book concentrates on issues of energy, infrastructure, communities/cities, and climates. It contributes to several SDGs, such as Affordable and Clean Energy (Goal 7), Industry, Innovation and Infrastructure (Goal 9), Sustainable Cities and Communities (Goal 11), and Climate Action (Goal 13). Figure 1.1 illustrates the main contributions from this book towards these SDGs.

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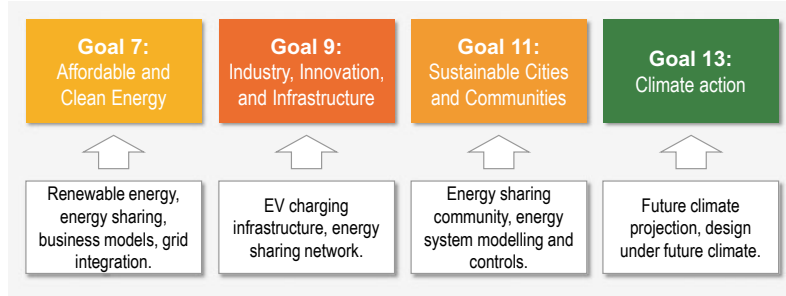

Fig. 1.1 Connection of this book with the UN's sustainable development goals

During the journey to achieve 'Affordable and Clean Energy' goal, many challenges are being brought forward to the existing paradigm of energy systems at building, district, and city scales, such as the emergence of renewable-energy-source (RES) solutions, energy sharing among different buildings/districts and even cities, as well as new business models and the ways to integrate RES with grid infrastructure. This change is predominantly due to the success and popularity of photovoltaics (PV), electrical vehicles (EV), microgrid, advanced control, and new regulations for energy sharing. These solutions entail an evolution in city energy planning and operation, modeling techniques, control intelligence, and management schemes for matching of energy supply and demand across various system scales.

This book is closely related to the goal of 'Industry, Innovation, and Infrastructure' from the following aspects: the planning of renewable powered EV charging infrastructure, data centers integration in the urban energy system, energy sharing microgrid network, COVID-19 impacts on the energy system, as well as novel business models for facilitating the implementation of new services such as peer-to-peer energy sharing, electromobility, and EV smart charging. Advanced control and design optimization methods are impacts tools for bringing these technologies into practice.

As a result, buildings and districts are becoming prosumers when they have more and more RES solutions within their boundaries, and the possibility to share energy in different ways. They not only import from the grid but also produce and sell energy to the grid or the neighbors. These activities are increasingly influencing the energy network by consuming, producing, storing, and supplying energy in different ways at various system scales. Thus, they are reforming the energy market towards more resource-efficient system. Such transition then accelerates the achievements in the goal of 'Sustainable Cities and Communities.'

This book also has a strong connection with the 'Climate Action' goal. Because it not only contains the solutions on how to mitigate the climate changes from building and cities points of view but also considers the future climate change's impact on the performance of buildings and cities. Most buildings nowadays are equipped by RES that will last for long period. Recently, European Commission is even proposing plans for mandatory solar PV panels on all new buildings by 2029 (EU Set to Make Solar Panels Mandatory on All New Buildings). This will be one of the main focuses in this book on how to maximize the harvesting of local RES especially on building. On the other hand, buildings/cities and their energy systems are vulnerable to the climate challenges. When climate changes, the energy demand and production will be different. Although we are adopting many strategies to mitigate the tread of climate change, it is changing now and will change in the future. We believe it is important to consider future climate projections when studying energy system of building and cities.

1.1.2 The Importance of Urban Energy System for Buildings

In principle, urban energy systems are not fundamentally different from other energy systems in that they need both to satisfy a suite of energy-service demands and to mobilize a portfolio of technological options and resources (Grubler et al. n.d.). Urban energy systems not only include demand and supply sides but also refer to broader distribution network and the integrated infrastructures. In this book, we mainly focus on a limited scope of urban energy system and we define urban energy system as follows (Alpagut et al. 2022):

> An urban energy system represents all functional processes related to the provision and use of energy service for demands from an urban population. The energy system comprises from primary energy supply, through conversion, distribution, storage to final use in different sectors (such as buildings and mobility). It requires an increasingly integration of planning, implementation, operation, and management towards an overall sustainable impact, with interactions between a larger number of components and actors.

Currently, buildings, communities, and cities are undergoing an accelerated transition to achieve the goals of sustainability, security, and resilience. The main drivers for such transition are the emergence of RES solutions, EVs, energy-efficient buildings, digitalization, automation, smart mobility, carbon emission reduction, and increased social awareness. However, the climate change, pandemic crisis, and socioeconomic uncertainties aggravate the challenges of transition.

The urban energy landscape is experiencing a major change in which the commonly centralized energy generation is increasingly replaced by a distributed system with dispersed energy recourses, actors, management structures, data sources, and software entities (Howell et al. 2017). This transition requires a coupled research in a wide variety of fields: distributed resources and infrastructures, energy efficiency renovation, RES solutions, distributed generation performance, energy storage behavior and economics, demand side management and virtual power plants, microgrids, energy hubs, and plug-in vehicles, as well as a growing penetration of information and communications technology (ICT), artificial intelligence, and data-driven management (Howell et al. 2017).

Urban energy system for buildings is regarded as an entry point to tackle these challenges in the current city energy paradigm. Buildings are connected with almost every single element of urban energy system, which influence greatly on the overall city energy planning. It is a sector that allows the systematic aggregation of energy information for different types of vectors, such as construction (buildings, infrastructure), operation (heat, electricity, domestic hot water, and networks), and transportation (commutes, mobility).

In addition, urban energy system for buildings could foster the local economic effectiveness and the operation feasibility to maximize the distributed RES harvesting and match with the respective energy demand and supply. It is important to investigate which RES solutions and their sharing strategies/business models are synergic in order to fully utilize the potential of distributed energy harvesting, storage, distribution, and load aggregation. The shift from the energy system at single building scale to district/city scale is crucial for the improvement of local and overall energy resource efficiency, through the interaction between the buildings, communities, and the energy infrastructure domain (Hu et al. 2012).

1.2 Aim and Objectives

This book explores the interdisciplinary fields of urban energy systems, urban planning, mobility, economic/business modeling across levels of buildings, communities, and cities. The aim is to provide solutions for a smooth transition of urban energy systems to be more flexible, resilient, energy-efficient, cost-effective, and resource-efficient.

This book mainly investigates three main characteristics of future urban energy system for buildings, including flexibility, resilience, and

optimization. It explores the energy flexibility by considering renewable energy integration with buildings, sector coupling, and energy trading in the local energy market. Energy resilience is addressed from aspects of future climate change, pandemic crisis, and operational uncertainties. Approaches for system design, dynamic pricing, and advanced control are discussed for the optimization of urban energy system.

Knowledge from this book contributes to the effective means in the future urban energy paradigm to closely integrate multiple energy systems (i.e., production, distribution, mobility, and storage) with different energy carriers (i.e., heat, electricity) in an optimal manner for energy use. It would facilitate the envision of next-generation urban energy systems, towards sustainability, resilience, and prosperity.

1.3 Motivations and Novelties

1.3.1 Motivations

There are a few books that start to address the similar topic. For instance, in the book of '*Urban Energy Systems for Low Carbon Cities*' (Eicker 2019), the authors introduce the indicators of urban energy performance, and they focus on a bottom-up modeling approach for the simulation of energy consumption, energy conversion systems, and distribution networks based on tradition engineering methods. Another book, '*Urban Energy Systems: An Integrated Approach*' (Keirstead and Shah 2013), analyzes the technical and social systems that satisfy these needs and asks how methods can be put into practice to achieve this from aspects of technologies, resources, and people.

A few other books also partly cover the urban energy systems. '*Digital Urban Modeling and Simulation*' (Arisona et al. 2012) focuses on the simulation on urban design and tools for the related planning. '*Guidelines for Community Energy Planning*' (Yu et al. 2020) discusses a range of methods and models for energy planning in different urban planning stages. '*Solar Buildings and Neighborhoods*' (Hachem-Vermette 2020) covers the main guidelines for designing buildings and neighborhoods on how to harvest solar energy. It discusses a wide range of design considerations, from building components (e.g., the building envelope) to urban planning issues (e.g., density and street layouts). '*Energy Sustainability in Built and Urban Environments*' (Motoasca et al. n.d.) presents different aspects of energy sustainability in residential buildings and neighborhoods, including the construction and design, heating ventilation and air-conditioning (HVAC) systems, lighting, and renewable energy aspects, from the aspects of technological, economic, social, and environmental point of views. '*Urban Energy Transition*' (Droege 2018) presents the planning of future cities, design of novel systems and technologies, as well as business models. '*Energy Positive Neighborhoods and Smart Energy Districts*' (Monti 2017) provides methods, tools, and experiences from the field for promoting the transition of the existing building communities towards more sustainable neighborhoods.

It is clear that these books are mainly from the perspective of traditional engineering modeling, RES-building design, energy system at building level, and urban planning, while few evidence was found by highlighting the three main characteristics of future urban energy system for buildings, such as flexibility, resilience, and optimization across different levels of building, communities, and cities. Thus, there is a great need to have a book to further cover the suitable methods and case studies in this area.

1.3.2 Novelties

The book investigates the flexibility of urban energy system by considering renewable integration, sector coupling, energy sharing, and trading in buildings and districts. These novel concepts and services, which are newly developed in recent decades with the development of new technologies, are effective ways to help the urban energy system become more resource-efficient and energy-efficient. This is the first book to present these technologies in a systematic way.

The book discusses the resilience of urban energy system when it faces challenges from future climate change, pandemic crisis, and system uncertainty. Due to the complexity in the modeling of uncertainty, climate change, and human behavior, it is challenging to investigate the resilience of the urban energy systems. This book fills in such research gap and provide validated research methodology and case studies.

The book covers the comprehensive optimization of urban energy system in aspects of energy system design and energy system control. The design optimization covers not only energy systems at both the single-building-level and building-community-level but also the energy sharing community itself for maximized resource efficiency. The control optimization also include system at both the single-building-level and building-community-level.

1.4 Structure and Contents

Apart from the first chapter, this book consists of three main parts. Part I introduces the energy flexibility, which relates to the integration of renewable energy, and electricity vehicles into the future urban energy system, as well as the waste heat recovery. Part II introduces the resilience of energy systems, especially related to the analysis and optimization considering challenges from future climate change, pandemic crisis, and system uncertainty. Part III is about the optimization of energy systems, which will cover a set of novel design and control optimization methods for energy systems in the built environment.

In Part I, there are seven chapters. Chapter 2 reviews the urban energy systems at the cluster level that incorporate building integrated renewable energy system solutions. A set of influential factors related to energy planning at cluster scale are discussed. Chapter 3 proposes the concept of solar mobility (i.e., produce solar power from one location and use it in another location) and reviews the existing technology (such as smart charging of electric vehicles (EVs), control of distributed energy systems, etc.) for promoting the solar mobility. Then, in Chap. 4, novel ways of integrating data centers as prosumers (i.e., as electricity consumers and the heat producers) in urban energy system are reviewed. Chapter 5 reviews a large number of positive energy district (PED) projects and analyzed their main characteristics, including geographical information, spatial–temporal scale, energy concepts, and building archetypes, which forms basis for the standardization of the PED technology. Chapter 6 introduces novel business models for facilitating the peer-to-peer electricity energy sharing within building communities based on agent-based modeling. Chapter 7 analyzes how different forms of charging EVs can be used as demand response for regulating the local power balances and enhancing the energy flexibility. In Chap. 8, the commonly used control algorithms for EV smart charging are reviewed and systematically compared in perspectives of control performances and computation efficiency.

Part II includes six chapters. Chapter 9 analyzes the impacts of future climate change on the peer-to-peer energy sharing performances in the building community. To make the building energy system more adaptable to the future climate, in Chap. 10, a differential evolution-based method is developed and presented, which can design energy systems with better performances under the future climate. In Chap. 11, the impacts of the COVID-19 pandemic crisis on electrical and thermal energy demands of a building mix at a district are systematically analyzed via simulation in the urban modeling interface tool. Chapter 12 investigates the impacts of uncertainty on the energy system design and develops novel designs which considers such uncertainty. Chapter 13 investigates the impact of the demand profile and the normative framework on the residential photovoltaic system and develops a novel design method for the residential PV system. Chapter 14 proposes a digitalization technique, i.e., a Gaussian mixture model (GMM) with an expectation–maximization (EM) algorithm, to produce synthetic building energy data, which can overcome the challenges of data insufficiency in urban energy system analysis.

Part III has six chapters. Chapter 15 optimizes the design of distributed energy storage systems

in energy sharing building communities, which can minimize the total investment with reduced energy losses. In Chap. 16, the design optimization of EV charging stations in the urban environment is presented considering the distributed renewable power production using the GIS technique. Chapter 17 develops advanced planning method to optimize the selection of buildings to form energy sharing communities with the maximized energy sharing potentials. In Chap. 18, a dynamic pricing strategy is developed, which can lead the different buildings to conduct demand response to approach the expected power production. Chapter 19 tacks the challenges of complex coordination of multiple complex energy systems and proposes a hierarchical demand response control. In Chap. 20, a similar top-down control method is presented for optimizing the energy sharing performances in energy sharing communities using the genetic algorithm.

All these chapters are interlinked and well-categorized to address the challenges in different aspects related to the future urban energy systems for the buildings.

1.5 Conclusion

This book contributes to the interdisciplinary study in the areas of urban energy systems, urban planning, mobility, economic/business modeling across levels of buildings, communities, and cities. This is the first book to report a systematic study which addresses 'flexibility,' 'resilience,' and 'optimization' of urban energy system, by considering renewable integration, sector coupling, smart charging, energy storage, energy sharing, and trading in buildings and districts.

The whole book totally contains 20 chapters that are divided into three main parts. Each part responds to issues of 'flexibility,' 'resilience,' and 'optimization' of urban energy system, respectively. The book covers the development related to several SDGs, such as Affordable and Clean Energy (Goal 7), Industry, Innovation and Infrastructure (Goal 9), Sustainable Cities and Communities (Goal 11), and Climate Action (Goal 13).

Those novel concepts and services highlighted in this book are expected to provide solutions for a smooth transition of urban energy systems towards more flexible, resilient, energy-efficient, cost-effective, and resource-efficient goals. This book will contribute to transferring such specific knowledge to various stakeholders to achieve smart and climate-neutral society in the future.

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2 Integration of Urban Energy Systems with Renewable Envelope Solutions at Building Cluster Level

Xingxing Zhang and Marco Lovati

Abstract

The emergence of renewable energy source (RES) envelope solutions, building retrofit requirements and advanced energy technologies brought about challenges to the existing paradigm of urban energy systems. It is envisioned that the building cluster approach —that can maximize the synergies of RES harvesting, building performance and distributed energy management—will deliver the breakthrough to these challenges. Thus, this chapter aims to critically review urban energy systems at the cluster level that incorporate building integrated RES solutions. We begin with defining cluster approach and the associated boundaries. Several factors influencing energy planning at cluster scale are identified, whilst the most important ones are discussed in detail. The closely reviewed factors include RES envelope solutions, solar energy potential, density of buildings, energy demand, integrated cluster-scale energy systems and energy hub. The examined categories of RES envelope solutions are (i) the solar power, (ii) the solar thermal and (iii) the energy-efficient ones, out of which solar energy is the most prevalent RES. As a result, methods assessing the solar energy potentials of building envelopes are reviewed in detail. Building density and the associated energy use are also identified as key factors since they affect the type and the energy harvesting potentials of RES envelopes. Modelling techniques for building energy demand at cluster level and their coupling with complex integrated energy systems or an energy hub are reviewed in a comprehensive way. In addition, the chapter discusses control and operational methods as well as related optimization algorithms for the energy hub concept. Based on the findings of the chapter, we put forward a matrix of recommendations for cluster-level energy system simulations aiming to maximize the direct and indirect benefits of RES envelope solutions. By reviewing key factors and modelling approaches for characterizing RES-envelope-solutions-based urban energy systems at cluster level, this chapter hopes to foster the transition towards more sustainable urban energy systems.

Keywords

RES · Building cluster · Energy system · Energy hub · Modelling · Optimization

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Abbreviations

ANN	Artificial neutral network
BES	Building energy simulation
BIPV	Building integrated photovoltaic
BIPV/T	Building integrated photovoltaics/thermal
CABS	Climate adaptive building shell
CdTe	Cadmium telluride
CES	City energy simulation
CFD	Computer fluid dynamics
CHCP	Combined heat, cool and power
CHP	Combined heat and power
DES	Distributed energy source
DEROP	Distributed energy resource optimisation algorithm
DSF	Double skin façade
Energy-Matching	H2020 Energy Matching project: https://www.energymatching.eu/
EV	Electric vehicles
FAR	Floor area ratio
GDP	Gross domestic product
GHG	Greenhouse gas
GIGS	Copper indium gallium selenide
GIS	Geographic information system
H2020	European Commission Horizontal 2020 research and innovation programme
LES	Large eddy simulation
LiDAR	Light detection and ranging
MAS	Multi-agents system
MILP	Mixed integer linear program
MINLP	Mixed integer non-linear programming
MPC	Model predictive control
NZEB	Net-zero energy buildings
ICT	Information and communications technology
PCM	Phase change material
PV	Photovoltaics
RES	Renewable energy sources
SAL-TVAC-GSA	Self-adoptive learning with time varying acceleration coefficient-gravitational search algorithm
STF	Solar thermal facade
SVM	Support vector machine
TMY	Typical meteorological year
UBES	Urban building energy simulation
WVM	Wavelet variability model

2.1 Introduction

In order to deliver urban sustainability, security and resilience, the urban energy system is undergoing an accelerated transition from a predominantly centralized to the highly distributed one. One of the driving forces is the significant growth of integrated distributed renewable energy sources (RES) within the built environment. This growth is predominantly due to the success and popularity of adaptive building envelope solutions, such as building integrated photovoltaics (BIPV) (Shukla et al. 2017) or building integrated photovoltaics/thermal (BIPV/T) (Debbarma et al. 2017), solar thermal façade (STF) (Zhang et al. 2015), heat pump components (Poppi et al. 2018) and their accompanying power storage (O'Shaughnessy et al. 2018) or thermal storage systems (Alva et al. 2018). The emergence of these RES

envelope solutions not only indicates a shift in the energy landscape towards more sustainable and resilient practices, but also entails an evolution in urban energy planning, modelling techniques, operation/control intelligence and management schemes for matching of energy supply and demand across various system scales. Buildings are becoming prosumers, rather than purely stand-alone energy consuming units of the grid. They are increasingly turning into active elements of the energy network by consuming, producing, storing and supplying energy. Thus, they transform the energy market characterized by centralized, fossil fuel-based national systems to a decentralized, renewable, interconnected and viable system.

Within the context of the European Union (EU), building retrofit provides a great opportunity to meet EU policy goals related to net-zero energy buildings (NZEB) (2010/31/EU 2016) and building integrated RES (2009/28/CE 2016). Current EU policies promote the reduction of building energy demand by 80% by 2050 by means of building retrofit (www.renovate-europe.eu, accessed on 25 Feb 2018 n.d.). The emerging challenges lead to the development of novel approaches that address buildings and their energy systems at different scales: from single buildings to cluster, district and urban levels. It is envisioned that energy planning at the building cluster scale is an effective strategy to combine energy efficiency retrofit and local RES supply, through the enhancement of district energy systems and decentralized energy supply (Vigna et al. 2018). Similarly to micro-communities in the society, neighbouring buildings will have the tendency to form a building cluster with an open cyber-physical system to exploit the economic opportunities provided by distributed RES systems (Li and Wen 2017). The cluster scale enables a systematic approach to reduce the unit cost of investment and reach cost optimality in energy planning by considering factors, such as retrofitting and adoption of technologies/strategies for increasing energy efficiency and minimizing carbon emissions (Koch and Girard 2013). Several benefits of a shared RES-distribution network at cluster level have been demonstrated in a number of existing case studies (e.g. the BedZED eco-community in London, Vauban in Freiburg, and Hammarby Sjöstad in Stockholm (Williams 2016)), such as increased energy efficiency, higher feasibility of storage and load complementarity due to building function differences (e.g. commercial and residential).

As a result, energy planning at building cluster scale fosters the economic effectiveness and the operation feasibility to maximize the distributed RES harvesting and match with the respective energy demand and supply. It is essential to determine which RES solutions are synergic when clustered, and what modelling methodologies should be implemented for operation in order to fully utilize the potential of distributed RES harvesting, storage, distribution, load aggregation and demand side management. The shift from the single building to the building cluster is crucial for the improvement of local energy resource efficiency, through the interaction between the buildings and the energy infrastructure domain (Hu et al. 2012). Thus, this chapter focuses on the building cluster approach for urban energy systems when considering the incorporation of RES envelope solutions. First, it aims to define the cluster method and its boundaries. Then, it discusses major influencing factors and modelling methodologies. Therefore, the scope of this chapter is limited by the boundary dimensions, methodologies and major influencing factors of RES envelope-based energy systems for a group of buildings (referred to as 'cluster' in the remainder of the document). Since in the existing studies, modelling is the dominant methodology for the evaluation of the energy systems at such level, this chapter focuses on the modelling methods that have been applied in the related assessments.

This chapter is motivated by answering the research question, illustrated in Fig. 2.1, of how is energy matched in terms of demand and supply in the cluster with RES envelope solutions? In order to find an answer, a knowledge-based matrix was structured through a literature review by answering the following two questions:

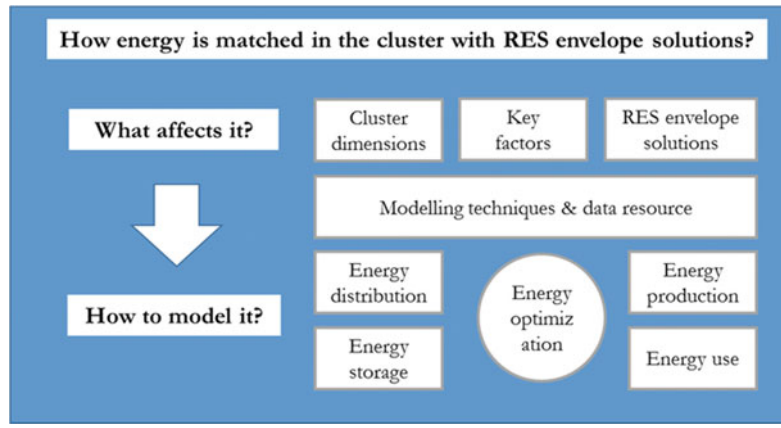

Fig. 2.1 Scheme of the research question and research tasks

- what affects the energy matching in the building cluster by defining cluster dimensions and identifying key influencing factors and RES envelope solutions;
- how to model energy systems by observing existing modelling and optimization techniques.

A comprehensive critical review was conducted based on academic literature, research reports, legislation and key data bases for RES envelopes and energy systems. The essential body of literature was broken down into thematic categories. The important influencing factors for energy matching in building cluster were either brainstormed by partners in H2020 EnergyMatching project or extracted from the literature. The existing RES envelope solutions at building cluster scale and the related modelling techniques, as well as optimization methods, were observed in the literature and summarized in tables and figures. The remainder of this section describes the scope of the chapter and delivers our insights.

After clarifying the chapter's scope and review method, we proceed with defining building cluster from energy system point of view. Then, the dimensions of the cluster (e.g. size of cluster area and energy performance resolution) and its influencing factors are introduce. Subsequently, we discuss the most important factors in detail and present the categorization of main RES envelope solutions based on the existing literature. Afterwards, we describe promising modelling techniques for assessing the potential of common RES at cluster scale utilizing solar energy. The density of buildings is then discussed, as it affects both solar energy potential and energy demand. Considering the importance of energy demand estimation within an increasingly varied and sophisticated urban energy system, we critically review a set of emerging modelling techniques. Next, we discuss the modelling and optimization techniques for complex, RES-based cluster-level energy systems and the energy hub concept in detail. Finally, the chapter lays out a number of suggestions for future research directions.

There are many existing review papers that address different aspects of urban energy systems, such as the impact of occupants' behaviour (Happle et al. 2018), energy tools/models at different scales (single building scale (Harish and Kumar 2016), district scale (Allegrini et al. 2015), urban scale (Manfren et al. 2011), regional/national scale (Hall and Buckley 2016)), energy demand (electricity (Torriti 2014), heating and cooling (Frayssinet et al. 2018)), demand response (Wang et al. 2017), micro-grid (Agüera-Pérez et al. 2018), solar PV (Orioli and Gangi 2014), electric vehicles (Sulaiman et al. 2018), energy storage (Yang et al. 2018), control strategies (Arul et al. 2015), energy hub (Mohammadi et al. 2018), water-energy nexus (Wakeel et al. 2016; Lee et al. 2017), planning and policy (Adil and Ko 2016). However, none

of these studies address RES envelope solutions at the building cluster scale and the corresponding modelling methodologies for the integrated energy systems. Therefore, this chapter aims to deliver a comprehensive literature review to fill this gap. The novelty of this chapter lies in (1) defining the concept of building cluster and its boundaries for modelling and assessment; (2) highlighting the main influencing factors across three aspects of urban energy system, including supply, demand and operation; (3) summarizing RES envelope solutions suitable for building clusters and (4) identifying modelling methodologies for integrated urban energy systems at the cluster level. The findings of the chapter can provide guidance to utilizing RES envelope solutions in the design or retrofitting of building clusters. The boundaries will help to improve the resolution and accuracy of the complex modelling. The modelling and optimization approaches shall facilitate the maximization of RES harvesting and socio-economic benefits of urban energy systems. Figure 2.2 illustrates the chapter scope and the contents of this chapter.

2.1.1 Building Cluster and Its Influencing Factors

2.1.1.1 Definition of Building Cluster

The building cluster scale, also known as 'building block or neighbourhood', represents an intermediate level between a single building and district or urban scale. It could be defined depending on different criteria, such as energy system, archetypes, location, building size, density (low, medium, high), function (residential, offices, mixed), number of stories (low, high), year of construction, geographical boundary and so on. In this chapter, we focus on the definition from energy system point of view. As a result, a building cluster is regarded as a group of buildings systemically interconnected to the same energy infrastructure, so that a change of energy performance of a single building affects both the energy infrastructure and other buildings of the cluster either in a synergic or a disruptive way (Vigna, et al. 2018). At this scale, urban building energy simulation (UBES) is a common strategy applied for modelling the interactions of energy structures, urban climate and building energy performance, while building energy simulation (BES) and city energy simulation (CES) are developed, respectively, for the scales from single building to district/city level or above.

2.1.1.2 Why Building Cluster?

The urban energy landscape is experiencing a major change in which the commonly centralized energy generation is increasingly replaced by a distributed system with dispersed energy recourses, actors, management structures, data sources and software entities (Howell et al. 2017). This transition requires and stimulates a large amount of research in a wide variety of fields: distributed resources and infrastructures, energy efficiency renovation, RES solutions, distributed generation performance, energy storage behaviour and economics, demand side management and virtual power plants, microgrids, energy hubs and plug-in vehicles, as well as a growing penetration of ICT, artificial intelligence and data-driven management (Howell

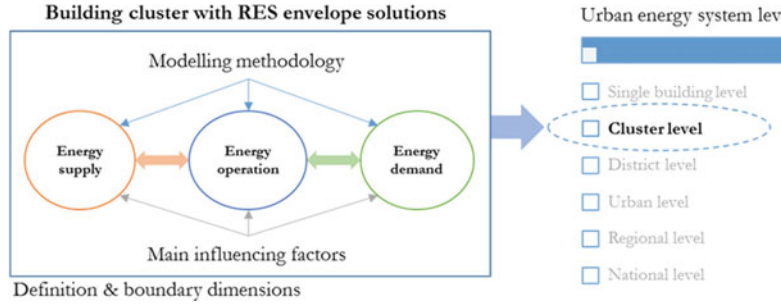

Fig. 2.2 Review scope and main contents

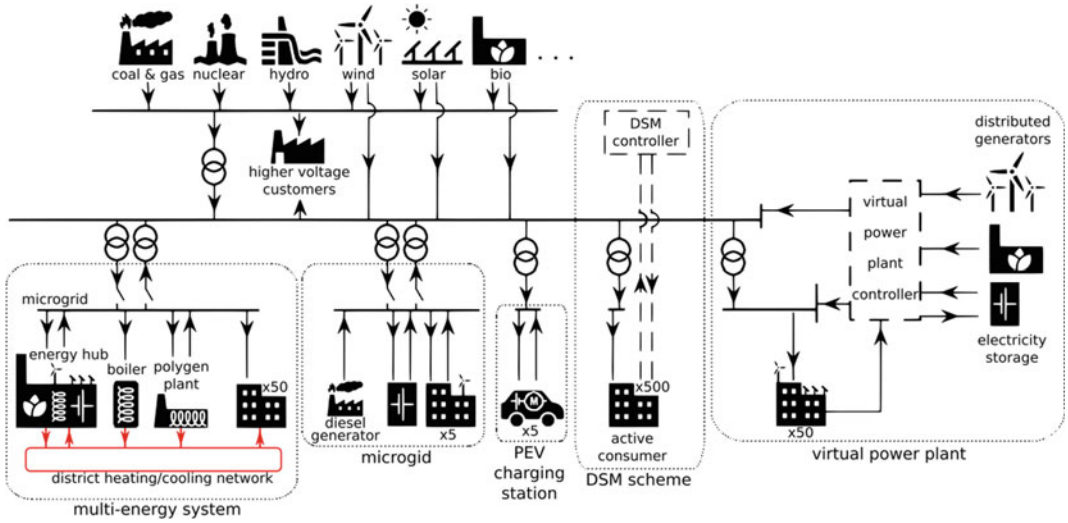

Fig. 2.3 Energy landscape emerging through smart grid and urban energy system concepts (Howell et al. 2017)

et al. 2017), as shown in Fig. 2.3. As stated in Sect. 2.1.1, energy planning at building cluster scale is regarded as an effective way to tackle these challenges in the current urban energy paradigm. The cluster scale is large enough to address energy matching better than in a single building, but remains small enough to allow concrete examination. It is a scale that allows the systematic aggregation of energy information for different types of vectors, such as construction (buildings, infrastructure), operation (heat, electricity, domestic hot water and networks), and transportation (commutes, shopping) (Rey et al. 2013). It is a realistic scale for RES envelope solutions because not all the buildings in practice are possible to integrate RES solutions, and those RES integrated buildings can then be defined as a cluster (though not a physical district). Fonseca and Schlueter (2015) also pointed out it is at cluster scale where most urban transformations in EU take place and where the newest instruments for financing energy efficiency strategies in the building sector exist.

In addition, according to Frayssinet et al. (2018), urban energy systems are now very complex to simulate at the city scale, due to the required large amount of input data/computation, the uncertainties of occupant behaviour, and the necessary involvement of complex urban environment. On the other hand, simulation at a single building scale is not accurate enough to respond to urban energy system, since buildings are not standing-alone units. Thus, building cluster presents itself as a possible intermediate scale to assess the interaction between buildings and urban energy infrastructures in detail, while also taking into current computational capacity and intelligence. At building cluster level, scenarios, such as energy sharing and competition, can be modelled and studied. With the increase in adoption of RES envelope solutions, research endeavours in building cluster modelling are gaining importance. The aim is to shift from single energy efficient unit to interconnected prosumers, and therefore maximizing the synergies among buildings, RES application, storage systems, and existing heating/electric girds. Some degree in the energy matching ability is required by buildings in order to gain resilience (building performance coupled with grid interaction) (Vigna et al. 2018). We hypothesize that the study of energy landscapes through the lens of building clusters will result in cost-effective RES solutions, which in turn will well equipped to cope with disruptive new technologies and alterations in the energy system. Figure 2.4 interprets the undergoing transformation of buildings into cluster aware units.

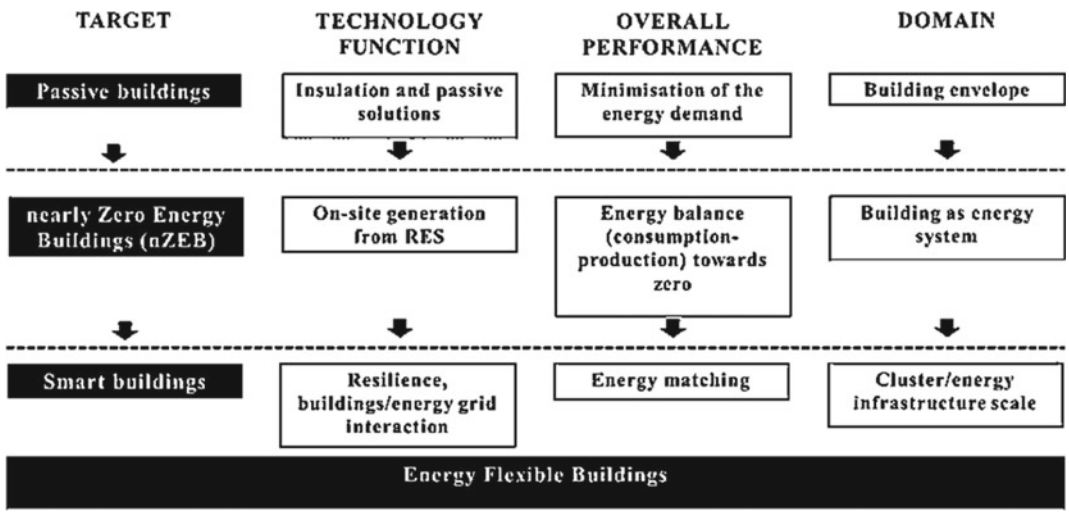

Fig. 2.4 Evolutionary path of building transformation (Vigna et al. 2018)

2.1.1.3 Spatio-Temporal Dimension of Building Cluster

Understanding spatio-temporal patterns of energy supply and demand is essential to assess the retrofitting strategies for stochastic RES envelope solutions and storage systems within buildings cluster. This is usually achieved though UBES approach, by either top-down or bottom-up ways (Swan and Ugursal 2009). However, such UBES approach is generally too computationally expensive to simulate building clusters (Frayssinet et al. 2018). In order to simulate energy systems in an efficient way, a trade-off front in spatio-temporal dimensions is delineating for the energy simulation at cluster scale, as depicted in Fig. 2.5.

Spatial scale is used in this chapter for describing the size of a cluster area for energy planning/simulation purpose. In Britter and Hanna's research (Britter and Hanna 2003), they classified studies in urban areas into four spatial scales, i.e. the regional scale (less than 100 or 200 km), the city scale (less than 10 or 20 km), the neighbourhood scale (less than 1 or 2 km) and the street scale (less than 100–200 m). Other studies, such as Srebric et al. (2015) and Huang's study (Hang and Li 2012), indicated that the impacts of urban neighbourhoods on the buildings, and associated modelling should be resolved within 1 km. The spatial unit of a cluster is usually equivalent with or less than a neighbourhood. We hereby recommend that the

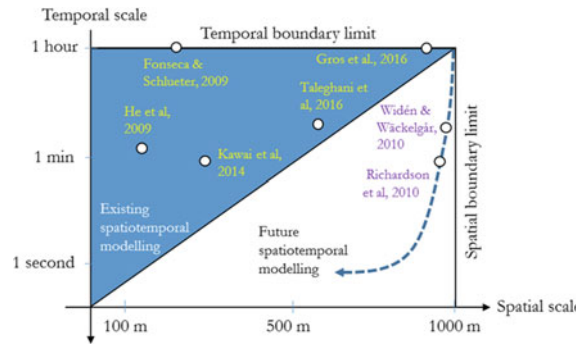

Fig. 2.5 Spatio-temporal dimension of building cluster

spatial dimension of a building cluster to be between 100 m and 1 km for energy system simulation purposes, which should be computationally viable in next a few years. The cluster territory is not strictly limited to a specific geometry but indicates a rough area, for instance, a circular territory has the cluster diameter between 100 m and 1 km, a square territory has the cluster side edge between the same two thresholds, etc. Currently, detailed computational studies at massing model level for gross parameterization of the energy flow within buildings are feasible at this spatial scale. This is also a scale at which some statistical homogeneity of energy systems may be anticipated. Accordingly, a city can be then regarded as a collection of clusters. Nevertheless, the cluster scale is likely to be regarded ineffective from other points of view, such as social or policy targeting (Harris and Johnston 2003). A fine-scale cluster geography for whole-city urban purposes is still confined to the future in terms of research.

Temporal scale is applied in this chapter to describe the energy performance resolution of buildings and systems within a cluster. The time required for building components/envelopes to respond and achieve a steady-state condition may take from hours to days. One another hand, the time for energy system/flow to respond a condition could be within seconds or minutes (Srebric et al. 2015). These different response times suggest that the time steps required to solve the energy matching at a comparable level of detail may differ in their orders of magnitude. At the moment, many studies choose the hourly energy demand for UBES as the minimal temporal resolution to estimate the energy load profiles (thermal load (Fonseca and Schlueter 2015; He et al. 2009) and electric load (Widén and Wäckelgård 2010; Richardson et al. 2010)). A good knowledge of the transient energy flow and a more accurate energy matching scenario in building cluster requires the order of magnitude of time scale to be reduced down towards minutes or even second level. This shift will only happen if minute-resolution data becomes the standard in UBES and BPS and would nevertheless increase the computational cost of UBES simulations.

2.1.1.4 Influencing Factors

The factors that influence the energy landscape are diverse at the cluster scale. Urban morphology parameters, such as plan area density, frontal area density, geometry of the buildings, and topographical features, influence energy use and available resource at the cluster scale (Jurelionis and Bouris 2016). Climate zone, construction period and building type are usually the parameters that serve as selection criteria for the building stock segmentation and thus affect the energy scenario (Monteiro et al. 2017). There are many other impacting parameters that can be divided into four different groups (Deru 2011), such as geometry (form), construction (fabric), systems (equipment) and operation (program). These parameters are dependent on the energy planning without compromising to each other. As different building clusters may require different parameters to access energy matching strategy, there is the fundamental need for the generalized key factors being able to adapt to each country/city characteristics.

A brainstorming session was firstly conducted for the key parameters among well-varied experts from diverse fields, in the H2020 project —'Energy-Matching'. After that, a literature analysis into these primary parameters was performed to describe their importance to building cluster concept. The results of the brainstorming session are presented in Fig. 2.6. The parameters defined as important for building cluster characterization are grouped in three main area of interest: grid, RES production and building. Among these, we recognize the main interesting parameters as mostly influencing the cluster energy performance, which may include energy supply side (RES envelope solutions, solar power potential, density of building), energy demand side and energy operation side (integrated energy systems and energy hub). Each of these factors is discussed in sequence following the logics represented in Fig. 2.7 in next sections.

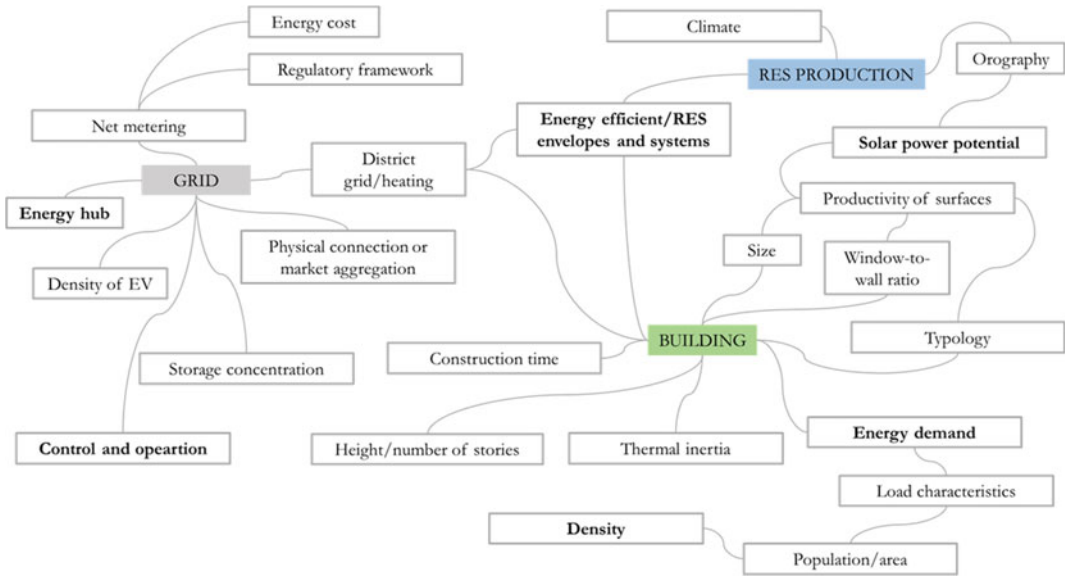

Fig. 2.6 Scheme of main parameters affecting energy characteristic in a building cluster

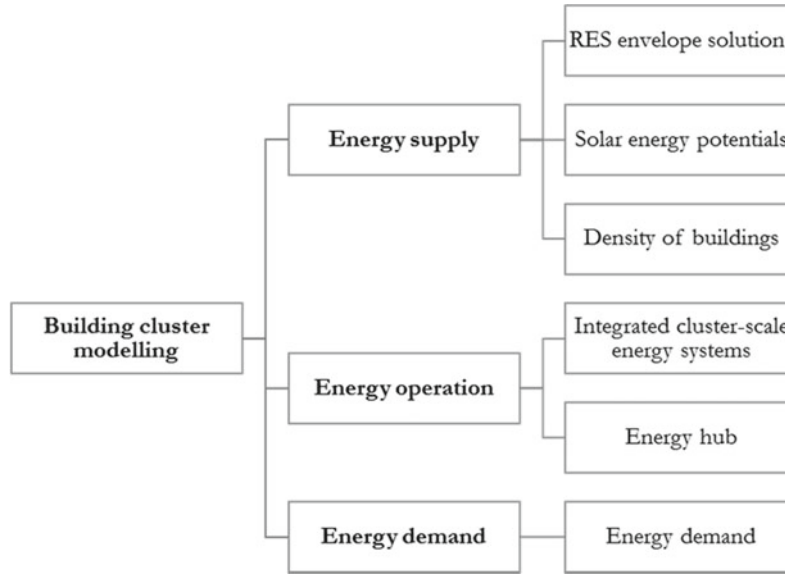

Fig. 2.7 Logics for discussion of the influencing parameters on energy systems at building cluster scale

2.1.1.5 RES Envelope Solutions

The range of RES solutions is very broad, which may be categorized in different ways. Within this work, the categorization is mainly performed based on the energy resources and the way those solutions contribute to building energy. The overall RES solutions have been categorized into the following groups as solar power solutions, solar thermal solutions and energy-efficient solutions, illustrated in Fig. 2.8. The whole framework fits well in the concept of 'Climate adaptive building shell' (CABS), according to Loonen et al. (2013). They defined CABS as 'A climate adaptive building shell has the ability to repeatedly and reversibly change some of its functions, features or behaviour over time in

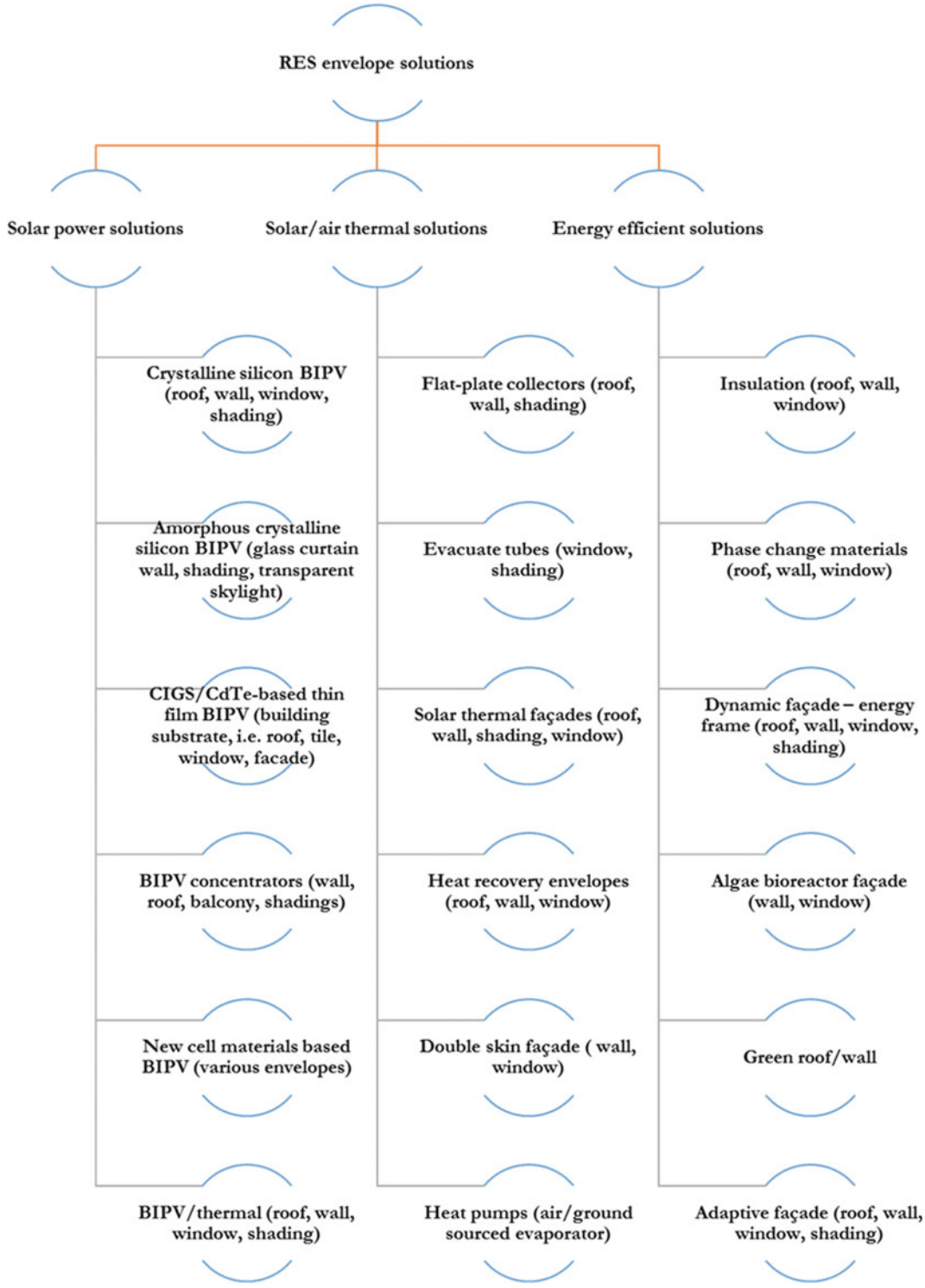

Fig. 2.8 Categorization of RES envelope solutions

response to changing performance requirements and variable boundary conditions, and does this with the aim of improving overall building performance'. As a result, modern RES solutions shall be able to offer potential opportunities for energy savings and improvement of indoor environmental quality, by drawing upon the concepts of adaptability, multi-ability and evolve-ability, in order to combine the complementary beneficial aspects of both active and passive building technologies into the building envelope (Loonen et al. 2013). The solar power and solar heat solutions are usually energy generators for buildings, while energy-efficient solutions contribute to reduction in energy use in buildings.

BIPVs are regarded the most important solutions in solar powered envelopes. They offer an aesthetical, economic and technical solution to integrate solar cells harvesting solar radiation to produce energy within the climate envelopes of buildings. The main stream of current BIPVs are crystalline silicon, amorphous crystalline silicon, and copper indium gallium selenide (GIGS)/cadmium telluride (CdTe) thin films. In future, new cell materials will steer BIPV into a more competitive era, which may include adaptive low-medium efficiency organic-based modules (Solar Cells Absorbing Non-Visible Solar Radiation, Polymer Solar Cells, Dye sensitized solar cells), ultra-high efficiency modules (sandwich solar cells, antenna-sensitizer solar cells, quantum dot solar cells), solar trapping systems embedded in solar cell structure (solar cell concentrators, inverted pyramid texturing), material beneath (PV Integration in concrete) and flexible lightweight inorganic thin film (solar cell paint, hybrid solar cells) (Jelle 2016). BIPVs are also flexible to be applied as a concentrator (Gu et al. 2018) and BIPV/T solutions for both electricity and heat generation (Debbarma et al. 2017).

In the group of solar/air sourced thermal solutions, flat-plate and evacuate tubes are the most common technologies applied in the past and existing period. Solar thermal façade (Zhang et al. 2015), heat recovery envelopes (O'Connor et al. 2016) and double skin façade (DSF) (Ghaffarianhoseini et al. 2016) are more adaptable to buildings, which are often connected by heat pumps for upgrade of heat generation (Poppi et al. 2018). In terms of energy-efficient solutions, green roof/wall systems (Besir and Cuce 2018), thermal insulations (Schiavoni et al. 2016) and phase change materials (PCM) (Kasaeian et al. 2017) are widely applied for either new buildings or building retrofit. Dynamic façade (also known as energy frame) (Johnsen and Winther 2015) and adaptive façade (López et al. 2017) are newly developed concepts by changing the façade properties, or tracking with solar radiation, or controlling daylight/humidity, depending on various climate conditions, etc. (Matteo 2018). In recent years, algae photo-biological facades (Mohammad 2018) are developed to reduce energy use by shading, but meanwhile generate heat and biomass for buildings.

It is observed that most of the existing RES envelopes are derived from solar (air) resource. Some of them converts solar energy directly into useful electricity and heat, such as categories of solar power solutions and solar/air thermal solutions, as well as photo-biological facades, while the other types either make uses of sensible heat from solar (air), e.g. thermal insulation, green roof/wall or the latent heat from solar (air), e.g. PCM; others still indirectly gain advantages from solar (air), such as dynamic façade. As a result, solar energy is regarded as the dominant renewable energy resource for envelope solutions in the building cluster.

2.1.2 Solar Energy Potential

Modelling the energy output of a large set of spatially distributed and building-applied photovoltaic (PV) or thermal systems, as in the case of building clusters, typically requires inclusion of three main components, as outlined in Shepero et al. (2018): (1) the solar irradiance over the systems, in sufficient spatio-temporal detail, (2) a method for identifying and representing the building areas on which the PV or the thermal systems are mounted and (3) suitable models for solar irradiance on tilted planes and for PV or

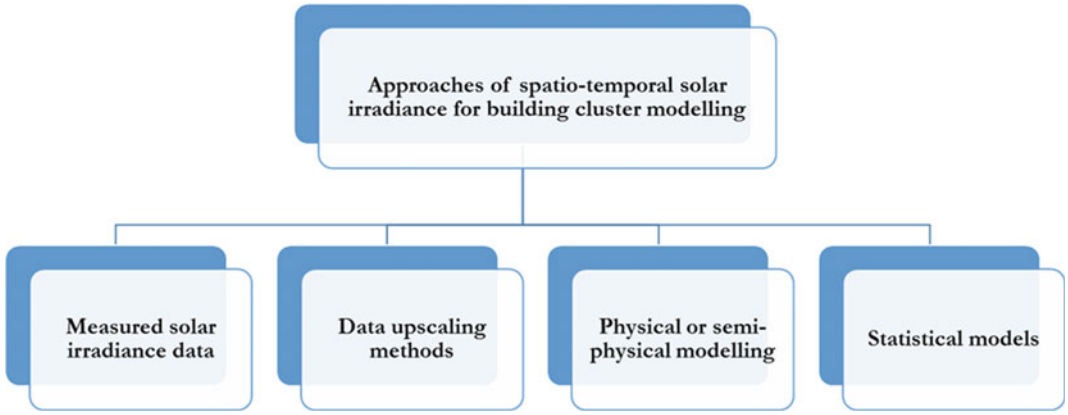

Fig. 2.9 Categories of approaches of spatio-temporal solar irradiance for building cluster modelling

thermal systems. As component (2) was covered in Sect. 2.3 and there are standard approaches for component (3), reviews of which can be found elsewhere (e.g. Klise and Stein 2009), this section therefore mainly focuses on component (1), which is also the most challenging part at the cluster level.

Business as usual when incorporating simulations of solar technologies in building modelling is to use hourly solar irradiance data for one representative site as input, often in the form of typical meteorological year (TMY) data. Modelling of solar technologies on the spatial and temporal scales proposed here (see Sect. 2.3) however requires more sophisticated approaches. On the minute and second scale, solar irradiance varies substantially due to variability in cloud patterns and their movements as well as to irradiance enhancement (Inman et al. 2016). For buildings dispersed over spatial scales of metres to kilometres, variations on these temporal scales do not occur simultaneously. As a consequence, correlations in power or thermal output between dispersed building-applied PV and thermal systems decrease characteristically over both space and time, effectively smoothing out the total solar power fluctuations to a degree that depends on the overall dispersion and the type of weather (Perez and Hoff 2013; Widén et al. 2015). For realistic building cluster simulations, the impact of these features should be measured and, if relevant, included in the data used as input.

We can identify four categories of approaches for obtaining spatio-temporal solar irradiance data suitable for building cluster modelling in available literature, as summarized in Fig. 2.9: (i) measured solar irradiance data, (ii) data upscaling methods, (iii) physical or semi-physical modelling and (iv) statistical models.

Measured solar irradiance data should be preferred when such exist for a studied site. The two most commonly used sources of solar irradiance data are ground sensors and satellite-derived data. For hourly data and over large spatial scales, radiometer network data are typically measured and made available by national or regional meteorological services. For these spatio-temporal scales, established methods for deriving irradiance from satellite images are also readily available (see, e.g. Perez et al. 2013). These types of data are unfortunately much more scarce on the spatio-temporal scales considered here. Dense networks of solar irradiance sensors have been constructed at various sites for studies of irradiance variability (for an overview see Widén et al. 2015). An example of state of the art is the Oahu solar irradiance grid on Hawaii, consisting of 17 pyranometers, dispersed up to 1 km, that measure global horizontal irradiance with a 1-s resolution (NREL 2018). The only available satellite data on the building cluster scale appears to be the Himawari-8 satellite, which covers Asia and the Pacific with down to 2.5-min temporal resolution and 0.5 km^2 pixel

resolution (Bright et al. 2017). Awaiting such high-resolution satellite imagery for wider regions, as well as validated methods for deriving irradiance from them, methods from the remaining three categories below could be used.

Data upscaling methods take data from a small set of reference irradiance sensors or PV (thermal) systems to generate data for a much larger set. One example is the Wavelet Variability Model (WVM), which uses irradiance data from one point sensor to simulate the smoothed-out profile for a larger set of sites (Lave et al. 2013). This method would be suitable for describing the aggregated profile from large numbers of buildings with PV systems, but for obtaining unique data for each building-applied system other methods would be required. Bright et al. (2017) proposed a method for generating 1-min, spatially resolved data from hourly observation data. In this method, a cloud field representative of each hour was generated based on general weather and cloud statistics and moved over an arbitrary set of dispersed sites. Bright et al. (2017) then provided an overview of other upscaling methods based on different spatial interpolation techniques, either through pure interpolation or in combination with system metadata and quality control routines. These methods have been used mainly for now casting of solar power in grids, but they should be able to provide irradiance data for building cluster simulation. Future research should be aimed at optimizing these methods in terms of interpolation technique, number and dispersion of reference sites and type and extent of metadata.

By physical or semi-physical modelling, we refer to models that do not use measured irradiance but instead derive spatio-temporal irradiance data by modelling the atmosphere and clouds in a physical sense (not purely statistical approaches). Typically, a clear-sky irradiance model is used to model the irradiance after passage through the atmosphere, and a cloud model is used to model attenuation due to clouds. Several established clear-sky models exist, varying in complexity but generally performing well (Ineichen 2016; Reno et al. 2012). Models of clouds and their development and movement over time also span a wide range of complexity. On the most complex extreme, but also among the most mature approaches, we find large eddy simulation (LES), where the microphysical details of clouds are simulated, down to a spatial scale of tens of metres, by solving the Navier Stokes equations (for applications specifically to solar irradiance, see Kurtz et al. 2017). Less complex are methods for generation of fractal cloud fields (for an overview of the most important studies see Lohmann et al. 2017), and even simpler are cloud fields made up of squares (Lave and Kleissl 2019) and circles (Arias-Castro et al. 2014). The idea behind all of these latter models is to generate spatial cloud fields that are moved over a set of PV systems to shade clear-sky irradiance, thereby generating realistic and spatio-temporally correlated time series at each system.

Finally, statistical models generate synthetic irradiance data using purely statistical methods, e.g. machine learning methods. These types of approaches are not yet very common for generation of spatio-temporal irradiance data, but are widely applied for solar forecasting (Meer et al. 2018). A statistical method for simulating instantaneous solar irradiance at arbitrary sets of dispersed sites has been proposed by Widén et al. (2017a, b), in which the irradiance at individual sites is sampled from probability distributions that are spatially correlated according to a correlation model, all of which are dependent only on the daily clear-sky index (degree of cloudiness). Full spatio-temporal statistical models of solar irradiance that allow generation of irradiance time series at multiple sites are yet to be developed.

This overview suggests that the preferred methods for obtaining reliable spatio-temporal solar irradiance data for building cluster simulations are either any of the data upscaling methods, which can be applied if at least irradiance or PV system data from one or a few sites in the cluster are available, or a semi-physical model, in which a synthetic cloud field is generated and moved over the cluster. Further research should also go into developing improved spatio-temporal statistical models for solar irradiance and PV systems.

2.1.2.1 Density of Buildings

Building density affects the energy planning at cluster scale, such as energy demand, energy transmission/distribution, distributed energy infrastructure, the quantity of RES technologies that can be installed and the degree of self-sufficiency. Different measures of building density are available in literature, such as plan area density (the ratio of built to total area (Macdonalda et al. 1998)), and frontal area density (the ratio of the windward-facing facade area to the area occupied by buildings (Cheung and Liu 2011)), as illustrated in Fig. 2.10. Existing studies at cluster level generally fall within the range of 0.11–0.69 plan area density, and within the range of 0.12–0.33 frontal area density. In extremely dense cities, like Hong Kong, the frontal area density frequently exceed 0.4 and can reach extreme heights, such as 1.07, in cases in which building blocks are attached to each other (Srebric et al. 2015). Existing literature suggest that higher building density leads to higher night-time urban air temperature, increasing therefore the urban heat island intensity. This in turn may increase cooling loads and decrease, often not significantly, the heating load of buildings (Li et al. 2018). For instance, Liu et al. (2015) utilizing CFD simulation found that when the plan area density increased from 0.04 (almost isolated building) to 0.44 (dense cities) the total energy use for cooling increased by more than twice the reduction in heating energy demand. The empirical study of Li et al. (2018) found a correlation between building density and household electricity consumption at a cluster level in summer months, but no correlation could be established for winter months. Furthermore, the study found that at higher building density, households in slab and tower apartments consume more electricity in the summer months, partly due to the increased heat island intensity.

However, there is disagreement in the literature regarding the effect of building density on building energy use and the magnitude of the effect. Some empirical studies found no significant increase in energy use at higher density (Ko and Radke 2014) (Kaza 2010). Ewing and Rong (2008) established three ways though which density can impact residential energy use: (1) energy losses through electric power transmission and distribution, (2) increased energy demand due to higher heat island intensity and (3) energy use variance owing to the size and type of the housing stocks. Li et al. (2018) postulated that differences between the numerical and empirical findings are owing to the fact that most simulation studies assess the building density-building energy use relationship on an annual basis, while this relationship might differ when simulated on a seasonal or on a higher-resolution basis. The authors also indicated that geographical and cultural contexts, such as demographic, socio-economic, behavioural and property-related characteristics, may also

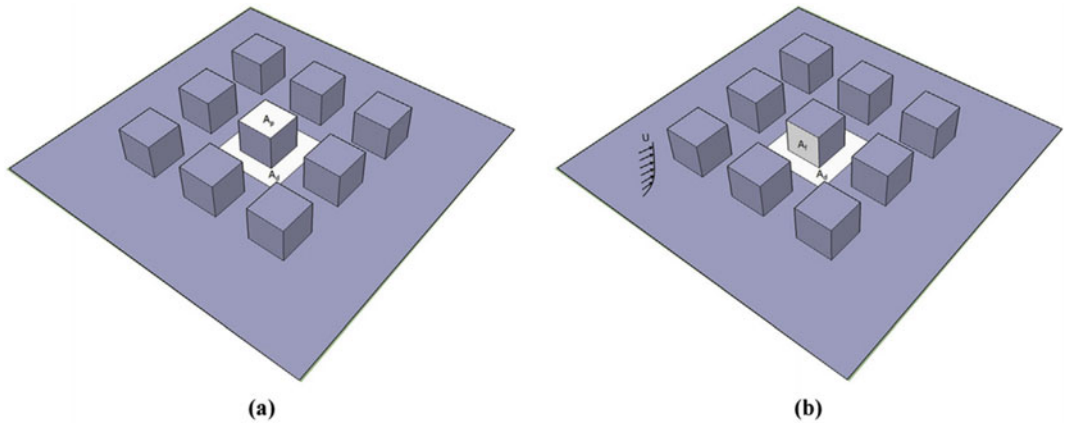

Fig. 2.10 Definition of **a** plan area density and **b** frontal area density for one building (Srebric et al. 2015)

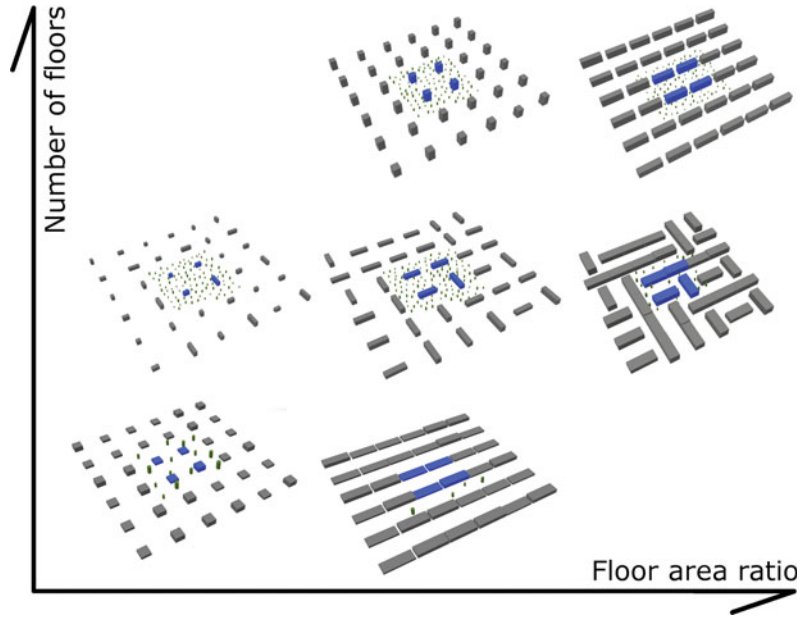

Fig. 2.11 Different types of cluster by number of floors and floor area ratio; an equal floor area ratio is achievable with different heights by varying the ground floor openness

influence the relation between density and energy use. Moreover, energy use relates to other planning and design factors even if the density is the same, such as buildings layout, street orientation, urban trees and building materials (Sosa et al. 2018).

Floor area ratio (FAR) is another important density parameter that influence RES at the cluster level. It is defined as the ratio of the gross floor area of all buildings to the total site area (Merriam 2004). Traditionally, FAR is obtained from site surveys from building shape and height data. However, this is a quite expensive and time-consuming approach. Light Detection and Ranging (LiDAR) is a novel, relatively quick and accurate method that besides the three-dimensional information of buildings also gathers topographic data (Priestnalla et al. 2000). The third method of obtaining building information data is from the remote-sensing images. However, good, high-resolution images may also be costly (Pan et al. 2008). FAR does not reflect the height or shape of the buildings, nor the open space between them (Alfirević 2016). The same FAR can be achieved with different building configurations, as illustrated in Fig. 2.11. Nonetheless, the three-dimensional characteristics of the built environment, as described by a variety of physical parameters, influence the availability of both direct solar radiation and daylight within the urban fabric (Strømann-Andersen and Sattrup 2011; Cheng et al. 2006). Hence, it affects both building energy use and RES power generations. Since at a given FAR, lower buildings have a higher relative roof surface and thus a lower relative façade area suitable for RES envelopes, information on the height of buildings is also of great importance. In contrast, taller buildings with the same FAR have greater distances between them, which allows for more direct solar radiation, hence for higher solar gains (Ayotunde and Ali 2017). In spatial planning, FAR between 1.5 and 2.5 has been identified as the optimal value for achieving high energy efficiency (LSE 2014). For instance, Yannas (1994) reported 40% heating energy savings in his comparative study of apartments and detached houses. The author concluded that a FAR of 2.5 might be the optimum density for neighbourhood development. Capuleto and Shaviv (2001) found that at 1.6–1.8 FAR it is possible to maintain solar access to all buildings within a neighbourhood. Ayotunde and Ali (2017) argued that a FAR of 1.0 is too low within the context of China, whose paper operates in the context of the major Chinese cities, argued that a

FAR of 1.0 is too low, while a FAR between 3.0 and 4.0 is too high for energy-use reductions at the cluster scale. FAR is a universal measure, which is also applied for other purposes, such as analysing urban spatial structure or proposing urban planning policies. For instance, Cao et al. (2016) applied FAR as a parameter for analysing the urban spatial structure of a diversified city in China. Joshi and Kono (2009) were able to optimize FAR regulation in a growing city as a practical alternative or supplement to the first-best policy against negative population externality. Barr and Cohen (2014) examined the FAR gradient in New York city over time and space, from the urban spatial structure point of view.

We hence argue that future studies should investigate the combined impacts of FAR on energy use, urban spatial structure, economic and environmental conditions at cluster scale. Since three-dimensional characteristics of the built environment affect the RES potential of a cluster (or even an entire city), it is necessary to identify an adequate building density measure that is capable to capture key characteristics. Based on the literature review and our understanding of the issue, we recommend that future studies adopt three relatively obtainable density measures: mean building height, plan area density and either the measure of façade area ratio or surface area ratio. The former is defined as ratio of all building facades over a given area, while the latter is the ratio of the total building envelope to site area. Table 2.1 summarizes the characteristics of density parameters in exiting literature from energy use point of view.

2.1.2.2 Energy Demand

Energy demand pattern at cluster level is crucial for planning RES harvesting envelopes because it is required to match capacity of energy infrastructures. It influences the stakeholders at various levels, from the development of regional strategies to the detailed design of buildings. Many models have already been developed to estimate the energy demand at the cluster level, categorized as 'top-down' and 'bottom-up' approaches, respectively (Frayssinet et al. 2018; Swan and Ugursal 2009). Top-down approaches, such as (Rey et al. 2013; Balaras et al. 2007), consider clusters as an entity by only describing the general characteristics of energy demand, rather than the explicit energy use profile of individual building. These approaches rely on statistical data and economic theory, correlating energy demand to macroeconomic parameters, such as energy price, income tax, GDP, greenhouse gas, population density and urban morphology. In contrast, bottom-up approaches detail the energy use profile of individual building/component using statistical/data-driven and engineering methods. Statistical (Robinson et al. 2017) and data-driven methods (Wei et al. 2018) relate the explicit energy demand and historical data depending on field historical measured data, utility metering, governmental statistics or surveys. Engineering methods for

Table 2.1 Summary of density parameters from energy aspect

Density parameters	Definition	Range	Relationship with energy	Potential impact factors
Plan area density	Plan (horizontal) area of buildings to site area	0.11–0.69	Directly proportional to energy use (modelling); No obvious impact (empiric)	Geographical and cultural contexts; Planning and design factors Urban spatial structure, policy, economic and environmental issues
Frontal area density	Windward building elevation to site area	0.12–0.33; 0.4–1.07 (high dense)		
Floor area ratio	Gross floor (accumulated horizontal) area of buildings to site area	1.5–2.5	Directly proportional to energy use; But need optimization for energy generation	

power load (Li and Wen 2017) and thermal load (Elci et al. 2018) calculate the explicit energy demand of each energy component of individual building, relying on the physical properties of buildings components and characteristics of systems. There are also many cases (Fonseca and Schlueter 2015) that combined statistical and engineering methods for estimation of energy demand.

Table 2.2 lists the examples of the main simulation models for energy demand at cluster level. It is observed that most existing models simply evaluate energy demand of buildings in an isolated manner, which don't include all major energy subsystems in one model, such as buildings, transports, electricity and heat networks. Energy demand is rarely evaluated in a comprehensive and systematic manner. Such narrow sectoral approaches would underestimate the energy demand for exclusive against shared energy resources and fail to identify the overall patterns of urban energy demand with respect to consumers. This would further result in the unreliable predictions and poor management decisions regarding the energy demand, which may lead to enormous waste in energy distribution and infrastructure investment. As a result, it is highly necessary to improve data collection and generate high-resolution spatio-temporal energy demand of both building and transportation activities or even industrial energy need. A few studies have included a broad range of energy demand, such as Fonseca and Schlueter (2015), Fichera et al. (2016).

2.1.2.3 Integrated Cluster-Scale Energy Systems

Most of these RES solutions have been extensively applied alone at building scale, while some researcher have yet started to explore a wider integrated application in energy generation and energy use reduction at cluster or district scale, as well as the corresponding influence on energy storage and grid distribution (Allegrini et al. 2015). Li and Wen (2017) proposed a net-zero building cluster emulator that can simulate energy behaviours of a cluster of buildings and their distributed energy devices as well as exchange operation data and control schemes with building systems; the emulator was developed for four simulation modules: building module (by EnergyPlus), ice tank module (EnergyPlus), PV-battery module (by TRNSYS) and operation module (by MATLAB and BCVTB); they demonstrated a proof-of-concept case to illustrate the possible ways for simulation of complex multi-energy systems at cluster level. The similar work was also conducted by Hachem et al. (2012) who applied EnergyPlus to assess energy demand of a cluster buildings and used TRNSYS to estimate electricity generation from BIPV. Protopapadaki and Saelens (2017) developed a model to assess the impact of heat pump and PV on residential low-voltage distribution grids as a function of building and district properties in a probabilistic way, though the combined approaches of Monte Carlo method, Modelica-based thermal-physical model and three-phase unbalanced loading of the grid network, as well as stochastic occupant behaviour model; they indicated that air-source heat pumps have a greater impact on the feeders than PV, in terms of loading and voltage magnitude, and building characteristics prove high correlations with the examined grid performance indicators. Figure 2.12 reveals the schematic of their modelling approach.

Hsieh et al. (2017) compared the solar thermal systems together with storage from building to a cluster scale of 11 buildings in Switzerland; all the relevant system components, including the buildings energy demand, solar thermal collectors, electrical heaters, storage tanks and district-heating network were modelled using EnergyPlus, the simulation results depict that the building-level long-term storage configurations perform best over all other system configurations, in terms of solar fraction and system efficiencies. The location of the thermal storage and the separation of short- and long-term storage are crucial that affect the performance of building-level renewable energy sources, and thus merit further investigation. Letellier-Duchesne et al. (2018) describe a simple three-step modelling workflow, illustrated in Fig. 2.13, to balance demand and supply, by integrating cluster-level

Table 2.2 Examples of simulation models for energy demand at cluster level

Types of models		Detailed methods	Cluster scale	Spatial/Temporal pattern	Data source	References
Top-down approaches		Statistical analysis	c.a. 1 km^2	Average annual spatio-temporal heat and electricity demand	Survey	Rey, et al. (2013)
Bottom-up approaches	Statistical model	K-means clustering	Two buildings	Temporal occupant-behaviour related electricity demand with 15 min interval	Field measurement	Pan, et al. (2017)
		Linear interactive and general optimizer (LINGO 15.0)	160 m as radius	Hourly temporal heat and electricity demand	Survey and research committee	Wu, et al. (2018)
		Artificial neutral network (ANN) heating degree day, cooling degree day	Seven buildings	Yearly temporal heating and cooling demand	Simulated data (DeST software)	Cheng-wen and Jian (2010)
		Support vector machine (SVM)	Seven buildings	Hourly temporal cooling demand	Field measurement	Li et al. (2009)
		Decision tree	80 buildings	Average temporal heat and electricity demand	Survey and research committee: energy use for residential buildings in Japan	Yu et al. (2010)
		Markov chain	200 detached houses and 200 apartments	Temporal electricity demand with 10 min interval	Data sets of TU-SCB-1996/TU/EL-SEA-2007/EL-SEA-2007; diaries	Widén and Wäckelgård (2010)
		Machine learning, including linear regressor, ridge regressor, support vector regressor, elastic net regressor, linear kernel support vector	Various scales up to 73,388 commercial buildings	Yearly temporal energy demand	Commercial Building Energy Consumption Survey; Augmented Local Law 84 data set (LL84)	Robinson, et al. (2017)

(continued)

Table 2.2 (continued)

Types of models		Detailed methods	Cluster scale	Spatial/Temporal pattern	Data source	References
		regressor, AdaBoost regressor, bagging regressor, gradient boosting regressor, random forest regressor, extra trees regressor, multi-layer perceptron regressor and k-nearest neighbour regressor				
	Engineering physical model	Simplified thermal models, and Energy Plus, TRNSYS	29 buildings	Hourly temporal heat and electricity demand	Simulation and measurement	Orehounig et al. (2015)
		Aggregation model in Modelica	35 buildings	Hourly temporal heat demand	German Meteorological Service	Elci et al. (2018)
		Simplified thermal model and Energy Plus	11 buildings	Monthly temporal heat and electricity demand with 15 min interval	Swiss standard SIA 380/1; SIA 2024; DHW demand profile for medium-load European households	Hsieh et al. (2017)
	Combined model	Statistical clustering and simplified engineering model (EN13790:2007/EN15316:2007), under GIS framework	172 building archetypes	Hourly Spatio-temporal heat and electricity demand	Weather database from software Meteonorm 7.0; urban GIS database from official database and open street maps; archetypes database, distributions database and measurement database from local collection	Fonseca and Schlueter (2015)
		UBEM, a mixed integer linear program, and thermal plant generation model under the GIS framework	0.5 km^2	Annual temporal heat and electricity demand	Measurement database from local collection	Letellier-Duchesne et al. (2018)

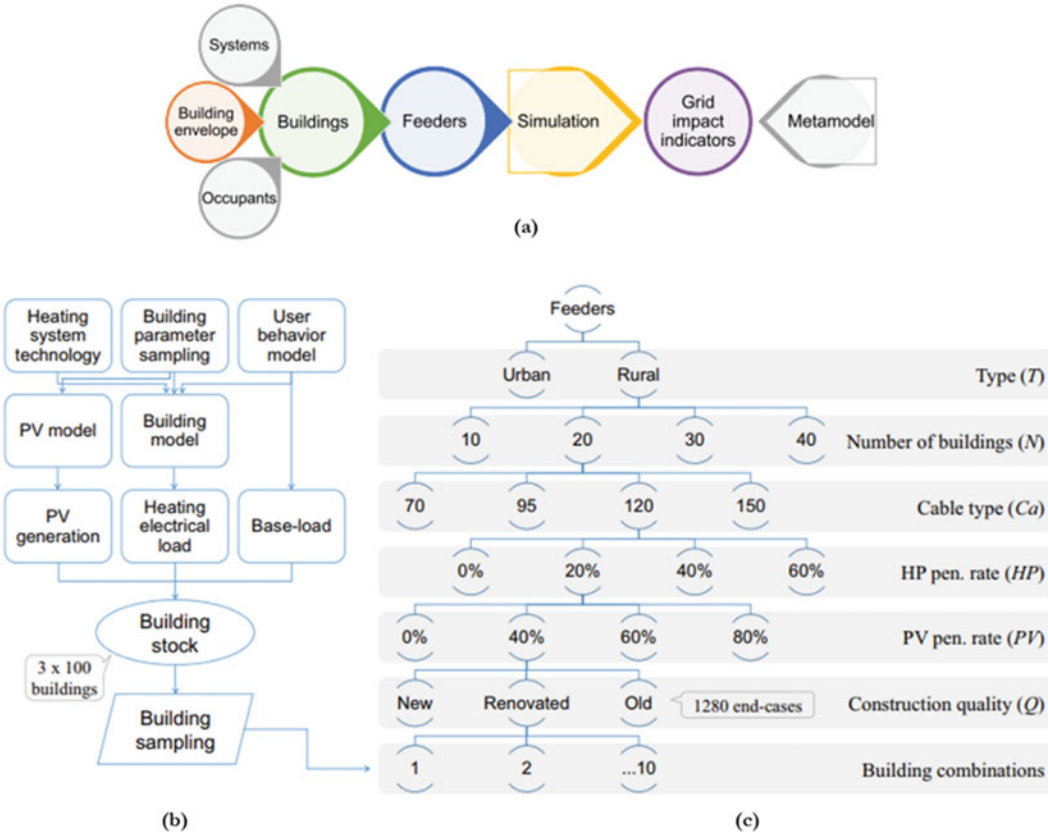

Fig. 2.12 **a** Schematic approach for integrated modelling of heat pump and PV in building cluster; **b** building cluster model; **c** feeder scenario definition (Protopapadaki and Saelens 2017)

building load calculations with detailed district energy network analysis models. In their study, they considered a comprehensive heat plants, including solar thermal collectors, heat pump, combined heat and power (CHP), natural gas boilers, heat network and hot water storage. Their model was depending on a Rhinoceros-based plugin (based on Radiance and Energy-Plus) and TRNSYS that targets a network topology optimization, a heat cogeneration scenario and economic analysis. They foresee this methodology demonstrate a new way of designing for future 4th generation district energy systems in accordance with the concept of RES solutions.

Pinto and da Graça (2018) present a study of energy refurbishment measures and a direct geothermal powered district heating system for a cluster of existing residential buildings in Groningen, Netherlands; in the study, they considered the retrofit measures including the improved envelope thermal insulation (walls, roof and windows), the reduced infiltration heat losses and the upgraded boiler. The study uses detailed thermal simulation models in Energy-Plus that rely on accurate building typologies and thermal characteristics, outdoor air infiltration data and occupant behaviour profiles. The predicted energy savings and costs show that both the geothermal and the energy refurbishment approaches are economically viable and result in large reductions in the environmental impact of space heating. Applying all refurbishment measures results in an 86% reduction in yearly gas consumption for heating with an investment payback time of fifteen years.

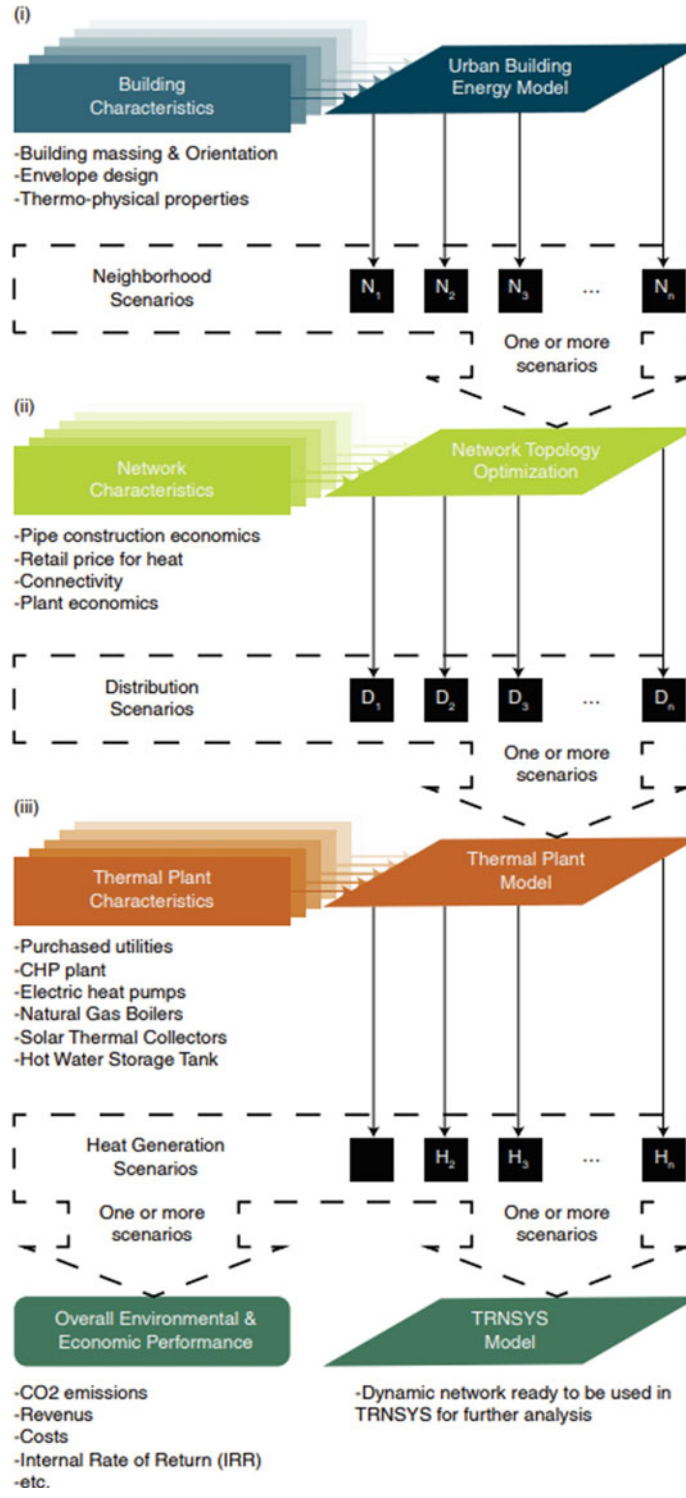

Fig. 2.13 Flow chart showing the three steps of the methodology: **i** An UBEM model is defined, **ii** a suited network topology is determined, **iii** a thermal plant scheme is analysed. Performance metrics of the cluster and a detailed TRNSYS model serve as outputs of the workflow (Letellier-Duchesne et al. 2018)

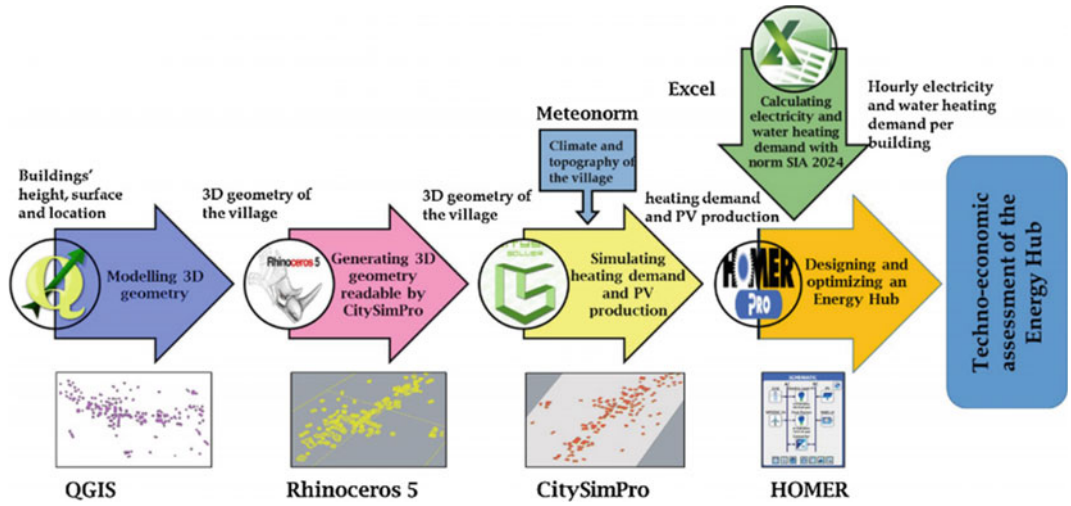

Fig. 2.14 Overview of the approach for assessing building retrofit and energy system improvements (Le Guen et al. 2018)

Le Guen et al. (2018) simultaneously optimized the procedures of the integration of renewable energy technologies and building retrofit at a cluster scale in Hemberg, Switzerland; they developed a computational platform, displayed in Fig. 2.14, by combining software CitySim, HOMER Pro, QGIS and Rhinoceros. The study began with collecting basic information for the buildings using QGIS which is an open-source geographic information system (GIS). The 3D geometries of the buildings in the village are modelled using Rhinoceros, based on the information from QGIS. This is done to prepare the DXF data files as input for CitySim Pro, a building/urban energy simulation tool (citysim.epfl.ch). CitySim Pro is then used to simulate the energy flow of the building stock in the village, including physical properties of the buildings, infiltration rate, occupancy profile, outdoor materials etc. CitySim considers the interaction among buildings, i.e. mutual shadings, and the outdoor radiative environment. HOMER is then used to analyse the renewable energy integration and the energy system improvements. Renewable energy potential, demand for multiple energy services, technical details for energy conversion measures (e.g. insulation of roof, floor walls and windows), market prices of system components, etc., are the input data. The energy demand of buildings (heating and cooling), as well as the electricity produced by renewable energy sources (e.g. BIPV, heat pumps) are the inputs for HOMER. The results show that retrofitting of all buildings after retrofit reduces the space heating demand by 70–85% and reduces the fluctuations in energy demand, thereby allowing the integration of more renewable energy. According to the simulations, BIPV panels have potential to cover the total annual energy demand of the village. However, the energy system assessment shows that it is difficult to reach beyond 60% when integrating non-dispatchable renewable energy.

Chen et al. (2017) developed a City Building Energy Saver (CityBES) platform, in order to simulate urban building energy system during large-scale building retrofitting, using EnergyPlus based on cities' building data sets and user-selected energy conservation measures, such as energy-efficient windows. CityBES is a bottom-up physics-based detailed energy modelling of every individual building retrofit in a city or district/cluster. There are three layers in the platform: the data layer, the simulation engine (algorithms) and software tools layer and the use-cases layer. It also provides a 3D visualization with GIS including colour-coded simulated site energy use intensity (EUI), which facilitates the

energy planning of different stakeholders at the early stage.

Wu et al. (2017) conducted a multi-objective optimization of energy systems and building envelope retrofit in a residential community. In their work, building energy systems and envelope retrofit are optimized simultaneously in a bottom-up approach. Dynamic building energy demand is simulated in EnergyPlus, combined with a mixed-integer linear program (MILP) optimization to select retrofit strategies and size and simulate the operation of different types of energy systems. Interactions between retrofit and building systems, such as the need to replace the windows, insulations, heat distribution system for low-temperature heating technologies at low retrofit levels, are taken into account. The life cycle GHG approach includes embodied GHG emissions in retrofit materials and differentiates between PV and grid electricity impacts for all electric conversion systems, including heat pumps. Promising retrofit and energy system strategies are explored by scaling typical building strategies to the cluster level. The proposed method can be divided into four steps.

Wu et al. (2018) present a non-linear model for the optimization of a neighbourhood-scale distributed energy system considering both supply and demand sides. They developed the specific mathematical model by considering four modules, namely energy demand simulation, energy supply characterizing and dispatch, constraint analysis as well as optimization objectives of primary energy saving ratio, energy cost and CO_2 emissions. The optimization is based on the commercially available solver linear interactive and general optimizer (LINGO 15.0), which aims at solving the following issues: how will the system combination and building mix be best suited to each other from the energy saving viewpoint; and when land use cannot be changed in an existing district, what will be the best system combination and the optimal heating/cooling transmission network for the building mix.

Table 2.3 presents an example of the existing modelling approaches for RES solutions and their complex energy systems in cluster level. From these studies, we observe that at cluster level (1) most studies reply on the existing bottom-up engineering-physical simulation tool/approaches (i.e. EnergyPlus, TRNSYS, Modelica) for estimation of energy demand, owing to in the limitation in obtaining reliable energy load profile and the complexity in prediction required by high-capacity computation; (2) most of studies focus on single objective, i.e. energy saving, while a few of them start to propose multi-objectives functions, such as energy use, economic and environmental indexes, in which optimization algorithms are necessary; (3) most studies only assess part of the energy systems, such as PV and battery, solar heat and thermal storage, RES supply and grid. An integrated evaluation of all the four layers of energy systems in a cluster level, i.e. supply, demand, storage and distribution, is therefore required. A solution for this, 'energy hub', has been proposed by several researchers, which will be discussed more in the following section. In addition, upon literature searching, we have not yet found any research addressing the overall spatio-temporal energy system in a cluster that including both building and transportation, which are usually investigated in a separate way (Rey et al. 2013; Samsatli and Samsatli 2015). These restrict a comprehensive characterization of energy systems in cluster as a whole. According to Sect. 2.5, there are a few studies that start to consider spatio-temporal energy demand (Rey et al. 2013; Fonseca and Schlueter 2015). Thus, one of future challenge in cluster level will be the integrated assessment of spatio-temporal energy systems. An energy hub concept could be the breakthrough point to this challenge.

2.2 Energy Hub

2.2.1 General Concept

Energy hub is considered here as an effective means to closely integrate multi-energy systems of different energy carriers through RES/DES convertors, energy distribution and storing components in an optimal manner for various energy use within building clusters (Howell et al. 2017).

Table 2.3 Examples of the existing modelling approaches for RES solutions and their complex energy system in cluster level

Modelling approaches	Solar power solutions	Solar thermal solutions	Energy-efficient solutions	Cluster-level energy systems
EnergyPlus + TRNSYS + MATLAB BCVTB Connection (Li and Wen 2017)	√	Possible	Possible	√
EnergyPlus + TRNSYS (Hachema et al. 2012)	√	Possible	Possible	√
Mont Carlo method + Modelica-based thermal-physical and gird models + stochastic occupant behaviour model (Protopapadaki and Saelens 2017)	√	Possible	Possible	√
EnergyPlus only (Hsieh et al. 2017)	Possible	√	Possible	√
Rhinoceros-based plugin (based on Radiance and EnergyPlus) + TRNSYS (Letellier-Duchesne et al. 2018)	Possible	√	Possible	√
EnergyPlus + measured data + statistical occupant behaviour data (Pinto and da Graça 2018)	Possible	Possible	√	√
A computational platform combining software CitySim, HOMER, QGIS and Rhinoceros (Le Guen et al. 2018)	√	√	√	√
CityBES platform based on EnergyPlus (Inman et al. 2016)	-	-	√	√
EnergyPlus + MILP with optimization (Wu et al. 2017)	√	√	√	√
Mathematical model + linear interactive and general optimizer (LINGO 15.0) (Wu et al. 2018)	–	–	√	√

Note '√' means that there are existing examples in the literature; 'possible' means it is possible to use the dedicated models in the respective field, even there is no existing example in the literature

The energy hub could also become the 'filling station' for individual or collective autonomous, shared electric or biogas vehicles; within a cluster, the vehicle could be better used and play several roles: mobility, energy transport, office, even living room (Quénard 2017). Energy hub is a node in overall urban energy system with multiple input and output energy vectors and typically consist of a more elaborate and complex internal arrangement of components, as shown in Fig. 2.15. The benefits of this close integration are identified as increased reliability, load flexibility and efficiency gains through synergistic effects (van Dam et al. 2008), which suits well in building cluster. Energy hub is also regarded as a practical way to offer more services by sharing and interconnecting household devices so as to reduce the carbon impact of new systems (Rey et al. 2013). Thus, energy hub is not a single entity containing all necessary systems for transformation, conversion and storing of energy, but an amalgam of individual energy consumers and producers distributed over an area. This allows to take into account variable loads, systems and energy sources of multiple buildings in diverse alternative paths (Kaza 2010).

2.2.2 Modelling and Optimization

The modelling concept of an energy hub describes the interactive relation between input and output energy flows, which can be applied to optimize the energy use and local generation during planning and operation. Existing efforts towards the optimal management of energy hubs have been observed in several studies. Geidl

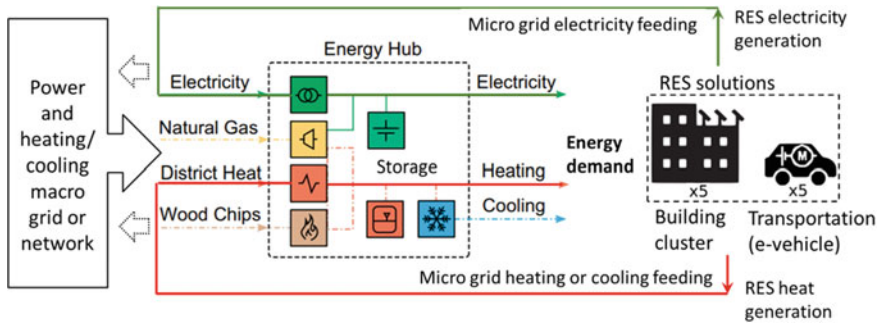

Fig. 2.15 Example of energy hub concept at building cluster level (by modifying figure in Howell et al. 2017)

et al. (2007) firstly proposed energy hub concept in 2007 by illustrating essential components and main functions; they envisioned energy hub will be a key element for future complex urban energy network. Orehounig et al. (2015) integrated the decentralized energy systems based on the energy hub concept in cluster of 29 buildings, including decentralized and local energy technologies such as PV, biomass or small hydro power, together with district heating systems, building and district conversion and storage technologies; as a result, RES generation, energy supply systems and local energy storage systems can be evaluated in a combined way. The mathematical model of an energy hub is combined with optimization techniques and balances energy supply and demand in the system boundaries with different design objectives, such as energy use, life cycle CO_2 emission and cost. The method requires a three-step approach, shown in Fig. 2.16: (1) estimation of demand, (2) estimation of renewable potential, (3) matching of demand and supply. The energy balance between inputs and outputs within certain constrains is the key principle, defined by Eq. (2.1).

$$\begin{bmatrix} L_\alpha \\ L_\beta \\ \vdots \\ L_\omega \end{bmatrix} = \begin{bmatrix} C_{\alpha\alpha} & C_{\beta\alpha} & \cdots & C_{\omega\alpha} \\ C_{\alpha\beta} & C_{\beta\beta} & \cdots & C_{\omega\beta} \\ \vdots & \vdots & \ddots & \vdots \\ C_{\alpha\omega} & C_{\beta\omega} & \cdots & C_{\omega\omega} \end{bmatrix} \begin{bmatrix} P_\alpha \\ P_\beta \\ \vdots \\ P_\omega \end{bmatrix} \tag{2.1}$$

In this equation, $[L_\alpha, L_\alpha, \ldots, L\omega]^T$ denotes the hub-output vector, $[P_\alpha, P_\beta, \ldots, P_\omega]^T$ the hub-input vector and the C terms make up the

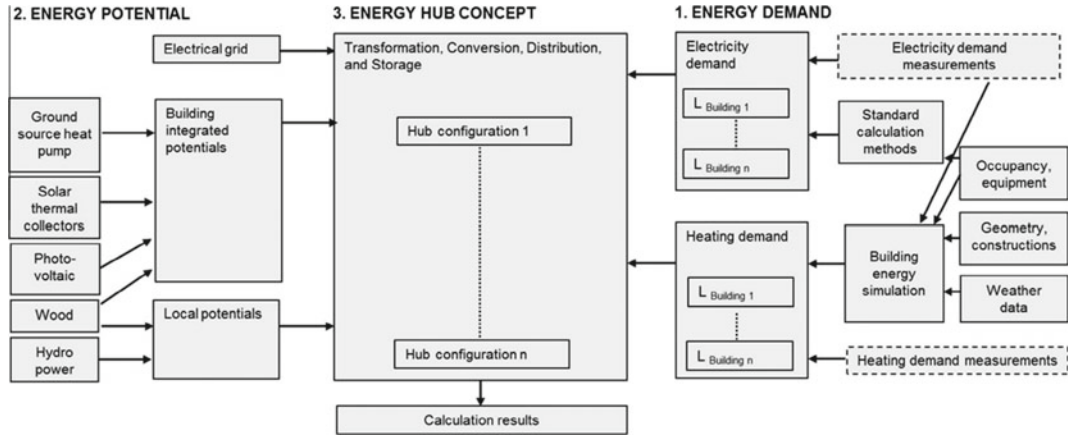

Fig. 2.16 Simulation approach for an energy hub with RES envelope solutions at cluster level (Orehounig et al. 2015)

converter coupling matrix, where α, β, …, ω are the different energy carriers and T is the time. The models of different energy carriers were then developed by Orehounig et al. (2015) using bottom-up approaches, respectively, and resolved/optimized them together using the Eq. (2.1) for defined objectives. They simplified the optimization problem as a linear programming problem and the optimization was carried by optimisation toolbox in MATLAB. Similar work has been done based on the energy hub concept in cluster, by integrating RES solutions, energy systems and building envelope retrofit, through engineering-physical simulation tool/approaches, operation/control strategies and dedicated optimisation solvers, such as CitySim/HOMER (Le Guen et al. 2018), MILP framework (Wu et al. 2017), mixed integer nonlinear programming (MINLP) framework (Setlhaolo et al. 2017) and linear coupling matrix (Wang, et al. 2018) etc.

Kuang et al. (2017) proposed a collaborative decision model to cooperatively operate building and electric vehicles (EV), based on a similar energy hub concept displayed in Fig. 2.17, which consists of thermal and electric storage system, combine cooling, heating and power system, PV panel and a EV charging station. A bi-objective MILP problem was then formulated to study the energy exchange between the building and the EV charging station, in order to minimize the operational cost for the building and the EV charging station simultaneously. They employed a weighted sum approach to solve the multi-objective MILP to obtain Pareto operation decisions for trade-off analysis between the building and the charging station.

Financial and environmental benefits of energy hub have also been investigated (Guler et al. 2018; Mohammadi et al. 2018). For instance, Moghaddam et al. (2016) set up the optimization objective as total profit made by energy hub to supply cooling, heating and electricity to building, indicated in Fig. 2.18; they implemented the MINLP model in GAMS optimization software and solved using the 'DICOPT solver for MINLP problems. Similarly, Taşcıkaraoğlu (2018) considered the objective of the optimization problem in a cluster-level energy hub from the perspective of the household owners' benefit, by minimizing the total cluster energy cost based on a net-metering approach. Davatgaran et al. (2018) developed a MILP model to maximize the profit of an energy hub in

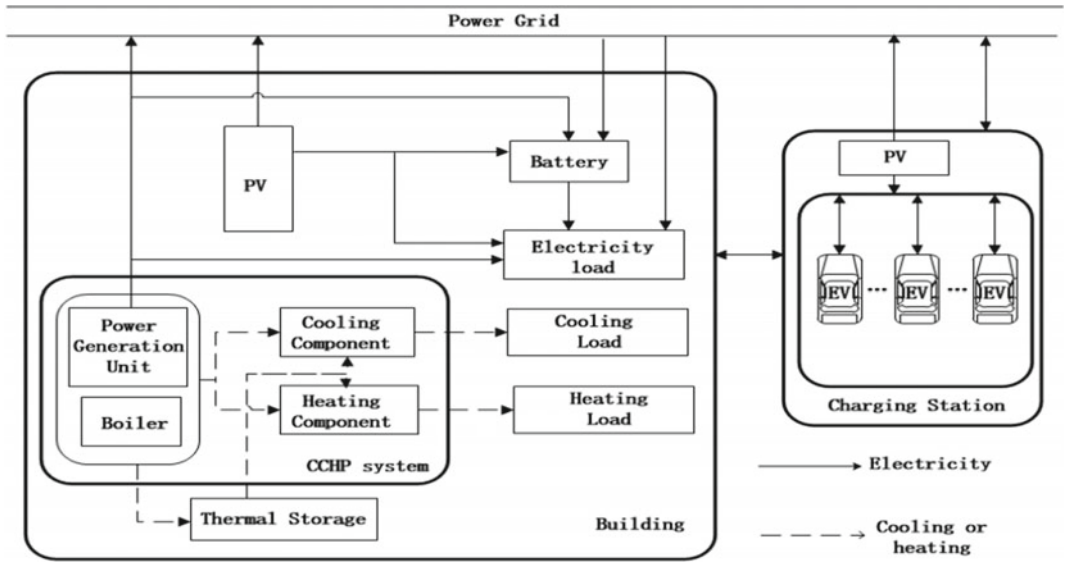

Fig. 2.17 Example of energy hub consisting of EVs (Kuang et al. 2017)

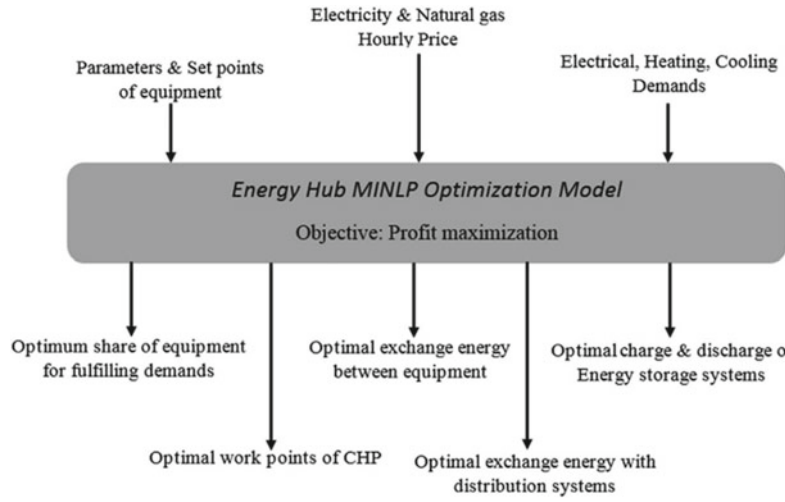

Fig. 2.18 Optimization of profit of an energy (Moghaddam et al. 2016)

day-ahead energy market, including electricity selling/buying and the operational cost, using model predictive control (MPC). Roldan-Blay et al. (2017) proposed a new distributed energy resource optimisation algorithm (DEROP) for energy hub to minimize energy costs by maximizing RESs generation and optimizing the management of energy storage system by non-linear functions; the DEROP algorithm, connecting SQL databases with real-time data, was executed by VBA code, and Microsoft Excel Worksheets were applied to show graphical results.

The optimization problem is usually set up to minimize the total energy cost in the system, within a deterministic framework of load demands, prices, efficiencies and constraints (Parisio et al. 2012). However, above studies mostly used steady-state parameters as the performance characteristic of energy components in energy hub. Off-design condition performance and non-steady-state condition performance have seldom been considered. But optimization of an energy hub with multi-energy systems and multi-energy carriers are complicated in practice, which has a considerable number of variables that makes a non-linear, non-convex, non-smooth and high-dimension optimization problem and the optimal solution cannot be achieved by conventional numerical techniques. Therefore, evolutionary algorithms are proposed, such as fuzzy decision making and teaching–learning-based optimization algorithm (Shabanpour-Haghighi and Seif 2015; Shabanpour-Haghighi et al. 2014), multi-agents system (MAS) (see Fig. 2.19) (Skarvelis-Kazakos et al. 2016), self-adoptive learning with time varying acceleration coefficient-gravitational search algorithm (SAL-TVAC-GSA) (Beigvand et al. 2017), robust optimization (Parisio et al. 2012) and memetic algorithm (Hu et al. 2012).

The energy hub concept is fairly new; it represents an interesting avenue for managing the complexity of multi-energy systems at the cluster level. Studies that give attention to this with a futuristic view on multi-carrier energy systems and achieving energy matching are still lacking. Figure 2.20 lists the modelling process of an energy hub, and Table 2.4 summarizes the examples of modelling, control and optimization of an energy hub. It is observed that most studies simplified the energy models of components within the energy hub and formed non-liner functions under dedicated control strategies (i.e. non-linear control, MPC, scheduling, optimal control, fuzzy logic control, multi-agent control, etc.) for single or multi-objectives, i.e. energy, cost and carbon emissions. MILP is found as the most common way to define steady-state energy hub operation, which can be solved, for instance, by the optimization toolbox in MATLAB. While for dynamic energy flows in an energy hub,

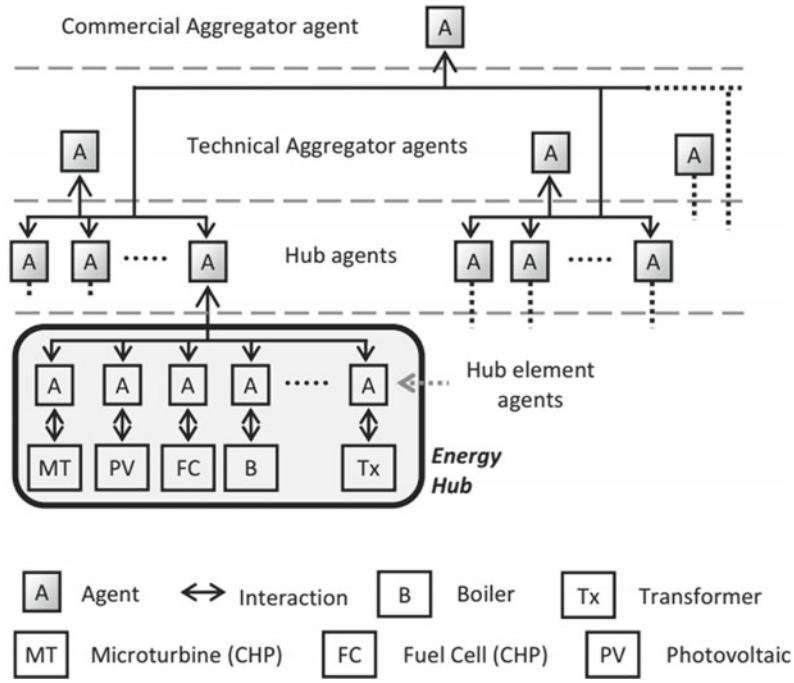

Fig. 2.19 Intelligent agent control architecture for energy hubs (Skarvelis-Kazakos et al. 2016)

advanced optimization algorithms have been proposed for operation and control with complex interactions among components, such as multi-agent systems. Very few studies (Kuang, et al. 2017; Hafez and Bhattacharya 2017) integrated EVs as part of the energy flow within an energy hub. Energy demand estimation in existing studies was unfortunately over simplified, such as ignore of aggregated demand (Beraldi et al. 2018). Future integration of detailed energy demand models described in Sects. 2.4 and 2.5 is strongly expected. In addition, future studies of energy hub at building cluster level, as planed in Fig. 2.21, must consider beyond existing energy systems/carriers, such as different architypes of buildings, RES envelope solutions, EV spatial demand and circular economic, etc., maximizing the synergies of all these components.

2.3 Discussion

This chapter work concentrates on building cluster modelling technique. It explains the importance of such method in current urban energy system and characterizes the corresponding features by addressing the main influencing factors. The factors defined as important for building cluster characterization could be grouped in three main areas of interest: grid, RES production and building. Among these, we recognize the main interesting factors as mostly influencing the cluster energy performance, including RES envelope solutions, solar energy potentials, density of building, energy demand, integrated cluster-scale systems and energy hub. Figure 2.22 highlights the most important findings of this chapter by a knowledge based matrix, and they are elaborated as below.

Based on the chapter work, the building cluster modelling is regarded as one of the most important approaches to assess contemporary transit of urban energy system. At this level, a group of buildings can systemically exchange the energy information either in a synergic or a disruptive way, where the existing modelling techniques and capacity are sufficient. Thus, simulations at this level can be applied to evaluate the detailed interaction between buildings and energy infrastructures, as well as the impact

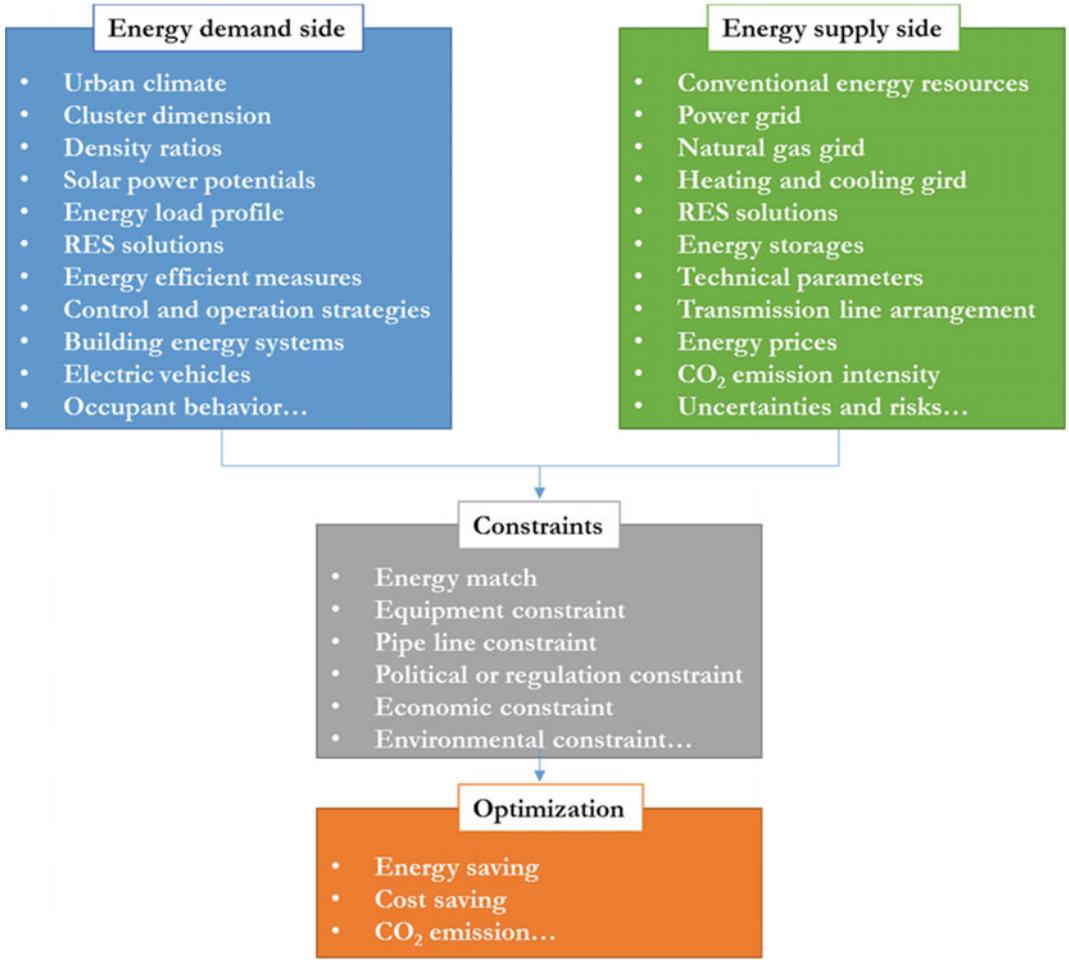

Fig. 2.20 Typical modelling process for energy hub at building scale

of adoption of RES envelope solutions. The building cluster modelling allows a potential shifting of building from a single energy efficient unit to an interconnected prosumer, therefore maximizing the synergies among RES application in buildings and energy systems. This will further reduce the operation cost of RES solutions and result in a wider application.

The spatio-temporal dimension is recommended for energy simulation at cluster scale in the next a few years. Owing to the complexity of energy systems and the limitation of computations, the optimal spatial dimension may range between 100 m and 1 km, while time scale should be reduced down towards minutes or even seconds level if both electricity and heat networks are integrated in the same model. Density of building blocks is generally in the range of: 0.11–0.69 (plan area density), 0.12–0.33 (frontal area density), 1.5–2.5 (floor area ratio). In general, higher density of building results in greater energy use, but it also depends on specific planning and design, urban spatial structure, policy, economic and environmental issues, geographical and cultural contexts, such as demographic, socio-economic, behavioural and property-related characteristics.

Solar energy is the most important available renewable resource at the building cluster level, especially in EU building retrofit context, which

Table 2.4 Examples of modelling and optimization of energy hub

Main input	Essential models	Control and optimization method/tool	Objectives	References
Building geometry Building details Efficiencies Weather conditions	Energy demand model PV generation model Wind power model	A design platform consisting of several existing (commercial and open source) tools, such as QGIS, Rhinoceros, CitySim HOMER	Energy Cost	Le Guen et al. (2018)
Occupancy Equipment Building details Weather data Electricity demand measurement Efficiencies Prices Carbon generation	Conversion model, i.e. heat pump, boiler, CHP, PV, wood District heating network model Battery and thermal storage Energy demand models Energy potential model	Model predictive control: mixed integer linear program (MILP), such as optimisation toolbox in MATLAB, McCormick relaxation	Energy Cost CO_2 emissions	Orehounig et al. (2015), Dai and Charkhgard (2018)
Prices Share ratio Demands	CHP generation system Electric heat pump Absorption chiller Electrical energy storage Thermal energy storage Natural gas	Non-linear operational scheduling: mixed integer non-linear program (MINLP) optimized in GAMS software using 'DICOPT solver	Profit (cost)	Moghaddam et al. (2016)
Electrical line data Natural gas and heat pipelines CHPs, boilers, pumps, and load levels	Natural gas sub-network CHPs and boilers Electric power plant	Fuzzy logic control and teaching–learning based optimization algorithm (based self-adaptive mutation wavelet) using IEEE 30-Bus and 57-Bus systems	Cost CO_2 emissions	Shabanpour-Haghighi and Seif (2015), Shabanpour-Haghighi et al. (2014)
Efficiencies Types of energy Device ratings Carbon generation	Hub element agents: micro-generators, electric vehicles, energy storage devices, boilers, controllable loads, converters, reformers Hub agent Technical aggregator agent Commercial aggregator agent	Multi-agent systems (MAS): multi-agent control with optimization algorithm of agent-based objective of maximization of social welfare on a Java-based platform—Java agent development framework (JADE)	Energy Cost CO_2 emissions	Skarvelis-Kazakos et al. (2016)
Efficiencies Prices Operation scheduling	Converter models: hydrogen plant, CHP, furnace Storage models: hydrogen storage, heat storage	Optimal control: mixed integer linear program (MILP) + Robust optimization in operation scheduling problem using CPLEX 12.0	Cost Energy demand	Parisio et al. (2012)

(continued)

Table 2.4 (continued)

Main input	Essential models	Control and optimization method/tool	Objectives	References
Efficiencies Prices	Transformer model CHP model Combined heat, cool and power (CHCP) model Gas furnace model Heater exchanger model Compressor air storage model	Non-linear control and Self-Adoptive Learning with Time Varying Acceleration Coefficient-Gravitational Search Algorithm (SAL-TVAC-GSA) in MATLAB	Energy Cost	Beigvand et al. (2017)
Temperature Solar radiation Price	Energy demand model Chiller model Ice storage model Battery model PV generation model	Pareto optimal control and Memetic algorithm in MATLAB	Cost	Hu et al. (2012)

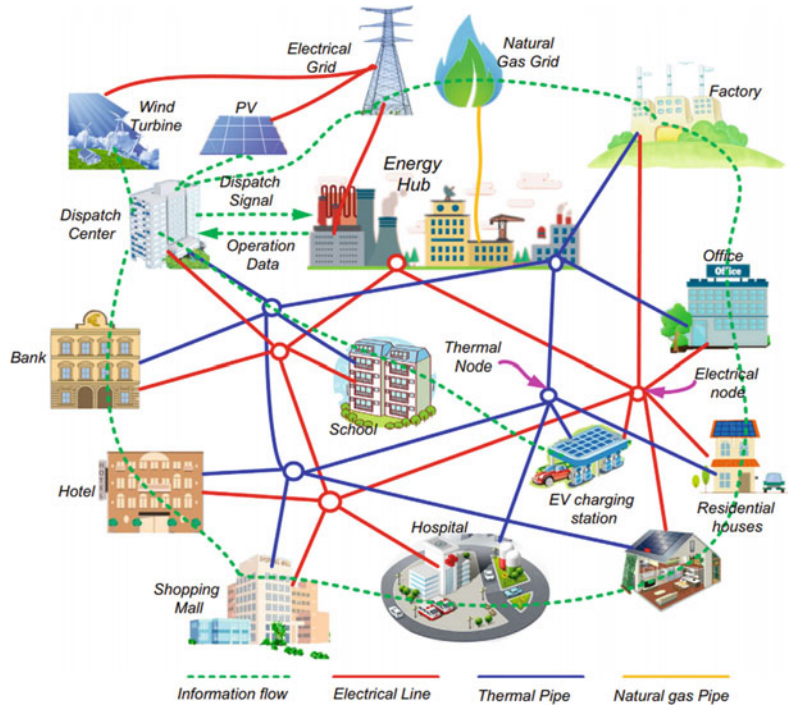

Fig. 2.21 Example of future energy hub at building cluster level (Ma et al. 2017)

is directly affected by the density of buildings in the city. Most of the RES envelope solutions are derived from solar and air resources, such as BIPV, BIPV/T, STF, heat pump, heat recovery, DSF, insulation, PCM, green roof/wall, energy frame, algae bioreactor façade and adaptive facades. The preferred methods available for obtaining reliable spatio-temporal solar irradiance data in building cluster simulations are either any of the data upscaling methods, which can be applied if at least irradiance or PV system data from one or a few sites in the cluster are available, or a semi-physical model, in which a synthetic cloud field is generated and moved over the cluster.

The modelling techniques for energy demand of building clusters can be categorized as top-down (statistical methods) and bottom-up

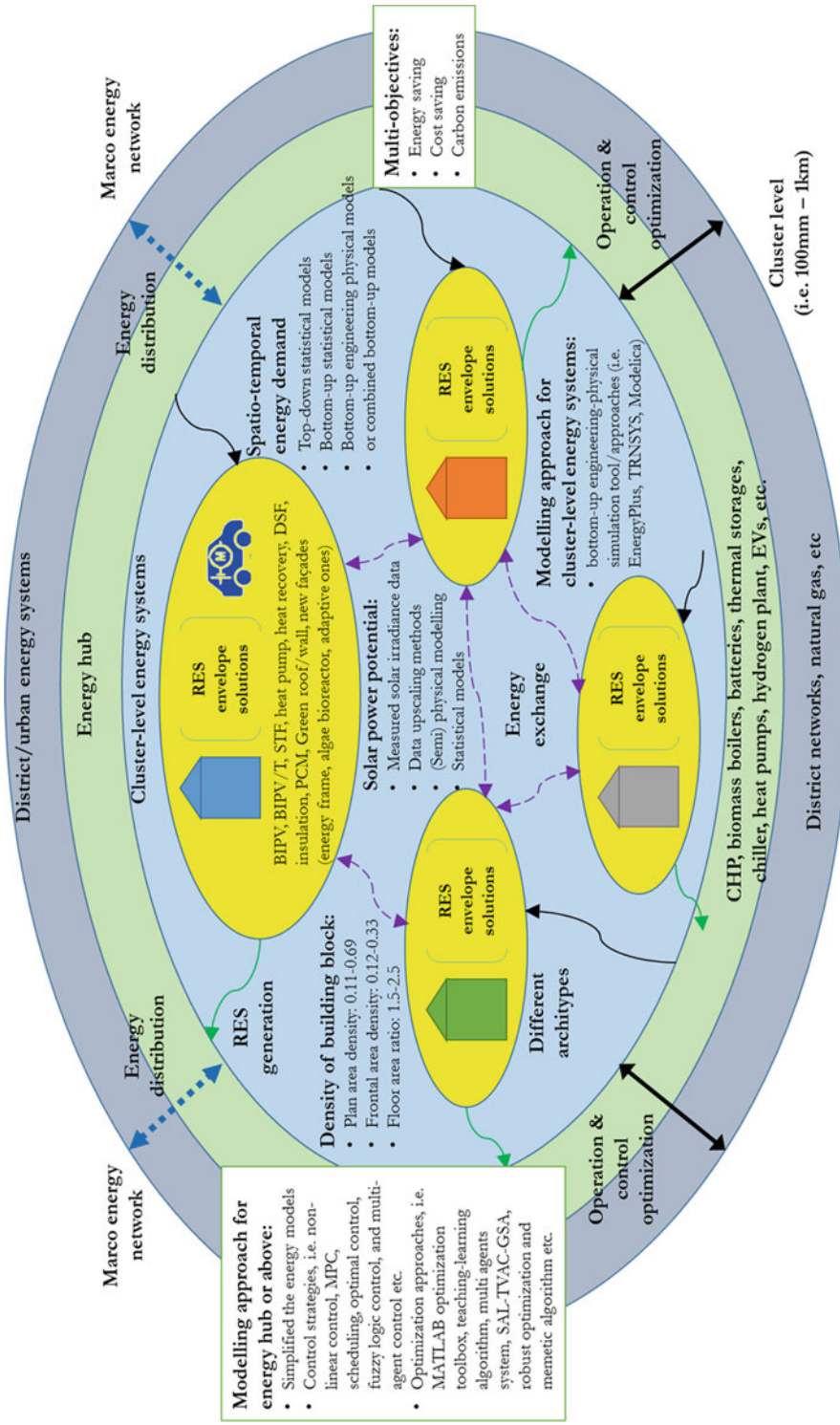

Fig. 2.22 RES envelope solutions-based energy systems modelling at building cluster level

approaches (statistical/data-driven methods, engineering/physical methods or combined methods). Most of them simply evaluate energy demand of buildings in an isolated manner, without comprehensive consideration in transports and synergies of energy exchange among architype/EVs. When coming to the complex energy system or energy hub level, commercialized simulation tool (i.e. EnergyPlus, TRNSYS, Modelica) or simplified model is the most common way for estimation of building energy demand, where most studies focus on (1) control/operational strategies, i.e. non-linear control, MPC, scheduling, optimal control, fuzzy logic control, multi-agent control, etc., and (2) optimization approaches, i.e. MATLAB optimization toolbox, teaching–learning algorithm, multi-agents system, SAL-TVAC-GSA, robust optimization and memetic algorithm. These studies intent to carry out the integrated investigation of various energy resources and energy carriers within cluster, such as RES envelope solutions, CHP, biomass boilers, batteries, thermal storages, chiller, heat pumps, hydrogen plant, EVs, district networks, and natural gas. The most common objectives for evaluation of these integrated energy systems are reduction of energy use, carbon emissions and costs.

2.4 Future Work

RES share is increasing rapidly when urban energy systems incorporate multiple energy sources. The successful integration of multiple RES envelope solutions relies not only on the energy performance of individual buildings (especially retrofitted ones), but also on optimal technologies for conversion, storage and distribution. The modern urban energy systems consist of different levels of complexity. At the moment, it is difficult to conduct a comprehensive assessment of energy efficiency, renewable energy integration and energy system improvements for the entire city. Build cluster approach is therefore regarded as one of the breakthroughs. Figure 2.23 depicts the basic steps for such approach based on the work in this chapter. A physical cluster scale must be initially determined so that the density of buildings can be estimated for assessing RES potentials. By doing so, appropriate RES envelope solutions can be finalized according to their energy generation potentials, which is further linked to energy demand, cluster-level complex energy systems/energy hub and district/urban energy systems. However, available studies are generally limited in their scope as they do not consider all the steps

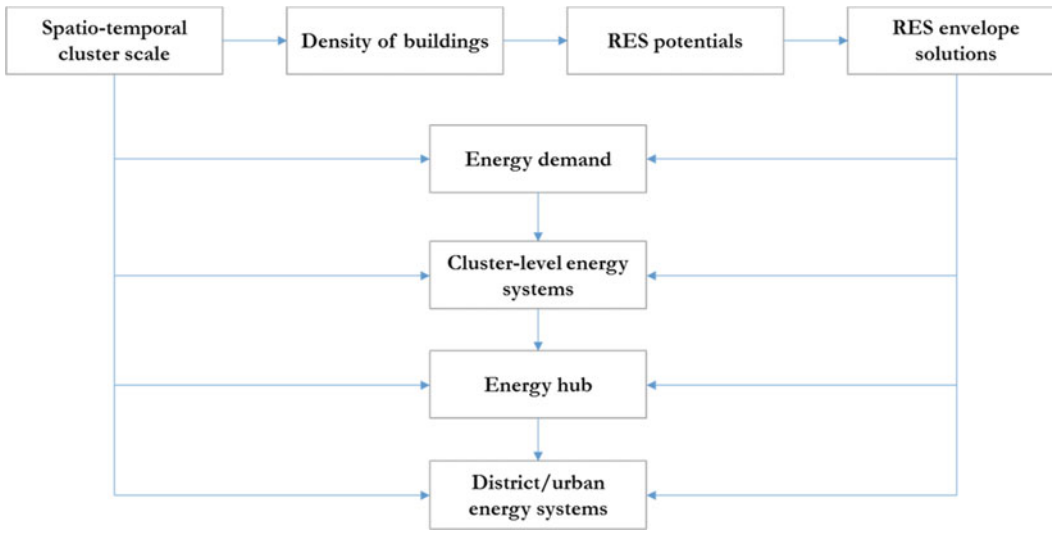

Fig. 2.23 Flow chart of RES envelope solutions based building cluster approach

and components together. In the following paragraph, we put forth a few suggestions for incorporating RES envelope solutions, in order to decrease energy cost and reduce the carbon footprint of the neighbourhood.

The combined method of facade area ratio and full spatio-temporal statistical model for solar irradiation would be effective way to estimate the potential power generation of RES envelope, such as BIPV. A surface area ratio is more straightforward to estimate the total available façade areas at building cluster level, which, in return, results in a more accurate solar mapping of RES envelopes. The full spatio-temporal statistical model, such as machine learning, allows the generation of solar irradiance time series at multiple site scale, as a cost-effective and reliable way suiting for building cluster.

High-resolution of energy flow profile: (1) owing to the difference in time required for building components/envelopes and energy system/flow, it is suggested that the time steps required to solve the energy matching should be reduced down towards minutes or even seconds level; (2) the basic model is often used for the optimal energy trade off among group of buildings or from an energy hub for multi-energy carriers, in which the energy flows through the hub are optimized for a single specific period with peak energy demand or annual energy use; in future, more advanced problems must be defined in terms of transient time series of energy flow capacities in cluster-level energy systems or energy hub.

Spatio-temporal energy demand: most existing models simply evaluate energy demand of buildings in an isolated manner, without comprehensive consideration in transports and synergies of energy exchange among architypes/EVs; there is a need to include building function/location, urban inhabitant behaviour in terms of activities and mobility in the models.

Detailed engineering/physical and statistical models in complex energy systems or energy hub using new optimization algorithms: (1) most studies reply on the existing commercialized simulation tool or simplified model for estimation of energy demand in complex energy systems and energy hub, where energy demand estimation is unfortunately over simplified and it needs more detailed models, such as the combined engineering/physical and statistical models; (2) this, however, will result in more complicated non-liner functions that requires new optimization algorithms under dedicated control and operation framework, such as multi-agent systems.

Multi-objective optimization of cluster-level energy systems or energy hub: (1) the direct benefits, i.e. reduction of energy use, carbon emissions and costs, are usually the three key indicators for cluster-level energy systems and energy hub; co-benefits are also important to evaluate, such as indoor air quality, thermal comfort, less risk exposure to future energy price increases and so on, represented in Fig. 2.24; (2) in addition, further scenario-based optimization of energy systems is expected to address uncertainty and risk of energy exchange within the cluster and semantic-interoperability-based energy patterns, as well as geographical and cultural contexts, such as demographic, socio economic, behavioural and property-related characteristics.

2.5 Summary

The wide-spread implementation of renewable energy source (RES) envelope solutions brought about new challenges to urban energy systems. It not only delivered a new paradigm of energy flow profiles and new requirements for energy matching within complex energy systems, but also uncertainties and risks in energy supply. We conclude that energy planning at building cluster scale has a potential to deliver a breakthrough in this regard. This approach fosters the economic effectiveness and the operation feasibility to maximize distributed renewable energy harvesting, while at the same time match the respective energy demand and supply.

The suitable spatial boundary of a building cluster modelling is between 0.1 and 1 km in

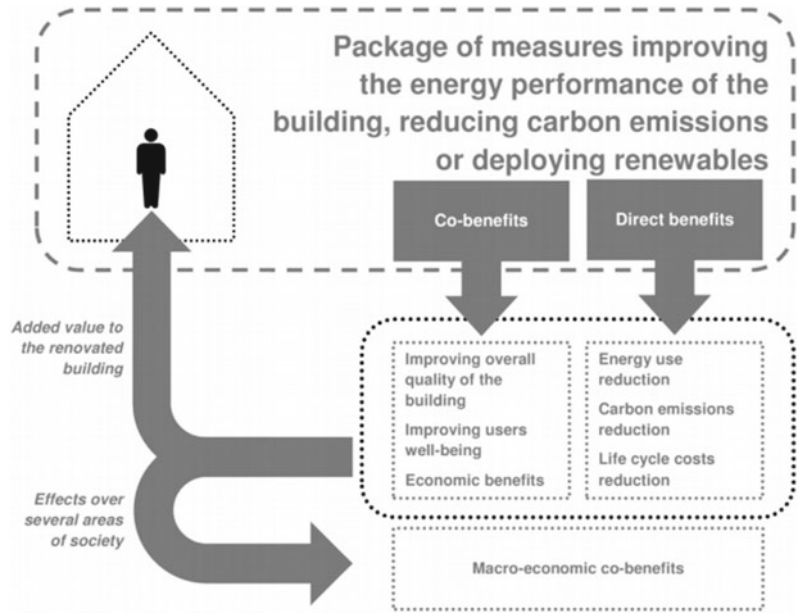

Fig. 2.24 RES envelope solutions-based energy systems modelling at building cluster level (Ferreira et al. 2017)

diameter, while a temporal scale of minute or second should be realistic within the next a few years. The methodology for building cluster modelling should be comprehensive, consisting of the following three aspects:

- Energy supply: assessing the solar energy generation potential within the boundary by combing density of the buildings and upscaling methods or semi-physical model, since most of the existing RES envelope solutions are derived from solar (air) resources at building cluster level.
- Energy demand: considering integrated energy demand in transports and synergies of energy exchange between buildings and energy infrastructures by using tools (software) such as EnergyPlus, TRNSYS, Modelica or self-developed ones.
- Energy operation: optimizing the operational performance of the integrated energy systems at cluster level by focusing on control strategies (i.e. non-linear control, model predictive control, scheduling, optimal control, fuzzy logic control, multi-agent control, etc.) and multi-objective optimization (i.e. MATLAB optimization toolbox, teaching–learning algorithm, multi-agents system, self-adoptive learning, robust optimization, memetic algorithm, etc.) for reduction of energy use, carbon emissions and operational costs.

The chapter raised important research questions and identified important factors influencing urban energy systems at the building cluster scale. It finally put forward several directions for future research development: (1) combining facade area ratio and full spatio-temporal statistical model for solar irradiation, (2) increasing the resolution of energy flow profile, (3) acquiring spatio-temporal energy demand, (4) developing detailed physical and statistical models in complex energy systems/energy hub using new optimization algorithms and (5) proposing multi-objective optimization of the cluster-level energy systems or the energy hub.

The outputs of the chapter are useful the simulation and optimization of urban energy systems incorporating RES envelope solutions, which facilitate the transition to sustainable and resilient urban energy systems.

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Urban Solar Mobility: From Solar to Buildings, Vehicles, and Storage

3

Xingxing Zhang and Pei Huang

Abstract

The deployment of solar photovoltaics (PV) and electric vehicles (EV) is continuously increasing during urban energy transition. With the increasing deployment of energy storages, the development of the energy sharing concept, and the associated advanced controls, the conventional solar mobility model (i.e., S2V, solar-to-electric vehicles, using solar energy in a different location) and context are becoming less compatible and limited for the future scenario. For instance, energy sharing within a building cluster enables buildings to share the surplus PV power generations with other buildings of insufficient PV power generations, thereby improving the overall PV power utilization and reducing the grid power dependence. But, such energy sharing techniques are not considered in the conventional solar mobility models, which limits the potentials in performance improvements. Therefore, this chapter conducts a systematic review of the solar mobility-related studies as well as the newly developed energy concepts and techniques. Based on the reviews, this chapter extends the conventional solar mobility scope from S2V to S2BVS (i.e., solar-to-buildings, vehicles, and storage). The detailed modeling of each sub-system in the S2BVS model and related advanced controls is presented, and the research gaps that need future investigation for promoting solar mobility are identified. The aim is to provide an up-to-date review of the existing studies related to solar mobility to decision-makers, so as to help enhance the solar power utilization, reduce the buildings' and EV's dependence and impacts on the power grid as well as reduce the carbon emission.

Keywords

Solar mobility · Electric vehicles · Building cluster · Energy storage · Energy sharing · Advanced control

3.1 Introduction

3.1.1 Background

Building energy use currently accounts for over 40% of total primary energy consumption in the USA and E.U. (Cao et al. 2016). Transportation sector also represents a large energy end-user and consumes approximately 25% of the primary energy worldwide (Administration 2016). To meet the large energy needs in both the building

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sector and transportation sector, renewable energy, which has much less carbon emissions and relatively lower costs compared with the conventional fossil fuel-based energy, offers a promising solution (Merrill et al. 2017). In this regard, many countries and associations have established regulations or targets to promote the deployment of renewable energy. For instance, the European '20-20-20 targets' aim to achieve 20% reduction in CO_2 emissions (compared to 1990 levels), 20% energy coming from renewables, and 20% increase in the energy efficiency by 2020. The E.U. also sets a target of 32% of energy generation from renewables by 2030, and a minimum share of at least 14% of fuel for transport purposes must come from renewable sources by 2030 (Cao et al. 2016; European Parliament 2018). In different states of the USA, different renewable energy targets have also been defined. For instance, Connecticut sets a target of 48% renewable generation share of electricity sales by 2030, and New Jersey sets a target to increase its renewable portfolio standards target to 50%. Among all the states, California is the most ambitious and sets a goal to achieve 100% carbon-free power by 2045 (U.S. Energy Information Administration 2019). China, as the world's biggest energy consumer, also aims for a 35% of renewable based electricity generations by 2030 (Bloomberg News Editors 2018). In order to achieve these renewable energy targets, two important aspects, the way how renewable energy is used and the renewable energy self-utilization, should be carefully determined.

3.1.1.1 Market Trends of PVs

The global PV market is increasing in an approximate exponential trend (Europe 2018). In 2018, a total of 102.4 GW PV panels were installed globally, representing a 4% year-on-year growth over the 98.5 GW installed in 2017. This led to a total global solar power capacity of over 500 GW. The Asia–Pacific region (including China) was leading the global PV market, and it owned more than half (55%) of the global solar power generation capacity. In this region, China alone operated nearly one-third of the world's solar power generation capacities. The European solar pioneers ranked second, but its share slipped to 25% based on a cumulative PV capacity of 125.8 GW. The Americas were the world's third largest solar region in 2018—with a cumulative installed capacity of 78.2 GW and a 15% stake.

3.1.1.2 Market Trends of Electric Vehicles

EVs, which are powered by electricity, are considered as a promising solution to the roadside air pollution and associated health damages since it directly cuts off the pollutions from the source. When EVs are charged by the renewable energy, such as PV and wind turbine, their operations are totally carbon-free, and thus, EVs can make substantial contributions to the greenhouse gas emission. Until now, a lot of governments have established policy or goals to promote the deployment of EVs. For instance, the Swedish government has set a goal that the 100% of the national energy used in vehicle fleets should be independent of fossil fuel by 2030 (Xylia and Silveira 2017). The US Federal Government has enacted policies and legislations to promote the US market for EVs, such as improvements of tax credits in current law, and competitive programs to encourage communities to invest in infrastructure supporting these vehicles (United States International Energy Agency 2010). The Hong Kong government has made efforts in promoting its practical applications, e.g., tax reductions, one-for-one replacement scheme, subsidies for EV purchase and EV licenses (Department 2019) (Huang et al. 2019b). The Singapore government has shown great interest in adopting EVs (Lokesh and Hui Min 2017). The Energy Market Authority (EMA) and the Land Transport Authority (LTA) of Singapore launched the EV testbed in 2011 to decide on the mass adoption of EV. Following the positive results, they approved BlueSG Pte Ltd (News 2011), a subsidiary of Bolloré Group, to launch the EV car-sharing program by 2017 (Ahamioje and Krishnaswami 2017). The French government set a target of 2 million EVs in 2020 (Merten et al. 2012). The stock of EVs (i.e., the number of EVs on the road) is projected to reach 18.7 million in

2030 (Cooper and Schefter 2018). Texts (II 2010) stipulated the introduction of a Building Plan stating that, with effect from 2020, the deployment of rechargeable electric and hybrid vehicles (HV) should represent 30% of all vehicle sales by 2020. Hence, the public acceptance of EVs and its swift global market penetrations because of that are imposing increased impacts on the deployment of PV and smart grid.

3.1.1.3 Market Trends of Storages

As reported by the KPMG, in 2016 the USA has the largest market for energy storage, both by number of projects and installed capacity. In some states, innovative energy storage incentive programs have been implemented. A total capacity of 62 MW energy storage has been installed by 2014, and the country has set a target of 1325 MW energy storage capacity by 2020. Japan has set an ambitious target to produce half of the world's batteries by 2020. The country has a subsidy program for 66% of the cost for homes and business that installs lithium-ion batteries. Japan also has a target of 30% renewables' implementation by 2030. India has a target of 40 GW of renewable energy capacity by 2030. Among the different energy storages, electricity storage is an economic solution off-grid in solar home systems and mini-grids where it can also increase the fraction of renewable energy in the system to as high as 100% (IRENA 2017). The International Renewable Energy Agency reported a total battery capacity in stationary application of 11 GW·h in 2017. The total capacity is expected to reach 100–167 GW·h in 2030 in the reference case. The large battery storage market is contributed by pairing the battery storage system with the installation of new small-scale solar PV systems. For example, motivated by the financial support for battery storage, nearly 40% of small-scale solar PV systems in Germany have been installed with battery systems in the last few years. In Australia, although there is not any financial support, nearly 7000 small-scale battery systems were still installed in 2016. The economics of battery storage in such application is also projected to increase significantly in the future. The growing capacity of energy storage provides great opportunities for renewable energy utilization and promoting the solar mobility.

3.1.1.4 Building Prosumers Role

Buildings consume about 40% energy worldwide (ASHRAE 2011), and this percentage is even larger in high-density cities (e.g., over 90% in Hong Kong) (EMSD 2017). To reduce the energy usage in the building sector, renewable energy systems, such as PV panels and wind turbines, are widely installed in buildings (Huang et al. 2018b). By such renewable energy systems integration, buildings are transferring their roles from conventional electricity consumers to electricity prosumers. As defined in (Tongsopit et al. 2019; Huang et al. 2020), electricity prosumers are electricity consumers who produce electricity for their own consumption using distributed energy technologies. From the perspective of buildings, the transformation can help cut down the grid power usage, thereby reducing the electricity costs as well as reducing carbon emissions if the grid power is produced from fossil fuels (Huang et al. 2018a). From the perspective of the power grid, the reduced demands at the building side can help alleviate the grid stress and thereby promoting the power grid's stable and reliable operation (Huang and Sun 2019). A common type of electricity prosumers is the zero-energy buildings (ZEBs), which produce the same amount of energy it consumes.

To facilitate the application of ZEBs, many governments have established policies and goals. For instance, Directive 2010/31/EU, the Energy Performance of Buildings Directive (EPBD), sets the goal that all new buildings built from the beginning of 2021 must be nearly zero-energy and cost-optimal (The European Parliament and the Council of the European Union (EPBD), 2010). ZEBRA 2020 (nearly zero-energy building strategy 2020) was launched by 17 countries in 2014, for the purpose of creating an observatory for nZEBs based on market studies and various data tools, and therefore generates data and evidences for optimization and policy evaluation (ZEBRA2020 2014). The US government sets a target that 50% of commercial buildings

achieve zero-energy by 2040, and all commercial buildings achieve zero-energy by 2050 (Huang et al. 2018b). The California Public Utilities Commission of the USA has set a net zero-energy target for all new residential buildings by 2020 and for all new commercial buildings by 2030 (Crawley et al. 2009). Similar promotion policies and targets can be also observed in China, Korea, Japan, and Australia (IPEEC Building Energy Efficiency Taskgroup 2018). As reported by Tronchin et al. built environment represents a suitable intermediate scale of analysis in Multi-Level Perspective planning, collocated among infrastructures and users (Tronchin et al. 2018). Temporal and spatial decoupling of supply and demand is an important element that should be considered for the evolution of built environment, especially when creating sectorial level planning strategies and policies. They also concluded the need of research for developing innovation pathways for the co-evolution of built environment and infrastructures. In such context, the analysis of complementarities is particularly powerful and should receive more attention. Solar mobility development, which seeks complementarities in multiple systems (i.e., buildings, EVs, PVs, and energy storage), is in line with this context.

3.1.2 Defining the Concept of Solar Mobility

As proposed by CEA-INES (Vu et al. 2008), the concept of solar mobility seeks the synergy between the three following systems: EVs, PV systems, and electricity network. The basic idea is to combine a standard grid-connected PV system with standard EVs, also connected to the grid (Popiolek and Thais 2016). In order to ensure the solar charging of the vehicles and minimize the grid impact, a local Energy Management System decides on the energy flows. Such process is called as Solar-to-Vehicles (S2V) in some studies (Birnie 2009), representing charging EVs directly using electricity from PVs. Vehicles can be charged at home using residential charging stations or at public charging stations in private business or public car parks. They consider such process reasonable since the average car is parked 95% of the time and charging can take a long time based on current usage models. In their proposed concept, the electricity produced by the residential PV panels is firstly used to supply the home electrical equipment (e.g., household appliances, multimedia devices, etc.) and then to charge the EV battery, as shown in Fig. 3.1. If there is any surplus generation, such amount is fed into the power grid. The smart grid will collect data on the grid loads and power needs and redistribute the surplus generation to meet these loads/needs. In such context, buildings, equipped with PV systems and energy storage systems, are becoming energy production sites where EVs can be charged. This convergence between buildings and transport will enable EV batteries to be used as a means of storage and supply of low-carbon electricity to meet fluctuations in production and consumption. Note that the EV battery is also allowed to discharge electricity back to the building/power grid in such model. This is advantageous for grid management and especially for peak smoothing. This concept has been extensively studied from a technological viewpoint in Department (2019), EMSD (2017), Europe (2018), European Parliament (2018), Fan et al. (2018), FERROAMP (2018) (Kempton and Letendre 1997; Schwan et al. 2013; Zhang et al.

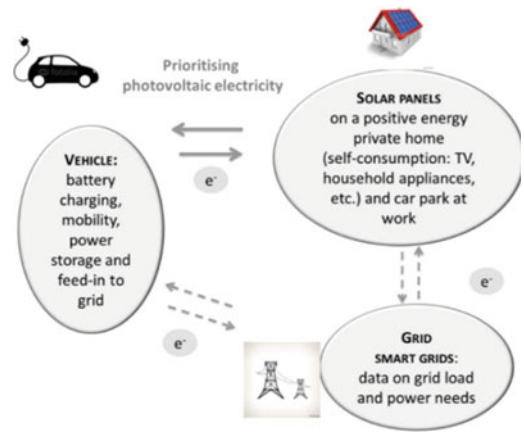

Fig. 3.1 Concept of solar mobility with self-consumption (Popiolek and Thais 2016)

2012; Nunes et al. 2015; Codani et al. 2016; Mesarić and Krajcar 2015).

3.1.3 Values, Problems, and Challenges to the Solar Mobility

Values: With the increasing deployment of PVs, EVs, and energy storage systems, it is important to smartly integrate them to maximize the energy efficiency and cost benefits and meanwhile minimize the impacts on the power grid. Under such context, the solar mobility can help improve the three values: autonomy, sustainability, and affordability. Specifically, autonomy indicates reducing the dependence and impacts on the public power grid. Ideally, within a building community/microgrid, by acting as an electricity prosumer and considering the EV demand, the buildings can be largely covered by their own PV system. Sustainability indicates increasing the self-consumption of locally-produced PV power and thus the needs of power from the public grid which largely depends on fossil energy in many countries. Affordability indicates increasing the economic benefits of the whole systems.

Problems: Although the importance of PVs, EVs, and energy storage has been well recognized globally, how to integrate and manage them in a holistic way, simultaneously taking into considerations of building loads and occupants' living requirements, needs to be addressed. Problems such as large power penetration on the power grid, low energy efficiency, low economic performances urgently need to be solved. Moreover, with development of the energy sharing concept and associated advanced controls, the conventional solar mobility concept and context are becoming less compatible and limited. For instance, energy sharing within a building cluster enables buildings to share their surplus generations with other buildings (including their EV demands) with insufficient supply, thereby helping improve the overall renewable energy utilization and reducing the grid power dependence. However, such energy sharing networks, including the system architecture and associated advanced controls, are not considered in the conventional solar mobility models. This will limit the potentials in performance improvements that can be otherwise be articulated and demonstrated by the newly developed concepts and methods. Another example is the lack of energy storage integration in the existing solar mobility model (note in this chapter, EVs are considered as a separate role from energy storage).

Challenges: Intermittency is one of the major shortfalls of solar power which has a direct influence on the voltage stability and the overall power system security. While for EVs, their charging loads are difficult to predict as highly affected by the driving patterns and driving distances. Both the PV power production and EV power demand are highly uncertain. For instance, the EV can be used for commuting purpose during the day where access to the charging facilities is not available. This means that charging could only happen in EV owner's premise during the night time when no electricity could be generated by the PV. Such time mismatch will limit the deployment of solar energy in the mobility sector. Thus, it is challenging to bridge the temporal and spatial demand–supply mismatch to facilitate solar mobility. When buildings and energy storage are integrated into the solar mobility context, another challenge is the proper management of the different types of systems with various response characteristics and operating constrains. This chapter will aim to propose some useful solutions from the existing studies to address these challenges.

3.1.4 Aim and Contributions of This Chapter

This chapter conducts a technical review of the solar mobility-related studies as well as the newly developed energy concepts and techniques. By reviewing the existing studies (Taşcıkaraoğlu 2018, Zhang et al. 2018; Huang et al. 2019a), this chapter extends the conventional solar mobility model from S2V to S2BVS (i.e., solar-to-buildings, vehicles, and storage). In the extended context, solar mobility involves solar energy

flow and exchange through buildings, vehicles, and storages and the renewable energy sharing network. The electricity generated by PV panels is applied to provide electricity to buildings and charge batteries/thermal energy storage (e.g., by heat pump), while EVs can be further charged at residential charging stations or at public charging stations in private business or public car parks. The associated S2BVS system architecture and models are proposed, and the associated advanced controls in existing literatures are reviewed. The aim is to help improve the solar mobility concept by introducing the up-to-date S2BVS models, to enhance the renewable energy utilization, reduce the dependence and impacts of buildings and EVs on the power grid, and reduce the carbon emission, in response to the future scenario with increased PV capacity, EV number, and storage capacities. The major contributions of this chapter are summarized as following:

- Assess values, problems, and challenges of solar power deployment to promote the solar mobility.
- Extend the existing S2V (i.e., solar to vehicles) concept to the S2BVS (i.e., solar-to-buildings, vehicles, and storage) with the integration of renewable energy system, buildings, energy storage system, EVs, and more importantly, the renewable energy sharing network.
- Identify the research gaps that need future investigations for promoting solar power utilization and solar mobility.

3.2 Overview of the Existing Studies on Solar Mobility

Extensive efforts have been devoted in promoting the solar mobility, and many papers have been published regarding this topic. Figure 3.2 defines the extended scope of solar mobility (as compared with Fig. 3.1) and highlights the location of reviews for each sub-system. Table 3.1 summarizes the main existing studies within such scope.

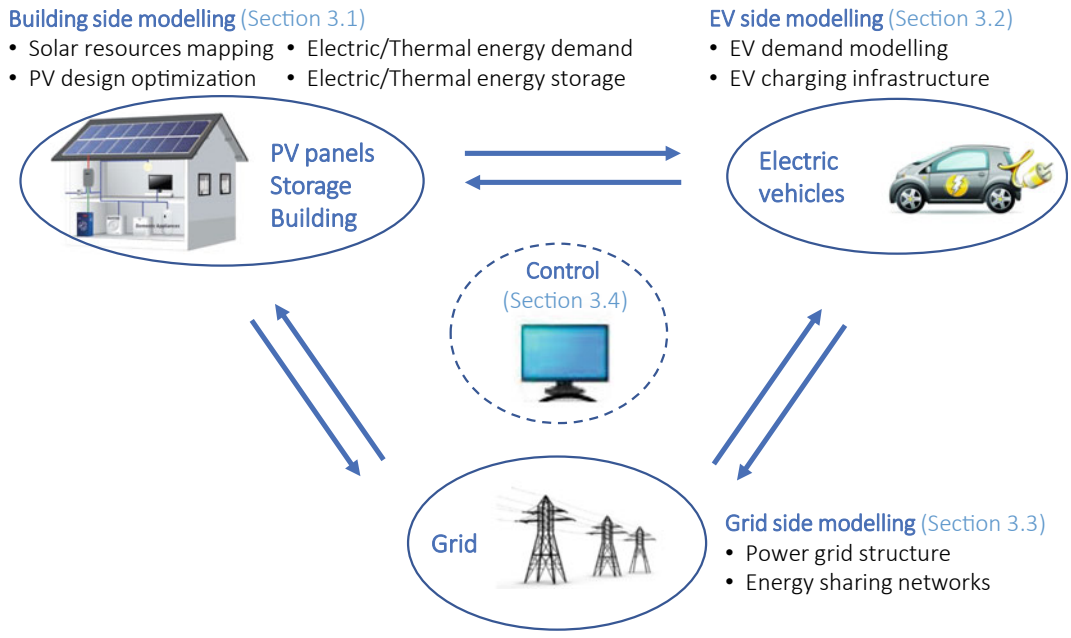

Fig. 3.2 Scope of solar mobility and the major sub-systems

Table 3.1 Summary of the existing studies on solar mobility

Authors	Region	System					Modeling tools	Main work done	Special points
		Building	PV	Storage	EVs	Other techniques			
Berthold et al. (2011)	France	Y	Y	N	Y	Wind energy	—	Develop a general control strategy which aims at minimizing the building's total energy costs by optimizing the charging/discharging of the PHEVs batteries. The considered systems include the power grid, local production from renewable, and vehicle. Using the EV battery to power the home appliances is allowed	Unlike conventional home-to-vehicle (H2V), this study also enables vehicle-to-home (V2H) for demand response control
Querini et al. (2012)	Germany, Denmark, Sweden, Spain, France, UK, Italy	N	Y	N	Y	Wind energy	Gabi4	1. Study the life cycle GHG emissions linked with EVs using PV and wind electricity in different regions of European countries. 2. Analyze the life cycle GHG emissions using wind energy, solar energy, and conventional fossil fuel	When using PV electricity, GHG emissions are always lower than conventional thermal vehicles
Zhang et al. (2012)	Kansai, Japan	N	Y	N	Y	Heat pump	Visual Studio C#.net 2008	1. Develop an hour-by-hour simulation model to derive a real-time supply–demand balance. 2. Evaluate the impacts of integrating PV power into future electricity system with EVs and HPs under smart control strategies in	The EVs are considered as virtual batteries similar to electricity storage, which can not only be charged by the grid power (G2V) but also discharge electricity to the power grid (V2G)

(continued)

Table 3.1 (continued)

Authors	Region	System					Modeling tools	Main work done	Special points
		Building	PV	Storage	EVs	Other techniques			
								Kansa. 3. Analyze a set of scenarios with different penetrations of PV, EV, and HP	
Dallinger et al. (2013)	Germany and USA (California)	N	Y	N	Y	N	PowerACE	1. Develop a method to characterize the fluctuating electricity generation of renewable energy sources and compare the difference for California and Germany. 2. Analyze the potential contribution of grid-connected vehicles to balancing generation from renewable energy sources for a 2030 scenario in California and Germany based on the developed method	1. Correlation between RES generation and the load curve affects the integrating of RES. 2. EVs play an important role in reducing residual load fluctuation if smart charging is used
Su et al. (2014)	Illinois, USA	N	Y	Y	Y	Wind energy	CPLEX + MATLAB + OpenDSS	1. Formulate a stochastic problem for microgrid energy scheduling, which aims at minimizing the expected operational cost of the microgrid and power losses by optimally dispatching the EV charging load and scheduling DGs and distributed energy storage devices. 2. Investigate the impact of PEVs on microgrid energy	Combined scheduling of EV charging loads and the energy storage systems

(continued)

Table 3.1 (continued)

Authors	Region	System					Modeling tools	Main work done	Special points
		Building	PV	Storage	EVs	Other techniques			
								scheduling under various charging schemes	
Chaouachi et al. (2016)	Italy	N	Y	N	Y	N	Matpower open-source package	1. Propose a conceptual smart grid framework and assessment methodology, to enable decentralized operational synergy between intermittent PV generation and EVs, based on coordinated EVs charging. 2. Test proposed methodology with a real distribution system, where different PV and EV penetration scenarios are assessed against charging behavior variants	Impacts of coordinated charging on the PV penetration and carbon emission reduction
Islam and Mithulananthan (2018)	Australia	N	Y	N	Y	N	MATLAB	1. Develop a non-iterative PV output model that does not require additional measurements or meteorological data, which saves money and time. 2. Develop a combined SOC-based, fair charging strategy which simultaneously reduces the runtime by lowering the number of variables involved and increases the charging fairness. The reduced runtime makes it suitable for more frequent	Consider grid-side parameters, such as voltage topology, structure The PV output model and charging strategy together lessen the probability of voltage limit violations and enhance PV harvest

(continued)

Table 3.1 (continued)

Authors	Region	System					Modeling tools	Main work done	Special points
		Building	PV	Storage	EVs	Other techniques			
								control of charging of a large EV population	
Sun et al. (2019)	Glasgow, UK	Y	Y	Y	Y	N	MATLAB (GA optimization tool)	1. Develop a model for minimizing energy cost of a residential household with an EV, an ESS, and other residential loads, where the EV's usage patterns are described by probability levels. 2. Conduct a practical survey of EV daily usage including driving purposes and usage at different time periods. 3. Investigate the total cost saving through case studies for various scenarios under fixed and time of use (TOU) tariffs	Optimization results based on this model can be used to determine whether V2G is beneficial for the EV owners under the optimal charging and discharging strategy
Taşcıkaraoğlu (2018)	Austin, TX, USA	Y	Y	Y	N	Energy sharing network	General Algebraic Modeling System (GAMS) with solver CPLEX	1. Develop the concept of energy sharing enabled neighborhood area networks, which is composed of a shared energy storage system and multiple consumer premises. 2. Develop a novel energy management strategy based on the implementation and scheduling the use of this shared energy storage system (ESS) with the	Energy sharing enabled neighborhood area networks (NANs) for cluster-level performance improvements. Such energy sharing can reduce the energy costs and peak demand significantly

(continued)

Table 3.1 (continued)

Authors	Region	System					Modeling tools	Main work done	Special points
		Building	PV	Storage	EVs	Other techniques			
								objective of exploiting the ESS unit in the context of an energy credit-based DR program	
Huang et al. (2019d)	Yuxi, China	Y	Y	N	Y	N	HOMER (for demand/supply calculation) MATLAB (for control)	1. Proposed a retired EV battery (REVB) model based on the mathematical model of capacity fade of Li-ion battery cells to simulate REVB's capacity loss. 2. Develop a power management system to mitigate the degradation of REVB and protect other system components. 3. Construct a triobjective optimization model considering reliability, energy waste, and cost	Use retired EV batteries as energy storage in buildings
Barone et al. (2019)	Naples, Italy	Y	Y	N	Y	N	MATLAB	Propose a novel energy management system for buildings connected in a microgrid, by considering EVs as active storage components of such energy scheme	Such B2V2B integration enables renewable sharing among different buildings

In the extended solar mobility scope, the energy prosumers are equipped with their own renewable energy systems, electrical storage, EVs, and other electrical appliances. The buildings are connected into a renewable energy sharing microgrid, in which the surplus renewable production can be delivered from one building to another. Such energy sharing network provides a platform for the buildings in a microgrid to share their surplus renewable energy generations with other buildings, thus helping enhance the overall cluster-level performances. The microgrid for energy sharing is also connected to the main power grid, in case there is a surplus or insufficient cluster-level renewable generation. The power exchange of the building cluster with the power grid will be metered by advanced metering facilities.

3.2.1 PV and EV Interaction via the Public Grid

Making use of the charging/discharging capability of EV battery, the EVs can be used as flexible electricity storage in the power grid and interact directly with the power grid. For instance, Zhang et al. investigated the energy and environmental impacts of integrating PV power into electricity systems in Kensai of Japan, under various scenarios with different EV penetrations and heat pump capacities (Zhang et al. 2012). It was found that EV and heat pump were helpful for keeping more PV power in the smart electricity systems. In their study, the EVs were considered as virtual batteries similar to electricity storage, which can not only be charged by the grid power (G2V) but also discharge electricity to the power grid (V2G). Sun et al. investigated the economy viability of discharging EV power back to the grid, which is called vehicle-to-grid (V2G) (Sun et al. 2019). They developed a model for minimization of the energy cost of a residential household with residential loads, an ESS, and an EV with its usage patterns described by probability levels. Using the developed model, they studied the total cost saving for various scenarios under fixed and time of use (TOU) tariffs. Their study results reveal that certain threshold levels of feed-in tariffs are expected to allow users' benefit from V2G technology. Noussan and Neirotti compared three archetypal charging profiles (i.e., home, public and work) evaluated on ten European countries over four years, to investigate the effects of national electricity mixes and of the type of charging location on the average emission factor of the electricity supplied to electric vehicles (Noussan and Neirotti 2020). Their study results show that the variability related to charging profiles is generally limited (with an average variation range of 6%) in all the selected countries, while in several countries, the variability in different years is much larger (with an average range of 18%).

3.2.2 PV and EV Interaction via the Buildings

Besides using as an electricity storage in the power grid, the EVs can also be used as flexible electricity storage in the buildings. For instance, Berthold et al. developed a control strategy, which aims at minimizing the building's total energy costs by optimizing the charging/discharging of the PHEVs' batteries (Berthold et al. 2011). The considered systems include the power grid, local production from renewable energy systems, and vehicles. Unlike the conventional controls which only enable home-to-EV (H2E) power transmission, this chapter also enables EV-to-home (V2H) power transmission, which extended the utilization of EVs in building demand response. In order to maximize the value of EV battery, Huang et al. proposed a retired EV battery (REVB) model based on the model of capacity fading of lithium battery cells (Huang et al. 2019d). Using the developed REVB model, a power management strategy (PMS), which considers multiple objectives including minimizing loss of power supply, system cost, and potential energy waste, was developed to regulate the energy flow for protecting the REVB and other system components. A multi-objective evolutionary algorithm NSGA-II is used for

generating the Pareto set of the optimal solution. The application of the developed method in a residential building indicates that PV-hydrogen-REVB hybrid energy system is a promising way to exploit REVBs' residual capacities. Similarly, Barone et al. proposed the concept 'Building-to-Vehicle-to-Building' (B2V2B), which enables bidirectional electricity exchange of EV batteries with the buildings (Barone et al. 2019). They also developed a novel energy management system for buildings connected in a microgrid, by considering EVs as active components of such energy scheme. Renewable energy sources (i.e., PV), energy storage systems, and bidirectional electricity exchange with the buildings and the grid were taken into account. A highlight of their proposed system is that PV power sharing is enabled among different buildings by applying bidirectional EV charging/discharging, as shown in Fig. 3.3. Such energy sharing can significantly improve the PV power utilization and thus bring economic and environmental benefits.

In Fig. 3.3, Case 1 represents the conventional unidirectional Building-to-Vehicle (B2V) system operation. Here, the plug-in EV is linked with the power grid by acting as a power load. The EV is charged through a home charger, and no renewable energy systems and batteries are installed on-site. Case 2 represents a novel concept of bidirectional Building-to-Vehicle-to-Building (B2V2B) system operation. The plug-in EV is linked with the power grid acting as a power load as well as a source for the House building and as a source for the Office space. A renewable energy system, consisting of PV panels, is on-site installed on the tilted roof of the House building. The House is also equipped with a stationary battery (HSB), which can also feed the EV battery (in case of available stored energy, otherwise the EV battery is conventionally supplied by the grid). An additional novelty is here represented by the transfer to Office through EV battery of the electricity potentially produced by the House PV panels. The EV battery can be also charged at Office, if necessary. Case 3 represents a novel concept as well of bidirectional Building-to-Vehicle-to-Building (B2V2B) system operation, based on swappable batteries. The system operation follows that of Case 2. The difference with Case 2 relies on the batteries; specifically, in Case 3, the House is equipped with a battery identical to the EV one and a quick swap of batteries is allowed between the EV and the House. The swapping option

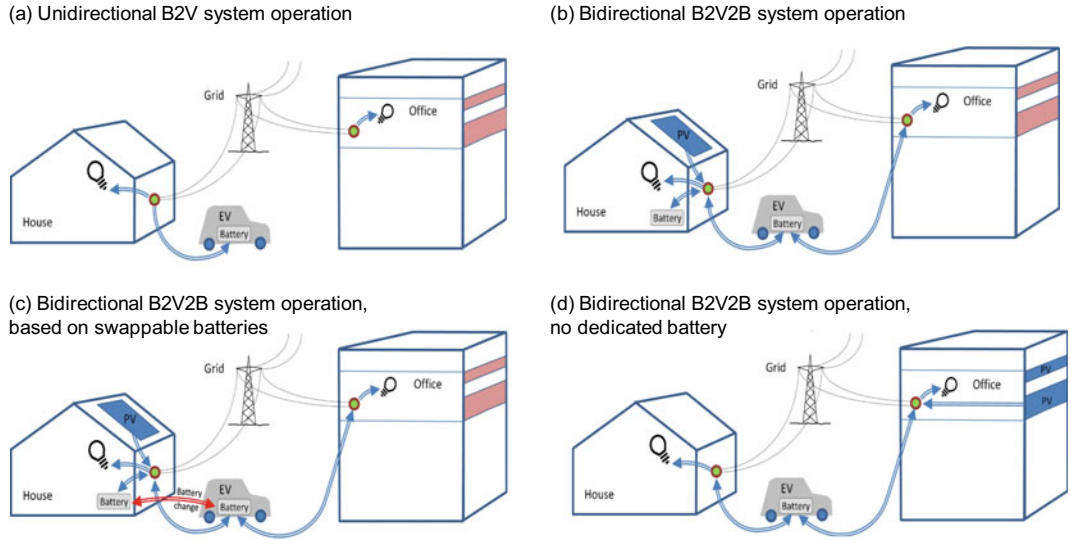

Fig. 3.3 Four different ways of building-EV-PV-grid integrations (Barone et al. 2019)

prevents the need of energy transfer from the HSB to the EV battery (when EV battery charge is required) and thus the related losses. Case 4 represents a different novel concept of bidirectional V2B system operation. The main difference with the previous Cases 1 and 2 regards the site of the PV panels, which are installed on the façade of the Office space, where no dedicated battery (i.e., HSB) is considered (solar energy is stored directly into the EV battery). The plug-in EV communicates with the power grid acting in this case as a power load as well as a source for the Office space and as a source only for the House building. In Case 4, the novelty is represented by the possible transfer to House through EV battery of electricity produced by the Office PV panels. The EV can be charged both at the House and Office buildings.

3.2.3 PV and EV Interaction via the Energy Sharing Network Considering Buildings and Energy Storage

In addition to making use of the flexibility charging/discharging capability of EVs to enable energy sharing, a more direct way (i.e., micro power grid) can also be used for large amount of energy sharing. For instance, Taşcıkaraoğlu developed a system structure for shared energy storage system (ESS) in the neighbor community, as shown in Fig. 3.4 (Taşcıkaraoğlu 2018). In their study, each building is equipped with a top PV-based distributed generation system and is connected to a shared ESS. The shared ESS, the transformer, and all the households are connected to a common point, which is named point of common coupling (PCC). Bidirectional power flow is enabled between PCC and the power grid via a neighborhood transfer, between PCC and the shared ESS, and also between buildings and PCC. Such system configuration makes three types of power exchanges available, i.e., the internal local power exchanges among the neighborhood buildings, the power exchanges between the neighborhood buildings and the grid, and the power exchanges between the shared ESS and grid/neighborhood buildings. In other words, the power consumed by a building can be produced by its PV system, be produced locally by the PV system of other buildings, and/or be drawn from the power grid/shared ESS. The EVs' charging load is considered as normal electricity load in this chapter. By sharing the ESS, the buildings can deliver their surplus renewable energy to other buildings with insufficient supply, thus increasing the overall renewable self-utilization rates and meanwhile

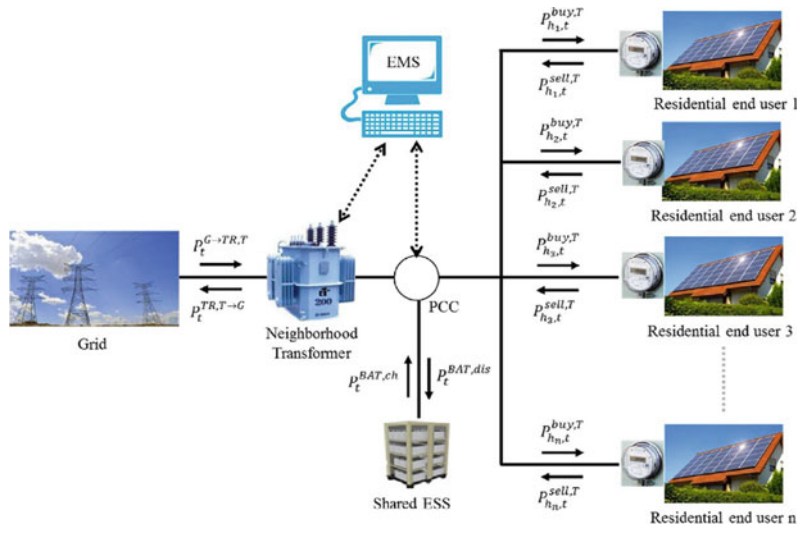

Fig. 3.4 Schemes of the shared ESS unit serving a neighborhood with multiple households (Taşcıkaraoğlu 2018)

reducing the interactions with the power grid. Their study results show that the shared storage can help decrease the peak electricity demand of the building cluster by as much as 30% and meanwhile reduce the electricity costs of the building cluster by over 10%. However, the investment of a shared ESS with large capacity will be much higher than distributed ESS with small capacities. In addition, the energy losses may be large due to the long transmission distance from the buildings to the shared ESS.

Zhang et al. also developed a novel structure for promoting solar mobility in the residential buildings, as shown in Fig. 3.5 (Zhang et al. 2018). In their study, each residential building has a microgrid, which connects the electricity production facilities (e.g., PV panels), electricity consumption devices (e.g., lighting, washing machine, EVs, etc.). The EVs are used as flexible electricity storage which can be charged by the PV system/grid electricity in periods with sufficient supply or discharge power to the building/power grid in periods with insufficient supply. An aggregator is utilized to connect multiple microgrids, which coordinates the energy sharing among different microgrids as well as their interactions with the power grid. Based on the developed system structure, they developed a two-stage control method to control the EV charging/discharging rates as well as the energy trades within the multiple microgrids and with the power grid. Compared with the concept proposed by Barone et al. (2019), which uses EV batteries with limited capacity as the medium for energy sharing among different buildings, Zhang et al.'s developed structure is much more flexible and enables larger amount of energy sharing. However, the stochastic charging behavior of EVs, the integration of thermal energy storage systems, and the dynamic electricity prices are not considered in their study. Also, the initial costs for constructing such microgrid will be high, which may hurdle its large-scale application.

Similarly, Huang et al. applied advanced energy concepts for retrofitting a residential building cluster in Sweden (Huang et al. 2019a). The studied system includes PV panels (installed in individual buildings), centralized thermal energy storage, heat pump, and EVs. A DC microgrid-based energy sharing network, which is developed by Ferroamp, is constructed in the building cluster. The excessive PV production

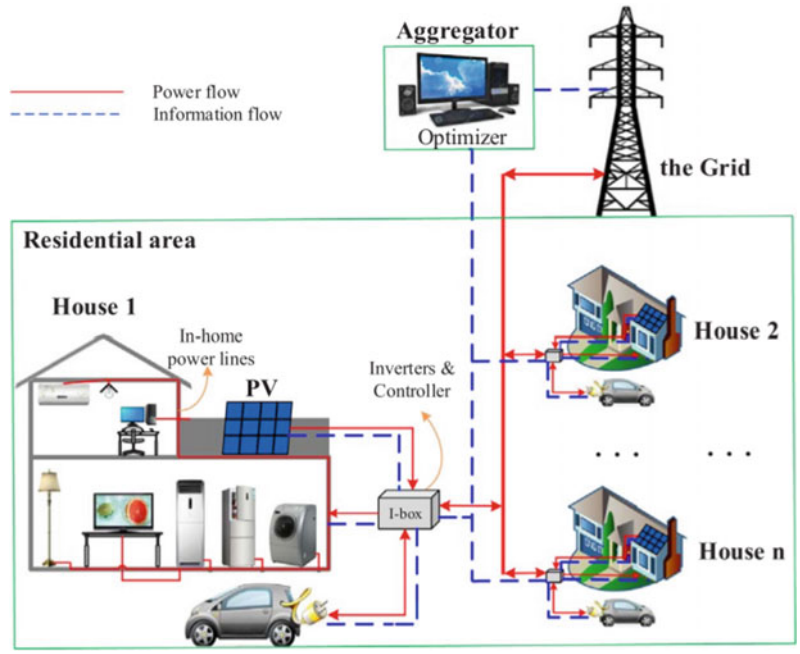

Fig. 3.5 Structure of a residential multiple microgrids (Zhang et al. 2018)

can be stored in the form of heat energy in the thermal storage by powering the heat pump to work. Their study results show that by enabling renewable energy sharing and integrating energy storage and EVs, the PV power self-utilization can be as high as 77% in the baseline case.

To sum up, existing studies have developed three approaches for deploying PV power in the EVs, i.e., via public power grid, via buildings, and via energy sharing networks considering buildings and energy storage. Among these three approaches, the third approach is superior to the other two approaches, as the energy sharing network makes the EV charging more flexible (i.e., from different PV power sources) and efficient (i.e., with enhanced PV power local usage) and allows integration of two other important players (i.e., buildings and energy storage) in the energy systems. Such extended scope of solar mobility, with PVs, EVs, energy storages, buildings (i.e., S2BVS), and energy sharing networks, is the future development trend.

3.3 Modeling of Sub-systems

This section reviews the modeling techniques in the essential components related to solar mobility, including the building side modeling, EV side modeling, grid modeling, and advanced control. Design and control optimization are two main means to help improve the deployment and utilization of solar energy. This section will review from these two aspects.

3.3.1 Building Side Modeling

3.3.1.1 Solar Resource Mapping

There are several commercial databases available for solar resource mapping such as Meteonorm (Meteotest) and global solar Atlas (Group). These tools make use of data inputs from geostationary satellites and meteorological models such as air temperature model, clear-sky model to predict the incident energy on earth surface at a defined spatial–temporal resolution. Zhang et al. carried a critical review and compared various models which are used to estimate the solar irradiation on basis of the time scale and estimation methods (Zhang et al. 2017). Basharat et al. (2013) compared 78 different models used for global solar irradiation estimation. They proposed a systematic classification of these models based on the input parameters which can be used to do similar analysis. The solar resource data obtained from various tools are often detached from surface topography and spatial distribution of the building stocks. However, while estimating the solar resource potential in an urban context, it is important to consider the effect of various objects such as neighboring building on total incident surface irradiation.

Most of the commercially available databases do not consider the effect of urban climate on the solar resources. The atmospheric thermodynamics in an urban climate is affected by several factors such as topography, shading objects, vegetation, urban infrastructure, and heat island effect (Sola et al. 2018). The simulation of an energy system based on a weather database, which does not consider these factors, can result in mismatch between the simulated and real energy system performances. To address such issue, researchers have proposed to couple the geographic information system (GIS) tool and meteorological databases to assess the solar potential in existing urban context. For instance, Quan et al. (2015) proposed a GIS-based energy modeling system for urban energy context which integrates building energy modeling and solar resources modeling using a three-dimensional urban environmental engine. Bergamasco et al. proposed and applied a hierarchical procedure which makes use of GIS data, available solar radiation maps, and statistical data on energy consumption, to determine the PV energy potential for an Italian climatic location (Bergamasco and Asinari 2011). With the development in the availability of high-quality LIDAR data, there is a significant interest from various stakeholders to integrate the urban scale 3D models and LIDAR data for high accuracy energy potential estimation on urban scale (Lingfors 2017). Jochem et al. (2009) proposed a methodology for solar potential estimation for

urban climates using airborne LIDAR data and 3D information of the point cloud.

3.3.1.2 PV Design Optimization

The installation of PV arrays (e.g., the position on the building facades and the tilt angles) has significant impacts on the PV power production. Existing studies have investigated the impacts of these factors and developed proper methods to optimize them. For instance, Abdul-Wahab et al. employed Hybrid Optimization Model for Electric Renewable (HOMER) to find the best PV system among 15 available alternatives and the best location for installing PV arrays for Oman's conditions by analyzing and comparing their costs and the carbon emissions reductions (Abdul-Wahab et al. 2019). Ning et al. (2017) developed a genetic algorithm-based optimization method to design the position, title angles, and azimuth of PV panels, with factors such as shapes and orientations of building exteriors and the surrounding obstacles considered. Their method can effectively improve the PV system power output by 36.1% and reduce the capital investment per unit power output by 4.5% and meanwhile significantly reduce the human labor. Similarly, Magnor and Sauer also developed a genetic algorithm-based optimization method to optimize the PV system installation including tilt angle and azimuth angle of the PV generator under various boundary conditions. Ullah et al. (2019) developed a method to optimize the PV tilt angle under different scenarios (fixed, seasonal, monthly, daily) for Lahore and some of the other major cities in Pakistan. They also proposed a model to estimate the upper/lower bounds of soiling losses and explored the tilt angle effect on the soiling losses by doing soiling experiments. Shirazi et al. (2019) proposed an integrated techno-economic evaluation tool to identify the most appropriate PV installation façades in urban areas in Tehran of Iran. They found that proper selection of the angles and building façades for installing PV panels could significantly increase the solar power production and internal rate of return. Huang et al. developed an iterative method based on genetic algorithm to optimize the capacity and positions of PV modules at cluster level, with the purpose of maximizing the self-consumed electricity under a non-negative net present value during the economic lifetime (Huang et al. 2019a). Boeckl and Kienberger (2019) developed a Fourier series approximation-based method for sizing the grid-connected PV storage system to maximize the solar energy self-utilization.

3.3.1.3 Electric and Thermal Energy Demand

The energy demand modeling approaches for buildings can be roughly divided into physical modeling (Fonseca and Schlueter 2015) and statistical modeling (Tardioli et al. 2017). The physical models are mathematical representations of heat and mass transfer phenomena between buildings, people, and the environment. For instance, Palacios-Garcia et al. (2018) developed a high-resolution model for calculating the electricity demand of heating and cooling appliances considering variables such as the number of residents, location, type of day (weekday or weekend), and date. In Palacios-Garcia et al. (2015), a model for simulating lighting power consumption profiles in Spain was developed considering the number of household residents and differentiating between weekdays and weekends. In Widén and Wäckelgård (2010), Widén developed a model for computing the occupancy and electricity load in Sweden. Physics-based models usually report high accuracy at the expense of high degrees of complexity and data requirements. The statistical models are mathematical representations of the relationship between an observed set of historical variables. In Fumo and Rafe Biswas (2015), a systematic review of the regression analysis-based statistical modeling approach was conducted. The simple and multiple linear regression analyses along with a quadratic regression analysis were analyzed and compared. In Ahmad et al. (2018), a systematic review of the data-driven-based statistical modeling approach was conducted. The data-driven-based statistical modeling approach was further classified as artificial neural network-based approaches, clustering-based approaches, statistical and machine learning-based approaches, and support vector

machine-based approaches. The statistical models offer a more parsimonious modeling alternative at the expense of detailed physical explanatory power.

3.3.1.4 Electric and Thermal Energy Storage

Generally, there are two main potential storage technologies: electrical and thermal energy storages. Specifically, the electrical energy storage is able to provide operational flexibility among a building cluster, regulating the power to fit the buildings demand and enhancing the energy self-sufficiency. The thermal energy storage stores the energy in form of thermal energy (e.g., heat). Similarly, the thermal energy storage can overcome, in a short-term, the hourly, daily, or weekly mismatch while in a long-term, seasonal variations between renewable energy supply and demand, maximizing the synergies among the building cluster.

Some of the studies have been focused on thermal energy storage. As an example, Hsieh et al. evaluated the performance of different storage configurations to a cluster scale of 11 buildings in Switzerland (Hsieh et al. 2017). Specifically, the energy deriving by the solar thermal collectors is assumed to cover the energy demand (hot water and space heating) of the buildings, while different configurations of thermal energy storage (long and short term) are included to overcome period during which there is not enough solar thermal energy available. Rodríguez et al. explored the potential of alleviating the energy poverty for a low-income housing district in Spain, by using a PV-heat pump thermal mass storage system (Romero Rodríguez et al. 2018). The surplus PV power is used to power the heat pumps, which provides heating/cooling to the whole dwellings, increasing the occupant's thermal comfort. In this chapter, the thermal mass storage capacity of the buildings themselves was used. Huang et al. developed an advanced energy-matching concept to improve building cluster performance (Huang et al. 2019a). In this chapter, a hot water storage is supposed to store the excess of PV energy in the form of heat. Specifically, the excess PV electricity power is transmitted to the heat pump to produce heating energy, and the produced heat is stored as the hot water.

Some of the studies have been focused on electrical energy storage. For instance, Roberts et al. analyzed the impacts of applying a Battery Energy Storage Systems (BESSs) in terms of increased PV self-consumption, maximized self-sufficiency, and reduced peak demand of several apartments in Australia (Roberts et al. 2019). It was pointed out that the embedded networks with PV-BESS in the building cluster could have a beneficial effect for the network distribution due to the reduced daytime export and evening peak demand. However, due to the current battery energy storage systems tariffs and costs, the usage of thermal energy storage systems could be a more financially attractive method to store the excess of energy produced by the PV at a building cluster level. Koskela et al. (2019) analyzed from an economic perspective the profitability of a PV system with an associated electrical energy storage for apartment buildings forming an energy community. The study highlighted the necessity of firstly sizing the electrical storage in order to have a profitable PV system size, leading to an increased amount of PV production in the residential sector. In the UK, Parra et al. (2016) found that the application of a community energy storage resulted a good solution to facilitate the usage of distributed renewable energy generation and manage the demand loads. Specifically, the authors quantified the performance of lead-acid (PbA) and lithium-ion (Li-ion) batteries performing demand load as a function of the size of the community by using simulation-based optimization. In a similar study, Parra et al. (2015) found that the community energy storage reduced the levelized cost of energy storage by 37% by performing PV energy time shift. Finally, Sardi et al. (2017) presented an analytical framework for community energy storage integration in an existing residential system with rooftop PV unit.

3.3.2 EV Side Modeling

3.3.2.1 EV Demand Modeling

The EVs adopt high-density lithium-ion batteries, which require relatively the least maintenance, less susceptible to memory effect and flexible cycling (Hu 2011). The subcompact EVs use batteries with power capacity 12–18 kW·h, mid-sized family sedans use batteries with capacity 22–50 kW·h, and luxury models, such as those from Tesla, employ batteries with capacity 60–85 kW·h (Battery University 2019). Battery capacity usually degrades over charging/discharging cycles. Factors such as charging rate, environment temperature, battery management, and charging behavior will affect the battery life and hence the EV range. As reported by Hall et al. (2006) 100% discharge of battery should be avoided to ensure maximum battery cycle life, otherwise, battery wear, and calendar fade (i.e., battery performance deteriorates over time whether the battery is used or not) which can occur due to a high state of charge (SOC).

Many studies have been conducted to investigate the EV energy usage patterns and estimate the EV load profiles. Some of the studies extract the EV usage patterns from the real data. For instance, based on a Dutch mobility study, Geth et al. (2010) rebuilt the probability distributions for work traffic, work shifts, and population activity for Belgium. Information about driving trips, such as motivation, time of day, and distance, are collected and used for generating driving profiles, which have a one-minute resolution and are available for both workdays and weekends. Lee et al. (2011) synthesized representative naturalistic cycles of EV behavior through a stochastic process utilizing transition probability matrices extracted from naturalistic driving data collected in the Midwest region of the USA. Using the real data on mobility behavior in Germany 'Mobilität in Deutschland', Fischer et al. (2019) first analyzed the impacts of a set of factors (e.g., the household type household economy status, place of residence, driver occupation, week day, trip index, and trip purpose) on the driving behavior (e.g., car use pattern, car trips per day, trip purpose, distance and driving time, first departure time and parking time). After identifying the most influential factors, they modeled the EV use with an inhomogeneous Markov chain to sample a sequence of destinations of each car trip, depending (among other factors) on the occupation of the driver, the weekday and the time of the day. Lojowska et al. (2011) derived the stochastic characteristics of the EV behavior of vehicles (including start/end-time of each trip, the respective traveled distance and battery state of charge) using a detailed transportation dataset for the Netherlands. Then, a Monte Carlo simulation-based approach is used for calculating the power demand of EVs under the scenario of uncontrolled domestic charging.

Based on the charging behavior of drivers, researchers have also developed advanced decision models of EV charging, such as Markov chain decision model and fuzzy-logic inference model (Shahidinejad et al. 2012). For instance, Moreira et al. simulated the EV movement in one-year period by using a discrete-state, discrete-time Markov chain to define the states of an EV with a time step of 30 min (Soares et al. 2011). At every unit of time, the model assumes the EV to be in one of the four event states: in movement, parked in a residential area, parked in a commercial area, and parked in an industrial area. By combining EV usage with synthetic activity generation of occupants' electricity-dependent activities, Grahn et al. (2013) used a Markov chain model to generate plug-in hybrid electric vehicles' (PHEVs' home charging patterns. The synthetic activity data are simulated based on time-use data collected in time diaries, and it defines the basis for calculating the PHEV home-charging behavior as well as the resident's electricity consumption. Shahidinejad et al. (2012) adopted a fuzzy-logic inference system to emulate the EV battery charging based on a large field-recorded driving database. The fuzzy inference assumes that the state-of-charge (SOC) of the battery and estimated parking duration are the two main factors that govern a driver's decision whether or not to charge when a plug-in vehicle is to be parked, as it is considered that the driver needs to be confident that for the next trip, the battery has adequate SOC.

Researchers have also developed some statistical models based on probability distributions derived from real data. For instance, based on the same EV home-charging model as Grahn et al. (2013), Munkhammar et al. (2015) developed a probability distribution model by merging it with another two separate existing probability distribution models (i.e., for calculating household power consumption and PV power production). Using the probability distribution model, the distributions of the power consumption/production mismatch were investigated on both the household level and aggregated level of multiple households. Darabi and Ferdowsi (2012) extracted probability density functions of the start time of EV charging, the required electrical energy, and required power using the available data from American national household travel surveys. A similar modeling approach is conducted by Remco et al. for the Netherlands scenario (Verzijlbergh et al. 2011) and by David et al. for the Germany scenario (Fischer et al. 2015).

For a large number of EV units, Islam et al. developed a combined SOC-based methodology to calculate day-ahead combined probabilistic charging loads (Islam et al. 2018). Instead of managing the charging rate of every EV separately, their proposed model charges EVs with a lower SOC level at a higher rate, and vice versa. Sharigul Islam et al. (2019) also proposed a correlated probabilistic model for EV charging loads in coordination with a solar PV-rich commercial grid. In their proposed model, correlated samples are first generated from uncorrelated samples containing a wide range of random variables associated with EV loads, PV outputs, battery energy storage powers, and grid loads.

3.3.2.2 Design/Plan of EV Charging Stations

Existing studies have developed a number of methods to optimize the design and plan of renewable powered charging stations in cities. For instance, Luo et al. (2019) developed a comprehensive optimization model concerning the joint planning of distributed generators and EV charging stations, which involves spatially dispatchable characteristic at the planning stage of distribution systems and deploys power devices in a cost-effective way. The proposed model is embedded with the spatial scheduling problem of EV charging demands and focuses on the optimum of relevant social costs. Notably, Huang et al. (2019b) developed a Geographic Information System (GIS)-assisted optimal design method for renewable (i.e., generated by roof-mounted PV systems)-powered EV charging stations in high-density cities. Using the GIS technique, their method first discretizes the studied district into equally sized grids and collects the geographic information of massive buildings in each grid. Then, the roof-based solar energy potential in each grid is estimated. Next, the optimal locations and optimal number of the renewable powered charging stations are searched by the genetic algorithm with the consideration of the existing charging stations and renewable potentials. The design method can be used in practice to help high-density cities build their renewable powered public charging networks with cost-effectiveness. Notably, Morrissey et al. (2016) performed an analysis of the charge event data for public charging infrastructure and household charging stations in Ireland. They found that EV users prefer to carry out the majority of their charging at home in the evening during the period of highest demand on the electrical grid, thus implying that incentivization may be required to shift charging away from this peak grid demand period. They also reported the popularity of using fast chargers in the car park locations and thus concluded that priority should be given to developing a highly connected network of strategically located fast chargers.

3.3.3 Grid Modeling

3.3.3.1 Overall Power Grid Architecture

The renewable energy systems, buildings (as electricity end-users), energy storage system, and EVs are essential players in the smart grid. Based on an extended NIST model (Greer et al. 2014), Chaouachi et al. (2016) proposed a conceptual

architecture of the modern smart grid, from interoperability and functionalities prospective, to enable decentralized operational synergy between intermittent PV generation and EVs, based on coordinated EV charging, as shown in Fig. 3.6. The conceptual architecture consists of several domains (i.e., distribution, distributed energy resource (DER), and consumption) and zones (i.e., enterprise, operation, station, field, and process). The domain dimension is partitioned by the electric distribution conversion chain that includes the distribution operation, DER, and prosumers. The zone dimension refers to the hierarchical system aspects spanning the whole smart grid (Aymen et al. 2016).

- The process zone contains the primary set of equipment associated to the physical layer of the electricity network.
- The field zone contains the auxiliary equipment that are committed to the control and monitoring of the electricity networks.
- The station zone is mainly characterized by the station controllers and Customer Energy Manager (CEM). The CEM optimizes its client's energy consumption and/or production based on signals received from the grid, consumer's settings, HAN, and contracts.
- The operation zone includes the energy management aggregators (EMA) offering services to aggregate energy production and controllable loads.
- The enterprise zone is at the highest hierarchical level. It includes the commercial and organizational processes, as utilities power scheduling, service providers, and energy traders. There can be some overlapping between the operational and the enterprise zones.

In the enterprise and operation zones, all the equipment can interact with each other using the existing technology that is compliant with the relevant communication standards (IEC61968-11 2013) (IEC61970-301:2011 2011). While for the local sub-networks in the field and station zones, which directly integrate the communication channels between EVs, DERs, smart appliances, and controllable loads, decentralized bidirectional communication and more interfacing flexibility are required to coordinate with other entities of the smart grid via high-level

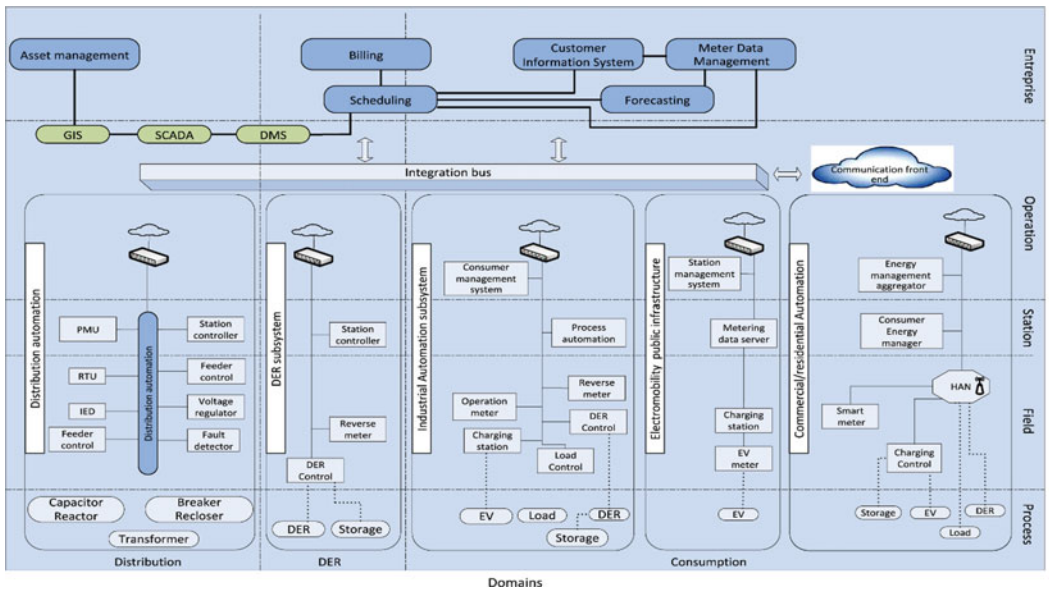

Fig. 3.6 Smart grid conceptual architecture for enhanced solar mobility (Chaouachi et al. 2016)

communication protocols. Such communication architecture would have the potential to convey and relay data through intermediate devices, creating a meshed network without requirement for centralized control (Chaouachi et al. 2016; Aymen et al. 2016).

3.3.3.2 Local Microgrid Structure

Considering the complexity of the future system, a more decentralized operation could simplify the tasks of grid operation by seamless prosumers involvement. In fact, the emerging distribution systems are planned to integrate plug-and-play devices, enabling a set of new functionalities related to the various actors (retailers, DSO, aggregator, prosumers), and implemented through various technologies (generation devices, loads, communication) which will be possible to efficiently exchange information and commands (Bruinenberg et al. 2012). The end-users specifically need to incorporate smart appliances capable of communicating their status and auto-adjusting their operation based on their requirement and/or the DSO/retailers provided set points (when relevant) (Chaouachi et al. 2016).

With reference to a number of studies introducing the new electricity network (Huang et al. 2018c, 2019a; Odonkor and Lewis 2015), a generic power distribution network, which represents the future development trend of micro power network for multiple energy prosumers and the public charging stations, has been summarized in this chapter, as shown in Fig. 3.7. The energy prosumers are equipped with their own renewable energy systems, electrical storage, EVs, and other electrical appliances. The buildings are connected into a renewable energy sharing microgrid, in which the surplus renewable production can be delivered from one building to another. Such renewable energy sharing microgrid provides platform for the buildings in the cluster to share their surplus renewable energy generations with other buildings, thus helping enhance the overall cluster-level performances. The central systems, such as central electrical battery/thermal energy storage and district heating system, are also connected into the energy sharing micro power grid. The microgrid for energy sharing is also connected to the main power grid, in case there is a surplus or insufficient cluster-level renewable generation. The power exchange of the building cluster with the power grid will be metered by advanced metering facilities. The micro power grid can be either a direct current (DC) power grid or an alternating current (AC) power grid. Proper

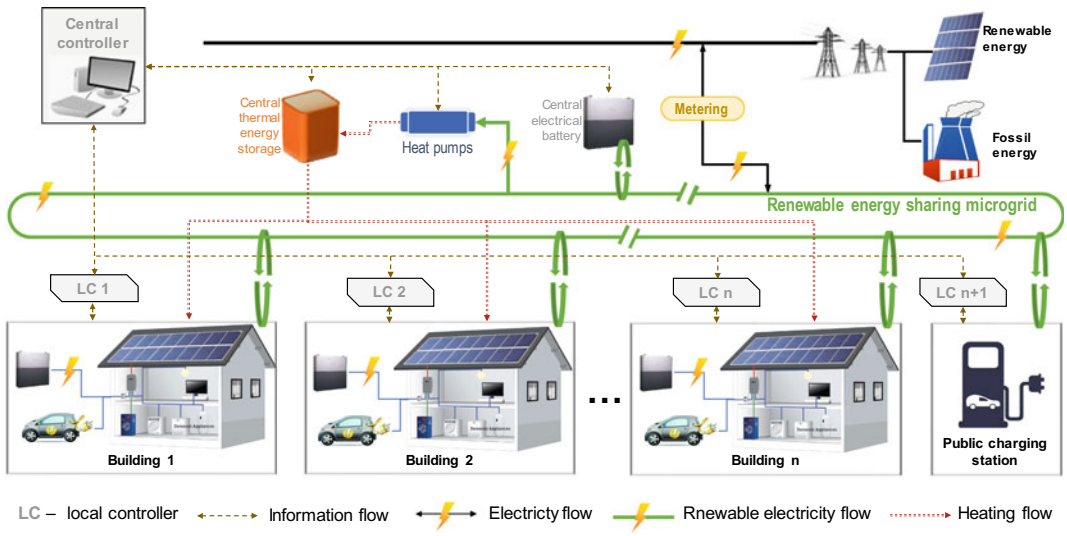

Fig. 3.7 Structure of energy sharing network

inverters/converters are needed corresponding to the different microgrids used.

Each building has a local controller to manage its own storage charging/discharging as well as the flexible electrical loads. The public charging station also has a local controller, which is able to manage and optimize its power flow. All the local controls are connected to a central controller for coordinating the operation of each building and the public charging station and regulating the renewable energy sharing within the micro power grid.

3.3.3.3 Energy Sharing Networks

Renewable energy sharing inside the building cluster is a promising way to enhance the building-cluster-level renewable energy self-utilization rates and reduce the impacts of large building electricity demand penetration on the power grid. To enable such energy sharing, specific energy sharing network is needed for the power transmission between different buildings. The PV panels, the battery storage, and many modern large loads, such as pumps, compressors, fans, servers, and EVs, are often operating with DC power. The DC/AC converting at both the supply side and demand side not only causes dramatic electricity losses but also reduces the system reliability due to increased complexity (Huang et al. 2019a). To address these issues, some researchers have recommended using DC microgrid for renewable energy sharing between buildings, instead of using AC microgrid. Chen et al. proposed a DC microgrid connecting a number of facilities including PV panels, wind turbine, battery storage, and other electrical loads such as EVs (Chen et al. 2013). A bidirectional inverter is used for connecting the microgrid with the public AC network. The installed DC bus has a voltage of 380 V. Each module in the system communicates with the EMS based on RS-485 or ZigBee communication protocol. The EMS will command the modules when to operate and collect operational status.

Ayai et al. (2012) proposed a DC microgrid system as a power network for introducing a large amount of solar energy using distributed PV generation units. The installed DC bus has a voltage of 350 V. They summarized the benefits of using DC microgrid as follows: (1) increase the introduction of distributed PV units; (2) reduce energy dissipation and facility costs resulting from AC/DC conversion by integrating the junction between a commercial grid and DC bus which connects PV units and accumulators; and (3) supply power to loads via regular distribution lines (not exclusive lines for emergency) even during the blackout of commercial grids. Notably, Ferroamp (2018) developed Energy Hub for DC microgrid power sharing. The Energy Hub converts and controls the energy flow in both directions between the DC grid and the facility AC grid. The operating voltage of the DC microgrid is 760 V. Loads that support a nominal DC voltage of 760 V can be powered directly from the DC grid. A minibus DC/DC converter has also been developed to step down the 760 V DC grid voltage to the output voltage required by other DC loads (120–400 V). The communication of Energy Hub is based on TCP/IP protocol.

3.3.4 Advanced Controls

Proper control is essential for the building energy system to achieve good performances. Existing studies have developed a lot for advanced controls for improving the building energy systems' performance and promoting solar mobility. The existing controls can be clarified into two categories: individual controls and coordinated controls. The individual controls focus on single-building's operation and aim to optimize the single-building-level performance, while the coordinated controls focus on coordinating of multiple buildings' operation and take the building-cluster-level performance as the optimization objects. The coordinated controls can be further classified into bottom-up approach and top-down approach, as summarized in Table 3.2. This section introduces the up-to-date controls for promoting solar mobility from these two aspects.

Table 3.2 Classification of the existing energy storage and EV controls related to solar mobility

Category		Principal	Pros	Cons	Examples
Individual controls	Individual approach	The operations of individual energy storage/the charging rates of individual EVs are first optimized separately. Then, the individual energy storage performances/EV charging loads are aggregated to obtain the aggregated-level performance	Easy to implement	1. The aggregated-level performance is not optimized 2. New demand peaks may occur when individual energy storage/EV takes the same action (e.g., shift demand to the same period)	Energy storage controls: Zhao et al. (2015), Lu et al. (2015), Allison, (2017); EV controls: Islam et al. (2018), Shariful Islam et al. (2019), Cai et al. (2019), Wu et al. (2017)
Coordinated control	Bottom-up approach	The operations of individual energy storage/the charging rates of individual EVs are optimized one by one in a sequence, and the optimization of each single energy storage's/EV's operation is performed based on the aggregated results of the earlier optimized energy storage/EVs	Better performance at aggregated level than individual approach	1. The aggregated magnitude becomes increasingly large after many optimizations, leading to the subsequent optimization less effective in improving the aggregated-level performance and non-optimal solution 2. High computational load	Energy storage controls: Odonkor and Lewis (2015), Fan et al. (2018), Prasad and Dusparic (2019); EV controls: Ma et al. (2016), Usman et al. (2016)
	Top-down approach	The aggregated-level performance is directly used as the optimization objective. The operations of individual energy storage/the charging rates of individual EVs are coordinated to achieve the obtained performance at the aggregated level	Optimized performance at aggregated level	The computation complexity increases with the number of energy storage systems/EVs	Energy storage controls: Huang and Sun (2019), Zhang et al. (2018), Huang et al. (2018c); EV controls: Geth et al. (2010), Dallinger et al. (2013), Islam and Mithulananthan (2018)

3.3.4.1 Individual Controls

Battery/thermal storage control: Aiming at reducing the carbon dioxide emission, primary energy consumptions, and operation cost, Zhao et al. developed a model predictive control-based strategy to schedule the operation of the energy systems in a grid-connected low energy building, which include stratified chilled water storage tank, PV system, and distributed power generation units (Zhao et al. 2015). At the start of each

day, the developed method uses nonlinear programming algorithm to schedule the operation of a combined cooling and power system and the thermal energy storage in the coming 24 h under day-ahead electricity prices. Similarly, Lu et al. (2015) proposed an optimal scheduling method of the building energy systems integrated with PV power generation, combined cooling and power system and a thermal storage tank. In each hour, the operation of whole system in the coming N hours (i.e., the time horizon) is optimized by a mixed-integer nonlinear programming algorithm, with the purpose of minimizing the operational costs. With parameter uncertainty (e.g., heat transfer coefficients, operational efficiency) considered, Allison developed a robust multi-objective nonlinear inversion-based control strategy for NZEBs equipped with micro CHP unit, PV, and battery storage, to minimize the NZEB's grid power utilization while fulfilling the thermal demands (Allison 2017). Their developed controller combines the inverse dynamics of the building, servicing systems, and energy storage with a robust control method. The inverse dynamics provides the controller with knowledge of the complex cause-and-effect relationships between the system, controlled inputs, and the external disturbances, while an outer-loop control ensures robust, stable control in the presence of modeling deficiencies/uncertainty and unknown disturbances.

EV charging control: Cai et al. (2019) derived a convex battery capacity loss model from a physically based degradation model, to capture the battery aging cost. Based on the developed model, they further proposed an aging-aware model predictive control approach, which takes account of the battery capacity degradation and its negative impacts on the costs, for optimized operation of sustainable buildings with on-site PV and battery systems. Considering the stochastic charging behavior, Shariful Islam et al. (2019) developed a coordinated EV charging method based on a correlated probabilistic model of EV charging loads. The charging control optimizes the power factors of PV and battery energy storage system to enhance the quality of service (QoS) while minimizing the probability of voltage and current noncompliance (PVCN). The developed control strategy was tested on a three-phase IEEE 37-bus unbalanced distribution system using the real data of vehicles and solar PV. Their developed control is effective in providing more quality of service.

These non-coordinated battery/thermal storage/EV controls can effectively increase the individual building's renewable energy self-consumption, reduce the peak demands as well as electricity costs. However, since these controls focused on individual building's performance optimization, the aggregated performances at building-cluster-level are not optimized.

3.3.4.2 Coordinated Controls

Battery/thermal storage control: Regarding renewable energy sharing among different buildings, Prasad and Dusparic developed a Deep Reinforcement Learning (i.e., a machine learning approach that enables intelligent agents to learn the optimal behavior via trail-and-error)-based method for ZEB community, with the purpose of reducing energy losses due to transmission and storage, and achieving economic gains (Prasad and Dusparic 2019). Fan et al. also developed a collaborative DR control of zero-energy buildings for building group performance improvements, in which the control of each building was conducted in sequence, and the optimization of one building's operation was based on the previously optimized buildings' operation, i.e., the optimization of $(k + 1)$th building's operation is based on the aggregated operation of the 1st to kth buildings (Fan et al. 2018). The genetic algorithm is used in the optimization of each individual building: the daily hourly charging/discharging rates of the battery are set as variables to be optimized, and the economic cost and grid friendliness are set as the objective function. The abovementioned controls can significantly improve the building-cluster performances by proper coordination of individual buildings and enabling energy sharing among them. However, these controls optimize the building cluster performance in a bottom-up way, and they merely perform very limited collaborations among buildings. Aiming at maximizing the benefits of

collaboration among buildings, Huang et al. developed a two-level top-down control for a cluster of buildings equipped with renewable energy systems and electrical energy storage system (Huang et al. 2018c). In their study, the whole building cluster is first considered as a 'virtual' building, and its electrical battery charging/discharging rates is optimized using the genetic algorithm. Then, based on the optimized performances at the building-cluster-level, the operation of every single building inside the cluster is coordinated using non-linear programming algorithm. Similarly, Odonkor and Lewis developed a control method of NZEBs using genetic algorithm and Pareto decision-making based on an adaptive bilevel decision model (with a facilitator agent at cluster level and local systems at single NZEB level) (2015). Considering the dynamic pricing, in Huang and Sun (2019), a similar three-step iterative demand response control algorithm is developed. Taking into account the uncertainty in PV production and renewable energy sharing among different buildings, Zhang et al. (2018) developed a two-stage adaptive robust optimization-based collaborative operation approach for a residential multi-microgrid to derive the scheduling scheme, with the purpose of minimizing the multiple microgrids' operating cost under the worst realization of uncertain PV output.

EV charging control: Regarding EV-coordinated charging control, Ma et al. (2016) proposed a multi-party energy management method for NZEB cluster based on non-cooperative game theory. They proved the existence of Nash equilibrium in the game model and modeled the process for solving the Nash equilibrium strategy as a multi-objective optimization problem (MOP). Their study results show that the proposed method can reduce the total cost of smart buildings by 4.6% and improve the load factor of the smart building cluster from 0.68 to 0.76. Geth et al. (2010) developed a coordinated charging for a number of EVs. In the developed coordinated control, a vehicle owner first indicates the point in time when the batteries should be fully charged. Then, the aggregator collects this information and calculates when each PHEV can start charging. Usman et al. proposed an automated coordinated control of EV fleets, which can plan the charging strategy at the cheaper moments and keep the vehicle charged enough to complete its scheduled trips (Usman et al. 2016). Islam and Mithulananthan proposed a method, which utilizes the measured PV output of a given sample and the supplied historical ramp to predict the PV output of the immediate next sample based on a non-iterative method (Islam and Mithulananthan 2018). Based on the predicted PV output, they developed an SOC-based charging strategy for EVs, which adjust the charging rates of the EV population in the interval between the successive samples.

Taking advantage of its charging/discharging ability, the EVs can be used as flexible voltage or frequency regulation services. Zhong et al. (2014) proposed a coordinated control strategy for large-scale EVs, battery energy storage stations (BESSs), and traditional frequency regulation resources involved in automatic generation control. According to the magnitude of area control error (ACE), i.e., difference between the scheduled and actual power generations within a control area on the power grid and ACE duration, different actions (e.g., operating the BESSs or operating the EVs) will be taken by the controller. With the purpose of regulating the power network voltage, Li et al. (2019b) developed a model predictive control method for a number of EVs. Their study results show that the developed method can effectively regulate the grid power voltage. Meanwhile, according to the size of the connected EVs, the EVs can assist or even replace the traditional reactive power compensation device to maintain the grid voltage within a stable range while satisfying its own charging requirements. Similarly, Jia et al. (2018) developed a coordinated control strategy for EVs and power plants in frequency regulation. They defined a robust stability criterion to determine delay margin of frequency control system. Su et al. (2014) developed a stochastic microgrid energy scheduling method, which aims at minimizing the expected operational cost of the microgrid and power losses by optimally dispatching the EV charging load and scheduling

distributed generators (DGs) and distributed energy storage devices (DESDs). Chaouachi et al. (2016) proposed a conceptual architecture of the smart grid to enable decentralized operational synergy between intermittent PV generation and EVs, based on coordinated EVs charging. Relying upon the proposed smart grid conceptual architecture, they also developed an assessment framework to maximize the renewable electricity and EV penetration for given electricity and transport systems. Dallinger et al. (2013) developed a method to characterize the fluctuating electricity generation of renewable energy sources (RESs) in a power system and compared the different parameters for California and Germany.

3.4 Simulation Platforms and Performance Metrics

This section reviews the simulation platforms and performance metrics for solar mobility studies.

3.4.1 Potential Modeling Platform for S2BVS

This section reviews some modeling platforms for S2VBS systems and categorizes them into three aspects based on their functions: modeling software for the demand/supply of buildings, for powerline/power grid (related to distribution network), and for advanced controls.

3.4.1.1 Modeling Platforms for the Demand/Supply of Buildings

Buildings and PV systems are two vital components in S2VBS systems. This subsection mainly introduces several modeling platforms for the building load profile and PV-generated energy.

- HOMER: Hybrid Optimization Model for Electric Renewable (HOMER) software is a widely used software for both grid-connected and off-grid energy systems for various applications. Based on HOMER, researchers can conduct sensitivity analysis and optimization for systems. They can also process operation and maintenance, capital and interest costs' calculations. In recent years, many studies have adopted HOMER for analyzing and optimizing renewable energy systems. Kumar et al. (2016) applied HOMER to study the feasibility of a stand-alone solar–wind–diesel hybrid power system for an ATM machine in remote areas. Abdul-Wahab et al. employed HOMER to find the best PV system for Oman's conditions (2019). Li et al. (2019a) assessed the feasibility of a hybrid PV/diesel/battery power system in the suburb of Harbin in China using HOMER.
- TRNSYS: Typically, Transient System Simulation (TRNSYS) is adopted for modeling the supply of renewable energy sources, like PVs or wind power, and the use of multi-zone buildings, including HVAC systems or other systems. An advantageous characteristic of TRNSYS is that submodules can be added in programming languages by users themselves, e.g., by Fortran (The university of Wisconsin 2006). Another advantage of TRNSYS is the flexibly to couple with other tools such as MATLAB for co-simulation, which enables optimization (Bava and Furbo 2017). In a recent study, Jonas et al. developed a user-friendly simulation model of solar and heat pump (SHP) systems for people with less professional knowledge (Jonas et al. 2017). Saleem et al. (2019) simulated a solar water heating system and a solar-hydrogen hybrid energy system through TRNSYS. Furthermore, TRNSYS was also used for modeling a building integrated solar thermal system with seasonal thermal energy storage by Antoniadis and Martinopoulos (Antoniadis and Martinopoulos 2019).
- EnergyPlus: EnergyPlus is a whole building energy simulation program that is used for both energy consumption and water use in buildings, especially for the loads of buildings with HVAC systems or multiple thermal zones. It is a stand-alone software that reads input and completes output to text files.

Uygun et al. (2018) studied the influence of building integrated PV solutions on the performance of a residential building in Anatolia by EnergyPlus. Apart from dwelling buildings, Boyano et al. (2013) used EnergyPlus to model the energy consumption traits of office buildings in different climate zones and advised some feasible ways for energy savings of those main energy consumption end-uses. Bingham et al. (2019) studied the optimization of a grid-connected residential building with PV and battery storage systems in EnergyPlus.

3.4.1.2 Modeling Platform for Powerline/Power Grid

This subsection primarily lists some modeling tools related to electricity and distribution.

- OpenDSS: Open Distribution System Simulator (OpenDSS) is a commonly used simulation tool for multiphase distribution systems. Users can define their required components into OpenDSS by setting certain parameters and then gain an extensive range of simulating conditions. Ke et al. (2015) carried out a study by adopting OpenDSS to establish the power distribution system model and the charge/discharge of a future battery energy storage system. Another research studied the power quality measurement of EV battery charging in Finland (Supponen et al. 2016). From the distribution network point of view, they chose OpenDSS to simulate the harmonic current flow of EV charging and form the network. Ahamioje and Krishnaswami (2017) considered the voltage fluctuations and frequency variations generated by high penetration of PV on gird power. Then, they employed OpenDSS with MATLAB to implement advanced inverter functionalities.

3.4.1.3 Modeling Platform for Advanced Controls

This subsection introduces some modeling platforms for implementing solar mobility related controls.

- MATLAB: MATLAB is one of the developed graphical tools in computer technology. Now, the applications of it extends extensively, including the simulation of whole system or a small part. Several different applications of MATLAB, especially, for controlling modules are discussed below. In Papas et al.'s (2018) study, MATLAB was used to simulate the global functioning of a BIPV building with HVAC systems. Another study modeled a stand-alone PV system with the combination of two software, PSIM and MATLAB (Lei et al. 2017). MATLAB was used for simulating the control circuit. Bava and Furbo (2017) developed a co-simulation between TRNSYS and MATLAB for a solar collector field.
- CPLEX: CPLEX is usually for linear programming (Barros and Casquilho 2019). The efficiency and robustness of its algorithm have been demonstrated in solving mathematic problems of multiple areas. In relevant studies, CPLEX has been widely adopted when researchers solve EV routing problem and simulate stochastic distribution of EVs. Zuo et al. (2017) took charge station into account and tried to develop a mathematical model for solving the EV routing problem based on CPLEX. Zakaria et al. (2014) conducted a study to compare CPLEX and a greedy algorithm when given the problem of modeling car relocation. In S2VBS systems, when there is a need to simulate the routes of EVs between buildings and network map of whole system, CPLEX has its advantage in linear programming and processing stochastic data.

There is also other software available (GAMS, Pyomo, etc.) for modeling and optimization. These types of software are not set up specifically for solar mobility problems, so they have significant drawbacks in the amount of effort required in setting up simulations for that purpose. For software that are specific designed for building energy demand/supply modeling, such as EnergyPlus, TRNSYS, and HOMER, there are typically built-in libraries and data available,

which can reduce the model construction efforts. While for software such as MATLAB and CPLEX, they do not have built-in libraries and data available. However, the internal solver, such as genetic algorithm, nonlinear programming, can be used to reduce the efforts in programming the algorithm.

3.4.2 Metrics as Optimization Objectives of S2BVS Models

This section reviews the popular metrics that can be used in the assessment of S2BVS models from three aspects: energy, economy, and environment. Table 3.3 summarizes the major performance metrics for performance assessment of the S2BVS model.

Regarding energy performance assessment, loss of power supply possibility (LPSP) is one of the most commonly used metrics when assessing the possibility of loss power for the whole energy system. Huang et al. (2019d) used LPSP to assess the energy system's reliability. Huang et al. also used a metric, potential energy waste possibility (PEWP), to evaluate the energy usage efficiency of the modeling energy system. Capacity factor (CF) is the most widely used metric for assessing the PV power generation. Adewuyi et al. (2019) analyzed the feasibility of implementing solar power system in Nigerian by assessing the capacity factor. Capacity factor ratio ($r_{cf0.8}$), which is derived from capacity factor, allows for a more detailed analysis of the energy availability. Using this indicator, Dallinger et al. (2013) compared the integration of renewable energy sources (i.e., solar thermal collector, PV systems, and wind turbines) in both California, USA, and Germany. The self-consumption (SC) is the annual average of the rate at which the electricity produced by the PV system is consumed on-site (Luthander et al. 2015). Huang et al. (2019a) used the SC to evaluate the performance of a building cluster with shared PV system and centralized heating system. Munkhammar et al. (2013) analyzed the PV power self-consumption rates under different levels of EV penetrations. Another indicator closely related to the SC is self-sufficiency (SS) (Munkhammar et al. 2013). Boeckl and Kienberger (2019) used SS to evaluate PV-storage systems and tried to maximize the SS with the smallest possible system capacity, with the aim of maximizing the usage of resources.

Regarding economic performance assessment, cost of energy (COE) is also the simplest economic indicator of PV systems. Sabah et al. calculated the COE of 15 types of PV system in 25 different locations inside a region in Oman and then determined the best PV system site and the best type of PV system for a village based on the calculated COE (Abdul-Wahab et al. 2019). Net present value (NPV) evaluates the profitability of the investment (Barone et al. 2019). O'Shaughnessy et al. (2018) analyzed the improvements in PV system NPV using a renewable energy optimization model. Heine et al. (2019) used NPV as the indicator to search the optimal capacity of battery in achieving the best economic performance. Based on NPV, it is further possible to calculate the internal rate of return (IRR), being a discount rate that makes NPV of all cash flows equal to zero (Murphy and Fung 2019). Profitability index (PI) also helps to assess the profitability of the investment and even rank the proposed layouts by quantifying the amount of value created per unit of investment (Barone et al. 2019). Barone et al. adopted PI to assess the profitability of making profits for an investment. To compare electricity prices across different technologies, such as fossil fuel-based grid power, stand-alone PV systems, levelized cost of electricity (LCOE) is commonly applied. Kästel and Gilroy-Scott (2015) compared electricity prices for wind and PV technologies based on LCOE. For a system including EVs, a separate energy storage system and renewable energy supply, total operation cost (C_{total}) is a common parameter in economic aspect. In view of the total operation cost, Sun et al. (2019) developed an operational cost minimization model for a residential energy system consisting of an EV, an energy storage system, a PV system, and other residential loads. Another

Table 3.3 Summary of the key performance metrics for performance assessment of the S2BVS model

Name	Calculation	Remarks in calculation	Representation/Meaning
Energy-related indicators			
Loss of power supply possibility (LPSP) (Huang et al. 2019d)	$\mathrm{LPSP} = \frac{\sum_{t=1}^{T} \mathrm{LPS}(t)}{\sum_{t=1}^{T} P_{\mathrm{load}}(t)}$	LPS: loss of power supply; P_{load}: the load demand power	It is defined as the ratio of the loss of energy supply to the load demanding during the considered period
Potential energy waste possibility (PEWP) (Huang et al. 2019d)	$\mathrm{PEWP} = \frac{\sum_{t=1}^{T} [P_{\mathrm{PV}}(t) - P_{\mathrm{load}}(t) - P_{\mathrm{bat_c}}(t) - P_{\mathrm{ele}}(t)]}{\sum_{t=1}^{T} P_{\mathrm{PV}}(t)}$	P_{PV}: the power of PV array; P_{load}: the load demand power; P_{bat}: the power of retailed EV batteries; P_{ele}: the power of the electrolyzer	It is defined as the ratio of the excess power to the potential PV output power during the considered period
Capacity factor (Cf) (Adewuyi et al. 2019)	$\mathrm{Cf_{pv}} = \left\{ \frac{\sum_{t=1}^{8760} P_{\mathrm{pv}}(t)}{8760 \times P_{\mathrm{pvrated}}} \right\}$	$P_{\mathrm{PV}(t)}$: the generated power at time t; P_{pvrated}: assumed PV generator capacity	It measures the energy production efficiency of the facility over a period based on the solar potentials in local site
Capacity factor ratio (r_{cf}) (Dallinger et al. 2013)	$r_{\mathrm{cf0.8}} = \frac{\mathrm{cf}_{\mathrm{Quantile}<0.8}}{\mathrm{cf}_{\mathrm{Quantile}\geq 0.8}}$	$\mathrm{Cf}_Q < 0.8$: sorted power values smaller than the 0.8 quantile; $\mathrm{Cf}_Q \geq 0.8$: sorted power values equal to and bigger than 0.8 quantile	It allows for a more detailed analysis of the energy availability
Self-sufficiency (SS) (Boeckl and Kienberger, 2019)	$\mathrm{SC} = \frac{E_{\mathrm{pv,onsite}}}{E_{\mathrm{pv,onsite}} + E_{\mathrm{pv,offsite}}}$	$E_{\mathrm{pv,onsit}}$: the aggregated PV power consumed on-site during a period; $E_{\mathrm{pv,offsit}}$: the aggregated PV power consumed off-site (e.g., exported to grid)	It shows how much less energy is needed from the conventional power mix from the grid
Self-consumption (SC) (Barone et al. 2019)	$\mathrm{SS} = \frac{E_{\mathrm{pv,onsite}}}{E_{d,\mathrm{whole}}} = \frac{E_{d,\mathrm{pv}}}{E_{d,\mathrm{pv}} + E_{d,\mathrm{grid}}}$	$E_{\mathrm{d,pv}}$: the aggregated electricity demand supplied by the PV system during a period, and it is equal to $E_{\mathrm{pv,onsite}}$; $E_{d,\mathrm{grid}}$: the aggregated electricity demand supplied by the power grid. The sum of them equals $E_{d,\mathrm{whole}}$	It represents the percentage electricity produced by on-site PV system used within the building

(continued)

Table 3.3 (continued)

Name	Calculation	Remarks in calculation	Representation/Meaning
Economy-related indicators			
Cost of energy (COE) (Abdul-Wahab et al. 2019)	$\mathrm{COE} = \frac{\mathrm{Cost_{tot}}}{E_{\mathrm{tot}}}$	$\mathrm{Cost_{tot}}$: Total cost for generation of energy for one year; E_{tot}: Total energy generated in one year	It calculates the unit cost of generated energy
Net present value (NPV) (Barone et al. 2019)	$\mathrm{NPV} = \sum_{i=1}^{N} \frac{\mathrm{EcS}}{(1+d)^i} - \mathrm{IC}$	EcS: yearly economic savings; IC: the total capital investment cost; d: discount rate	It is calculated as difference between the present value of cash inflows and the outflows over a period
Internal rate of return (IRR)	$\mathrm{NPV}\vert_{d=\mathrm{IRR}} = 0 \rightarrow \mathrm{IRR}$	NPV: Net present value; d: discount rate	It is a discount rate that makes the NPV of all cash flows from a project equal to zero
Levelized cost of electricity (LCOE) (Kästel and Gilroy-Scott 2015)	$\mathrm{LCOE} = \frac{I_0 + \sum_{t=1}^{n} \frac{A_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{M_e}{(1+i)^t}}$	I_0: the investment; M_e: the electricity output in year; A_t: the annual total costs; i: the interest rate (discount rate); n: the economic lifetime in years; t: the year of operation (1, 2, …, n)	It is based on the concept that all costs over the lifetime of an energy project are discounted to their net present value in a money unit divided by the discounted energy production
Total operation cost (C_{total}) (Sun et al. 2019)	$C_{\mathrm{total}} = C_{\mathrm{purchase}} + C_{\mathrm{EV}} + C_{\mathrm{ESS}} + C_{\mathrm{EV-outside}} - C_{\mathrm{income}}$	C_{purchase}: cost to purchase electricity from the grid; C_{EV}: degradation cost of the EV battery; C_{ESS}: cost of ESS battery; $C_{\mathrm{EV\text{-}outside}}$: degradation cost of EV battery due to driving; C_{income}: income from selling electricity to the grid	It calculates the residential household energy costs by considering EV with driving usage, EV and ESS battery degradation, and PV energy supply

(continued)

Table 3.3 (continued)

Name	Calculation	Remarks in calculation	Representation/Meaning
Annualized cost of system (ACS) (Huang et al. 2019d)	$\mathrm{ACS} = C_{\mathrm{acap}} + C_{\mathrm{amain}} + C_{\mathrm{arep}}$	C_{acap}: the annualized capital costs of PV panels, retailed EV battery (REVBs), electrolyzer, and hydrogen tank; C_{amain}: the annualized maintenance costs of the above components; C_{arep}: the annualized replacement cost of REVBs, FC, electrolyzer	It calculates the total cost of the whole systems per year
Profitability Index (PI) (Barone et al. 2019)	$\mathrm{PI} = \frac{\sum_{i=1}^{N} \frac{\mathrm{EcS}}{(1+d)^i}}{\mathrm{IC}}$	EcS: yearly Economic Savings; IC: the total capital investment cost; d: discount rate	It assesses the profitability of the investment and rank the proposed layouts by quantifying the amount of value created per unit of investment
Environment-related indicators			
Well to wheels (WTW) (Querini et al. 2012)	$\mathrm{WTW} = \left[\begin{array}{c} \left(\text{energy prod. GHG}\left(\mathrm{MT}^{-1}\right) \times \text{energy cons.}\left(\mathrm{MJ.distance}^{-1}\right)\right)_{\mathrm{WTT}} \\ + \left(\text{exhaust GHG}\left(\mathrm{distance}^{-1}\right)\right)_{\mathrm{TTW}} \end{array} \right]_{\mathrm{WTW}}$	*Energy prod. GHG*: greenhouse gas emission during energy production; *Energy cons.:* Energy used for traveling a specific amount of distance; *Exhaust GHG:* pollutant emissions by a car on a given distance	It calculates the well-to-tank (WTT) stage which covers the production of required energy and the tank-to-wheels (TTW) stage covering the consumption of energy and pollutant emissions by a car on a given distance
Carbon intensity (CI) (Miller et al. 2019)	$\mathrm{CI} = \frac{e_{\mathrm{total}}}{\mathrm{CF} \times c_{\mathrm{DC}} \times t_{\mathrm{hr}}}$	e_{total}: life cycle GHG emissions (gCO_2e); CF: capacity factor; c_{DC}: rated DC capacity; t_{hr}: PV system lifetime	It calculates the carbon emission of unit energy consumption regarding the life cycle assessment

commonly used index for economic analysis is the simple payback period (SPB). Barone et al. (2019) calculated SPB for a system renovation under different scenarios, including Vehicle-to-Building system (V2B) and Building-to-Vehicle-to-Building (V2B2).

Regarding environmental performance assessment, for EVs with different electricity technologies, greenhouse gas (GHG) emission is the major concern. The same as the concept in life cycle assessment, well-to-wheels analysis (WTW) is an environmental metrics, especially for EVs (Querini et al. 2012). Querini et al. used WTW analysis to calculate GHG emissions of EVs, and the results showed that EVs with PV electricity always released less GHG than conventional thermal vehicles (Querini et al. 2012). Another commonly used environmental footprint indicator is carbon intensity (CI). In a recent study, Miller et al. built a performance model to estimate PV power's CI under diverse scenarios and analysis the impacts of distinct influencing factors (Miller et al. 2019).

3.5 Future Directions

Advanced building coordinated controls with EV regulating considered: As reviewed in Sect. 3.4, existing studies have developed a number of controls for both the electrical/thermal energy storages and the EVs. These controls can be divided into individual controls (i.e., focusing on single-level performance) and coordinated controls (i.e., focusing on multiple-level performance). However, with the concern of complexity, the existing controls rarely integrate the EV operating controls in the whole building energy system management. The flexible charging/discharging capability of EV batteries are not fully exploited, which restrains the cluster-level performance in renewable energy utilization. Future work is needed to develop more comprehensive and advanced controls, which can fully deploy both the electrical/thermal energy storages and the EVs.

Alternating current (AC) power or direct current (DC) power: In the recent decades, the modern solutions have raised a number of DC loads (e.g., pumps, compressors, servers, EVs, etc.), and most of the renewable-based distributed generation units directly produce DC output power. Considering the increased energy efficiency, the improved system reliability, and reduced complexity in DC microgrid, researchers have reconsidered the DC power grid for application, instead of AC power. As introduced in Sect. 3.3, most of the existing energy sharing networks are based on DC power. However, not all these redesigning procedures of DC power grid are accomplished until now. More researches are needed to make such advanced systems a reality in a large scale. In addition to the technical development of DC microgrid and related controls, great efforts are also needed in legislation to promote the real implementation of the energy sharing DC microgrid, since the legislation on DC power microgrid and the energy sharing among different buildings are still unclear in many countries.

Proper plan of energy sharing building clusters: The benefits brought by renewable energy sharing among different buildings, e.g., increased renewable energy self-utilization and reduced energy storage capacity required, are greatly affected by the type of buildings in the cluster (Huang et al. 2019b). For instance, there can be more renewable energy sharing between a residential building an office building, compared with the sharing between two residential buildings. Therefore, the building cluster, in which energy sharing is enabled, should be well planned to maximize the benefits brought by renewable energy sharing. Besides the energy characteristics, geographical locations of the buildings should also be considered to avoid large energy loss due to the long-distance power transmission. Future work is needed to develop such building cluster plan methods.

Proper pricing strategy for promoting solar mobility: The electricity pricing strategy has large impacts on the renewable energy flow, as well as the solar mobility (Huang et al. 2019c). There are two sets of electricity prices involved in this context: the prices of electricity purchasing from or selling to the power grid and the

prices of electricity purchasing/selling within the energy sharing building neighborhood. The latter price is proposed due to the developing trend of building cluster and energy sharing concept. A proper pricing strategy should be able to provide incentives for households when they share their surplus solar power generations (Lovati et al. 2020). Future work is needed to develop such proper pricing strategy.

Communication protocols: The imbalance between power demand and supply may cause lines congestion, overload, reverse power flow, voltage-VAR deviations, and excessive phase unbalances. To mitigate such conditions and guarantee system integrity within its technical limits, clear monitoring structure and management protocols need to be defined and implemented. In fact, as illustrated in Fig. 2.6, as EV and DER can belong to different domains and subdomains (i.e., DER, consumption, industrial, commercial and residential), the domain affiliation implies different energy management strategies and communications support. To ensure reliable and safe information exchanges between the facilities and demands in different domains and fields, future work is also needed to develop proper communication protocols.

3.6 Summary

This chapter has conducted a systematic review of the existing studies related to the solar energy, building, EVs, energy storage system, and energy sharing concept for promoting the renewable energy utilization and solar mobility. The up-to-date studies about the solar-to-building, vehicle, and storage (S2BVS) in essential components, including the building side modeling, EV modeling, grid modeling, and advanced controls, have been reviewed and summarized. Such solar mobility model can fully exploit the potentials of PV system, energy storage system, EVs, and energy sharing network as well as advanced controls, to achieve an optimized performance at microgrid level (e.g., increased renewable self-utilization and autonomy). Next, the platforms that can be used in the S2BVS modeling framework have been reviewed from aspects of demand/supply modeling, powerline/power grid modeling, and control implementation. The related performance indicators in the S2BVS modeling framework have been reviewed from aspects of energy, economy, and environment. Last, the remaining research gap in S2BVS techniques and improving solar mobility have been identified. Future work is needed to develop advanced building coordinated controls with EV regulating considered, to make a selection between the AC power or DC power microgrids, to develop proper plan method of energy sharing building clusters, to develop proper pricing strategy for promoting solar mobility, and to develop proper communication protocols. The study can help improve the existing solar mobility concept, so as to enhance the renewable energy utilization efficiency corresponding to the future scenario with increased PV capacity, EV number and storage capacities.

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Data Centers as Prosumers in Urban Energy Systems

4

Xingxing Zhang and Pei Huang

Abstract

As large energy prosumers in district energy systems, on the one hand, data centers consume a large amount of electricity to ensure the Information Technologies (IT) facilities, ancillary power supply, and cooling systems' work properly; on the other hand, data centers produce a large quantity of waste heat due to the high heat dissipation rates of the IT facilities. To date, a systematic review of data centers from the perspective of energy prosumers, which considers both integration of the upstream green energy supply and downstream waste heat reuse, is still lacking. As a result, the potentials for improving data centers' performances are limited due to a lack of global optimization of the upstream renewable energy integration and downstream waste heat utilization. This chapter is intended to fill in this gap and provides such a review. In this regard, the advancements in different cooling techniques, integration of renewable energy and advanced controls, waste heat utilization and connections for district heating, real projects, performance metrics and economic, energy, and environmental analyses are reviewed. Based on the enormous amount of research on data centers in district energy systems, it has been found that: (1) global controls, which can manage the upstream renewable production, data centers' operation, and waste heat generation and downstream waste heat utilization are still lacking; (2) regional climate studies represent an effective way to find the optimal integration of renewable energy and waste heat recovery technologies for improving the data centers' energy efficiency; (3) the development of global energy metrics will help to appropriately quantify the data center performances.

Keywords

Data center · District energy system · Renewable energy · Waste heat recovery · Energy efficiency

4.1 Introduction

The rapid increase of needs for data processing, data storage, and digital telecommunications has led to dramatic increase in the data center industry (Whitehead et al. 2014). Data centers are buildings, dedicated spaces inside a building or a group of buildings that house the Information Technologies (IT) equipment used for processing

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and storage of data and communication networking (EPA 2007) and the associated extensive supporting infrastructures to power and cool the IT equipment. Specifically, IT and HVAC are the two major energy end-user equipment. IT equipment needs to operate continuously without a stop nearly every day, consuming a huge energy quantity and producing very high internal loads. The HVAC systems in return consume energy to maintain the proper working environment temperature for normal functioning (Ashrae 2011). The statistics indicate that data centers now consume about 3% of the global electricity supply and account for about 4% of total greenhouse gas (GHG) emissions (Koomey 2011). Therefore, with the growing needs of Internet, storage, and communication, the energy consumption of data centers is expected to increase, leading to high operational costs and environmental problems in the near future (Aksanli et al. 2011). As an example, a recent report of 2018 from Cushman and Wakefield predicted annual growth of data centers to be 12–14% over the next two to five years, resulting in 1/5 of global electricity consumption by 2025 (Andrae 2017). Therefore, with the concerns of environmental pollution, fossil fuels' shortage, and increasing gray energy costs, applying renewable energy has gain popularity in data centers in the past decades.

Some studies have been conducted to explore the integration of renewable energy in data centers in order to reduce their carbon footprint and costs. For instance, Sheme et al. (2018) investigated the feasibility of using renewable energy to power data centers in 60° north latitude in terms of energy and cost savings. In fact, even though the higher latitudes can assure low cooling costs due to the cold climate, the high variance in solar output makes data centers more fossil fuel than renewable energy-based grid power dependent. By that end, the authors developed a method for determining the optimal ratio of wind turbine and photovoltaic (PV) panel capacities that can maximize the on-site renewable energy generations in a high latitude location. Moreover, in order to maximize the utilization of renewable energy, many advanced demand controllers and schedulers have been developed. It was found that the combination of solar energy and wind sources provided greater surplus hours compared to using only one source of renewable energy. Regarding the costs' saving, the use of only solar energy source cannot produce significant saving values, as fossil fuel-based energy is provided to the data center when the solar energy is lacking. By mixing solar and wind energy to compensate the lack of wind, a less but a more stable income can be produced. With the aim to overcome the green energy dependency to the environmental changes, Aksanli et al. (2011) developed a data center demand response strategy which is able to optimize the usage of green energy sources. This new job scheduling methodology will cancel or reschedule jobs whenever the instant green energy availability is low and thus is reducing data center dependency form gray energy during scarce green energy availability. Similarly, Goiri et al. (2015) developed a scheduler named GreenSlot, which predicts the near-future solar energy generations and then schedules the data center workload to maximize the utilization of renewable energy while meeting the job's deadlines. By using this scheduler, the data center consumed significantly more green energy by lowering gray energy costs. Therefore, it was found that thanks to GreenSlot scheduler, the datacenter's solar array could be amortized in 10–11 years, compared to 18–22 years required to amortize those cost under the conventional or even energy-aware schedulers. Integrating renewables with data center has been put into practice nowadays, and there are already some well-known IT companies building new data centers that are partly or fully powered by renewable energy. For instance, Apple built a 40 MW solar array for its North Carolina data center in order to provide it additional 17.5 MW of power (Apple). Facebook constructed a solar-powered data center in Oregon, and other three new utility-scale solar projects in Utah and New Mexico are expected to be realized in the near future (Anonymous).This new capacity will help Facebook to entirely power its operation with renewable energy, by reducing GHG emissions (related to its operation) by 75% by 2020. HP has

used a biofuel-based gas turbine to supply the demand in its net-zero data center making the entire system less fossil fuel-based grid power (Arlitt et al. 2012b).

Together with the increasing effort to integrate renewable energy sources to produce electricity in order to drive IT and HVAC equipment, there is a growing effort to capture and reuse waste heat of electronic facilities in all types of energy conversion systems. In fact, the amount of data centers' waste heat is large, and it has been estimated that 68% of it can be recovered. However, due to a low cooling temperature, the grade of data centers' waste heat is usually low, which is a major hurdle to its large-scale applications. By that end, the utilization of the waste heat from data centers is becoming easier with the advancement of heat pump and other low-temperature heat recovery technologies.

In this regard, many studies have been conducted to explore new systems and ways to recover and reuse the waste heat from data centers. For instance, through conducting numerical simulations in TRNSYS, Oró et al. (2019) analyzed the energy and economic feasibility of applying an air-cooled data center's waste heat in district heating (DH) networks for improved energy efficiency. Specifically, two types of cooling solutions (CRAH + chiller and rear-door technology) for air-cooled data centers were numerically evaluated, and for both of them, different waste heat recovery solutions were provided. It was found that the Energy Reuse Factor (i.e., reused energy divided by power consumed by the data center) resulted the best metric to quantify the heat reuse integration in data centers. This metric resulted equal to 55% for heat recovery in the condenser of the vapor-compression chiller, while between 25 and 45% for heat recovery in the return hot aisle. Similarly, Davies et al. (2016) explored the use of heat pumps to boost a data center's waste heat temperature to meet the DH requirements and also analyzed the feasibility of applying the data center's waste heat for DH in London. It was found that coupling 3.5 MW data center with a heat recovery system could lead to savings of over 4000 tons of CO_2e and nearly £1 million per annum. Finally, also MikkoWahlroos et al. investigated the utilization of a data center's waste heat for district heating in Espoo of Finland and analyzed the overall system efficiency from the perspectives of both the data center and district heating network operators (by considering them as one system). Owning to the waste heat utilization, the operating hours of both the combined heat and power plants and heat-only boils were largely reduced. Yu et al. (2019) conducted a simulation study on a novel heat recovery system from a data center in Harbin of China to serve the subsidiary buildings. The main novelty of this system relies on the fact that it is able not only to recover the waste heat but also to shift between space cooling and heating for secondary buildings such as apartment, offices, fitness centers). They found that this new heat recovery system had a better economic viability, with total operational costs 458.3 thousand yuan lower than the one with air source heat pump. Haywood et al. (2012) investigated the thermodynamic feasibility of recovering a data center's waste heat, by using water as heat transfer fluid, to drive an absorption chiller. The main feature of this technology is its capability to relive the cooling load on conventional date center's air conditioner, thanks to the utilization of the waste heat to drive an abortion system. By this way, a very efficient power usage effectiveness (PUE) ratio (less than 1) can be achieved. Marcinichen et al. (2012) developed a novel hybrid two-phase cooling cycle for direct cooling of the chips and auxiliary electronics in data centers. The main advantage of using two-phase microchannel flow relies in the fact that the latent heat capacity of the fluid is more effective, in the heat removal process, than using sensible heat of single-phase fluid. Based on the developed cycles, they explored applying the waste heat recovered from the condenser in a feed-water heater of a coal power plant and found that over 2.2% improvements in the power plant thermal efficiency can be achieved. Deymi-Dashtebayaz and Valipour-Namanlo (2019) investigated the feasibility of reusing the waste heat from a data center by employing an air source heat pump for space heating of an adjoining office building in

Mashhad city of Iran. Specifically, the heat pump has the dual benefit of providing space heating and removing the heat generated by the data center. Their analysis results showed that the waste heat recovery system could achieve significant reduction in natural gas and electrical energy usage, thus reducing the economic costs and CO_2 emission. However, it was found that the main limitation affecting this system is related to its capital cost that could be reduced by utilizing more efficient heat transfer techniques.

To date, a number of studies have been conducted to review the interaction of data centers with the surrounding district energy systems. This means that data centers actively participate as energy consumer and producer to the district energy metabolism. Considering data centers as energy producers in the district energy systems, most of the review papers have been focused on their capability of providing energy (e.g., domestic space and water heating) which is extracted and recovered during cooling process. As an example, Davies et al. (2016) reviewed the main available cooling techniques (including air-cooled systems and liquid cooled systems) which produce a large amount of waste heat and the potential technologies available for its recovery and reuse. Similarly, Ebrahimi et al. (2014) conducted a more comprehensive review of the data center cooling techniques (including air-cooled systems, liquid cooled systems, and two-phase cooled systems). The operating conditions and the associated waste heat temperatures of different systems were analyzed, and various heat recovery technologies were introduced and compared. Nadjahi et al. (2018) conducted a systematic review of thermal management and innovative cooling strategies in data centers. They classified the data centers' waste heart reuse as passive cooling techniques and summarized several of the most promising applications, including domestic heating, organic ranking cycles, and absorption chillers. Similarly, Zhang et al. (2018) reviewed the recent advancements on thermal management and evaluation metrics for data centers, as well as the energy conservation solutions including free cooling and waste heat recovery. Considering the data centers as consumers in the district energy system, most of the review papers focused on the analysis of all the studies aimed at studying the implementation of renewable energy resources to decrease the total energy operational demand. As an example, Oró et al. (2015) conducted a comprehensive review with the goal of analyzing the integration of renewable energy in different data center infrastructures, providing an instrument that could be used by researcher and investors in order to find the best combination based on renewable energy source and capital value. Rong et al. (2016) reviewed the progress of energy-saving technologies in high-performance computing, energy conservation technologies for computer rooms, renewable energy applications, and performance metrics for data centers. With energy consumption, cost reduction, and environment protection considered, they proposed a set of strategies to maximize data centers' efficiency and minimize the environmental impact. It was found that together with the optimization of resources scheduling algorithm and management strategies, renewable energy could effectively reduce the overall energy consumption of data centers. Shuja et al. (2016) reviewed cloud data centers from various aspects, including renewable energy integration, virtual machine migration, etc., to survey the enabling techniques and technologies. They also presented case studies that demonstrate favorable results for sustainability measures in cloud data centers.

To sum up, through the analysis of the available literature, it has been found that most of data centers infrastructure are located in urban areas (Oró et al. 2015). This means that they interact with the surrounding district energy system as both energy consumer and energy producer. In the first case, most of the research work focused on reducing the power usage of cooling systems and powering data centers by using renewable energy sources. In the second case, most of the effort was put on finding the most efficient way to recover and reuse the waste heat in different types of energy conversion systems. However, the potentials in performance improvements are limited due to a lack of global

consideration and optimization of the upstream renewable energy supply and downstream waste heat utilization. For example, most of the developed controls schedule the data centers' energy usage profile to match the renewable energy generations (e.g., large productions during daytime when PV panels are used), producing large amount of waste heat during daytime, which may not match the heating needs in the downstream heat consumers (e.g., district heating network, see Fig. 4.1) (Wahlroos et al. 2017).

Therefore, more efforts should be put on considering data centers as energy prosumers, which can trade off energy demand and supply within a district energy system while ensuring energy efficiency at facility level and reducing operational costs and GHGs emissions (Balaras et al. 2017). This chapter conducts a systematic review of data centers from the perspective of energy prosumers with both the upstream renewable energy integration and downstream waste heat utilization considered. By providing the engineers/researchers a full picture of data centers in the district energy systems, this chapter aims to seek new opportunities for improving data centers' overall energy efficiency and reducing GHG emissions.

This chapter first introduces the basics of data centers, including the major IT components, required indoor environments, and IT facility heat dispassion rates. Then, the various cooling technologies used in data centers are presented. Next, the integration of renewable energy in data centers is reviewed, including different ways of integration, and the up-to-date advanced control algorithms associated with the utilization of renewable energy. After that, the data centers' waste heat recovery and reuse technologies are reviewed. The different locations for data centers' waste heat recovery and various ways of waste heat reuse are reviewed. As this chapter reviews the data center as part of the district energy systems, utilization of the data centers' waste heat for DH is reviewed separately from other waste heat reuse technologies. In fact, the authors believe that waste heat recovery for DH is a promising way to connect the data centers with district energy systems, especially in Nordic countries, which are characterized by a high share of renewables in their DH network due to the use of wood fuels. Some project examples that integrate renewable energy utilization or waste heat utilization are presented. Last, the related economic and environmental analyses for data centers are summarized. Challenges and future work for data center study are finally provided. The major contributions of this chapter are highlighted as follows.

- Review the up-to-date techniques in the data centers' demand response control for increasing renewable energy utilization.
- Review the application of data centers' waste heat for district heating, including the locations for waste heat recovery, the connection architecture, and the connection at both sides.
- Review some successful projects that consider either upstream renewable energy integration

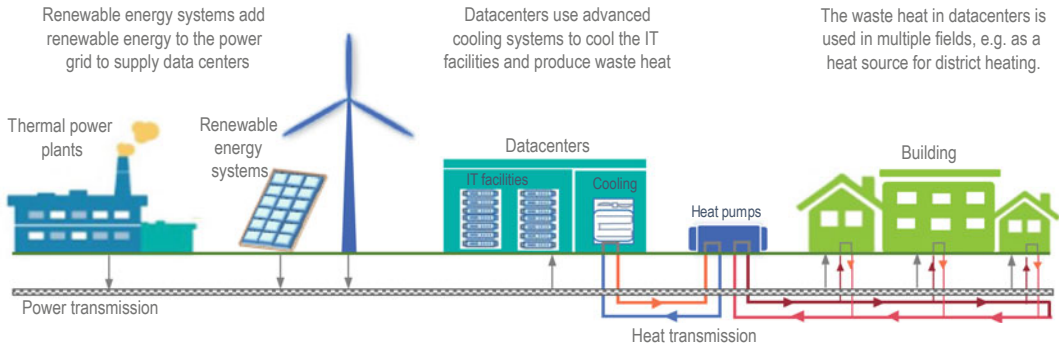


Fig. 4.1 Data centers as prosumers in the district energy systems

or downstream waste heat reuse in district heating or consider both of them.
- Analysis the performance of data centers as prosumers in the district energy system from aspects of energy, economy, and environment.
- Identify the future research directions, which can help to improve the data center's overall performances as energy prosumers in district energy systems.

The review method is briefly introduced as follows. The references, which are directly or indirectly relevant for the data centers as prosumers in district energy systems, have been selected. These references include journal papers, conference papers, design manuals, handbooks and reports. The reviewed literatures were all written in English. Through preliminary searches, we have compared databases and chosen two that we found to have generated, especially relevant results: ScienceDirect and IEEE. Combined searches have been conducted based on two or more of the following key words: data center, energy, green, sustainable, waste heat recovery, DH. The references in closely related literature have also been checked.

4.2 Data Center Overviews

4.2.1 Physical Organization

A data center is a repository for data and information storage, management, and dissemination organized around a particular body of knowledge or related to a particular business (Uddin et al. 2013). A data center is a huge building placing various minor physical components such as racks, storage devices, and switches, while the major one is the server, which is used to store, analyze, and transmit enormous data. In a typical data center, the space is divided into three main areas: IT rooms, IT support areas, and the ancillary spaces. The IT rooms house the equipment and cabling directly related to the computer and telecommunication systems. The IT support area houses the power distributing systems (e.g., UPS), switch boards, and the cooling systems. The ancillary spaces are mainly the offices, lobby, and restrooms.

4.2.2 Environmental Requirements

Data centers contain large amounts of IT equipment, such as the computers, storages, racks, network equipment, as well as supplementary devices such as monitors, workstations. The normal operation of these IT devices imposes high requirements on the temperatures, humidity, and air quality of the thermal environment. A poor control of the indoor environment can cause low computing efficiency or even severe faults/outages. A high humidity could cause condensation on the equipment surfaces, while a low humidity can lead to electrostatic discharges. Table 4.1 summarizes the four classes of thermal environment specified by the 2015 ASHRAE thermal guidelines (American Society of Heating et al. 2015), and Fig. 4.2 presents the associated

Table 4.1 Summary of the 2015 ASHRAE thermal guidelines for data centers (American Society of Heating et al. 2015)

Class	Dry-bulb temperature	Humidity range	Maximum Dew Point
Recommended (suitable to all four classes)			
A1–A4	18–27 °C	−9 °C DP–15 °C DP (60% rh)	
Allowable			
A1	15–32 °C	−12 °C DP (8% rh)–17 °C DP (80% rh)	17 °C
A2	10–35 °C	−12 °C DP (8% rh)–21 °C DP (80% rh)	21 °C
A3	5–40 °C	−12 °C DP (8% rh)–24 °C DP (85% rh)	24 °C
A4	5–45 °C	−12 °C DP (8% rh)–24 °C DP (9s0% rh)	24 °C

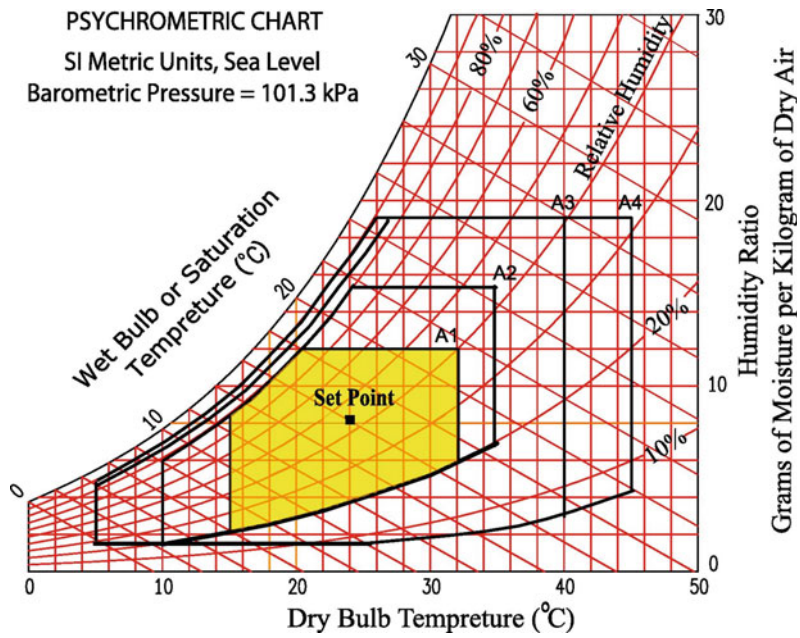


Fig. 4.2 ASHRAE thermal guides for data center operating environments (Ashrae 2011)

area of each class in the psychrometric chart. The inlet air into the IT equipment room should meet the values in the table and figure.

Besides temperature and humidity, air pollution also has significant impacts on the stable and reliable operation of the data centers. The air pollutants can accelerate the corrosion of metals together with moisture. The detailed guidelines for data center air pollutants can be found in (ASHRAE whitepaper 2009).

4.2.3 Heat Dissipation Rates of Components

There are significant differences in temperature between different electronic components held in the IT server racks inside the data center. Consequently, the heat dissipation rates of different electronic components are different. A summary of the heat and temperature distribution of different components in the server is presented in Table 4.2 (Brunschwiler et al. 2009). Different types of servers have different temperatures and dissipate different proportions of waste heat, leading to different waste heat temperatures and heat densities. Note that, with the advancement of materials and techniques, the temperatures and proportions of total heat can vary.

4.3 Cooling Systems in Data Centers

For the conventional data centers, their heat dissipation rates are in the range of 430–861 W/m^2. With the development of more compact high power modules, in the newer generations of data centers, the heat dissipation rates have increased at least 10 times and reaches 6458–10,764 W/m^2 (Rasmussen 2005). The heat dissipated inside data centers should be removed by the cooling systems. Due to the large variance in data centers' heat dissipation rates, different cooling techniques have been developed to meet the different cooling needs.

4.3.1 Air-Cooled Systems

Air-cooled systems make up most of the cooling systems in existing data centers. They typically arrange server racks into cold aisles and hot aisles. The cold aisles provide cool intake air to

Table 4.2 Distribution of heat and temperatures within IT servers

	Component	Proportion of total heat	Temperature
For standard server	Microprocessors	30%	85 °C
	DC/DC conversion	10%	50 °C
	I/O processor	3%	40 °C
	AC/DC conversion	25%	55 °C
	Memory chips	11%	70 °C
	Fans	9%	30 °C
	Disk drives	6%	45 °C
	Motherboard	3%	40 °C
For high performance cluster (HPC)	Microprocessors	63%	85 °C
	DC/DC conversion	13%	115 °C
	I/O processor	10%	100 °C
	Memory chips	14%	40 °C

each server, while the hot exhaust air exits the servers in the hot aisles (Ebrahimi et al. 2014). There are four typical configurations in air-cooled data centers, i.e., computer room air conditioner units (CRACs) (see Fig. 4.3a), computer room air handler units (CRAHs) (see Fig. 4.3b), in-row cooling (see Fig. 4.3c), and rear-door cooling (see Fig. 4.3d) (Oró et al. 2019). The main difference between CRAC and CRAH systems is related to size of the data center in which they are applied. The first one is usually utilized for small data center (< 100 kW), while the second one for medium-big size data centers (> 100 kW). Finally, row cooling and rear-door cooling are normally used for medium–high (> 10 kW per rack) and high (up to > 35 Kw) energy density facilities, respectively.

4.3.2 Water-Cooled Systems

In many newly built data centers, the power load increases to the level that can hardly be removed by air-cooled systems. Therefore, to satisfy the higher heat removal needs, liquid cooling systems have been developed, see Fig. 4.4. In fact, thanks to water high heat storage capacity, convective heat transfer coefficient, and direct contact with server components, higher heat transfer rates can be obtained with respect to air-cooled systems (Ebrahimi et al. 2014). This allows low-temperature differences between the cooling liquid and the server components, and thus, liquid coolant with significantly high temperature (e.g., 60 °C) can be used (Habibi Khalaj and Halgamuge 2017). Such high temperatures produce high-quality waste heat that can be easily recovered (Brunschwiler et al. 2009). Moreover, since liquid cooling provides more efficient heat transfer, substantial energy savings can be achieved in liquid cooling.

4.3.3 Two-Phase Cooled Systems

With the increased needs for Internet and computing resources, the scales and power densities have exceeded 1000 W/cm^2 in some data centers. Two-phase cooling has been developed to deal with such high-density heat dissipation (Ebrahimi et al. 2014). It cools the racks by taking advantage of the high convection heat transfer efficiency associated with the nucleate boiling. In two-phase cooled systems, the coolant has two phases: liquid and vapor. In the cooling process, the liquid coolant near saturation is pumped into the cold plate (contacting the electronics that needs to be cooled), where it starts to boil and evaporate, cooling the electronics and storing the

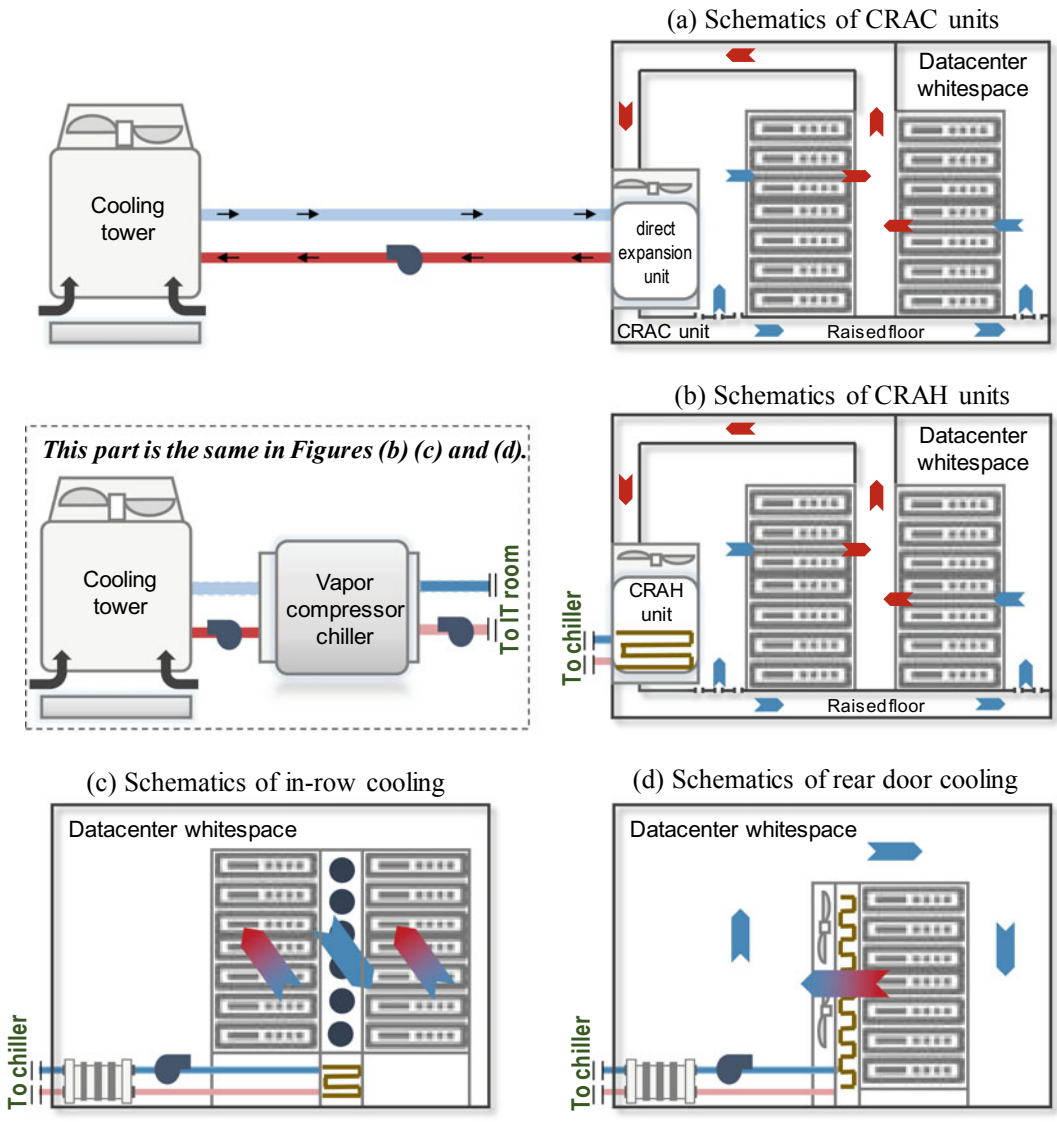


Fig. 4.3 Schematics of different air-cooled systems **a** schematics of CRAC units, **b** schematics of CRAH units **c** schematics of in-row cooling, **d** schematics of rear-door cooling

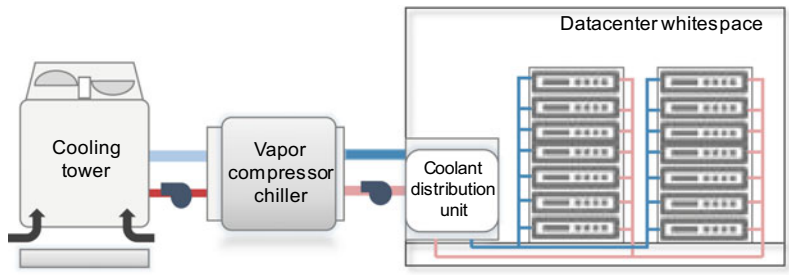


Fig. 4.4 Schematics of a water-cooled system for data centers (Zimmermann et al. 2012)

energy in the latent heat. The vapor then flows to a condenser where it is condensed back into liquid phase and heat is released from the fluid to ambient or a vapor sink.

Compared with water-cooled systems, the two-phase cooling can remove higher heat fluxes (between 790 and 27,000 W/cm^2 (Bowers and Mudawar 1994)) while working with smaller mass flow rates and thus lower pumping power (Capozzoli and Primiceri 2015). Moreover, the temperature of coolant in the two-phase cooling system can reach as high as 80 °C, which increases the quality of waste heat and thus allows an easy recovery of waste heat (Capozzoli and Primiceri 2015). Due to refrigerant two-phase status, there are two ways to drive the refrigerant circulations, i.e., liquid-pump driven and vapor-compressor driven. Figure 4.5 presents the two-phase cooling cycle (i.e., Fig. 4.5a) and the schematics of liquid-pump cooling cycle (i.e., Fig. 4.5b) and vapor-compression cooling cycle (i.e., Fig. 4.5c).

4.3.4 Comparison of Different Cooling Systems

In the present section, a comparison of the performances of the three cooling systems is carried out. The quality of waste heat in two-phase cooled systems is the highest among the three types of cooling systems, due to the higher heat transfer efficiencies. For air-cooled systems, the quality of waste heat is low, and thus, heat pumps are usually added to the system when harvesting waste heat (Table 4.3).

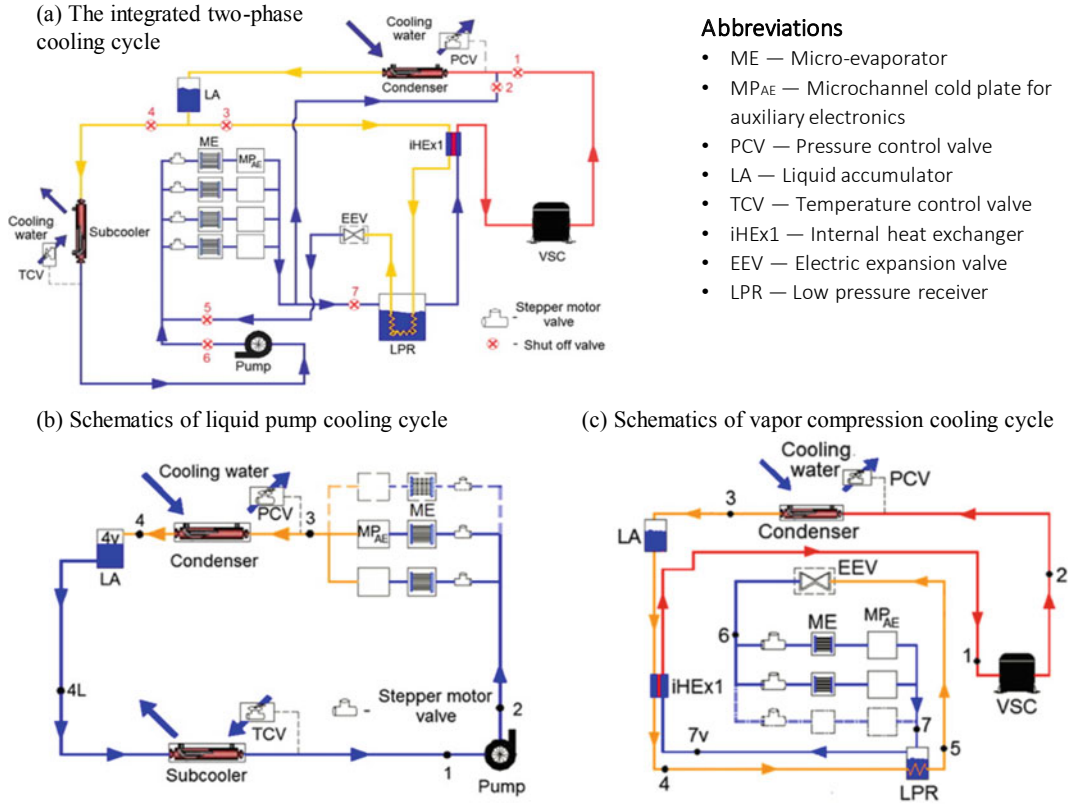


Fig. 4.5 Principles of a hybrid two-phase cooling cycle **a** the integrated two-phase cooling cycle, **b** schematics of liquid pump cooling cycle, **c** schematics of vapor-compression cooling cycle (Marcinichen et al. 2012)

Table 4.3 Summary of 'typical' data center heat sources and streams (Ebrahimi et al. 2014)

	Parameter	Value
Air cooling	Cold aisle (CRAC supply) temp	10–32 °C
	Hot aisle (CRAC return) temp	50–60 °C
	Temp. rise over servers	10–20 °C
	Airflow per rack	200–2500 CFM
	Chiller water supply to CRAC	7–10 °C
	Chilled water return from CRAC	35 °C
Water cooling	Water supply to server	20–60 °C (std) 70–75 °C (max)
	Water exit from server	2–5 °C temp. rise over servers
	Water flow rate per rack	5–10 GPM
	ΔT from water to lid	5–18 °C
	Buffer heat exchanger flow rate	5–10 GPM
	Buffer heat exchanger supply temp	3–5 °C above ambient
Two-phase cooling with liquid pump	Coolant supply to evaporator	60 °C saturated liquid (std.) 70–75 °C (max)
	Coolant exit from evaporator	62 °C at 30% quality (std.) 75–80 °C (max)
	Condenser cooling fluid inlet	30 °C
	Condenser cooling fluid outlet	45–90 °C
Two-phase cooling with vapor compressor	Coolant supply to evaporator	60 °C saturated liquid (std.) 70–75 °C (max)
	Coolant exit from evaporator	60 °C saturated liquid (std.) 70–75 °C (max)
	Coolant temperature at the exit of vapor compressor	~ 90 °C
	Condenser cooling fluid inlet	30 °C
	Condenser cooling fluid outlet	~ 90 °C

4.4 Data Centers as Consumers—integration with Renewable energy Generations

Most of the existing data centers use the fossil-based grid power to supply their electricity demand. Since the fossil-based fuels are the primary source of GHG emissions and they are expected to be run out in the near future, renewable energy has drawn much attention to replace the carbon-based energy for powering the data centers.

4.4.1 Different Ways of Integration

There are four typical ways of integrating renewable energy into data centers, as depicted in Fig. 4.6. The details of these integration ways are introduced below.

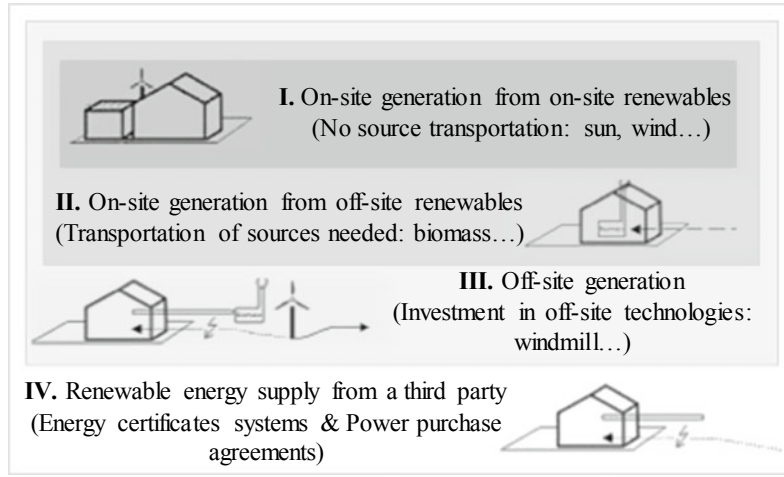


Fig. 4.6 Overview of possible renewable supply options for data centers (Depoorter et al. 2015)

4.4.1.1 Data Centers with Generation of Renewable Energy

In this way of renewable integration, the data centers generate their own renewable energy and have a direct control or influence of the energy resources. The renewable energy can be generated either 'on-site' or 'off-site' (Oró et al. 2015). The integration of renewable generations will lead to increase in the capital costs but can continuously reduce the consumption of conventional fossil fuels during its life cycles. There are three kinds of renewable generations (Marszal et al. 2011).

On-site generation from on-site renewables: Data centers install renewable energy systems within their own facilities. The generation of usable form of energy takes place within the infrastructure footprint or site. The on-site renewable energy includes thermal energy produced by solar collector and electrical energy by PV panels, wind, or hydro turbines. The renewable energy sources are directly available on the site, such as the solar energy, wind energy. Among the research landscape, some research studies and IT companies focused on the development of different on-site green energy into data centers. As an example, Sharma et al. (2011) presented Blink, a cluster of ten laptops which is intermittently powered by on-site micro wind turbines and two solar panels. Similarly, HP labs developed a data center partially powered by solar panels (Arlitt et al. 2012a). Goiri et al. (Í et al. 2014) developed Parasol, an on-site solar-powered datacenter, and combined it with GreenSwitch. The latter is a system for scheduling workloads, selecting which source of energy to use (renewable, battery, and/or grid), and choosing the renewable energy storage medium (battery or grid) at each point in time. Finally, the first 100% on-site wind power data center has been developed in Illinois (US) (DataCenter 2010).

On-site generation with off-site renewables: Data centers install renewable energy generation systems within their own facilities. But, the data centers have to rely on renewable energy sources from off-site places. The generation of usable form of renewable energy takes place on the project site. Example of this type of integration is the transportation of biomass or biogas from outside the data centers to the data centers for producing the needed heat or electricity. As an example, Apple, together with Aarhus University (Viborg campus) in Denmark, is now co-developing an agricultural waste biomass project. By this way, the methane produced from biomass digester reaction will be used to power Apple's data centers. A 3 MW biomass co-generation plant is now powering a data center in the Grand Duchy of Luxemburg, which total energy requirement is equal to 5 MW (Supponen et al. 2016).

Off-site generation: Due to factors such as insufficient renewable energy potentials and limited spaces available, the on-site renewable generations are not suitable in some places. In this case, off-site renewable generation is preferable for data centers. The data center owners invest in a renewable energy plant in places with sufficient renewable sources or in a community system. The power grid or heat/cooling networks are used as the carrier of produced energy. When the energy transmission is through the existing general infrastructures, availability of legislation to connect new renewable generation capacity to the buildings is required, and a mechanism that records the off-site energy production and incorporates this part of energy into data center energy bills should be established. Among the research landscape, some companies implemented off-site green energy to power data centers. As an example, Facebook's new Papillion Data Center is supported by renewable energy from Enel Green Power's Rattlesnake Creek Wind Farm in Nebraska. The wind farm will generate about 1300 GWh of sustainable energy each year and part of it will be sold to Facebook (2019). Part of the Rattlesnake Creek Wind Farm energy will be sold to Adobe as well. Salt River Project and Apple defined an agreement to purchase power from the Apple-owned 50-megawatt photovoltaic solar power plant in Pinal County (east of its data center in Mesa, Arizona) (Apple 2018). Finally, Google is going to purchase the output of new solar farms built in Alabama and Tennessee (around 150 megawatts respectively) in order to match the energy consumption of its upcoming data centers (Google 2019).

The data centers may need to be connected with the power or thermal networks or equip storage systems in case there is renewable energy shortage. However, by generating their own usable forms of energy, the data centers can significantly reduce the dependence on the grid or district cooling networks. The on-site generation can significantly lower the energy losses as the generated power undergoes less conversions and is not transmitted over long distances. However, the generation capacity may be limited for the on-site generation, since the optimal location of a data center is planned with the consideration of factors such as network latencies, labor force availability, and tax structure, which may not match the locations with the maximum renewable potentials. The off-site generation is much more flexible in the site planning of renewable energy systems, and thus, the capacity can be larger. However, the losses due to energy transmission are inevitable, and the penetration of renewable energy may reduce the reliability and efficiency of the existing energy networks.

4.4.1.2 Data Centers with Renewable Energy Provided by a Third Party

Besides generating their own renewable energy, the data center operators can also purchase renewable energy from other entities to reduce the carbon footprint. There are two important mechanisms for this type of renewable energy integration.

Energy certificate systems: Electricity certificates provide an efficient and reliable tracking mechanism for the energy origin of electricity system. An energy certificate includes the information of the generation attributes (e.g., renewable fuel type, capacity and age of the plant, etc.) of the related electricity production (Hulshof et al. 2019). Typically, one energy certificate is assigned for 1 MW · h electricity from a verified production plant. There are two types of certificates: quota and tracking (Ltd., 2014). In the quota system, an obligation to buy energy certificate is imposed on a suitable party such as electricity suppliers and large electricity consumers. In the tracking certificates, the consumers and electricity suppliers are voluntary to guarantee their energy sources of consumed or sold electricity. Tracking certificates provide proof about the energy source of electricity to the end-consumers. Examples of the certificate systems include the Guarantees of Origin in the EU and Renewable Energy Credits in the USA (EPA 2018).

Power purchase agreements (PPAs): A PPA is a contract between a supplier and a consumer which specifies how much electricity the supplier has promised to place on the power gird and how much the consumer will take off (Bruck et al. 2018). A PPA also specifies an electricity price. However, it does not deliver electricity attributes that are different from the grid-average unless tracking certificates are transferred in combination with the electricity (Oró et al. 2015). An examples of PPA is that Google contracted to buy 114 MW · h wind power from a wind project in Ames to supply its data center in Council (Anonymous 2013). Another example is that Microsoft purchased wind power to supply part of its data center in Dublin (D.A.Ireland 2012).

4.4.2 Advanced Controls to Maximize the Use of Renewable Energy

This section introduces the advanced controls of data centers for maximizing the renewable energy usage in data centers. The principals of the different controls are introduced first (see Sect. 4.2.1). Then, the existing data center demand controls for maximizing the renewable energy utilization are reviewed (see Sect. 4.2.2). Finally, the integration of energy storage in data centers is introduced (see Sect. 4.2.3).

4.4.2.1 Principals of Controls for Maximizing the Renewable Energy Usage

Unlike other energy sources, the renewable energy is available only when there is wind or solar irradiation, and it is always varying. The intermittency and varying characteristics make it hard for data centers to efficiently use the renewable energy (Yang et al. 2018). Thus, data centers are usually tied into the power grid to achieve high availability and ensure that all the harvested renewable energies can be used to power the data centers. Table 4.4 summarizes the typical data center management measures to improve the self-consumption of renewable energy. These measures are based on two basic principles to deal with the power intermittency and variance: either rescheduling or migrating the workloads based on the renewable energy availability.

Rescheduling workloads: The workload can be classified into critical workloads (which need immediate execution) and non-critical workloads (which can be delayed). The data centers can reschedule the non-critical workloads to the periods when the renewable generation is sufficient (e.g., noontime for solar energy) (Arlitt et al. 2012b).

Migrating workload: Migration is conducted between different data centers. For instance, in

Table 4.4 Summary of data center management to improve renewables utilization (Stewart and Shen 2009)

Types of data center management	Measures to use renewables more efficiently	Measures during intermittent outages
Capacity planning	Power certain machines only when renewables are available	Turn off some machines
	Plan the geographic location of data center sites according to intermittency patterns	Connect some sites to the grid
Load balancing	Route requests to data centers with unused renewables	Route fewer requests
	Move entire services to data centers that expect long periods of renewable power	Re-migrate services to other data centers
Job scheduling	Aggressively prefetch from hard disk drive (HDD) before expected outages	Prefetch less data
System maintenance	Delay solid-state drive (SSD) erasures until renewables are available	Further delay erasures

cloud computing, the data centers with insufficient renewable energy generations can migrate the workloads to other data centers that have sufficient renewable production (Stewart and Shen 2009).

4.4.2.2 Examples of Advanced Controls for Maximizing the Renewable Energy Usage

How to reschedule or migrate the data center workload are challenging the effective utilization of renewables in data centers. To date, some control methods have been developed to address such challenges (Deng et al. 2014). Table 4.5 summarizes the literatures on the data center controls to improve self-consumption of renewables, including the location of the data centers, the types of renewable supply, the targeted problem, the considered factorsin the control optimization, and their major contributions. Some control platforms have been developed, such as the Blink, SolarCore, GreenHadoop, iSwitch, GreenSlot (see Table 4.5). These controllers all share the same goal to increase the match between the data center electricity demand profiles and the renewable energy generation profiles. These methods typically follow a two-step framework. In Step 1, the renewable energy generations are predicted by advanced modeling tools. Based on the predicted renewable supply, in Step 2, the controllers adjust the workload to match the renewable generations.

Most of the considered renewables are solar energy and wind energy, and only a limited studies considered other renewable solutions, such as solid waste (Gmach et al. 2010). This is because the solar energy and wind energy have larger intermittency and variance compared to other renewable energy, such as hydro power and biomass fuels, and thus, their effective utilization and control are more difficult. Most of the developed controls focused on the single data center's workload scheduling, only a few studied the cooperation of geo-distributed data centers. The considered metrics typically include the amount of utilized renewable energy and the costs. The wind and solar resources are highly affected by the locations and climate regions. Li et al. (2011) investigated the solar power production variance in different locations and climate regions of the USA. Similarly, Li et al. (2012) studied the wind power generation variance in different locations of the USA. Although some efforts have been devoted to reducing the data centers' dependence on the power grid (Arlitt et al. 2012b), currently, the data centers still cannot be completely independent of the power grid. In some studies, the data centers are equipped with renewable energy systems of small sizes, and the grid power is used as the second energy source to power the data centers. In some studies, the annual production of renewable energy is equal to the annual electricity demand (the data center is therefore called a zero-energy building), the power grid acts as a huge virtual electrical energy storage, which stores the excessive renewable energy and powers the data centers when there is renewable insufficiency.

There are also unsolved issues associated with the data center control to improve self-consumption of renewable energy. Large uncertainties exist in the generation of renewable energy and the data center workload. For instance, the weather parameters, such as solar irradiation, ambient air temperature, and wind speed, have large uncertainties and it is difficult to predict them accurately. The workload of data centers is affected by factors such as the date (e.g., holidays), time (e.g., day or night), which also have large uncertainties. The existing controls are in a deterministic framework, assuming that the predicted renewables and workloads are accurate. Consequently, once calculated solution could be highly infeasible in practice.

4.4.2.3 Integration of Energy Storage in Data Centers

The renewable energy, such as wind and solar energy, is highly intermittent and always varying. The intermittency and varying characteristics make it hard for data centers to efficiently use the renewable energy. To address the issue of power intermittence, energy storage systems offer potential solutions.

Table 4.5 Summary of the literatures on the data center controls to improve efficiency of using renewables

Controller name	Location	IT facilities power	Types of renewable supply	Power of renewables	Considered factors	Target problem	Work done
A data center profiling method, 2010 (Gmach et al. 2010)	Houston	1013 W (peak) 695 W (idle) × 8	On-site solar panels and solid waste	4 kW PV 15 kW from a municipal solid waste facility	1. Net energy balance 2. Sustainability and quality of service metrics 3. Cooling system energy usage	To co-manage data center's energy supply and demand side	Use a dynamic migration controller that continuously monitors resource consumption, migrates workloads off overloaded servers, and consolidates workloads from under-loaded servers (The control of demand to match renewable supply is not considered.)
An adaptive job scheduler, 2011 (Aksanli et al. 2011)	San Diego, USA	251 W	On-site PV panels and wind turbines	—	1. MapReduce 2. Quality of Service (QoS) 3. Batch load and web request	Schedule the batch jobs without significantly affecting the performance of latency sensitive web requests to match the renewables	1. Predict the solar and wind energy production by a developed green energy predictor 2. Apply the scheduler which ensures the required response time targets for services are met while maximizing the completion times and the number of batch tasks run
Blink, 2011 (Sharma et al. 2011)	Amherst, USA	Under 100 W	On-site wind turbines and solar panels	100–140 W	1. Battery storage 2. Workload distributions, throughput and latency	To make the servers and applications gracefully handle intermittent constraints in their power renewable supply	1. Blinking applies a duty cycle to servers that controls the fraction of time they are in the active state, e.g., by activating and deactivating them in succession, to gracefully vary their energy footprint 2. Make the case for blinking systems design a blinking hardware/software platform; design, implement, and evaluate BlinkCache 3. Consider three policies: activation policy, synchronous policy, and load-proportional policy

(continued)

Table 4.5 (continued)

Controller name	Location	IT facilities power	Types of renewable supply	Power of renewables	Considered factors	Target problem	Work done
SolarCore, 2011 (Li et al. 2011)	Phoenix, Golden, Elizabeth City, Oak Ridge, USA	—	On-site PV array	—	1. MPP tracking efficiency 2. PV supply current, voltage, and power 3. Performance in different geographic locations and seasons	To harvest the maximal renewable energy and reduce the impact of supply variation; To intelligently allocate the dynamically varied power budget across multiple cores to maximize workload performance	1. Develop a MPPT method to maximize the solar energy harvest to maximize the power budget without utilization of batteries 2. Conduct load optimization based on the workload throughput-power ratio to ensure that the dynamic load tuning across multiple cores achieves the optimal performance
GreenHadoop, 2012 (Goiri et al. 2012)	Rutgers solar farm, USA	2375 W	On-site solar energy	3220 W	1. Computational workload 2. Brown energy prices 3. Jobs' time bounds	To manage a data center's computational workload to match the green energy supply	1. GreenHadoop predicts the amount of solar energy that will be available in the near future and schedules the MapReduce jobs to maximize the green energy consumption within the jobs' time bounds 2. If brown energy must be used to avoid time bound violations, GreenHadoop selects times when brown energy is cheap, while also managing the cost of peak brown power consumption
Policy-based energy balancing, 2012 (Arlitt et al. 2012b)	Palo Alto, CA., USA	— (four BL465c G7 servers)	On-site PV panels	134 kW (peak power)	1. Workload schedule 2. Design for achieving net-zero energy/cost 3. Prediction error of PV power	To minimize the dependence on the grid power while reducing the capital costs	1. Classify the workloads into two types: critical workloads (need immediate execution) and non-critical workloads (can be delayed) 2. Schedule the non-critical workloads to the periods with large renewable supply

(continued)

Table 4.5 (continued)

Controller name	Location	IT facilities power	Types of renewable supply	Power of renewables	Considered factors	Target problem	Work done
iSwitch, 2012 (Li et al. 2012)	California, Arizona, Colorado, Texas, Wyoming, Utah USA	186W (peak) and 62W (idle) × 4800	On-site wind power	Different values in different locations	1. Wind power intermittency and variability 2. Control overhead	To minimize the performance overhead and increase the renewable utilization with the intermittent and varying wind energy characteristics considered	1. iSwitch is a supply-aware power management scheme. It applies the appropriate power management strategy for wind variation scenarios to achieve the best design tradeoff 2. As an alternative to load power throttling, iSwitch intelligently shifts the computing load from one energy source to another to achieve best load power matching
A workload scheduling and capacity management approach, 2012 (Liu et al. 2012)	California, USA	100 kW (500 servers)	On-site PV array	130 kW (Peak generation)	1. Different cooling sources (chiller cooling/outside air cooling) 2. Dynamic energy pricing	To minimize the data center overall costs considering multiple factors	To schedule non-critical IT workload and allocate IT resources within a data center according to the availability of renewable power supply and the efficiency of cooling
Carbon-aware cloud provisioning policy, 2012 (Deng et al. 2012)	—	42W	On-site solar panels	Varying in different sets of experiments	1. Actual integration of renewable energy 2. Carbon emissions 3. Cloud applications	To reduce the carton footprint by integrating renewable energy in real cloud applications	1. A carbon emission limit is assigned to each cloud applications 2. Carbon-aware policy decides which cloud instances to provision such that (1) the grid energy limit is not exceeded and (2) performance goals are met
A grid-aware scheduler, 2012 (Krioukov et al. 2012)	California, USA	—	Off-site wind power	30 MW	1. Load proportionality 2. Dynamic electricity pricing	To improve the data center performances by matching a load's demand profile to a supply profile, with the consideration of demand slack, available renewable power, and real-time market price	1. Determine slack for a workload. In the interactive case, slack is expressed as a level of performance degradation; in the batch case, slack is expressed as the time to a deadline 2. By shifting and scaling the demands to match the demand profile with the supply profile

(continued)

Table 4.5 (continued)

Controller name	Location	IT facilities power	Types of renewable supply	Power of renewables	Considered factors	Target problem	Work done
GreenSlot, 2015 (Goiri et al. 2015)	Rutgers solar farm, USA	235 W	PV solar panels	2.3 kW	1. Parallel batch jobs 2. Varying brown energy prices 3. Renewable energy utilization and costs	GreenSlot seeks to minimize brown energy consumption by instead using solar energy, while avoiding excessive performance degradation	1. GreenSlot delays some jobs (within their deadlines) to guarantee that they will use green energy 2. GreenSlot may delay certain jobs to use cheaper brown energy
A green-aware power management strategy, 2015 (Wang et al. 2015)	Qinghai, China	259 W × 40	Off-site PV panels and wind turbines	—	1. Virtual machine migration 2. Cooling system energy usage 3. Costs	Minimize the total revenue of data centers (including IT facility power and cooling power) for the virtual machine migration problem with the utilization of renewable energy	1. Use k-nearest neighbor (k-NN)-based algorithm to predict the solar energy generation for the next day 2. Use genetic algorithm to search the best virtual machine placement scheme to minimize the total revenue (When green energy is sufficient, the racks will be supplied by green power first. Grid energy is used only when the renewable energy is not enough to maintain the operation of the running devices)
A heuristic method, 2018 (Lu et al. 2018)	Different locations in the USA	Randomly selected within kWh	Wind energy	—	1. Geo-distributed data centers (cloud application) 2. Bulk data transfer 3. Varying wind power and electricity prices	To maximize the renewable energy use and minimize grid energy costs for bulk data transfers between green data centers	1. If the available wind power can be used to transfer all the bulk data, it is used to complete the inter-GD BDTs; 2. If the available wind power is insufficient to accomplish, the inter-GD BDTs, the optimal demand division, and routing selection are used to minimize the energy cost caused by grid power

For the electrical batteries, they can be used to store the renewable power when there is surplus generation (e.g., excessive solar power generation in daytime) and then power the data centers when there is insufficient renewable generation (e.g., at night when there is no solar power generation). For instance, a battery system with storage capacity of 32 kW•h is integrated into a small solar-powered data center (Í et al. 2014), which houses 64 Atom-based servers with a total power of 1.92 kW). Their installed PV system has a capacity of 3.2 kW. The economic analysis of this data center shows a daily 9% saving of the grid electricity cost which can be achieved and the investments in PV system and batteries can be returned in 7.6 years, with the implementation of advanced controls. However, according to reference (Li et al. 2011), electrical storage is not recommended to be applied in large data centers with high electricity demands because of the following reasons: (1) The large current drawn by data center IT facilities requires batteries with large capacity, which increases the demanded space for hosting batteries as well as the investments. (2) The turn-around efficiency of battery is low due to internal resistance and self-discharge. (3) Due to the frequent charging/discharging and self-discharging, the lifetime of the existing batteries is very limited. The maintenance costs to ensure batteries' normal operations can even surpass the investments of renewable energy systems.

Conversely, thermal energy storage is much cheaper and can have larger capacity with relatively smaller required space. There are already some datacenters using the thermal energy storage. For example, The Fortress International Group (2009) investigated the application of ice storage system in data centers for reducing the electricity bills. The building cooling system operates to produce ice stored in a thermal containment system during non-business hours when electricity is cheaper. At peak hours when electricity is expensive, the cooling stored in ice storage tanks will be released to the data center for reducing the chiller power demands. Their study results show that 15.9% electricity savings can be achieved in Austin Texas. Phoenix ONE (Miller 2009) adopted a thermal storage system using a solution of water and 28% glycol as the coolant and Cryogel ice balls (i.e., 4-inch polyethylene spheres filled with water) as cooling storage. ARANER (ARANER, 2018) developed a cooling system with thermal energy storage for the Al Ashghal Data Center. They used a two stratified water tanks for the cooling system with a capacity of 210 TR. In both the Phoenix ONE (Miller 2009) and ARANER (2018) projects, the cooling system operates in a similar manner as (Fortress International Group 2009). Most of these applications aim to reduce operational costs of data centers via shifting electricity demanded by the cooling system from high-electricity-price periods (e.g., daytime) to low-electricity-price periods (e.g., nighttime). As a result, the energy storage system can only help reduce electricity costs, while the amount of electricity drawn from the power grid is not reduced. Alternatively, when the data centers are equipped with renewable energy systems, they can use the energy storage system to store the excessive renewable power generations, i.e., charging the thermal energy storage system by operating the cooling system in the periods when there are surplus renewable power generations than the data center's electricity needs, thus helping increase the renewable energy self-utilization and reduce the amount of grid electricity imports. However, this type of application is rarely considered in existing data centers. More future work is needed to enhance such applications.

4.5 Data Centers as Producers—Waste Heat Recovery

This section first introduces the best locations for capturing the data centers' waste heat. Then, the different techniques for reusing waste heat from data centers are presented and compared. With the consideration that the waste heat recovery for

DH is a promising way to connect the data centers with district energy systems, this technique is introduced in detail in this section.

4.5.1 Locations for Waste Heat Recovery

For the data centers' waste heat recovery, the quality and quantity of the captured waste heat are strongly affected by the type of cooling systems and the locations of waste heat recovery. This section introduces the best locations for waste heat recovery in each type of cooling systems.

Air-cooled systems: In air-cooled data centers, the heat in the hot air returning from the racks is rejected to the outdoor environment typically using a chiller and sometimes cooling tower loop. As summarized in Table 4.3, the average temperature in the chilled water return is lower than the return air, which limits the effectiveness of waste heat capture. Therefore, the optimal location to capture waste heat in air-cooled data centers is the rack exhaust (35–45 °C) prior to room air mixing which causes exergy losses. Alternatively, waste heat can be collected at the air return to CRAC (30–40 °C) or at the condensate water return, but the grade of heat will be lower (Ebrahimi et al. 2015).

Water-cooled systems: In water-cooled systems, the temperature of water exit from servers/racks is the highest (75–80 °C maximum). Thus, the optimal location to recover waste heat is in the water exit from server/racks. This can be done by adding a water-to-water heat exchanger at the water exit from servers/racks. Alternatively, waste heat can be collected at the condensate water return, but the grade of heat will be lower.

Two-phase cooled systems: In two-phase cooled systems, the coolant contacts the racks directly, which increases the complexity of heat recovery. For both liquid-pumped and vapor-compression systems, the optimal location for collecting waste heat is at the condenser (45–90 °C) by using a secondary working fluid (water in most cases) to cool the primary coolant of the systems (Ebrahimi et al. 2014).

4.5.2 Waste Heat Reuse for District Heating Networks

Data centers' waste heat reuse in DH network is an effective and efficient way to connect the data centers with the district energy systems, especially in the Nordic countries, which are characterized by a high share of renewables in their DH network due to the use of wood fuels. Most of the existing data centers adopt air-cooled systems, which produces low-grade waste heat. Heat pumps are usually equipped for upgrading the low-temperature waste heat. This section first introduces different thermodynamic cycles in heat pumps for waste heat upgrading. Then, the connections of data centers with DH networks are reviewed and introduced. At the end of this section, some prototypes of data centers' integration with DH systems are introduced.

4.5.2.1 Different Thermodynamic Cycles in Heat Pumps for Upgrading Waste Heat

Heat pumps can be used for effectively transforming low-grade heat into high-grade heat using a two-phase refrigerant. There are four main components in a heat pump: evaporator, compressor, condenser, and expansion valve. In the evaporator, the low-grade heat is extracted from a waste heat source. In the condenser, the high-grade heat will be released. The core operating principle of a heat pump is based on the thermal property that the boiling point of a fluid increases with the pressure. For a single-stage single-compressor cycle (see Fig. 4.7a), the low-temperature heat is absorbed by the liquid refrigerant in the evaporator, making it vaporize. The refrigerant vapor is then drawn into a compressor. Both the pressure and temperature of the refrigerant vapor will increase after compression. Then, at a high temperature and pressure, the refrigerant vapor flows to a condenser and is condensed there. Next, the liquid refrigerant flows through an expansion valve to reduce its pressure and temperature before returning into the evaporator.

Depending on the real application, there can be a wide range of thermodynamic cycles that

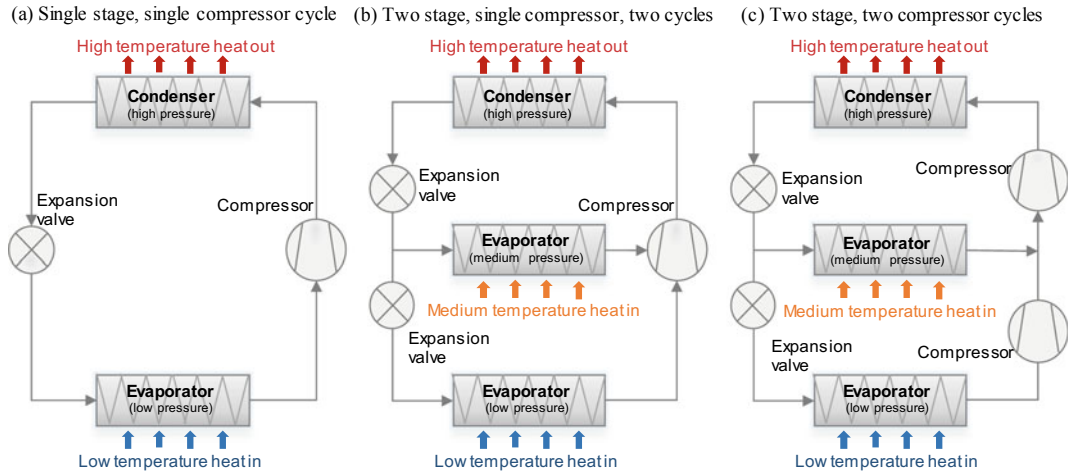


Fig. 4.7 Potential thermodynamic cycles for upgrading data center waste heat **a** single-stage, single-compressor cycle, **b** two-stage, single-compressor, two cycles, **c** two-stage, two-compressor cycles

could be adopted, such as including two or more stages and one or more compressors and refrigerants (see Figs. 4.7b,c), to achieve the desired heat upgrading. While selecting an appropriate cycle for a specific heat pump application, the considered factors should include the followings: (1) the involved values and ranges of temperature and (2) whether more than one waste heat sources with different temperatures are available, e.g., different locations in data centers permit waste heat to be captured at different temperatures (see Table 4.3). By applying multiple-stage cycles with two or more evaporators operating at different temperatures, the overall system efficiency can be dramatically enhanced compared with single-stage cycles. The multiple-stage cycles also provide a means to capture all the waste heat from data centers. Figure 4.7 presents three examples of different thermodynamic cycles that can be applied for upgrading the data centers' waste heat. Note that, depending on the practical situation, other configurations are also possible.

4.5.2.2 Different Prototypes of Integration Systems

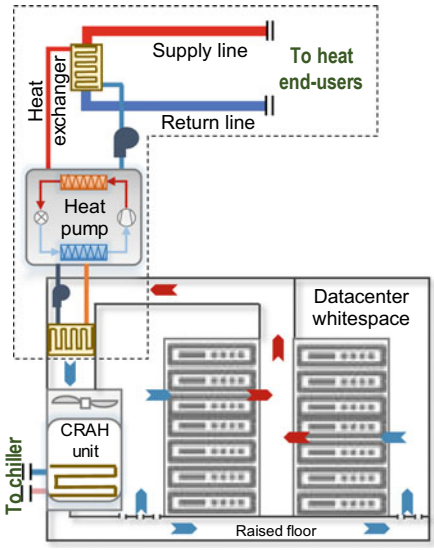
This section reviews the different prototypes of integrating data centers' waste heat with DH systems. The chapter is conducted from three aspects: the connection at data center side for waste heat recovery, the connection at the DH system side for injecting heat, and architecture of the overall system connection.

Connection at Data Center Side for Waste Heat Recovery

There are two locations where the waste heat is typically captured: return hot aisle and the chiller condenser. Figure 4.8 presents schematics of the waste heat recovery in these locations. Note that the heat recovery in CRAC cooling systems is not considered, as this type of system is normally equipped in small data centers. The potential heat that can be recovered is small and non-profitable for DH networks.

Waste heat recovery in the return hot aisle is unique to the CRAH cooling systems, since the hot air streams in IT facility rooms are gathered at a relatively high temperature and delivered in a common duct to the air-handling units (Deymi-Dashtebayaz and Valipour-Namanlo 2019). For in-row cooling systems and rear-door cooling systems, the operational air temperature is very low, and thus, it is inefficient to recover heat at the air side. For heat recovery from the return hot aisle, a water-to-air heat exchanger is installed in the return hot stream air from the white space. Depending on the working conditions of the servers and racks, the return air temperature can

(a) Schematics of waste heat recovery at air side for CRAH cooling technology



(b) Schematics of waste heat recovery at condensate water side (applicable to air-cooled system e.g., CRAH, rear door cooling and in-row cooling, water cooled systems and two-phase cooled systems)

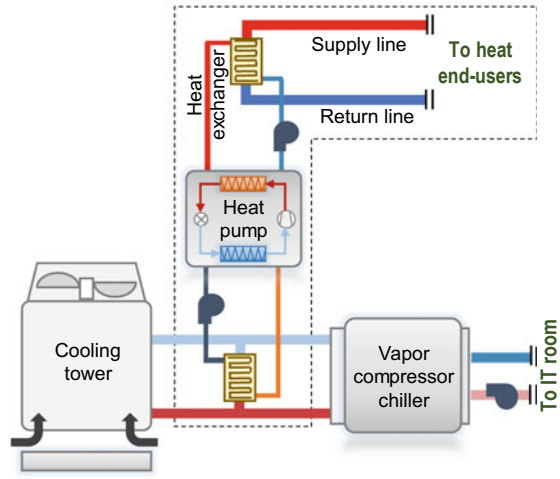


Fig. 4.8 Schematic diagram of the heat reuse solutions **a** schematics of waste heat recovery at air side for CRAH cooling technology, **b** schematics of waste heat recovery at condensate water side (applicable to air-cooled system, e.g., CRAH, rear-door cooling and in-row cooling, water-cooled systems, and two-phase cooled systems)

reach as high as 47 °C (Oró et al. 2019). The return hot air will be cooled down by the water in the heat exchanger, thereby reducing the cooling needs from the chillers in CRAH cooling systems. The low-grade heat in the warmed water is fed into a heat pump for upgrading the heat to the temperature that can be directly used by the DH network. The process is presented in Fig. 4.8a.

In many cooling systems, the return water temperature of the coolant circuit is usually low, and thus, it is also not viable for heat recovery. A feasible solution is to capture heat from the chiller condenser, as depicted in Fig. 4.8b. Waste heat recovery in the chiller condenser is applicable to the air-cooled systems (e.g., CRAH, rear-door cooling, and in-row cooling), water-cooled systems, and two-phase cooled systems. For heat recovery from the chiller condenser of CRAH systems, a water-to-refrigerant heat exchange is installed in parallel with the condenser (or dry cooler) of the chiller. Part of the rejected heat from the chiller goes to the surrounding environment, and the remaining heat is captured by a secondary water circuit. The temperature can reach up to 50 °C. The low-grade heat in the warmed water is fed into a heat pump for upgrading to the temperature that can be used by the DH network. For waste heat recovery from in-row cooling systems, rear-door cooling systems, water cooled systems, and two-phase cooled systems, the design and operation of this connection are similar to the CRAH systems described above by installing a water-to-refrigerant heat exchanger in parallel with the condenser (or dry cooler) of the chiller. However, the detailed mass flow rates and temperatures may vary.

Connection at the District Heating Network Side for Injecting Heat

As reported by (Lennermo and Lauenburg 2016), there are four ways of connection between the distributed heat sources and the DH systems.

Return/supply: In a return/supply system, the water takes heat from the return pipe. The water is heated to a proper temperature and then fed into the supply pipe of the DH system.

Return/return: In a return/return system, the water takes heat from the return pipe. The water

is heated to any temperature as it already has a higher temperature than the return water in the DH system. The water is fed into the return pipe of the DH system.

Supply/return: In a supply/return system, the water takes heat from the supply pipe. The water is heated to any temperature as it already has a higher temperature than the return water in the districted heating system. The water is fed into the return pipe of the DH system.

Supply/supply: In a supply/supply system, the water takes heat from the supply pipe. It is heated to any temperature higher than the water temperature in the DH system supply line and fed into the supply pipe.

Note that supply/return and supply/supply systems are rarely used, for the data centers. This is because the water temperature in the return pipe is usually higher than the water in the supply pipe due to the absorption of heat from the IT facilities. It is not economic to use the heat from the supply water, as extra heat sources are needed to increase the temperature of water fed into the DH system. The most beneficial system is the return/supply system since it has the least impacts on the DH system (Oró et al. 2019). The return/supply configuration does not affect the return line temperature, and thus, it is able to transfer the same power without enhancing pumping consumption.

Architecture of the Overall System Connection

The simplest architecture connecting a data center and DH loads is to add an energy sharing heat pump, as depicted in Fig. 4.9. The low-grade waste heat recovered in the data center is fed into the heat pump. After being upgraded to the temperature required by the DH network, the heat is delivered directly to the DH end-users. This connection is easy to implement, and the initial investments for heat recovery are relatively low. But, such connection requires the waste heat amount to be consistent with the heating demands in the DH network as there is no thermal storage in the system. Otherwise, the excessive waste heat will be released to the atmosphere. The COP of this connection can be as high as 4.3 (Murphy and Fung 2019).

To increase the utilization of waste heat when there is mismatch between the data centers' waste heat and the DH heating demands, a ground source heat pump integrated with a borefield system can be added, as shown in Fig. 4.10. The energy sharing heat pump is used only when the simultaneous cooling loads (i.e.,

Fig. 4.9 Schematics of the energy sharing system (Murphy and Fung 2019)

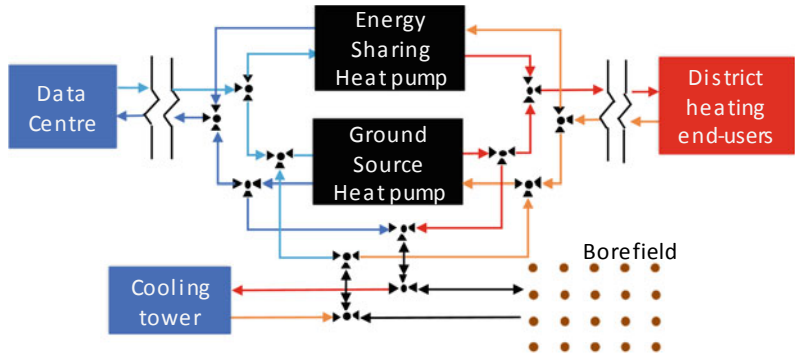


Fig. 4.10 Schematic diagram of the one-borefield system (Murphy and Fung 2019)

cooling needs to remove waste heat in data centers) and heating loads (i.e., heating requirements at the DH network) are equal. When there is mismatch between the heating and cooling loads, the ground source heat pump will operate in cooling/heating mode to control the temperature leaving its condenser/evaporator. The borefield acts as a large thermal energy storage to alleviate the mismatch. Here, two heat pumps are needed, since one heat pump cannot maintain the temperatures either leaving the evaporator when the condenser is receiving varying temperatures from the borefield or leaving the condenser when the evaporator is receiving varying temperatures from the borefield. The thermal load of the borefield should be balanced between the annual injected and extracted thermal energies. As the annual waste heat from the data center may not be equal to the annual heating demand of the DH end-users, a cooling tower is installed to offset the heat injected to the borefield. The COP of this connection can reach as high as 8.2 for the cooling mode and 3.5 for the heating mode.

To improve the ground source heat pump heating efficiency and provide free cooling for the data centers, one more borefield can be added, as shown in Fig. 4.11. One of the two borefields acts as a 'hot' thermal storage and the other acts as a 'cold' thermal storage. The water capturing the waste heat from the data center is first fed into the 'hot' borefield, warming up the 'hot' borefield. After exiting the first borefield, the cool water is then fed into a second 'cold' borefield to further release heat. The 'cold' borefield is also cooled by using dry cooler to capture the coldness of the outdoor air in winter. This dry cooler circuit runs when the outdoor air temperature is low. The cold water is then delivered into the data center to meet the cooling demand. In this system, due to the application of free cooling and removal of cooling tower (and thus more recovered heat), the COP of such connection can reach as high as 40 for the cooling mode and 4.1 for the heating mode. However, due to the deployment of two borefields, the capital investments of such connection is very high.

4.6 Data Center Projects with Renewable Energy Integrated or Waste Heat Reused

Table 4.6 summarizes examples of some data center projects with renewable energy integrated or waste heat reuse. While selecting the sites for installing data centers, the countries and regions with sufficient renewable generations serve good choices for IT companies to should their social responsibilities in environmental protection and energy savings, as they can use the renewables from the local energy markets. Many green data centers are located in the Nordic countries, e.g., Finland and Denmark. This is because these regions have many advantages, such as cold

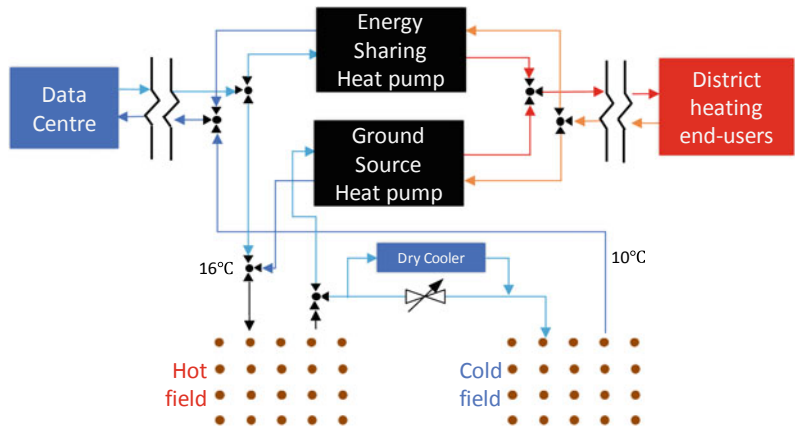


Fig. 4.11 Schematic diagram of the two-borefield system (Murphy and Fung 2019)

Table 4.6 Data center projects considering renewable energy integration or waste heat reuse

Company	Location	Year	Cooling system	Renewable energy?	Waste heat reuse?	Heat pump for waste heat reuse?
Apple (2015)	Viborg, Denmark	2015	Free cooling	Yes (hydropower and wind energy PPA, 100% renewable energy coverage)	Yes (DH)	—*
Aquasar, Swiss Federal Institute of Technology (IBM 2015)	Zurich, Switzerland	2010	Direct water-cooled: using hot water to cool the system	No	Yes (warm dorms)	No
Bahnhof (Pionen 2007)	Stockholm, Sweden	2007	Air-cooled	No	Yes (DH)	Yes
CSC (2018)	Kajaani, Finland	2010	Free cooling	Yes (hydropower PPA)	Yes (DH for local households)	—
Ericsson (Fotum 2016)	Kirkkonummi, Finland,	2016	Heat pumps	No	Yes (DH for 1000 single-family homes)	Yes
Facebook (2017)	Odense, Denmark	2017	Outdoor air through indirect evaporative cooling technology	Yes (wind energy)	Yes (recover 100,000 MWh of energy per year—enough to warm 6,900 homes)	Yes
GIB-Services (AG 2008)	Uitikon, Switzerland	2008	—	Yes (Hydropower PPA)	Yes (recover 2800 MW · h of energy per year, used for heating the swimming pool)	No
Google (Centers 2009)	Hamina, Finland	2017	Natural water-cooled: seawater from the Bay of Finland	Yes (wind energy PPA)	No	No
Google (Economics 2018)	North Carolina, USA	2007	Air-cooled	Yes (solar energy PPA)	No	No
HP Net-Zero Energy Data Center (Group 2012)	Palo Alto, CA, USA	2012	Air-cooled	Yes (PV array)	No	No
Mashhad municipality data center (Deymi-Dashtebayaz and Valipour-Namanlo 2019)	Mashhad, Iran	2019	Air-cooled	No	Yes (used for heating an adjoining office building)	Yes

(continued)

Table 4.6 (continued)

Company	Location	Year	Cooling system	Renewable energy?	Waste heat reuse?	Heat pump for waste heat reuse?
Quebecor (Fontecchio 2008)	Winnipeg, Canada	2008	—	No	Yes (used for heating the adjoining rooms)	No
Telecity Group (Harri 2016)	Helsinki, Finland	2013	District cooling + free cooling	—	Yes (DH for 4500 block apartments and 500 detached houses)	—
Telehouse West (West 2010)	London, UK	2010	Air-cooled and free air cooling	Yes (on-site roof-mounted PV cells)	Yes (DH)	—
Telia Company (Telia 2018)	Helsinki, Finland	2016	—	—	Yes (DH)	—
Tieto (Korhonen 2018)	Espoo, Finland	2011	Water-cooled systems and CRARs	No	Yes (DH for 1500 detached houses)	Yes
Yandex (Mäntsälä 2015)	Mäntsälä, Finland	2015	Free cooling	—	Yes (DH for 5,000 private houses)	—

*Not mentioned

climate, cheap electricity, stable political situation, efficient electrical grid, and most importantly, large share of renewables in the electricity portfolio (Wahlroos et al. 2018). Specifically, data center cooling is a fundamental part of data center efficiency and environmental friendliness. The mostly known cooling technologies are mechanical and free cooling, where the first one is considered less profitable (due to the high-energy consumption) than the second one. As an example, when the data centers are located in cold regions, they can adopt free air cooling techniques to reduce the electricity consumption for cooling the IT facilities (Apple 2015; CSC 2018; Harri 2016; West 2010; Mäntsälä 2015). When the data centers are located near lakes or sea, they can also apply natural water cooling techniques (Centers 2009). In this regard, the power usage effectiveness (PUE) metric is considered the most popular benchmark metric which is able to measure the power usage effectiveness of a specific data center. A PUE equal to 1.0 means that an efficient cooling system is implemented and 100% of the energy is used by the data center to power the IT equipment. PUE depends largely on ambient conditions, and therefore, a data center placed in a cold environment will have a better (lower) PUE than data centers in warmer climates. For instance, the Google created an innovative seawater cooling system to reduce the cooling energy required to run a data center located in Hamina Bay, in Finland. The cool seawater exchanges heat with the coolant in a heat exchange, and then, the coolant is distributed to the IT facilities to cool them down. After cooling the coolant, the warm seawater is mixed with some fresh water to reduce its temperature down to approximate the temperature in the bay, where it is delivered (IBM 2015). According to Google, this represents the most advanced cooling system all over the world, boasting an outstanding PUE of 1.14. In order to establish all these facilities, an investment of EUR 650 million has been done. The operational cost has been accounted to be equal to EUR 150 million. The analysis of the

economic impact of the investment estimated that Google facilities supported a gross domestic products' (GDPs) contribution of EUR 60 million on average per year since 2009 (Thelle et al. 2017). Therefore, it can be stated that the large Google investment has the potential to transform regions in economic recruiting by offering new jobs and increasing investment capital. Wind powers the Hamina center. Therefore, in order to use as much as renewable energy as possible, Google entered long-term agreements with wind farm developers. In fact, due to the intermittence and varying characteristics of renewable energy, most of the data centers cannot fully depend on the renewable energy to supply their IT loads. Therefore, in order to improve their greenness, data centers can apply the power purchase agreements (PPAs). Details of the PPA are presented in Sect. 4.1.2. These PPAs result in the best way to integrate renewable energy like wind energy and hydropower in order to obtain green data centers in the Nordic countries. In this regard, it has been estimated that the Google data center (Hamina, Finland) can achieve 97% renewable energy hourly coverage of electricity usage by using wind energy PPAs (Centers 2009). A few data centers with outstanding renewable energy coverage include the Apple data center in Viborg, Denmark (Apple 2015). The latter resulted the best location due to the perspective of satisfying the entire data center energy demand by using wind power PPAs. Moreover, the surplus of the heating produced will be recovered and recycled in the local district heating system, minimizing the environmental impact. According to Apple, this resulted in the biggest European investment, exceeding one billion euro.

The ability to reuse excess heat from servers is helping to improve the energy efficiency profile of those facilities. In northern districts with large heating requirements, the waste heat from data centers is mostly used for DH. For instance, Stockholm Exergi (formerly Fortum Värme) and Datacenter in Rosersberg AB (subsidiary owned by Ericsson) signed a new contract for cooling supply and heat recovery for a data center located in Stockholm. Specifically, by using the chilled water supplied (10 MW per year) by Stockholm Exergi, Ericsson can reduce the capital cost and obtaining environmental benefits. In fact, the heat recovered (80 GWh) thanks to the cooling process can be used to heat homes and offices by district heating. The excess heat results are able to heat 15 000 apartments with an expected carbon dioxide emissions reduction of about 4800 tons per year. Similarly, the internet supplier Bahnhof, which built an innovative data center by using an old rock cavity, estimated a potential heat recover equal to 112 GW·h. Moreover, the 'Open District Heating project', managed by Stockholm Exergi, gave Bahnhof company the possibility to sell the surplus heat to the city's district heating network with relevant economic and environmental benefits. As an example, during cold winter days, the megawatt hours recovered are ten times more than the megawatt produced during warm summer days. Currently, in Finland, the Telecity Group operates five data centers and three of them are using waste heat in the district heating system in order to provide heat to 500 single houses and 4500 block apartments. Specifically, in Espoo, the Tieto data center resulted is able to supply Fortum's DH network with 30 GWh of waste heat annually, winning the Uptime Institute 2011 Green Enterprise IT Beyond the Data Center granted by U.S.

Until now, the potential usage of data centers recovered heat in district heating has been evaluated. However, for small-sized data centers or in regions that are not too cold, the waste heat can also be recovered and reused for other purposes. As an example, an IBM data center in Switzerland is being used to heat a nearby swimming pool. Specifically, the hot air produced by the data center is sent to a heat exchanger where it is transferred to the water that is then pumped to the swimming pool. It has been found that the IBM data center is able to produce 2800 MWh of waste heat per year operating at full capacity, resulting a good solution in terms of not only security and energy efficiency data center but also as a technology that also the town can benefit. Finally, in Canada, the Quebecor Media, in Winnipeg, uses its data centers' waste heat in

order to heat the nearby offices of a local newspaper. Specifically, the entire project is expected to make use of outside cold air by installing air-side economizers that draw in the outside air based on the data centers' cooling needs. Once the air circulated throughout the servers is warmed up, 60% of it is used for editorial office heating and the remaining 30% is for adjacent warehouse.

4.7 Economic, Energy, and Environmental Analysis for Data Centers as Prosumers

This section reviews the economic and environmental analysis of data centers as prosumers in the existing studies.

4.7.1 Economic Analysis for Data Centers as Prosumers

4.7.1.1 Economic Analysis for Data Centers as Energy Consumer

The payback periods of renewable energy systems are affected by a number of factors, such as the renewable resources availability (Sheme et al. 2018), local electricity market (Lu et al. 2018), and renewable energy utilization efficiency (Goiri et al. 2015). For data centers located in sites with sufficient renewable resources, the payback periods are usually shorter. For instance, Sheme et al. (2018) compared the economic performances of PV panels in three regions, and their study showed that installing PV panels was more economic in Nigeria, followed by Greece, and the least economic in Finland. Due to the intermittency and variance of renewable sources, there can be large mismatch between the produced renewable power and the data center's power demand. Such mismatch will lead to reduced renewable energy self-consumption rates and thus causing economic losses. For instance, a team led by NREL conducted assessment of the Salem ARC data centers and evaluated the incorporation of renewable energy technologies (Metzger and Van Geet 2014). By installing rooftop PV panels, an annual cost saving of $15,338 can be achieved. However, due to the large implementation costs (i.e., $592,000), the payback periods reached as long as 46 years. To avoid waste of excessive renewable energy, advanced demand response controls have been developed (see Sect. 4.2), which can improve the economic performances of integrating renewable energy in data centers. For instance, by increasing renewable utilization, the Green BDT, a scheduling tool of bulk data transfers between geo-distributed sustainable datacenters, could reduce the energy costs by at least 48% (Lu et al. 2018). By scheduling the MapReduce jobs within the jobs' time bounds to maximize the green energy consumption and shifting the brown energy usage to low-price periods, the electricity cost could be saved by up to 39% (Goiri et al. 2012). Goiri et al. (2015) proposed a scheduler for parallel batch jobs in a data center equipped with photovoltaic solar arrays. By proper load shifting, their controller could achieve a cost saving of 28–31%. The capital cost of the studied datacenter's solar array can be amortized in 10–11 years, whereas it would take 18–22 years to amortize those costs under the conventional or even energy-aware schedulers.

4.7.1.2 Economic Analysis for Data Centers as Energy Producer

Many studies have also been conducted to analyze the economic benefits by recovering and reusing the data centers' waste heat. Oró et al. (2019) studied the economic benefits of reusing the air-cooled data centers' waste heat. They compared the cost benefits of different types of cooling systems (i.e., CRAH, CRAH + heat exchanger, CRAH + chiller, rear-door, and rear-door + chiller) and different locations of waste heat recovery (i.e., CRAH return air ducts and condenser) and concluded that the solution with water-to-air heat exchangers in the CRAH return air-stream is the most economic one. This is because the other solutions, which recover waste heat at the condenser side, need to replace vapor-compression chillers with new ones

incorporating heat reuse integration in the condenser. The average discounted payback period (DPBP, i.e., the period of time that is required to refund the capital investments with the consideration of discount rates (Pärssinen et al. 2019)) of the solution with water-to-air heat exchangers in the CRAH return air-stream was around 10 years, while the DPBP of the other solutions is over 15 years. Their study also indicates that the working temperature of air and fluids also has significant impacts on the economic performances of waste heat recovery solution. Marcinichen et al. (2012) investigated the cost benefits of waste heat recovering from a two-phase cooled data center. Their study showed that for liquid-pump-driven cooling, 30–40 million dollars could be saved when the condensing temperature varied between 45 °C and 100 °C. While for compressor-driven cooling, cost savings of 36–45 million dollars could be achieved when the condensing temperature varied between 45 °C and 60 °C. Murphy et al. (Murphy and Fung 2019) analyzed the feasibility of utilizing a 4000 kW cooling load data center's waste heat to supply multi-unit residential buildings in Canada and concluded that such waste heat reuse can achieve a 25% reduction in cooling costs. They also compared three different ways of connection between data centers and the DH network, i.e., using a heat pump, using a heat pump and one borefield, and using a heat pump and two borefields. The comparison showed that the scenario using a heat pump was the most profitable with a 11.9% 30-year after-tax internal rate of return (IRR, i.e., a discount rate that makes the net present value of all cash flows from a particular project equal to zero (Murphy and Fung 2019)), the scenario using a heat pump and one borefield was the second most profitable with an 7.8% 30-year after-tax IRR, and the scenario using a heat pump and two borefields was the least profitable with an 8% 30-year after-tax IRR for 15% of the total capital cost. Based on the actual plant and heat demand data, Wahlroos concluded that applying the data center's waste heat in Espoo of Finland could achieve a cost savings of 0.6–7.3% for the DH supplier, depending on the oil price and the electricity prices (Wahlroos et al. 2017). Using computational fluid dynamic (CFD) tools, Antal et al. (2019) investigated the waste heat reuse in a virtual data center for heating three neighborhood buildings. Their estimation indicated that the return on investment time was around 5.7–11.4 years, depending on heat demand/supply and price of heat.

The economic performances of waste heat recovery and reuse in data centers are greatly affected by the locations of waste heat recovered, the temperatures of recovered waste heat, and the connections with the downstream waste heat reuse facilities. In general, the waste heat recovery solutions without replacing the existing systems (e.g., waste heat recovery in CRAH return air ducts) will have better economic performances. A higher temperature of recovered waste heat will produce more cost savings. A simple connection (e.g., only heat pump is used) will reduce the payback periods. However, due to the reduction in the initial investments, in most cases, the waste heat reuse solutions with good economic performances do not have very good environmental performances.

4.7.2 Energy and Environmental Analysis for Data Centers as Prosumers

4.7.2.1 Energy and Environmental Analysis for Data Centers as Energy Consumer

Integrating renewable energy into data centers brings direct environmental benefits by replacing the high-GHG-emission fossil fuel with green and clean renewable energy. As the world's largest corporate buyer of renewable energy, Google is moving toward the 24 × 7 carbon-free energy at its data centers (Google 2018). Remarkably, in Hamina of Finland, as high as 97% of the Google's data center's electricity usage was matched by renewable energy on an hourly basis, which achieves substantial CO_2 reductions. Sheme et al. (2018) investigated using renewable energy to power data centers in 60° north latitude, and their simulation indicated that the annual value of minimum percentage

supply (MPS, i.e., the ratio of the renewable energy generated by 1 wind turbine and 1 m^2 PV panel to the data center's energy consumption, which describes whether applying renewable energy to supply data center is economical in a specific location (Sheme et al. 2018)) could change dramatically from 1.1% to 0.21% under different configurations of PV panels and wind turbines. With the development of advanced renewable power prediction and demand response controls (refer to Sect. 4.2), the renewable energy utilization can be further enhanced, thus producing more environmental benefits. For instance, Aksanli developed a mixed batch and service job scheduler in data centers based on renewable energy prediction, which could reduce the need of brown energy from 4.6 kW · h by more than 7 times to 0.64 kW · h (Aksanli et al. 2011). The parallel batch job scheduler developed by Goiri et al. (2015) was able to increase the green energy consumption by up to 117% compared with the case no job scheduling is conducted. The GreenHadoop, a MapReduce framework for a datacenter powered by solar energy and the electrical grid, could increase the renewable energy consumption by up to 31% (Goiri et al. 2012).

4.7.2.2 Energy and Environmental Analysis for Data Centers as Energy Producer

Regarding the environmental benefits brought by recovering and reusing the data centers' waste heat, Oró et al.'s (2019) study showed that the energy reuse factor (ERF, i.e., a metric for quantifying the percentage of reused energy to the data center total energy usage) metric could reach values about 55% for the heat reuse solutions integrated in the condenser of the vapor-compression chiller, while the ERF value was between 25 and 45% for the heat recovery solutions integrated in the return air flow. More waste heat can be recovered at the condenser side, since higher temperature waste heat can be captured. Marcinichen et al.'s (2012) study on waste heat recovery of two-phase cooled data center showed that 25% of the total potential savings in CO_2 can be achieved by using a liquid pumping cycle for the datacenter, equal to approximately 17,000 tons per year for a 173 MW power plant. While by using a vapor-compression cycle, the potential reduction in CO_2 emission can reach 70,000 tons per year. They concluded that the vapor-compression cycle has a larger impact on the secondary application making use of the waste heat than the liquid pumping cycle, due to the higher temperatures achievable. Murphy et al.'s (Murphy and Fung 2019) analysis showed that the scenario using a heat pump could reduce the buildings' annual heating-related GHG emissions by 2220 tons (i.e., 53%) and reduce the data center's annual cooling-related GHG emissions by 67 tons (i.e., 51%). Due to the deployment of thermal energy storage, the scenario using a heat pump and one borefield could achieve a GHG emission reduction of 3306 tons (i.e., 79%) for the buildings and a reduction of 74 tons (i.e., 56%) for the data center. Further, the deployment of two borefields could accomplish a GHG emission reduction of 3,522 tons (i.e., 84%) and 108 tons (i.e., 82%) for the buildings and data center, respectively. Wahlroos et al.'s (2018) study indicated that data centers in Finland would consume 5 TW · h power, and most of the consumed power could be used for DH. For a HP with COP of 2.6, the potential heat output could reach as high as 8 TW · h, accounting for 20% of the total DH production. This amount of reused waste heat would also cut down the current oil use from 10% of the total heat supply to 4%. He et al.'s (2018) study indicated that the waste heat recovered from a data center located in Hohhot of China could save 18,000 more tons of coal each year compared with the coal-fired boiler heating system, and meanwhile, nearly 10% of the annual power saving can be achieved for the data center.

To sum up, recovering waste heat from chiller condensing side, increasing captured waste heat temperatures, and using thermal energy storage are three effective ways to improve the efficiency of reusing data center's waste heat and thus enhancing the environmental benefits. However, the increased environmental benefits will usually

require high investments (e.g., replacement of the existing chillers, installation of thermal energy storage). Thus, a balance between the economic and environmental performances should be made by the decision-makers for data center's waste heat recovery and reuse in practice.

4.8 Challenges and Future Work Discussion

For integration of renewable energy, the existing controls optimize data centers' operation to match its demand with the renewable energy generations, while neglecting the impacts on the production of waste heat and its impacts on the waste heat utilization systems. For instance, many existing controls shift the flexible workload to the daytime when there is sufficient solar energy. Correspondingly, large amount of waste heat is generated in the daytime period. If the waste heat is used for heating the adjacent rooms in the data center or heating swimming pool, there will be a large mismatch between the waste heat supply and end-users' heating demand, which will result in reduced heating efficiency. Global controls, which can manage the upstream renewable production, data centers' operation, and waste heat generation, and downstream waste heat utilization are lacking. Thus, future work is needed to develop such supervisory controls that can globally optimize the data centers' operations. Meanwhile, large uncertainties exist in the renewable energy production, data center workload, and the end-users' heading needs. Most of the existing studies do not consider these uncertainties and merely perform controls in a deterministic framework. As a consequence, the optimal solutions obtained from the deterministic scenario tend not to be the optimal in the real situation with large uncertainties existed. Thus, the uncertainties in the prediction of renewable energy generation and data center workload should be integrated, and robust optimization methods should be introduced for improving the robustness and reliability of the controls.

As introduced in Sect. 4.2.3, different ways of connecting data centers with the downstream waste heat utilization facilities can lead to very different overall system energy efficiencies. Factors such as the temperature of recovered heat, number of used heat pumps, heat pump cycles, and whether there is energy storage or not, all have significant impacts on the overall heat reuse efficiency. More investigations are needed to develop proper means of data center connections (e.g., the ancillary components, connection topology) in the district energy systems, especially in the waste heat utilization side, to increase the system overall energy efficiencies. The developed connection means can be used as prototypes for application of the waste hear reuse solutions in data centers.

The utilization of renewable energy and waste heat reuse for data centers have large dependence on the climate regions and the locations of data center. For instance, in Nordic countries, there are rich wind resources, and thus, many data centers use the wind power to supply their IT facilities. While in the USA since there are rich solar resources, the commonly used renewable energy resources for powering the data centers are the solar energy. Similar to the renewable energy integration, waste heat utilization is also region specific. For instance, utilization of waste heat in DH is only applicable to the cold regions with large heating demands. For warm or hot regions which do not have large heating needs, the waste heat from data centers has to be used for other purposes other than DH. In different locations, the optimal way of integrating renewable energy and waste heat recovery could be totally different. Thus, there is a need to fully investigate such diversities, caused by climates and renewable resources, for promoting the renewable energy utilization and waste heat reuse in data centers and thus improving the data centers' energy efficiency.

The existing metrics evaluate the data centers' performances either from the perspective of energy consumers (e.g., PUE) or from the perspective of energy producers (e.g., ERE, ERF). An evaluation mechanism that assesses data

centers as prosumers with the overall performances (i.e., considering renewable integration and waste heat utilization simultaneously) considered is still lacking. This is hindering the performance improvements of data centers from a higher and global level (with both the upstream renewable integration and downstream waste heat reuse considered). Future work is needed to develop such global metrics for more appropriate quantification of the data center performances.

4.9 Summary

This chapter has presented a comprehensive review of data center from the perspective of energy prosumers in district energy systems. As an energy consumer, data centers can use renewable energy to replace the fossil fuel-based energy to play a significant role in reducing carbon footprint. As an energy producer, the waste heat in data centers can be utilized in multiple ways to improve the overall energy efficiency. The purpose of this chapter is to seek new opportunities for improving data centers' overall energy efficiency and reducing carbon emission by providing engineers/ researchers a full picture of data centers as a role in the district energy systems.

This chapter has systematically reviewed the data center technologies in aspects of cooling systems, integration with renewable energy and advanced controls, waste heat recovery and reuse (especially district heating) techniques, some real green data center projects, and the commonly used metrics for assessing data centers' performances. In aspect of cooling systems, three types of cooling technologies for data centers, namely air-cooled systems, water-cooled systems, and two-phase cooled systems, have been reviewed, and their different temperatures and performances have been compared. In aspect of integration with renewable energy, the various ways of integration and the up-to-date advanced controls have been presented and discussed. In aspect of waste heat utilization, different locations in data centers for waste heat recovery and various ways of waste heat reuse solutions have been summarized. Special attentions have been given to waste heat utilization in district heating, due to its relative high efficiency and easy implementation in district energy system application. Then, some real green data center projects, with either renewable energy integrated or waste heat reused, have been introduced and discussed. Last, the metrics and performances indicators related to data centers as a part of district energy systems have been reviewed.

Future work is needed to develop advanced control methods that can coordinate the operation of the whole systems, to develop new system topology that can maximize the operation energy efficiency, and to develop new metrics that can evaluate the data center overall performances (with upstream renewable energy integration and downstream waste heat utilization considered).

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Characteristics of Urban Energy System in Positive Energy Districts

5

Xingxing Zhang

Abstract

Positive energy district (PED) is recently proposed to be an integral part of a district/urban energy system with a corresponding positive influence. Thus, the PED concept could become the key solution to energy system transition toward carbon neutrality. This chapter intends to report and visualize the initial analytical results of 60 existing PED projects in Europe about their main characteristics, including geographical information, spatial–temporal scale, energy concepts, building archetypes, finance source, keywords, finance model, and challenges/barriers. As a result, a dedicated database is developed, and it could be further expanded/interoperated through an interactive dashboard. It is found that Norway and Italy have the most PED projects so far. Many PED projects state a 'yearly' time scale while nearly 1/3 projects have less than 0.2 km^2 area in terms of spatial scale. The private investment together with regional/national grants is commonly observed. A mixture of residential, commercial, and office/social buildings are found. The most common renewable energy systems include solar energy, district heating/cooling, wind and geothermal energy. Challenges and barriers for PED-related projects vary from the planning stage to the implementation stage. Furthermore, the text mining approach is applied to examine the keywords or concentrations of PED-related projects at different stages. These preliminary results are expected to give useful guidance for future PED definitions and proposals of 'reference PED.'

Keywords

PED · Characterization · Review · Text mining

5.1 Introduction

Recently, the positive energy district (PED) concept has been discussed substantially as it could become the key solution to energy systems in transition toward carbon neutrality. According to European Strategic Energy Technology (SET) Plan Action 3.2 (SET-Plan ACTION N° 3.2 Implementation Plan.), PED could be defined as an energy-efficient and energy-flexible urban area with surplus renewable energy production and net-zero greenhouse gas emission in a certain time frame. Some PED initiatives aim to create a

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knowledge base and a roadmap to achieve the energy transition of cities according to established time horizons (Civiero et al. 2019).

Most of the studies and practical experiences about PEDs are based on newly built districts or planning of future districts. Monti (2017) described the process of adaption and the challenges/barriers faced by the PED decision makers. They also proposed how simulation, optimization, ICT approaches, and business models are combined in a holistic and pragmatic way. Lindholm et al. (2021) defined three types of PEDs (i.e., PED autonomous, PED dynamic, and PED virtual), depending on the system boundary and energy import/export conditions. They also pointed out that PED is highly dependent on local context with many impacting factors, such as the available renewable energy sources, energy storage potential, population, energy consumption behavior, and costs and regulations, which affect the design and operation of PEDs in different regions. A series of technical solutions, such as the integration of batteries, electric vehicles (EVs), and grid-responsive control, were discussed to promote the development of PEDs (Zhou et al. 2021). Samadzadegan et al. (2021) developed a framework to design energy systems for PED or zero-carbon districts, by focusing on estimating heating and cooling demand and sizing related renewable energy systems, e.g., solar photovoltaic (PV) and heat pumps. Shnapp et al. (SHNAPP et al. n.d.) proposed handling the energy performance targets by transferring to the district level the minimum energy requirements imposed by the energy performance of buildings directives to individual buildings. Moreno et al. (2021) proposed a methodology for calculating the energy balance at the district level and energy performance of those districts with the potentials to become PEDs. A 'double density' simulation scenario was studied further by Bambara et al. (2021) to test residential densification potential for PED, where each existing detached house in a community is replaced with two energy-efficient houses of equal living area on the same land lot. From economical and technical points of view, Laitinen et al. (2021) concluded that it is more feasible to achieve PED or net-zero-energy district, rather than full energy self-sufficiency after they studied a series of technologies (e.g., local centralized wind power, solar PV, battery, heat storage, and heat pump), using Helsinki as a case study. Moreover, Soutullo et al. (2020) suggested that urban living labs could be a driver to achieve PED. Fatima et al. (2021) studied PED's implementation potential from a citizen engagement aspect. Uspenskaia et al. (2021) recommended planning and modeling the replication of PED at the very early stage because it is important to find tailor-made solutions to fit spatial, legislative, socioeconomic conditions, and historical growth of the cities.

Apart from the newly built districts, an explanatory study was carried out as the first step to support the complex planning urban refurbishment, in order to achieve PED (Nzengue et al. n.d.). In their study, the key information on the different district types (e.g., energy consumption) was simulated to identify the districts with the highest potential for energy refurbishment. Civiero et al. (2021b) provide a view of a district simulation model able to analyze a reliable prediction of potential business scenarios on large-scale retrofitting actions and to evaluate a set of parameters and co-benefits resulting from the renovation process of a cluster of buildings. Gouveia et al. (2021) also argued that the transformation of the existing districts is essential, including historic districts, which present common challenges across EU cities, such as degraded dwellings, low-income families, and gentrification processes due to massive tourism flows. In their report, they discussed how the PED model can be an opportunity for historic districts to reduce their emissions and mitigate energy poverty. Moreover, a methodology for the evaluation of positive energy buildings and neighborhoods is proposed in the report (Salom et al. 2021), where a set of key performance indicators (KPIs) are defined with details on the calculation procedure for categories of energy and environmental, economic, indoor environmental quality (IEQ), social, smartness, and energy flexibility.

A research gap is thus observed that there are many studies starting to address technical,

economic, social aspects of PED, but very limited studies are found in characterizing PED. The Joint Programme Initiative Urban Europe (JPI UE) (JPI Urban Europe—ERA-LEARN) plays an important role in coordinating PED projects across Europe; it actively engages the interests of different stakeholders, particularly, cities in PEDs. To accomplish its objectives, only Bossi et al. (2020) summarized part of PED's characteristics in aspects of geographic distribution, implementation status, building structure, land use, energy typology, success factors/challenges, and barriers while Brozovsky et al. (2021) identified different terminologies of PED, and related focused aspects (i.e., energy, social, climate). JPI UE needs more comprehensive scientific advice on the knowledge and methods for guiding the design, monitoring the operation, and evaluating the performance of PED projects. Therefore, many other PED characteristics need to be abstracted and categorized for further development of PED, such as district size, finance source, energy concepts, building archetypes, spatial/temporal scale, and keywords. Moreover, as PED projects are expanding all the time, it is necessary to use a common tool/database to increase the semantic interoperability among different stakeholders, for an updated summary of PED's main characteristics.

In the framework of both International Energy Agency—Energy in Buildings and Communities (IEA EBC) Programme Annex 83 (Hedman et al. 2021) and EU Cost action CA19126 (Action CA19126—COST n.d.), the working groups are now collecting data of PEDs and characterizing them for potential proposal of reference and replication of PEDs in different contexts. This chapter, therefore, reviews the existing 60 projects within the European area from the JPI Urban Europe PED booklet, establishes the database, and further analyze/visualizes them for the main characteristics. The chapter aims to illustrate the basic characteristics of existing PED projects in the EU and then deliver the information to the targeted stakeholders, such as municipality, urban planner, real estate developer, utility company, policy/regulation maker, renewable energy provider, and energy engineer, for them to further define, design, promote, and implement potential PED projects. As the PED concept is new to most of the stakeholders, this chapter intends to transfer the knowledge to the targeted groups through the review/analysis and the development of a database. The result will be also used for the iterative definition of PED in the two initiatives of IEA and EU cost action.

5.2 Data Source and Research Methods

5.2.1 Data Source

The data of PED-related projects are collected from the PED booklet (Gollner et al.) by JPI UE updated latest on 2019. JPI Urban Europe is conducting a program on 'Positive Energy Districts and Neighbourhoods (Positive Energy Districts (PED) | JPI Urban Europe for Sustainable Urban Development' with an implementation plan, SET (Strategic Energy Technology) Plan Action 3.2 (SET-Plan ACTION N°3.2 Implementation Plan.), participated by about 20 European member states, in the context of Europe commitment toward clean energy transition and carbon neutrality. The total databank consists of 60 projects' data that have similar goals to PED projects in Europe. These projects have been identified and updated by the participated cities of workshops conducted by JPI Urban Europe. The database is divided into several key parameters shown in Table 5.1.

However, it has been challenging to understand the energy typology and detailed strategies due to unclear/insufficient information for many projects from the JPI Urban Europe booklet. The data for the temporal scale of the projects are only available for very few projects. Due to this insufficient information, external sources, such as the Web site/publication of the specific project, have been studied and reviewed in order to collect more detailed information [25–42] (32. RHC—2050 Vision for 100% Renewable Heating and Cooling in Europe., n.d.; American Football in Oulu, Finland—Oulu Northern Lights, n.d.; Aulahunosa.Es, n.d.; Évora—POCITYF—

Table 5.1 Table parameters for data collection

Key parameters	Type of Data
Project characteristics	Location, initiated year, development stage, project area, finance model, etc.
Type of buildings involved	Residential, commercial, social, industry, etc.
Common energy technologies	Solar thermal, geothermal, PV, heat pumps, etc.
Key energy concepts	Energy combinations and strategies to meet the goals
Keywords	Positive energy district, smart city, etc.
EV/E-mobility	Included/excluded in energy strategies
Temporal scale	Hourly/monthly/yearly, etc.
Driving stakeholders	Municipality, citizens, real estate developers, etc.
Others	Supporting regulations, barriers, key success factors, etc.

POCITYF, n.d.; La Fleuriaye, Nature d'avance—Carquefou (44) | La Fleuriaye, Nature d'avance—Carquefou (44), n.d.; Medicon Village, E.ON Ectogrid., n.d.; Municipal District Heating Company of Amindeo (D.H.C.A.)—ΔΕΤΕΠΑ, n.d.; Om Smart Energy Åland | Smart Energy Åland, n.d.; Samen Werken Aan Minder CO2: Sustainable Energy and Environment, n.d.; Sinfonia Smartcities—Home, n.d.; Stadtteil Dietenbach—www.Freiburg.de—Planen Und Bauen/Aktuelle Projekte/Stadtteil Dietenbach, n.d.; Stadtwerke Hennigsdorf—Research Projects, n.d.; Stardust, n.d.; Urban Innovation Lab—Competence and Network Regarding Sustainable Social Development, n.d.; Werksviertel—Werksviertel München, n.d.; ZERO EMISSION NEIGHBOURHOODS IN SMART CITIES Definition, Key Performance Indicators and Assessment Criteria: Version 1.0. Bilingual Version, n.d.; Giourka et al. 2020; Olivadese et al. 2021).

5.2.2 Research Methods

5.2.2.1 Development of Database

A comprehensive critical review was conducted based on the JPI Urban Europe booklet and the related academic literature. The essential data of literature were broken down into thematic categories as shown in Table 5.1. The important characteristics for PED were either discussed by experts in IEA EBC Annex 83 (Hedman et al. 2021) and EU Cost action CA19126 (Action CA19126—COST) or extracted from the literature. All the information was observed, recorded, and summarized in the Excel sheet, which forms up the basic database for this review.

The key thematic parameters for the database are described in detail as below:

- Project characteristics include the location of the project, initiation year, the status of the project in 2019, which is further divided into stages 'in planning,' 'in implementation,' 'implemented/in operation.' Such categorization refers to the projects where construction of the energy systems is completed and yet to be commissioned or integrated into the existing energy networks. The amount of area is being consumed by the cumulative of all energy systems installed with this project implementation. The appropriate financing source of each project is also checked.
- The type of buildings involved in the PEDs consists of residential, commercial, and industrial. In most cases, renewable energy systems are installed on building components (e.g., roofs, envelopes) to reduce local energy demands and further supply excess energy generation to the neighborhoods.
- The common energy technologies used in PED are reviewed, including energy supply and storage.
- Key energy concepts are examined with strategies and detailed planning to reach the

project goals. The selection of energy system combinations with different technologies is crucial, which needs intensive investigation and planning.

- The keywords used in the projects are identified, and the most common keywords are abstracted. These keywords vary between the projects with different names, comparing to PED, such as smart city, positive energy blocks, zero-energy building, smart grid, zero-energy district, and urban energy transition.
- Inclusive strategies of EV/e-mobility are identified and included in the data collection. The strategies aim to encourage clean transport solutions within PED scope and integrate with energy systems to provide energy flexibility.
- The temporal scale of the project refers to achieving the project goals, relative to the time period in a day/month/year scale. Since most of the projects are still under planning and implementation stages and due to insufficient information from the sources, the data for temporal scale are only available for less than 50% of the identified projects.
- Stakeholders in each project are summarized, such as a regional municipality, citizens, and real estate developers. They are involved in a different stage of project development. The key drivers vary between every project and have analyzed the common driving stakeholders to understand the trends.
- The key success factors with supporting regulations along with challenges are collected. Every project would come across challenges/barriers or have key success factors while planning and implementing the project.

5.2.2.2 Text Extraction and Mining Method for Keywords Abstraction

The data used for extracting word clouds and sentiments are collected from the JPI Urban Europe booklet available in.pdf (portable document format) format. The projects are grouped according to the PED ambition and the development phase they are in, as shown in Table 5.2.

Step 1: Text extraction and mining methods were firstly applied in Python with the aid of Pandas library (version 1.2.4, GitHub, Inc., San Francisco, USA) (Reback et al. 2021) to transform this data from an unstructured mix of tables and text into clean and structured data frames. These cleaning methods involved extracting the data from '.pdf' format into '.txt' (Text) format (since it is more friendly for running analysis), setting up of the text as structured data frames, removal of extra spaces, special characters, line breaks, Web site protocols, formatting the cases, stemming (Anjali and Jivani 2011), and removal of stop words. Hence, the resultant is a data frame consisting of 6 cleaned records (belonging to the 6 groups of projects mentioned in Table 5.2), each record containing consolidated transcripts of all the project descriptions belonging to the respective groups.

Step 2: Natural language processing (NLP) method using text mining in Python with the aid of the Natural Language Toolkit (NLTK)

Table 5.2 Project groups according to PED ambitions and their development phase

Project phase	Description
PED implemented	Indicate PED ambition and are implemented
PED in implementation	Indicate PED ambition and are amidst implementation
PED planning	Indicate PED ambition and are still being planned
Toward-PED implemented	Did not declare a PED ambition but present interesting features for the PED Program and are implemented
Toward-PED in implementation	Did not declare a PED ambition but presents interesting features for the PED Program and are amidst implementations
Toward-PED planning	Did not declare a PED ambition but presents interesting features for the PED Program and are still being planned

libraries (Version 3.5, O'Reilly Media Inc., California, USA) [45] was subsequently used to extract the most used words from the 60 projects. Each word from each of the 6 records of the cleaned data frame is tokenized into its own variable, and the number of times the word repeats itself is the count value of that token. A new data frame is created to capture the tokenized word and its count value. This is repeated for each of the 6 groups, and the top 50 words from each group are extracted along with their count value and plotted on a word cloud. A word cloud is a method of visualizing the most used words in transcripts of text data by using the count value of the tokenized words for the sorting. The words in a word cloud are displayed in a specific spatial format: the font size of the words indicates relevance to the magnitude of their use, and colors vary for esthetic reasons.

Step 3: TextBlob library (Version 0.16.0, Steven Loria, New York, USA) (Loria n.d.) was then used to carry out a sentiment analysis study (Loper and Bird 2002) on the dataset in order to determine the polarity and subjectivity of the groups of projects. The polarity value is used to indicate the positive or negative sentiments of a sentence, for example, 'happy,' 'nice,' 'sad,' 'bad,' and such. Each word has a certain polarity value (positive or negative), and aggregated results of the values of words in an entire transcript are used as the key indicator of the opinion of that transcript (Agarwal and Mittal 2016). Subjectivity and objectivity are the next measures determined wherein subjectivity is the expression of opinion in a text, and objectivity is the expression of facts.

5.2.2.3 Data Visualization

Given that the dataset contains several projects across different cities in Europe, a spatial visualization of the location of these projects was deemed vital. QGIS software (Version 3.10, Open Source Geospatial Foundation, Beaverton, USA) (References—Citing QGIS in Formal Publications—Geographic Information Systems Stack Exchange) is a Geographic Information System (GIS)-based open-source software used here to display the cities on a map. Each project is appended with the latitude and longitude of the city it lies in, and these latitudes and longitudes are wrapped over a European base map.

Another visualization technique used to plot the dataset in this project is an interactive dashboard (for non-spatial variables only) developed using the open-source Konstanz Information Miner Analytics Platform (Knime) (Version 4.3.2, KNIME AG, Zurich, Switzerland) (FAQ | KNIME). Variables across the dataset are plotted against each other using interactive graphs and charts, for example, for visualizing the type of financing against the year of initiation of the project and other such co-relations. Interactive means that a user can click on a project in one plot to highlight characteristics about that specific project in other plots across the dashboard as well.

5.3 Results

5.3.1 Characteristics of Existing PED Projects

5.3.1.1 Initiation Year

The section shows the year of initiation of the first phase of all the 60 collected PED-related projects in Europe. From Fig. 5.1, the first project was initiated in 1970 and the second project in 1995, both in France. There have been very few projects, less than one project each year until before 2014, where 5 projects took place in that year. The momentum has increased from then with 8 projects in 2016, 9 projects in 2017, 11 projects in 2018, 6 projects in 2019, 4 projects in 2020, and no data for 5 projects.

5.3.1.2 Location of Identified 60 PED-Related Projects

This location of the identified 60 PED-related projects is displayed in Fig. 5.2. The most amount of projects are located in Norway, i.e., 9 projects, followed by 8 projects, 7 projects, 6 projects, 5 projects in Italy, Finland, Sweden and The Netherlands, respectively. There are 4 projects in Spain, Germany, and Austria, 2 projects

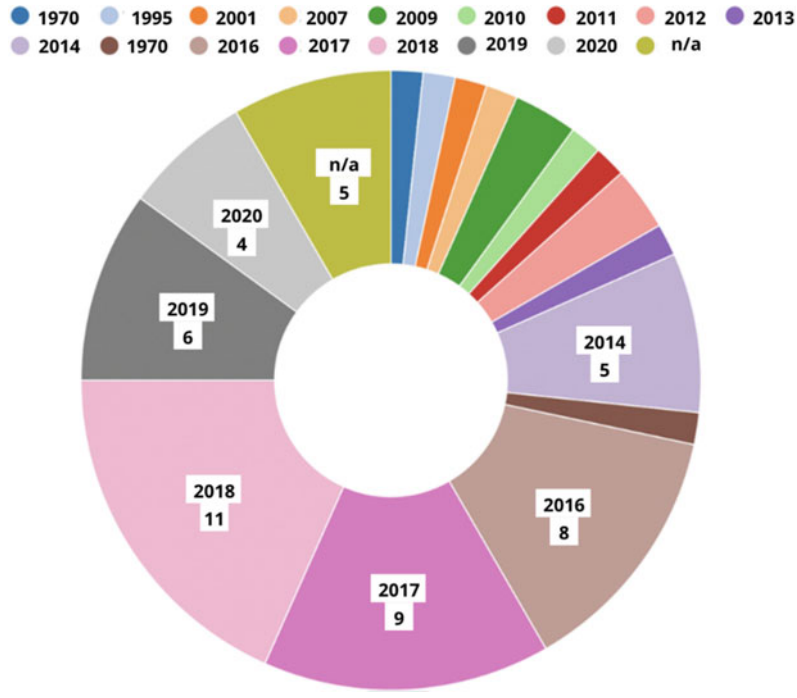

Fig. 5.1 Initiated year of PED-related projects

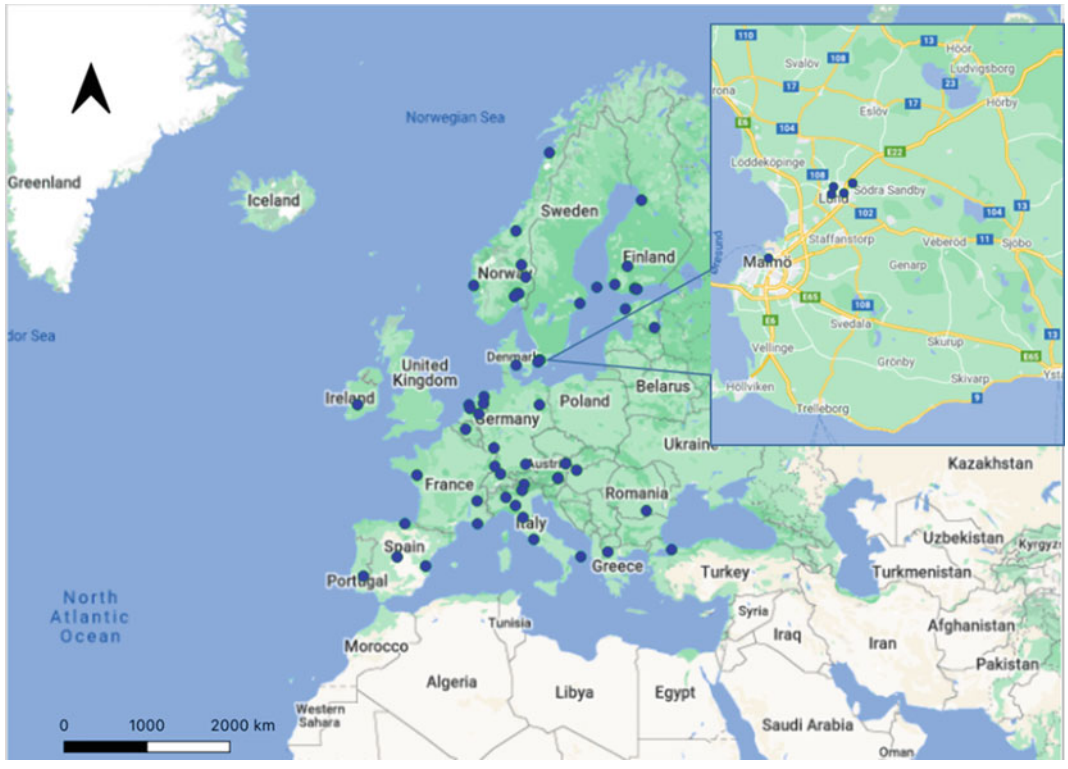

Fig. 5.2 Locations of 60 PED-related projects

in both France and Denmark. There is one project in each of the remaining countries, Portugal, Turkey, Ireland, Belgium, Hungary, Switzerland, Greece, Estonia, and Romania.

5.3.1.3 Status of the Identified Projects

This section reports the current development stage of 60 PED projects divided into categories mentioned in the development of the database. From Fig. 5.3, the results clearly indicate that majority of the projects are under the implementation stage, i.e., 26 projects. There are 11 projects under the planning stage and 6 projects under both the planning and implementation stages. In total, 16 PED-related projects are already implemented or in operation, among which 5 projects have completed implementation but have yet to integrate the energy systems into the existing local energy networks of the specific projects, while 11 projects are finally in operation stage. Information is not available for one project.

5.3.1.4 Project Area (Spatial Scale)

The amount of project area (spatial scale) is counted by considering the installation of the planned energy systems in their locality. These energy systems might be installed on the residential, commercial or industrial roofs, or flat ground-mounted in open fields, or even through the virtual presence of an energy system. From Fig. 5.4, most of the projects, i.e., 19 projects are claimed to be using less than 0.2 km^2 area, 7 projects between 0.21 and 0.4 km^2 area, 8 projects consuming area between 0.81 and 3.0 km^2, and there is one project claim to be consuming more than 25 km^2 area.

5.3.1.5 Finance Models Used in PED Projects

In order to meet the project goals and bring clean energy transition, the finance model plays a vital role whereas this section demonstrates the common trends being deployed in 60 PED-related projects shown in Fig. 5.5. The combination of public, private, and others, such as national or regional grants, has been the most common strategy in 20 projects. Only public financing in terms of EU grants or municipality funding is observed in 14 projects out of 60 projects in Europe, 5 projects which solely depend on

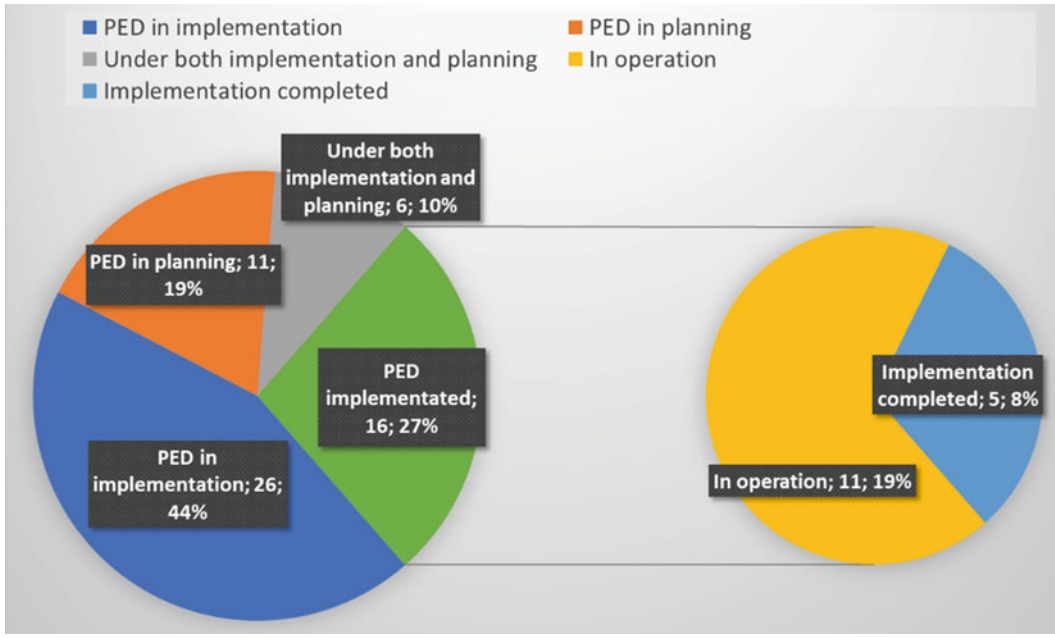

Fig. 5.3 Development stage of collected 60 PED-related projects

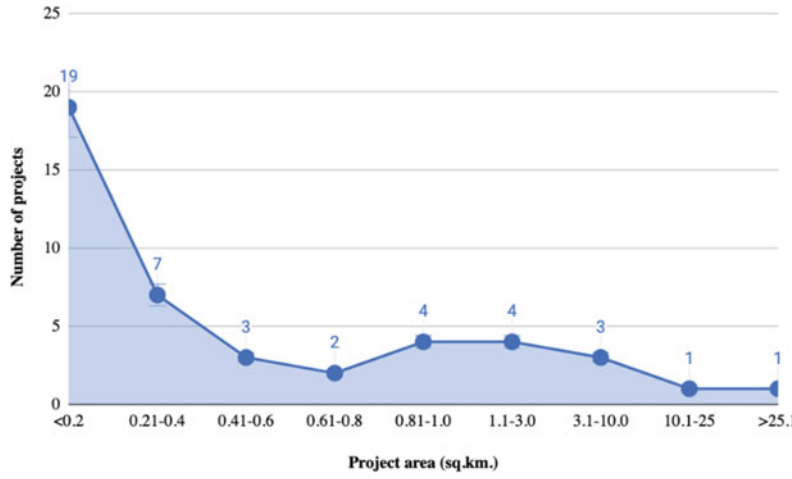

Fig. 5.4 Project area of the 60 PED-related projects

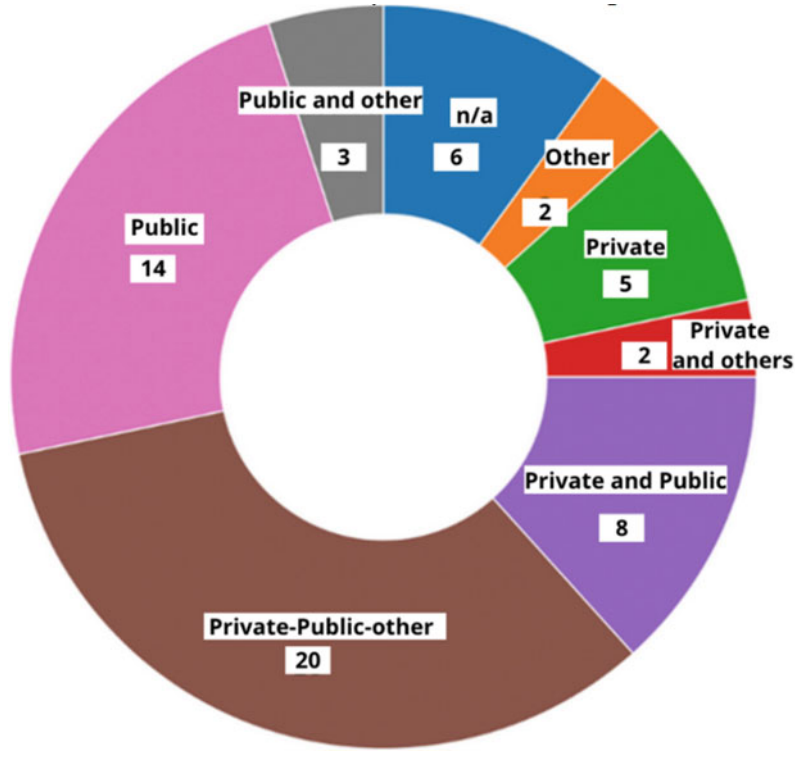

Fig. 5.5 Commonly used type of finance models

private financing strategy, and there are 8 projects forwarding with private and public finance combination. However, there are more than 6 projects which do not have proper information about the financial model in the PED booklet by JPI Urban Europe.

5.3.1.6 Type of Buildings Involved

This section presents the commonly involved building types for installation of energy systems to supply local energy demand and also to generate excess energy to increase energy flexibility according to the specific project goals. Figure 5.6

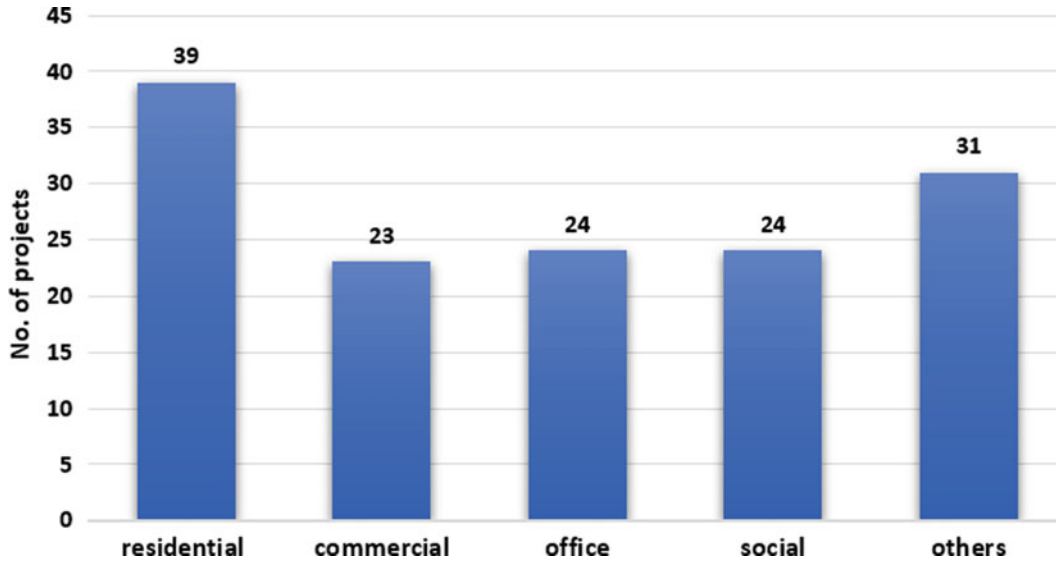

Fig. 5.6 Type of buildings involved for space utilized by energy systems

illustrates that the residential sector appears to be predominantly used in the majority of the projects to install energy systems on available roof areas as it is being the primary focus for 39 projects. Office and social buildings are identified to be the main focus in around 24 projects and also followed by commercial buildings spaces for more than 20 projects. Other types of buildings such as institutional and cultural are utilized as secondary spaces for implementing the energy systems.

It is also observed that almost all the projects have considered a mixture of different building types, depending on the major type of buildings existing in the locality. However, the overall trend focuses on involving the citizens as key drivers with the right motivating strategies which eventually address the spatial challenges to install energy systems required for local energy demand.

5.3.1.7 Major Energy Technologies

The commonly used energy technologies in these PED projects are examined and referred to as the three pillars of Energy Generation Energy Flexibility Energy Efficiency.

These energy technologies are divided into categories as solar, district heating/cooling, heat pumps, geothermal energy, combined heat and power (CHP), energy storage, wind, e-mobility, and others present in the inner circle of the pie chart shown in Fig. 5.7. Solar energy technology is identified to be the primary source of energy supply in almost all projects, specifically photovoltaics (PV) and thermal are the main contributors for producing electricity and heating applications, respectively. There are five situations where projects claimed to use solar technology but have not been specific about the type of solar energy. Other new/innovative forms of solar such as hybrid photovoltaic/thermal (PVT), building integrated photovoltaics (BIPV), floating solar, and solar roads technologies also have been considered in few projects.

District heating/cooling has been founded in 45 projects, in which heating is used in 43 projects and cooling in 2 projects. Heat pumps, geothermal energy, and CHP plant used in 37 projects, 27 projects, and 21 projects, respectively. Electro-chemical energy battery technology storage for electricity application and seasonal thermal energy storage technology for heating/cooling application are explored as under the energy storage category. Wind energy and E-mobility technologies are identified using in 6 projects and 8 projects, respectively. Other

Fig. 5.7 Commonly used energy technologies

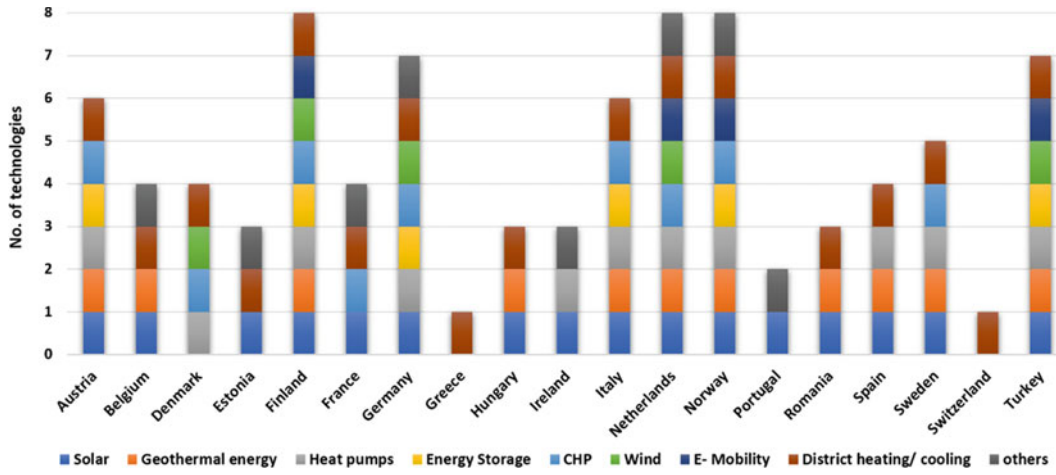

Fig. 5.8 Country-wise approach of energy typology

technologies, such as bioenergy, green hydrogen, hydropower, and natural/mechanical ventilation, have also been integrated partly in few PED-related projects in Europe.

Figure 5.8 represents the diversity of energy technologies in each country. Solar energy, district heating/cooling, and heat pumps technologies are commonly considered in almost of the

countries; geothermal energy and CHP plant are being used in nearly half of the countries as represented in Fig. 5.8. Wind energy is integrated in a smaller number of countries such as Denmark, Finland, Germany, The Netherlands, and Turkey, and energy storage is only seen in few countries such as Austria, Finland, Germany, Italy, Norway, and Turkey.

Furthermore, the results indicate that Finland, The Netherlands, and Norway have high diversity of using more types of energy technologies, followed by Germany, Austria, Italy, and Turkey.

5.3.1.8 Challenges Under Different Implementation Stage

The data collection focuses on challenges/barriers that are categorized into 'under planning,' 'under implementation stage,' and 'implemented/in operation' stages shown in Table 5.3. The gathered information on challenges/barriers reveals the following main topics: Administrative and policy (A&P), Legal and Regulatory (L&R), Technical. Environmental, Social and Cultural, Information and Awareness, Economical and Financial, and Stakeholders interest perspective (Civiero et al. 2021a).

Challenges associated with stakeholders' involvement, administrative, and technical issues had great relevance in all PED stages. The economic and financial feasibility was crucial in both planning and implementation stages as well as supporting studies or knowledge. However, legal and regulatory barriers were important in the implementation and operation stages. Finally, only in the operation stage environmental and social and cultural aspects were considered possible barriers.

5.3.1.9 Most Commonly Used Words and Sentiment Analysis

Figure 5.9 shows the most commonly used words in the project description transcripts according to their classification from Table 5.2. As seen from the figure, projects that are already implemented (both PED and toward PED) show high use of words like 'consumption,' 'passive,' 'heating,' and 'industry.' On the other hand, projects that are yet planning (both PED and toward PED) use words such as 'urban,' 'solutions,' 'quarter,' 'research,' and 'residential.' Projects that are in implementation (both PED and toward PED) mostly repeat words like 'citizen,' 'planning,' 'urban,' heating,' 'supply,' and 'cost.' Finally, both implemented and in implementation toward PED projects use heating, cost, and supply words.

Figure 5.10 displays the sentiments portrayed by the 6 groups of projects in the context of polarity (positivity and negativity) and subjectivity-objectivity (opinions-facts). In general, PED implemented projects have very positive feedback, reflecting by the text. We see both PED and toward PED implemented projects have higher subjectivity than objectivity, compared to their planning phase counterparts. This could be interpreted as the implemented projects are mostly influenced by diverse factors, such as dynamic data, citizens, and other stakeholders, while those projects in planning stages emphasize more on objective learning experience from literature, simulation data, and the related estimations.

5.3.2 Interactive Dashboard

The interactive dashboard consists of five visualization charts in total (as shown in Fig. 5.11). The display begins with a pie chart that visualizes the proportions of projects initiated across the years. The respective color scheme index displays the corresponding year in which the project was initiated. The displayed values across the pie chart can be toggled between the number of projects and proportions in the form of a percentage. Below the yearly distribution chart, on the left, is a horizontal bar chart that shows the proportions of the projects based on their grouping from Table 5.2 (i.e., PED ambition and phase of implementation). On the right, a second pie chart visualizes the types of investments received by the projects and their respective proportions. Finally, two scatter plot charts are displayed at the bottom of the dashboard. The left chart shows the

Table 5.3 Challenges and barriers in different stages of PED projects according to the main topics

Topic	PED in planning	PED in implementation	PED implemented/in operation
Administrative and policy	Conflicts between different authorities involved in the project	Political management	Approvals and permits from municipality and other entities might lead to project timeline extension
Legal and regulatory		Regulatory framework which governs involved actors throughout Europe	Regulatory barriers for piloting/testing
Technical	System boundary conditions defined	Identification and deployment of local feasible clean energy systems	Analysis required for hybrid energy system operations
	Coping with rapid growth of new technologies		Analysis required for underground seasonal energy storage
			Energy generation system is far away from the consumers
			Thermal mining challenges in the urban areas to reduce the distance from energy generation system far away
			The electricity supply examined properly above 90 degrees
Environmental			Disallowing inefficient and high polluting energy generation systems
Social and cultural			Cultural differences between different cities involved in the partnership
Information and awareness		Local citizen acceptance toward new things in rural areas	
Economical and financial	Economic feasibility	Finance dependence on private investors	
	Finance availing according to the project timeline	Local finance	
	Overlapping implementation with local ongoing constructions		
Stakeholders interest Others	Encouragement of project drivers like real estate developers	Stakeholders and involved actor's commitment toward project goals	Conflicts due to lack of common interest between different landowners
	Uncertainty in stakeholder's commitment	Creating interest in project drivers like building owners and landlords	Strong collaborations needed between energy companies and real estate developers for fast implementation
	Active consideration of local knowledge	Lack of supporting studies/knowledge for implementation	
	Lack of supporting studies/knowledge for planning		

Fig. 5.9 Most commonly used words for PED

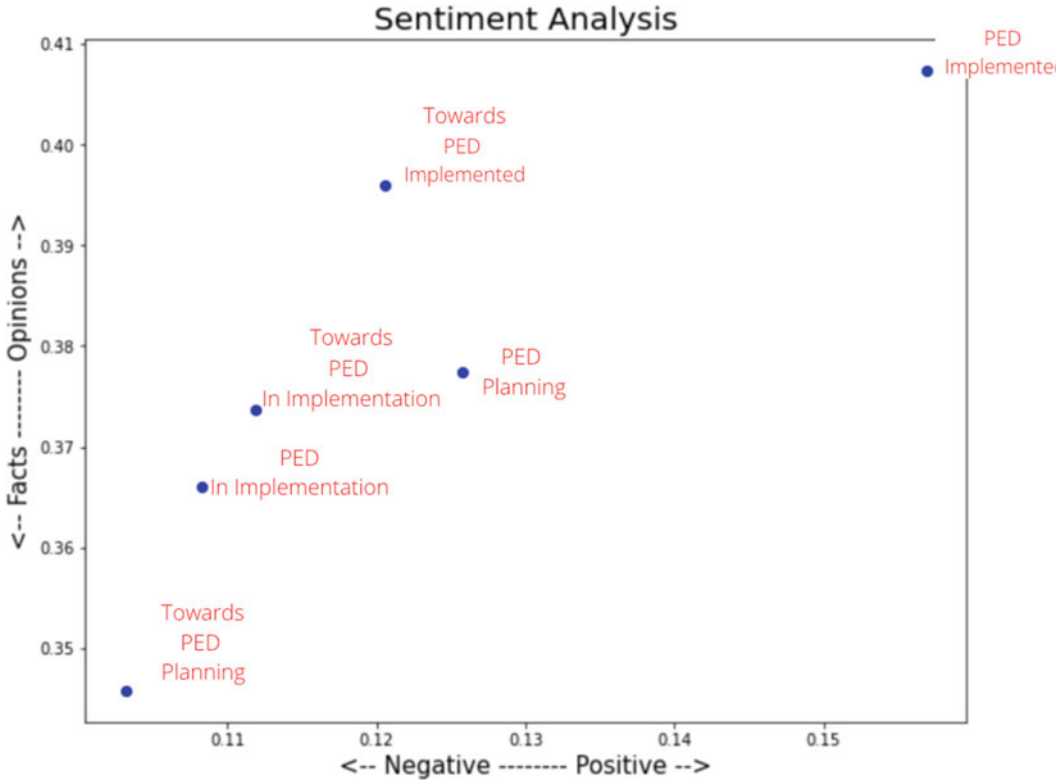

Fig. 5.10 Sentiment analysis

co-relation between the initiation year of the projects and the phase it is in today, and the right chart displays the co-relation between the initiation year of the projects and the financial model it observes. Multiple colors for the data points across the y-axis on these two charts are for ease of visualization for the viewer. Selecting any segment or data point from any of the plots highlights all the characteristics covered by those selected projects in the remaining 4 plots.

Such a dashboard is built upon the database developed in Sect. 3.1 and can be further extended and updated automatically once there is new project information in the database. It is also possible to upload the dashboard online, to increase the ease of sharing the knowledge, data, and experience in PED-related projects, as well as to enable interoperable interaction with different stakeholders when they plan or implement PED projects.

5.4 Discussion

In this study, the projects have been taken from the PED book by JPI Urban Europe, which invited voluntary input data over the project experience and knowledge. It should also be noted that this is not an overview of the PEDs in Europe, as countries have contributed unequally to the development of the book. Since most of the projects are still under planning and

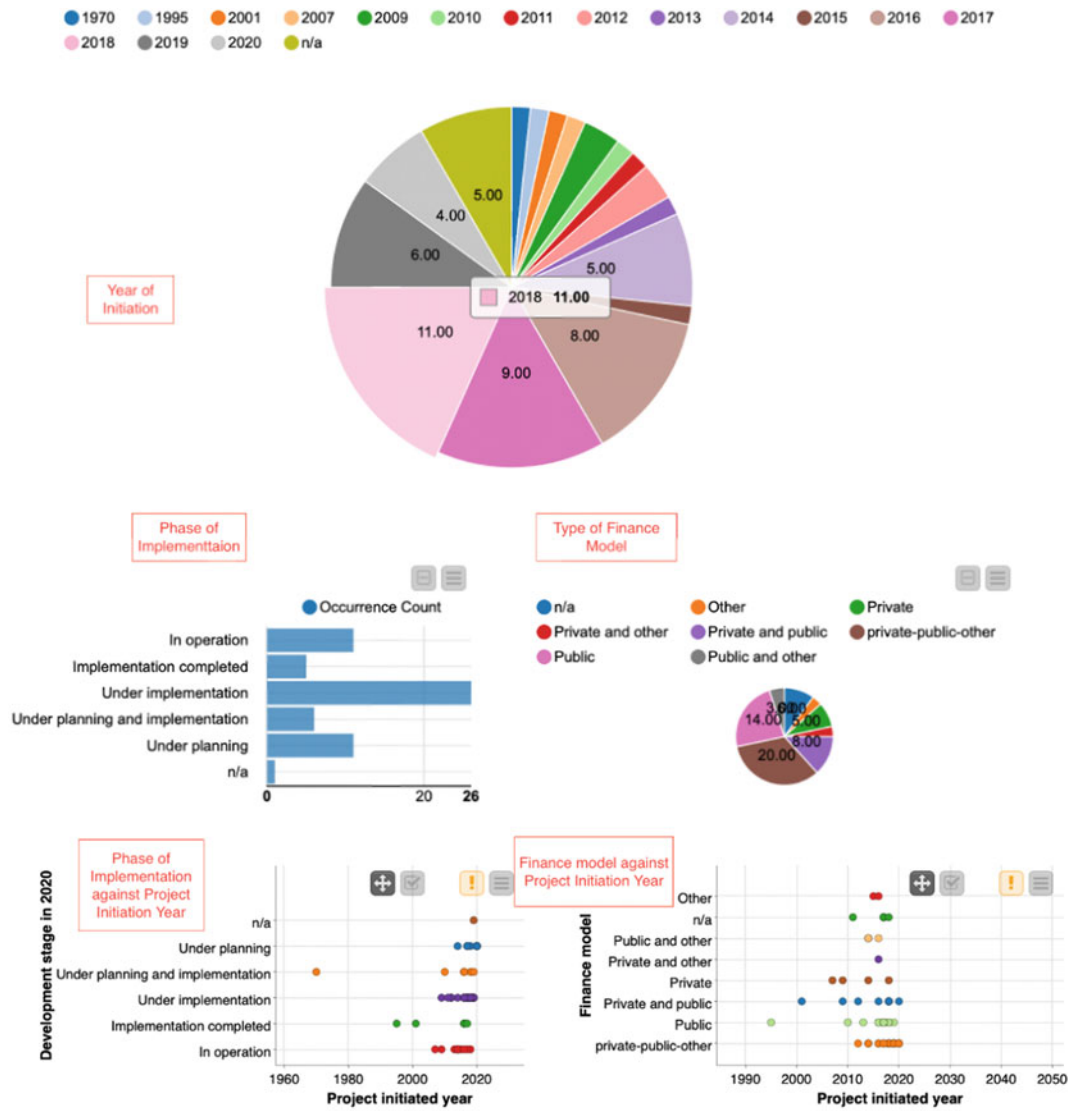

Fig. 5.11 Interactive Knime dashboard

implementation stages, it has been challenging to understand the updated information/data of many projects. In addition, due to the insufficient information, there are little data, such as energy technologies for PED, which is unclear during data collection. These bring certain uncertainty to the analysis result.

However, it is interesting to examine the main characteristics of the collected 60 PED-related projects, and the results shall have certain guidelines for the final PED definition and the proposal of 'reference PED.' The non-existence of a standard and consolidated definition of the PED concept is in fact one of the main limitations to its development and deployment in European cities, so as to boost the energy transition within a common reference framework (Reference Framework for Positive Energy Districts and Neighbourhoods Key Lessons from National Consultations.) for sustainable urban development. So, different approaches and aspects related to the realization of PEDs will be aligned taking into account European cities diversity.

According to results, the identified 60 projects are constituted in Europe with a large number of projects in Norway (9 projects) and Italy (8 projects), respectively. Although the first project took place in 1970, the momentum for such climate-neutral goals has started in 2014.

According to the database, most PED-related projects choose 'yearly' as the time scale. However, it is not possible to identify the temporal scale for many projects since they are still under the planning stage. Regarding the project area (spatial scale), the general trend is to include residential, commercial, and industrial buildings for installation of renewable energy systems in a city or district, which is to avoid the deployment of large energy systems in open fields. This might need supporting policies that support direct consumers to involve in adapting implementation on their premises. However, this strategy would need to consider providing economic feasibility or encouraging policies that attract private investments. The analysis observes that public, private with regional/national grants is a commonly used financial model which reflects active involvement from the private sector. In addition, there are some projects that do not have many local renewable energy sources, but they purchase energy from outside of the district boundary (so-called 'virtual PED').

Based on the results, residential, commercial, and office/social buildings are highly involved in the installation of energy systems, which depends on citizens commitment toward project goals (but the goals might deviate from the designed timeframe of the project). Meanwhile, the stakeholders, such as the municipality, would need to address overcoming the policy restrictions to further ease the process of adapting the energy system, and also need to conduct necessary activities to bring awareness in consumers and motivate for participation.

The energy mix for project goals includes solar energy, district heating/cooling, wind and geothermal energy are primary technologies, where solar technologies show dominance because of its potential. However, due to the unavailability of solar energy during most half of the day and during winter seasons, exploration toward other forms of renewable energy sources, such as geothermal energy and wind, yet may not be totally reliably options during peak demands. In this context, energy storage might be the alternative way. Apparently, energy storage has not been part of the major energy strategies, which might be due to the unavailability of enough planning, economic feasibility, high maintenance, etc. This also might be part of the reason for PED-related projects choosing a yearly temporal scale rather than daily/monthly or seasonally.

In terms of the most used words in the project descriptions, it is observed that projects that are already 'implemented' (both PED and toward PED) tend to concentrate highly on 'consumption,' 'production,' 'heating'—characteristics that are generally repeatedly showed interest in when the project is implemented and running. On the other hand, projects that are yet 'planning' (both PED and toward PED) tend to concentrate on 'solutions,' 'research'—characteristics that are generally repeatedly discussed when a project is being planned. Projects that are in the middle,

i.e., 'In Implementation' (both PED and toward PED) mostly repeat words like 'planning' and 'solution,' like the 'planning' stage projects, but given they are closer to 'implementation' they also display interest in 'heating' and 'supply,' In the sentiment analysis plot, we deduce that while the X-axis does not reflect a particular pattern, it is observed that projects that are still in the planning phase are more akin to depend on established facts for their documentation, whereas the implemented projects lean toward expressing more opinions (that hint their documentation is developed through experience) and do not have to depend solely on facts. The lessons learned from the preliminary analysis of these PED projects provide a starting point for achieving the objective of reducing the existing research gap in the characterization of PEDs. A key aspect is facing the complexity of the urban system and the resulting interrelationships between social inclusion, energy systems, infrastructure, circular economy, and mobility for sustainable urbanization. This calls up building or PED-related simulation tools or platforms to tackle such challenges (Harkouss et al. 2018; Hong et al. 2018).

Moreover, a short summary of a few PED projects with a good level of detailed data has been further analyzed in terms of their energy balance/flows. Table 5.4 provides the main energy concept/flows, and some of them in the implementation/operation stage have clear energy flows, such as Åland Island in Finland, Stor-Elvdal and Drammen in Norway. The annual energy flows in the year 2030 for two scenarios (2030—100% sustainable mobility: (1) 2030 SM Syn scenario—Domestic production of sustainable fuels 2030, (2) 2030 SM EI scenario—High Electrification 2030) at Åland Island are illustrated in Fig. 5.12 (Child et al. 2016, 2017).

It is observed the major energy contributions varies from district to district. For instance, Åland Island replies on biomass and wind power a lot, while Stor-Elvdal municipality prefers CHP plant, and in Drammen municipalities, a heat pump is used mostly. However, these districts are not fully self-sufficient, and they have to import energy to cover peaks. For instance, as shown in Fig. 5.12, Åland Island has to import 4 or 7 GWh of electricity in 2030. It is not easy to judge whether they are PEDs or not at this stage since there is no standard and KPIs available now. According to the mentioned work from EERA JPSC and JPI UE, four categories of PEDs have been established based on two main aspects: the boundaries and limits of the PED in order to reach a net positive yearly energy balance and the energy exchanges (import/export) in order to compensate for energy surpluses and shortages between the buildings or the external grid. All the four described categories of PEDs (PED autonomous, PED dynamic, and PED virtual, Candidate-PED) are based on the accomplishment of a yearly positive energy balance, measured in greenhouse gas emissions, with use of renewables within the defined boundaries, and considering both building energy use and non-building energy use in a neighborhood. Auto- and dynamic-PEDs are the only categories where a net positive energy balance is achieved and candidate-PED should compensate the energy difference with imported certified energy from outside the boundary. According to the boundaries descriptions aligned to the draft definition of PEDs from EERA JPSC working group and JPI Urban Europe, the net positive yearly energy balance is assessed within the functional or virtual boundaries. Thus, PEDs will achieve a net positive energy balance and dynamic exchanges within the functional/virtual boundaries, but in addition, it may provide a connection between buildings within the virtual boundaries of the neighborhood.

It is necessary to pay specific attention to the differences between cities across different regions when promoting the development of PEDs. This is because cities differ from each other at the local, national, and international levels from the perspectives of geography, resources, social, economy, culture, infrastructure, and progress for the carbon–neutral target. This would bring a difference in planning, technology selection/implementation, investment portfolio, stakeholders involvement, regulations, keywords, etc., during the PED

Table 5.4 Summary of major energy concepts and flows of a few PED projects

City/District	Country	Development stage in 2020	Temporal scale	Major energy flows
Åland Island	Finland	Under implementation	Yearly	Target: 100% self-sufficient and 100% fossil-free Solar PV now: 1.7–0.7% of power demand Wind now: about 20% of total power demand Other sources, such as waste heat and CHP, bioenergy, and wave power are still under implementation
Stor-Elvdal Municipality	Norway	In operation	n/a	The demand for heat on the campus is covered by on-site heat production through the CHP plant One-third of the electricity demand is covered The rest is supplied by solar PV with batteries
Drammen	Norway	In operation	Yearly	85% of the heating needs are met by the large-scale fjord source heat pump (13 MW). The rest of the 15% heating needs are met by gas fired boiler The average annual energy supply is 67 GWh The heat pump is significantly cheaper than a gas heating system, saving the city around €2.7 m a year 1.5 million tons of CO_2 have already been saved by switching from gas to the ammonia heat pump
Oulu	Finland	Under implementation	Yearly	District heating system supplemented with solar PV and geothermal energy technologies PV installations on the roof and geothermal heat pump and thermal borehole storage underneath the shopping mall Surplus heat shall be used for refrigeration and seasonal energy storage tanks increasing self-reliance during peak loads
Turku	Finland	Under planning	n/a	Aim to become carbon neutral by 2029 515 solar PV panels installed on new residential buildings will supply energy more than consumption in summer Utilizing the ground source heat with waste heat recovery extracted from 30 other buildings nearby 1 MW solar park is installed in the district by energy company, where the company rents out solar panels and reduces consumer electricity bills Solar thermal collectors are used to produce heat and store underground to use for winter needs Further two-way heat trading facility is provided
Tampere	Finland	Under implementation	n/a	Solar PV farm installed outside the city will be used for energy needs inside the city along with geothermal local district heating and heat pumps
Bodø	Norway	Under planning	Yearly	Although this municipality has excess power production capacity, distribution networks is the main drawback in several places. Therefore, smart city goals are focused on achieving energy efficiency, creation of stable and sustainable energy systems, and reducing of peak demands This energy system uses local renewable energy productions, supply and optimization with

(continued)

Table 5.4 (continued)

City/District	Country	Development stage in 2020	Temporal scale	Major energy flows
				regional, national, Nordic and EU electricity networks
Elverum	Norway	Under planning and implementation	n/a	Firstly, reducing the energy demand in buildings and depending energy production on local renewable energy sources Energy storage in the form of batteries or thermal energy storage
Trondheim	Norway	Under implementation	n/a	Conventional electricity is being provided by largely hydropower with 21 g CO_{2eq}/kWh and district heating through burning local waste Installation of solar PV arrays, heat pumps integration Large 1500 kWh battery storage would attribute to reaching the energy peak demands and surplus energy supply
Bergen	Norway	Under planning	n/a	Primarily improving energy efficiency to reduce energy demand Individual energy systems based on renewable energy sources such as PV and thermal technologies are developed Further surplus power will be supplied to EV mobility solutions
Odense	Denmark	Under implementation	Yearly	To eliminate fossil fuels by 2025 and reach to top 3 cheapest district heating prices in Denmark District heating supply with waste heat, energy power production from renewables such as wind power Further strategically investing in smaller energy units which include 10–20 MW heat pumps, 30–50 MW biomass boilers, and + 50 MW electric boilers
Osterby	Denmark	Under implementation	Yearly	The project aims to reduce the heating costs from district heating with other networks Connecting and sharing energy with the large district heating facilities with neighborhoods reflecting energy flexibility 2.07 MWp PV roof mounted installation that will operate the cooling machines in the mall
Lund	Sweden	Under implementation	n/a	Producing heat through local waste is enough to provide heating for the whole area Large-scale district heating is installed to provide low temperature applications with renewable energy systems integration
Lund (Brunnshög)	Sweden	Under implementation	Yearly	Existing district heating used by biomass will be replaced by large-scale biofuel CHP plant along with geothermal energy unit, waste heat combustion, and district cooling heat pumps
Lund (Medicon Village)	Sweden	Implementation completed	Yearly	Primarily trying to reduce the energy needs yearly by improving energy efficiency Installing solar power on rooftop of buildings for more sustainability

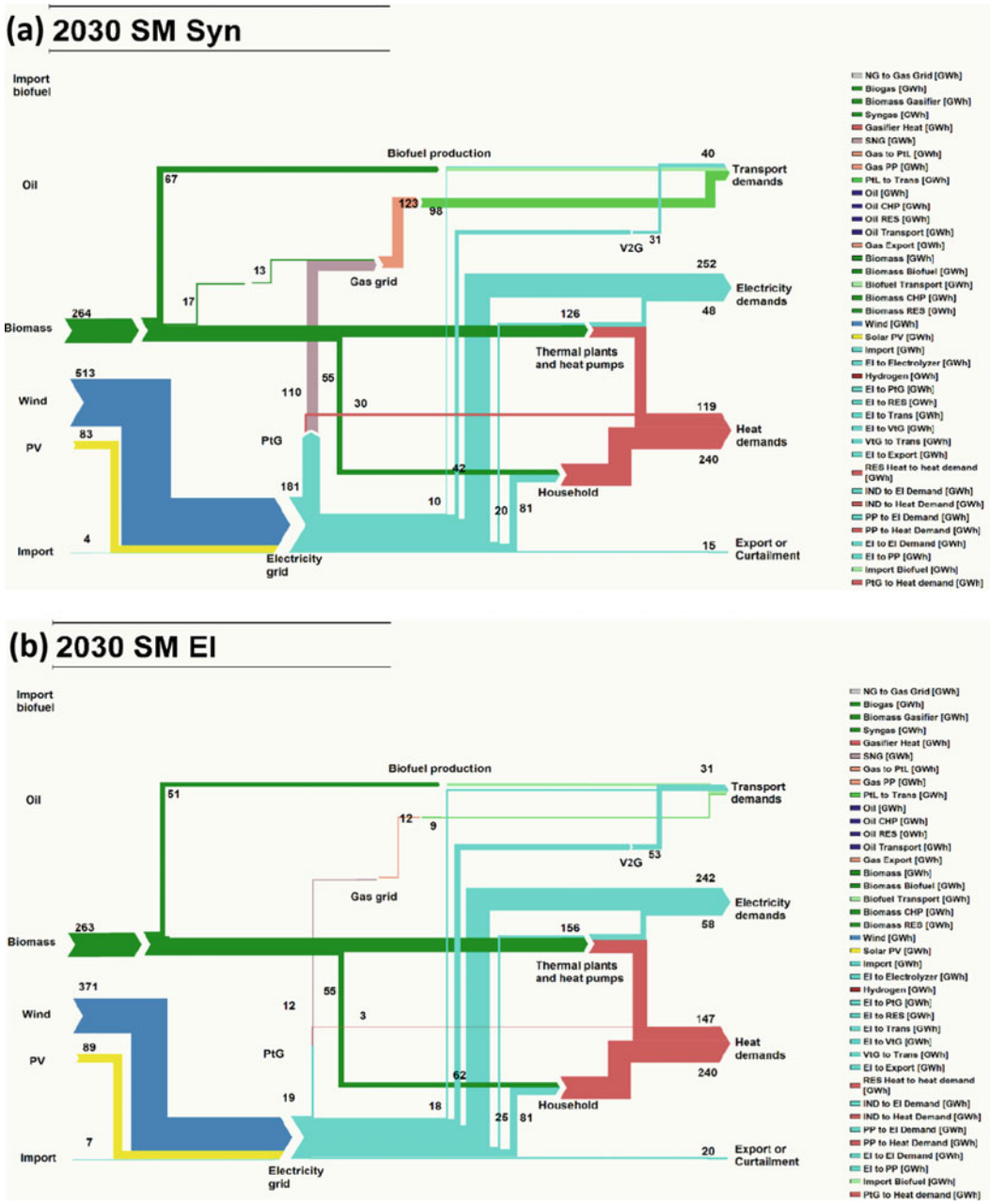

Fig. 5.12 Annual energy flows in the year 2030 at Åland Island toward 2030 target—100% sustainable mobility for two scenarios: **a** SM Syn scenario—domestic production of sustainable fuels and **b** SM EI scenario—high electrification. Reprinted from (Child et al. 2016)

development. However, it is important to have a commonly recognized definition of PED and its related KPI framework for evaluation. By learning the main characteristics from those existing PED projects in the EU, it is helpful to define PED or propose 'reference PED' in other cultural and geographical contexts, which will bring significant common values in terms of replicability and potential generalization of PED across the globe.

5.5 Future Work

This chapter focuses on preliminary analysis of identified PED projects, including projects with insufficient information. In order to understand the detailed analysis, the number of projects might be filtered based on projects with sufficient information to conduct the detailed analysis. Given that only 11 of the evaluated projects are at an advanced (operational) stage, a continuous evaluation of the progress of the PEDs currently in the planning and implementation phase is foreseen in order to update the initial database in subsequent stages. Collecting this additional information will extend and improve the PED characterization, especially in aspects such as energy technologies and boundaries definition. Besides, more PED-related projects have to be identified with sufficient data to support more comprehensive analysis. Such a task is ongoing in both IEA EBC Annex 83 and EU Cost action CA19126. This preliminary study of PED characteristics based on key parameters will be deepening and widening with a particular focus on key energy concepts, EV mobility, driving stakeholders, and temporal scale. Furthermore, it is necessary to identify the potential projects with daily or monthly temporal scales, in order to discover the energy combinations to achieve a net positive energy balance and dynamic exchanges within the functional/virtual boundaries. In addition, a PED may provide a connection between buildings within the virtual boundaries of the neighborhood.

In the context of text mining, the current analysis is developed using the cleaned dataset for the transcripts. However, when it comes to data cleaning, there are several more layers of refining and cleaning that can be carried out on the current transcripts to gain results that are even more accurate and finely assessed. To narrow down the uncertainty of the overall word cloud results, a deeper and multi-layered approach to designing the most used word cloud along with other clouds, such frequency and unique words used, can provide deeper insights. It is also planned to expand the scale of text mining, from the current PED booklet to comprehensive literature, project Web sites/reports, and so on. Furthermore, the Knime dashboard can include multi-variate plots across more than two variables (as is currently), allowing more significant insights on patterns of co-relation between the variables. An online version of such a dashboard will further enhance the interoperable interaction with different stakeholders when they plan or implement PED projects.

Additionally, within the same framework of developing a PED, different areas across the globe must not only take into account specificities at the local level but also have a common definition of PED for standardized assessment. Ongoing works in the EU Cost action CA19126 also consider the integration of PED-Labs characteristics in mapping PEDs projects and initiatives framework. The PED mapping activities are also related to providing a very practical tool needed to guide PEDs implementation as well as to exchange knowledge and information. Potential integration of such a GIS data-driven platform with the Knime dashboard could greatly support the involvement of cities stakeholders and show the feasibility and impact of certain strategies that can pave the way to PED and climate-neutral cities. The alignment of these pilot initiatives could enhance the knowledge not only in the planning and deployment of PEDs in all aspects such as social, technical, financial, and regulatory, but also in the PED characterization/definition/KPIs, as well as showing ground for new methodologies, technical solutions and services to be developed in the future implementation of PEDs. These databases thus constitute an integrated approach to deploy an optimal integration in the technical, evaluation, and management infrastructures of the city in different contexts.

5.6 Summary

This chapter conducts a preliminary analysis of the main characteristics for 60 identified PED projects in Europe. A dedicated database is

developed by considering a series of key parameters. It is found that a large number of PED projects locates in Norway and Italy. Although the first PED project took place in 1970, the momentum for such climate-neutral goals started in 2014. Most PED-related projects choose 'yearly' as the time scale. Nearly, 1/3 of projects have less than 0.2 km^2 area as their spatial scale. In this case, the definition of the project area and the information regarding its boundaries calculation are both very relevant to evaluate the PEDs features of the projects and the business model adopted. Different financing mechanisms and innovative procurement solutions are required to support different large-scale actions. The private investment together with regional/national grants is a commonly used financial model which reflects active involvement from the private sector. Residential, commercial, and office/social buildings are mostly involved in the installation of renewable energy systems, which includes solar energy, district heating/cooling, wind and geothermal energy are primary technologies, where solar technologies show dominance. Substantial challenges and barriers for PED-related projects vary from planning stage to implementation stage.

The non-technological PED solutions (e.g., solution for Governance, Economic, Social, Environmental, Spatial, Legal/Regulatory) are not clearly considered in the Booklet analysis. This is why the next interactive PEDs mapping tools will take into account those aspects that could help to share information and boost the PEDs replication within the main target groups and according to a local broader perspective.

In addition to the development of the database, the text mining approach is applied to further examine the keywords of PED-related projects. It is observed that projects that are already 'implemented' (both PED and toward PED) concentrate highly on 'consumption,' 'production,' 'heating.' While the projects that are yet 'planning' (both PED and toward PED) focus on 'solutions,' 'research.' Projects that are 'In Implementation' (both PED and toward PED), mostly repeat words of 'planning' and 'solution,' but given they are closer to 'implementation' they also display interest in 'heating' and 'supply.' We also deduce that the projects that are still in the planning phase are more akin to depend on established facts for their documentation, whereas the implemented projects lean toward expressing more opinions by high involvement of stakeholders.

Although there is uncertainty due to limited data at the initial stage, the results are expected to give useful guidance for the final PED definition and proposal of 'reference PED.' It is confident that the alignment among ongoing initiatives will represent the best way and very practical solution to step forward and facilitate the PEDs implementation in the next years, with more useful guidance and tools.

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6 Economic Interactions Between Autonomous Photovoltaic Owners in a Local Energy Market

Xingxing Zhang, Pei Huang, and Marco Lovati

Abstract

Solar photovoltaic (PV) is becoming one of the most significant renewable sources for positive energy district (PED) in Sweden. The lacks of innovative business models and financing mechanisms are the main constraints for PV's deployment installed in local community. This chapter therefore proposes a peer-to-peer (P2P) business model for 48 individual building prosumers with PV installed in a Swedish community. It considers energy use behaviour, electricity/financial flows, ownerships and trading rules in a local electricity market. Different local electricity markets are designed and studied using agent-based modelling technique, with different energy demands, cost–benefit schemes, and financial hypotheses for an optimal evaluation. This chapter provides an early insight into a vast research space, i.e. the operation of an energy system through the constrained interaction of its constituting agents. The agents (48 households) show varying abilities in exploiting the common PV resource, as they achieve very heterogeneous self-sufficiency levels (from ca. 15% to 30%). The lack of demand side management suggests that social and lifestyle differences generate huge impacts on the ability to be self-sufficient with a shared, limited PV resource. Despite the differences in self-sufficiency, the sheer energy amount obtained from the shared PV correlates mainly with annual cumulative demand.

Keywords

Microgrid · PV · Peer to peer · Self-consumption · Energy community · Local market

6.1 Introduction

6.1.1 Background and Literature Review

Positive energy districts (PEDs) are defined as energy-efficient and energy-flexible building areas with surplus renewable energy production and net zero greenhouse gas emissions (IEA n. d.). Solar photovoltaic (PV) is ideally a leading renewable source in PEDs due to its easy scalability, simple installation and relatively low maintenance. Distributed PV systems are the main driver in the Swedish PV markets, due to

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smaller size and distributed ownership, which are better adapted to permeate the urban environment. The installed capacity of PV systems in Sweden is expected to continuously soar in the future, mainly driven by homeowners and private or public companies at relatively small or medium scales, according to its particular market setup and subsidy (e.g. SOLROT deduction, tax reduction, etc.) (Energimyndigheten 2016). However, relying on the subsidy is not sustainable for PV deployment in the long term. At the moment, there is still limited access to capital and appropriate financing mechanisms, resulting in a slow uptake of PV under traditional business models (i.e. power purchase agreements and the net-metering mechanism), which are no longer applicable for small PV systems (Huijben and Verbong 2013). The existing business models may need to be further developed to exploit the full potentials generated by distributed energy supply, demand and energy sharing. Thus, in a future without subsidies, prosumers (i.e. small PV owners) will have to sell their excess production at market price back to the grid. This scenario would be unprofitable for PV owners and also strain grid stability and reduce its reliability.

Fortunately, the possibility to form energy communities, where energy can be locally shared, has been regulated at European level in the Clean Energy package presented by the European Commission (Commission n.d.) and at Swedish level under § 22 (a) of the IKN Regulation 2007: 215 (Riksdag n.d.). This can be an opportunity for a new business model development within the energy sector, e.g. peer-to-peer (P2P) trading. In such business model, consumers and prosumers organise in energy communities, in which the excess production could be sold to other members (Parag and Sovacool 2016). The benefits are threefold as the prosumers could make an additional margin on their sale; consumers could buy electricity at a more advantageous price, and the grid could be more stable and resilient. This can be a potential solution to promoting PV installation in a sustainable way, while reducing the reliance on subsidies.

To support new regulations, careful design and optimal modelling of P2P business models for PV penetration is necessary by analysing current scenarios and proposing future ways of exchanging energy. Huijben and Verbong (2013) summarised three possible ownerships of PV systems: customer-owned (single ownership), community shares (multiple ownership) and third-party ownership. Based on these possibilities, Lettner et al. (2008) further described three different system boundaries of a PV prosumer business concept (as illustrated in Fig. 6.1): Group (1) single direct use (one consumer directly uses the generated PV electricity on site); Group (2) local collective use of PV in one building (several consumers share the generated PV electricity with or without the public grid); and Group (3) district power model (PVs are installed in several buildings, where those prosumers directly consume locally generated PV power, and the PV electricity is further shared using public or private microgrid). It is possible to have different ownerships in each category of these boundary conditions, resulting in many possibilities and uncertainties in the practical business operation. Learning and mapping (i.e. testing) a wide array of these possible designs and combinations are necessary. There are a few existing regulatory and modelling studies about the P2P PV electricity trading. Community-owned PV system was surveyed as an innovative business model in Switzerland, where it can seemingly be a successful distribution channel for the further adoption of PV (Stauch and Vuichard 2019). Roberts et al. tested a range of financial scenarios in Australia, based on the P2P concept, to increase PV self-consumption and electricity self-efficiency by applying PVs to aggregated building loads (Roberts et al. 2019). Zhang et al. (2018) established a four-layer system architecture of P2P energy trading (as shown in Fig. 6.2, i.e. power grid layer, ICT layer, control layer and business layer), during which they focused on the bidding process on business layer using non-cooperative game theory in a microgrid with 10 peers. A price mechanism for the aggregated PV electricity exchange amongst peer buildings was also developed using either

Lagrangian relaxation-based decentralised algorithm (Xu et al. 2017) or mixed integer linear programming (Nguyen et al. 2018). Jing et al. (2020) then applied the non-cooperative game theory to modelling the aggregated energy trading between residential and commercial buildings by considering fair energy pricing mechanism for both PV electricity and thermal energy simultaneously. Lüth et al. (2018) designed two local markets for decentralised storage (flexi user market-individually owned batteries) and centralised storage (pool hub market-commonly owned battery), based on a multi-period linear programming. It focused on the evaluation of two different ownerships of batteries and optimised P2P energy trading local markets. They indicated that the end users can save up to 31% electricity bills in the Flexi User Market and 24% in Pool Hub Market. Furthermore, two different ownership structures, namely the third-party owned structure and the user owned structure, were investigated in a P2P energy sharing network with PV and battery storage (Rodrigues et al. 2020). These existing studies almost cover all four layers of a P2P network. The impact of other system and market components on the economic performance of PV P2P business models has been investigated, such as electric vehicle (EV) batteries (Tang et al. 2018), gas storage (Basnet and Zhong 2020), heat pump/hot water storage (Huang et al. 2019), advanced control (Thomas et al. 2019), energy cost optimisation (Alam et al. 2019), bidding strategies for local free market (El-Baz et al. 2019), double auction market (Chen et al. 2019), local market designs (Sousa et al. 2019), integration of local electricity market into wholesale multi-market (Zepter et al. 2019), microgrid ICT architecture (Cornélusse et al. 2019) and grid operation (Almasalma et al. 2019).

According to the above studies, a research gap is found in the lack of examination on full P2P energy trading process at the business layer in a local market for individual participant, which, in time sequence, consists of bidding, exchanging and settlement, under different local market conditions with various ownerships of PV systems and market rules. Bidding is often the first process when energy players (generators, consumers and prosumers) agree to trade energy with each other at a certain price for a specific amount of energy. Energy exchanging is the second process, during which energy is generated, transmitted and consumed. Settlement is the last process when bills and transactions are finally settled via settlement arrangements and payment (Zhang et al. 2018), which results in the final economic benefits. In cases of the physical network constraints, due to the varying energy demand and the intermittent generation of PVs, there are always mismatches between sellers and buyers. Such difference between electricity generation and demand are to be evaluated and charged/discharged during settlement stage.

6.1.2 Novelty and Contribution

Several studies have focused on the technical or economic aspects of the microgrids and shared RES, but the endeavour has been tackled in a segmented way analysing a narrow sample of possibilities amongst the vast search space of the business models. The existing studies have not yet fully tested the effectiveness and compared the characteristics of various P2P business models, in the case of heterogeneous peer (individual) energy supply/demand and dynamic market rules for the full trading process on the business layer. There is a lack of a concise and efficient method yet to model.

Although this chapter only analyses three different setups, it attempts to lay the groundwork for a systematic study of the subject. In other words, the results and the discussion presented in this chapter, although not conclusive by themselves, are part of a well-defined search space. This allows the outcomes to be interpreted from the perspective a larger systematic endeavour.

In summary, the elements of novelty of this chapter are described as the following:

1. The particular result of the study: To the knowledge of the authors, no study has linked the price of the electricity offered within a

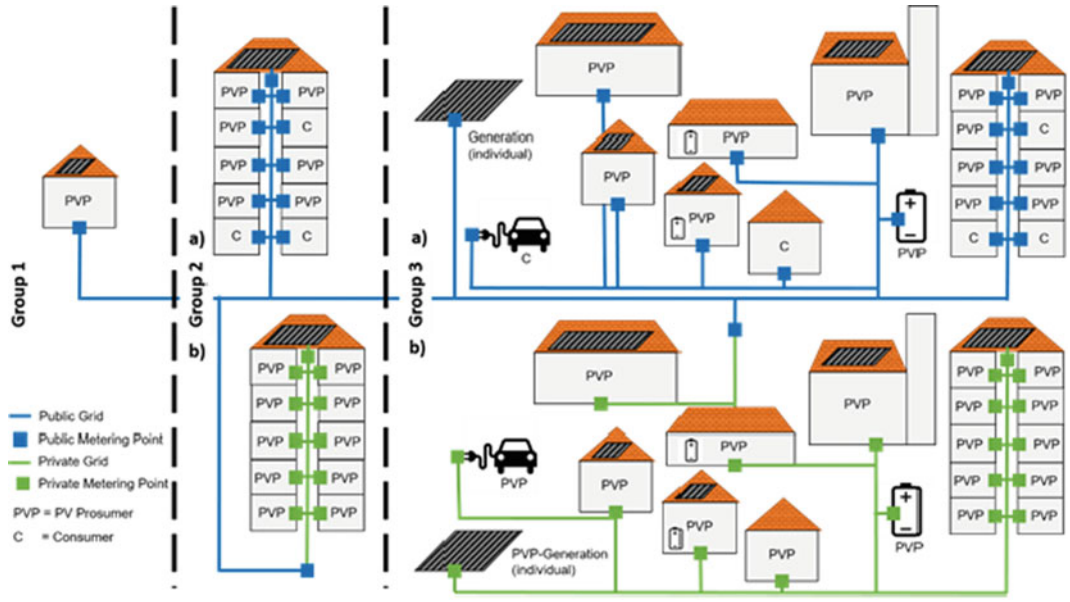


Fig. 6.1 Classification of integration concepts (Lettner et al. 2008)

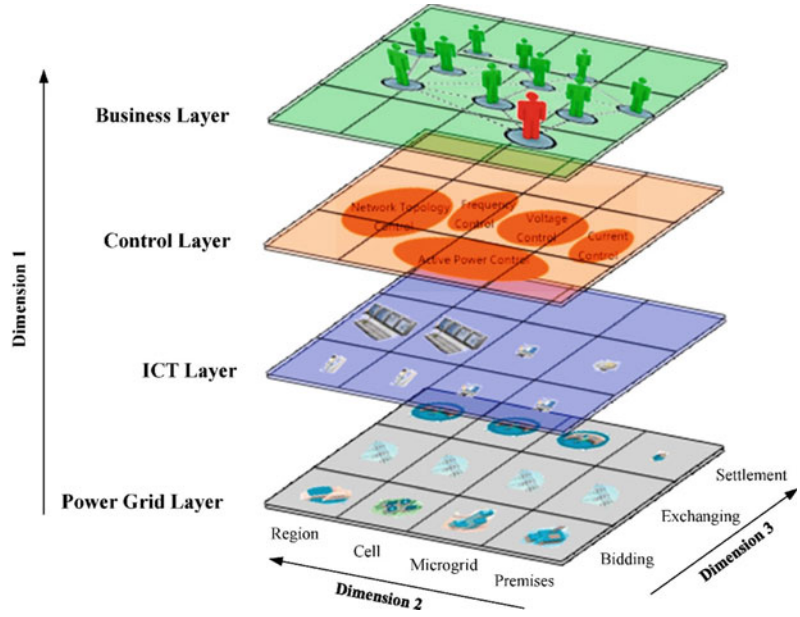


Fig. 6.2 Four-layered system architecture of P2P energy trading from (Zhang et al. 2018)

shared RES to both the risk of economic loss and the potentials for earning amongst the individual households within the shared microgrid. Furthermore, the dominance of shear annual cumulative consumption over self-sufficiency in determining the earning potential in a shared RES is an unknown phenomenon. It deserves to be further analysed (i.e. tested under different datasets) to be proven.

2. The examples of business models presented in the study are included in a well-defined

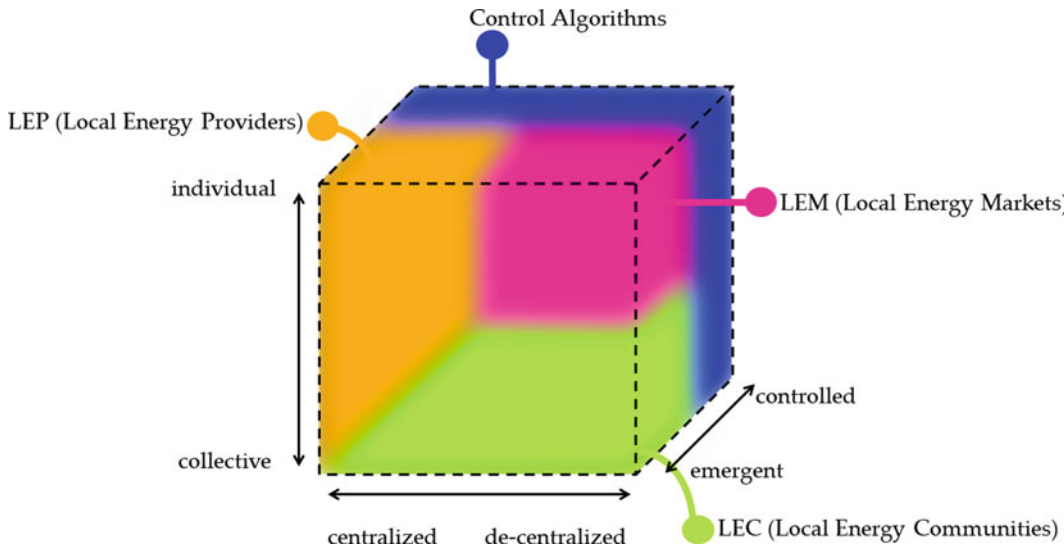


Fig. 6.3 District scale renewable energy systems behaviour map

search space map (see Fig. 6.3). This facilitates a systematic inquiry and offers a way to organise the results presented in this chapter and in the follow-ups.

This chapter studies the P2P business model for 48 individual building prosumers with PV installed in a Swedish community. This chapter discovers "latent opportunities" that were previously unknown and optimises the market design and its variables for the best benefit. It has significant influence that integrates energy needs, supply and market rules. This chapter is expected to provide knowledge for policymakers to design a fair, effective and economical P2P energy framework. The research results will useful be to optimise PED's three functions (energy efficiency, energy production and flexibility) towards energy surplus and climate neutrality.

6.2 Materials and Methods

The definition of ownership structures from (Huijben and Verbong 2013) distinguishes amongst customers, communities and third parties. In general, a similar distinction could be applied to the behaviour of the local grid instead to the ownership. In this way, the concept of ownership is not associated with the functioning of the grid, and it is easier to describe hybrid forms (e.g. some shareholder of an energy provider, or more providers, which form a market although not prosumers). Thinking about the behaviour of the shared system, a space can be defined according to three dimensions (see Fig. 6.3):

1. The **controlled versus emergent** dimension describes how much there are rules or a controller that directs the exchanges, versus an emergent behaviour from the interactions between agents.
2. The **centralised versus decentralised** dimension describes how much the agents are equivalent amongst each other, versus the presence of few (potentially one) agents that concentrate some functions for a larger number of others.
3. The **individual versus collective** dimension describes how much each agent controls and directs its own resources (i.e. PV, storage, demand-response resources, etc.), versus having larger pools of agents who share some common resources.

The behaviour map does not refer to any specific levels (Zhang et al. 2018), although the last two (i.e. controls and business) are particularly affected from the volume of the map, in which they are located. In fact, the control of the energy and monetary flows between generation and demand points can be decided by a controller, which can be assigned by the internal rules of a community or emerged as the result of an auction.

6.2.1 Agent-Based Model

Given the number and nature of the emergent behaviours in the behaviour map (i.e. Fig. 6.3), an agent-based model (ABM) simulation was developed to get insight into the energy and economic fluxes exchanged between the different actors in the local grid. Usually, every agent of the simulation represents one household in the local grid (i.e. a consumer or a prosumer), but producers are not excluded. An example of a producer is an energy provider. For instance, companies or investor interacts with the local grid without necessarily being served by it or the parent grid, i.e. the larger grid in which the local grid is embedded. The local grid could be a microgrid but also a secondary network, where the prosumers are allowed to have a certain level of control of the network.

In an ABM, each agent can interact with all the other agents by trading energy. Thus, it can send energy in exchange for money or vice versa. The movement of energy in the microgrid is an emergent behaviour, which results from the interaction of a number of independent actors. This is opposed to a control algorithm, where the behaviour is set by a series of rules or conditions. Naturally, the freedom of the agents can be limited by the introduction of rules. For instance, a producer could be forced to prioritise the sale of renewable electricity to those consumers that have used the least of it in a given period. If the rules become tighter, the freedom of each individual agent is reduced. While if the rules are as tight as to completely limit any possibility of choice for the agents, the ABM degenerates into a control algorithm.

In the present study, the behaviour of the agents is extremely simplified: the consumers prioritise the purchase of electricity from the cheapest source available at any given time; on the other end, the producers have the ability to set the price, and they do so according to the case as explained in the following section (i.e. ownership structures and business models). Figure 6.4 presents the possible ownership structures arranged into three main families; these are slightly different from those in (Huijben and Verbong 2013) for the purpose of this chapter.

1. Local Energy Provider (LEP) (Fig. 6.4a): It occurs when a single agent owns the totality of the production or storage capacity of the entire local network, and the other agents are strictly consumers. The owner of the plant can be either a producer or a prosumer.
2. Local Energy Community (LEC) (Fig. 6.4b): It is the case in which a communal plant is shared amongst all or a group of agents; the shares could be equally distributed or according to other principles such as energy

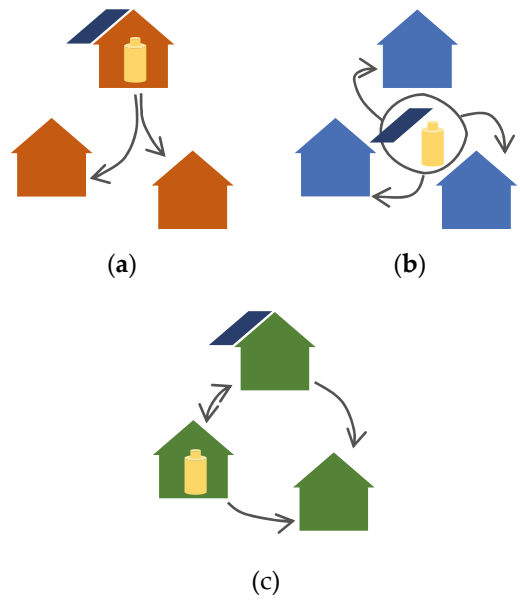


Fig. 6.4 Possible ownership structures organised in three main families: local energy provider (LEP) (**a**), local energy community (LEC) (**b**), and local energy market (**c**)

used from the plant or the share of the initial investment.

3. Local Energy Market (LEM) (Fig. 6.4c): It is the most complex and free-form of all the structures; it is characterised by the presence of multiple producers, consumers and prosumers. In this arrangement, the interaction between agents can reach significant complexity, and the agents could achieve higher earnings by engaging in intelligent behaviours.

6.2.2 Ownership Structures and Business Models

In the case study examined (see Sect. 6.2.3), a communal PV plant is shared amongst the different households in the building. This allows for two of the three basic ownership structures in Fig. 6.4 (i.e. LEP and LEC) to be applied. The ownership structure is intertwined with the business model and the rules of the market. In the following studies, the same communal PV plant is shared between the households in the local grid in three different scenarios:

1. **LEC gratis**: In this arrangement, the electricity from the communal PV plant is given for free when available. All the households participate in the initial investment and in the operation and maintenance (O&M) costs of the plant according to equal shares.
2. **LEC LCOE**: In this arrangement, the electricity from the communal PV is given at production cost (i.e. without profit) and the revenues are divided amongst the shareholders. Although variable shares are possible, in this chapter, all the households are equal sharers in the LEC (i.e. initial investment and O&M costs, and the revenues are shared equally).
3. **LEP n%**: This arrangement is a pure form of LEP. Thus, the production plant is owned by a single provider who can set the price at its own will. Obviously, the provider cannot set the price higher than that of the parent grid (i.e. the average price for Swedish household consumer as assumed in Sect. 6.2.3) as the consumers retain the right to purchase electricity from the cheapest source. In this chapter, the provider sets the price as half-way between the minimum of the local LCOE and the maximum of the consumer price from the parent grid. More precisely, the provider sets a price at a percentage n so that $n = 0$ is the LCOE, $n = 100$ is the price offered by the parent grid and $n = 50$ is half-way. This setup is valid under the assumption that the LCOE of the system is lower than the price of the electricity for the consumer. Of course, if this assumption does not hold true, the provider will not be able to charge above market price and will thus operate at the minimum loss.

In all arrangements, the consumer is programmed to buy electricity from the cheapest source. However, by having a single source in the local grid, the choice is only between the local source and the parent grid. This implies that the price of electricity in the local grid must be at any time below the Swedish consumer price. If the local production is absent or insufficient (i.e. local consumption > local production), the demand shall be covered partially or totally by the parent grid. If the local production is not sufficient, at a given point in time, to cover entirely the demand, all the households will be served equally in terms of percentage of their demands as shown in the system of relations in Eq. (6.1).

$$\begin{cases} E_{\text{local}} = \eta \times D_{\text{local}} \\ E_{\text{house}} = \eta \times D_{\text{house}} \times \forall_{\text{house}} \\ D_{\text{local}} = \sum D_{\text{house}} \end{cases} \tag{6.1}$$

where E_{local} and E_{house} are the amount of electricity available in a given time for the aggregated local grid and for a specific household, respectively. η is the self-sufficiency: a number between 0 and 1 that represents the share of the demand covered by locally produced electricity; note that it is the same globally and for each household. D_{local} and D_{house} represent the aggregated demand and the demand of each single household, respectively.

Equation (6.1) implies that having a larger consumption when the local electricity production is scarce guarantees access to a larger amount of local energy, although equal in percentage. Another consequence of the relation in (6.1) involves the price of the electricity for each household: the price results from the weighted average (weighted on energy) of the prices from the different sources of electricity purchased. In the specific case of this chapter, the price can be calculated with the relation Eq. (6.2):

$$P_{\text{house}} = P_{\text{local}} \cdot \eta + P_{\text{parent}} \cdot (1 - \eta) \qquad (6.2)$$

where P_{house}, P_{local} and P_{parent} represent the electricity price for the individual household, the price for the energy produced locally and the price for the energy bought from the parent grid, respectively. η is the self-sufficiency as defined for (6.1).

Considering that η is the same for every household in the local grid as shown in (6.1), Eq. (6.2) implies that at any given time there is a unique price of the electricity within the local grid, which depends on the relation between the aggregated energy demand (D_{local}) and the aggregate energy production (E_{local}). Thus, the price for the electricity is solely function of the Hour of the Year (HOY) and is not a function of any given household.

6.2.3 Case Study

The agent-based model is tested on a digital representation of a moderate size residential district (see Fig. 6.5) equipped with a shared PV system + DC microgrid as described in (Huang et al. 2019). The group of three buildings with three stories is located in Sunnansjö, Ludvika, Dalarna region, Sweden. The common PV system is formed by the arrays shown in Table 6.1. In total, there are three arrays on the roof and one on the southern façade (total 65.5 kWp).

The system capacity and the position of the arrays over the building resulted from an optimisation process, presented in (Huang et al. 2019), in order to maximise the self-sufficiency while maintaining a positive NPV over the lifetime. In this system, no electric storage was installed. The levelised cost of electricity (LCOE) of the system was calculated to be about 0.83 SEK/kWh (0.077 €/kWh) under the following assumptions:

- Local initial price of the turn-key system without taxation: 10,000 SEK/kWp (935 €/kWp).
- Price of the inverter: 2500 SEK/kWp (234 €/kWp) (changed two times over the lifetime). The number of changes was retrieved as the expected value assuming a lifetime of the inverter between 12 and 15 years.
- Planned lifetime of the system: 30 years.
- Maintenance costs for the system (substitutions, cleaning and inspection): 5109 SEK/year (477 €/year). This value was calculated as the expected value out of 100 stochastic simulations.
- Degradation of the performance of the system: ca. − 1.15%/year.

The weather file and the production of the diverse arrays of PV were calculated from PVGIS (Šúri et al. 2005). The load profile of the 48 households could not be published for privacy concerns. Thus, the study is presented using data generated by load profile generator (LPG) software (Pflugradt and Muntwyler 2017). Load profile generator is a tool that simulates the electric demand for residential light and appliances. The variability of the aggregated curve according to the number of households has been validated against a real low voltage grid consumption (Pflugradt et al. 2013). The electric demand is generated by simulating every household component as an agent. Its demand is determined by the power absorption and duration of use of devices amongst an available selection (see Fig. 6.6). These are chosen by the household components according to a set of activities and needs. The needs are modelled as counters that grow at each time step: a high counter represents something that is in urgent need of satisfaction. Different needs have different growth rates for each time step, which means that some needs are to be satisfied more often than others.

Fig. 6.5 Bird view of the small district in the case study (Huang et al. 2019)

Table 6.1 Characteristics of the shared PV system

Block	Facing	Tilt (degree)	Capacity (kW_p)	Production (MWh)
B	South	18	28.4	22
C	East	18	15.9	10.4
A	West	18	15.9	10.3
A	South	90	5.3	3.4

The parent grid (i.e. the Swedish national grid) was assumed to offer electricity for 1.8 SEK/kWh (0.17 €/kWh) from October to March and 1.2 SEK/kWh (0.11 €/kWh) from March to October. These prices were assumed as a reasonable price for each single household at the annual cumulative level of consumption observed. According to Eurostat (2007–2019), the average price for household electricity in 2019 was 1.39 SEK/kWh (0.1297 €/kWh) for electricity transmission, system services, distribution and other necessary services. If VAT and levies are added, the average price would reach 2.2 SEK/kWh (0.2058 €/kWh) (Eurostat, 2007–2019). It is not clear what taxes can be avoided consuming locally produced electricity, but it is reasonable to believe that VAT can be avoided in both the LEC cases explored as the electricity is offered for free or at a price equal to production cost. Conversely, it is not possible to estimate how much of the base 1.39 SEK can be reduced thanks to the aggregation of the loads. The price of the electricity is not static but is projected to grow linearly over the next 30 years at a rate of + 1%/year. This is under the assumption that the national grid will need liquidity to invest in the energy transition. Conversely, the revenues for the energy sold to the grid are set to be worth 0.3 SEK/kWh (0.028 €/kWh), but are assumed to shrink by 1.67%/year under the assumption that the increase in installation of PV will gradually discount the energy during sunny hours.

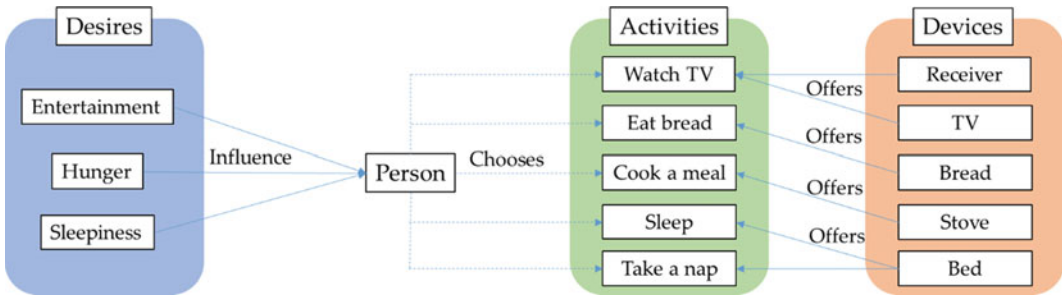


Fig. 6.6 Workflow diagram of the load electricity generation (Pflugradt and Muntwyler 2017)

6.3 Results

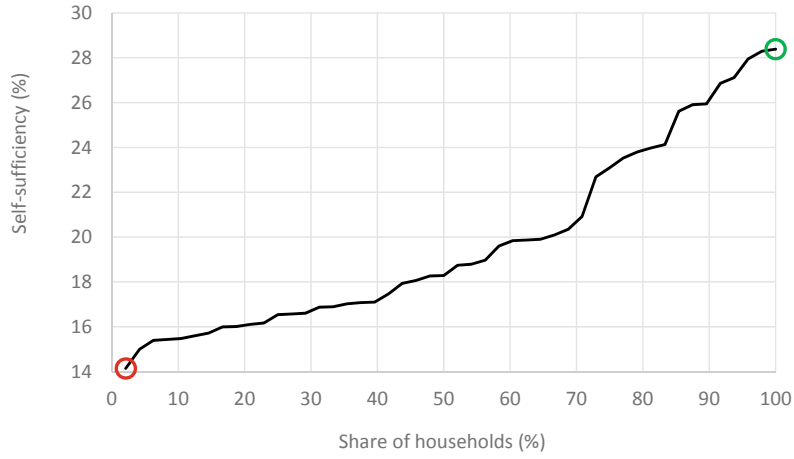
This section begins with a discussion about the self-sufficiency of the different households in the local network. It then proceeds with a techno-economic analysis of each arrangement to establish its features and behaviour (i.e. distribution of risk and profit amongst stakeholders). Given that the local PV plant is unique, the movement of energy in the network is the same in all the arrangements, thus the self-sufficiency is a static figure throughout the arrangements.

6.3.1 Self-Sufficiency of the Households

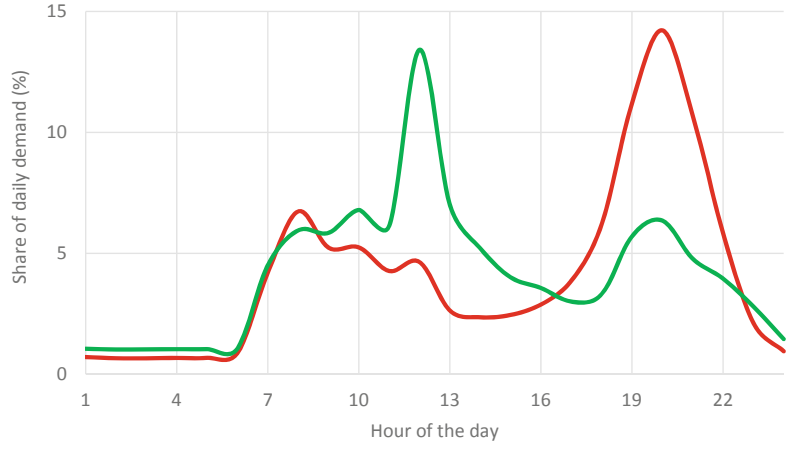
PV self-sufficiency is defined as the share of total demand in a household that is being supplied by locally generated electricity from PV system (Luthander et al. 2015). In this chapter, the system, as it is designed, allows to cover an estimated 20.2% of the annual cumulative demand of the district. This result is satisfactory for a system without any electric storage (see (IEA 2020)). The country, with the most electricity production from PV (i.e. Honduras), has an estimate PV self-sufficiency of 14.8% with the EU on average having 4.9%. It has been calculated in references Lovati et al. (2019) and Huang et al. (2019) that the economically optimal self-sufficiency of a conveniently aggregated system, even in absence of electric storage, is comfortably above any penetration level we see today (i.e. often above 20%). The economically optimal self-sufficiency sets a conservative limit of hosting capacity in an electrical system in a regime of self-sufficiency. The P50 (i.e. 50th percentile or median) household has a self-sufficiency of 18.5% as shown in Fig. 6.7a: this value is below the average value of the aggregated district because the slope of the increase is higher to the right of P50 (see Fig. 6.7a). The P50 household has a relatively low self-sufficiency because there is a positive correlation between annual cumulative demand and self-sufficiency (see discussion about Fig. 6.8). In general, the variability in self-sufficiency between the households in the microgrid is high. The most self-sufficient household possesses in fact a value double of the lesser one (14.1–28.4%). This strong variability suggests that, even without any deliberate attempt for demand control, some households show habits, or a way of life, that can take the most from the available PV energy.

Figure 6.7b, c shows the share of the annual demand in different hours of the day or month of the year, respectively, i.e. how much of the total annual demand is concentrated during a specific hour of every day or month along the year. In the household with the highest self-sufficiency, the electricity demand around 12:00 is particularly prevalent (see Fig. 6.7b). It indicates that its inhabitants cook at home for lunch. On the other end, the evening peak of the most self-sufficient household is way less prominent than in the lowest one. Looking at the prevalence throughout the months of the year (Fig. 6.7c), the difference is less marked compared to the daily average: both households present a steep drop in sunny months, which seems to indicate an absence due to summer holidays. The most self-sufficient household appears to have had an absence for holidays during May instead of June, as shown in Fig. 6.7c. This might be advantageous as it allows to use more PV electricity when the overall electricity demand of the district is lower and the radiation from the sun is higher. It should be noted that, in general, the best performing household presents a smaller dip in demand for the summer holidays; it is unknown whether it is due to a shorter holiday or the presence of some household components at home.

The examples shown in Fig. 6.8 highlight the two apartments that are extreme in terms of self-sufficiency. To infer more generalised information on the time of high consumption that favours high self-sufficiency (see Fig. 6.3), the following formula was used:



(a)



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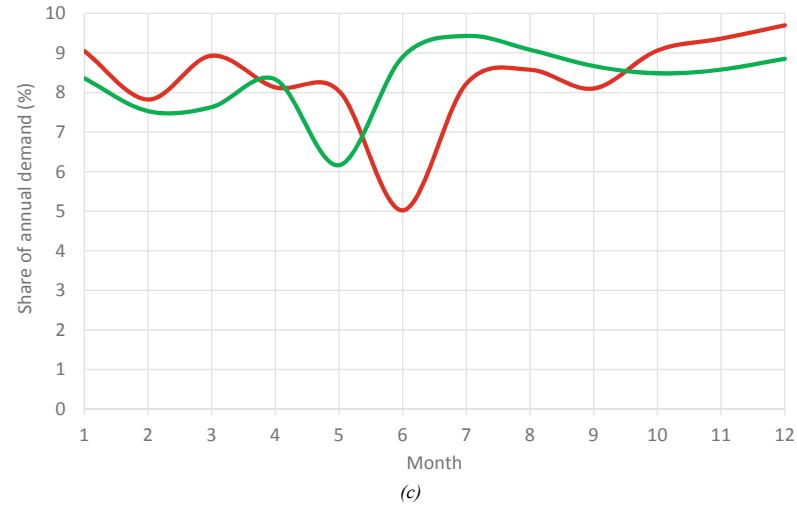
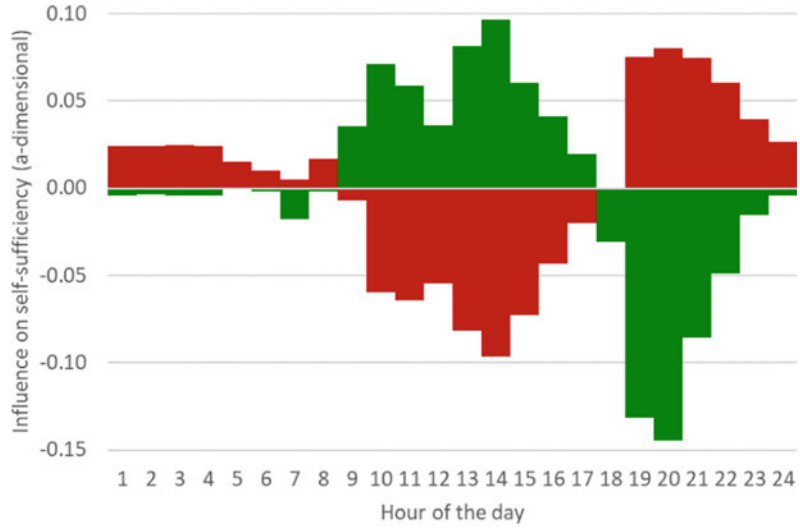
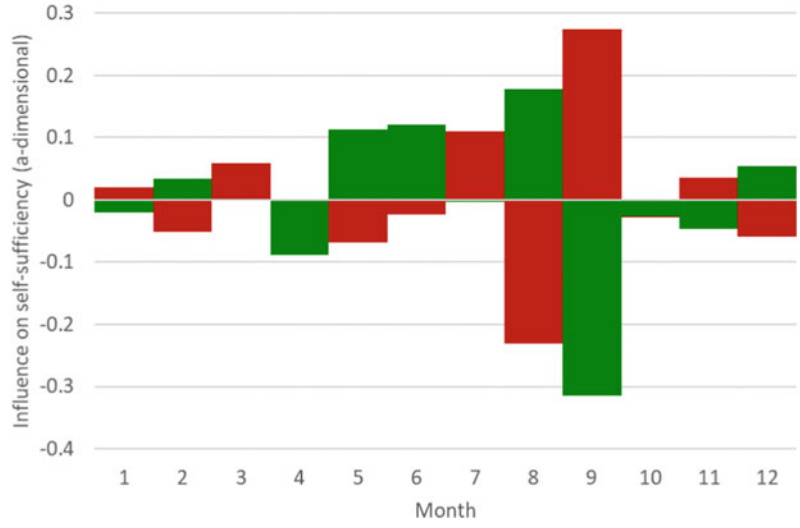


(c)

Fig. 6.7 Self-sufficiency of the apartments in the local grid: **a** the distribution of self-sufficiencies across the 48 households, **b** the hourly average of the extreme households, and **c** the monthly average consumption of the extreme households



(a)



(b)

Fig. 6.8 Influence on self-sufficiency of high demand in: **a** each hour of an average day, and **b** each month of the year. The value is a-dimensional, but it expresses the positive (or negative) influence of a high electric demand at a given time step compared to all the others (see Eq. (6.3))

$$\mathrm{ISelfS}_{\text{time step}} = \sum_{\mathrm{HH}=1}^{48} \begin{cases} \mathrm{TP}_{\text{time step, HH}} - \mathrm{TP}_{\text{time step, tot}} & \forall \mathrm{SelfS}_{\mathrm{HH}} > \mathrm{SelfS}_{\mathrm{tot}} \\ 0 & \forall \mathrm{SelfS}_{\mathrm{HH}} \leq \mathrm{SelfS}_{\mathrm{tot}} \end{cases} \tag{6.3}$$

where $\mathrm{ISelfS}_{\text{time step}}$ is the influence of high energy demand in a given time step (which could be an hour of the day or a month of the year). *HH* stands for household as the curve results from the sum of all the individual households. $\mathrm{TP}_{\text{time step, HH}}$ and $\mathrm{TP}_{\text{time step, tot}}$ are the typical power demand (W) of said time step for the nth household (HH) and the whole district (tot), respectively. The sum of all time steps is then rescaled so that it is equal to 1.

In practice, the curve is influenced only by the households that have a self-sufficiency above average. It represents the influence (positive or negative) that the demand in each time step has on the overall self-sufficiency. Unsurprisingly, Fig. 6.3 a shows that a lower average demand in the evening and early morning hours is

associated with high self-sufficiency. On the contrary, the central hours of the day are generally above average in highly self-sufficient households. It is interesting to notice how the electric demand at 12:00 is in general less beneficial for self-sufficiency than the hours around it: this is somewhat counterintuitive, but it makes sense since at 12:00 the high general consumption due to lunch causes scarcity of renewable energy more often than in the hours immediately before or after. The signal on a monthly basis is not so easy to interpret. It appears to be beneficial to have above-average consumption in August and below-average in September: this is possibly due to a fraction of the households that went on holiday later in any given year. Given the sharp drop in irradiation in the month of September compared to July and August, it seems reasonable that going on holiday in September increases the self-sufficiency over the year.

6.3.2 Exploitation of the Common Renewable Resources: Sheer Cumulative Consumption Versus Self-Sufficiency

Figure 6.9 shows the relation between the annual cumulative demand and the annual cumulative energy received from the shared PV system. These two variables are strongly correlated ($R > 0.9$); thus, the quantity of energy consumed from the PV system can be assumed with good confidence from the annual cumulative demand alone (i.e. regardless of the self-sufficiency).

This aspect, although counterintuitive, is a consequence of the highest variability in annual cumulative demand compared to the variability in self-sufficiency: if in fact the highest self-sufficiency is two times the lowest one, the highest cumulative demand is almost five times the lowest one (excluding the highest value as an outlier; otherwise, it is more than seven times). The strong prominence in variability of cumulative demand compared to self-sufficiency reduces the variation in self-sufficiency as a mere noise compared to the other variable (as visible in Fig. 6.9). Furthermore, as self-sufficiency is a share of the demand, it does not have much importance in absolute terms when applied to households with low cumulative demand. This fact represents somewhat a hindrance as it implies that increasing overall consumption works better than improving self-sufficiency to seize larger quantities of scarce local renewable resources. Nevertheless, it is not clear what power an individual household has to change its cumulative energy demand. Further investigation on the aspects that influence the cumulative energy demand (e.g. number of people in the household, cooking habits, holiday habits, etc.) is needed to assess whether it is something that the inhabitants can change. If each household has significant power on the cumulative energy consumption, it is reasonable to fear a sharp increase in the overall consumption after the installation of the communal PV system. It should be acknowledged that the lack of data with respect to other households might focus the attention of the inhabitants on their own energy demand advising them to increase the self-sufficiency. Another interesting aspect, shown in Fig. 6.9, is that the linear interpolation of the household data points has a steeper slope than the average self-sufficiency of the 48 households. This means that the household with the highest annual cumulative consumption also has, on average, a highest self-sufficiency. The highest slope of the interpolation implies that at low consumption the self-sufficiency of a household tends to be lower than average, while at higher consumption tends to be higher. A correlation analysis between annual cumulative consumption and self-sufficiency found a positive, albeit weak, correlation ($R \approx 0.2$). Although it is weak and thus uncertain, the correlation suggests that highly consuming households might have more contemporaneity with the production from PV. This might be due to larger households having some members who stay at home during daytime or to electric consumption by people who spend daytime at home being larger overall.

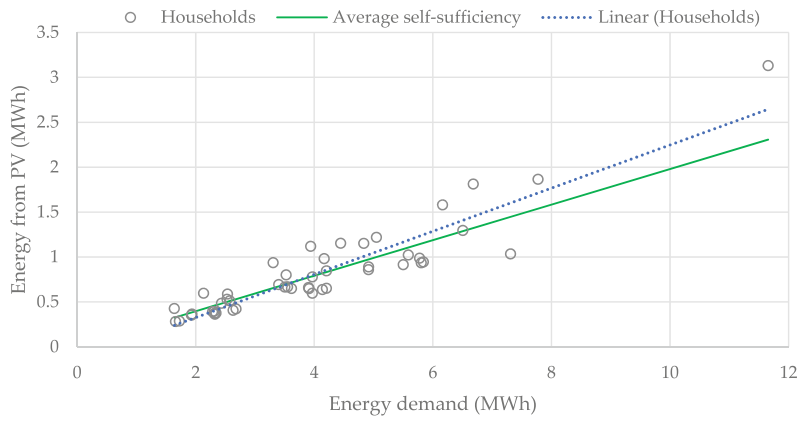


Fig. 6.9 Annual cumulative energy demand and annual cumulative energy used from the PV system for every household in the local grid

6.3.3 LEC Gratis

In this arrangement, the households in the district are shareholders of the system. Thus, they can use the electricity produced by the system for free when available. In this chapter, the shares of the PV system are equal. Each household will therefore have to pay 13,646 SEK (1275 €) of initial investment plus ca. 342 SEK/year (32 €/year) for maintenance and substitution of the inverter. Different ownership structures are possible, but the business model should be modified to avoid loopholes in the risk–benefit balance. For instance, equal shares could be distributed to a sub-group of the households (i.e. there are consumers who do not hold shares). In this case, an electricity price for non-owners should be established (see Sect. 6.2.2 LEP n%).

Figure 6.10 shows the difference in price between the energy offered by the parent grid and the energy available within the local system. The chart shows monthly values, which refer to the average cost of the electricity that month in the grid. We know from the 6.2.1 "Ownership structures and business models" that at any given time the price of the electricity is unique within the microgrid and depends on the relationship between production of PV and demand (see Eqs. (6.1) and (6.2)). The bars in Fig. 6.10 are the average of all electricity prices of the respective month weighted by the aggregated electric consumption in that month. Obviously, since the energy not met by the local production is bought from the parent grid, the external price has an influence on the internal one. In simpler terms, the internal price of the electric energy in one month, according to Eq. (6.2) with Plocal = 0, is proportional to the residual demand. Notice that, due to the higher external price, the drop in cost of electricity during the months of March (Month 3) is similar to that in April (Month 4) despite a lower self-sufficiency.

Even if the price of the electricity is the same within the microgrid at any given point in time, the average price paid by each household varies according to the time patterns of consumption. A household will enjoy a lower average price when they consumed a large share of its annual consumption at times when the electricity was free (or at least cheaper). This is to say, a higher self-sufficiency will lower the average price. However, in terms of gross economic benefit (i.e. the sum that can be saved), it is not the average price that matters but the cumulative energy received for free. In this sense, the conclusion from the results in Fig. 6.9 is troublesome as the earnings are not due to the ability to obtain a higher self-sufficiency but simply to the sheer cumulative consumption. In Fig. 6.11, the households in the microgrid are divided into three groups of 16 elements each according to their annual cumulative consumption. As in Fig. 6.9, the correlation of the key performance indicator (KPI) with annual cumulative consumption is evident. In fact, the lifetime economic balance is determined solely by the savings, thus by the

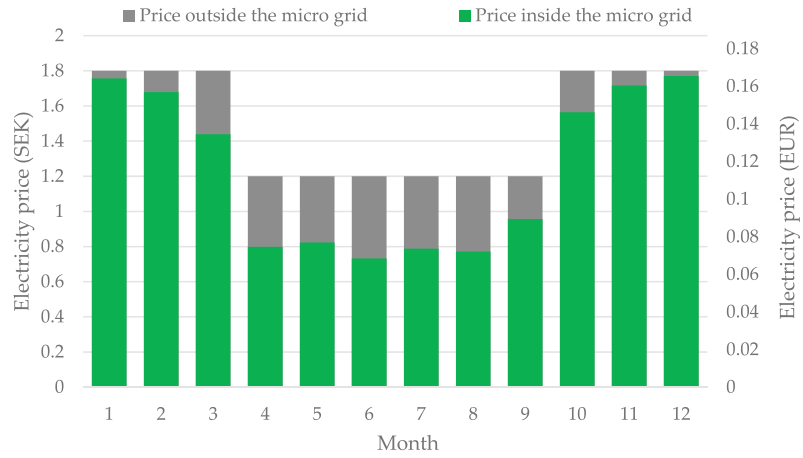


Fig. 6.10 Monthly difference in price between the energy offered by the parent grid and the average paid by the shareholders in a LEC gratis arrangement

sheer quantity of energy that is received by each household. In Fig. 6.11a, it is visible how being in the upper third of the cumulative consumption charts guarantees substantial earnings (IRR: internal rate of return from 1.9% to 6%), in the case of the initial investment about 13,646 SEK (1275 €/household). Conversely, the low-consumption households are doomed to economic losses, which means they are unable to recover the investment itself.

If the relation between annual cumulative consumption and lifetime earnings would become known by the households in the local grid, there is a risk that there would be a considerable increase of the cumulative demand after the installation of the communal system. This fact, although potentially reducing the risk for those investing in the system (especially in a LEP case), would counteract the purpose of reducing consumption of electricity from the grid.

6.3.4 LCOE of LEC

If the energy is sold at production cost (LCOE), instead of being given for free, the difference in lifetime balance from the different households is greatly reduced, but they persist. In this case, the advantage associated with the use of energy from the system is influenced by the stake of ownership of the system. In general, it can be noted that the lifetime earnings (i.e. Figure 6.11a, b) follow a linear transformation from the extreme inequality (as in Fig. 6.11a), to a situation of complete equality of earnings (if a LEC grid-price is hypothesised), where no benefit is obtained by the use of on-site electricity. In the hypothesis, a benefit for self-consumed electricity would spur increased self-sufficiency. A balance should be found between risk for the low consumption households and reward for the consumption of local renewable energy.

6.3.5 LEP N%

In this arrangement, the PV system is owned by a single provider who has the right to set the price. Obviously, since the parent grid can supply 100% of the demand of the district, the owner cannot set the price higher than the electric grid lest being completely out-bid (e.g. no household would use the owner's energy). In this chapter, the provider sets the price as half-way between the minimum of the local LCOE and the maximum of the consumer price from the parent grid. More precisely, the provider sets a price at a percentage n so that $n = 0$ is the LCOE, $n = 100$ is the price offered by the parent grid and $n = 50$ is exactly half-way in between.

Table 6.2 shows how the annual revenues, the balance over the lifetime and the real IRR change according to the price at which the electricity is sold.

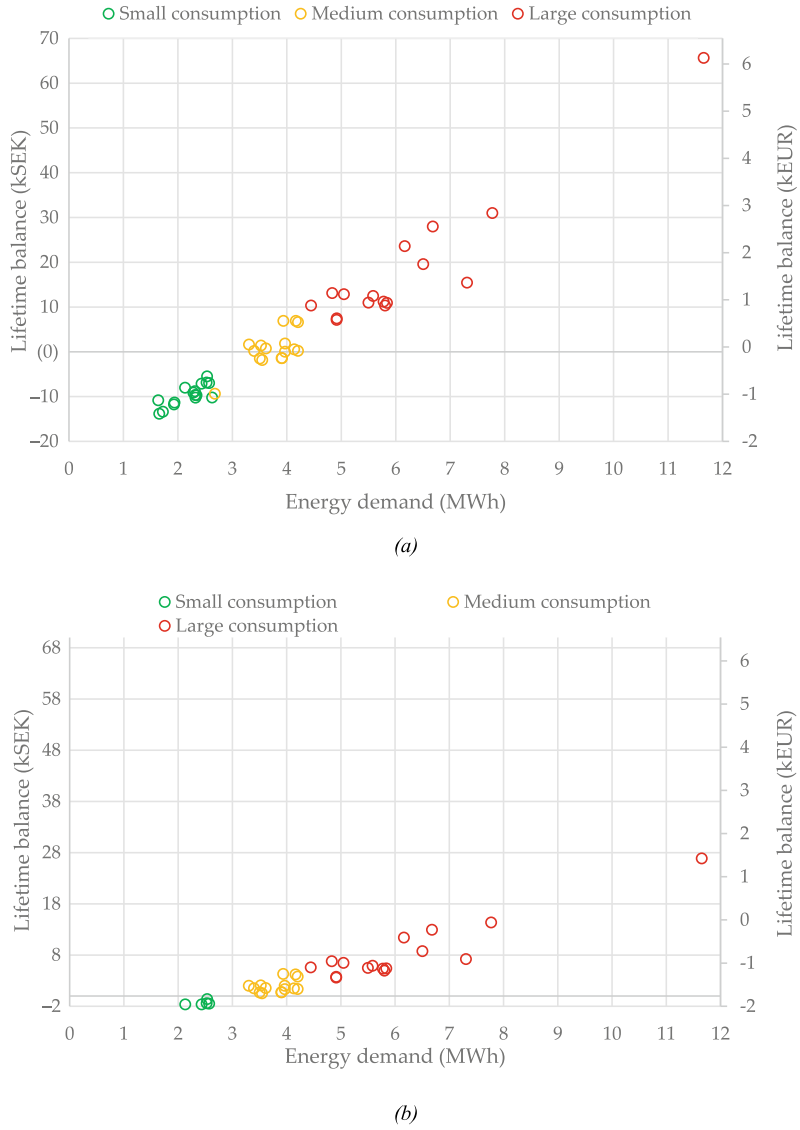


Fig. 6.11 Cumulative balance over the lifetime of the system against the annual energy demand. The households have been divided in three groups, each of 16 specimens, according to their cumulative consumption: **a** LEC Gratis, and **b** LEC LCOE

Table 6.2 Annual revenues, lifetime balance and internal rate of return (real) of the investment by different prices set by the owner

1 (%)	N	2 (SEK)	Revenues	3 (SEK)	Balance	4 (€)	Balance	5 (%)	IRR
6	0	7	34,553	8	− 94,058	9	− 8790	10	− 0.5
11	9.43	12	37,689	13	0	14	0	15	0.0
16	25	17	42,864	18	155,247	19	14,509	20	0.7
21	50	22	51,174	23	404,553	24	37,809	25	1.6
26	75	27	59,484	28	653,859	29	61,108	30	2.3
31	100	32	67,794	33	903,165	34	84,408	35	2.9

Notice how with $n = 0\%$ (i.e. the electricity sold at production cost of 0.83 SEK/kWh), the balance and thus the IRR result are negative. This is because the self-consumption of the system is not 100% (it is in fact ca. 85%). In other words, not all the energy produced by the PV system is consumed by the households in the local grid. Therefore, part of the production is sold to the grid below LCOE and results in a moderate loss over the lifetime. The existence of this loss justifies the use of a LCOE adjusted for self-consumption, as described in (Huang et al. 2019). This loss also explains why, under LEC LCOE arrangement, some households experience economic losses over the lifetime when the electricity by the communal system is given at price of cost (see Fig. 6.11b). When the electricity is sold at LCOE, the IRR of the PV system is negative; thus holding its shares leads to a loss unless the benefit for cheaper energy outweighs the costs.

Applying an $n = 9.43\%$ does not result in any loss or gain over the lifetime of the system. It can be argued that no investor would like to take any risk to have an expected net present value (NPV) of 0 at the end of the lifetime with a discount rate of 0. Nevertheless, there are potential business models for large homeowners such as general contractors or municipalities who could substitute part of the roof and façade cladding with BIPV thus avoiding the cost of an alternative material. Furthermore, this price tag is extremely interesting as price of sale from LEC. It in fact presents the advantage of expected lifetime economic balance in positive ground for each household.

A good business opportunity is finally offered by the $n = 100\%$. This price, while suggesting a real IRR around 3% for the LEP, offers the occupants the opportunity to largely increase their share of renewable energy use without having to pay any upfront cost. In this case, the households have no economic benefit in installing the PV, but they have no risk or upfront investment and could receive information about their own self-sufficiency by the provider, e.g. with a monthly email.

6.4 Discussion

6.4.1 Social and Cultural Differences Amongst Households Have a Huge Impact on Self-Sufficiency

In the local grid, if the renewable energy is not enough to cover the electric demand during a specific hour, the aggregated self-sufficiency is assigned to each household regardless of its demand (see Eqs. (6.1) and (6.2)). A large difference in terms of self-sufficiency has been observed within the 48 households, with the individual self-sufficiencies spanning from ca. 14% to more than 28% (see Fig. 6.7a). Considering the absence of active strategies to increase the self-sufficiency in the cluster, such large differences can be attributed only to sociocultural factors and spontaneous lifestyle choices. In Fig. 6.7b, it appears that the most self-sufficient household has on average the peak of energy consumption at noon (possibly due to home cooking), while the least self-sufficient one has usually its peak consumption at 20:00. Differences are visible also over the different months of the year, but their effect is not as clear as in the hours of the day. The large differences observed in self-sufficiency, having no active engagement or use of demand-shifting technologies, invite a deeper analysis and understanding of the existing electric demand and the factors which affect self-sufficiency.

6.4.2 High Cumulative Energy Demand is More Effective Than High Self-Sufficiency in Exploiting the Shared Renewable Resource

Despite the large variation in self-sufficiency, it has been observed that the sheer amount of energy used from the system is mainly determined by the annual cumulative demand (see Fig. 6.9). This phenomenon, albeit

counterintuitive, is due to the fact that the variability of cumulative demand far outweighs the variability in self-sufficiency (the largest being five or even seven times the smallest one). In other words, the fraction self-consumed is not significant when applied to a group of households whose entire demand is hardly significant compared to others. This fact is problematic because the energy savings (i.e. the main earning mechanism of the investment in some market designs) come from the amount of PV energy consumed and not from the self-sufficiency reached. The relation between annual cumulative consumption and cumulative energy from PV is transposed in the relation between energy consumption and lifetime balance (see Fig. 6.11). The balance in a LEC gratis arrangement (Fig. 6.11a) is almost completely determined by the cumulative consumption, with the self-sufficiency being reduced to a noise in the linear relation. Moreover, if the households are divided into three groups according to their cumulative consumption, the biggest consumers all have positive balance, and the smallest consumers all have a negative one. This aspect suggests that, if the communal PV system is installed under a LEC gratis arrangement, the shareholders might increase their electric demand in a bid to outdo each other's energy consumption. This behaviour would possibly defeat the purpose of installing on-site renewables in the first place. It should also be considered that, due to privacy laws and standard practice, each individual household is likely only aware of its own electric demand and self-sufficiency. This lack of data might drive each household to work on improving self-sufficiency instead of annual cumulative demand. It should also be remembered that the earnings are savings, thus increasing the cumulative demand would lead to an increase in the energy bill. In this sense, the increased exploitation of the common electricity through increased cumulative demand would happen only if increased consumption is perceived as a value, for example through the purchase or increased use of energy hungry appliances for cooking or Do It Yourself (DIY) purposes. How easy or difficult it is to change self-sufficiency compared to cumulative demand should also be considered to assess the likelihood of one scenario over the other. For example, cumulative demand might be strongly constrained by working schedule or number of household members. These aspects reiterate the need for a deeper study on the aspect of demand that influence self-sufficiency. From the perspective of the investment in PV, both the changes in behaviour envisioned would increase self-consumption, hence earning potential.

6.4.3 *Different Selling Prices Generates Various Business Opportunities*

Assuming that the shared PV system is owned by a single entity in a local energy provider (LEP) arrangement, this entity enjoys freedom in setting the price for the sale of electricity. This freedom is nevertheless constrained by the LCOE of the PV system and by the price offered by the parent grid. If the LEP sells electricity at a higher price than the parent grid, it will have no purchaser amongst the households. This happens because the grid has the capacity to satisfy 100% of the demand of the whole district at any time. For this reason, a coefficient "n" has been devised so that: $n = 0$ is the LCOE of the local system and $n = 100$ is the sale of energy at the exact same price as from the parent grid. It has been shown that at $n = 0$, despite selling at production cost, the lifetime balance is < 0. This is due to the self-consumption being below 100% (i.e. ca 85%), hence ca. 15% of the energy produced being sold at spot price (i.e. 0.3–0.15 SEK/kWh or 3–1.5 € cent/kWh). This loss also explains why in the LEC LCOE arrangement some households still have a negative lifetime balance, as demonstrated in Fig. 6.11b. Another interesting selling price is the one obtained with $n = 9.43\%$ because this is the price at which no profit or loss is made from the LEP. This price tag, albeit unattractive as an investment for a third-party PV owner, presents an interesting way for building owners to substitute other claddings on their properties. Using

this selling price offers in fact a building material that, contrary to every other, does not cost anything over its lifetime. If applied as common price in a LEC, it allows all households to have a positive lifetime economic balance, yet to have individual differences in earnings. It should be said that this price was determined at the end of a previous run when the overall self-consumption was already known. In a real case, to obtain such an equilibrium, the price should be updated at any point in time according to the evolution of self-consumption and energy prices. Selling energy at the price of the parent grid ($n = 100$) could be an interesting investment as it guarantees the LEP with a real IRR of around 3%; it provides no economic benefits for the household consumers, but it gives them the ability to boost their reliance on renewable without any upfront cost or risk. Furthermore, the possibility for the households to buy voluntarily sized shares of the LEP could kick start a set of tantalising business opportunities.

6.5 Summary

In the study, a newly developed agent-based model was tested on a shared PV system serving a small district comprising 48 apartments in a local community. Different ownership structures were explored. The LEC arrangement was studied both with the electricity given for free to all the equal shareholders or given at a price (in the study the LCOE). For the LEP, because the free offering would make no sense, an array of different prices was tried (see Table 6.2).

Key Findings

The main findings of the study are reported as follows and interpreted in the corresponding paragraphs in the discussion Sect. 6.4:

- **Social and cultural differences amongst households have a huge impact on self-sufficiency**: The households were simulated without introducing any demand-response measure or smart control. However, some households achieved a self-sufficiency of almost 30% using the common PV system while others stopped short of 15%.
- **High cumulative energy demand is more effective than high self-sufficiency in exploiting the shared renewable resource**: Despite the large differences observed in self-sufficiency amongst households, the quantity of energy received from the shared system has been determined almost completely by the annual cumulative demand rather than by self-sufficiency.
- **Different selling prices generates various business opportunities**: Different values of n %, as defined in Sect. 6.2.2, generate advantage and interesting features for diverse stakeholders. For instance, a very low n% (i.e. < 10%) generates a strong drive for the shareholders to self-consume as much PV energy as possible, but it contains a risk for the least consuming ones. Higher n% (i.e. from ca. 10–100%) are interesting for building owners and BIPV solutions and, amid increasing n%, become more and more interesting for third-party energy providers.

6.5.1 Follow-Up Studies

The present study shows a plain setup and a narrow set of possibilities, but it sets the stage for a broader class of studies. In principle, some of the simplifying assumptions employed in this chapter should be removed in favour of a higher realism and a more complex modelling; nevertheless, models that are too complex for the level of uncertainty and for the input data available should be avoided.

For instance, it is tempting to change the present model for the prices from the parent grid (i.e. static seasonal price and long-term linear trends for sold and bought electricity) into a spot price with distribution costs. However, while the change reflects reality better, the long-term modelling of the spot price would be a daunting task and affected by huge uncertainty. Thus, it might pay off to just maintain a simplified

model for the prices (i.e. two seasonal prices for purchase and sale and time of day variation), but to perform a stochastic simulation with variability in the time-evolution of the prices. In other words, any further complexity addition should only be determined by the use case of the model. Furthermore, for this model, the use case is the market design to finance and maintain a fair and remunerative local electric energy system.

On the other end, there are several low hanging fruits that can be easily harvested: for example, while in this chapter the price was always set by a unique actor (be it a community or a provider), it would be interesting to explore the effect of different prosumer setting each an arbitrary price and explore their interaction. In this sense, one more step could be to endow the agents with some level of intelligence and let them adjust the price reacting to the environment to maximise potential economic gains.

In the present study, there are devices and loads that have not been investigated, such as EVs and electric storages, in the local grid. These features, given a simplified enough model, are extremely easy to be implemented and can constitute a game-changer in the effectiveness of a business model.

Another interesting and potentially prolific research direction would be the study of the demand itself. Given the large variation of self-sufficiency found amongst the different agents participating in the microgrid, it is possible to find correlation with socioeconomic and lifestyle parameters such as median age, work–home schedules, and number of members in an household. This does not constitute information in itself, but it can lead to different results according to the different shared renewable systems. In other words, each social mix might demand a different system (capacity of PV capacity of electric storage).

Regarding the demand, it is of paramount importance to consider how often a house remains vacant due to change or death of the owner. These aspects should be investigated in terms of impact over each business model, but also in terms of risk-mitigating effect of larger local grids. It shall not be forgotten that lower risk can allow lower IRR for the investment, thus unlock wider market niches. The vacancy of the households is also affected by socioeconomic parameters and median age of the households; these aspects likely present spatial variability in different parts of the city and the world.

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7 Electric Vehicle Smart Charging Characteristics on the Power Regulation Abilities

Pei Huang and Linfeng Zhang

Abstract

Electric vehicle (EV) smart charging, which regulates the charging rates of EVs in response to the availability of surplus solar photovoltaics (PV) power or the electricity prices, can effectively enhance the local power demand–supply balance and help improve the PV power local utilization. In this regard, researchers have developed two categories of EV charging controls: (i) B2V or G2V ((i.e., building-to-vehicle or grid-to-vehicle) power flow in which the EV can only be charged; and (ii) B2V2B or G2V2G (i.e., building to vehicle to building or grid-vehicle-to-grid) in which the vehicles can be both charged and discharged. The frequent charging/discharging could potentially accelerate the EV battery degradation, which might make such applications not economical. However, systematic investigation has rarely been conducted for the impact of various EV usage and charging factors (including the EV charging strategy, different EV charging forms, EV charging limits, and commuting distance) on the power balancing performances and EV battery cycling degradation. As a result, the EV owners may not be willing to join the smart charging demand response due to the concerns of accelerated battery degradation, and this hinders the applications of EVs in the power regulation in the future energy system. This chapter aims to investigate the effect of different ways of using EVs on the demand response performances and the EV battery degradation. A parametric study considering a set of different scenarios combining various EV charging forms, EV charging limits and commuting distances will be conducted in Sweden. A smart charging control method of the EV will be developed, which can optimize the EV charging and discharging rates to minimize the grid interactions. A degradation model, which can evaluate the EV battery degradation due to charging/discharging cycling, will be constructed to investigate the EV battery degradation under typical scenarios. The performances of each scenario will be analyzed and compared to draw conclusions. The study results can help improve researchers' understanding of the impacts of smart EV charging in the building community performances. The obtained impacts on the battery degradation can also support decision makers in selecting suitable EV charging and usage strategies.

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Electric vehicle (EV) · Charging strategies · Degradation · Demand response

7.1 Introduction

With the increasing environmental concerns, many governments have established targets and policy to replace the fossil fuel-based vehicles with the electric vehicles (EVs) (Huang et al. 2019b). For instance, the Swedish government has set a target that the vehicle fleets should be completely fossil-free by 2030, while a large percentage should be achieved by EV deployment (Xylia and Silveira 2017). The Norwegian Parliament has made a national goal that all new cars sold by 2025 should be zero-emission (via either electric vehicles or hydrogen vehicles) (Association 2021). To promote the markets for EVs, the U.S. Federal government has enacted policies and legislation, such as improvements of tax credits in current law and competitive programs, to encourage investment in infrastructure supporting EVs (United States International Energy Agency 2010). As a result, the number of EVs is continuously increasing in recent years. The Swedish market statistics reported a 310% increase of the full battery EVs in 2020 (Kane 2021). The number of EVs on the road is projected to reach 18.7 million in 2030, up from slightly more than 1 million at the end of 2018 (Cooper and Schefter 2018). In the future, due to the large penetration, EVs are expected to have substantial impacts on the grid power demands. The large increase of electric vehicles numbers poses new challenges to the existing grid systems, as the existing grid infrastructure are not designed to host the large shares of new electric loads (Jang et al. 2020). As a result, problems such as the voltage deviations and overloading of components may arise. Considering the climate change impacts on the future energy demand (Zou et al. 2021), the increased EV charging loads, if not well regulated, will cause great challenges to the existing grid system.

To mitigate the negative impacts of the large EV penetration, existing studies have proposed various smart controls of EV charging (Yu et al. 2022a, b). For instance, Dallinger and Wietschel (2012) conducted a systematic study to investigate the potentials of EV smart charging in balancing the power in Germany. They first proposed a model to evaluate the stochastic EV mobility behavior. An agent-based equilibrium model was then developed to decide the market electricity prices according to the marginal generation costs of electricity, which is affected by renewable power productions. Based on the EV mobility model and market electricity price model, they further developed an optimization control method to optimize EV charging with the target of minimizing the charging costs. Their study results showed that suitable electricity pricing strategy can help incentivize EV owners to participate in the grid power regulation to balance the intermittent renewable power generation. Huang et al. (Huang et al. 2020) developed a smart EV charging control based on genetic algorithm integrated with the building community energy sharing controls. The proposed control optimizes the charging rates of all EVs in the community to obtain the best performances at the community aggregated level. Their study results showed that combining EV smart charging and community energy sharing can further improve the performances at the building community level. Fachrizal and Munkhammar (2020) developed a centralized charging control method of EV fleet for a residential building cluster, which sequentially optimizes the charging loads of EVs one by one based on the arrival time and departure time considering the interaction of individual EVs and PV power production. The study showed that proper coordination of the charging among EV fleet can also enhance the power regulation performances and the PV power self-utilization rates at the aggregated level. The abovementioned controls only consider unidirectional power flow, i.e., from buildings/grid to the vehicles (B2V or G2V). In such power flow modes, the EVs can only be charged, and its charging rates can vary in different time intervals.

In recent years, researchers have also proposed bidirectional power flow, called B2V2B (i.e., building to vehicle to building), V2G (i.e., vehicle to grid), or V2H (i.e., vehicle to home) (Alirezaei et al. 2016). In the B2V2B mode, the EVs are used as battery storage systems, which can be both charged and discharged, when they are parked and plugged into the charging cords. Compared to the unidirectional power flow, the bidirectional power flow can achieve better performances in aspects of enhancing the match between power supply and demand, reducing the infrastructure of power transmission, improving the power grid stability, and reducing the electricity costs (Liu et al. 2013). Based on the B2V2B concept, Barone et al. (2019) developed a dynamic simulation model in Matlab for regulating the energy demand and PV power production balance as a function of the considered electric vehicles energy use patterns. Their study considered the EV to be charged in one building and then discharged power back to a different building. The study results showed that the B2V2B energy management systems can improve the building grid reliance and meanwhile significantly reduce the grid electricity consumption up to 77%. Nayak et al. (2019) designed an adaptive controller to optimize the EV charging/discharging rates at the behest of no system information available at the controller end for grid frequency regulation during the grid restoration. Regarding the economic performances, Noori et al. (2016) conducted a prediction of the future net revenue and life cycle emissions savings of V2G technologies for use in ancillary (regulation) services in different regions of the U.S. They designed an agent-based model to investigate the various uncertain actions and used exploratory modeling and analysis to evaluate the future net revenue and emissions savings from the V2G technologies. The abovementioned studies consider the demand response from the perspective of buildings and the power grid, while the degradation of EV battery due to the charging/discharging is rarely considered. The frequent charging/discharging of EV battery will accelerate the battery degradation, which may affect the application of such bidirectional power flow.

To understand how EV charging affects the battery degradation, researchers have conducted some studies of the degradation under different scenarios. For instance, Bishop et al. (2013) conduced sensitivity analysis to study the impact of V2G services on battery degradation for EV and PHEV considering different battery capacities, charging regimes, and battery depth of discharge. They found that the EV battery degradation is most dependent on energy throughput. When providing ancillary services, the degradation is the most sensitive to the depth of discharge (DoD). In this chapter, however, the EV smart charging for balancing the grid power was not considered. Wang et al. (2016) developed a method to quantify EV battery degradation from driving and several V2G services based on a semi-empirical lithium-ion battery capacity fade model. Using the developed model, they studied the EV battery degradation performances under three V2G services: peak load shaving, frequency regulation, and net load shaping. They also compared the degradation with baseline cases of driving only and uncontrolled charging. Their study concluded that the EV battery degradation from V2G is inconsequential compared to the naturally occurring battery wear (i.e., from driving and calendar aging) when V2G services are offered only on days of the greatest grid need (20 days/year in this chapter). Similarly, González-Garrido et al. (2019) investigated the grid performances, the potential savings on the charging cost and EV battery degradation when applying EV V2G services in an island in Denmark. They concluded that the intensive operation of V2G strategy may reduce the lifetime of the battery and increase the total EV costs due to potential needs of battery replacement. Notably, Bui et al. (2021) conducted a systematic investigation of the EV battery degradation under various charging strategies and driving behaviors. Their study adopted a comprehensive EV battery considered detailed two sets of different operational profiles. But, the impacts of EV smart charging on the building community grid interaction performances are not fully studied. The impacts of other factors, such as EV

charging limits and EV commuting distances, are not studied, neither.

To sum up, until now a systematic study of the various EV usage and charging factors' impacts on the power balancing performances and EV battery cycling degradation is still lacking. As a result, it is still unclear how the EV charging strategy, different EV charging forms, EV charging limits, and commuting distance will affect the grid interaction and EVs. The EV owners may not be willing to join the smart charging demand response due to the concern of large battery degradation. This will hinder the application of EVs in the power regulation in the future energy system. Therefore, this chapter conducts parametric studies to investigate how the different ways of using EVs will affect the demand response performances and the EV battery degradation. First, a set of different scenarios combining different EV charging forms, EV charging limits and commuting distances will be defined. Then, a smart EV charging control method will be developed, which can optimize the EV charging and discharging rates to minimize the grid interaction (the peak energy exchanges with the grid in this chapter). Next, a degradation model, which can evaluate the EV battery degradation due to cycling, will be constructed to simulate the EV battery degradation under different scenarios. Finally, the performances of each scenario will be analyzed and compared to draw conclusions. The findings from this chapter can help researchers understand the impacts of smart EV charging in the building community performances. The studied impacts on battery degradation can also support decision makers in selecting suitable EV charging strategies.

This chapter is structured as follows: Sect. 7.2 presents the full methodology of this chapter. The detailed modeling of different systems and EV usage scenario definition are provided in Sect. 7.3. Section 7.4 presents and discusses case studies and results of energy performances and EV battery degradation under different scenarios. A conclusion is given in Sect. 7.5.

7.2 Methodology

Figure 7.1 shows the overall methodology of the proposed study for investigating the impacts of different EV charging strategies and usage scenarios on the EV battery degradation and energy performances. The proposed study consists of four steps. In Step 1, the different scenarios of EV charging forms, EV usage, and EV charging limits will be defined. In Step 2, the daily charging/discharging rates of EVs will be optimized using the particle swarm optimizer for different scenarios defined in Step 1. The optimizer will derive the optimal EV charging/discharging profiles according to a predefined optimization target. Based on the optimized EV charging/discharging profiles in each day (obtained from Step 2) and EV information such as daily initial state of charge (SOC), arrival and departure time, in Step 3, a profile of the full-year SOC will be derived for each scenario. Then, the full-year SOC profile will be used as inputs in the Rainflow cycle counting algorithm for evaluating the EV battery cycling degradation (Huang et al. 2018a). In Step 4, the performances of each scenario will be investigated and compared. The considered performances indicators include the annual EV battery degradation and peak energy exchanges with the grid. The analysis will reveal the impacts of charging forms, charging limits and EV usage on the EV and building energy performances. The details of each step are introduced in the following subsections.

7.2.1 Step 1: Define Various Scenarios of EV Usage and Charging Limits

This chapter will study the impacts of various scenarios of EV usage and charging limits on the EV battery degradation and demand response performances. The considered factors include the EV charging forms, EV driving distance, and EV charging limits. Three EV charging forms will be

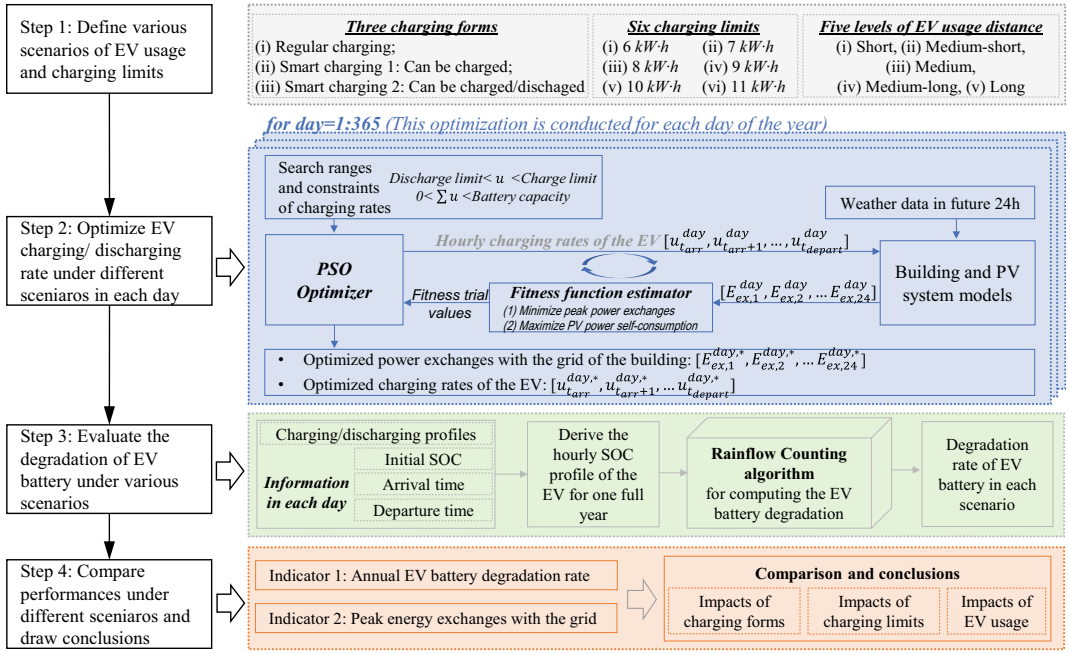

Fig. 7.1 Flowchart of the proposed study for investigating impacts of different factors on EV battery degradation and performances

studied: (i) Regular Charging (RC), in which an EV is charged at a constant charging rate (e.g., the rated charging rate); (ii) Smart Charging 1 (SC1), in which an EV can be charged at varying charging rates; and (iii) Smart Charging 2 (SC2), in which an EV can be charged and discharged at varying rates (i.e., the EV is used like a battery storage system). Five EV commuting distances scenarios will be considered: (i) short (e.g., around 20 km); (ii) medium-short (e.g., around 33 km); (iii) medium (e.g., around 45 km); (iv) medium-long (e.g., around 58 km); and (v) long (e.g., around 70 km). In this chapter, it is assumed that the EVs have fixed routes on workdays and weekends. The five different commuting distances are selected to represent how far the EV owners are from their workplaces. In general, the longer the distance an EV commutes, the faster the EV battery is expected to degrade due to the large depth of discharge (DoD) cycle. Six EV charging limits will be selected for the second and third charging forms. While for the first charging form, i.e., regular charging, the charging rates of the EV will be the rated values. The larger the charging limit is, the more flexible energy demand shifting ability the EV has. This can potentially improve the energy performances of the building. But, on the other hand, the EV battery may suffer faster degradation. In total, there are 65 scenarios (= 5 cases of distances × (1 charging forms + 6 cases of limits × 2 charging forms)) considered. Table 7.1 summarizes the considered 65 scenarios.

7.2.2 Step 2: Optimize EV Charging/discharging Rate Under Various Scenarios in Each day

After the scenarios are decided, the EV charging/discharging will be optimized in the second step. Figure 7.1 presents the selected control method for the individual approach. This control is implemented for the EV considering the available solar power in a building community with PV panels installed. After meeting the power needs of the community, the remailing surplus PV power from the community can be used to supply the charging station. If the surplus PV power is

Table 7.1 Considered scenarios for the proposed study

Charging forms		Regular (**RC**)	Smart Charging 1 (**SC1**)						Smart Charging 2 (**SC2**)					
Charging limits		1 fixed charging rate	6 case of charging limits						6 cases of charging limits					
Distance	A: short	1A	2A	3A	4A	5A	6A	7A	8A	9A	10A	11A	12A	13A
	B: medium-short	1B	2B	3B	4B	5B	6B	7B	8B	9B	10B	11B	12B	13B
	C: medium	1C	2C	3C	4C	5C	6C	7C	8C	9C	10C	11C	12C	13C
	D: medium-long	1D	2D	3D	4D	5D	6D	7D	8D	9D	10D	11D	12D	13D
	E: long	1E	2E	3E	4E	5E	6E	7E	8E	9E	10E	11E	12E	13E

not sufficient to meet the EV charging load, the grid electricity will be purchased to charging the EV. Considering the fast computation speed and good ability to search the optimal solutions, the particle swarm optimization (PSO) will be used in the control optimization. In each iteration, a set of alternatives for EV charging/discharging rates will be produced by the algorithm. These alternatives will be used, together with the energy demand and PV power production data of the community, for calculating the hourly energy exchanges with the grid. Then, the hourly energy exchanges will be imported into the fitness function to evaluate the performances of the alternatives. The iteration will continue until the optimal solutions are found. The optimization is conducted in each day, and it will be repeated for a full year for each scenario.

The mathematical modeling is introduced as follows: The hourly charging rates of the EV in each day should be in the range of a lower limit (u_{lower}) and an upper limit (u_{upper}) as seen in Eq. (7.1):

$$u_{\text{lower}} \le u_{t_{\text{arr}}}^{\text{day}}, u_{t_{\text{arr}}+1}^{\text{day}}, \ldots, u_{t_{\text{depart}}}^{\text{day}} \le u_{\text{upper}}, \quad (7.1)$$

where u_{lower} (kW) is 0 for the SC1 (Smart Charging 1), while for SC2 (Smart Charging 2) it varies in each charging limit scenario. Meanwhile, the charging/discharging rates should follow the battery constraints, as explained by Eq. (7.2)

$$0 \le \text{SOC}_0 \times \text{CAP} + \left(u_{t_{\text{arr}}}^{\text{day}} + u_{t_{\text{arr}}+1}^{\text{day}} + u_{t_{\text{arr}}+i}^{\text{day}} + \ldots\right) \times \tau \le \text{CAP} \quad \text{where } i = 1, 2, \ldots, t_{\text{dep}} + 1 - t_{\text{arr}} \quad (7.2)$$

where SOC_0 (%) is the initial state of charge of EV battery when the EV arrives at the charging location. CAP (kW·h) is the capacity of the EV battery. t_{arr} is the EV arrival time at the charging port, and t_{depart} is the departure time from the charging port. τ is the charging duration (1 h in this chapter).

In order to meet the traveling needs of the next trip, the EV battery should be charged to a

user-specified target level (SOC_{taget}) before they depart the charging port. This constraint is expressed by Eq. (7.3). When SOC_{taget} equals 1, it represents the EV users require the EV battery to be fully charged before the next trip:

$$\begin{aligned} &SOC_{taget} \times CAP \leq SOC_0 \\ &\quad \times CAP + \left(u^{day}_{t_{arr}} + u^{day}_{t_{arr}+1} + \ldots + u^{day}_{t_{depart}}\right) \\ &\quad \times \tau \leq CAP. \end{aligned} \tag{7.3}$$

This chapter considers the strategy to minimize the peak energy exchanges with the grid. Following this control strategy, a fitness function is determined, as expressed by Eq. (7.4) (Salom et al. 2011)

$$J_{grid} = \min\left(E^{day}_{peak}\right), \tag{7.4}$$

where E^{day}_{peak} (kW·h) is the peak power exchange of the community with the grid. It is calculated by Eq. (7.5)

$$E^{b}_{peak} = \max\left(\left|E^{day}_{ex,1}\right|, \left|E^{day}_{ex,2}\right|, \ldots \left|E^{day}_{ex,24}\right|\right), \tag{7.5}$$

where $E^{day}_{ex,i}$ (kW·h) is the power exchange with the grid of the building community in the ith hour in a specific day considering the EV charging load. It is calculated by Eq. (7.6):

$$E^{day}_{ex,i} = E^{day}_{d,i} + u^{day}_{i} \times \tau - E^{day}_{s,i}, \tag{7.6}$$

where $E^{b}_{d,i}$ (kW·h) and $E^{b}_{s,i}$ (kW·h) are the hourly energy demand and renewable energy production of the building community. u^{k}_{i} (kW) is the charging load of the EV parked in the building.

7.2.3 Step 3: Evaluate the Degradation of EV Battery Under Various Scenarios

After producing the optimized charging profiles of the EV in different scenarios, a simplified hourly SOC profile for a full year will be derived for each scenario. It is assumed that the SOC of the EV will first decrease in a short period after its departure, then it becomes stable, and the SOC will decrease again in a short period before its arrival at the charging location in the next day. Considering that the simulation interval to be 1 h in this chapter and the commuting distances in various scenarios varies from 20 to 70 km, this short period is set as 1 h. Such simplification does not consider the detailed EV driving patterns while being used. But, it will not have large impacts on the comparative studies in Step 4, as it is kept the same for all the scenarios.

The degradation of the EV battery due to the cycling charging/discharging will be then simulated using the Rainflow cycle counting algorithm. The Rainflow cycle counting algorithm derives two parameters for the degradation estimation: the *ranges* of the DoD (depth of discharge) and *cycles* of each range (Shi et al. 2018). The principle of the Rainflow cycle counting algorithm is shown in Fig. 7.2 (Laestander 2017). The implementation of this algorithm is explained as follows: First, the time-series depth of discharge (DoD) file is reduced to include only valleys and peaks through identifying slop reversals. All the 4-data-point and 3-data-point full cycles are then identified, counted, and removed following the steps in Fig. 7.2. Finally, the remaining half cycles are counted. Using these ranges and cycles information, the degradation rate will then be determined by Eq. (7.7) (Alam and Saha 2016):

$$D_{CL} = \sum_{k=1}^{m} \frac{\text{Cycle of } R_k}{a \times (R_k)^b}, \tag{7.7}$$

where a and b are coefficients that determine DoD-cycle curve. Note that these two coefficients are specific for each type of battery. In this chapter, a is set to be 512.2, and b is set to be − 1.9928, following a recent study (Electropaedia 2020).

7.2.4 Step 4: Compare Performances Under Different Scenarios and Draw Conclusions

In this step, the performances of each EV charging and usage scenario will be collected from the previous steps. The considered performances will

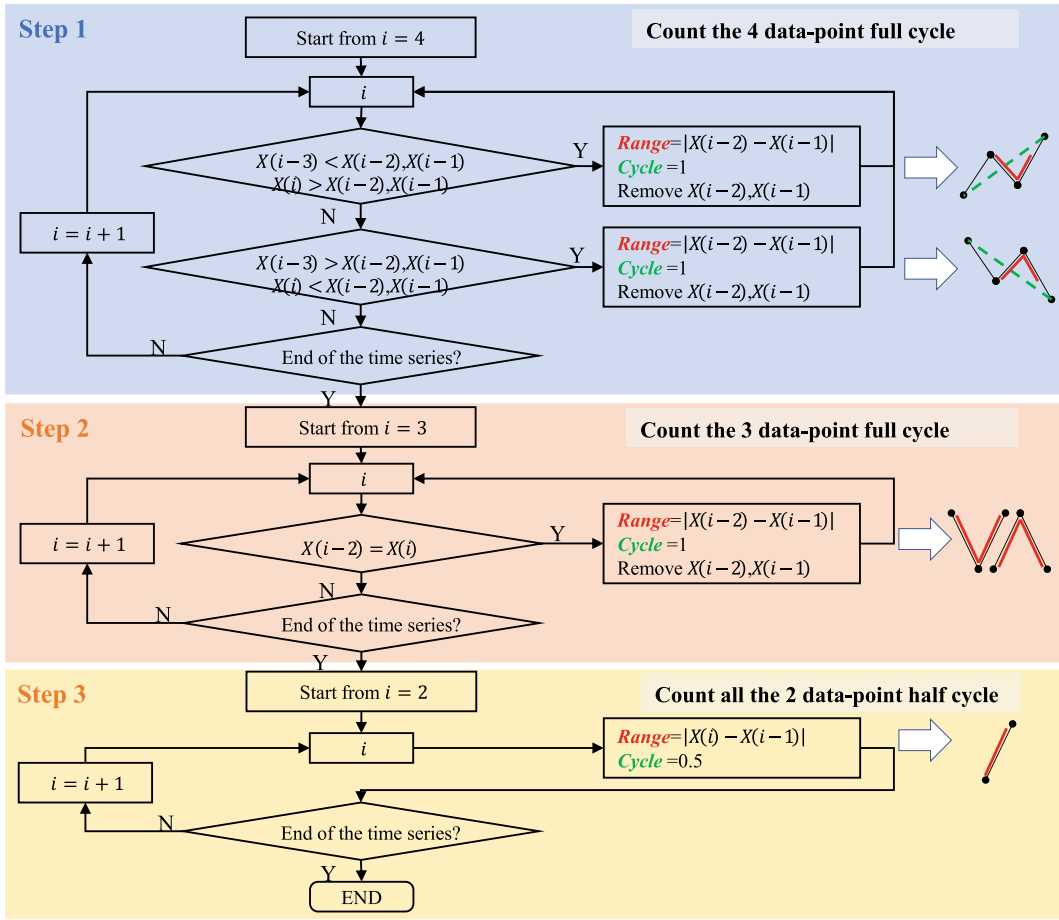

Fig. 7.2 Flowchart of the Rainflow counting algorithm for simulating the EV battery cycling degradation

include the EV battery degradation rate (calculated by Eq. (7.7)) and the building community peak energy exchanges with the grid (calculated by Eq. (7.5)). Then, the performance results in different scenarios will be compared to investigate the impacts of different factors (i.e., EV charging forms, charging limits, and EV driving distance) on the EV battery degradation and the building community energy performances.

7.3 Simulation Configuration

This section introduces the modeling of the building community power demand, PV power production, and electric vehicles, and the configuration of the studied scenarios.

7.3.1 Modeling of the Building Community Electricity Demand

A building community located in Ludvika of Dalarna County, Sweden, is considered in this chapter. This community has three separate buildings, which is connected with a direct current (DC) microgrid as described in (Huang et al. 2019a) for energy sharing. The whole community can be considered as one 'virtual' building: Its power demand equals the aggregation of individual buildings' demand, and its PV power supply equals the aggregation of the individual buildings' PV power production. For more details about this building community, please refer to (Huang et al. 2021) and the

EnergyMatching project (Matching 2021). The power demand of the building community is modeled using the LoadProfileGenerator (LPG), which is a behavior-based tool used to simulate the energy consumption in time intervals down to one minute (Pflugradt 2017).

7.3.2 Modeling of the Building Community Electricity Production

The PV power production P_{PV} (kW) is calculated by Eq. (7.8) (Huang et al. 2018b; Klein et al. 2004):

$$P_{\mathrm{PV}} = \tau_{t-a} \times I_{\mathrm{AM}} \times I_T \times \eta \times \mathrm{CAP}_{\mathrm{PV}}, \quad (7.8)$$

where τ_{t-a} is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T ($\mathrm{W/m^2}$) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; $\mathrm{CAP}_{\mathrm{PV}}$ ($\mathrm{m^2}$) is the PV surface area. The PV system is sized to have an annual PV power production equivalent to the annual power demand of the building community (without considering the EV loads).

7.3.3 Modeling of the Electrical Vehicle

The EV battery is calculated using a simplified model, as expressed by Eqs. (7.1–7.3). It is estimated as the aggregation of the hourly charging rates and discharging rates. This chapter considers the Tesla Model 3 Standard Range Plus (Database 2021), with a battery capacity of 55 kW·h and a usable battery capacity of 50 kW·h. Its energy consumption is set as 204 Wh/km considering the cold weather in Sweden. The arrival and departure time of the EV to the charging location (assumed to be in the building community, near the EV owner's workplace) are decided from the National Travel Survey (Analysis 2007). Considering that the daily routine for an EV owner can be very similar (e.g., drive from home to workplaces at around 8:00 and drive from workplaces back to home at around 16:00), this chapter uses a normal distribution to quantify the arrival and departure time to the case building community in each day. The daily commuting distance is also assumed to follow a normal distribution:

$$t_{\mathrm{arr}} \sim N\left(\mu_1, \sigma_1^2\right),\ t_{\mathrm{depart}} \sim N\left(\mu_2, \sigma_2^2\right),\ d \sim N\left(\mu_d, \sigma_d^2\right), \quad (7.9)$$

For the distributions of the arrival and departure time, μ_1 and μ_2 are set as 08:30 and 16:30, respectively. σ_1 and σ_2 are both set as 30 min. For the distribution of the distance, five different base values are considered, representing the different scenarios of traveling distances (see Sect. 7.3.4 for the detailed values). The standard deviation σ_d in each day for all the scenarios is set as 10 km. The traveling distances will affect the EV battery energy usage, which will further affect the initial SOC at the arrival time in the next day. The target SOC at the departure time is set as 0.95 in each day for all the scenarios. This chapter only considers the charging of EV on workdays, as they can potentially be parked in the workplace community. The charging of EV and performances on weekends are not considered.

7.3.4 Configuration of the EV Charging and Usage Scenarios

As described in Sect. 7.2.1, this chapter will consider 65 scenarios in total, considering different charging forms, charging limits, and EV driving distances. Table 7.2 summarizes the setting of EV charging limits and average commuting distances in the parametric study. For the five different commuting distances, this chapter selects 20, 33, 45, 58, and 70 km. In the regular charging, a fixed charging rate of 10 kW is used. In the Smart Charging 1, since the EV can only be charged, the hourly charging rate is larger than

Table 7.2 Configuration of the EV parameters in each scenario

Five level of distances						
A: short	B: medium-short	C: medium		D: medium-long		E: long
20 km	33 km	45 km		58 km		70 km
Six levels of charging limits (kW)						
Regular Charging (RC)	10					
Smart Charging 1 (SC1)	[0 ~ 6]	[0 ~ 7]	[0 ~ 8]	[0 ~ 9]	[0 ~ 10]	[0 ~ 11]
Smart Charging 2 (SC2)	[−6 ~ 6]	[−7 ~ 7]	[−8 ~ 8]	[−9 ~ 9]	[−10 ~ 10]	[−11 ~ 11]
Other EV parameters						
EV battery capacity (kW·h)	55					
Energy consumption (Wh/km)	204					
Target SOC when departure in each day	95%					
Arrival time at the charging location on workdays	Mean—08:30			Standard deviation—30 min		
Departure time at the charging location on workdays	Mean—16:30			Standard deviation—30 min		

0 all the time. While in the Smart Charging 2, the EV can also discharge power back to the building community, and thus the hourly charging rates can be negative. If the charging limits are too small, the EV may risk not reaching the target SOC when it departs the charging location. While if the charging limits are too large, the potential improvements in demand response will be marginal, and thus not so necessary. Thus, this chapter selects six different charging limits that are neither too small nor too large (i.e., from 6 to 11 kW at an interval of 1 kW) considering the EV battery capacity.

7.4 Case Studies and Results

This section first presents the building community power demand and PV power production results, which will be used as the inputs in implementing the EV smart charging. Then, the performances of different scenarios (as specified in Sect. 7.2.1) will be analyzed and compared. Section 7.4.2 presents the results under the control strategy of minimizing the peak energy exchanges with the grid.

7.4.1 Building Community Power Demand and PV Power Production Results

Using the TMY weather data in Ludvika, Sweden, the PV power production is calculated. The case community hourly power demand is in the range of 10.3 ∼ 32.7 kW, and its hourly PV power production is the range of 0 ∼ 63.5 kW. Figure 7.3 shows the PV power production and power demand of the case building community in four separate weeks (each week is in one season). Since the EVs are assumed to be parked only on workdays, the power demand and supply profiles of five days are presented in each week. As can be seen, in winter weeks, the hourly PV power production is always insufficient to meet the hourly power demand due to the low solar radiation level. Meanwhile, the period with solar radiation is also very short, since the studied case building community is located in high-latitude region. In summer weeks, it has much larger PV power production, and the period with PV production is also much longer than the winter period. During most time of the day, the hourly PV power production can meet the hourly power demand. In spring and autumn, the PV power production is more equivalent to the power demand.

7.4.2 Analysis of the Detailed Operation for a Typical week

This section presents and analyzes the hourly performances under the charging strategy of minimizing the peak energy exchanges with the grid. The EV charging profiles and the community energy flow in the spring week (see subplot 2 in Fig. 7.3) are presented and analyzed in detail.

Figure 7.4 shows the EV SOC changes in the spring week (see inputs in subplot 2 in Fig. 7.3) under different charging form and EV commuting distance scenarios. The charging limits are set to be 10 kW in both the smart charging forms. In *RC*, the EV battery is charged at the rated charging rate 10 kW once being connected to the charging station. In *SC1*, the EV battery can be charged at the any value within the charging limits, i.e., in the range [0 ∼ 10] kW. In SC2, the EV battery can also discharge power back to the building community; the hourly charging rates are in the range [−10, 10] kW. A negative value indicates discharging power back to the community. Each column of Fig. 7.4 shows the results from one charging form, and each row shows the results from one commuting distance. The red hollow circles represent the arrival state of the EV, including the arrival time and the initial SOC in each day. The black-filled circles represent the departure state of the EV, including the departure time and the target SOC in each day. As can be seen, all the charging controls can meet the charging needs by achieving the target SOC (i.e., 95%) when the EV departures. In RC, the charging SOC

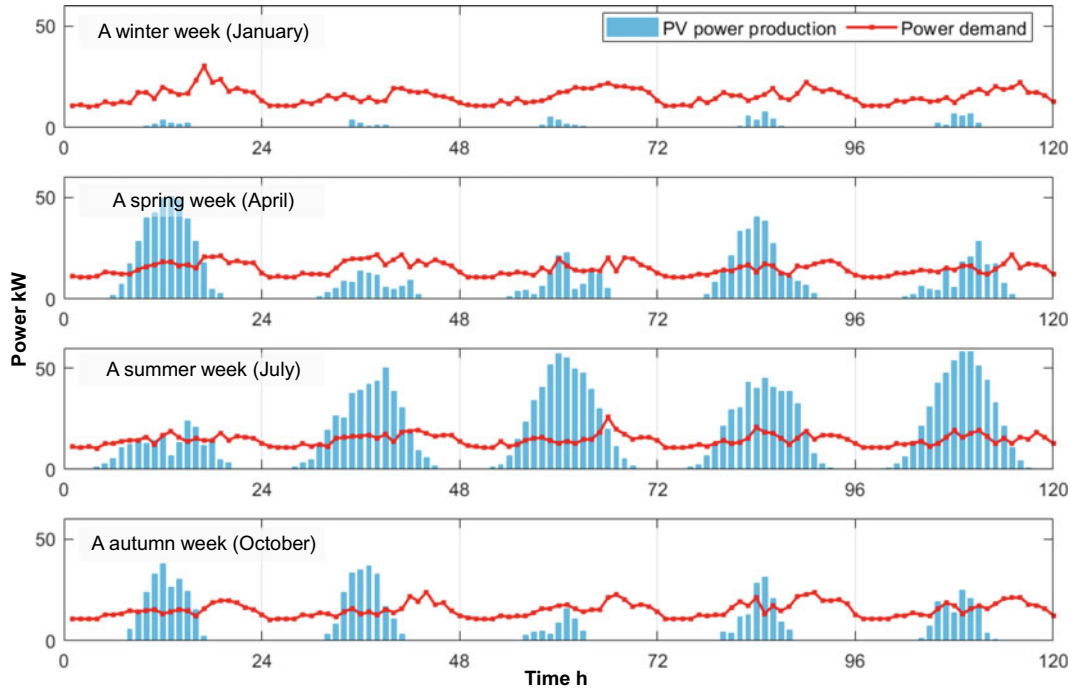

Fig. 7.3 Illustration of the case building community power demand and PV power production in different seasons

increases dramatically soon after the EV's arrival at the charging location. In most cases, the SOC can reach the target within two to three hours. While for SC1, the increase of SOC is more stable than RC, depending on the mismatch between the community power demand and PV power production. For SC2, since the EVs can also discharge power back to the community, the SOC decreases in some hours (see the parts highlighted by dashed circles), but eventually it can reach the target at the departure time. With the increase of commuting distances, the average hourly charging rates increases in SC1. While in SC2, with the increase of commuting distances, the occurrence of EV battery discharging power back to the community decreases. This is because more amount of electricity needs to be charged into the EV battery. Under the same charging limits, the EV battery has less time for discharging power, otherwise, it cannot reach the SOC target at the departure time.

To investigate how each form of charging works, the power flow of the building community, including the EV charging/discharging, is analyzed. Figure 7.5 shows the power flow of the building community in the same spring week (see inputs in subplot 2 in Fig. 7.3) for the three charging forms under the charging limits of 10 kW and medium-short commuting distances (corresponds to the second row in Fig. 7.4). The green bars represent the PV power production in each hour; the orange bars represent the building community power demand, the blue bars represent the EV charging/discharging rates, and the blue dashed curve represents the deviation between the total demand and the total supply, which will be adjusted by the grid, i.e., the power exchanges with the grid. Note in this figure, a positive power value indicates the power usage, while a negative value indicates the power supply. The peak energy exchanges with the grid can either be the large power imports from the grid or large power exports to the grid. In RC, the EVs charging occurs mostly in the first two hours. As a result, in some days, it produced large peak energy exchanges with the grid, see Day-2, Day-3, and Day-5. In SC1, the EV charging loads are scheduled in the best timeslots to minimize the

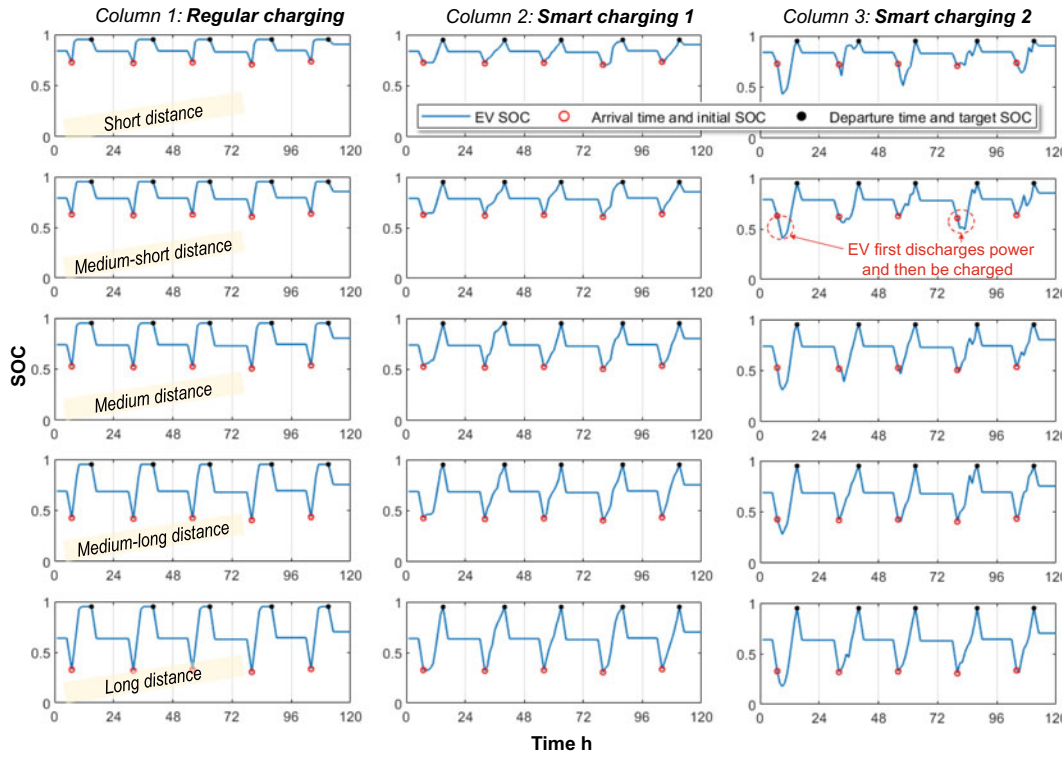

Fig. 7.4 SOC of the EV battery under different scenarios of charging forms and driving distances in the spring week. (The rated charging rate is 10 kW in regular charging. The charging limits are [0 ∼ 10] and [−10 ∼ 10] kW for Smart Charging 1 and Smart Charging 2, respectively.)

peak energy exchanges. For instance, in Day-1 and Day-4, the EV charging loads are planned in the middle of the day, when the PV power production reaches the peak. Thus, it effectively reduced the peak energy exports to the grid. In Day-2 to Day-5, since the PV power production is very limited, the EV charging loads are planned evenly during the whole charging period. SC2 is more effective in reducing the peak energy exchanges with the grid compared to SC1. As can be seen in Day-1, in order to reduce the large PV power export to the grid, the EV first discharges some power to the community in an earlier time, which leads to more storage capacity in a later period with large PV power production. As a result, the peak power production can be further reduced in this charging form.

7.5 Summary

This chapter has investigated the impacts of three EV charging forms, six EV charging limits, and five EV commuting distances on the EV demand response energy performances and EV battery degradations. The three different charging forms considered in this chapter include regular charging (RC), smart charging 1 (i.e., SC1: the EV only can be charged at varying rates), and smart charging 2 (i.e., SC2: the EV can be charged and discharged at varying rates). In total, 65 scenarios have been analyzed and studied. The PSO algorithm has been adopted for control optimization of the EV charging/discharging rates in different scenarios, and the Rainflow

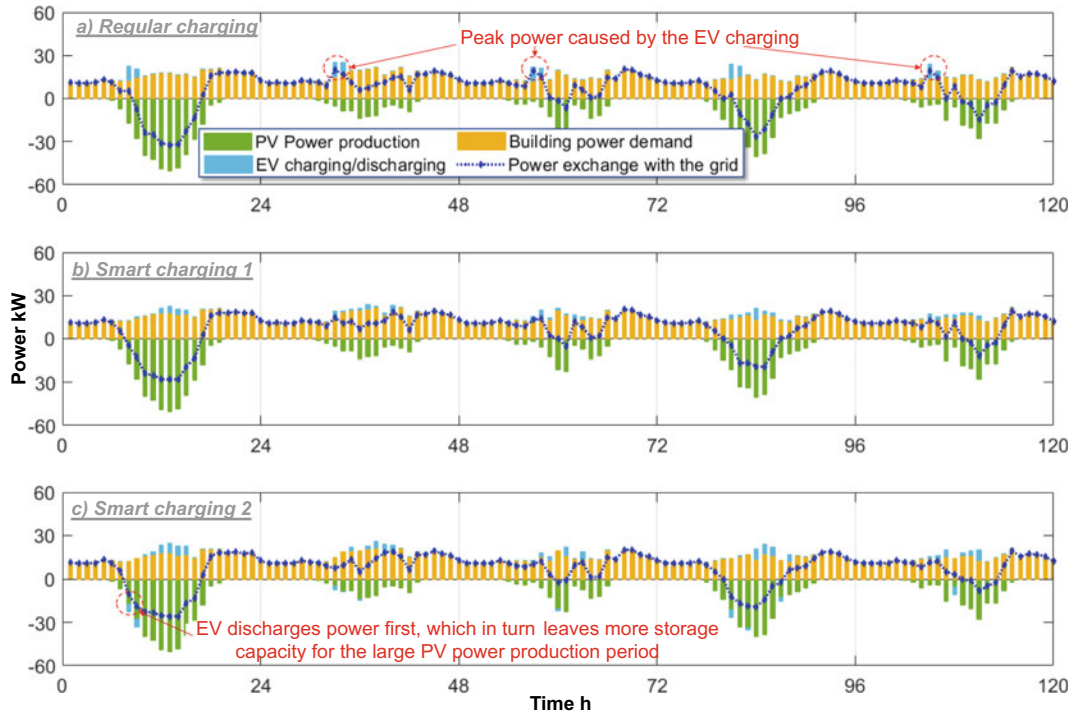

Fig. 7.5 Power flow of the EV battery under different scenarios of charging forms. (The rated charging rate is 10 kW in regular charging. The charging limits are [0, 10] and [-10, 10] kW for Smart Charging 1 and Smart Charging 2, respectively. The driving distance is medium-short.)

cycle counting algorithm has been used to evaluate the degradation of EV battery due to the charging/discharging cycling. The major finds are summarized as follows:

- *Impacts of charging forms:* Regarding the impacts on peak energy exchanges, both the smart charging forms can effectively reduce the peak energy exchanges compared to the regular charging because the smart charging can optimally schedule the charging loads in power 'valley' to avoid large power exchanges with the grid. SC2 is more effective in reducing the peak energy exchanges compared to SC1, since it can discharge power in an earlier period to prepare more storage capacity for peak shaving in later periods. Regarding the impacts on EV battery cycling degradation, SC1 produces the same level of battery degradation as the regular charging, since the ranges of DoD are the same (i.e., from initial SOC to target SOC in each day). SC2 leads to more battery degradation than the other two charging forms since the EV has increased charging-discharging cycles in the charging period. But, if an EV is used for long-distance commuting every day, the increase of battery degradation will become very small in SC2 compared to SC1 (need to brief explanation).
- *Impacts of commuting distances:* Regarding the impacts on peak energy exchanges, with the increase of the commuting distances, the peak energy exchanges increase, since the power demands increase. Regarding the impacts on EV battery degradation, the increased commuting distances would lead to increased EV battery degradation, due to the larger DoD ranges.
- *Impacts of charging limits:* Regarding the impacts on peak energy exchanges, with the increase of charging limits, the ability to reduce the peak energy exchanges increases. This is because the larger charging limits result in larger storage capacities that can

better balance the large surplus PV power production. Regarding the impacts on EV battery degradation, for SC1, the charging limits have limited impacts on the EV battery degradation. While for SC2, with the increase of charging limits, the EV degradation rate increases. This is because the EV has more flexibility in charging and discharging during the parking period.

- Different factors lead to different battery cycling degradations. If the EV owners are concerned about their EV battery degradation, they can participate in either SC1 or SC2 with small charging limits, which will contribute to significant peak energy exchanges shaving (i.e., 9.3 $\sim$ 13%) but will not increase the battery degradation dramatically.

The findings from this chapter can help researchers understand the how various smart EV charging scenarios affect the building community demand response performances and battery degradations, and thus help support decision makers in selecting suitable EV charging strategies in practice. This chapter did not consider the calendar degradation of the EV battery and energy losses. The detailed driving patterns of the EV are also simplified. It is assumed that the EVs are parked in the community within a specified period on workdays, which might affect the degradation slightly. Future work will consider more detailed EV mobility patterns and more detailed EV battery model. The economic benefits of EV smart charging for both the grids and EV owners, which is dependent on the system operating conditions and market mechanisms, are not considered. Future work will evaluate the economic benefits and profit allocations among different stakeholders and considers more grid services.

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8 Three Fleet Smart Charging Categories of Electric Vehicles for the Grid Power Regulation

Pei Huang and Yongjun Sun

Abstract

The use of electric vehicles (EVs) has been on the rise during the past decade, and the number is expected to keep increasing in the future. The large EV charging loads, if not well regulated, will cause great stress on the existing grid infrastructures, as they are not designed to host such large power flow. The use of smart EV charging can be a resource-efficient and cost-effective way to enhance the local power balance, and it can mitigate power issues while avoiding expensive upgrading of the existing infrastructure. In this regard, some studies have developed dedicated EV smart charging controls. Most of the existing EV smart charging controls can be categorized into three approaches according to their optimization principles: individual, bottom-up, and top-down. Until now, systematic comparison and analysis of the different approaches are still lacking. It is still unknown whether a control approach performs better than others and, if yes, why is it so. This chapter aims to fill in such knowledge gaps by conducting a systematic comparison of these three different control approaches and analyzing their performances in depth. A representative control algorithm will be selected from each control approach; then, the selected algorithms will be applied for optimizing EV charging loads in a building community in Sweden. Their power regulation performances will be comparatively investigated in two perspectives: minimize peak power exchanges with the grid and maximize PV power self-consumption. The computational performances are also investigated. The study results show that the top-down coordinated approach is superior to bottom-up coordinated approach and individual independent approach in terms of demand response performances, nevertheless the computational loads are much higher. This may make the convergence difficult. Control strategies also have large impacts on the power regulation performances. This chapter will help pave the way for the developments of more sophisticated control algorithms for EV smart charging in the future.

Keywords

PV · Electric vehicle · Grid power regulation · Top-down coordinated approach · Bottom-up coordinated approach · Individual independent approach

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8.1 Introduction

To reduce the roadside carbon emission and greenhouse gas emissions, many governments and organizations have set targets and policies for promoting the utilization of electric vehicles (EVs) to replace the fossil fuel-based vehicles. For instance, the electric vehicles initiative (EVI), i.e., a multi-governmental policy forum, launched the EV30@30 campaign and set a collective aspirational goal for EVs (excluding two/three-wheelers) to reach 30% sales share by 2030 across all signatory countries (IEA 2021). The European Union has set an ambitious target to have at least 30 million zero-emission vehicles (mostly achieved by EVs and hydrogen-powered vehicles) on its roads by 2030, with a long-term goal of steering countries away from fossil fuel-based transport (economy 2020). The Norwegian Parliament has set a national target that all new cars sold by 2025 should produce zero emission (Association 2021). The U.S. Federal government has enacted policies and legislations to incentivize the investments in infrastructure supporting EVs, such as competitive programs and increase of tax credits (United States International Energy Agency 2010). In response to these targets and policies, a boost of the EV deployment has been witnessed in the past decades. The market statistics still showed strong momentum in the EV deployment despite the Covid pandemic. The number of EVs on the roads worldwide reached 10 million by the end of 2020 (IEA 2021). In Sweden, there was a 310% increase in the full battery EV number in 2020 (Kane 2021). With the large number of EV deployment, the massive EV charging loads are expected to impose great stress on the existing grid infrastructures.

On the other hand, the deployments of renewable energy systems (RES), such as photovoltaic (PV) systems and small wind turbines, are also increasing in Nordic countries. The market statistics in 2019 reported an 88% increase of distributed PV system capacity in Sweden and a 100% increase of the distributed PV system capacity in Norway (programme 2019). The intermittent renewable energy, if not well regulated, will also cause severe problems to the grid, such as overvoltage and frequency fluctuation. The large increase of EV numbers and intermittent PV capacity poses new challenges to the society; as in many countries, the existing infrastructures are not designed to host the large shares of new EV loads and intermittent PV power feed-in (Fachrizal et al. 2021). As a result, problems such as the voltage deviations and overloading of components can arise. In some existing urban areas, the local grid operators have to impose strict limits on the power flow, which severely affects the resilience of the built environment (e.g., no power for cooking or lighting) (IEA 2018).

To mitigate these problems and avoid the expensive upgrading of the existing power infrastructure, a resource-efficient and cost-effective way is to use the smart EV charging (i.e., charge EV in the periods with large renewable power production) to enhance the local power balance and electrical energy matching (Chaouachi et al. 2016). In other words, by properly scheduling the EV charging loads, the EV battery can be used as flexible energy storage to help regulate the electricity demands to match the renewable power production in the electric grid and thus improve the power supply quality (e.g., frequency regulation). Existing studies have proposed a number of smart EV charging controls for such application (Mohamed et al. 2020). A comprehensive review of the various EV smart charging schemes, including the objectives, configurations, and control algorithms, is provided in (Fachrizal et al. 2020). According to the optimization logic, most of these controls can be roughly categories into three approaches: individual independent approach, bottom-up coordinated approach, and top-down coordinated approach (Huang et al. 2020b). Both the bottom-up and top-down coordinated approaches are coordinated controls, which typically considers the optimization of multiple EVs' charging loads and takes the aggregated-level performance optimization as the optimization target. The top-

down coordinated approach typically conducts coordination from the aggregated-level and optimize multiple EVs' charging load simultaneously, while the bottom-up coordinated approach typically optimizes single EV's charging load sequentially to achieve the aggregated-level optimum. The individual independent approach typically considers the optimization of only one EV's charging load and takes the single-level performance optimization as the target. Details about each approach are presented in Sect. 8.2.1. In the existing studies, the bottom-up coordinated approach is sometimes called decentralized approach, while the top-down coordinated approach is sometimes termed centralized approach (Mou et al. 2015). These two approaches are also called global scheduling optimization and local scheduling optimization in (He et al. 2012). In some studies, the individual independent approach is also called 'non-cooperative' scenario, and the two coordinated approaches are called 'cooperative scenario' (Hu et al. 2016).

Regarding individual independent approach, Cai et al. (2019) developed an aging-aware model predictive control method. In their method, a convex battery capacity loss model is first built from a physical-based degradation model to capture the battery aging cost. Based on the developed model, an aging-aware model predictive control approach is proposed, which considers the battery capacity degradation and its negative impacts on the costs, for optimized operation of sustainable buildings with on-site PV and battery systems. Dallinger and Wietschel (2012) proposed a stochastic model to determine the vehicle mobility behavior, an optimization model to minimize vehicle charging costs, and an agent-based electricity market equilibrium model to estimate variable electricity prices. The variable electricity prices are calculated based on marginal generation costs. In the proposed method, the optimizer calculates a cost-minimized charging schedule of EVs based on the price signals and mobility behavior (e.g., state of charge (SOC) and charging periods). Such control is able to balance the fluctuation of renewable energy sources in the grid.

Regarding the bottom-up coordinated approach, in (Fachrizal and Munkhammar 2020), a centralized EV fleet charging control method is proposed for a residential building cluster, which optimizes the charging loads of EVs sequentially one by one based on the arrival time and departure time considering the interaction of individual EVs and PV power production. The control aims at minimizing the building community-level power demand supply mismatch. Similarly, in Geth et al. (2010), a coordinated charging control for a number of EVs is proposed. In this control, a vehicle owner first indicates the point in time when the batteries should be fully charged. Then, the aggregator collects this information and calculates when each EV can start charging. The charging is optimized to be scheduled to the most economical period when the total demand (including the residential, industrial, and EV consumption) is low. Case study shows that the coordinated charging can effectively decrease the peak load, as the coordination makes the aggregated-level charging load profile much flatter. Usman et al. (2016) also proposed an automated coordinated control mechanism among EV fleet and the grid operator that plans a charging strategy for electric vehicles while sustaining the grid capacity constraints. Such mechanism adopts a grid agent to grant tokens to EVs in the idle state for starting an optimization. By shifting charging loads to low electricity price period (usually with low aggregated electricity demands in the power grid), this control can effectively schedule the charging load at the cheaper moments while fulfilling the future travel needs. Similarly, Dorokhova et al. (2021) developed a deep reinforcement learning control of EV charging for coordinating the charging loads of multiple EVs in a sequential way.

Regarding the top-down coordinated approach, Islam et al. (2019) developed a correlated probabilistic model for EV charging loads. In this correlated probabilistic model, correlated samples are first generated, which contain various random variables associated with EVs, battery energy storage, and PV. The generated samples are subsequently incorporated in the coordinated

EV charging method to maximize quality of service while minimizing probability of voltage and current noncompliance. Nayak et al. (2019) developed an adaptive optimal controller of EV fleets for power grid frequency compensation and regulation during the generation-load imbalance while ongoing restoration process. The proposed method leverages the bidirectional power flow capabilities of EVs. It can provide the EVs with optimized charging/discharging solutions when no system information is available at controller end. Notably, van der Kam and van Sark (2015) developed a linear programming-based optimization control of EV fleets for maximize the PV power self-consumption rates. The proposed method considers the bidirectional power flow of EVs (i.e., discharging power back to the grid is also allowed) and optimizes the charging/discharging rates of all EVs simultaneously. They also compared this method with three simple charging scenarios: no control; real-time charging based on PV power sufficiency; real-time charging based on PV power sufficiency and with vehicle-to-grid enabled. Their study shows that coordinated EV controls can achieve much higher self-consumption rates and meanwhile reduce the peak electricity demand compared to the other scenarios.

Although existing studies have developed different control approaches for implementing smart EV charging, there is a lack of a systematic classification and comparison of these different approaches. As a result, researchers do not fully understand how each approach works and what their advantages and disadvantages are. This can affect the decision-making in selecting optimization approaches, as well as the optimization performances and computation complexity, especially when a large number of EVs are considered. Therefore, this chapter conducts an overview of different control approaches, systematically compares their advantages and disadvantages, and analyzes their performances in depth (see Sect. 8.2.1). The purpose is to have a deep understanding of the various control approaches, which will pave way for the resource-efficient integration of the large EV load and PV power penetration in the future. In this chapter, a representative control algorithm will be selected for each control approach. The different control approaches are then compared based on a case building community in Sweden, and their performances are compared and analyzed systematically. Note in this chapter, the control approaches refer to the control logic (i.e., how optimization is conducted and achieved), while control algorithms refer to the detailed control implementation. Several different controls algorithms can belong to a same control approach. The following contributions are made in this chapter:

- Comparative studies of three control approaches (i.e., individual independent approach, bottom-up coordinated approach, and top-down coordinated approach) for EV smart charging.
- Investigation of how various optimization objectives (i.e., minimize peak power exchanges with the grid and maximize renewable energy self-utilization) affect the performance of different control approaches.
- In-depth analysis of the advantages and disadvantages of different control approaches as well as the reasons behind the performance differences.

The structure of the paper is as followings. In Sect. 8.2, different approaches of EV fleet charging are presented, and a control algorithm is selected and introduced for each approach. Section 8.3 presents the detailed building model and energy system models for testing each control algorithm. In Sect. 8.4, the different control approaches are applied on a case building community located in Sweden, and the detailed performances under the two control strategies are presented. The conclusions are given in Sect. 8.5.

8.2 Control Approaches of EV Fleets

This section first introduces the basic idea of different control approaches. Then, one control method will be proposed for each control

approach, and the detailed implementation of each control method will be described.

8.2.1 Basic Idea of Different Control Approaches

Figure 8.1 presents the schematics of three typical EV fleet control approaches for balancing power in the building sector: individual independent approach, bottom-up coordinated approach, and top-down coordinated approach. Both the bottom-up and top-down approaches are coordinated controls, which take the aggregated-level performance optimization as the target while the individual independent approach takes the single-level performance optimization as the target. In the individual independent approach, the optimization of individual EV's charging rates is conducted independently considering neither the other EVs' charging load nor the aggregated one (see Fig. 8.1a). As a result, different EVs may charge at a same period, leading to the 'avalanche effect (Kühnbach et al. 2021)' (i.e., creating new peak power loads for the power grid). In the bottom-up coordinated approach, the charging rates of individual EVs are optimized one by one in a sequence, and the optimization of each single EV's operation is performed based on the aggregated results of the earlier optimized EVs (see Fig. 8.1b). The optimization is based on the arrival time. Since the bottom-up coordinated approach takes the aggregated-level performances as the optimization target, the performances at the aggregated-level are superior to the individual independent approach. However, due to a lack of global coordination, the optimization from bottom-up coordinated approaches may not produce the global optimum solution. Similar to the bottom-up coordinated approach, the top-down coordinated approach takes the aggregated-level performance directly as the optimization objective. The charging rates of individual EVs are coordinated simultaneously to achieve the optimized performance at the aggregated level (see Fig. 8.1c). Since all the EVs are optimized at the same time, the computational load could be much higher compared to the other two approaches. In some existing controls (Usman et al. 2016), to reduce the number of variables to be optimized, the charging rates of EVs are fixed. The optimizer only needs to search for the charging time slots and thus reduces associated computational complexity. With fixed charging rates, the optimization performances might not be the optimal. Table 8.1 summarizes the three different approaches and compares their pros and cons.

8.2.2 Representative Control Algorithm for Each Approach

This section presents the selected EV charging control for each approach, including the control logic and detailed mathematical modeling. Note this chapter only considers the unidirectional power flow, i.e., the G2V (grid to vehicle) mode. The EVs can only be charged. The bidirectional power flow, i.e., G2V2G (i.e., grid to vehicle to grid) is not considered.

8.2.2.1 Control Algorithm for Individual Control

Figure 8.2 presents the selected control method for the individual independent approach. This control is implemented for each individual EV considering the single building's (i.e., the building whether the EV gets power from) power demand and renewable power production.

Considering the computation speed and ability to find optimal solutions, genetic algorithm (GA), an evolutionary optimization algorithm, is used in the control. The GA reflects the process of natural selection where the fittest individuals are selected for reproduction to produce offspring of the next generation. In each iteration, a set of alternatives (in this chapter, the alternatives refer to EV charging loads) will be produced by the GA optimizer. These alternatives will then be imported into the building and PV system model for calculating the power exchanges with the grid in each timestep. Then, the power exchanges in each timestep will be imported into the fitness

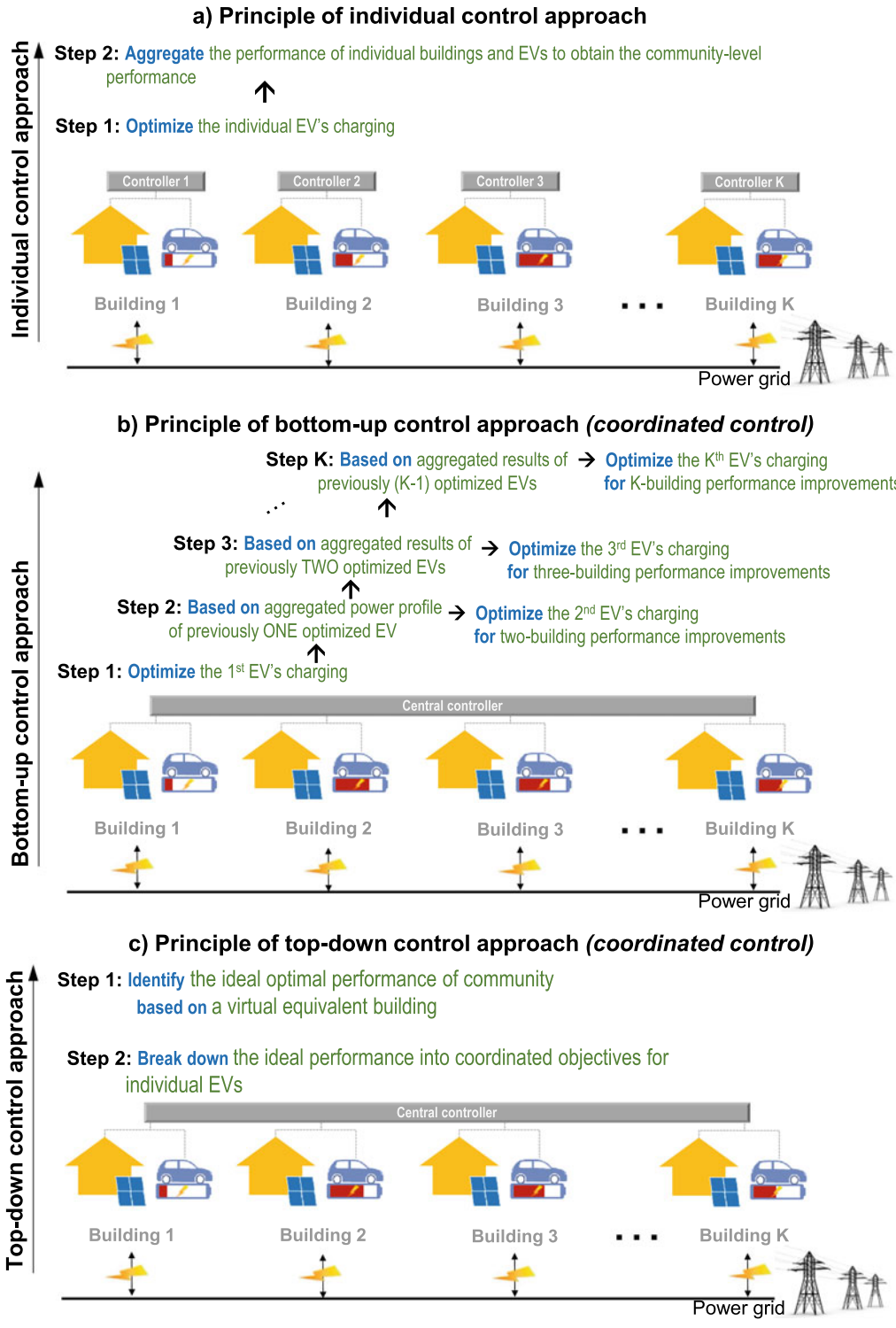

Fig. 8.1 Schematics of three different control approaches for EV fleets in the building sector (*Note* in the top-down and bottom-up control approaches, the optimization is at community level, instead of single building level. So the optimization is done considering the community-level performance)

Table 8.1 Comparison of different EV fleet charging approaches

Category	Principal	Pros	Cons
Individual independent approach (Cai et al. 2019; Dallinger and Wietschel 2012)	The optimization of individual EV charging rates is conducted separately without considering the other EVs' charging load. Then, the individual EV charging loads are aggregated to obtain the aggregated-level performance	Easy to implement	1. The aggregated-level performance is not optimized 2. New demand peaks may occur when individual EVs take the same action (e.g., shift demand to the same period)
Bottom-up coordinated approach (Fachrizal and Munkhammar 2020; Usman et al. 2016)	The charging rates of individual EVs are optimized one by one in a sequence, and the optimization of each single EV's operation is performed based on the aggregated results of the earlier optimized EVs	Better performance at aggregated level than individual independent approach	1. Due to a lack of global coordination, the optimization from bottom-up coordinated approaches may not produce the global optimal solution 2. High computational load
Top-down coordinated approach (Shariful Islam et al. 2019; Nayak et al. 2019; van der Kam and van Sark 2015; Huang et al. 2020a; Weckx and Driesen 2015)	The aggregated-level performance is used as the optimization objective. The charging rates of individual EVs are coordinated to achieve the best performances at the aggregated level	Optimized performance at aggregated level	1. It requires the controller to know/forecast the expected future arrival time in the building for all EVs, as the scheduling also considers the EVs that have not arrived yet 2. The computational complexity increases with the number of EVs

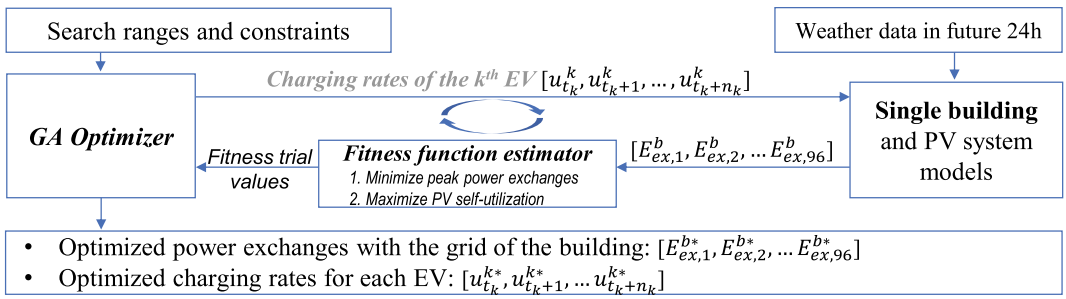

Fig. 8.2 Selected optimization logic for the individual control approach

function to evaluate the fitness values of the alternatives. The alternatives with good fitness values will be selected and used for producing the alternatives in the next generation. The iteration will continue until the optimal solutions are found. Since there are existing toolbox available for implementing the GA optimization in Matlab and Python, this chapter will not go into details about the mathematics of GA. For more details about the GA, please refer to (Yang 2021).

The detailed mathematical modeling is introduced below. In each timestep, the charging rates (u^k (kW), $0 < u^k < u^k_{\text{limit}}$, in which u^k_{limit} (kW) is the maximum charging rate) of the kth EV should meet the following two constraints:

$$0 \le \mathrm{SOC}_{0,k} \times \mathrm{CAP}_k + \left(u_{t_k}^k + u_{t_k+1}^k + \cdots + u_{t_k+i}^k\right) \times \tau \le \mathrm{CAP}_k \quad \text{where } i = 1, 2, \ldots, n_k, \tag{8.1}$$

where $\mathrm{SOC}_{0,k}$ (%) is the initial state of charge when the kth EV arrives at the charging port; CAP_k (kW·h) is the capacity of the kth EV battery; t_k is the arrival time of the kth EV at the charging port; n_k is the parking duration; and τ is the simulation interval (15 min in this chapter).

In order to meet the traveling needs of the next trip, the EV battery should be charged to a user-specified level ($\mathrm{SOC}_{1,k}$) before they depart the charging port. This constraint is expressed by the equation below,

$$\mathrm{SOC}_{0,k} \times \mathrm{CAP}_k + \left(u_{t_k}^k + u_{t_k+1}^k + \cdots + u_{t_k+n_k}^k\right) \times \tau \ge \mathrm{SOC}_{1,k} \times \mathrm{CAP}_k. \tag{8.2}$$

In Eq. (8.2), when $\mathrm{SOC}_{1,k}$ equals 1, it represents that the EV users require the EV battery to be fully charged before they depart the charging port. This chapter considers two strategies. The first strategy aims to minimize the grid power interaction. In this strategy, minimizing the peak power exchanges with the grid is used as the optimization target. Following this control goal, a fitness function is determined, as expressed by (Salom et al. 2011)

$$J_{\mathrm{grid}} = \min\left(E_{\mathrm{peak}}^b\right), \tag{8.3}$$

where E_{peak}^b (kW) is the peak power exchange of the building with the grid. E_{peak}^b (kW) in Eq. (8.3) is calculated by the two equations below.

$$E_{\mathrm{ex},i}^b = E_{d,i}^b + u_i^k - E_{s,i}^b. \tag{8.4}$$

$$E_{\mathrm{peak}}^b = \max\left(\left|E_{\mathrm{ex},1}^b\right|, \left|E_{\mathrm{ex},2}^b\right|, \ldots \left|E_{\mathrm{ex},96}^b\right|\right). \tag{8.5}$$

In Eq. (8.4), $E_{d,i}^b$ (kW) and $E_{s,i}^b$ (kW) are the power demand and renewable power supply of the individual building in each simulation interval (i.e., 15 min), respectively. u_i^k (kW) is the charging load of the kth EV parked in the building in the ith simulation step. $E_{\mathrm{ex},i}^b$ (kW) is the power exchange with the grid of the building in each simulation interval considering the EV charging load. Note there are 96 simulation steps in each day (96 = 24 h × 4 simulation steps/hour), as the simulation timestep is 15 min (see Eq. (8.5)).

The second strategy aims to maximize the renewable energy self-utilization rate, as described by (Huang et al. 2019)

$$J_{\mathrm{self-utilization}} = \max\left(\sum_{i=1}^{96} E_{\mathrm{onsite},i}^b / \sum_{i=1}^{96} E_{d,i}^b\right), \tag{8.6}$$

where $E_{\mathrm{onsite},i}^b$ (kW) is the amount of load covered by the power generated on-site in the ith simulation step. It is calculated by

$$E_{\mathrm{onsite},i}^b = \min\left(E_{s,i}^b, E_{d,i}^b + u_i^k\right). \tag{8.7}$$

The outputs of GA search are the optimal charging loads ($[u_{t_k}^{k*}, u_{t_k+1}^{k*}, \ldots u_{t_k+n_k}^{k*}]$ (kW)) for the kth EV to meet the control targets.

8.2.2.2 Control Algorithm for Bottom-Up Control

Figure 8.3 presents the selected control method for the bottom-up coordinated approach. This control is implemented for each individual EV in a sequence considering the aggregated-level power demand and renewable power production. Again, the GA algorithm will be used in the control optimization. The detailed mathematical modeling is introduced below.

In the bottom-up coordinated approach, the optimization is based on the aggregated-level power demand and PV power production. The power demand ($E_{d,i}^c$ (kW)) and PV power production ($E_{s,i}^c$ (kW)) of the community in the ith simulation step are calculated by aggregating single building's power demand and PV power production.

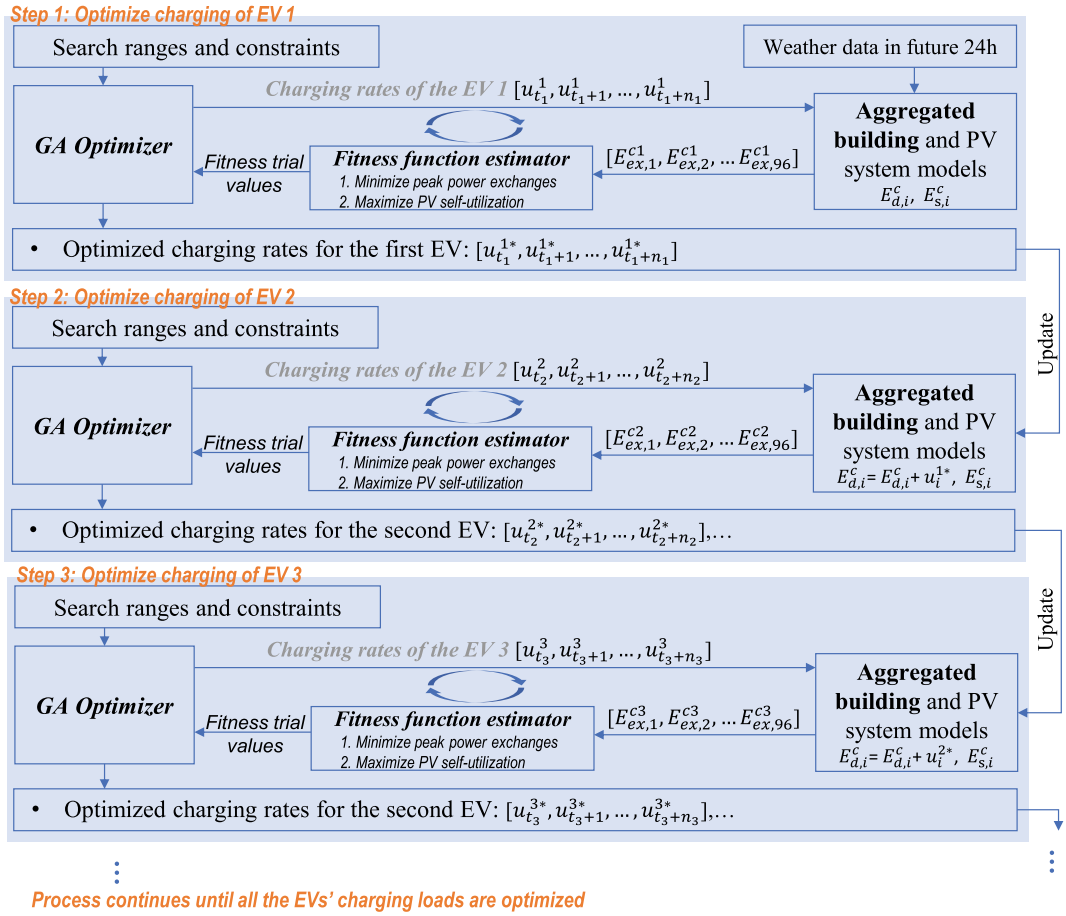

Fig. 8.3 Selected optimization logic for the bottom-up control approach

$$E^c_{d,i} = \sum_{j=1}^{n} E^{b,j}_{d,i}. \tag{8.8}$$

$$E^c_{s,i} = \sum_{j=1}^{n} E^{b,j}_{s,i}. \tag{8.9}$$

Equations (8.8) and (8.9) show the calculation of the aggregated-level power demand and PV power production of all the n buildings. The peak power exchange of the whole building community with the grid E^c_{peak} (kW) is calculated by

$$E^c_{\text{peak}} = \max\left(\left|E^c_{\text{ex},1}\right|, \left|E^c_{\text{ex},2}\right|, \dots \left|E^c_{\text{ex},96}\right|\right), \tag{8.10}$$

where $E^c_{\text{ex},i}$ (kW) is the power exchange with the grid of the whole building community considering the EV charging load in the ith simulation step. In each iteration, the calculation of $E^c_{\text{ex},i}$ is updated by adding the previously optimized EV charging load.

1st step—optimize EV 1:

$$E^c_{\text{ex},i} = E^c_{d,i} - E^b_{s,i}. \tag{8.11}$$

2nd step—optimize EV 2:

$$E^c_{\text{ex},i} = E^c_{d,i} + u^{1*}_i - E^b_{s,i}. \tag{8.12}$$

3rd step—optimize EV 3:

$$E^c_{\mathrm{ex},i} = E^c_{d,i} + \left(u_i^{1*} + u_i^{2*}\right) - E^b_{s,i}. \tag{8.13}$$

$$\vdots$$

nth step—optimize EV n:

$$E^c_{\mathrm{ex},i} = E^c_{d,i} + \left(u_i^{1*} + u_i^{2*} + \cdots + u_i^{n*}\right) - E^b_{s,i}. \tag{8.14}$$

Equations (8.11–8.14) show the calculating of the power exchange with the grid ($E^c_{\mathrm{ex},i}$ (kW)) in each iteration. The fitness function for the first optimization strategy (i.e., minimize the peak power exchanges with the grid) is depicted by

$$J_{\mathrm{grid}} = \min\left(E^c_{\mathrm{peak}}\right) \tag{8.15}$$

The fitness function for the second optimization strategy (i.e., maximize the PV power self-utilization) is depicted by

$$J_{\mathrm{self-utilization}} = \max\left(\sum_{i=1}^{96} E^c_{\mathrm{onsite},i} / \sum_{i=1}^{96} E^c_{d,i}\right). \tag{8.16}$$

The process will continue until all the EVs being optimized in the building community.

8.2.2.3 Control Algorithm for Top-Down Control

Figure 8.4 presents the selected control method for the top-down coordinated approach. This control is implemented for all the EV simultaneously considering the building-aggregated-level power demand and renewable power production. Again, the GA algorithm will be used in the control optimization.

The fitness functions for the two control strategies are the same as the bottom-up coordinated approach, as depicted by Eqs. (8.10) and (8.16) while the calculation of the power exchange with the grid of the building community ($E^c_{\mathrm{ex},i}$ (kW)) in each timestep should consider all the EV charging load, see the equation below,

$$E^c_{\mathrm{ex},i} = E^c_{d,i} + \left(u_i^1 + u_i^2 + u_i^3 + \ldots\right) - E^b_{s,i}. \tag{8.17}$$

The top-down control requires the controller to know/forecast the expected future arrival time in the building for all EVs, as the scheduling also considers the EVs that has not arrived yet. It then produces the optimal charging loads of all the EVs in one simulation. The outputs are the optimized charging loads of all the EVs $[u_{t_1}^{1*}, u_{t_1+1}^{1*}, \ldots, u_{t_1+n_1}^{1*}]$, $[u_{t_2}^{2*}, u_{t_2+1}^{2*}, \ldots, u_{t_2+n_2}^{2*}]$, …

This chapter will compare the above-mentioned three control algorithms, as well as a benchmark scenario without any smart EV charging control, i.e., the EVs are charged at rated charging rate once they are plugged into the charging station, as mostly used nowadays.

8.3 Buildings and System Modeling

This section presents the building information, energy demand, and renewable energy system modeling. Each building is installed with a renewable energy system (i.e., PV panels). One EV is assumed to be parked and charged in one building. Considering the traveling patterns, only the workday in a typical week is selected for the analysis.

8.3.1 Building Modeling

A building community located in Ludvika, Dalarna region of Sweden, is used for analysis in this chapter. There are three separate buildings in the studied building community. The total façade surface gross area of the community is 2146 m^2; the total roof surface gross area is 1750 m^2. The heating is provided by municipality district heating system. So, the PV panels will only need to meet the power needs of the electricity devices (including lighting, computers, and EVs). For more details about the studied building community, please refer to (Huang

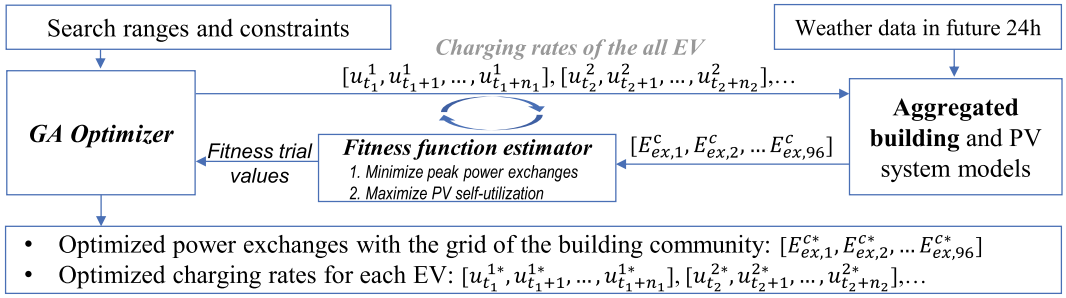

Fig. 8.4 Selected optimization logic for the top-down control approach

et al. 2021) and the EnergyMatching project (Matching 2021). The electrical demand profiles of the building community were produced using the stochastic model developed by Widén and Wäckelgård (Widén and Wäckelgård 2010). This model was derived based on the real occupancy schedules and lighting usage in Swedish scenario. Meanwhile, to achieve acceptable accuracy, the measured data about the annual electricity usage were used to calibrate the model.

8.3.2 Renewable Energy System Modeling

The power generation from the PV panel P_{PV} (kW) is calculated by (Huang et al. 2018; Klein et al. 2004)

$$P_{\mathrm{PV}} = \gamma \times I_{\mathrm{AM}} \times I_T \times \eta \times \mathrm{CAP}_{\mathrm{PV}}, \quad (8.18)$$

where γ *(dimensionless)* is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} *(dimensionless)* is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T $(\mathrm{W/m^2})$ is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; $\mathrm{CAP}_{\mathrm{PV}}$ $(\mathrm{m^2})$ is the PV surface area.

8.3.3 EV Battery Modeling

A simplified EV battery model is used in this chapter, as has also been used for EV charging modeling in (Shepero and Munkhammar 2018).

$$E_{\mathrm{EV},k,i} = \mathrm{SOC}_{0k} \times \mathrm{CAP}_k + \left(u_{t_k}^k + u_{t_k+1}^k + \cdots + u_{t_k+i}^k\right) \times \tau, \quad (8.19)$$

The electricity stored ($E_{\mathrm{EV},k,i}$ (kW·h)) in the battery of the kth EV is calculated by Eq. (8.19). SOC_{0k} is the initial state of charge of the kth EV. CAP_k is the capacity of the kth EV battery. $u_{t_k}^k$ (kW) is the charging rates in each simulation step, and τ is the charging duration (15 min).

This chapter considers three EVs. The capacity, maximum charging rates as well as the parking periods of each EV are summarized in Table 8.3 in case studies. EV 1, EV 2, and EV 3 are assumed to be charged in Building A, B, and C, respectively. The arrival and departure time of the three EVs were decided from statistical distributions with reference to the Swedish Travel Survey (Analysis 2007). The EV battery capacity and maximum charging rates are referred from the available EV models in the market in (Ustun et al. 2013). In all the three scenarios, the EVs are required to achieve 95% SOC before they leave the charging station (i.e., before departure time). When the EVs arrive in a charging station in the building community, the initial SOC (between 0 and 1) is calculated based on SOC in the previous departure time and a random traveling distance. Note in each day, a small random variance (within 45 min) was added to the arrival time and departure time considering the uncertainty in traveling patterns.

8.4 Case Studies and Results Analysis

This section first presents the individual building's electricity demand, renewable energy production, and the EV information. Then, the performances of the different charging approaches under the two optimization strategies (i.e., minimize the peak power exchanges with the grid and maximize the PV power self-utilization) are presented. An overall comparison of the different control approaches is given and discussed in the end.

In the case studies, a summer week was selected to test the different control approaches. Considering the different driving patterns in workdays and weekends (the EVs might not be parked in the community during weekend daytime), this chapter only considers the five workdays for simplicity. The weather data of Ludvika were used for modeling the local PV power productions. Table 8.2 summarizes the input parameters used in the case studies. Table 8.3 summarizes the EV information used for the simulation. Considering the uncertainty in the EV usage (e.g., the EVs could be used more in some days), the initial SOCs in each day are set different. The time interval of simulation was set as 15 min with the purpose of improved accuracy.

8.4.1 Building Electricity Demand, Renewable Generation, and Electricity Mismatch

Figure 8.5 displays the power demand, PV power generation, and electricity mismatch in each timestep (i.e., 15 min interval) during the workdays in a typical summer week for the three buildings. The trends of PV power production of the three buildings are similar, since the solar irradiation is nearly the same for the three buildings. As Building C has the largest roof area, more PV panels can be installed on its roof. Thus, it has the largest average PV production. Power mismatch of each building is calculated as the deviation between its power demand and renewable generation in each timestep. A positive value of power mismatch indicates insufficient renewable generation (and thus grid power is needed), while a negative value of power mismatch indicates excessive renewable generation (and thus selling electricity to the grid is needed).

Note this chapter only conducts simulation for five days. This is because of two reasons: (1) One week simulation can already reveal the characteristics of each control approach. (2) The computational load will increase exponentially if the simulation is conducted for a full year. To increase the optimization performance, this chapter used a simulation timestep of 15 min. In other words, there are 4 variables to be optimized in each hour. In the optimization, the average number of variables to be optimized for each EV is 32 (i.e., 8 h parking period on average in each day). Considering the large number of variables to be optimized, in the GA optimization, the number of generations is set as 8000, and the population size is set as 10,000.

8.4.2 Performances Comparative Investigation Under Objective of Minimizing Peak Power Exchanges with the Grid

This section presents the performance comparison of the three control approaches under the strategy to minimize the peak power exchanges

Table 8.2 Configuration of the PV system (Huang et al. 2020a)

Input parameter	Value
Area of PV panel in Building A (m^2)	100
Area of PV panel in Building B (m^2)	200
Area of PV panel in Building C (m^2)	300

Table 8.3 Configuration of EVs for the simulation in the workdays of the selected week (Ustun et al., 2013)

	Capacity (kW·h)	Charging limit	Day	1	2	3	4	5
EV 1	53	4 kW per time step (15 min)	Arrival time	06:39	06:19	06:10	05:58	05:30
			Departure time	13:54	14:24	13:50	13:50	14:12
			Initial SOC	49%	48%	38%	36%	39%
EV 2	39	4 kW per time step (15 min)	Arrival time	07:47	08:04	07:42	07:55	07:46
			Departure time	15:59	16:29	16:07	16:12	16:23
			Initial SOC	45%	43%	31%	33%	32%
EV 3	32	4 kW per time step (15 min)	Arrival time	10:11	10:16	10:10	09:46	10:47
			Departure time	18:59	18:48	19:00	18:50	19:19
			Initial SOC	42%	42%	31%	31%	27%

Note the EV charging rates will vary within the charging limits

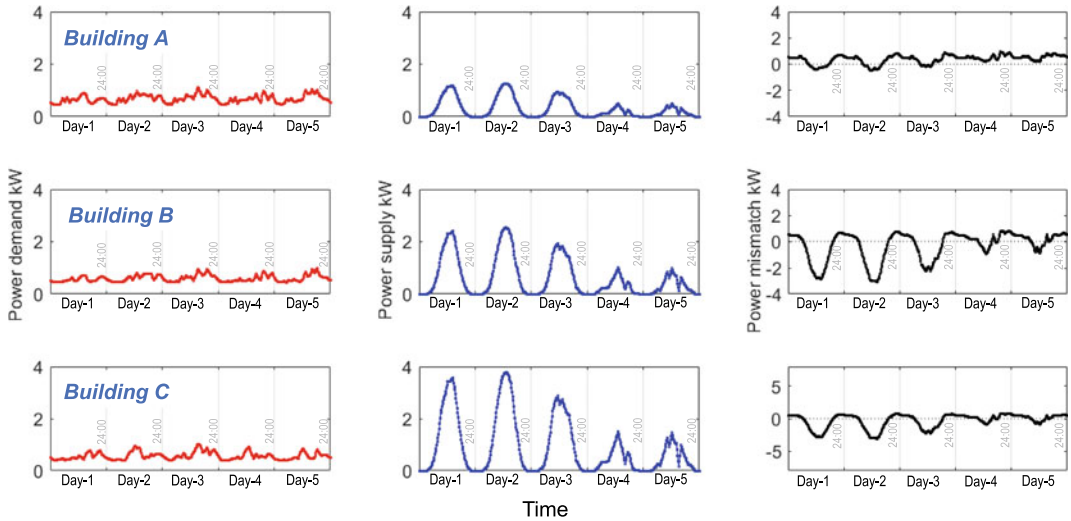

Fig. 8.5 Power demand, PV power generation, and power mismatch in each timestep during the weekdays in the selected summer week for the three buildings (*Note* the analysis is conducted with a timestep of 15 min only for the five workdays, not for a full year)

with the grid. Figure 8.6 compares the energy flow of the whole building community under different EV charging approaches. The bars represent the power flow of PV system, building demand, and EVs, respectively. And the blue curve with marks represents the power mismatch, which is deviation between the total power demand and total power production. The power mismatch will be adjusted by the power grid and thus can be interpreted as the energy exchanges with the grid. When EV charging loads are not included, in Day-1 to Day-3, since the total PV power production is much larger than the community electricity demand, the peak power exchanges are caused by the large renewable power exports to the grid. While in Day-4 and Day-5, the peak power exchanges are caused by the large power demand due to the insufficient PV power production.

Figure 8.6(a) is the case when there is no EV smart charging. Since the EVs are charged immediately at the rated charging rates after they are connected to the charging ports, large EV charging loads are observed near the arrival time

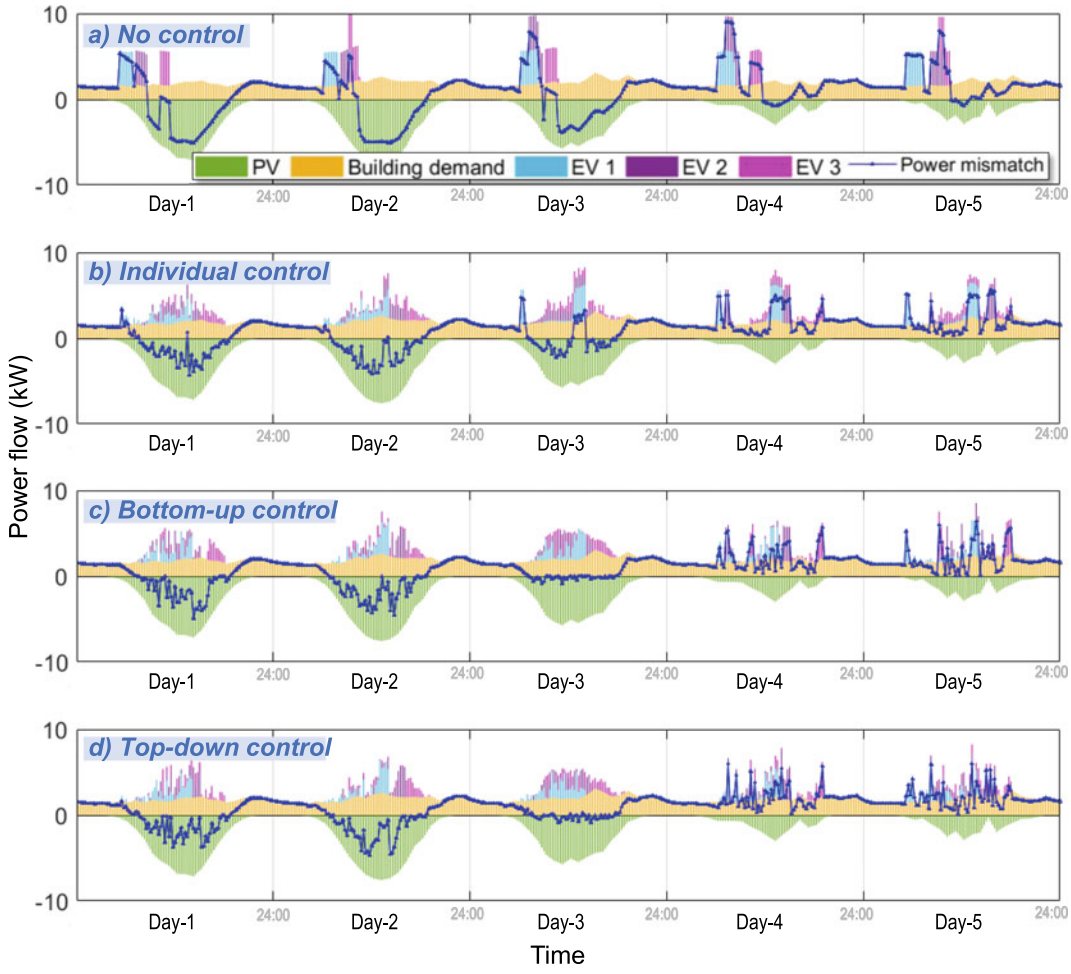

Fig. 8.6 Power flow of the building community and EVs under different EV control approaches for the goal of minimizing peak power exchanges with the grid (with a simulation timestep of 15 min)

of each EV (e.g., see Day-1 to Day-5). While when there are multiple EVs being charged simultaneously, large peak demands will be produced (e.g., see the large peak loads in Day-3 and Day-4). As can be seen in Day-1, Day-2, and Day-3, there are both large PV power exports to the grid and large demand penetration on the grid. Figure 8.6b shows the performance under the individual control approach. Compared to the no EV smart charging case, the large charging loads are more evenly distributed in periods with large PV power production, and thus, the large power demands are avoided. As can be seen in Day-1, Day-2, and Day-3, the peak power exchanges are caused by the PV power exports. The large power demands in Day-4 and Day-5 are also significantly shaved. Since there is a lack of coordination among different EVs' charging loads, there are large fluctuations in the power exchanges (e.g., see the large PV power exports in Day-1 and Day-2). Figure 8.6c presents the performance under the bottom-up control approach. The bottom-up control implements coordination of the EV charging loads, i.e., shift the charging load of an EV to avoid overlapping with the charging load of previously optimized EVs, and thus, the peak power exchanges are reduced compared to the individual control approach. The fluctuation of peak power exchanges is also smaller (e.g., see the large PV power exports in Day-1 and Day-2).

Figure 8.6d shows the performance under the top-down control approach. There are obvious limits of the peak power exchanges with the grid. As can be seen in Day-1 and Day-2, the peak energy exports are well limited to specific values in a long period, and in Day-4, the peak power demand is well limited to a specific value during the daytime. This is because the top-down control implements a global coordination of all the EVs' charging load and takes the community-level aggregated performances directly as the control target. It is capable of identifying the global optimum.

To have a more straightforward look into the performances of different control approaches, the daily peak power exchanges with the grid are calculated and compared, as shown in Fig. 8.7. The performance improvements (i.e., the reduction in the peak power exchanges) compared with the benchmark case (i.e., no smart charging) are also calculated in each day and presented in Fig. 8.7. As can be seen, all the three controls can significantly reduce the daily peak power exchanges with the grid. The individual control reduces the daily peak power exchanges by 15% to 65% compared to the benchmark case. The bottom-up control performs better than the individual control in reducing the peak power exchanges. It can reduce the daily peak power exchanges by 38% to 71% compared to the benchmark case. The top-down control performs the best among all the control approaches, and it can reduce the daily peak power exchanges by 48% to 73% compared to the benchmark case in the studied week.

8.4.3 Performances Comparative Investigation Under Objective of Maximizing PV Power Self-Consumption

This section presents the performance comparison of the three control approaches under the strategy to maximize the PV power self-consumption. The same week as Sects. 8.4.1 and 8.4.2 is used as the input for the analysis. Figure 8.8 compares the energy flow of the whole building community under different EV charging approaches. In order to maximize the PV power self-consumption, the power exports to the grid should be avoided as much as possible. In other words, the controls would try to decrease the amount of negative power mismatch values. In Day-1 and Day-2, the aggregated PV power production is much larger than the aggregated power demand (including the EV charging loads), and thus, the negative power mismatches (i.e., power exports to the grid) could not avoided. In Day-3, the aggregated PV power production is more equivalent to the aggregated demand (including the EV charging loads), and thus, the coordinated controls (i.e., both the bottom-up and top-down controls) could achieve good performances and successfully avoided the surplus PV power export. While in the individual control (see Fig. 8.8b), the power regulation performances are not as good as the coordinated controls, as the controllers take individual actions based on the single building's power mismatches without considering the aggregated-level performance. In Day-4 and Day-5, since the aggregated PV power production is much smaller than the aggregated power demand (including the EV charging loads), there is no surplus PV power which could be exported to the grid when implementing smart EV charging. The benchmark case still has PV power exports in the middle of the day as the EVs are already fully charged in the morning. As can be seen from Fig. 8.8, the performance of the bottom-up control is very close to the performance of the top-down control when the control target is to maximize the PV power self-consumption.

Figure 8.9 compares the daily PV power self-consumption rates under the three different control approaches, as well as the relative performance improvements compared to the benchmark case. Since there is more PV power production in the first two days, the self-consumption rates are much higher than the last two days. In the first two days, all the three controls significantly improved the daily PV power self-consumption, and the final performances are very close. In Day-1, the self-

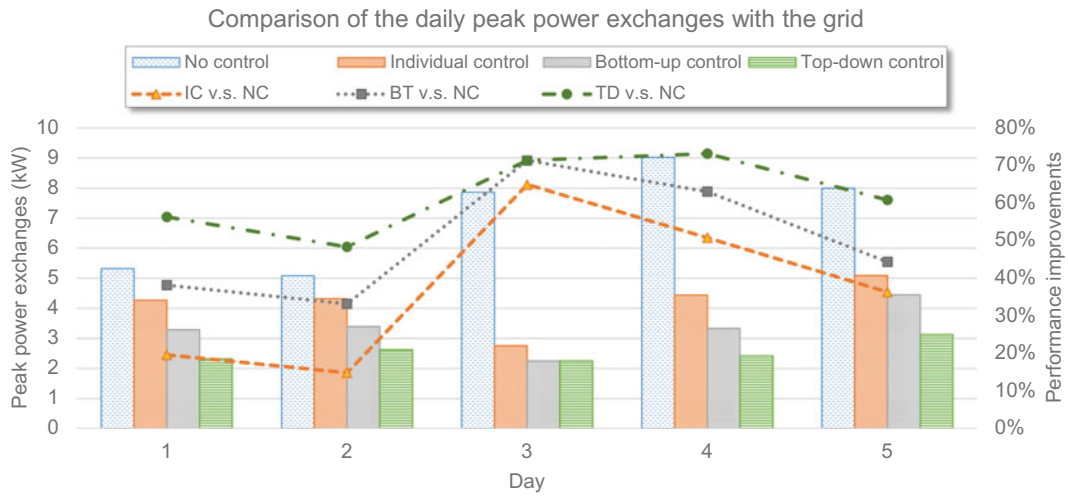

Fig. 8.7 Comparison of the daily peak power exchanges with the grid and the relative performance improvement under different control approaches (IC: individual control; BT: bottom-up control; TD: top-down control)

consumption rates are 69.4%, 72.2%, and 72.6% for individual control, bottom-up control, and top-down control, respectively. In Day-2, the self-consumption rates are 72.2%, 73.6%, and 74.4%, respectively. This is because most of the power demand during daytime can be met by the PV power production. In the last two days, due to the insufficient PV power production, all the PV power are used on-site, and thus the three controls produced the same performances. In Day-3, since the aggregated PV power production and aggregated power demand are more equivalent, proper coordination of single EV's charging loads is required to improve the load-supply match. The performances of the coordinated controls become much better than the individual control. It can also be noted that when the control target is maximizing the PV power self-utilization rates, the bottom-up control will lead to similar but slightly worse performances compared to the top-down control.

Figure 8.10 illustrates why the two coordinated control approaches have different performances in aspects of PV power self-consumption rates. In bottom-up control, the optimization of single EV's charging rates is conducted in a sequence based on the arrival time. Since the optimization of the EV1's charging rates does not consider the aggregated-level performances, it will tend to produce a non-optimal start for the subsequent optimizations, as can be seen in the white region in Fig. 8.10a. In the optimization of the EV2, the power balance in the period between the EV1 arrival and EV2 arrival is not regulated. And in the optimization of the EV3, the power balance in the period between the EV2 arrival and EV3 arrival is not regulated. As a result, with the combined effects of a non-optimal start and the accumulation of the 'biased' subsequent optimization, bottom-up control approach cannot lead to global optimum like the top-down control approach. The top-down coordinated approach takes the aggregated-level performances as the optimization target and considers the full periods (i.e., from the EV1 arrival to EV3 departure), and thus it can achieve the global optimum. However, since the scheduling also considers the EVs that has not arrived yet, it requires the controller to know/forecast the expected future arrival time in the building for all EVs.

8.4.4 Computational Performances Comparative Analysis

The computational loads of the three controls are also compared. Table 8.6 compares the total

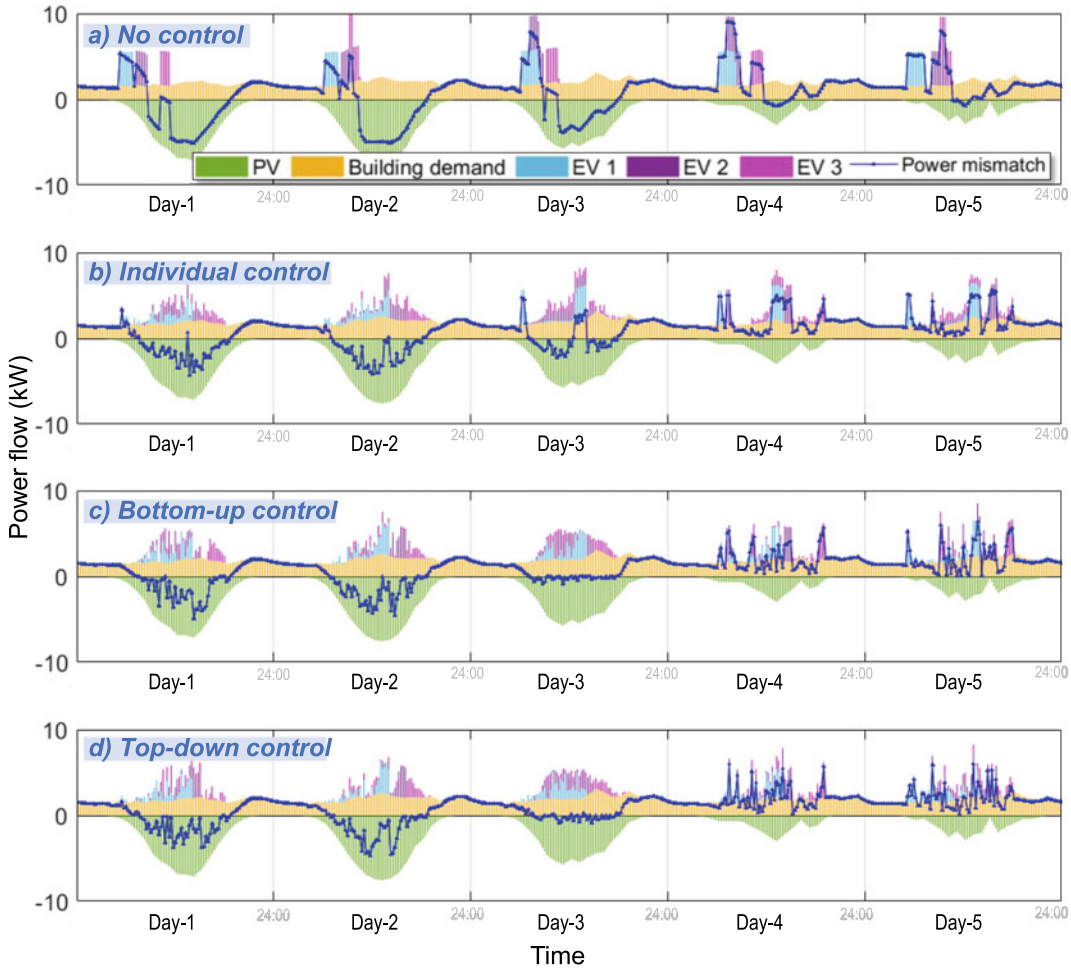

Fig. 8.8 Power flow of the building community and EVs under different EV control approaches for the goal of maximizing PV power self-consumption (with a simulation timestep of 15 min)

computational performances for the five-day simulation considering three EVs with a simulation timestep of 15 min. The simulation was conducted on a computer with 128.0 GB installed RAM and AMD Ryzen Threadripper 3990X 64-Core Processor 2.90 GHz. In the GA search engine, the number of generations is set as 8000, and the population size is set as 10,000. In the individual control and bottom-up control, as the EV charging loads are optimized one by one, the average number of variables to be optimized simultaneously is 32 (i.e., an average parking period of 8 h a day and 4 variables in each hour). The call times of GA engine of these two controls are both 15 (i.e., 3 EVs in each day and 5 days in total). While for the top-down control, the average number of variables to be optimized in each optimization increased to 96 (i.e., an average parking period of 8 h a day, 4 variables in each hour and 3 EVs), but the call times of GA engine reduced to 5 (i.e., 5 days). The total computing time for the individual control, bottom-up control, and top-down control are 8 h, 6.5 h, and 10.8 h, respectively. The ability to converge is assessed by checking while the optimized solutions become stable after several repeated optimization. If the required number of repetitions to reach a stable solution is large, the convergence ability is poor. The ability to converge is poorer in top-down control compared to

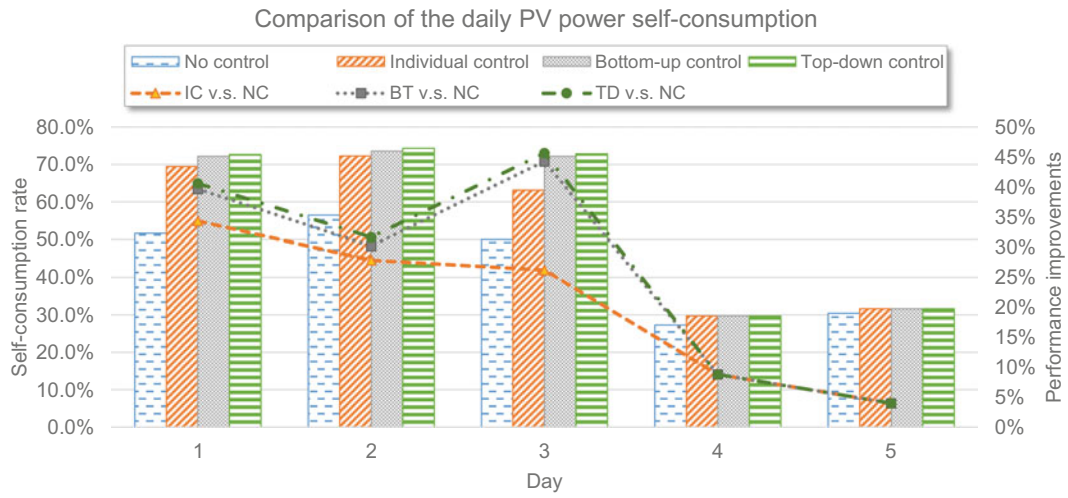

Fig. 8.9 Comparison of the daily PV power self-consumption and the relative performance improvement under different control approaches

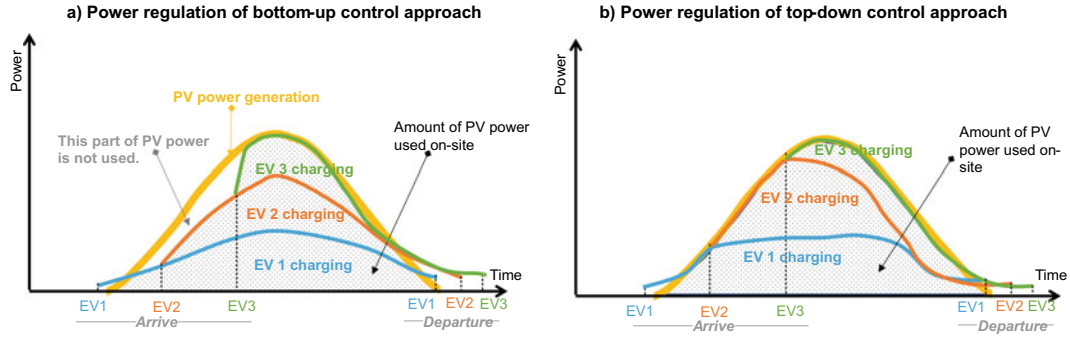

Fig. 8.10 Illustration of how the performances differ in the two coordinated control approaches **a** Bottom-up coordinated control approach **b** Top-down coordinated control approach. Note in the two coordinated controls, the PV power production and electricity demand are the community aggregated values

the other two controls, since the total number of variables to be optimized is three times the number in the other two controls. This means the optimization might need to be implemented multiple times to obtain a converged global optimum. Note with the increase of the EV number, the computing complexity of top-down control will increase dramatically in each call of the GA engine, which may deteriorate the optimization results. While for the individual control and bottom-up control, the computing load in each call of GA engine will not change much.

8.5 Summary

This chapter has classified existing EV smart charging controls into three categories according to the optimization principle: individual independent approach, bottom-up coordinated approach, and top-down coordinated approach. The principles, pros, and cons of these three approaches have been analyzed. For each control approach, a control method has been selected using the GA algorithm, and performances of the

Table 8.6 Comparison of the computational performances of the three approaches considering three EVs in a typical summer week (five workdays) at a simulation timestep of 15 min

Controls	Principles	Average number of variables to be simultaneously optimized	Call times of GA engine	Computational load (hour)	Ability to converge
Individual control	GA algorithm is used for searching the optimal charging rates of each EV independently	32	15	8	Good
Bottom-up control	The charging rates of the three EVs are optimized sequentially using GA	32	15	6.5	Good
Top-down control	The charging rates of the three EVs are optimized simultaneously using GA	96	5	10.8	Poor

three controls have been systematically compared in different aspects. Two control objectives, namely minimizing the peak power exchanges with the grid and maximizing the PV power self-consumption rates, have been considered in the comparison. A case building community located in Dalarna County of Sweden with three EVs has been used for the case studies and comparison. The major findings are summarized as follows:

- Both the two coordinated controls, i.e., bottom-up and top-down controls, can achieve much better control performances at the aggregated-level compared to the individual control. In aspect of minimizing the peak power exchanges, the individual control reduces the daily peak power exchanges by 15 $\sim$ 65% compared to the benchmark case, while the two coordinated controls reduce it by 38 $\sim$ 71% and 48 $\sim$ 73%, respectively. In aspect of maximizing the PV power self-utilization, the two coordinated controls also perform better than the individual control (e.g., the self-consumption rates are 29.7 $\sim$ 72.2% for individual control and 29.7 $\sim$ 74.4% for the coordinated controls), especially in days when the aggregated PV power production is close to the aggregated power demand.
- The top-down control approach can achieve better control performance than the bottom-up control under both the control strategies. This is because the top-down control is able to coordinate the operation of all EVs considering the constraints during the full optimization period. While with the combined effects of a non-optimal start and the accumulation of the 'biased' subsequent optimization, bottom-up control approach cannot lead to global optimum like the top-down control approach.
- The control strategies also have large impacts on the deviations of performances of each approach. When it comes to minimize the peak power exchanges with the grid, the top-down coordinated approach has much better performances compared to the bottom-up and individual independent approaches. While for the strategy of minimizing the PV power self-consumption rates, the three controls produced slightly different performances. This finding implies that depending on the control targets, bottom-up coordinated approach can sometimes produce acceptable performances as the top-down coordinated approach and thus can be a good option.
- Despite the good optimization performances, the computation complexity in singe call of GA search engine is much larger in the top-down control, as the large the number of

parameters to be optimized simultaneously. As a result, with the increase of EV number, the computing complexity in single optimization increases significantly. This will make it difficult to converge and identify the global optimum.

The results from this chapter suggest that decision makers need to make a balance between the control performances and the computational complexity. If the number of EVs is substantially large and the control strategy is not so dependent on the control approach (like maximize the PV power self-consumption), bottom-up coordinated approach can achieve near-optimum performances with good convergence ability. While if the number of EVs is relatively small, the top-down coordinated approach is a better alternative, which can converge and identify the global optimal solutions. The idea of top-down coordinated approach, which directly takes the aggregated-level performances as the optimization target, can find the global optimal solutions. More efforts can be devoted into this approach in the future research to develop more powerful controls which can achieve good convergence even when the number of EVs is large. This chapter considers simple scenarios of EV mobility patterns and did not consider the EV battery degradation due to the smart charging. Future work will use more sophisticated EV usage models to investigate the EV smart charging benefits.

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9 Peer-to-Peer Energy Trading in a Local Community Under the Future Climate Change Scenario

Pei Huang, Marco Lovati, and Xingxing Zhang

Abstract

Peer-to-peer (P2P) energy sharing among neighboring households is a promising solution to mitigating the difficulties of renewable power (such as solar photovoltaics (PV)) penetration on the power grid. Until now, there is still a lack of study on the impacts of future climate change on the P2P energy trading performances. The future climate change will cause variances in the renewable energy production and further lead to changes in the economic performances of households with various energy uses and affect the decision making in PV ownership and pricing strategies. Being unaware of these impacts could potentially hinder the P2P energy sharing application in practice. To bridge such knowledge gap, this chapter conducts a systematic investigation of the climate change impacts on the energy sharing performance in solar PV power shared communities. The future weather data is generated using the Morphine method, and an agent-based modeling method is used for simulating the energy trading behaviors of households. Four comparative scenarios of different PV ownerships and pricing strategies are designed. The detailed energy trading performances (including the PV power self-sufficiency, cost saving, revenues, and compound annual growth rate) for the four comparative scenarios are analyzed under both the present and future climates and compared. The study results of a building community located in Sweden show that the future climate change is more beneficial to large energy use households while less beneficial to small households. High price of energy trading can improve the fairness of the economic performances in the community, especially when some of the households do not have any PV ownership. This chapter can help understand the future climate impacts on the energy sharing performances of building communities, which can in turn guide decision making in PV ownership and price setting for different households under the future climate change to facilitate real applications.

Keywords

Photovoltaics (PV) · Energy sharing · Peer-to-Peer (P2P) · Building community · Agent-based modeling · Future climate change

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9.1 Introduction

The latest report from International Energy Agency (IEA) (Agency 2021) on Roadmap to Net Zero by 2050 calls for scaling up solar and wind rapidly this decade, reaching annual additions of 630 gigawatts of solar photovoltaics (PV) and 390 GW of wind by 2030, four times the record levels set in 2020. In such context, increased number of renewable energy systems are now installed worldwide. A commonly used way for mitigating the negative impacts of large intermittent renewable energy penetration on the power grid is to install energy storage system, which can store the renewable energy in large production periods and discharge back power in low production periods. The integration of energy storage systems has been proven effective. However, such integration requires high investments on the energy storage system. Moreover, the energy storage system capacity will inevitably decrease with time due to calendar degradation and frequent charging/discharging. Another cost-efficient and effective solution is to implement peer-to-peer (P2P) energy sharing among peer buildings (Soto et al. 2021). By P2P energy sharing, the large surplus renewable energy production can compensate with the large electricity demand in a same microgrid, thus leading to flattened power profiles at the aggregated level. A study conducted by Luthander et al. (2016) reported that a simple energy sharing among 21 residential buildings in Sweden, i.e., aggregate the electricity demand and supply of all the buildings, can easily improve the PV power self-consumption by over 15%.

Regarding P2P energy sharing, a lot of studies have been conducted in aspects of *design of community-level energy systems*, *advanced energy sharing controls*, *energy sharing community designs*, and *energy sharing microgrid technology*. Regarding the *design of community-level energy systems*, Huang et al. (2021) proposed a hierarchical design optimization method for the distributed energy storage system in PV-power-sharing neighborhoods considering energy sharing. The developed design method first optimizes the aggregated capacity of the distributed energy storage using genetic algorithm, and then coordinates the single storage capacity using non-linear programming. Case studies showed that the design optimization can not only reduce the required energy storage capacity at the aggregated level but also reduce the energy loss due to energy sharing and power transmission. Regarding *advance energy sharing controls*, Fan et al. (2018) developed a bottom-up control optimization of the distributed energy storage systems for an energy sharing community, which conducts the sequential optimization of the charging/discharging of distributed energy storage system considering the energy sharing between neighboring buildings. Compared to the individual control, such coordinated control can achieve better performances at the community-level as energy sharing enabled. Regarding the *energy sharing community design*, Jafari-Marandi et al. (2016) designed a homogeneity index for assessing the diversity of power demand and renewable power supply in an energy sharing community. The developed homogeneity index can be used for assessing whether a building community design is good or bad. In Huang and Sun (2019), a clustering-based planning method was developed for optimizing grouping of buildings in a neighborhood to form energy sharing community. The developed design optimization methods can identify and combine the buildings with the largest diversity of power profiles (similar to increasing the homogeneity index in Jafari-Marandi et al. (2016)), and thus maximizing the energy sharing potentials. Regarding the *energy sharing microgrid technology*, a Swedish company, Ferroamp, developed EnergyHub direct current (DC) microgrid system for energy sharing within a building community based on the Transmission Control Protocol/Internet Protocol (TCP/IP) protocol. The EnergyHub microgrid system was implemented in a real building community in Sweden (Ferroamp 2018). Similarly, in Ayai et al. (2012) designed a DC microgrid system for integrating distributed solar PV system. The abovementioned studies pave way for the

implementation of P2P energy trading within building communities in practice.

Another important aspect for P2P energy sharing is the P2P energy trading, which is related to business models and pricing strategies. Existing studies have also developed many business models for the P2P energy trading in building communities. For instance, Lovati et al. (2020) proposed a P2P business model for a group of building prosumers considering the user behavior, electricity/financial flows, ownerships of renewable energy systems and trading rules inside the local electricity market. They also defined a behavior map of the district-scale renewable energy systems from three dimensions: local energy providers (emergent or controlled), local energy communities (individual or collective), and control algorithms (centralized or decentralized). In another study (Lovati et al. 2021), an agent-based modeling method was developed to analyze the P2P energy trading performances under different scenarios of PV system ownerships (i.e., all households have ownerships or only part of households have ownerships) and pricing strategies (i.e., high selling price, low selling price, and dynamic selling price). Their study results showed that smaller households have higher revenues and lower savings compared to larger households. An et al. (2021) proposed a business feasibility evaluation model to optimize the price of P2P electricity trading using genetic algorithm and Pareto optimum. The developed business model can maximize the profits of both the market participants (i.e., buyer and seller) in the P2P trading. Their developed model was validated in seven cities in Korea. Inspired by the flocking behavior of birds, Bandara et al. (2021) proposed a flocking-based decentralized double auction method for P2P energy trading within a neighborhood. Besides the buildings with renewable energy systems installed, their study also considers the microgrid, electric vehicles, and charging stations as prosumers, which can participate in the P2P energy trading. The case study based on a residential PV data set from California, USA, showed that the developed method can achieve the highest energy trading as compared to the conventional distributed double auction and centralized double auction.

The P2P energy trading performances are greatly affected by the climate change, as the climate change has large impacts on the building energy demand and renewable energy productions. Under climate change, the raised outdoor temperature would result in increased cooling energy use but decreased heating energy use. This will have large impacts on the energy demand in the buildings. For instance, Olonscheck et al. (2011) pointed out the cooling energy demand of the residential buildings in Germany increased by up to 59% and heating energy demand decreased by up to 75% in the period of 2031–2060 in comparison with the period of 1961–1990. Santamouris et al. (2015) reviewed the recent studies of climate change impacts on building energy use and summarized that the building energy use increased at a rate of 0.5%–8.5%/°C in response to outdoor temperature rise under climate change. On the other hand, climate change also gives rise to renewable resource variations and thus affect the renewable energy generation. For instance, Lucena et al. (2009) pointed out both biodiesel and hydropower would decrease by 5–10% in the northeast of Brazil under climate change, and thus result in insufficient renewable energy generation. Robert and Kummert (2012) found that in 2050s the average wind speed increased by 7.4% in winter and decreased by 9.2% in summer in comparison with 1961–1990, thereby leading to surplus wind energy in winter but insufficient wind energy in summer. As climate change affects both building energy demand and supply, it should be carefully considered in P2P energy trading performances. However, in the existing studies, the impacts of climate change on the P2P energy trading performances are rarely studied quantitatively.

However, the impacts of future climate change on the P2P energy trading are rarely studied. The future climate change will cause variances in the renewable energy production and further lead to changes in the economic performances of households with various energy uses and affect the decision making in PV ownership and pricing strategies. Being unknown about these impacts

could potentially hinder the P2P energy sharing application in communities. Thus, this chapter conducts a systematic investigation of the future climate impacts on the P2P energy trading performances. The future weather data in Ludvika of Sweden is first derived from the present weather data using the Morphine method. Then, an agent-based modeling approach is presented for simulating the P2P PV power trading and interactions of households under different PV ownership scenarios and pricing scenarios. The performances of the P2P trading are then analyzed under both the present climate and future climate and compared. A case study is conducted using the data from a building community in Sweden. The major contributions of the present study are summarized as below.

- The impacts of future climate change on the P2P energy sharing performances of solar power shared building community, including the self-sufficiency, cost savings, revenues, and compound annual growth rate (CAGR), are investigated and analyzed.
- The households in the community are divided into different groups based on the power demands, and the changes of performances for each group of households due to the future climate change are revealed.
- Four comparative scenarios of different PV ownership and power sharing prices are defined, and the future climate change impacts under different scenarios are studied.
- This chapter reveals the impacts of future climate change on solar powered building community energy sharing performances, which can help guide decision making in PV ownership and price setting for different households under the future climate change to facilitate real applications.

The structure of the paper is as followings. Section 9.2 describes the method for deriving the future weather data and modeling the energy trading behaviors. Section 9.3 presents the detailed building demand modeling and energy system models. In Sect. 9.4, the case studies are conducted based on a real building community in Sweden. Section 9.5 further summarizes the impacts of future climate change on the P2P energy sharing performances and extends the discussions. The brief conclusions are given in Sect. 9.6.

9.2 Methodology for Investigating the Future Climate Impacts

This section first introduces the method for predicting future climate data using the present climate data. Then, the agent-based modeling for the P2P power trading is presented, and the performance indicators considered in this chapter are described.

9.2.1 Prediction of the Future Climate Using the Morphine Method

9.2.1.1 Climate Models

Concerning future climate conditions, the Intergovernmental Panel on Climate Change (IPCC) is devoted to assessing the current and expected state of climate system and its impacts on societies and ecosystems using *global-scale* and *regional-scale* climate models. These two are the established tools for investigating the climate system response to expected future Greenhouses Gas (GHG) concentrations. Global Climate Models (GCMs) are characterized by coarse spatial resolution and for regional-scale climate studies finer-resolution Regional Climate Models (RCMs) are generally used. These latter models are driven by initial and boundary conditions provided by the GCMs. RCMs produce a more detailed climate information considering more realistic representation of orography features of a certain region.

9.2.1.2 Future Climate Scenarios

In addition, future climate projections are produced considering different emission scenarios to which correspond different radiative forcing. The different emission scenarios will correspond different future climate conditions. Emission

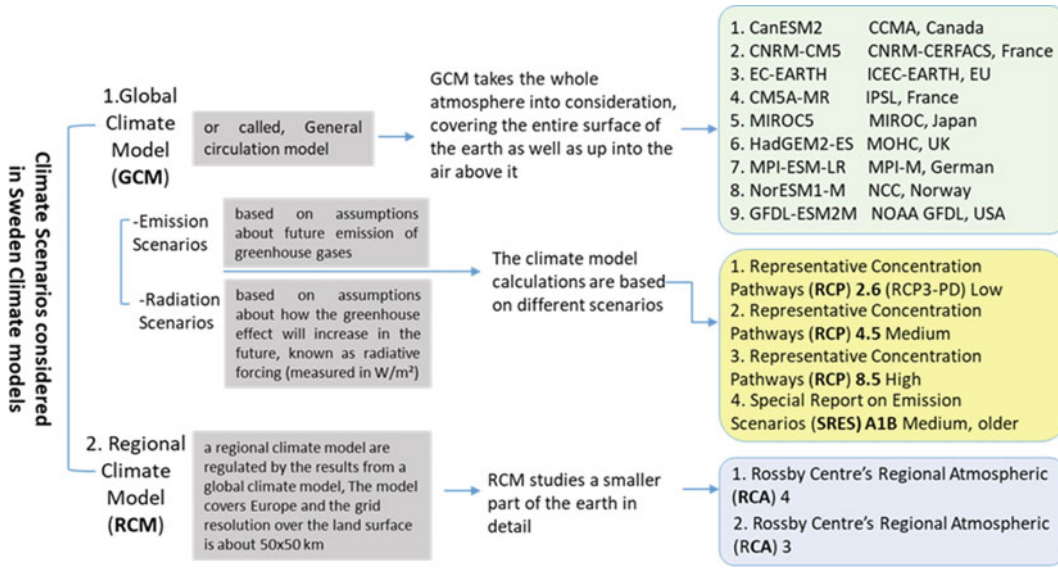

Fig. 9.1 Climate scenarios results produced by Swedish Meteorological and Hydrological Institute (SMHI)'s Climate Research

scenarios rely on assumptions about future GHG emissions, based on estimates of the development of the world economy, population growth, globalization, increasing use of green technology, etc. The amount of GHGs that are emitted depends on global evolution. Two main types of emission scenarios are considered in the current scientific literature. Special Report on Emission Scenarios (SRES) scenarios (Nebojsa and Rob 2000) which was superseded by Representative Concentration Pathways (RCPs) (Moss 2010) in 2014. These scenarios provide numerical information in terms of future radiative forcing (measured in W/m^2). If there is an increased emission of GHGs, then there will be more radiative forcing. RCPs that are named with the level of radiation drift achieved in 2100 with 2.6, 4.5, 6.0 or 8.5 W/m^2 [19]. Considering results from the most recent IPCC assessment reports (Collins 2013), extremes climate events magnitude and frequency are expected to globally increase. However, these changes are expected to largely varying according to spatial and temporal scales considered (Kovats et al. 2014).

Even taking into account substantial measures to limit future GHG emission, climatic changes projections for the next 30–60 years are already 'locked in,' as consequence of GHG emission (2019). This information is therefore essential in perspectives to plan and design future buildings and energy system. Future climate conditions will drive designers, engineers, and planners with significant opportunities to create or remodel more adaptive outdoor spaces, where will enhance the livability of, and quality of life for the future communities. In addition, given people usually spend around 90% time indoor nowadays (2018), a correct design strategy plays a vital role in the creation of adaptive, resilient, livable built environment that can manage health and wellness risk caused by climate change impacts in the coming decades. Figure 9.1 shows a subset of GCMs and RCMs in the climate scenario studies in Sweden. Considering combination of driving GCM and RCM, a multi-model ensemble is obtained. An ensemble approach is generally preferred to a one-model response (Knutti et al. 2010).

9.2.1.3 Morphed Method

In order to execute energy simulation under the future scenarios, an hourly dependent climate dataset is necessary for a dynamic simulation. Unfortunately, GCM/RCMs for limitation of data

storing facilities generally provide outputs at daily time-steps (only in few cases at 3–6 hourly time-step). For this reason, a morphing method was applied to produce hourly climate datasets for use in building energy simulation tools, such as the prediction of the impacts of climate change on energy consumption for a medium-size office building (Moazami et al. 2019), the investigation of climate change on an office building energy consumption with two climate models (Wang et al. 2017), the energy consumption variation due to climate change for an office building in Japan (Shibuya and Croxford 2016) and in Shanghai, China (Zhao et al. 2017), and the estimation of climate change impact on energy consumption in a residential building in Kaunas, Lithuania (Sabunas and Kanapickas 2017) to ensure a globally consistent, statistically stable and available database from 60 years back through the current time and into the future.

However, there are limitations in this approach. Firstly, it uses the previous emission scenario, the HadCM3 A2 experiment ensemble, instead of the newest generation of emission scenarios (RCPs). However, as previously shown, SRES scenario is largely consistent and comparable with results RCP climate projections. However, different from the yearly or monthly dependent variables under future scenarios, this method enables to project a free hourly climate datasets ready for use in energy simulation and are more practical for climate adaptive building/energy system design until RCM data becomes widely available for the public.

With the intention of the hourly future climate data projection, the authors therefore collect the latest 2020 climate data from SVEBY (data originally coming from SMHI) with the format of excel as baseline climate dataset. Each weather file in certain location around Sweden contains eight different climate parameters for all hours of the year adapted for energy calculation programs. In accordance with the announcement from SVEBY, it is recommended to use, especially when verifying energy use. In the next step, the excel-based climate file is further processed by the author and further transferred into the format of 'epw.' After then, the 'CCWorldWeatherGen' tool, developed by Energy and Climate Change Division, University of Southampton, UK, was processed for the 'present-day' weather files as the baseline data preparation for the future climate morphing in the next stage. The morphed weather data is compatible with the climate adaptive design assessment and the building performance simulation in the future study. To be mentioned here, the morphed future climate data using the RCM RCA4 that driven by the GCM MOHC-HadGEM2-ES for the period 2020, 2050, and 2080, respectively.

9.2.2 Agent-Based Modeling of the P2P Energy Sharing Under Different Scenarios

Figure 9.2 presents the schematics of the P2P energy sharing in a community. The energy sharing is implemented by connecting all the houses with one energy sharing microgrid. The surplus power from one house can be delivered to another house via the energy sharing network.

An agent-based model is a model of a complex system in which the behavior of individual player/system is not controlled by a single algorithm, but it comes/emerges from the interaction of a number of sub-systems (i.e., the agents). Following the classification of dimensions of different control and algorithm in (Lovati et al. 2020), the simulations presented in this chapter are for individual and de-centralized systems causing an emergent behavior in the microgrid. In other words, there are multiple PV owners in the microgrid and each owner can set the price according to its own independent will. In general, the behaviors of each agent in a time-step can be summarized as shown in Fig. 9.3. The behaviors of each agent are considered the same in different PV ownership and pricing scenarios.

The agent-based modeling is described by the following set of rules.

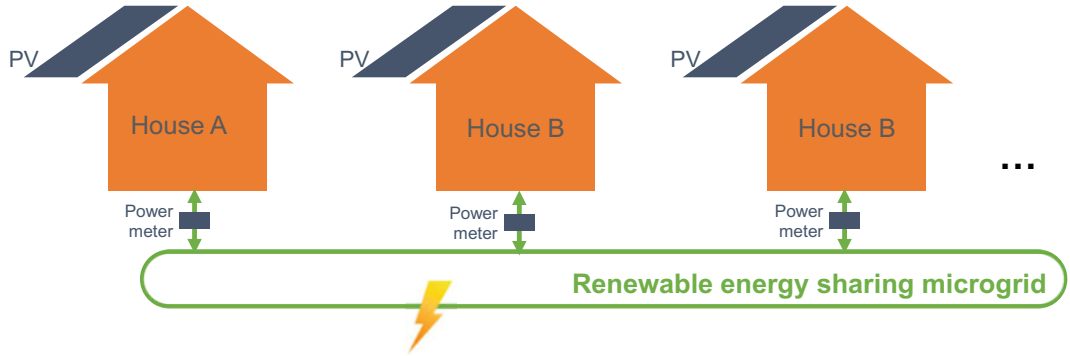

Fig. 9.2 Schematic of the P2P energy sharing in a community

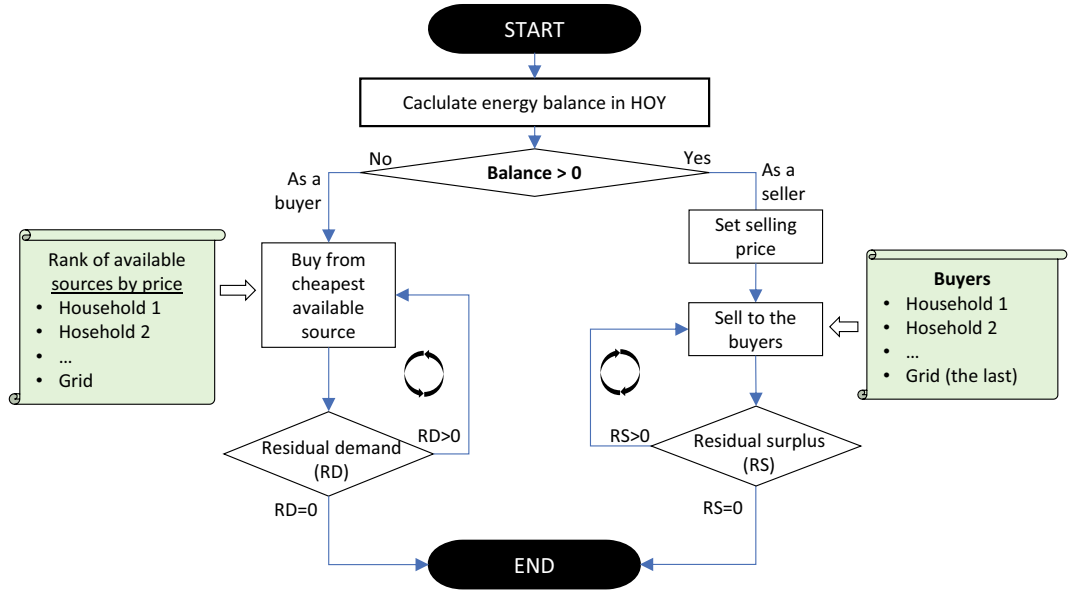

Fig. 9.3 Schematic of the agent-based modeling and behavior of each agent in every time-step of the simulation

- Every household is considered as one independent agent.
- Every agent has an energy balance in each Hour of the Year (HOY). The energy balance is calculated as the deviation between its hourly PV power production (if it owns a PV system) and its hourly power demand. If the balance is negative, the agent will be a net buyer in that HOY, otherwise it will be a seller. This rule assumes that each household can sell only the excess PV production (after meeting its own demand).
- Each seller can set the price for the surplus power to be sold.
- If the electricity is offered by multiple sellers, the buying agent will select from the cheapest source.
- If the aggregated demand of the district exceeds the offer of the cheapest source, the demand of each household is met proportionally by the cheapest source. For example, if the cheapest source covers 30% of the aggregated demand in that HOY, each household is provided 30% of its power demand by the cheapest source.

- If the on-site renewable power exceeds the power demand in a certain HOY, the cheapest sources are consumed preferentially. Thus, the more expensive sellers risk to be in excess of the demand and sell part or all their power to the grid. Those who sell to the grid cannot set the price but are simply valued the price paid by the grid (which is always way lower than that of the local sellers).

In this chapter, four different P2P energy trading scenarios are considered and the economic performances of all these scenarios under both the present and future climate are studied and compared. The details for each scenario are explained below. The percentage of ownership and price settings are also summarized in Table 9.1. The summer grid price is 1.2 SEK/(kW·h) and the winter grid price is 1.8 SEK/(kW·h). Note the abovementioned rules apply to all the considered scenarios.

Scenario 1: All households have an ownership of the PV system. Every household invests an equal share of the whole PV system. The price for the sale within the micro-grid is agreed for the long term as the 83% of summer grid price (thus 1 SEK/(kW·h) in both winter and summer at the year 0).

Scenario 2: Similar to Scenario 1, all residents have an ownership of the PV system. The price for the sale within the micro-grid is agreed for the long term as 99% of the grid price, therefore whoever buys electricity from another household has almost no savings compared to the grid.

Scenario 3: Only half of the households agree to purchase the PV system. Each PV equipped household has an equal share of the total system. The price for the sale within the microgrid is agreed for the long term as the 83% of the summer grid price, like Scenario 1.

Scenario 4: Like Scenario 3, only half of the residents agree to purchase the PV system. Each PV equipped household has an equal share of the total system. The price for the selling surplus power within the microgrid is agreed for the long term as 99% of the grid price, like in scenario 2.

9.2.3 Performance Indicators for Analysis

This chapter will investigate three economic performance indicators: namely the cost savings, the revenues, and Compound Annual Growth Rate (CAGR). The savings (SEK) represent the reduction in the electricity costs due to the avoided purchase of the electricity from the external grid. The revenues (SEK) indicate the incomes obtained by each shareholder of the PV panels for selling the surplus PV power from their shares. The CAGR (%) is the average rate of return that would be required for an investment to grow from its beginning balance to its ending balance. The calculation of the three indicators is introduced below.

The cost savings includes two parts: (i) the savings from using the self-produced PV power and (ii) the savings from purchasing power at a

Table 9.1 PV capacities per household and prices in the four different scenarios

Scenario	Ownership of PV panels	Household PV capacity	Electricity price (SEK/(kW·h))
1	100% of households have PV ownership	$Capacity_{PV,tot}/N_{household}$	1
2	100% of households have PV ownership	$Capacity_{PV.tot}/N_{household}$	1.19 (summer), 1.78 (winter)
3	Only 50% of households have PV ownership	$2 \times Capacity_{PV.tot}/N_{household}$ or 0	1
4	Only 50% of households have PV ownership	$2 \times Capacity_{PV.tot}/N_{household}$ or 0	1.19 (summer), 1.78 (winter)

cheaper price within the community. The calculation of cost savings $\text{Cost}_{\text{save}}$ (SEK) is shown by Eq. (9.1).

$$\text{Cost}_{\text{save}} = \sum_{\text{Ts}=1}^{8760} \left(\left(P_{\text{self.Ts}} \cdot d_{\text{grid.Ts}} \right) + P_{\text{peer.Ts}} \cdot \left(d_{\text{grid.Ts}} - d_{\text{peer.Ts}} \right) \right) \tag{9.1}$$

Ts (h) is the internal simulation time-step of the model. $P_{\text{self,Ts}}$ (kW·h) is the power self-consumed by a household in the specific time-step, which is calculated as the smaller one of the hourly electricity demand and hourly PV power production. $d_{\text{grid,Ts}}$ (SEK/(kW·h)) is the cost of electricity offered by the external grid in the specific time-step, i.e., the grid electricity price. $P_{\text{peer,Ts}}$ (kW·h) is the amount of electricity purchased from a peer household within the local community in the specific time-step. $d_{\text{peer,TS}}$ (SEK/(kW·h)) is the cost of electric power offered by a peer in a specific time-step.

The revenues are obtained either from selling power to the public grid or from selling power to the peers in the community. Note the price of selling power to the public grid is much lower than selling to the peers as feed-in-tariff can increase the grid stress. The calculation of revenues $\text{Cost}_{\text{revenue}}$ (SEK) is expressed by Eq. (9.2).

$$\text{Cost}_{\text{revenue}} = \sum_{\text{Ts}=1}^{8760} \left(\left(P'_{\text{peer.Ts}} \cdot d'_{\text{peer.Ts}} \right) + \left(P'_{\text{grid.Ts}} \cdot d'_{\text{grid}} \right) \right) \tag{9.2}$$

$P'_{\text{peer.Ts}}$ (kW·h) is the amount of electricity sold by the selling household to all peer households in the specific time-step. $d'_{\text{peer.Ts}}$ (SEK/(kW·h)) is the electricity price set by the selling household to the peers in the specific time-step. $P'_{\text{grid.Ts}}$ (kW·h) is the amount of electricity sold by the selling household to the grid in the specific time-step. d'_{grid} (SEK/(kW·h)) is the feed-in-tariff. This price is static, thus is independent by the time-step.

The calculation of CAGR is expressed by Eq. (9.3).

$$\text{CAGR} = \left[\left(\frac{\text{Income} - (\text{CAPEX} + \text{OPEX})}{\text{CAPEX}} \right)^{\frac{1}{\text{lifetime}}} - 1 \right] \cdot 100 \tag{9.3}$$

Income (SEK) is the cumulative income of the household derived by the ownership of the share of the PV system during its lifetime. It is calculated according to Eq. (9.4). CAPEX (SEK) is the capital costs. It includes the turn-key cost of the system including design and installation costs. It can be calculated by multiplying the unitary cost by the installed capacity (see Table 9.3). OPEX (SEK) is the operational costs. The operational costs include a standard annual cost of 80 SEK/kWp each year for the substitution and cleaning of the modules, as well as the substitution of the inverter in case of rupture. The cost of inverters is set as 3.5 KSEK/kWp and is assumed to be changed at least once in the planned lifetime of the system. The *Lifetime* is assumed to be 30 years in the analysis.

$$\text{Income} = \sum_{T=0}^{\text{lifetime}} \left(\text{Cost}_{\text{save}} + \text{Cost}_{\text{revenue}} \right) \cdot (1 - \Delta\eta \cdot T) \cdot (1 + \Delta d \cdot T) \tag{9.4}$$

$\text{Cost}_{\text{save}}$ is the cost saving and calculated by Eq. (9.1), and $\text{Cost}_{\text{revenue}}$ is the revenue and calculated by Eq. (9.2). T is the number of years since the installation of the PV system. $\Delta\eta$ is the change of the PV production efficiency due to component degradation, which is assumed to be 1% per year. Δd is the change in the price of the electricity for the consumer, it is assumed to be + 2% per year in the design stage.

Besides these economic indicators, the PV power self-sufficiency (SS) is also calculated for the households within the energy sharing community. The self-sufficiency represents the percentage of power demand which is met by the on-site PV production as compared to the total demand. It is calculated by Eq. (9.5).

$$SS = \frac{E_{d,pv}}{E_{d,pv} + E_{d,grid}} \quad (9.5)$$

$E_{d,pv}$ (kW·h) is the aggregated electricity demand that is supplied by the PV system during the whole year. $E_{d,grid}$ (kW·h) is the aggregated electricity demand that is supplied by the power grid. A larger SS indicates a better performance since a building becomes less dependent on the power grid. Note for the households in the community, even though some of them do not have a PV ownership, they still have self-sufficiency as they can use the surplus PV power from the peers inside the community.

9.3 Buildings and System Modeling

This section introduces the building information for electricity demand modeling, as well as the modeling of PV systems.

9.3.1 Building Modeling

This chapter considered a real building cluster located in Ludvika, Dalarna region, Sweden. This building cluster consists of three separate buildings, as shown in Fig. 9.4. The building cluster (all the three buildings) includes 48 dwelling units over three floors, and most of the apartments have one or two bedrooms. The total façade surface gross area of the complex is 2146 m^2, and the total roof surface gross area is 1750 m^2. These buildings will be improved by a series of renovation plans including installation of PV and direct current (DC) microgrid. It is assumed the heating is provided by district heating system. So, the PV panels will only need to provide power supply to the domestic electricity demand (e.g., lighting, TVs, dish wash). In this chapter, the electric demand used for the study was generated using Load Profile Generator (LPG) (Lovati et al. 2021) assuming population characteristics as described in Table 9.2. In total, there are 48 households in the three multi-family apartment blocks.

Fig. 9.4 Case building cluster located in Ludvika, Sweden

9.3.2 Renewable Energy System Modeling

The power generation from the PV panel P_{PV} (kW) is calculated by Eq. (9.6) (Lovati et al. 2020) and simulated in TRNSYS (i.e., an energy simulation platform),

$$P_{PV} = \tau \times I_{AM} \times I_T \times \eta \times CAP_{PV} \quad (9.6)$$

where τ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (W/m^2) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; CAP_{PV} (m^2) is the PV surface area. Equation (9.6) shows the calculation of the PV power production for each hour. In each hour, the values of parameters (including the hourly solar radiation) in this equation are updated and then used for the calculation of the hourly PV power production in TRNSYS. This equation is calculated 8760 times to simulate the PV power production during a full year period.

In this chapter, the PV system capacity was sized under the present weather data using the design optimization tool developed in the Horizon 2020 EnergyMatching project. The capacity of the PV systems was optimized to maximize the self-sufficiency of the building community while meeting the constraint of keeping a positive net present value. For details about the design optimization of PV systems, please refer

Table 9.2 Configuration of the simulation households

Groups	Household ID	Number of households	Occupant number
1	1–2	2	Five
2	3–7	5	Four
3	8–12	5	Three
4	13–27	15	Two
5	28–48	21	One

Table 9.3 Input parameters for PV system capacity optimization

Parameters	Values
Unitary cost of the PV system	12,000 SEK/kWp (ca. 1175 €/kWp)
Planned lifetime of the system	30 years
Degradation of the PV system	− 1%/year (annual percentual efficiency losses)
Nominal efficiency of the system	16.5%
Performance ratio at standard test conditions	0.9
Price of the electricity from external grid	1.2 SEK/(kW·h) (Summer), 1.8 SEK/(kW·h) (Winter)
Price of the electricity sold to the external grid	0.3 SEK/(kW·h)
Annual discount rate	3%
Growth of electric price for consumer	+ 1.5%/year (annual percentual price increases)
Optimized capacity of the installed PV system	65.5 kWp (Huang et al. 2018)

to (Pflugradt and Muntwyler 2017). Table 9.3 summarizes the parameters used for optimizing the PV system capacity. These cost parameters are also used in the economic analysis of P2P trading.

9.4 Case Studies and Results Analysis

The case studies are conducted based on a case building community located in Ludvika, Sweden. In this section, the weather data and PV power production under both the present and future scenarios are first analyzed and compared. Then, the P2P energy trading performances under the two climates are compared and discussed.

There are 48 households considered in the case studies. The PV system capacity is optimized under the present climates, and the optimized capacity is 46.5 kWp, as calculated in (Huang et al. 2019). In the four considered scenarios, the PV system allocated for each household is 1.36 kWp in Scenarios 1 and 2, while is either 2.73 kWp or 0 in Scenarios 3 and 4. In the following analysis, the PV system capacity is kept the same under both the present climate and the future climate. In other words, the climate change will be the sole factor affecting the P2P energy trading performances. In the economic performance analysis, it is assumed a 2% increase in the electricity prices in each year.

9.4.1 Comparison of the Present and Future Climates

Statistical analysis is conducted to compare the solar radiation in the current and future climates. Figure 9.5 shows the comparison of the hourly solar radiation and associated PV power production of the system (as specified in Sect. 7.3.2). As can be seen from Fig. 9.5a, in the small solar

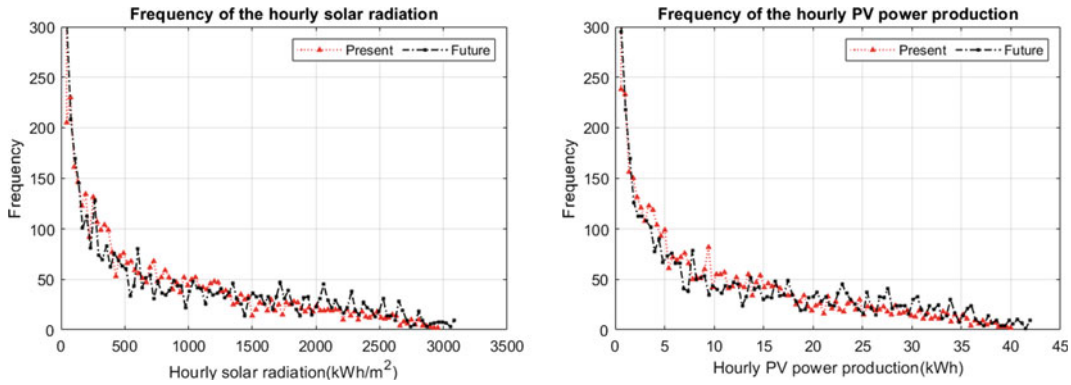

Fig. 9.5 Comparison of the frequency of **a** solar radiation and **b** PV power production under the present and future climates

radiation range 0 ~ 1000 kWh/m^2, the future scenario has lower occurrence as compared to the present scenario. While in the large solar radiation range 1500 ~ 3000 kWh/m^2, the future scenario has larger occurrence. The large solar radiation mostly occurs in summer while the small solar radiation mostly occurs in winter. This means that in the future there are more solar radiation in summer while less solar radiation in winter. The frequency analysis shows a similar trend for the PV power production: with an increased frequency in large summer period while decreased production in winter period. The maximum hourly PV power production is about 40 kW·h under the present climate, while the maximum production under the future climate is about 42 kW·h (about 5% increase).

Figure 9.6 compares the monthly PV power production under the present climate and future climate. As can be seen, during summer months in the future, i.e., from April to October, the PV system has more power production compared to the present scenario. The increase of power production reaches maximum in August at about 28.8%, followed by 24.8% in July. While in the future winter months, i.e., January, February, March, November, and December, the PV system has less power production compared to the present scenario. In total, the annual PV power production increased by 10.7% under the future weather compared to the present scenario.

9.4.2 Comparison P2P Energy Sharing Performances

9.4.2.1 Energy Performances

Figure 9.7 presents the cumulative probability distribution of PV power self-sufficiency under both the present and future climates with different PV ownerships. The blue curves show the performances under the present climate and the red curves show the performances under the future climate. Figure 9.7a shows the case in which 100% households have an equal PV ownership (for Scenarios 1 and 2), and Fig. 9.7b shows the case in which 50% households have an equal PV ownership (for Scenarios 3 and 4). Note in this chapter, the PV power self-sufficiency is calculated based on the community-produced PV power. In other words, even though a household does not have PV ownership, this household still can have a self-sufficiency, as its demand can be partly covered by PV power from its peers. Table 9.4 summarizes the mean and ranges of the self-sufficiency under different climates and different PV ownerships.

When 100% of the households have an equal PV ownership, the values of PV power self-sufficiency are evenly distributed within a narrow range (i.e., 0.2 ~ 0.34) for all the households. Due to the climate change, the distribution of PV power self-sufficiency shifted slightly to the right

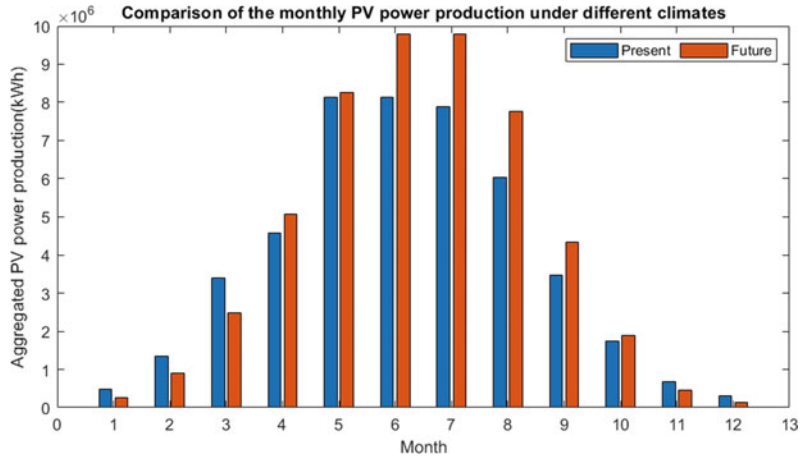

Fig. 9.6 Comparison of the monthly aggregated PV power production under the present and future climates

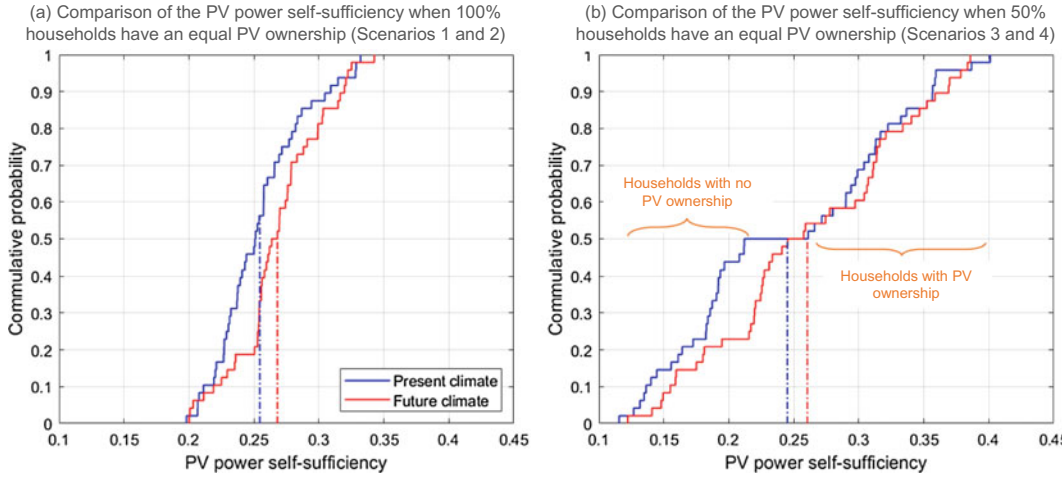

Fig. 9.7 Comparison of the PV power self-sufficiency distributions under both the present and future climates with different PV ownerships **a** 100% households have an equal PV ownership (for Scenarios 1 and 2); **b** 50% households have an equal PV ownership (for Scenarios 3 and 4)

Table 9.4 Comparison of the PV power self-sufficiency under

PV ownership		Climate	Ranges	Mean	Change (%)
100% households have PV ownership (Scenarios 1 and 2)	Figure 12.7 a	Present	(0.20 ~ 0.33)	0.25	
		Future	(0.20 ~ 0.34)	0.27	5.4
50% households have PV ownership (Scenarios 3 and 4)	Figure 12.7 b	Present	(0.11 ~ 0.40)	0.24	
		Future	(0.12 ~ 0.39)	0.26	6.2

side with larger values. As summarized in Table 9.4, the mean value of household PV power self-sufficiency increased by 5.4% in the future compared to the present climate. Meanwhile, the ranges of self-sufficiency shifted rightward. This is because of the increased PV power production under the future climate, which can help improve the overall self-consumption of the community.

When only 50% of the households have an equal PV ownership, the values of PV power self-sufficiency have wider ranges (i.e., 0.12 ~ 0.4). This is because for the households with a PV ownership, the allocated PV capacity is twice the value in the case with 100% household PV ownership. Consequently, the self-sufficiency is much larger for these households. For the households with no PV ownership, they can only purchase PV power from their peers to meet the power demand, and thus their self-sufficiency is much lower. A gap can be observed under both climates in the self-sufficiency distribution between the households with and without ownership. Again, the climate change has a positive impact on the self-sufficiency. The average self-sufficiency increased by 6.2% under the future climate. Note that the maximum value of self-sufficiency decreases under the future climate. This is because the increase of PV power production occurs in summer period (see Fig. 9.6), where most of the households can also be very self-sufficient. While in the winter period under the future climate, the PV system has reduced power production, leading to reduce self-sufficiency. Overall, the decrease of self-sufficiency in winter leads to decreased annual self-sufficiency.

Table 9.5 summarizes the amount of energy trading during a full year between different groups of households (see Table 9.2 for the details of groups) under both the present and future climates in Scenario 1. The values in the table indicate the amount of energy sold from the row-group households to the column-group households. For instance, the value in the first row and first column represents that households in Group 2 together sold 1776 kW·h electricity to households in Group 1. As can be seen, due to the climate change, the amount of shared energy will increase a lot under the future climate. The amount of shared energy increased by 11% ~ 21%. This is because of the larger PV power production in the future, which leads to more surplus PV power for sharing. Note that Group 1 households are large energy end-users with five occupants, while Group 5 households are small energy end-users. Thus, the aggregated energy sharing from small energy end-users to large energy end-users are positive, indicating more selling than purchasing.

9.4.2.2 Economic Performances

Figure 9.8 compares the cost saving (as calculated by Eq. (9.1)) and revenues (as calculated by Eq. (9.2)) of each group of households under both the present and future climates. The different colors represent different groups of households characterized by the number of occupants (see Table 9.2). In total, there are five colors corresponding to the five groups. The hollow markers represent the performances under the present weather data, and the filled markers represent the performances under the future weather data.

In all the four scenarios, the large energy end-users (i.e., Group 1 with five occupants) have larger savings in the electricity costs but smaller revenues. This is because these large energy end-users can use the PV production to meet more demands, and thus leading to larger electricity cost savings. While on the other hand, the smaller energy end-users (i.e., Group 5 with one occupant) have larger revenues but smaller savings in the electricity costs. This is because their power demand is already small. As a result, the amount of power demand which can be changed to be supplied by the household's own PV system or the community shared PV power is limited, which eventually leads to lower cost savings. But these small energy end-users can sell their surplus PV power to the community, and thus, the revenues are higher.

In Scenario 1, for large energy end-users, both the revenues and savings will increase slightly in the future, due to the increased PV power production in the future. For small energy end-users,

Table 9.5 Summary of energy trading (kW·h) among any two households within the community in the full year for Scenario 1

	Group 1	Group 2	Group 3	Group 4	Climate
Group 2	1776	–	–	–	Present
	2106	–	–	–	Future
	19%	–	–	–	*Increase*
Group 3	777	158	–	–	Present
	898	176	–	–	Future
	16%	*11%*	–	–	*Increase*
Group 4	1540	597	119	–	Present
	1791	694	139	–	Future
	16%	*16%*	*17%*	–	*Increase*
Group 5	1054	459	104	51	Present
	1233	540	123	61	Future
	17%	*18%*	*17%*	*21%*	*Increase*

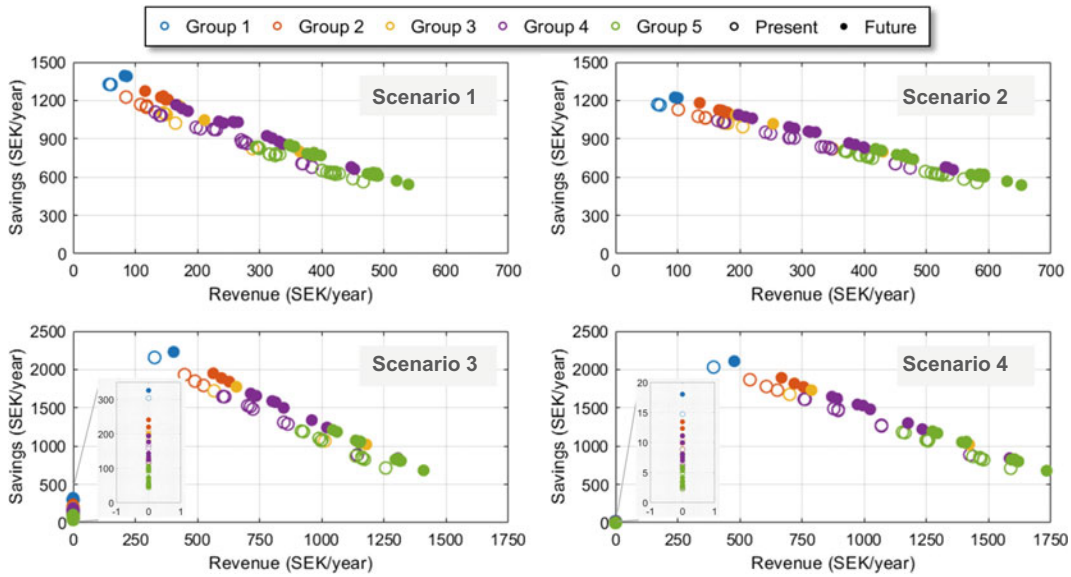

Fig. 9.8 Comparison of savings and revenues of each household under both the present and future climates for the four scenarios

the revenues will increase at a higher level compared to large energy end-users. For instance, for the end-user with the highest revenue (i.e., the rightmost hollow and filled circles), its annual revenue increased from 466 to 540 SEK (ca.15.9% increase). However, the savings in the electricity costs of small energy end-users are reduced slight. This is because the increase of PV power production due to climate change is distributed in summer months (i.e., from June to October, see Fig. 9.6). The small energy end-users can already achieve good self-sufficiency in these months, and thus, the PV power production increase in the future will not

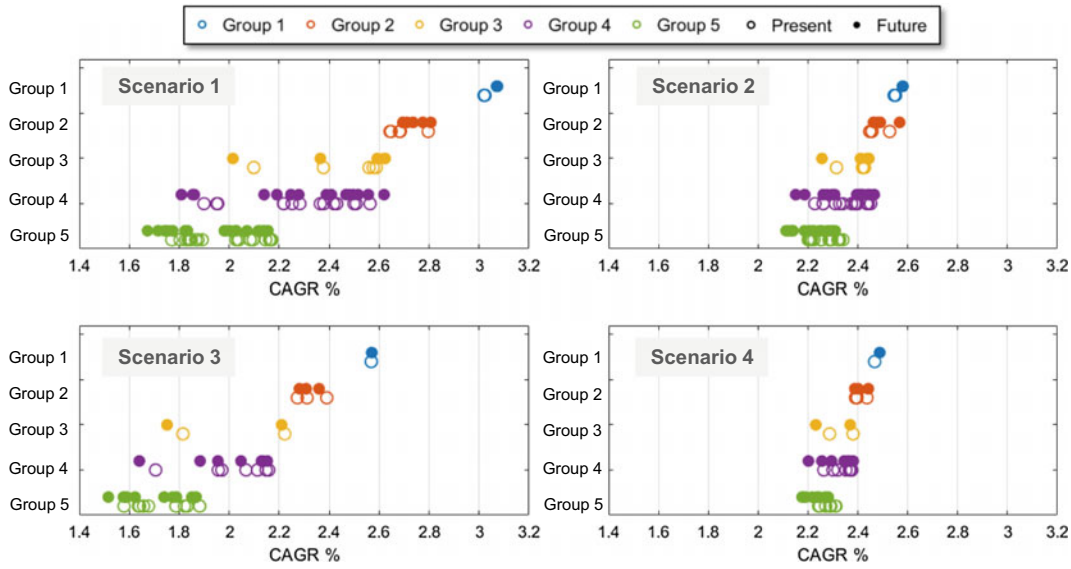

Fig. 9.9 Comparison of the CAGR in the four scenarios under both the present and future climates

contribute to the cost saving. While in winter months under the future climate, the PV power production will reduce, which will lead to reduce cost savings. As a whole, the small energy end-users have less electricity cost savings in the future climate.

In Scenario 2, due to a higher energy trading prices within the community (i.e., 99% of the grid price), the saving of electricity costs becomes smaller, but the revenue becomes larger as compared to Scenario 1. Again, for large energy use households (i.e., Group 1 with five occupants), both the savings and revenues will increase under the future climate. For instance, for the end-user with the lowest revenue (i.e., the leftmost hollow and filled circles), its annual revenue increased from 68 to 95 SEK (ca. 40% increase), and its annual cost saving decreased from 1169 to 1224 SEK (ca. 4.7% increase). But for small energy use households (i.e., Group 5 with one occupant), the savings will be reduced slightly, but the revenues will be increased a lot under the future climate. For the end-user with the highest revenue (i.e., the rightmost hollow and filled circles), its annual revenue increased from 580 to 652 SEK (ca. 12.4% increase), and its annual cost saving decreased from 557 to 537 SEK (ca. 3.6% decrease).

In Scenarios 3 and 4, since only 50% of the households have PV ownership, the revenues and savings for these households with PVs are much larger compared to Scenarios 1 and 2. Since in Scenario 4, the selling price of power within the community is higher than Scenario 3 (very close to the grid price), the average cost savings are relatively lower, but the revenues are higher. As can be seen, for the households with PV ownership, the impacts of climate change are very similar to Scenarios 1 and 2: The climate change leads to increased savings and revenues for large energy end-users. While for small energy end-users, it leads to reduced cost savings but larger revenues. Note for the households without PV ownership, the climate change also has positive impacts by increasing their savings in the electricity costs. This is because there is more power shared by the peer households in the community, which is contributed by the enhanced PV power production. In Scenario 4, the cost savings of these households are much smaller than Scenario 3.

Figure 9.9 compares the Compound Annual Growth Rate (CAGR) of each household in the four scenarios under both the present and future climates. Note in Scenarios 3 and 4, the CAGRs are calculated only for the households with a PV ownership. If a household has no investment in

Table 9.6 Comparison of CAGR (%) under both the present and future climates for different groups

Scenario	Climate	Group 1	Group 2	Group 3	Group 4	Group 5
1	Present	3.02	2.69	2.44	2.29	1.97
	Future	3.07	2.74	2.44	2.29	1.90
	Relative changes	*2%*	*2%*	*0%*	*0%*	*−3%*
2	Present	2.55	2.47	2.40	2.35	2.26
	Future	2.58	2.49	2.40	2.34	2.20
	Relative changes	*1%*	*1%*	*0%*	*−1%*	*−3%*
3	Present	2.57	2.33	2.02	2.03	1.74
	Future	2.57	2.32	1.98	2.01	1.69
	Relative changes	*0%*	*0%*	*−2%*	*−1%*	*−3%*
4	Present	2.47	2.41	2.33	2.34	2.28
	Future	2.49	2.41	2.30	2.32	2.23
	Relative changes	*1%*	*0%*	*−1%*	*−1%*	*−2%*

the PV system, the CAGR is 0. Table 9.6 compares the mean of CAGR for each group of households in all the scenarios. A larger CAGR indicates a better economic performance.

It can be observed in Fig. 9.9 that Scenario 2 is fairer than Scenario 1 in terms of CAGR, as well as Scenario 4 compared to Scenario 2. This is consequence of the difference in price: Scenarios where local electricity is sold at a very low price favors larger households (i.e., group 5 and 4) over smaller ones due to their larger annual cumulative consumption of local electricity. In general, savings produce a larger benefit compared to revenues because they amount to the whole price of the electricity, but their advantage becomes minor when local electricity is expensive. In other words, expensive local energy generates more revenues from the sale of electricity within the community, this consequentially reduces the savings potential for the receiver of this local energy. Contrary to what stated in a previous study, it can be seen in Fig. 9.9 that owners of larger shares of PV, in scenarios with uneven ownership (i.e., 3 and 4), cannot reach the same CAGR they obtain when having smaller shares. This is due as well to the fact that they have a larger PV system relative to their size, and therefore, the share of electricity self-consumed is minor. Unsurprisingly, the scenarios with higher prices of local electricity (2 and 4) are characterized by a smaller difference in CAGR between different ownership structures. In other words, there is less difference in Scenario 2 from 4 then Scenario 3 from 1. Despite having a lower CAGR, the cumulative earning of these households is almost double to what they had in Scenario 1 or 2. In a case in which the local electricity has the same price of the electricity from the grid the CAGR would be the same regardless of the ownership structure.

As can be seen from the relative changes of CAGR in the four scenarios in Table 9.6, in the future the large energy end-users (i.e., Group 1) will have $0 \sim 2\%$ higher CAGR values, as compared to the present scenario. This is because the large energy end-users will have increased savings (as more PV power can be used to meet their own power demand) and revenues (as more PV power can be sold), as can be seen from Fig. 9.8. While the CAGR values will decrease by $2 \sim 3\%$ for small energy end-users (i.e., Group 5) in the future scenario as compared to the present scenario. This is because the CAGR is more correlated to the cost savings contributed

by the PV system. Compared to being sold to the peer households, the PV power can make bring more benefits to the household economics if it is used by the household itself. Since in the future scenarios, the small energy end-users have reduced cost savings (see Fig. 9.8), their CAGR values are smaller.

9.5 Discussion of the Chapter Results

This section summarizes the performance changes under the future climate change and discuss findings from the analysis for the real application. Table 9.7 summarizes the performance variations under the future climate change for different energy use households considering different scenarios. From Table 9.7 and the case study results in Sect. 7.4.2, the following findings can be obtained.

Overall, the future climate change will increase the difference between the CAGR of different households. The households with large CAGR under the present climate will have even larger CAGR in the future (i.e., the economic performances become better), while the households with small CAGR under the present climate will have even smaller CAGR in the future (i.e., the economic performances become worse).

The future climate change is more beneficial to large energy use households, i.e., they can have increased cost savings, revenues, and CAGR. This is because the large energy use households can consume more of the increased PV power production under the future climate.

The future climate change is less beneficial to small energy use households. This is because the increased PV power production in the future summer will not help increase the self-consuming of the small energy use households (as they are already self-sufficient). On the contrary, the decrease of PV power production in the future winter will reduce the self-consuming of the small energy use households, which will reduce the cost savings and eventually the CAGR.

Table 9.7 Summary of the performance variations under the future climate change

Households	Scenarios		Performances			
			SS	Cost savings	Revenues	CAGR
Small energy users	**1:** Equal PV ownership, low trading price		↑	↓	↑	↓
	2: Equal PV ownership, high trading price		↑	↓	↑	↓
	3: Half PV ownership, low trading price	With PV	↓	↓	↑↑	↓
		Without PV	↑	↑	–	–
	4: Half PV ownership, high trading price	With PV	↓	↓	↑↑	↓
		Without PV	↑	↑	–	–
Large energy users	**1:** Equal PV ownership, low trading price		↑	↑	↑	↑↑
	2: Equal PV ownership, high trading price		↑	↑	↑	↑
	3: Half PV ownership, low trading price	With PV	↓	↑	↑↑	↑
		Without PV	↑	↑	–	–
	4: Half PV ownership, high trading price	With PV	↓	↑	↑↑	↑
		Without PV	↑	↑	–	–

Note '↑' indicates performance improving, i.e., the performance becomes better in the future. '↓' indicates performance deteriorating, i.e., the performance becomes worse in the future. '–' represents not applicable. Double symbols represent relatively more changes in the performances

Based on these findings, the following conclusions can be drawn to guide the decision making of the PV ownership and price setting under the future climate change to facilitate real applications.

It is not economical for a household to have a PV ownership larger than its demands, especially under the future climate change. When a household have surplus power frequently, it has to either sell the power to the community or to the grid, which makes the return of investment worse (especially when selling to the grid at a much cheaper price).

Specially, for small energy use households, if they want to improve the economic performances in the future, they can set their share of PV ownership equivalent to their demands in the community. In this chapter, despite their small demand, they have equal ownership as the large energy use households. As a result, they have surplus power production frequently and will have to sell it. If they can reduce the PV ownership to be equivalent to their demands, the unnecessary surplus PV power exports can be reduced and thus the CAGR values will be higher.

High price of energy trading can improve the fairness of the economic performances in the community, especially when the some of the households in the community do not have any ownership of the PV system. It can help keep the CAGR values of various households with a PV ownership in a narrow range, and thus leading to similar return of investment in a PV system. Therefore, if a community wants to incentivize all the households to have some share of the PV system, it is preferable to set the energy sharing price high.

Another way to improve the CAGR of PV ownership to mitigate the negative impacts of future climate change on the small energy use households is to install energy storage system. This can help keep more PV power to be used by the household itself. However, the investment of energy storage system could potentially increase the payback period of the total system (including PV and energy storage).

9.6 Summary

This chapter has conducted a systematic investigation of the impacts of climate change on the P2P energy trading performances under different pricing strategies and PV ownerships. Case studies have been conducted using the data from a building community located in Ludvika, Sweden. The future weather data of Ludvika was produced using the Morphine method. An agent-based modeling method was developed to simulate the household P2P trading behavior. Four different scenarios, i.e., two with different PV ownership (100% households have an ownership, or 50% households have an ownership) and two with different prices (a cheap price or prices close to the grid prices), were considered and the P2P performances under the four scenarios were studied and compared. The key findings from this chapter are summarized as below.

Due to the climate change, the annual PV power production will increase by 10.7% in Ludvika in the future scenario compared to the present scenario. The PV power production will increase dramatically in summer months (e.g., 24.8% in July and 28.8% in August) but decrease in winter months.

Overall, the future climate change has positive impacts on the self-sufficiency. The increased PV power production in the future scenario will lead to an increased in the household PV power self-sufficiency. For the case that 100% of the households have a PV ownership, the average PV power self-sufficiency will increase by 5.4% in the future scenario. For the case that 50% of the households have a PV ownership, the average PV power self-sufficiency will increase by 6.2% in the future scenario.

Due to the increased PV power production in the future scenario, the sum of cost savings and revenues will increase for all the households under all the pricing strategies and PV ownerships. For large energy end-users, both the cost savings and revenues will increase. While for small energy end-users, the cost savings will be reduced slightly, as in the future winter scenario,

the PV power production will be reduced, leading to reduced PV power self-usage.

Overall, under the equal PV ownership scenarios, the future climate change will increase the difference between the CAGR of different households. The households with large CAGR under the present climate will have even larger CAGR in the future (i.e., the economic performances become better), while the households with small CAGR under the present climate will have even smaller CAGR in the future (i.e., the economic performances become worse). This is because the large energy use households (which already have large CAGR under the present climate) can consume more of the increased PV power production under the future climate. While the small energy use households (which already have small CAGR under the present climate) have to sell more surplus PV power under the future climate.

It is not economical for a household to have a PV ownership larger than its demands, especially under the future climate change. When a household have surplus power frequently, it has to either sell the power to the community or to the grid, which makes the return of investment worse (especially when selling to the grid at a much cheaper price). The return worsens in relation to the investment, but in terms of shear earnings it improves (i.e., in terms of the gross SEK that a household earns).

High price of energy trading can improve the fairness of the economic performances in the community under both the present and future climates. It can help keep the CAGR values of various households with a PV ownership in a narrow range, and thus leading to similar return of investment in a PV system. If a community wants to incentivize all the households to have some share of the PV system, it is preferable to set the energy sharing price high.

It should be mentioned that the study has not considered the impacts of the uncertainties/errors of the weather prediction results from the morphing method. The associated uncertainty analysis will be considered as a part of our future studies. This chapter has discussed the ownership of PV systems, but for those households with an ownership, the PV capacity is the same. Future work will consider more diversified PV ownership considering the individual household power demand. Another potential factor affecting the P2P energy trading performances is the integration of energy storage. When households have their own energy storage, they may tend to store their surplus power in the storage, instead of sharing the surplus with the peer households in the community. Future work will also try to investigate the impacts of energy storage integration on the P2P energy trading performance.

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Differential Evolution-based System for Net-zero Energy Buildings Under Climate Change

10

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Abstract

Net-zero energy building (NZEB) is considered a solution to the increasing energy problems. A proper system design is crucial for a NZEB to achieve the desired performance during its lifecycle. Most conventional design methods utilize TMY (typical meteorological year) data or multi-year historical data for NZEB system sizing. Due to the climate change, future weather data may differ considerably from these utilized data. Consequently, these designs may not guarantee NZEBs to achieve the expected performance during their lifecycle. Therefore, this chapter proposes a differential evolution-based system design for NZEBs under climate change. Using the predicted future weather data, the proposed system design can optimize building system sizes for minimizing its lifecycle cost with user-defined performance constraints satisfied. Three performance constraints were considered and they were thermal comfort, energy balance, and grid interaction. Using the real future weather data, the proposed design has been validated by comparing with two conventional designs (i.e., TMY data-based design and multi-year historical data-based design). The results indicated that the proposed design can achieve better performance in terms of lifecycle cost and constraints satisfaction. With improved performance, the proposed design can be used in practice for NZEB system sizing especially as climate change is considered.

Keywords

Net-zero energy building · System design optimization · Climate change · Lifecycle cost · Multi-criteria constraints

10.1 Introduction

According to the U.S. Energy Information Administration (EIA), the global primary energy consumption and CO_2 emission have grown by 85% and 75% from 1980 to 2012, with annual average increases of 2.7% and 2.3%, respectively (Sieminski 2013). Problems of energy consumption and greenhouse gas emission have become primary concerns of the world. Net-zero energy building (NZEB), utilizing renewable energy generation to meet its own energy demand (Sartori et al. 2012), has been considered

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as an effective means to achieve energy-saving and carbon emission reduction. To promote NZEB's practical applications, different countries have legislated clear targets (Crawley et al. 2009; Recast 2010; Lu et al. 2015a, b).

A NZEB has a lifecycle of fifty years or even longer (Sartori, Napolitano et al. 2012; Cui et al. 2015). To achieve a NZEB's expected performance in lifecycle, its systems (e.g., air-conditioning system and renewable system) need to be well designed by considering influential factors' variations in building lifecycle. Outdoor weather conditions are crucial for the estimations of the building thermal/electrical loads as well as the renewable energy generation, and thus they will affect the design of NZEB systems (Handbook 2009; Hopfe 2009; Nema et al. 2009; Robert and Kummert 2012; Athienitis and O'Brien 2015). For instance, the extreme outdoor temperature directly influence a building's peak cooling/heating load and thus influence associated air-conditioning (AC) system size (Handbook 2009), while solar radiation and wind speed will influence the renewable generations and thus affect the sizing of renewable systems (Nema, Nema et al. 2009). For this reason, the associated weather data and their changes in NZEB lifecycle should be systematically considered in the system design.

Typical meteorological year (TMY) weather data, with significant convenience and reduced computation, have been widely used for existing system designs (Belcher et al. 2005; Iolova and Bernier 2007; Attia et al. 2012; Fong and Lee 2012; Todorović 2012; Eshraghi et al. 2014; Lu et al. 2015a, b). Using the local TMY data, Lu et al. (Lu, Wang et al. 2015a, b) optimized the renewable system design for a NZEB in Hong Kong. In North-America, the TMY data have been commonly used to assess the performance of the design systems in NZEBs (Iolova and Bernier 2007). Using the TMY data, a multi-stage simulation-based optimization method was proposed to identify the cost-effective system design for a NZEB in Finland (Belcher et al. 2005). In fact, TMY data are single-year data derived from a multi-year database, and they can well represent typical weather conditions over the last several decades (Herrera et al. 2017). But, ignoring the effects of the climate change, the TMY data cannot reveal the variations of the weather conditions in long terms. Existing studies have already shown such significant changes of weather conditions which cannot be neglected. For instance, it was found that in the Chinese cities, the variation of the annual average dry bulb temperature was up to 4.5 °C within 55 years in comparison with local TMY data (Cui et al. 2017). Consequently, conventional TMY-based design methods may lead to improper system designs which may not be able to achieve the expected performance in multiple perspectives (Athienitis and O'Brien 2015).

In recent years, researchers recognized the limitation of the TMY-based design methods, and they chose to use the multi-year weather data instead of the TMY data (Hong et al. 2013; Cui et al. 2017). The multi-year weather data can be the real historical ones or artificial ones that are generated based TMY data using sampling approaches (Macdonald 2002; Domínguez-Muñoz et al. 2010). Comparing with TMY data representing average weather conditions, the multi-year weather data contain more information on climate change (Kershaw et al. 2010; Pernigotto et al. 2014). Existing studies have demonstrated the improved performance of the multi-year weather-based design methods (Sun et al. 2014, 2015; Gang et al. 2015; Zhang et al. 2016; Lu et al. 2017; Huang et al. 2018). Zhang et al. (2016) conducted optimal system design for a NZEB using the 400-year sampled weather data. The overall performance of the proposed method was improved by 24% compared with the conventional TMY-based method. Sun et al. (2014) performed comparative analysis of the AC system design using the historical weather data and the TMY weather data and found that the risks of improper system sizing can be largely decreased when the multi-year weather data were used. Taking the multi-year sampled weather data as inputs, Yu et al. (2016) proposed a GA-based system design method for a NZEB considering practical constraints. The effectiveness of the proposed method was further validated using the 20-year real weather data.

However, the abovementioned design methods are still historical data based, i.e., based on either a typical historical year or multiple historical years. Weather data of a single typical year contain little information on climate changes, while weather data of multiple historical year contain information on climate changes of past instead of future. In fact, with complex climate change effects, future weather can be significantly different from not only TMY data but also multiple historical years' weather data. For example, the extreme weather conditions were found to occur more frequently and the air temperature could rise up more quickly in the future (Fouillet et al. 2006; Stocker 2014). It was also reported that the maximum outdoor temperature in France can exceed the seasonal norm by 11 °C for nine consecutive days due to the climate change (Fouillet et al. 2006). According to the latest report by the Intergovernmental Panel on Climate Change (IPCC), the global air temperature was projected to rise 3–4 °C in the 21th century in comparison with that in the late twentieth century (Stocker 2014). Meanwhile, Robert and Kummert (Robert and Kummert 2012) showed that in comparison with 1961–1990, the average wind speed increased by 7.4% in winter and decreased by 9.2% in summer in 2050s, thereby leading to surplus wind energy in winter but insufficient wind energy in summer.

With limited considerations of the climate change effects, the conventional historical data-based design methods may not be able to guarantee the desired performance during a NZEB's lifecycle in different aspects, e.g., thermal comfort, energy balance and grid interaction. Climate change can have direct impacts on a NZEB's indoor thermal comfort due to the variations of extreme weather conditions in the future. For instance, Gupta and Gregg (Gupta and Gregg 2012) found that because of more frequent extreme hot weather conditions, the TMY-based system design had much poorer thermal comfort performance in the future summers, i.e., the indoor temperature set-point unmet hours dramatically increased from 12 (under TMY weather) to 474 (under future 2030s' weather). Meanwhile, climate change can affect both building energy demand and renewable energy generation, thereby influencing annual energy balance. Santamouris et al. (2015) reported building energy demand increased at a rate of 0.5%–8.5%/°C in response to outdoor temperature rise under climate change. Shen and Lior (2016) showed that in comparison with 1976–2005, the annual mean solar radiation would decrease by 5.0% in 2040–2069, which led to reduced solar energy generation. For this reason, a solar powered NZEB may not be able to achieve the energy balance target in future years. Regarding grid interaction, climate change can influence a NZEB's power mismatch (i.e. difference between building power demand and renewable power generation), and thus further influence electrical power exchange between the NZEB and the grid. Salom et al. (2014) proved that the peak power delivered to the grid may not be able to be limited at a desired level due to climate change. A larger peak power delivery could increase the grid stress on power balance and even deteriorate the grid power supply quality.

With the abovementioned chapter and analysis, there is a need to systematically consider the climate change impacts on NZEB system design. Thus, this chapter proposes a differential evolution-based system design optimization method, which aims at improving NZEB lifecycle performance in different aspects. The major contributions of the chapter include:

- Considering climate change and associated impacts, the proposed design method is able to minimize a NZEB's lifecycle cost while meeting users' defined performance requirements in multi-criteria (e.g. thermal comfort, energy balance, and grid interaction).
- Using the real future weather data, the proposed method has been verified to have lifecycle performance improvements by comparing with two conventional design methods using TMY data and multi-year historical data respectively.

The chapter is organized as follows. In Sect. 10.2, the proposed differential evolution-based optimal system design method is introduced. Section 10.3 presents the dynamic NZEB

platform including building and energy system models. Case studies are conducted to validate the proposed method in Sect. 10.4. Comparative and conclusive remarks are drawn in the last section.

10.2 Methodology

10.2.1 Overview

Considering climate change, the proposed method aims at minimizing the lifecycle cost while satisfying the user-defined multi-criteria performance requirements/constraints including thermal comfort, energy balance, and grid interaction. Figure 10.1 shows the basic idea of the differential evolution-based NZEB system design under climate change. The proposed method contains three steps.

In the first step, the morphing method is employed to predict the future weather data. Due to its flexibility and accuracy, the morphing method has been commonly adopted for future weather predictions (Belcher, Hacker et al. 2005; Guan 2009; Wang and Chen 2014). The principle of the morphing method is to combine a baseline hourly weather data file with the future monthly weather data predicted by the proper global climate models (GCMs). In the second step, the differential evolution algorithm will be utilized to search the optimal system sizes with minimized lifecycle cost while satisfying the user-defined performance requirements in the aspects of the thermal comfort, energy balance, and grid interaction. The differential evolution algorithm is selected mainly because it is more robust and easier to be implemented than other evolution algorithms (e.g., genetic algorithm) (Storn and Price 1997; Wang et al. 2014; Shen et al. 2019). In the third step, the effectiveness of the proposed method will be validated by comparing with two conventional design methods using the real future weather data (i.e., TMY data-based design and multi-year historical data-based design).

10.2.2 Prediction of Future Weather Using the Morphing Method

10.2.2.1 Generation of Typical Meteorological year (TMY)

The single-year TMY weather data can be created by selecting twelve typical meteorological months (TMMs) from the multiple historical years (i.e., 1980–1997). The Finkelstein-Schafer (FS) method is a commonly adopted method for TMMs selection due to its convenience and simplicity. The FS method is briefly described as follows. First, a set of daily weather variables (e.g., daily mean and maximum outdoor temperatures) are generated from the raw weather datasets. Then, the Finkelstein-Schafer (FS) statistics is used to measure the closeness between the short-term and long-term CDFs of each daily weather variable, as shown in Eq. (10.1).

$$\mathrm{FS}_x(y,m) = \frac{1}{N}\sum_{i=1}^{N}\left|\mathrm{CDF}_m(x_i) - \mathrm{CDF}_{y,m}(x_i)\right| \tag{10.1}$$

where N is the number of values for daily weather variable x in the month m over the long-term period (18 years in this chapter); CDF_m and $\mathrm{CDF}_{y,m}$ are the long-term and short-term (for the year y) cumulative distribution functions of the daily weather variable x for month m, respectively.

Last, the FS statistics of each weather variable (i.e., FS_x (y,m)) is multiplied by the weighting factor WF_x of the weather variable x, and the associated results are summed to produce a cumulative FS, as shown in Eq. (10.2). Note that existing studies have proposed methods in selecting proper WF_x (Hall et al. 1978; Chan et al. 2006). The month with the smallest cumulative FS will be selected as the typical month. Details of TMMs selection procedures can be found in the existing studies (Hall et al. 1978; Chan et al. 2006).

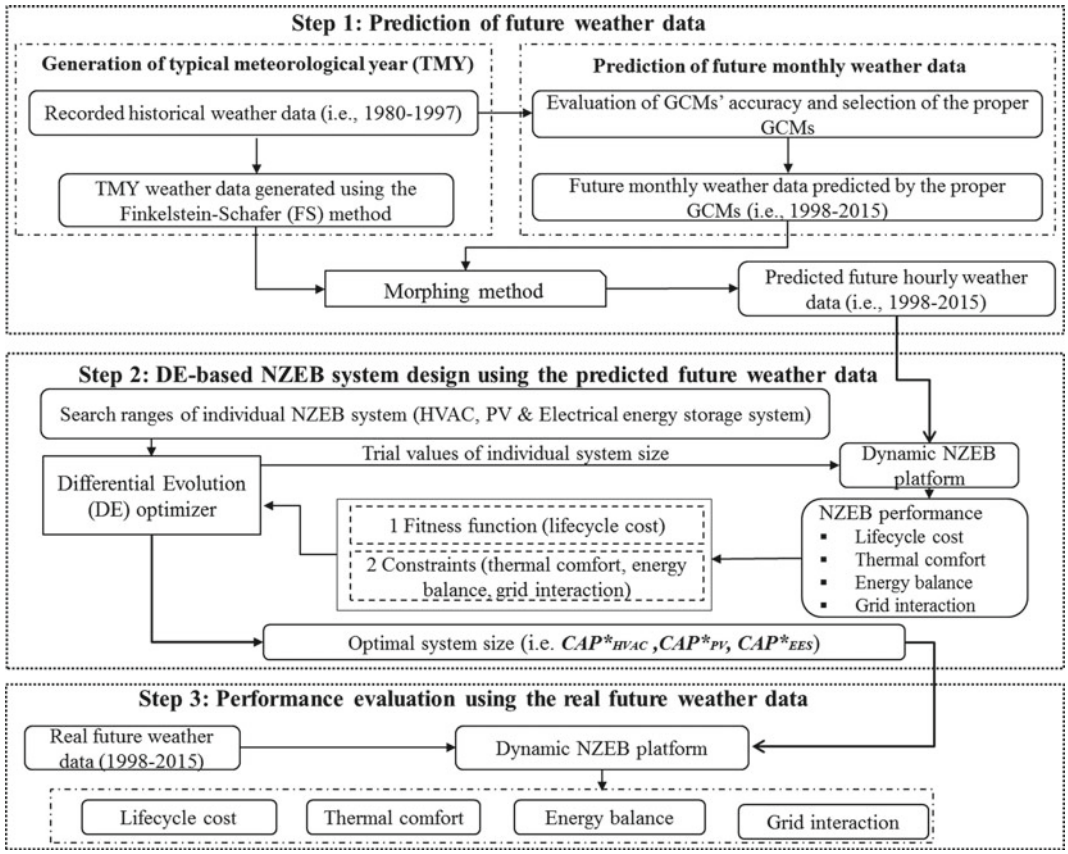

Fig. 10.1 Basic idea of differential evolution-based NZEB system design under climate change

$$\mathrm{WS}(y,m) = \sum_{x=1}^{M} \mathrm{FS}_x(y,m) \times \mathrm{WF}_x \qquad (10.2)$$

where WS is the cumulative FS; M is the number of daily weather variables.

10.2.2.2 Prediction of Future Monthly Weather Data Using the Identified GCMs

In this chapter, the global climate models (GCMs) are adopted to predict the long-term monthly weather data. Different GCMs have different horizontal resolutions (latitude × longitude), thereby resulting in uncertainties of future weather predictions (Stocker 2014). Thus, the GCMs need to be carefully selected. The main selection procedures are shown as follows. First, 10 GCMs in total with high horizontal resolutions are preliminarily selected from the multi-model datasets (Taylor et al. 2012). Second, to identify the top 3 GCMs with high accuracies, the prediction results of these models will be compared with the real historical weather data measured by the Hong Kong Observatory (HKO). In the identification process, the statistical metric MAPE is used to evaluate the prediction performance of these GCMs, as shown in Eq. (10.3). A smaller MAPE indicates higher prediction accuracy. Last, using the multimodal ensemble average method (Reichler and Kim 2008), the top 3 GCMs are used for the prediction of future monthly weather (X), as shown in Eq. (10.4).

$$\mathrm{MAPE} = \sum_{i=1}^{n} \left| \frac{P_i - M_i}{M_i} \right| \times \frac{100\%}{n} \qquad (10.3)$$

where MAPE is the mean absolute percentage error; P_i and M_i are the predicted and measured weather data in the ith month, and n is the total months within the evaluation period (i.e., 1980–1997).

$$X = \frac{1}{3} \times \sum_{i=1}^{3} X_i \tag{10.4}$$

where X_i (i = 1, 2, 3) is the predicted monthly weather from the top 3 GCMs, respectively.

10.2.2.3 Morphing Method

As the future weather data obtained from the identified GCM are in monthly interval, the morphing method (Belcher et al. 2005) is needed to downscale the predicted monthly weather data to the hourly baseline (i.e., TMY). For different weather variables, existing studies have proposed different downscale algorithms as below (Belcher et al. 2005; Guan 2009).

(1) A stretch of α_m is applied to generate the future hourly solar radiation y based on the TMY hourly solar radiation y_0.

$$y = \alpha_m y_0 \tag{10.5}$$

where α_m is the fractional monthly change of y from the selected GCM in the mth month.

(2) A combination of shift (Δz_m) and stretch (β_m) is applied to generate the future hourly outdoor temperature z based on the TMY hourly temperature z_0.

$$z = z_0 + \Delta z_m + \beta_m \left(z_0 - (z_0)_m\right) \tag{10.6}$$

where Δz_m and β_m are the absolute and fractional monthly changes of z from the selected GCM in the mth month respectively; $(z_0)_m$ is the monthly mean value of z under TMY in the mth month.

10.2.3 Differential Evolution-Based NZEB System Design Using the Predicted Weather Data

The inputs of the differential evolution-based system design include the system searching ranges and the predicted future weather. The considered systems include an air-conditioning system, a photovoltaic panel system, and an electrical energy storage system. During the differential evolution-based system sizing process, a set of individual system sizes is generated in each generation by the differential evolution optimizer, and these trials together with the predicted future weather are taken as inputs for lifecycle performance assessment via a dynamic NZEB platform. These lifecycle performances include lifecycle cost, thermal comfort, energy balance, and grid interaction. The lifecycle cost is considered as the fitness function, while the other three performances are regarded as constraints which represent the user-defined performance requirements in multi-criteria. The system trialed sizes that meet the constraints are taken by the differential evolution optimizer and further used to produce next generations with reduced lifecycle costs. The trial process will stop until the lifecycle cost is minimized (Wang et al. 2014).

10.2.3.1 Fitness Function of the Differential Evolution Optimizer

The lifecycle cost indicates the economic sustainability, and reduction of lifecycle cost is helpful to improve associated long term cost-effectiveness. This chapter takes the lifecycle cost as the fitness function (J) in the differential evolution optimizer with multi-criteria constraints considered, as shown in Eq. (10.7). The lifecycle cost is the sum of initial cost and operational cost, which are calculated by Eq. (10.8) and (10.9) respectively.

$$J = \min\left(\text{Cost}_{\text{initial}} + \sum_{i=1}^{N} \text{Cost}_{\text{annual},i}\right); \quad \text{subject to } C_{\text{comfort}},\ C_{\text{balance}},\ C_{\text{grid}} \tag{10.7}$$

$$\text{Cost}_{\text{initial}} = u_{\text{AC}} \times \text{CAP}_{\text{AC}} + u_{\text{PV}} \times \text{CAP}_{\text{PV}} + u_{\text{EES}} \times \text{CAP}_{\text{EES}} \tag{10.8}$$

$$\text{Cost}_{\text{annual}} = u_{\text{ele,imp}} \times E_{\text{imp}} - u_{\text{ele,exp}} \times E_{\text{exp}} \tag{10.9}$$

where $\text{Cost}_{\text{initial}}$ and $\text{Cost}_{\text{annual}}$ are the initial cost and annual operational cost, respectively; C_{comfort}, C_{balance} and C_{grid} are the practical constraints for thermal comfort, energy balance, and grid interaction, respectively (see Eq. 10.15–10.17); N represents the service life of the energy systems. u_{AC}, u_{PV} and u_{EES} are the unit prices of air-conditioning system, PV system and electrical energy storage system respectively; CAP_{AC}, CAP_{PV} and CAP_{EES} are the capacities of each energy system. $u_{\text{ele,imp}}$ and $u_{\text{ele,exp}}$ are the price of buying/selling electricity from/to the grid, respectively; E_{imp} and E_{exp} are the annual amount of imported/exported electricity, respectively.

10.2.3.2 *Search Constraints Based on User-Defined Performance Requirements*

With references to existing studies (Salom et al. 2014; Huang et al. 2015; Cubi et al. 2017), this section firstly introduces three indices to evaluate the performances in thermal comfort, energy balance, and grid interaction. Since users may have different performance requirements in the three aspects, the chapter considers the user-defined performance requirements as the search constraints for the system design optimization. In other words, the searched optimal design results must fulfill these user-defined performance requirements.

Thermal comfort represents the users' satisfaction with the indoor thermal environment. The adopted thermal-comfort index (Ψ_{comfort}) indicates the annual unmet hours in which the supplied heating/cooling cannot meet the actual heating/cooling load (Huang et al. 2015; Zhang et al. 2019), as shown in Eq. (10.10). Here, a smaller value of Ψ_{comfort} means improved thermal comfort.

$$\Psi_{\text{comfort}} = \sum_{j=1}^{8760} \tau_j \begin{cases} \tau_j = 1, \text{ if } L_{\text{sup},j} \le L_{\text{actual},j} \\ \tau_j = 0, \text{ if } L_{\text{sup},j} \ge L_{\text{actual},j} \end{cases} \tag{10.10}$$

where τ_j represents the value of unmet hour at the jth hour in a year; L_{sup} and L_{actual} are hourly heating/cooling supply and actual heating/cooling demand, respectively.

Regarding energy balance, an index (Ψ_{balance}) is used in the chapter to evaluate whether a NZEB's annual generation can meet its annual energy demand (Salom et al. 2014), as expressed in Eq. (10.11). Here, if the value of Ψ_{balance} is larger than 1 (i.e., $\Psi_{\text{balance}} \ge 1$), the NZEB is considered to achieve the energy balance target.

$$\Psi_{\text{balance}} = E_{\text{generation}} / E_{\text{demand}} \tag{10.11}$$

where $E_{\text{generation}}$ and E_{demand} are the building annual energy generation and demand, respectively.

Grid interaction assesses the exchange of energy between a NZEB and a power grid. The adopted grid-interaction index (Ψ_{grid}) evaluates the annual total energy exchange (Cubi et al. 2017), as shown in Eq. (10.12–10.14). Note that there are many different indices related with grid interactions as well as grid friendliness, and the chapter selects the simplest one in which the overall energy exchange is considered. A smaller value of Ψ_{grid} indicates improved grid friendliness.

$$\Psi_{\text{grid}} = \sum_{j=1}^{8760} \left|\text{pow}_{\text{exchange},j}\right| \tag{10.12}$$

$$\text{Pow}_{\text{exchange},j} = \text{Pow}_{\text{mismatch},j} - \text{Pow}_{\text{charge},j} \tag{10.13}$$

$$\text{Pow}_{\text{mismatch},j} = \text{pow}_{\text{generation},j} - \text{pow}_{\text{consumption},j} \tag{10.14}$$

where $pow_{exchange,j}$ is hourly power exchange between a building and a power grid; $pow_{mismatch,\ j}$ is the hourly power mismatch; $powcharge,\ _j$ is the hourly power charge; $pow_{generation,j}$ and $pow_{consumption,j}$ are the hourly power generation and consumption of a NZEB respectively.

According to the above performance indices, the search constraints based on the user-defined performance requirements are established as below. In the differential evolution-based system design, the NZEB lifecycle performances are evaluated within N years rather than a single year. Thus, *N-year* data are obtained for each performance index, and they are expressed as $\Psi_{comfort[1,2,\ldots,N]}$, $\Psi_{balance[1,2,\ldots,N]}$ and $\Psi_{grid[1,2,\ldots,N]}$. Based on these annual performance results, the constraints are as shown in Eq. (10.15–10.17). Note that a performance threshold ($\widehat{\Psi}$) and an expected cumulative probability ($\xi_{expected}$) are introduced to reveal the performance requirements of different users. For instance, regarding thermal comfort, if a user selects the threshold $\widehat{\Psi}_{comfort} = 50h$ and the probability $\xi_{comfort} = 80\%$ (i.e., probability $\left(\Psi_{comfort[1,2,\ldots N]} \leq 50 \text{ hours}\right) \geq 80\%$), it means the designed system must ensure that more than 80% years in a building lifecycle have failure time (i.e. indoor temperature unmet hour) less than 50 h.

$$C_{comfort} : \text{Probability}\left(\Psi_{comfort[1,2,\ldots N]} \leq \tilde{\Psi}_{comfort}\right) \geq \xi_{comfort,expected} \tag{10.15}$$

$$C_{balance} : \text{Probability}\left(\Psi_{balance[1,2,\ldots N]} \geq \tilde{\Psi}_{balance}\right) \geq \xi_{balance,expected} \tag{10.16}$$

$$C_{grid} : \text{Probability}\left(\Psi_{grid[1,2,\ldots N]} \geq \tilde{\Psi}_{grid}\right) \geq \xi_{grid,expected} \tag{10.17}$$

where $\tilde{\Psi}_{comfort}$, $\tilde{\Psi}_{balance}$ and $\tilde{\Psi}_{grid}$ are user-defined performance thresholds for thermal comfort, energy balance, and grid interaction, respectively; $\xi_{comfort,expected}$, $\xi_{balance,expected}$ and $\xi_{grid,expected}$ are the expected cumulative probabilities for thermal comfort, energy balance, and grid interaction, respectively; and N is the number of years.

10.2.4 Validation Through Performance Comparisons Between the Proposed Method and Two Conventional Ones

Using the real future weather data (1998–2015), the proposed method will be validated by comparing with other two conventional design methods (i.e., TMY data-based design and multi-year historical data-based design). The performance comparisons will be conducted in aspects of lifecycle cost, thermal comfort, energy balance, and grid interaction. For a fair performance comparison, same differential evolution algorithm is used in the two conventional methods for searching optimal system sizes. The slight differences in the optimal search between the conventional methods and the proposed one are described as below.

1. In the TMY data-based design, the input design weather data are the single-year TMY data. Since only one year weather data are available for the design, the $\sum_{i=1}^{N} Cost_{annual,i}$ representing the lifecycle operational cost cannot be calculated and thus operational cost has not been included in the fitness function. Meanwhile, the performance constraints should be satisfied in one year instead of the probability format for multiple years (i.e., $\text{Probability}\left(\Psi_{comfort[1,2,\ldots N]} \leq \tilde{\Psi}_{comfort}\right) \geq \xi_{comfort}$). It should be mentioned that in the chapter, this conventional design method does not consider the safety factor since it heavily depends on the designer's experience and expertise. It could be difficult for the designers to select proper safety factors especially when complex and multiple systems are involved (Djunaedy et al. 2011; Woradechjumroen et al. 2014).

2. In the multi-year historical data-based design, the input design weather data are the multiple historical years' weather data. The other parts are exactly the same as the proposed method.

10.3 Dynamic NZEB Platform

In this chapter, a dynamic NZEB platform is constructed using the Trnsys software (Klein 2007). The platform includes a building model and the associated energy systems, i.e., an air-conditioning system, a PV system, and an electrical energy storage system. The air-conditioning system is equipped to provide indoor thermal comfort. The PV system is mainly used to provide the renewable generation. The electrical energy storage system is adopted to mitigate the power exchange between a NZEB and a power grid and thus improve the grid friendliness.

10.3.1 Building Modeling

A three-story office building located in Hong Kong is built using the multi-zone model (i.e., Type 56) in Trnsys, and each floor has the same size (i.e., 20 m long and 10 m wide). The building has two windows facing north and south on each floor, and the window-to-wall ratio is 0.25. The room temperature cooling set point is 26 °C. The ventilation rate is set as 1 ACH (Air Change per Hour) and the lighting load is set as 10 W/m^2. The occupant density is set as 12 m^2/person. A computer with a power of 80 W is assigned to each person, and it is on–off controlled according to the regular occupancy schedule (i.e., from 8:00 am to 18:00 pm) (China 2008). Note that in Hong Kong, heating is not needed at all.

10.3.2 Building Energy System Modeling

10.3.2.1 Air-Conditioning System

The studied air-conditioning system includes a cooling coil, a chiller, a cooling tower, two pumps, and a fan. The schematic diagram of the air-conditioning system is shown in Fig. 10.2.

The power consumption of the chiller W_{chiller} can be calculated according to the fraction of full load power (FFLP), rated cooling capacity ($Q_{\text{rated,chiller}}$) and nominal coefficient of performance of the chiller (COP_{nom}), as shown in Eq. (10.18).

$$W_{\text{chiller}} = \text{FFLP} \times \frac{Q_{\text{rated,chiller}}}{\text{COP}_{\text{nom}}} \tag{10.18}$$

The power consumption of the variable/constant speed pump can be calculated according to the

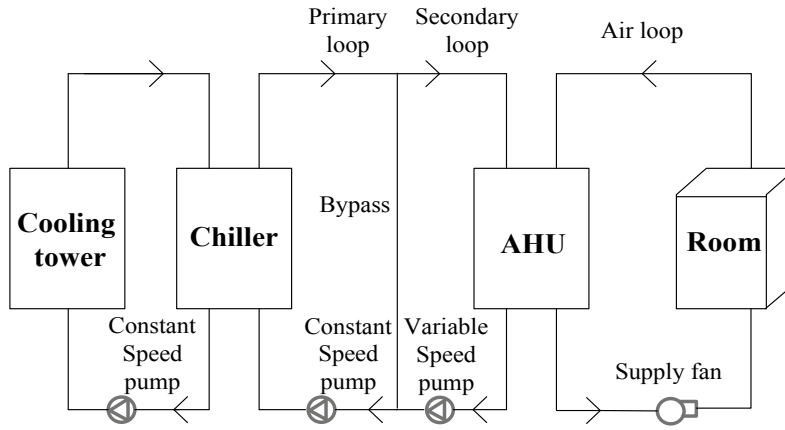

Fig. 10.2 Schematics of air-conditioning systems

water flow rate $\dot{m}_{r,w}$ and the associated water head ΔP_w, as shown in Eq. (10.19).

$$W_{\text{pump}} = \dot{m}_{r,w} \times \Delta P_w \tag{10.19}$$

The power consumption of the AHU fan can be estimated by the air flow rate $\dot{V}_{\text{air}}$ and the associated pressure drop of the air flow, as shown in Eq. (10.20).

$$W_{\text{fan}} = \dot{V}_{\text{air}} \times \Delta P_{\text{air}} \tag{10.20}$$

10.3.2.2 Renewable System

As solar radiation is considered as one of the most reliable renewable energy sources (Das et al. 2018), the PV panel is utilized to generate the renewable energy in this chapter. The PV panel model (i.e., Type 562) in TRNSYS is used to estimate the energy generation. Equation (10.21) shows the PV energy output (i.e., W_{PV}) with a selected size (i.e., CAP_{PV}).

$$W_{\text{PV}} = \tau \times \alpha \times \text{IAM} \times I_T \times \eta \times CAP_{\text{PV}} \tag{10.21}$$

where τ and α are the transmittance and absorptance coefficients of the PV cover respectively; IAM is the overall incidence angle modifier; I_T is the total incident solar radiation; and η is the overall efficiency of PV panel.

10.3.2.3 Electrical Energy Storage System

Using the electrical energy storage system, the power interaction between the building and the grid can be reduced through charging and discharging processes. In the charging process, the excessive renewable energy is firstly stored in the energy storage system, and the storage energy can be calculated by Eq. (10.22). Once the energy storage system is fully charged, the rest part of the excessive energy will be exported to the grid. While in the discharging process, the insufficient renewable energy is firstly supplemented by the energy storage system. Once the energy storage system is fully discharged, the rest part of the insufficient energy will be imported from the grid. The charging and discharging power (i.e., $\text{Pow}_{\text{charge},j}$) can be calculated by Eq. (10.23).

$$E_{\text{store},j} = \sum_{i=1}^{j-1} \text{Pow}_{\text{charge},i} \tag{10.22}$$

$$\text{Pow}_{\text{charge},j} = \begin{cases} \min((\text{CAP}_{\text{EES}} - E_{\text{store},j}), \text{Pow}_{\text{mismatch},j}) & \text{if } \text{Pow}_{\text{mismatch},j} > 0 \\ \max(-1 \times E_{\text{store},j}, \text{Pow}_{\text{mismatch},j}) & \text{if } \text{Pow}_{\text{mismatch},j} > 0 \end{cases} \tag{10.23}$$

where $E_{\text{store},j}$ is the energy stored in the energy storage system in the jth hour, CAP_{EES} is the capacity of energy storage system, and $\text{Pow}_{\text{mismatch},j}$ is the power mismatch in the jth hour (as shown in Eq. 10.10). A positive $\text{Powcharge},j$ indicates a charging process, while a negative one represents a discharging process.

10.4 Case Studies and Results Analysis

10.4.1 Future Weather Prediction and Validation

10.4.1.1 Selection of TMMs and GCMs for Future Weather Prediction

The values of the cumulative Finkelstein-Schafer (FS) statistics were calculated for each month in the 18-year period (1980–1997), and the results were used to select the twelve TMMs to form the TMY baseline data, as shown in Table 10.1. Meanwhile, the top 3 GCMs with high prediction accuracy were selected from the preliminary 10 high-resolution GCMs. The performance evaluations of the 10 GCMs are briefly summarized in Table 10.2. It reveals that the three models ACCESS1.3, MIROC4h, and CNRM-CM5 had the smaller MAPEs (less than 7.6%) than the other GCMs, and thus they were selected for future monthly weather predictions. Detailed selection results of the TMMs and the GCMs can be found in Appendix 1 and 2, respectively.

Table 10.1 Twelve TMMs selected from the multiple historical years (1980–1997)

TMM	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year	1995	1997	1984	1980	1997	1987	1990	1984	1993	1988	1988	1993

Table 10.2 Performance evaluation of the preliminary 10 GCMs

GCM	ACCESS1.3	MIROC4h	CNRM-CM5	MIROC5	HADGEM-ES	INM-CM4	GISS-E2-R	IPSL-CM5A-LR	GISS-E2-H	HadCM3
MAPE (%)	**6.878**	**7.192**	**7.589**	8.401	8.541	8.547	10.220	11.650	12.667	17.156

The bold indicates the associated information should draw more attention from the readers

10.4.1.2 Validation of the Predicted Future Weather

This chapter mainly considered the predictions of outdoor temperature and solar radiation, and the results were validated by comparing with the real data (1998–2015). The TMY weather data and multi-year historical weather data were also presented for comparisons.

Figure 10.3 shows the cumulative hours of outdoor temperature for different weather datasets scaled to one year. The cumulative hours were the product of 8760 h and the cumulative probability, which were generated by sorting the hourly outdoor temperature from the lowest to the highest in each dataset. The x axis was set from 26 °C (i.e., the outdoor temperatures in which cooling supply may need) to show the validation results of outdoor temperatures. In comparison with the TMY and historical ones, the predicted outdoor temperatures agreed better with the real ones due to its better revealing of climate change. Meanwhile, this figure also shows that the high outdoor temperatures occurred more frequently due to the climate change. For example, there were about 300 h (i.e., from 8460 to 8760) in which the outdoor temperatures were higher than 31 °C in the predicted case and real case, while the corresponding hours were only 210 h (i.e., from 8550 to 8760) in the TMY and historical years. Such differences of outdoor temperature can affect both indoor thermal comfort satisfaction and total energy use, thereby influencing the sizing results of the air-conditioning system and the renewable system.

Figure 10.4 presents the cumulative hours of hourly global solar radiation for different weather datasets scaled to one year. The cumulative hours were the product of 8760 h and the cumulative probability, which were generated by sorting the hourly solar radiation from zero to the highest in each dataset. It shows that the solar radiations of the four weather datasets were almost overlapped. In other words, the solar radiation varied little under climate change. It was in line with the findings from the existing studies (Guan 2009; Wan et al. 2011).

10.4.2 Optimal System Sizing Results and Validation of the Proposed Method

10.4.2.1 System Sizing Results from the Three Design Methods

The differential evolution algorithm was utilized to search the optimal system sizes in the proposed design and the two conventional designs (i.e., TMY data-based design and multi-year historical data-based design). The parameters of differential evolution algorithm were tuned in advance. In the differential evolution algorithm, the maximum step of interaction was set as 200 and the population size was set as 40 with a crossover probability of 0.5. Note that the mutation rate was 0.5. Regarding the practical constraints, the three performance thresholds (i.e., $\tilde{\Psi}_{\text{comfort}}$, $\tilde{\Psi}_{\text{balance}}$ and $\tilde{\Psi}_{\text{grid}}$) were set as the

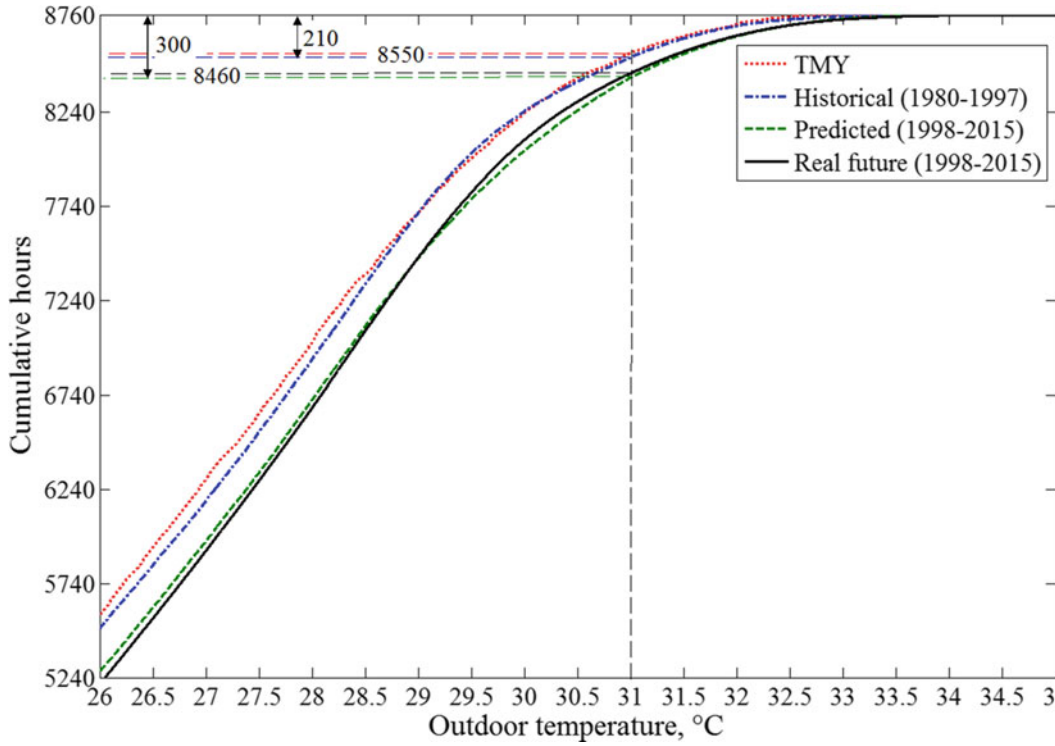

Fig. 10.3 Cumulative hours of outdoor temperature for different weather datasets scaled to one year

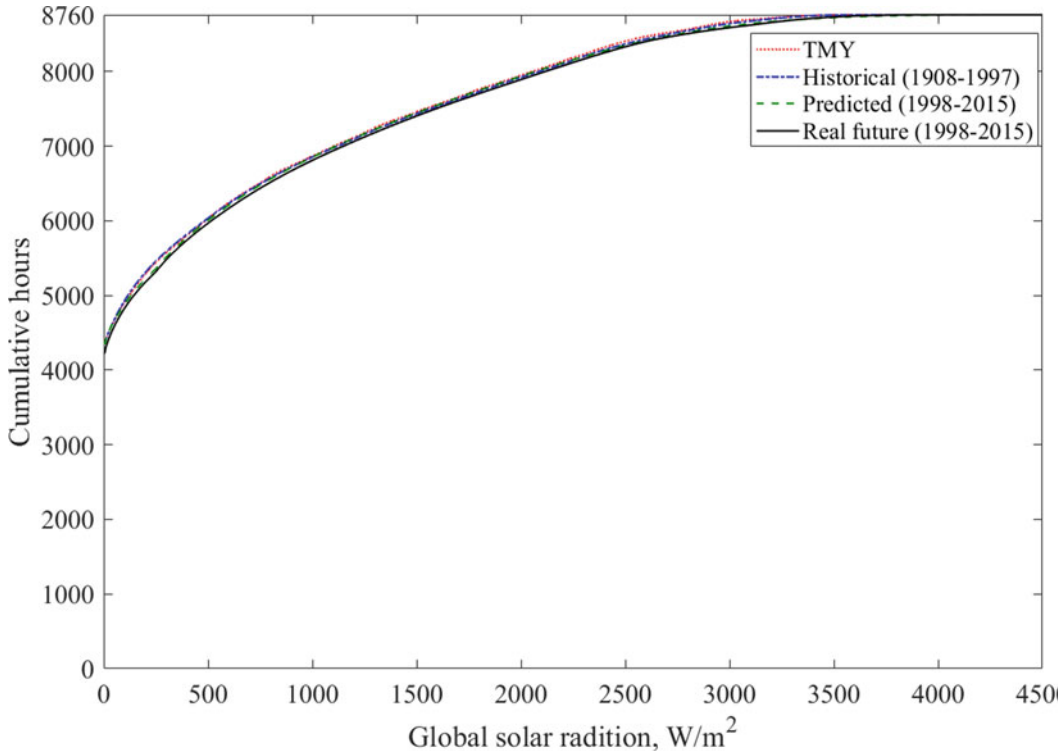

Fig. 10.4 Cumulative hours of global solar radiation for different weather datasets scaled to one year

Table 10.3 Summary of price information used in the fitness function

items	Unit	Price	References
Air-conditioning system	HKD/kW	1400	(Yu et al. 2016)
PV system	HKD/m^2	8000	(Eshraghi et al. 2014)
Electrical energy storage system	HKD/(kWh)	1600	(Yu et al. 2016)
Electricity (purchased)	HKD/(kWh)	2	(Shen et al. 2016)
Electricity (sold)	HKD/(kWh)	0.22	(Shen et al. 2016; Abdulla et al. 2018)

values of 35 h, 1 and 6000 kWh, respectively. The expected cumulative probability ($\xi_{expected}$) was set as 80% in the chapter. With reference to existing studies, the unit prices of the systems and electricity prices are presented in Table 10.3. It should be mentioned that these prices have been used for the life cycle cost calculation, i.e., the fitness function. Table 10.4 shows the optimal system sizes of the three design methods. The individual system sizes were the smallest in the TMY data-based design, while they were the largest in the proposed design. The performance of these system sizes designed from the three different methods was evaluated and compared in details in the following sections.

10.4.2.2 Method Validation by Performance Comparisons with the Two Conventional Designs

Using the systems sized by the three design methods (see Table 10.4), the NZEB performances were evaluated under the real future weather conditions (1998–2015) in aspects of lifecycle cost and multi-criteria constraints (i.e., thermal comfort, energy balance and grid interaction). The performance evaluation results are summarized in Table 10.5. The proposed design not only led to the minimum lifecycle cost but also resulted in a smaller error between the estimated and actual lifecycle costs (i.e., 2.2%) due to its better prediction of the future weather. In contrast, the multi-year historical data-based design had a larger error, i.e., 8.4% in the lifecycle cost estimation, while the TMY data-based design was unable to estimate its lifecycle cost since it involved only one single year weather data. It also should be mentioned that the TMY data-based design caused the maximum actual life cycle cost in comparison with the other two. Regarding the constraints' satisfactions, only the proposed method can meet all the practical constraints, while the two conventional designs fully failed to meet them. In fact, the system sizing results from the two conventional designs were far away from achieving the expectations concerning constraint satisfactions. For instance, it was expected that the multi-year historical data-based design would meet the thermal comfort constraint in the 80% of building life-cycle years. But, it turned out only 44% of the lifecycle years can meet the constraint.

The system sizes affect the initial cost and actual operational cost, and eventually influence the lifecycle cost (i.e., the sum of initial cost and operational cost). Figure 10.5 shows the variations of initial cost and actual operational cost

Table 10.4 System sizes from the three design methods

Design methods	Air-conditioning systems (kW)	PV system (m^2)	Electrical energy storage system (kWh)
Proposed design	**39.27**	**103.20**	**559.41**
Multi-year historical data-based design	37.56	96.23	435.22
TMY data-based design	36.58	88.60	373.24

The bold indicates the associated information should draw more attention from the readers

Table 10.5 Performance summary of the three design methods

			Proposed design	Multi-year historical data-based design	TMY data-based design
Lifecycle cost		Estimated (×10^4 HKD)	241.5	228.2	NA
		Actual (×10^4 HKD)	246.9	249.2	263.6
		Estimation error (%)	2.2	8.4	NA
Multi-criteria constraints	Thermal comfort	$\xi_{expected}$	80%	80%	80%
		ξ_{actual}	83%	44%	17%
		Constraint satisfied?	✓	×	×
	Energy balance	$\xi_{expected}$	80%	80%	80%
		ξ_{actual}	89%	61%	33%
		Constraint satisfied?	✓	×	×
	Grid interaction	$\xi_{expected}$	80%	80%	80%
		ξ_{actual}	83%	50%	17%
		Constraint satisfied?	✓	×	×

Notes
Estimation error = |Estimated lifecycle cost - Actual lifecycle cost|/Actual lifecycle cost × 100%;
$\xi_{expected}$ and ξ_{actual} are the expected and actual cumulative probabilities of constraint satisfaction respectively.

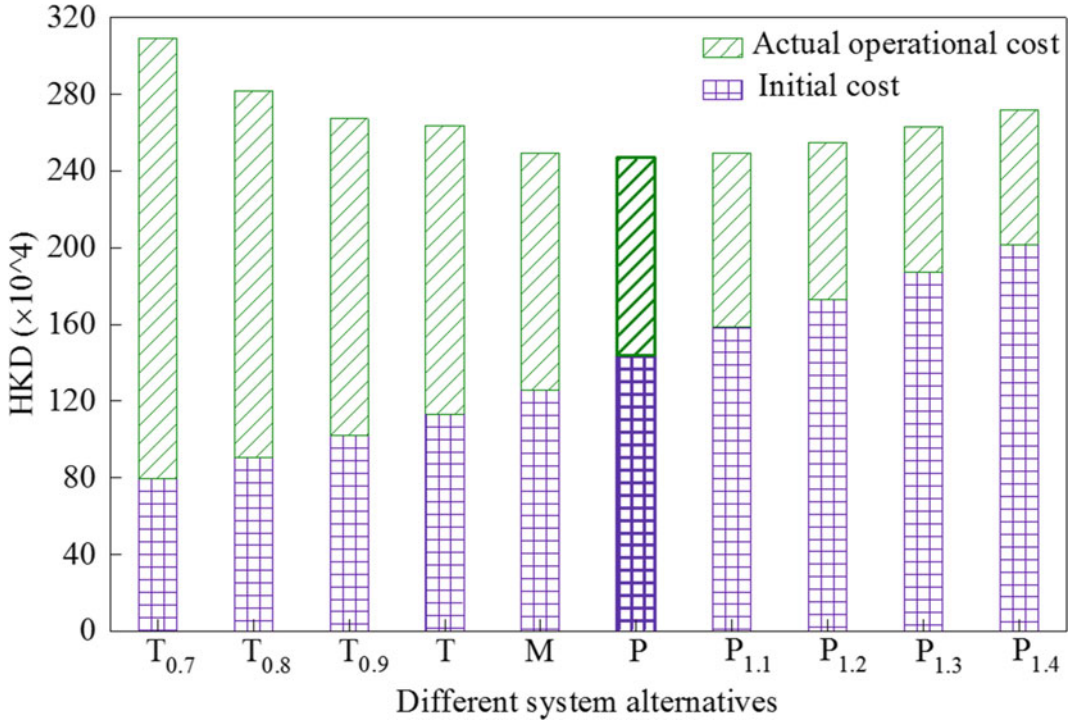

Fig. 10.5 Variations of initial costs and actual operational costs under different system alternatives

under different system alternatives. For simplicity, the three sets of designed systems in Table 10.4 were marked as T (TMY data-based design), M (multi-year historical data-based design), and P (proposed design), respectively. Meanwhile, other system alternatives were also introduced to systematically investigate the impacts of system sizes on lifecycle cost, and they were system T_x (x = 0.7, 0.8, 0.9) and system P_x (x = 1.1, 1.2, 1.3, 1.4). Here, x meant the corresponding system sizes were x times of those in the system T or P. As shown in the figure, from the system $T_{0.7}$ to system $P_{1.4}$ (i.e., with the increase of system sizes), the initial cost gradually increased while the actual operational cost gradually decreased. Here, the operational cost decreased mainly because a larger PV system can generate more energy to meet the building energy use and thus help decrease the imported energy. Consequently, the lifecycle cost firstly decreased and then increased, and it reached the minimum value using the system P (i.e., the system sized by the proposed design). In other words, the system P could well balance the tradeoff between the initial cost and operational cost and thus minimize the lifecycle cost. It also should be mentioned that all the three performance constraints (i.e., thermal comfort, energy balance, and grid interaction) can be satisfied using the system P, while they may not be fully satisfied using other systems. For instance, the constraints of thermal comfort and energy balance can be met using the system $P_{1.1}$, while that of grid interaction cannot be met due to the unsatisfactory energy shortage system.

Figure 10.6 shows the results of thermal comfort in the future years (1998–2015) using the systems sized by the three design methods. There were 15, 8, and 3 years (i.e., 83%, 44%, and 17% of total 18 years) that the annual unmet hours were smaller than 35 h (a user-defined thermal comfort threshold) for the proposed design, multi-year historical data-based design and TMY data-based design, respectively. Only the proposed design can meet the thermal comfort constraint (i.e., at least 80% of total years should have annual unmet hours smaller than 35 h). It also shows that the proposed design had better thermal comfort than the two conventional designs (especially the TMY data-based design). For instance, in the year 2015 the annual unmet hours were 149 h and 40 h for the proposed design and TMY data-based design respectively.

The thermal comfort result was mainly affected by the sized air-conditioning system. Based on the annual cumulative hours of the cooling load (i.e., the product of 8760 and the cumulative probability of cooling load) in the three designs, Fig. 10.7a shows the sizing of air-conditioning system according to the required thermal comfort in different design methods. The air-conditioning system sizes were 38.8 kW, 37.1 kW, and 36.5 kW for the proposed design, multi-year historical data-based design and TMY data-based design, respectively, when the same annual unmet hours (i.e., 35 h) were required. Figure 10.7b presents the evaluation results of the three sized air-conditioning systems on annual unmet hours in real future years. It shows that the annual unmet hours were the smallest (31 h), medium (60 h) and largest (86 h) for the proposed design, historical data-based design, and TMY data-based design, respectively. Only the proposed design can enable the annual unmet hours to be smaller than the required ones (i.e., 35 h). It also shows that the deterioration rates of unmet hours were different at different system sizes. The unmet hours were reduced by 26 h as the size increased from 36.5 kW to 37.1 kW (a 0.6 kW increase); while they were reduced by 29 h as the size increased from 37.1 kW to 38.8 kW (a 1.7 kW increase). Such inconsistent deterioration rates should be carefully considered in optimal system design; otherwise, undesirable results will be obtained. For instance, an increase of system with enough size can contribute little to the improvement of thermal comfort while it can cause unnecessary increase of initial cost and thus increase the lifecycle cost.

Figure 10.8 shows the results of energy balance in the future years (1998–2015) using the systems sized by the three designs. In this chapter the energy balance constraint was that in at least 80% of total 18 years (i.e., 14.4 years) the PV annual generation can meet the building annual

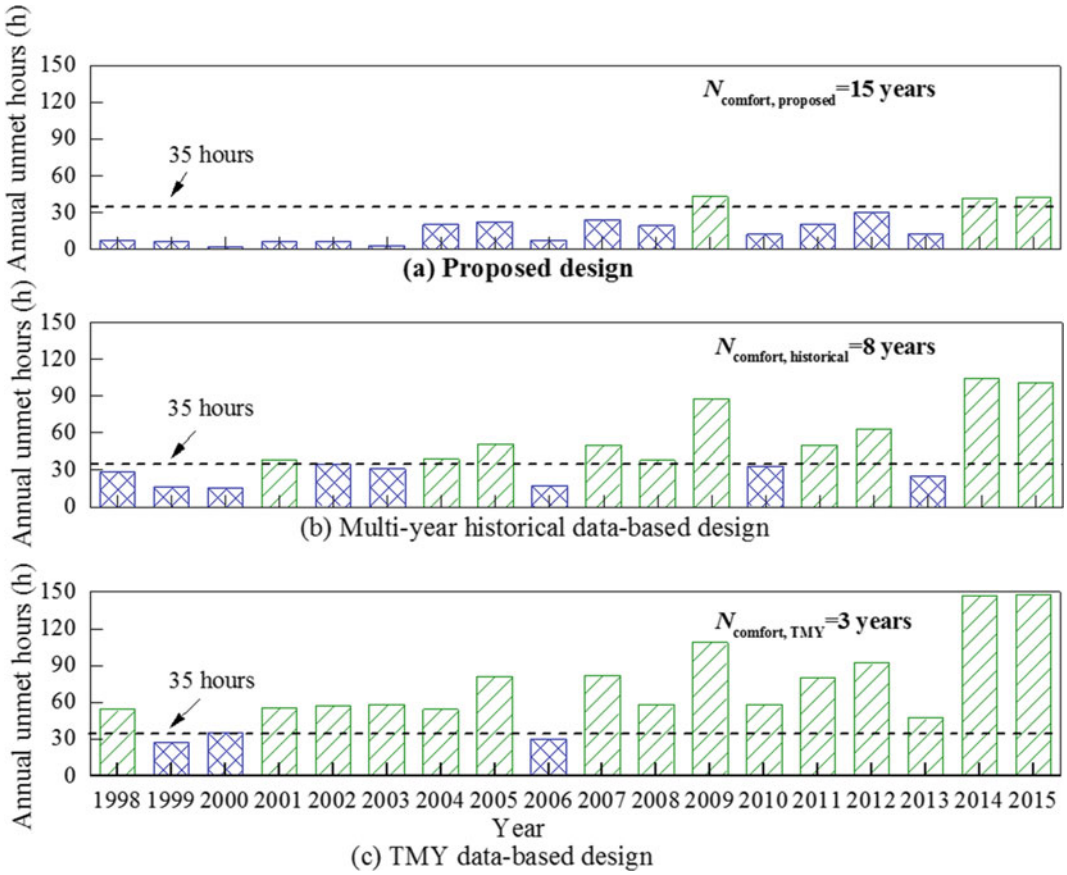

Fig. 10.6 Results of thermal comfort in the future years (1998–2015) using the systems sized by the **a** proposed design; **b** multi-year historical data-based design and **c** TMY data-based design (*note* The blue bar represents the years in which the thermal comfort constraint is satisfied)

energy demand. Only the proposed design can satisfy the energy balance constraint while the two conventional designs cannot fulfill such goals (especially the TMY data-based design). It also shows that the different energy balance results of the three designs are mainly attributed to the PV annual generations (i.e., PV system sizes). For instance, in 2002, the building annual energy demands of the three designs were similar (i.e., around 13,900 kWh), while the PV annual generations were 14,000 kWh, 13,100 kWh, and 12,000 kWh for the proposed design, historical data-based design, and TMY data-based design respectively. Thus, in 2002 the proposed design can achieve energy balance while the historical data-based design and TMY data-based design had 5.8% and 13.7% renewable energy shortages for achieving energy balance, respectively. The analysis for PV system sizing is shown below.

The sizing of PV system mainly affected by the building annual energy demand. In the three designs, since the electrical loads of lighting and equipment were relatively constant in NZEB's lifecycle, the annual energy demands were determined by the cooling energy uses. While the different cooling energy uses of the three designs were mainly influenced by the outdoor temperatures. Here, the annual accumulated temperature deviations (i.e., $T_{outdoor}$ - 26) were calculated to show the associated impacts on cooling energy use. For comparison, the temperature deviations of future years were also presented. Figure 10.9 shows the cumulative probabilities of annual accumulated temperature deviations for different

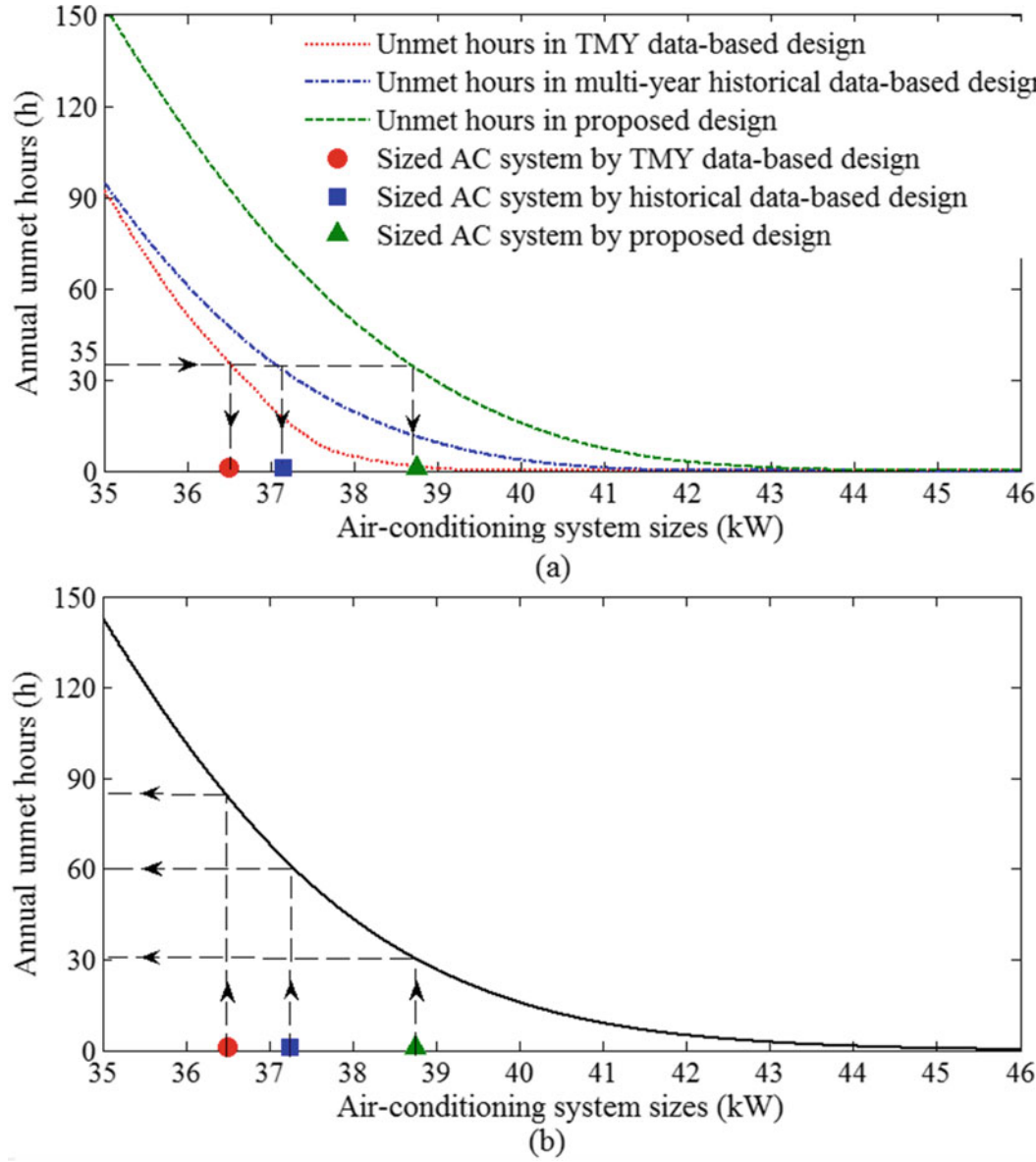

Fig. 10.7 **a** Sizing of air-conditioning system according to the required thermal comfort under different weather conditions; **b** Evaluation of the three-sized air-conditioning systems on annual unmet hours in real future years

weather datasets. The probability profile of predicted years stays close to that of future years. In comparison, due to the neglect of average temperature rise and the more occurrences of high temperatures, the probability profile of historical years stay far from that of the future years. For instance, at the 0.8 percentile, the accumulated temperature deviations were 8890 °C * h, 9951 °C * h and 10,060 °C * h for the historical years, predicted years, and future years, respectively. The temperature deviation of TMY was 7700 °C * h. Consequently, the associated cooling energy use of predicted years would be larger than that of historical years and TMY, thereby resulting in larger PV system sizes.

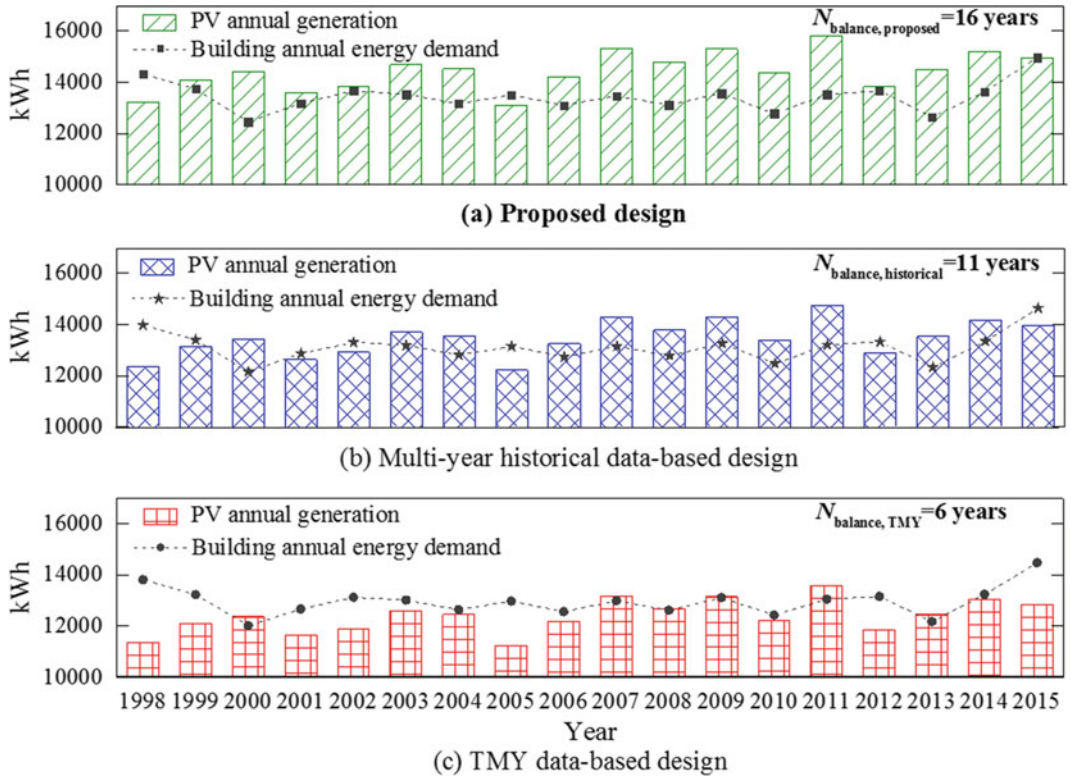

Fig. 10.8 Results of energy balance in the future years (1998–2015) using the systems sized by the **a** proposed design; **b** multi-year historical data-based design and **c** TMY data-based design (*Note* $N_{balance}$ is the number of years in which the energy balance was achieved)

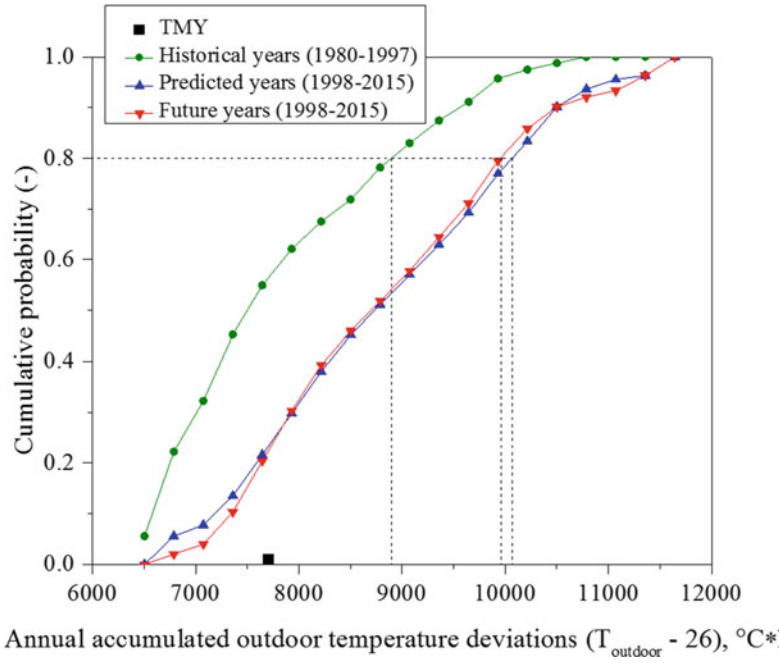

Fig. 10.9 Cumulative probabilities of annual accumulated outdoor temperature deviations for different weather datasets (*note* only the outdoor temperatures above 26 °C were considered)

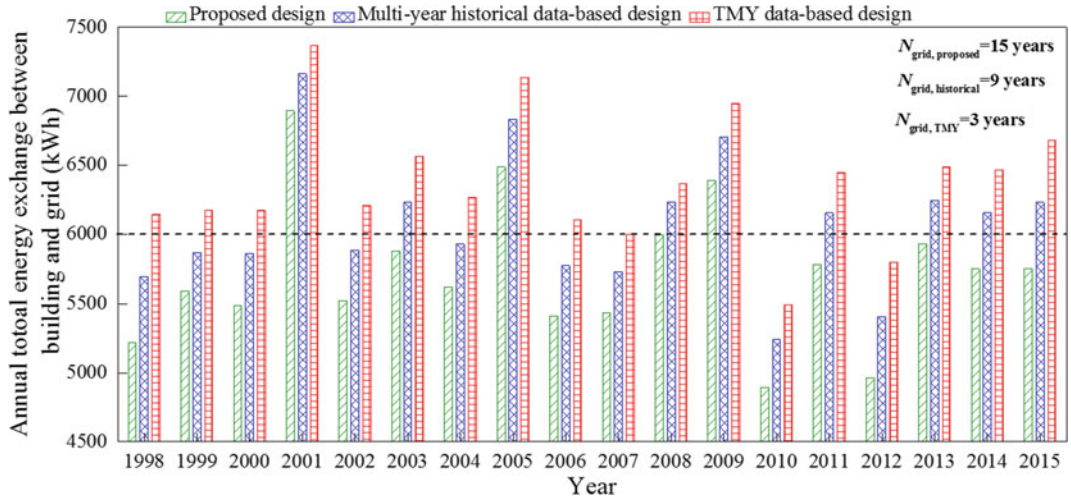

Fig. 10.10 Results of grid interaction in the future years (1998–2015) using the systems sized by the three designs (i.e., proposed design, multi-year historical data-based design, and TMY data-based design) (*Note* N_{grid} is the number of years in which the annual total energy exchange was smaller than 6000 kWh)

Figure 10.10 shows the results of grid interaction in the future years (1998–2015) using the systems sized by the three designs. In this chapter, the grid interaction constraint was that the total energy exchange should be smaller than 6000 kWh (a user-defined grid interaction threshold) in at least 80% of total 18 years (i.e., 14.4 years). Only the proposed design can satisfy the grid interaction constraint, while the two conventional designs failed to achieve the goals, especially the TMY data-based design. It also shows that in each year, the total energy exchange was the largest in the TMY data-based design while it was the smallest in the proposed design. Such results were mainly determined by the sized electrical energy storage system (i.e., a larger storage system resulted in better grid interaction) in each design. The analysis for grid interaction is shown below.

Figure 10.11a shows the hourly power mismatch (i.e., the difference between the hourly power generation and consumption) using the systems sized by the three designs in a future year, while Fig. 10.11b shows the power exchange between building and grid under the same hours. For a clear comparison, only 100 h results were displayed instead of whole year ones. The PV system of the proposed design can improve or deteriorate the grid friendliness as renewable energy was insufficient or surplus. As shown in Fig. 10.11a, when the renewable energy was insufficient, the larger PV system of the proposed design resulted in less insufficient renewable energy than two conventional designs (especially the TMY data-based design), thereby improving the grid friendliness. In comparison, when the renewable energy was surplus, the larger PV system of the proposed design resulted in more surplus renewable energy than two conventional designs, thereby deteriorating the grid friendliness. The electrical energy storage system can help mitigate the power mismatch, thereby reducing the power exchange between building and grid. As shown in Fig. 10.11b, the larger storage system of the proposed design can greatly reduce the surplus renewable energy and eventually result in less power exchange than the two conventional designs (especially the TMY data-based design), thereby improving the grid friendliness.

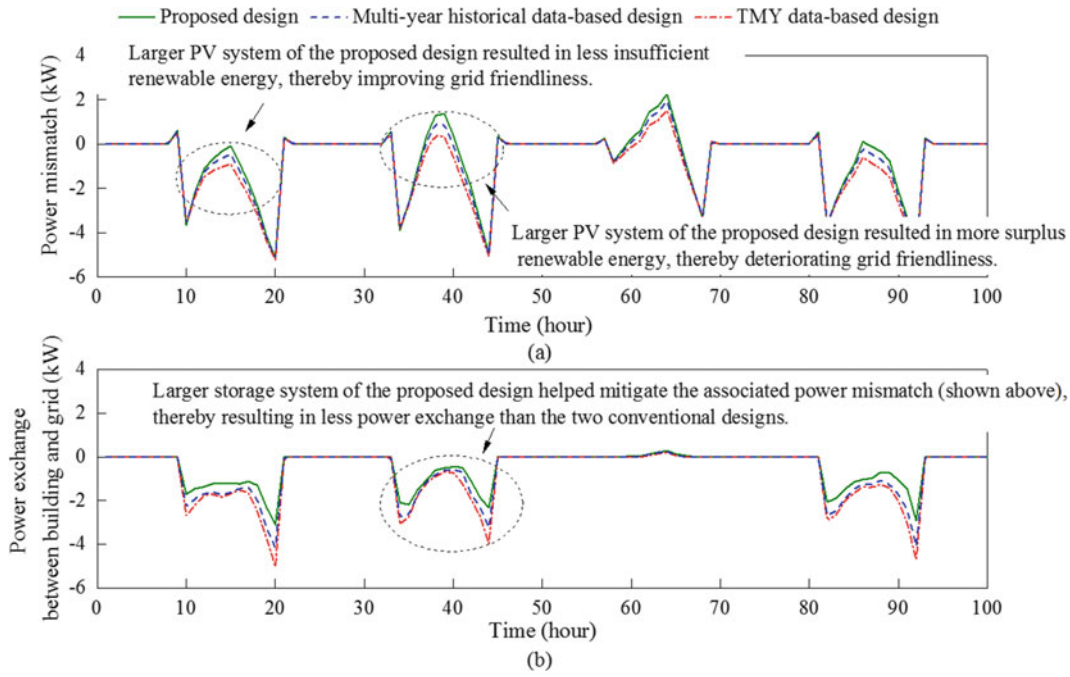

Fig. 10.11 **a** Hourly power mismatch using the systems sized by the three designs (i.e., proposed design, multi-year historical data-based design and TMY data-based design); **b** Power exchange between building and grid under the same hours

10.5 Summary

This chapter proposes a differential evolution-based system design for NZEB under climate change. With the multi-criteria performance constraints considered, the proposed design method is able to optimize NZEB system sizes under climate change for minimized lifecycle cost. Using the real future weather data, the proposed design has been validated by comparing with the two conventional designs (i.e., the multi-year historical data-based design and the TMY data-based design). Among the three designs, the proposed design method can achieve the best performance, while the TMY data-based design achieves the poorest performance in terms of lifecycle cost, thermal comfort, energy balance, and grid interaction. For instance, the cumulative probability of thermal comfort satisfaction was 83% for the proposed design, which was much larger than those of the historical data-based design (44%), and TMY data-based design (17%). Such performance differences are mainly caused by the deviations between their used weather data and future real weather data. In the proposed method, the predicted weather data from the morphing method can accurately reveal the future climate change, and thus it achieves much better performance than the other two designs. The multi-year historical weather data reveal the climate change already occurred in the past, and they can partially describe the climate change in the future. In contrast, the single year TMY data cannot describe the future climate change at all. The chapter also analyzes in detail the system size differences and their impacts on the building lifecycle performance. The proposed design can be used in practice to replace the conventional designs for system sizing of NZEB as climate change is considered. In the future work, system aging and performance degradation will be considered for NZEB lifecycle performance evaluations.

	January	February	March	April	May	June	July	August	September	October	November	December
1980	0.0644	0.0738	0.0606	0.0428	0.0696	0.1040	0.0630	0.0672	0.0808	0.0502	0.0992	0.1061
1981	0.1191	0.0748	0.0596	0.0895	0.0772	0.0683	0.0671	0.0995	0.0940	0.0576	0.0959	0.1447
1982	0.0907	0.0725	0.0781	0.0939	0.0511	0.0677	0.0731	0.0569	0.0541	0.0763	0.1442	0.0777
1983	0.1323	0.2130	0.1582	0.0793	0.0807	0.1061	0.1070	0.0581	0.0701	0.1124	0.1914	0.0811
1984	0.0931	0.0814	0.0514	0.1248	0.0601	0.1004	0.1712	0.0368	0.0770	0.0496	0.0522	0.0787
1985	0.0669	0.1624	0.0614	0.0472	0.1083	0.1046	0.0698	0.0890	0.0922	0.0822	0.0681	0.0559
1986	0.1659	0.0764	0.0874	0.0618	0.0589	0.0606	0.0628	0.0581	0.1112	0.0559	0.0728	0.1062
1987	0.0957	0.1225	0.0897	0.0725	0.1293	0.0475	0.0641	0.1075	0.0469	0.0994	0.1550	0.1170
1988	0.1095	0.0863	0.1004	0.0790	0.0870	0.1200	0.0796	0.1079	0.0796	0.0474	0.0351	0.0939
1989	0.1246	0.1561	0.1589	0.0833	0.0718	0.0595	0.0751	0.0514	0.0544	0.0699	0.0996	0.0706
1990	0.1435	0.0964	0.1106	0.0956	0.0612	0.0670	0.0546	0.1426	0.0428	0.0950	0.0754	0.0965
1991	0.1187	0.1105	0.0923	0.1005	0.1065	0.0532	0.0618	0.0594	0.0868	0.0834	0.0526	0.1197
1992	0.1197	0.0883	0.1409	0.0847	0.1028	0.0923	0.0585	0.0974	0.1070	0.1538	0.1199	0.1178
1993	0.0859	0.1335	0.0767	0.0607	0.0607	0.0737	0.1040	0.0456	0.0414	0.0707	0.1115	0.0539
1994	0.0607	0.1046	0.0962	0.1625	0.1491	0.0782	0.1773	0.1121	0.1356	0.1296	0.1207	0.2096
1995	0.0545	0.0682	0.0628	0.0899	0.0676	0.0734	0.0994	0.1473	0.0571	0.0910	0.0843	0.1030
1996	0.0786	0.0790	0.0551	0.0721	0.0628	0.1103	0.0685	0.0479	0.0495	0.0633	0.0703	0.0736
1997	0.0686	0.0495	0.1022	0.0607	0.0467	0.1188	0.1414	0.0951	0.0749	0.1297	0.0715	0.1400

Note The month with the smallest cumulative FS was shown in **bold**, and it was considered as the TMM.

Appendix 1

Values of cumulative FS statistics for each month of the 18-year period (1980–1997).

Appendix 2

Performance evaluation of GCMs for the three temperature indexes using MAPE (%).

GCMs	T_{mean}	T_{max}	T_{min}	Ag_MAPE
ACCESS1.3	6.377	7.358	6.900	6.878
MIROC4h	7.254	6.360	7.963	7.192
CNRM-CM5	7.366	6.560	8.842	7.589
MIROC5	8.213	7.651	9.341	8.401
HADGEM-ES	8.228	7.880	9.533	8.541
INM-CM4	8.322	8.624	8.678	8.547
GISS-E2-R	9.181	7.729	13.748	10.220
IPSL-CM5A-LR	6.457	7.996	20.497	11.650
GISS-E2-H	11.206	8.801	17.994	12.667
HadCM3	14.685	11.306	25.477	17.156

Note

1. MAPE is the mean absolute percentage error (see Eq. 10.1), and a smaller MAPE indicates high prediction accuracy of one GCM.
2. Ag_MAPE is the averaged MAPE values of the three temperature indices.
3. The ten models were ranked according to the Ag_MAPE.

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A Preliminary Simulation Study About the Impact of COVID-19 Crisis on Energy Demand of a Building Mix at a District in Sweden

11

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Abstract

The COVID-19 outbreak is exacerbating uncertainty in energy demand. This chapter aims to investigate the impact of the confined measures due to COVID-19 outbreak on energy demand of a building mix in a district. Three levels of confinement for occupant schedules are proposed based on a new district design in Sweden. The Urban Modeling Interface tool is applied to simulate the energy performance of the building mix. The boundary conditions and input parameters are set up according to the Swedish building standards and statistics. The district is at early design stage, which includes a mix of building functions, i.e., residential buildings, offices, schools, and retail shops. By comparing with the base case (normal life without confinement measures), the average delivered electricity demand of the entire district increases in a range of 14.3–18.7% under the three confinement scenarios. However, the mean system energy demands (sum of heating, cooling, and domestic hot water) decrease in a range of 7.1–12.0%. These two variation nearly cancel each other out, leaving the total energy demand almost unaffected. The result also shows that the delivered electricity demands in all cases have a relatively smooth variation across a year, while the system energy demands follow the principle trends for all the cases, which have peak demands in winter and much lower demands in transit seasons and summer. This chapter represents a first step in the understanding of the energy performance for community buildings when they confront with this kind of shock.

Keywords

COVID-19 · Building · District · Energy · Demand

11.1 Introduction

In the past few months, the COVID-19 crisis has significantly affected all aspects of our life, such as global economy, social connection, environment, and energy demand/supply. Different countries are trying various confinement measures in order to reduce the impact of such pandemic (Qarnain et al. 2021). These confined measures have subsequent influence on the energy sector, which is exacerbating the energy demand issue. Steffen et al (2020) therefore

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highlight that it is important to navigate the new situation without jeopardizing the imperative clean energy transition under the COVID-19 outbreak.

According to the recent report (European Power Demand Falls below Five-Year Average amid Covid-19 Outbreak—Energy Live News 2020), energy demand across Europe has fallen 'significantly' below the 2015–2019 average range in many major European markets. Unsurprisingly, the strictness of confinement measures correlates with drops in overall energy consumption at whole system level, which are about 25% in Italy, 20% in France, 12% in the UK (Impact of Covid-19 on the Global Energy Sector—Pv Magazine International 2020). Mcwilliams and Zachmann (Impact of Covid-19 on the Global Energy Sector—Pv Magazine International 2020) compare the average weekday hourly electricity demand for the last few weeks to the year before, in which they visualize the moment when the current crisis began to have an impact on national economies and how large that impact was, from each week in 2020 with the corresponding week from 2019 for peak hours (08:00–18:00). According to their report, the average daily electricity consumption in Sweden is about 99.57% related to the same periods in 2019 in the past a few months, which shows very limited changes due to the 'soft' confinement in practice. This can be explained that on a whole system level in Sweden, the overall effect of COVID-19 could be limited due to the varying needs in different sectors. For instance, electricity demand may be reduced in transportation, industries, commercial buildings, etc., but it would rise in residential buildings. The energy demand could be in an opposite direction in an industrial district than that in a residential district. It is therefore necessary to investigate new partners and varying trends in different types of districts, so that an overall understanding on the whole energy system level can be achieved.

Moreover, the impact of COVID-19 outbreak on energy demand is largely depending on confinement measures in different countries, and cultural contexts. In general, specific consideration needs to be paid to the differences between countries, which may have different energy infrastructures, urban energy systems, occupant behaviors, confinement measures, district functions and building performance in the perspectives of geography, climate, socio-economy, culture, infrastructure, and so on. A dedicated analysis of COVID-19's impact on local energy demand is thus necessary. In addition, there might be a few more waves of COVID-19 outbreak, which may lead to different closure levels in Sweden, where the corresponding impact on Swedish energy demands in buildings stays unknown. It is thus important to have a predictive study that will help mitigate the influence from COVID-19 with appropriate pre-preparations for new policy designs that can withstand future long-term shocks (Steffen et al. 2020).

Buildings at district level form up the minimum local energy infrastructure. Building performance simulation is then considered as the main approach to conduct such predictive study for energy demand in buildings. There are several classical tools that can simulate the building energy performance of a district (Abbasabadi and Ashayeri 2019; Zhang et al. 2018), such as CitySim, EnergyPLAN, E-GIS, Urban Building Energy Models (UBEMs), Urban Modeling Interface (UMI), and City Building Energy Saver (CityBES), and other data-driven models. Among these tools, UMI is a free multi-objective modeling tool based on Rhino developed by MIT Sustainable Design Lab 8[(PDF) UMI—An Urban Simulation Environment for Building Energy Use, Daylighting and Walkability, n.d.], which can evaluate the energy and environmental performance of buildings in districts and cities with respect to energy demand, walkability, daylighting potential, and so on. UMI includes an application programming interface (API) for researchers to add additional performance modules and metrics. As a result, UMI is determined in this chapter as the main tool to conduct the study, with validated accuracy and possibility in adding additional evaluations in the future.

The occupancy profile in buildings is a key parameter for simulation of energy demand. During the early design stage, the classical way

to estimate energy demand is based on the use of benchmarks or local standards. In this way, the occupancy density, behavior, and schedule usually influence significantly on energy use in lighting, equipment and heating, ventilation, and air conditioning (HVAC) systems. However, owing to the different confinement levels, the occupancy profile in buildings will be different to the conventional setup. For instance, residential energy consumptions are likely to rise when people stay longer time at home, because of both augmented conventional demand (lighting, space heating, hot water, cooking, and dishwashing) and new energy demand (online meetings, computation-related workings) (Measures to Tackle the COVID-19 Outbreak Impact on Energy Poverty 2020). While offices, schools, and retails will have different operation schedules by comparing to that on normal week days and weekends. As a result, the impact of COVID-19 on energy demand may be not simply similar to the difference between normal working days and weekends (Widén et al. 2009; Widén and Wäckelgård 2010). In fact, in most of normal cases, there are generally far fewer occupants in residential buildings during the weekdays than during weekends, so the energy demand is usually lower (Barthelmes et al. 2018; Richardson et al. 2008). This cannot be expected to be the same during the COVID-19 outbreak due to the containment measures, during which people are quarantined at home to couple living and work for longer time. This means that building energy and services systems (such as HVAC, lighting, and plug-loads) in residential buildings have to remain operating in order to provide energy, thermal comfort, and ventilation even during weekends (Lee et al. 2020). Conversely, offices and schools will have less internal heat gains and less operation time due to less occupants and by scaling back their activities, which will result in a drop of the energy demand especially during the weekdays. Although a few studies have started to investigate the impact of new home-working mode on the energy demand, most of them have a rather small scale of target buildings, and only concentrate on potentials of demand reduction, demand shifting, and 'smart' controls at homes (Hampton 2017) or in transportation (Cerqueira et al. 2020) through behavior change. To the best knowledge from the authors, there is no research so far in investigating the impact of different occupancy schedules on the coupled energy demands for a large group of archetypes in a neighborhood under different confinement levels due to crisis as COVID-19 pandemic. This is driven by the unknown impact on the energy demand, as there are different variation trends for mixed buildings in a district.

In order to bridge such a research gap, this chapter thus aims to **investigate the impact of COVID-19 outbreak on energy demand of different buildings in a community**, which consists of different archetypes, such as residential buildings, school, offices, and retail shops. It proposes different occupancy schedules that represent different levels of confinement, such as normal operation, 'soft', and full lock down. With the UMI tool, the energy demand of different buildings can be then predicted in the district under different scenarios. Detailed energy variation trends for different buildings and the coupled energy demand can thus be investigated. The research results would be useful for employers, researchers, energy suppliers, policy makers, and home workers themselves, to withstand future potential crisis as COVID-19 pandemic.

However, the authors would like to highlight that this chapter is only an exploratory and simulation-based pilot study to understand general trends on energy demand in a mixed-use neighborhood if occupant profiles changes due to confinements in crisis as COVID-19. It is a simulation study of a non-existing (virtual) community, which is still on design and planning stage. This chapter considers the key parameters for simulation of energy demand according to the setup requirement in UMI tool, such as thermal properties of construction layers, set point temperature, occupancy density/schedule, hot water demand, ventilation system, heating/cooling system, lighting, and equipment power density. While the detailed occupancy behavior changes (e.g., power increased for TV or computer) due to COVID-19 is not addressed in this chapter,

which is a limitation due to lacking of existing data. The underlying occupant schedules are based on practical building regulation, historical statistics and informed assumptions. This chapter is a first step in the understanding of energy performance of the buildings in such a residential district when they confront with this kind of shock. It cannot reflect the influence of COVID-19 on whole energy system level, but it opens up possibilities in investigating the energy performance of buildings in other types of districts and contexts.

The whole chapter is structured as followings: Sect. 11.2 briefs the research method about simulation process and definition of occupancy schedule due to COVID-19 outbreak; in Sect. 11.3, the case studied is described with boundary conditions and input parameters; Sect. 11.4 presents the simulation results and the related discussion; finally the summary is presented in Sect. 11.6.

11.2 Simulation Process and Definition of Occupancy Schedule Due to COVID-19 Outbreak

The simulation in this chapter is conducted by UMI tool with the assistance of Rhino 6. Figure 11.1 illustrates the simulation methodology. The first step is to import and adjust the 3D building models in Rhino 6 of the new district and then define different archetypes according to the design. The 3D building models are further imported into UMI for simulation. A climate file can be subsequently defined and imported into the tool. Different construction layers should be defined, by choosing the materials from the UMI library and adding them into the definition when they are missing. This process needs to be done for each type of building. Afterward, schedules for occupancy, lighting, household equipment, domestic hot water (DHW), heating, cooling, and ventilation are defined according to the Swedish national regulations on building and planning during normal use in a year. Finally, since this chapter focuses on the simulation at early design stage, it considers the standard simulation method and keeps the setups of other parameters (such as temperature set points, infiltration, and efficiency of the heater and cooler) the same in different conditions, according to the building regulations for each building type.

The key variable in this simulation is the occupancy profile, owing to the different closure policies in case of the COVID-19 spreading. The essential considerations are the occupants density and their varying schedules in different confinement scenarios, which will affect the energy load in return. In Table 11.1, different confinement levels for all types of buildings are defined according to the current/possible closure policy, statistics, and building regulation in Sweden. On the level 1 (no COVID-19), it is considered as the base case (normal schedule) with 10 h, 14 h, and 15 h unoccupied (outside of buildings), respectively, in residential, school/retail shop, and office buildings. In Sweden, the building standard recommends totally 14 occupied hours (inside building), 7 days per week, and 52 weeks, in a residential building per year (Brukarindata Bostäder 2012). Similar explanations can be applied to other types of buildings (Brukarindata Kontor Svebyprogrammet 2013). Confinement level 2 and level 3 are defined in this chapter as ‘soft’ measures to COVID-19 outbreak. Compared to level 1, half unoccupied hours are assumed in residential, but half occupied hours are defined in other buildings except retail shops, which are considered open longer to allow citizens to have essential purchase, delivery, and pickup of food, medicine, and commodity at the special periods, but the occupancy density is reduced in order to create enough social distance. On confinement level 2, it is assumed that the unoccupied hours in residential building become half of that in level 1, which are 5 h in total. This means that people are allowed to work outside of home or go out for 5 h per day (each 2.5 h in the morning and afternoon). Such confinement results in additional half hours reduce, respectively, in school and office building, based on the level 1. So there will be only 5 occupied hours in school, and 4.5 h in office building. The schedule of retail shops is 4 h

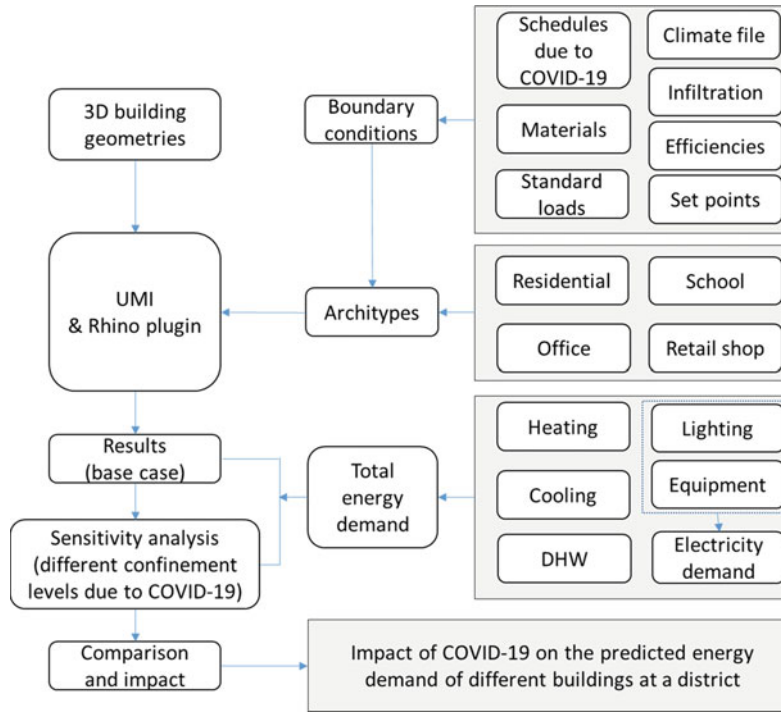


Fig. 11.1 Schematic of simulation method

longer than that in level 1. The same principle applies to the definition of level 3. While in level 4, it is considered as the full lock down scenario, where residential buildings are occupied in 24 h per day but citizens are still allowed to go out for urgent or necessary food, medicine, and commodity, while school and office buildings are closed totally.

11.3 Description of the New District

11.3.1 Archetype Design

The new district 'Jakobsgårdarna' locates in Borlänge city, Sweden (as shown in Fig. 11.2 with red color), which is mainly a residential area but mixed with offices, retail shops, and schools. All the facilities are mainly connected through walking and cycling routes (Johnson 2017). The total floor areas for this district are designed at about 199,016.9 m^2, where residential, school, office, and retail shops are 164,639.0 m^2, 5723.4 m^2, 16,089.5 m^2, and 12,565.1 m^2, respectively. There are various types of residential buildings, from single-family houses to small apartments. The residential buildings are divided in two subgroups, which differ from the structure: wood frame and masonry/concrete frame. While offices, retail shops, and school are grouped, and they have the same concrete frame. The archetypes are shown in Fig. 11.3. This chapter includes total 166 buildings, in which residential buildings are plotted in red and brown, offices are in blue, retail premises are in yellow, and the school is in cyan.

The structures for buildings in this district are defined as three groups as explained above. In UMI tool, the construction layers consist of façade, ground floor, interior floor, partition, and roof. Regarding the wood frame, plywood and softwood are the two main components, and standard fiberglass is selected as insulation layer for the façades and the roof, while the extruded polystyrene (XPS) panel is used for the ground floor. The façade has an air gap of 3 cm to prevent condensation issues and to remove excess moisture. The masonry structure is made by clay bricks, an air gap of 3 cm, and concrete block of 15 cm. In this case, the insulation layer is

Table 11.1 Definition of different occupancy schedules owing to the different closure policies in case of the COVID-19 spreading

Closures ratio	Residential building (Brukarindata Bostäder 2012)		Educational/Commercial building (Brukarindata Undervisningsbyggnader 2015)		Office building (Brukarindata Kontor Svebyprogrammet 2013)	
	Occupancy schedule	Scenario remark	Occupancy schedule	Scenario remark	Occupancy schedule	Scenario remark
Level 1 (base case—no COVID-19 case)	Normal schedule (e.g., full time working) (14 h/7 d/52 w)	10 h unoccupied	Normal schedule (e.g., full time working) (10 h/5 d/47 w)	14 h unoccupied	Normal schedule (e.g., full time working) (9 h/5d/47w)	15 h unoccupied
Confinement level 2 (compared to level 1, half unoccupied hours in residential, but half occupied hours in other buildings)	2.5 h free on the morning and 2.5 h free on the afternoon for part time work (19 h/7 d/52 w)	5 h unoccupied[a]	3.5 h free on the morning and 3.5 h free on the afternoon for part time work (5 h/5 d/47 w)	19 h unoccupied; retail shops open longer[b]	3.5 h free on the morning and 4 h free on the afternoon for part time work (4.5 h/5d/47w)	19.5 h unoccupied
Confinement level 3 (compared to level 2, half unoccupied hours in residential, but half occupied hours in other buildings)	1 h free on the morning and 1.5 h free on the afternoon (21.5 h/7 d/52 w)	2.5 h unoccupied[a]	1.5 h free on the morning and 2 h free on the afternoon (2.5 h/5 d/47 w)	21.5 h unoccupied; Retail shops open longer[b]	1.75 h free on the morning and 2 h free on the afternoon (2.25 h/5d/47w)	21.75 h unoccupied
Confinement level 4	24 h at home (24 h/7 d/52 w)	Fully stay at home[c]	School is closed	Retail shops open longer[b]	Fully closed	Full closure

Note

[a]During the confinement scenarios, all plugging equipment schedules are proportional to respective occupancy schedules

[b]Retail shops are supposed open longer for urgent and necessary purchase, delivery, and pickup of food, medicine, and commodity

[c]Citizens are still allowed to go out for occasionally necessary food, medicine, and commodity

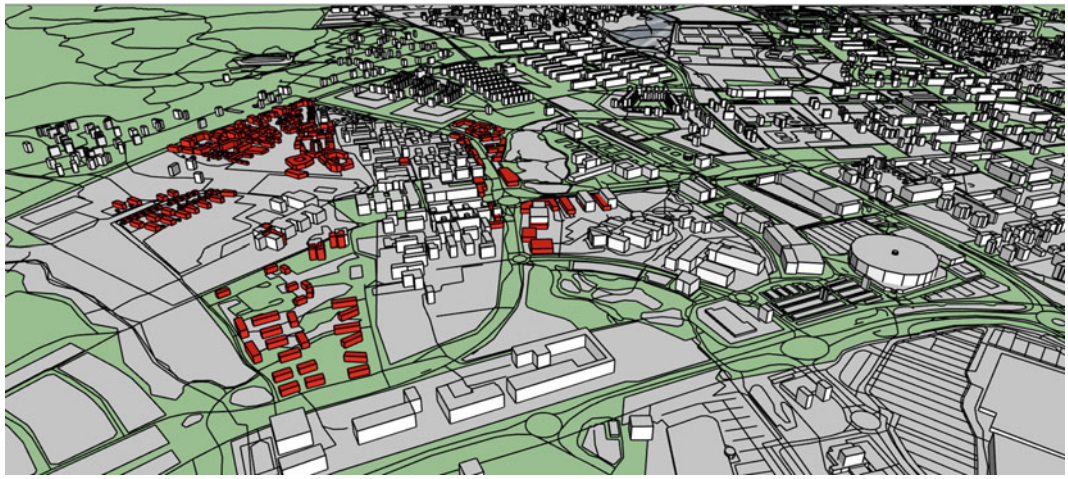


Fig. 11.2 3D view of Borlänge city, in which red blocks showing the studied Jakobsgårdarna area

Fig. 11.3 3D view of the investigated district with surrounding information

considered as XPS in all the structure except for the roof, where there is fiberglass that is lighter without too much weight load. The concrete structure for offices, retail shops, and the school is simply configured by concrete blocks, XPS as insulation layer and cladding inside and outside. In the latter case, the roof is designed with light concrete and fiberglass as insulation. The whole buildings in this district are designed by triple glazed windows with air between panels and treated with low-emission coating. The thermal properties and the related U-values are then calculated for all the structures to meet local building requirements (OVK—obligatory ventilation control—boverket), which are summarized in Tables 11.2 and 11.3 for residential buildings and rest buildings respectively. Figure 11.3 also illustrates the categorization of wooden (brown

Table 11.2 Thermal properties of construction layers in residential building according to standard (OVK—obligatory ventilation control—boverket)

Wood				Masonry			
Façade	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)	Façade	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)
Softwood_general	0.03	0.13	0.231	Clay_brick_H	0.06	0.41	0.146
Air_wall_3cm	0.03	0.1	0.3	Air_wall_3cm	0.03	0.1	0.3
Fiberglass_batts	0.3	0.043	6.977	XPS_board	0.3	0.037	8.108
Plywood_board	0.03	0.11	0.273	Concrete_block_H	0.15	1.25	0.12
Gypsum_board	0.015	0.16	0.094	Gypsum_plaster	0.003	0.42	0.007
Total	0.405	R_{tot}	8.044	Total	0.543	R_{tot}	8.852
		U-value (W/m^2K)	0.124			U-value (W/m^2K)	0.113
Roof	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)	Roof	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)
Ceramic_tile	0.02	0.8	0.025	Ceramic_tile	0.02	0.8	0.025
Plywood_board	0.03	0.11	0.273	Concrete_MC_light	0.05	1.65	0.030
Fiberglass_batts	0.35	0.043	8.140	Fiberglass_batts	0.35	0.043	8.140
Air_floor_15cm	0.15	0.7	0.214	Air_floor_15cm	0.15	0.7	0.214
Gypsum_board	0.015	0.16	0.094	Gypsum_board	0.015	0.16	0.094
Total	0.565	R_{tot}	8.915	Total	0.585	R_{tot}	8.673
		U-value (W/m^2K)	0.112			U-value (W/m^2K)	0.115
Ground	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)	Ground	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^2K/W)
Concrete_RC_dense	0.2	1.75	0.114	Concrete_RC_dense	0.2	1.75	0.114
XPS_board	0.25	0.037	6.757	XPS_board	0.25	0.037	6.757
Concrete_MC_light	0.07	1.65	0.042	Concrete_MC_light	0.07	1.65	0.042
Cement_mortar	0.08	0.8	0.1	Cement_mortar	0.08	0.8	0.1
Ceramic_tile	0.02	0.8	0.025	Ceramic_tile	0.02	0.8	0.025
Total	0.62	R_{tot}	7.168	Total	0.62	R_{tot}	7.168
		U-value (W/m^2K)	0.139			U-value (W/m^2K)	0.139

Table 11.3 Thermal properties of construction layers in offices/shops/school according to standard (Cerqueira et al. 2020)

Concrete			
Façade	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^{2}K/W)
Vinyl_cladding	0.005	0.16	0.031
XPS_board	0.3	0.037	8.108
Concrete_block_H	0.15	1.25	0.12
Gypsum_plaster	0.003	0.42	0.007
Total	0.458	R_{tot}	8.437
		U-value (W/m^{2}K)	0.119
Roof	Thickness, d (m)	Conductivity, k (W/mK)	R-value d/k (m^{2}K/W)
Ceramic_tile	0.02	0.8	0.025
Concrete_MC_light	0.05	1.65	0.030
Fiberglass_batts	0.35	0.043	8.140
Air_floor_15cm	0.15	0.7	0.214
Gypsum_board	0.015	0.16	0.094
Total	0.585	R_{tot}	8.673
		U-value (W/m^{2}K)	0.115
Ground	Thickness (m)	k (W/mK)	R d/k (m^{2}K/W)
Concrete_RC_dense	0.2	1.75	0.114
XPS_board	0.25	0.037	6.757
Concrete_MC_light	0.07	1.65	0.042
Cement_mortar	0.08	0.8	0.1
Ceramic_tile	0.02	0.8	0.025
Total	0.62	R_{tot}	7.168
		U-value (W/m^{2}K)	0.139

highlighted) and masonry buildings (red highlighted) in the district.

11.3.2 Climate Analysis

The simulation work is carried out using the climate file of Borlänge city, Sweden, which is derived from the national statistics (Boverkets föreskrifter och Allmänna Råd 2016:12). From weather information collected from Figs. 11.4, 11.5 and 11.6 the climate is characterized with freezing winter and pleasantly warm summers in general. During the winter, the average external air temperature drops below freezing (0 °C) from late October and reaches the lowest temperature in early spring. Summer starts from June to August, which is a mild season with more sunny days. Extreme hot temperatures are recorded rarely: Occasionally the external air temperature can reach 28 °C within historic records. There is an obvious diurnal temperature variation along with greater windy occasions. Sometimes, it can be very cool, or even cold at night, since the external air temperature can drop below 10 °C even in summer. Due to its northern latitude, the days are very short during winter season and the amount of solar radiation resource is valuable with the maximum daily total of direct normal radiation lower than 3000 Wh/m^2 per day. In terms of wind condition, the annual wind velocity is around 2.5 m/s, presenting pleasant light breeze. The prevalent wind directions are from 225° (Southwest) and 292.5° (Northwest).

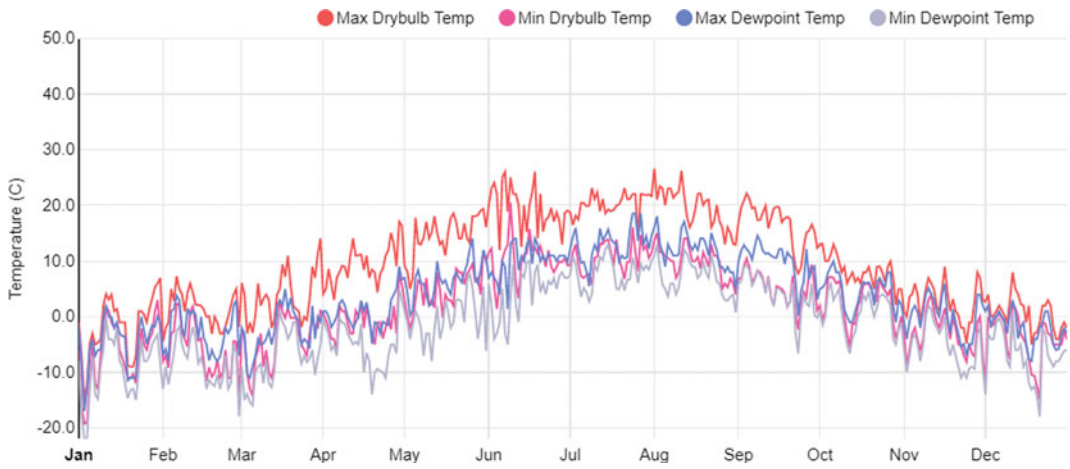


Fig. 11.4 Daily maximum and minimum external air temperature in Borlänge, Sweden (Boverkets föreskrifter och Allmänna Råd 2016:12)

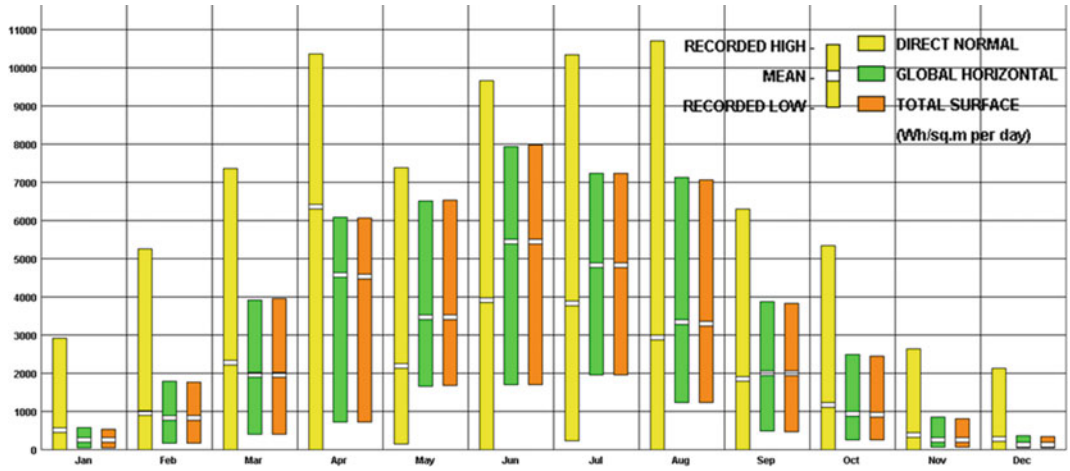


Fig. 11.5 Daily total solar radiation profile in Borlänge, Sweden (Boverkets föreskrifter och Allmänna Råd 2016:12)

11.3.3 Boundary Conditions and Parameters Setup

In order to perform the simulation as accurate as possible, the key parameters need to be setup in the tool. For each archetype, it is necessary to define standard electricity loads, air conditioning, windows type, and DHW load. These input data are chosen carefully according to the SVEBY ('standardize and verify energy performance' in Sweden) for residential building (Brukarindata Bostäder 2012), school (Brukarindata Undervisningsbyggnader 2015) and commercial building (Brukarindata Kontor Svebyprogrammet 2013), or with reasonable assumptions where needed as suggested from the national building standards (BBR[1] 26 and BFS[2] 2016:12 BEN[3] in Sweden (OVK—obligatory ventilation control—boverket) (Boverkets föreskrifter och Allmänna Råd

[1] BBR stands for 'Boverkets byggregler' in Swedish, which means the National Board of Housing, Building and Planning for building regulations.

[2] BFS represents the collection of statutes at the National Board of Housing, Building and Planning.

[3] BEN stands for 'Byggnadens Energianvändning' in Swedish, which means Building Energy Use.

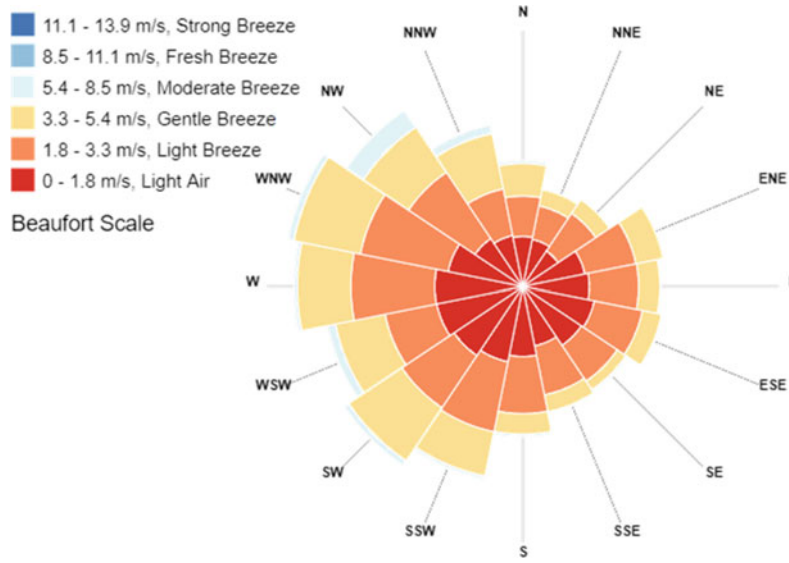


Fig. 11.6 Monthly mean wind speed profile and wind rose in Borlänge, Sweden (Boverkets föreskrifter och Allmänna Råd 2016:12)

2016:12), as well as the recommendation from industry. We assume the simulation is based on one year duration. It is practical because there might be different waves of the virus, and we may have to maintain the confinement measures for a long time. Table 11.4 summarizes of boundary conditions and key parameters setup.

11.3.3.1 Residential Buildings

The residential buildings account for a large proportion in this district and it is important to define the input data carefully. Both wood and masonry structures are considered the same parameters.

The standard electricity loads include with occupancy density, equipment power density, and lighting density per square meter. The occupancy density for residential buildings is defined by national building standard [BFS 2016:12 BEN (Boverkets föreskrifter och Allmänna Råd 2016:12)] based on how many rooms and kitchens there will be in the building. Since this design is still at the early stage, there is no detailed information about the interior design of each building. It is thus considered 20 m^2 for each room and 5 m^2 for the kitchen in total. So each person has 25 m^2. By converting it in person/m^2, the occupancy density will be 0.04 person/m^2. The equipment and lighting power density are considered to be 8 W/m^2 in total. The BFS 2016:12 BEN1 (OVK—obligatory ventilation control—boverket) suggests 30 kWh/(m^2year) for the household electricity (equipment and lighting) (Boverkets föreskrifter och Allmänna Råd 2016:12), which is about 3.5 W/m^2 that is smaller than the 8 W/m^2 considered. However, the power density will be depended on specific schedule as defined in Table 11.1, so the power system is not always on, which will result in a closer power density to the standard. Moreover, according to the Swedish Internet Foundation (Internet hjälper dig Att Jobba Hemifrån 2020), more than two thirds of Swedes already work online from home for certain time, with around a third doing this on a daily or weekly basis. Depending on the survey from 2009 to 2018, about 48% companies allowed employees working from home in average (Statista Research Department and 7 2020). The existing social and company policies in Sweden champion flexible and remote working culture as part of a balanced and gender-equal lifestyle. As a result, we assume that the energy demand due to telecommuting and distant working has been already considered in the existing Swedish building regulations and statistics.

The heating set point in all the residential buildings is assumed at 21 °C (OVK—obligatory ventilation control—boverket) and the heating is always ‘on’ for the entire heating season, which

Table 11.4 Summary of the boundary conditions and key parameters setup

	Residential	Office	School	Retail
Set point temperature (°C)	21	21	22	21
Occupancy density (person/m^2)	0.04	0.05	0.067	0.3
Occupancy schedule	Figure 11.7, a1 and a2	Figure 11.7, b1 and b2	Figure 11.7, d1 and d2	Figure 11.7, c1 and c2
DHW				
Hot water annual demand (kWh/m^2)	25	2	2	2
Ventilation				
Infiltration ACH	0.05	0.05	0.05	0.05
Mechanical ventilation	Yes	Yes	Yes	Yes
Min fresh air per area (l/s/m^2)	0.35	0.13	0.25	0.23
Min fresh air per person (l/s/person)	0.7	0.7	0.7	0.7
Heat recovery type sensible efficiency	0.8	0.8	0.8	0.8
Conditioning				
Heating set point (°C)	21	21	22	21
Heating COP	1	1	1	1
Heating schedule	All day on from 1st October to 30th April			
Cooling set point (°C)	–	26	26	26
Cooling COP	3	3	3	3
Cooling schedule	Cooling season is from 15th May. to 31th August			
	–	Figure 11.7, b7	Figure 11.7, d7	Figure 11.7, c7 and c8
Standard loads				
Equipment (W/m^2)	4	10	8	7
Equipment schedule	Figure 11.7, a5 and a6	Figure 11.7, b5 and b6	Figure 11.7, d5 and d6	Figure 11.7, c5 and c6
Lighting power (W/m^2)	4	6	4	8
Illuminance target (lux)	200	500	500	300
Lighting schedule	Figure 11.7, a3 and a4	Figure 11.7, b3 and b4	Figure 11.7, d3 and d4	Figure 11.7, c3 and c4
Windows to wall ratio (%, S–N–E–W)	50–40–40–40	60–40–40–40	60–40–40–40	60–50–50–50

lasts from October to April in Sweden. All the buildings are considered to be connected with the district heating network, where the heating coefficient of performance (COP) equals to 1.0 (OVK—obligatory ventilation control—boverket). The mechanical ventilation is implemented and it is set at 0.35 l/sm^2 (OVK—obligatory ventilation control—boverket), thus 0.00035 m^3/sm^2. In addition, an infiltration of 0.05 air change per hour (ACH) is considered as a typical new building in Sweden.

The window sizes and their orientations can determine how much solar radiation can come into the building, and hence how much free gain

that is possible to use during the winter season. On the other hand, during the summer season, a high share of solar radiation coming into the building could cause overheating problem and increase the cooling demand. Goia (2016) studied three different cities in different locations, in which Oslo (59° 57′) is northernmost city and its results can be useful for Borlänge since they are about at the same latitude. The optimal window-to-wall ratios (WWR) in buildings in different locations are considered carefully by referring the design in Oslo (59 °57′-same latitude of Borlänge). Different orientations have different optimal WWR. In particular, the optimal range for the south is 0.5–0.6, while for the other orientations is always 0.37–0.43 (Goia 2016). However, the recommended ranges are meant for office buildings and not for residential. We thus set the lower part of the range for the south direction, with a WWR of 0.5 toward south and 0.4 toward the other directions. Both SVEBY and BFS 2016:12 BEN1 suggest DHW load at $25/\eta_{syst}$ (kWh/m^2) for new apartments (OVK—obligatory ventilation control—boverket) (Boverkets föreskrifter och Allmänna Råd 2016:12), with η_{syst} the efficiency of the system (1.0 in case of district heating). As a result, DHW load is assumed at 25 kWh/m^2.

11.3.3.2 Office Buildings

The office buildings have an occupancy density of 0.05 person/m^2, which means 20 m^2 per person (OVK—obligatory ventilation control—boverket). The equipment power density is higher than the residential one and it is set at 10 W/m^2, while the lighting power density is set at 6 W/m^2. The heating set point is at 21 °C and the heating COP is still considered 1.0 with the district heating network. These buildings have a cooling system with set point at 26 °C and COP equals to 3.0. The mechanical ventilation is present and considers 1.3 l/sm^2 (OVK—obligatory ventilation control—boverket), thus 0.0013 m^3/sm^2. In addition, an infiltration of 0.05 ACH is considered. The WWR is equal to be 0.6 toward south and 0.4 to all the other directions in order to have the maximum advantage from the solar radiation all over the year (Goia 2016). DHW load is assumed at 2 kWh/m^2, according to the building regulation [BBR 26 (OVK—obligatory ventilation control—boverket)].

11.3.3.3 Retail Shops

The occupancy density is higher in the shops at about 0.3 p/m^2, around 3 m^2 per person. The activity power density is thought to be 15 W/m^2, precisely 7 W/m^2 for the equipment and 8 W/m^2 for the lighting. The heating set point for retail shops is at 21 °C, the COP is considered to be 1.0 and the cooling system is present. The cooling set point is at 26 °C with a COP of 3.0. In addition, there is a mechanical ventilation system with minimum fresh air per area equal to 0.0023 m^3/sm^2 (by considering the minimum of 7 l/s per person (OVK—obligatory ventilation control—boverket)). In addition, an infiltration of 0.05 ACH is considered. WWR is set at 0.6 toward south and 0.5 to all the other directions by considering the recommendation of Statista Research Department and 7 (2020) and practical showcase purpose. DHW load is assumed at 2 kWh/m^2, according to the building regulation [BBR 26 (OVK—obligatory ventilation control—boverket)].

11.3.3.4 School

BBR regulation suggests the occupancy density of school at 0.067 person/m^2 (OVK—obligatory ventilation control—boverket). Therefore, the activity power density is set at 12 W/m^2 with 4 W/m^2 for the lighting power and 8 W/m^2 for the equipment. In the school, the heating set point is set at 22 °C, and the COP is set at 1.0. The cooling system has a set point of 26 °C with a COP of 3.0. There is a mechanical ventilation system with minimum fresh air per area equal to 0.0025 m^3/sm^2 (2.5 l/sm^2) (OVK—obligatory ventilation control—boverket). In addition, an infiltration of 0.05 ACH is considered. The size and orientation of the windows in a school are considered to be similar to an office because of the similarity on daylight needed in the large spaces. Thus, the same values of WWR ratio are used: 0.6 toward south and 0.4 to all the other directions. DHW load is assumed at 25 kWh/m^2,

according to the building regulation [BBR 26 (OVK—obligatory ventilation control—boverket)].

11.3.3.5 Schedules

In this chapter, the schedules are defined according with BBR 26 regulation (OVK—obligatory ventilation control—boverket) or practical assumptions in Table 11.1. For instance, in a residential building of the base case, the lights are considered to be on during two times in the weekdays: one in the morning around 7/8 am and then during the evening. During the weekends, part of the lights are also considered to be on during the day. There are mainly two peaks for equipment during the weekdays and a larger use in the weekends. Part of equipment, such as fridge, is considered to be always on. The heating is always on in order to keep the set point temperature and protect the buildings from moisture damage during the heating season, which starts the first of October and ends the thirtieth of April. From May to September, the heating is considered to be totally off. The rest schedules for offices, retail shops, and the school are similar to each other. Retail shops are considered to be opened over the year in the base case, while offices have one month vacation in August and school has a lower occupancy density in July. The cooling season in all of these facilities lasts for three month and a half, which starts in middle May and finishes at the end of August. The rest schedules for other scenarios can be referred to Table 11.1.

Figure 11.7 illustrates an example of the schedules setup in confinement level 2 for occupancy, lighting, equipment, and cooling in each building type: (a) residential building, (b) office, (c) retail shop, and (d) school. The heating is assumed to be turned on all the time (24 h) during the heating season, from first of October and ends the thirtieth of April. There is no heating from May to September. DHW is setup at the same load across seasons on an annual base. There is no cooling system in residential buildings. On confinement level 2, it is assumed that the unoccupied hours in residential building are 5 h in total each day. For instance, as indicated in Fig. 11.7a, each 2.5 h in the morning and afternoon is assumed for outdoor activities during weekdays, while outdoor hours appear mostly in the afternoon during weekends. These changes in occupancy then influence the setups in school and office buildings. There will be 5 occupied hours in school, and 4.5 h in office building, respectively, as shown in Fig. 11.7b, d. The schedule of retail shops, as shown in Fig. 11.7c, is 4 h longer than that in base case, which are the same for both weekdays and weekends. All the associated lighting and equipment, as well as cooling schedule, are then changed according to the occupancy schedules in each building type, based on the default setup in the base case.

11.4 Results and Discussion

11.4.1 Detailed Simulation Results of Base Case (Level 1)

Figure 11.8 visualized the total energy demands of all the studied buildings in one year, which varying from 13,372 kWh/year to 483,056 kWh/year. Both Figs. 11.9 and 11.10 further break down the all energy consumed categories according to each archetype. In general, there are three dominant energy consumptions in the studied district, where heating, equipment, and lighting account for 40%, 30%, and 27%, respectively. Once looking into each archetype, the principal archetype is the residential building, which need 60% of electricity in lighting category and 94% in heating category, due to its greatest floor area percentage in the district (except for the cooling demand). From the aspect of area weighted average energy demand, the total energy demand of the district is simulated with the results of about 80.0 kWh/m^2/year in average. More details can be found in Table 11.5. It can be divided into the different archetypes as followings: 64.1 kWh/m^2/year for the school; 92.2 kWh/m^2/year for the retail shops; 72.1 kWh/m^2/year for the offices; 80.4 kWh/m^2/year for the residential buildings. Among these, retail shops have the highest energy demand due to the

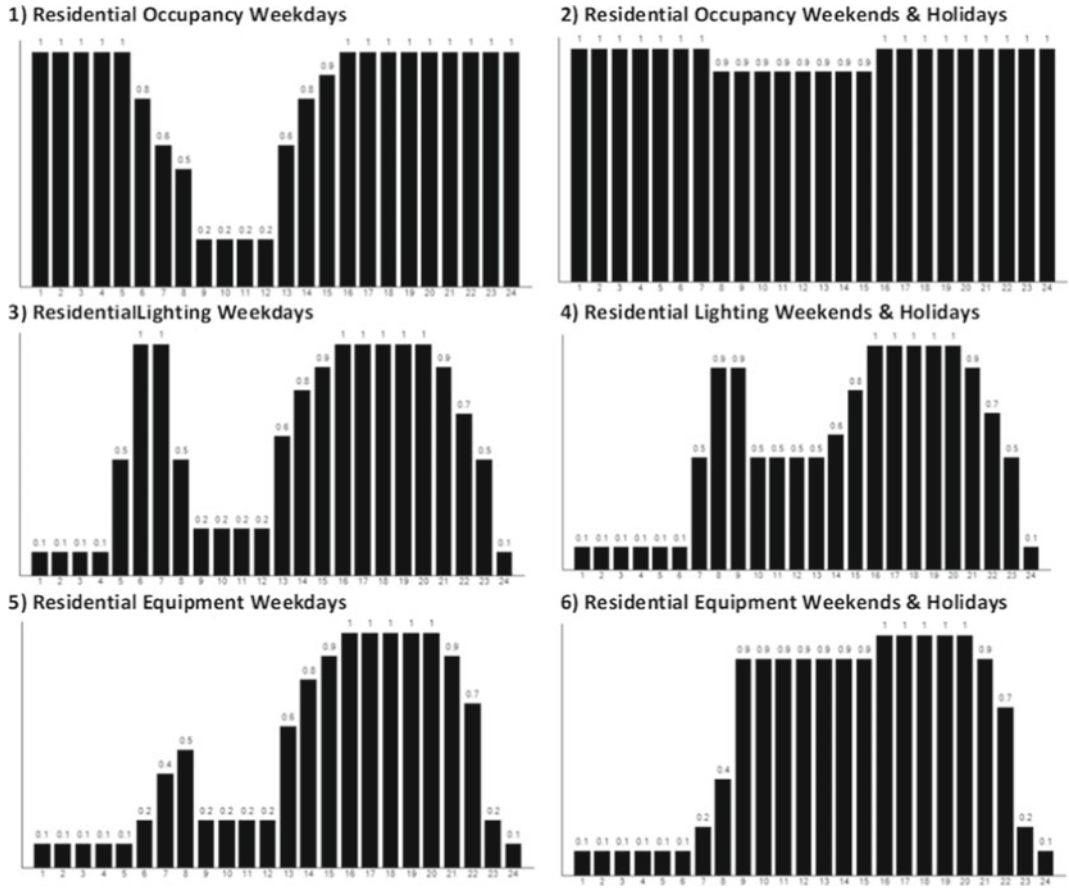
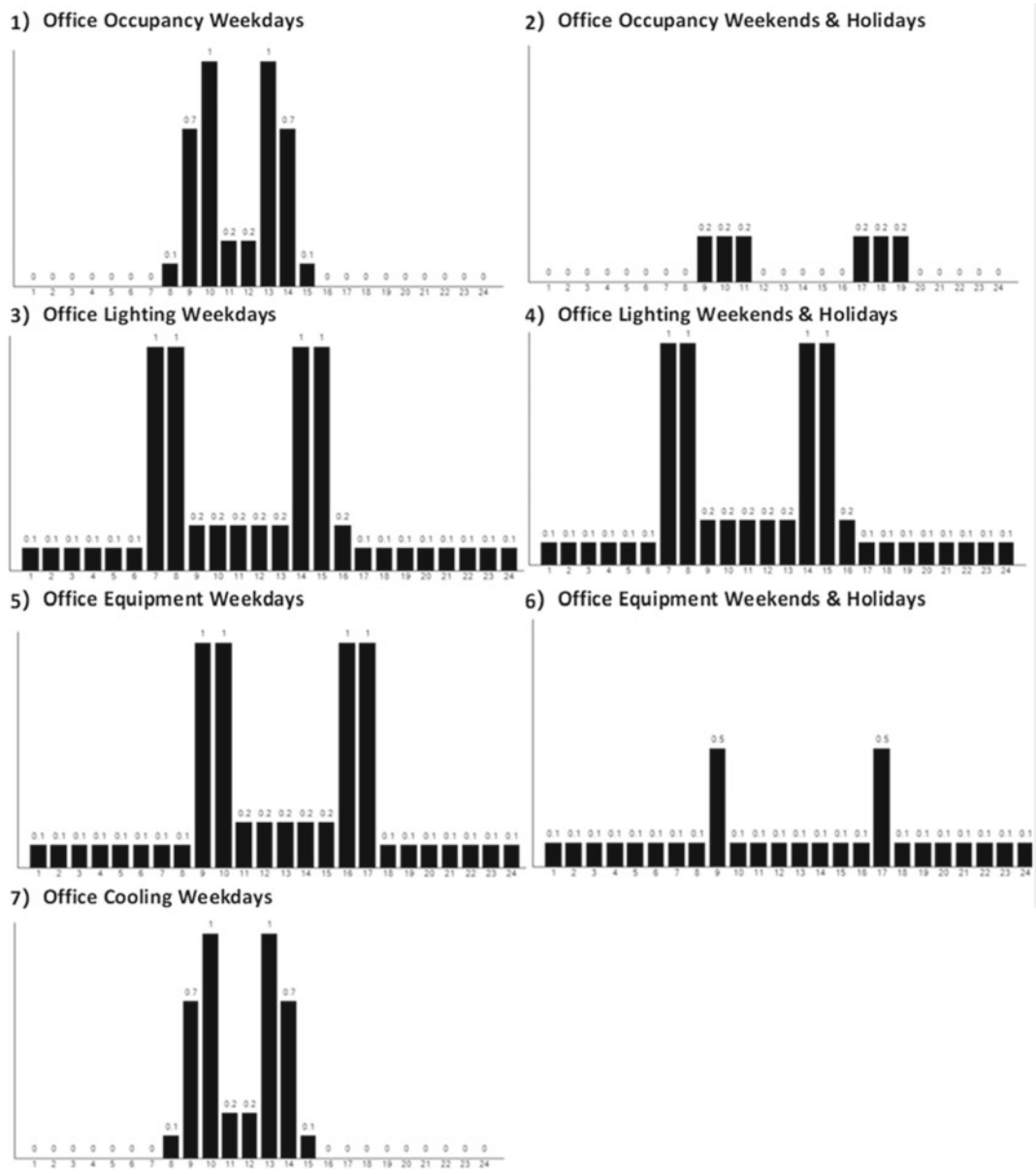


Fig. 11.7 Example of schedules setup in confinement level 2 for occupancy, lighting, equipment, and cooling for each building type: **a** residential building, **b** office, **c** retail shop, and **d** school

high needs in lighting and the equipment operation. Nevertheless, the retail shops have a relatively low heating need (2.4 kWh/m^2). Higher occupancy density, electrical energy demands, and higher WWR together contribute to useful heat gains in reducing the heating load accordingly. On the other hand, these heat gains are not so useful during the summer time. Therefore, the retail shops need most cooling demand in order to avoid overheating issues. Residential buildings ranks the second since the heating demand and the DHW loads are high. Offices and school have nearly the same energy demands, due to the similarity of the input parameters. Offices have a slightly greater consumption in lighting and equipment, hence in cooling consumption, but the heating need is a bit lower.

In this chapter, apart from the total energy demand, we also define the delivered electricity demand and the system energy demand. The delivered electricity demand is the sum of lighting and equipment power loads, while the system energy demand is the sum of other service system energy needs, such as heating, cooling, and DHW.

Figure 11.11 displays these two energy demands varying with floor areas. It is observed that the delivered electricity demand increases nearly linearly with the increasing of the floor area, since the key parameters are set according

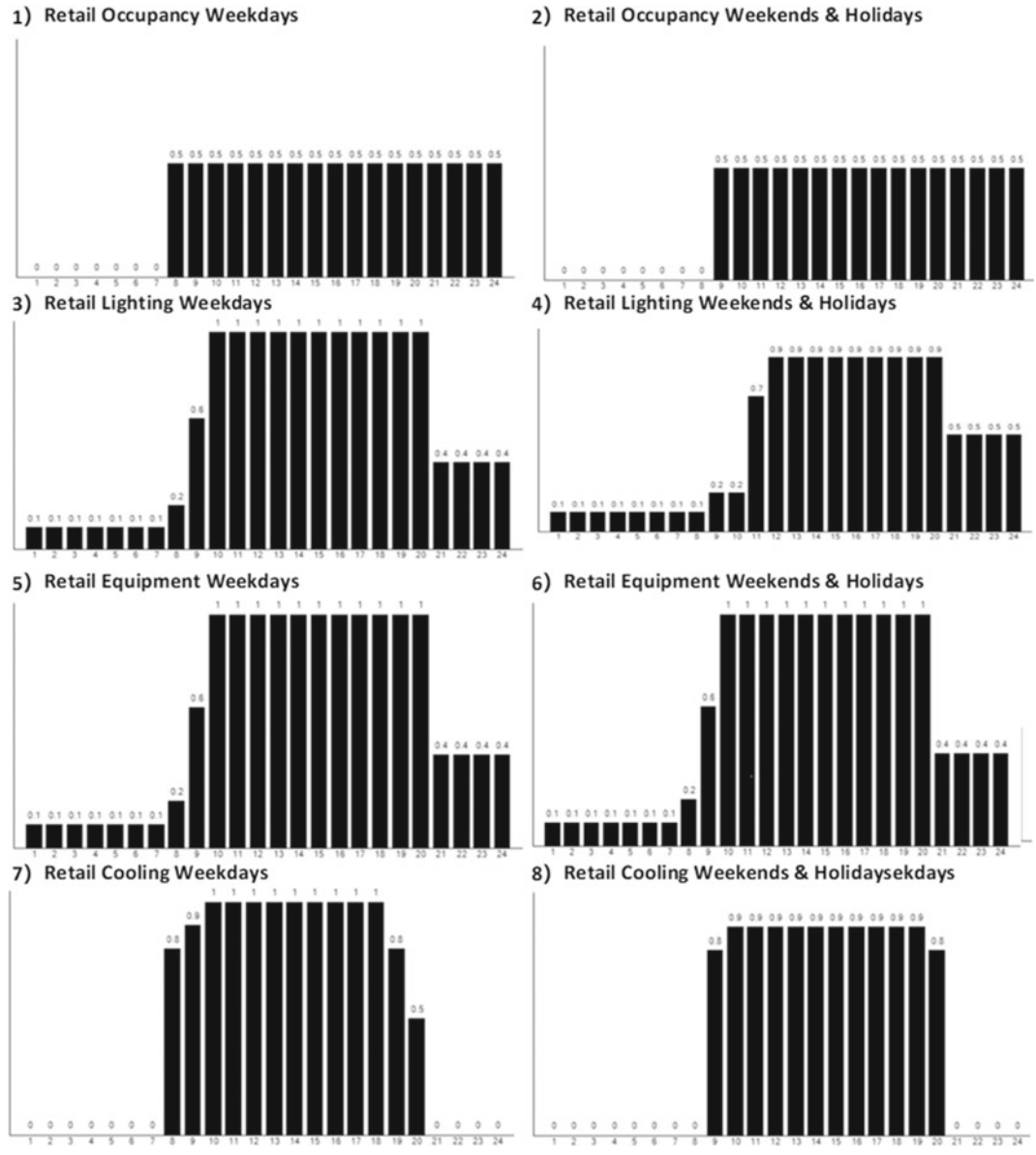


(b)

Fig. 11.7 (continued)

to the floor areas. In contrast, it is not obvious to find a similar trend in the system energy demand, which largely follows the trend of normalized heating demand. There may be two reasons for this: firstly, the DHW demands are set according to individual archetype instead of being dependent on the floor area; secondly, the cooling is not needed in all residential buildings, which account for the greatest total floor area in the district.

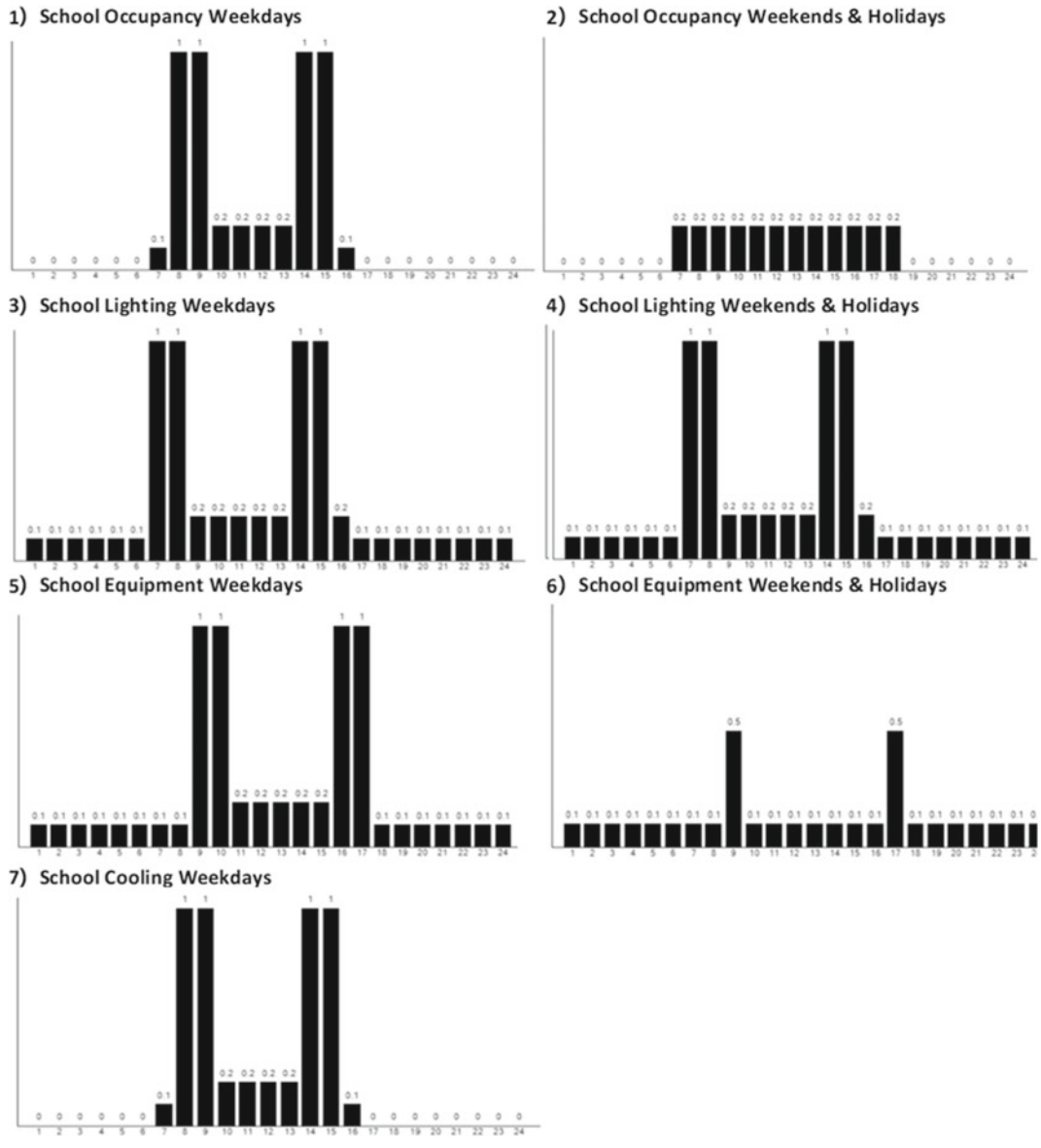
Figure 11.12 presents explicit monthly variations of the studied energy demands in



(c)

Fig. 11.7 (continued)

accordance with each archetype. The delivered electricity demand generally has a relatively smooth annual variation. On the contrary, the system energy possess a distinct seasonal deviation, where it is significantly high during the heating season (starts the first of October and ends the thirtieth of April), but much low from April to September. It is found that the district achieves the peak system energy demand in January when external air temperature is the lowest, dominating by heating demand; it requires least system energy demand in September, when heating and cooling are not needed so much during the transition season.



(d)

Fig. 11.7 (continued)

11.4.2 Uncertainty Analysis

Because this whole study is fully investigated through model setup and simulation, all the results are highly dependent on the assumptions and inputs. Therefore, it is necessary to have an uncertainty analysis about the input parameters based on the simulated results. It is conducted using the base case for the purpose of uncertainty analysis. In general, there are four main types of inputs into the model, which are occupancy profile, DHW load, electrical equipment power density, and lighting power density.

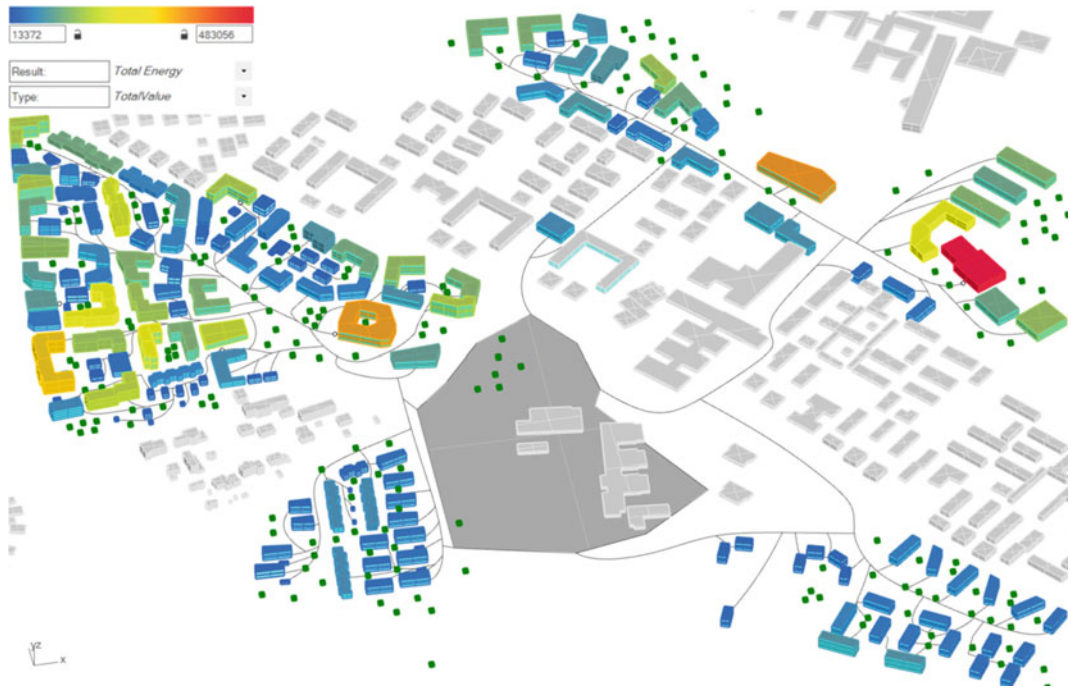


Fig. 11.8 Visualization of total energy demands in the residential district in one year

11.4.2.1 Occupancy Profile Input

According to Tables 11.1 and 11.4, the occupancy density (person/m^2) is set with 0.04, 0.05, 0.067, and 0.3 for residential buildings, office buildings, school, and retail shops, respectively, while the occupancy duration (hours/day/person) is set with 14, 9, 10, 10, and 10, respectively. The occupancy density inputs follow the lower design thresholds from Swedish standards of BBR 26 (OVK—obligatory ventilation control—boverket) and BFS 2016:12 BEN (Boverkets föreskrifter och Allmänna Råd 2016:12). At the same time, the occupancy durations are derived from reports in SVEBY for residential building (Brukarindata Bostäder 2012), school (Brukarindata Undervisningsbyggnader 2015), and commercial building (Brukarindata Kontor Svebyprogrammet 2013), as well as BBR 26 (OVK—obligatory ventilation control—boverket) and BFS 2016:12 BEN (Boverkets föreskrifter och Allmänna Råd 2016:12), especially supporting by several individual studies from Statistics Sweden. Among these, SVEBY stands for 'standardize and verify energy performance in buildings', which is a development program run by the construction and real estate industry, aiming for definition and verification of buildings' energy performance. In SVEBY's reports, user behaviors is continuously updated over time since 2012 in order to obtain continuity and clarity in verification. Thus, it is important that input data is reasonably setup according to the statistics in SVEBY's manuals. For instance, there were two main concluded average occupancy duration hours for residential buildings. The first average occupancy hours is 15.5 h/day person that is concluded from 179 households investigations from different types of places and parts of Sweden, while the another average value is popularly accepted with the average value of 14 h/day/person. In this chapter, we take the latter value as the input in the model.

11.4.2.2 DHW Information Input

Regarding the DHW load, the input (kWh/m^2 per year) is set with 25, 2, 2, 2 for residential buildings (taking majority type of multifamily building), office buildings, school, and retail

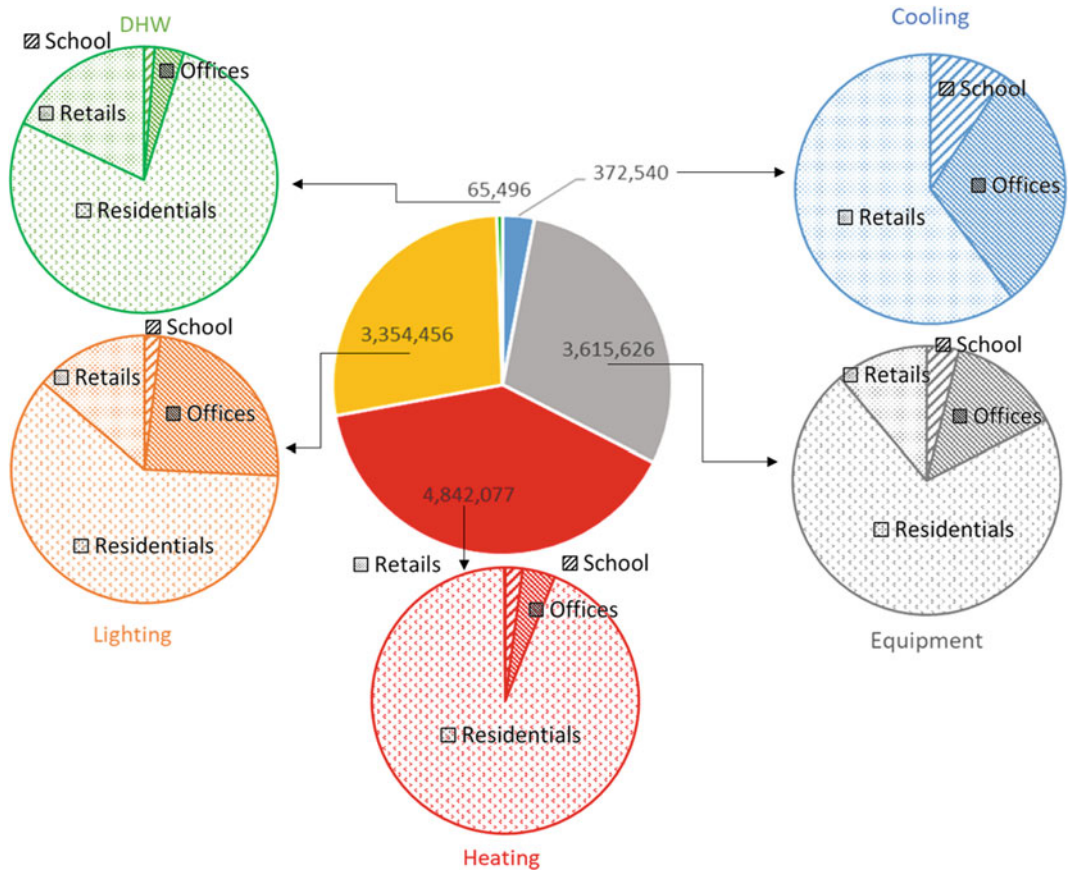


Fig. 11.9 Breakdown pie charts of all energy consumed items for each archetype in the district (kWh/annual)

shops, respectively, which are in accordance to the design suggestions from SVEBY for residential building (Brukarindata Bostäder 2012), school (Brukarindata Undervisningsbyggnader 2015), and commercial building (Brukarindata Kontor Svebyprogrammet 2013), as well as study (Ellegård 2002). Some of these values are also backed up by statistic studies in Sweden. The DHW load of residential buildings comes from a study of 1500 apartments in Stockholm between 1997 and 2003 (Elmroth 2015). The load of office buildings is derived from the average value from the statistics of several office properties (Levin 2013). The DHW load of school is selected for either the primary school or the high school in Sweden (Boverkets föreskrifter och Allmänna Råd 2016:12).

11.4.2.3 Input of Lighting and Equipment

Other than the inputs of occupancy and annual DHW load, the electrical inputs of residential buildings are more sensitive to the final outputs, for the reason that greater magnitude of amount and the following lock down modes investigation. According to the power density of lighting and equipment in Table 11.4, two input categories (parameter range) have been defined for lighting and equipment power density, respectively, using the value uniformly discrete within the suggested parameter range in the references of SVEBY for residential building (Brukarindata Bostäder 2012), and statistics in previous studies (Ellegård 2002) (Levin 2013) and surveys (Indata För Energiberäkningar i Kontor Och

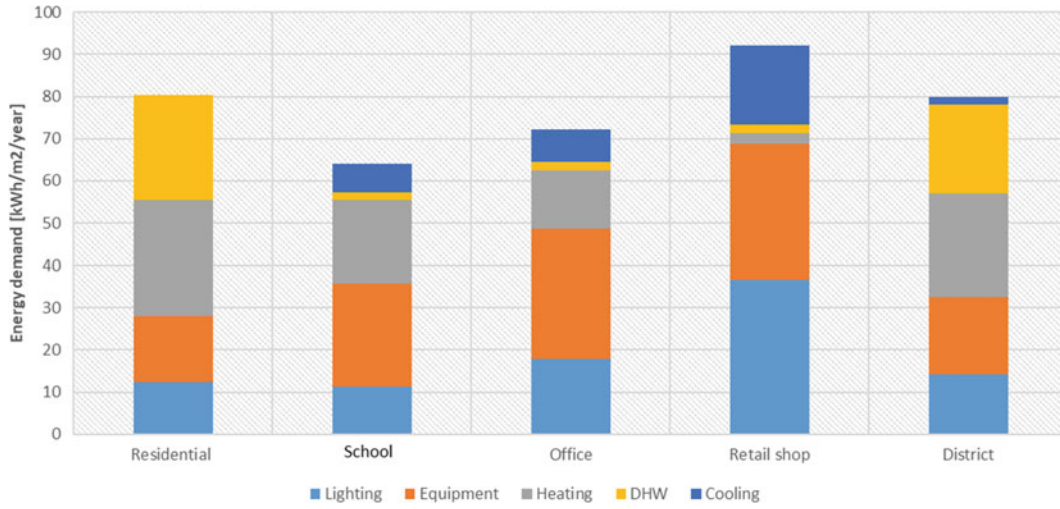


Fig. 11.10 Share of normalized individual energy demand in each archetype

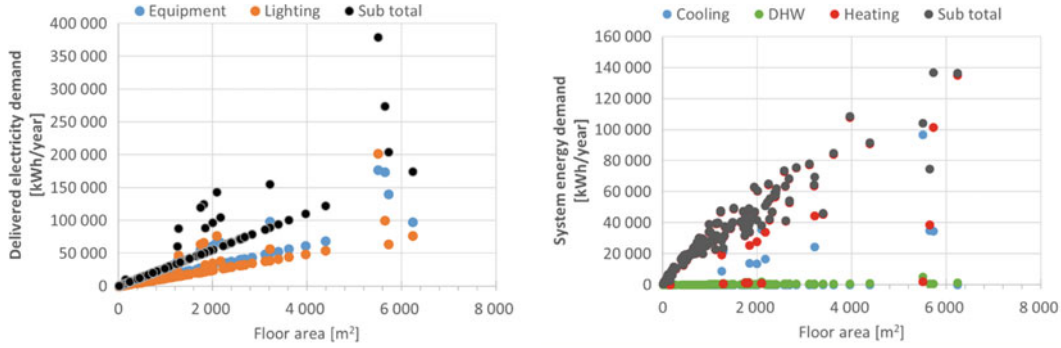


Fig. 11.11 Scatter plots of energy demand with floor area

Småhus 2007). The proposed sensitivity analyses cover all the archetypes with total 16 varied cases, for the purpose in exploring the potential impacts from these two critical inputs on the entire district energy performance in a year. In each case, only one value from the potential input category is tested on one building type, while other parameters keep their given values as same as that in the base case.

Figures 11.13 and 11.14 demonstrate the sensitive impacts of the two input categories on 'normalized delivered electricity demand' and 'normalized system energy demand' for this district in a year. In each diagrams, there are 5 dots in each archetype, including the default value existed in the base case and corresponding 4 parameters that vary with 0.5 W/m^2 each time and for comparison. For instance, the default equipment power density is 4 W/m^2 in residential buildings, so that the tested equipment power density values are set at 3 W/m^2, 3.5 W/m^2, 4.5 W/m^2, 5W/m^2, respectively, in the same residential buildings. It is clear that all the tested parameters have reasonable sensitivity performance by comparing to the base case. In terms of the normalized delivered electricity load, the deviations change in range of − 2.8% to 2.5% from the angle of equipment power density variation, and in the range of − 2.6% to 2.5% from the aspect of lighting power density variation. In terms of normalized system

Table 11.5 Energy demands for each archetype at base case in one year duration

Archetype	Lighting (kWh/m^2)	Equipment (kWh/m^2)	DHW (kWh/m^2)	Heating (kWh/m^2)	Cooling (kWh/m^2)	Delivered electricity (kWh/m^2)	System energy (kWh/m^2)	TOTAL energy demand (kWh/m^2)
	O;1	O;2	O;3	O;4	O;5	O;1 + O;2	O;3 + O;4 + O;5	O;1 + O;2 + O;3 + O;4 + O;5
Base case with no COVID-19 influence (Area weighted average: average values are calculated based on the respective area percentage of each archetype)								
Residential (82.7% of total area)	12.3	15.6	25.0	27.4	0.0	28.0	52.4	80.4
School (2.9% of total area)	11.2	24.5	2.0	19.6	6.7	35.7	28.4	64.1
Office (8.1% of total area)	17.7	30.8	2.0	14.0	7.6	48.5	23.6	72.1
Retail shop (6.3% of total area)	36.7	32.1	2.0	2.4	18.9	68.9	23.3	92.2
Area weighted average	14.3	18.2	21.0	24.5	2.0	32.4	47.6	80.0

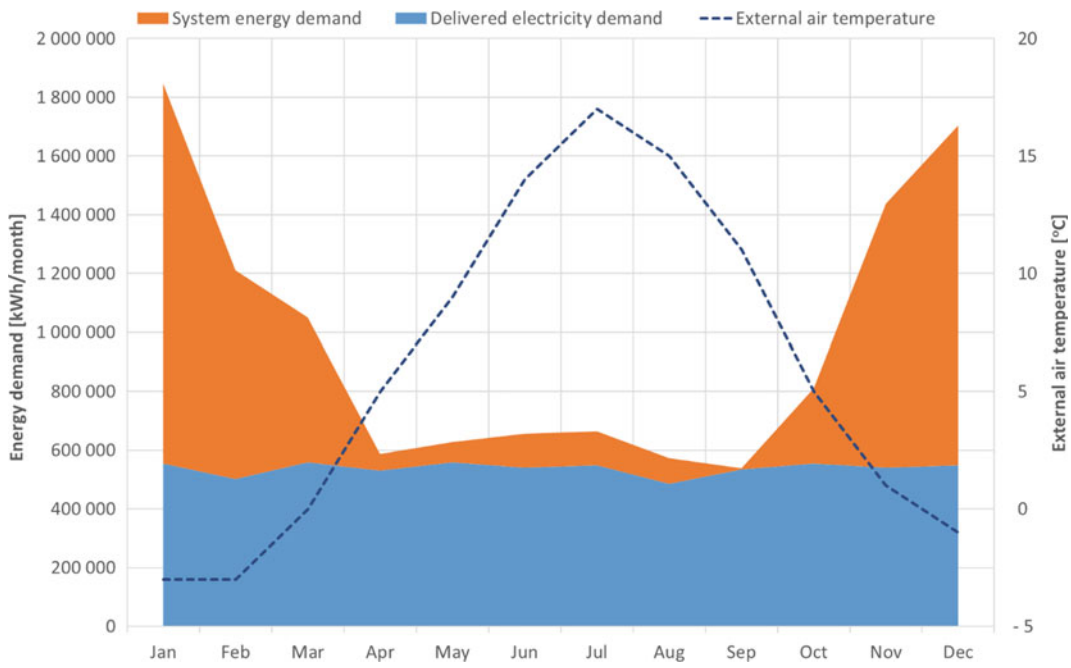


Fig. 11.12 Variation of energy demands and air temperature with months

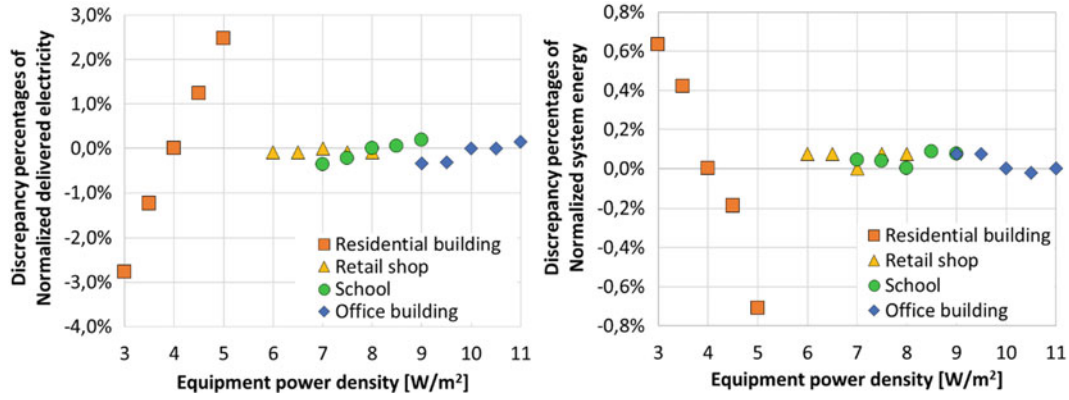


Fig. 11.13 Sensitive study for impacts of the equipment power density on energy demand

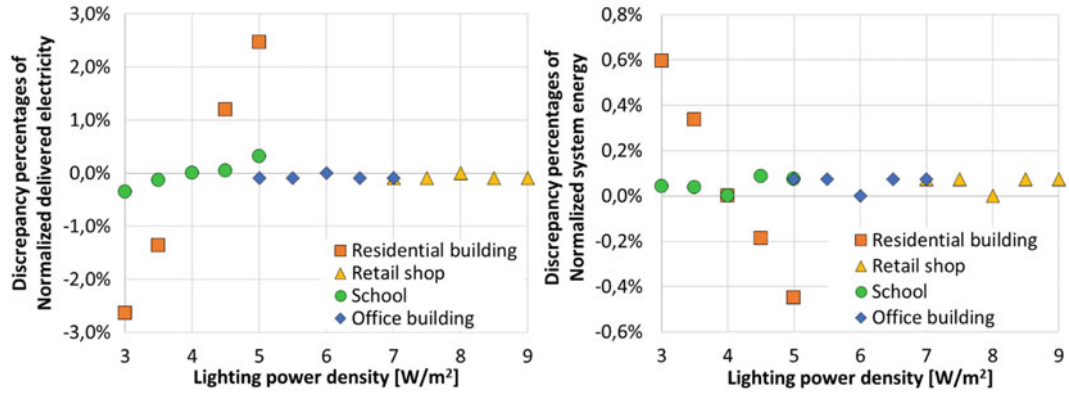


Fig. 11.14 Sensitive study for impacts of the lighting power density on energy demand

energy load, the maximum deviations are even less, which are in the range of − 0.7% and 0.6% from the aspect of equipment power density variation, and in the range of − 0.4% and 0.6% from the view of lighting power density variation. The majority of the discrepancy percentages appear around ± 0.1%. All the noteworthy deviations are only discovered in the residential buildings, because residential buildings account for the largest floor area in the whole district, leading this archetype become the most varying one. But the overall impact on the residential building is limited with the acceptable discrepancy ratios (i.e., ± 2.8%) in both varying cases.

11.4.2.4 Comparison to the Building Standards

In this section, the weighted average system energy demand from the base case is compared with the requirement at different local standards, including the Swedish Housing Agency's building rule (OVK—obligatory ventilation control—boverket), Passive house standard in Sweden (FEBY 2018), and Swedish green building standard (Miljöbyggnad). Table 11.6 presents the comparison results. It is found that the weighted average system energy demand is simulated at about 47.6 kWh/m^2/year, which is much lower the basic requirement in building regulation

Table 11.6 Comparison of the simulation result to different standards in Sweden

Area weighted average system energy demand (kWh/m^2/year)	Boverket's building regulations (BBR 26) (OVK—obligatory ventilation control—boverket) (kWh/m^2/year)	Passive house standard in Sweden (FEBY 2018) (kWh/m^2/year)	Swedish green building standard (Miljöbyggnad) (kWh/m^2/year)
Base case result	Basic requirement	Gold level	Gold level
47.6	$\leq$ 80	$\leq$ 52	$\leq$ 40

(BBR26), and it meets the gold level of passive house standard. But it is a bit higher that the requirement of gold level in green building standard. As a result, the simulation result is reasonable by comparing to local standards.

11.4.3 Simulation Results of Different Confinement Levels

Table 11.7 reports the summary of the simulation results in different confinement levels. In the level 2, confinement measures are considered as 'soft' when citizens can go out for work and activities in certain hours. It is observed that the average delivered electricity increase from 32.4 kWh/m^2/year at level 1 to 38.5 kWh/m^2/year (about 18.7% increase) at the level 2, contributed mainly by residential buildings and retail shops. This can be explained by the fact that the electricity demand in residential buildings rises when people stay longer time at home, with augmented conventional demand (lighting, cooking, and dishwashing) and more equipment demand (online meetings and computation-related workings). The retail shops are run longer for necessary purchase, delivery and pickup of food, medicine, and commodity, so a higher electricity demand is found for lighting and equipment. Heating demand varies differently in each archetype, but overall heating is needed less. For instance, heating is less required in residential buildings because of more internal heat gains when people stay longer at home; both school and offices need more heating due to less internal heat gains; in retail shops, heating is needed less when it operates with much lower occupant density and thus with less cold ventilation required. The general cooling demand reduces slightly. Offices and school decrease cooling need as they are open for shorter time with less occupants and less heat gain in summer. While retails shops need a bit more cooling since they are run for longer time, even though they have less internal heat gains due to less occupants. The DHW loads are less than that on level 1 by considering the less occupants in school/offices/shops. The average total system energy demand at this level is about 44.4 kWh/m^2/year, decreasing by 7.1% comparing to that at level 1 (base case). It is then observed that the increased electricity demand dominates the change in energy demand when DHW, heating and cooling demands are required less.

At the level 3, confinement is even harder than that on level 2. Citizens can still go out for work and activities in fewer hours. The varying trends of energy demands for archetypes, energy categories, and total amount are similar to that on level 2. Comparing to the base case, the average overall electricity demand increases by 17.7%, and the mean total energy demand drops by 10.0%. The increased ranges are slightly smaller than that on level 1. The electricity demand is slightly lower than that on level 2, because less power needed in offices and school, while residential and retails stays nearly unchanged. The overall heating, cooling, and DHW demands become less than that on level 2 since fewer occupants are expected at offices, school, and retail shops. This shows that the electricity demand still influences mostly, while the system energy demand is required less.

In the case of level 4, citizens stay at home 24 h per day, while offices and school are empty. Only the retail shops are open but with the least occupancy density. Compared to the base case, the average delivered electricity in the whole strict increases by 14.3%, which contributed

Table 11.7 Simulation results of different confinement levels in one year duration

Archetype	Lighting (kWh/m^2)	Equipment (kWh/m^2)	DHW (kWh/m^2)	Heating (kWh/m^2)	Cooling (kWh/m^2)	Delivered electricity demand (kWh/m^2)	System energy (kWh/m^2)	Total energy demand (kWh/m^2)
	O;1	O;2	O;3	O;4	O;5	O;1 + O;2	O;3 + O;4 + O;5	O;1 + O;2 + O;3 + O;4 + O;5
Level 2 (Area weighted average approach: average values are calculated based on the respective area percentage of each archetype)								
Residential (82.7% of total area)	18.1	19.0	25.0	23.0	0.0	37.1	48.0	85.1
School (2.9% of total area)	8.1	19.2	1.5	23.2	6.2	27.3	30.9	58.2
Office (8.1% of total area)	10.9	19.0	1.5	23.6	5.6	29.9	30.7	60.6
Retail shop (6.3% of total area)	39.1	34.2	1.5	1.1	19.2	73.3	21.9	95.1
Area weighted average	18.6	20.0	20.9	21.7	1.8	38.5	44.4	82.9
Level 3 (Area weighted average approach: average values are calculated based on the respective area percentage of each archetype)								
Residential (82.7% of total area)	18.1	19.0	25.0	21.5	0.0	37.1	46.5	83.6
School (2.9% of total area)	7.2	16.8	1.0	25.4	5.1	24.1	31.4	55.5
Office (8.1% of total area)	9.8	16.9	1.0	26.1	5.2	26.7	32.2	58.9
Retail shop (6.3% of total area)	39.1	34.2	1.0	1.2	19.1	73.3	21.3	94.5
Area weighted average	18.5	19.7	20.8	20.7	1.8	38.2	43.2	81.4
Level 4 (Area weighted average approach: average values are calculated based on the respective area percentage of each archetype)								
Residential (82.7% of total area)	18.1	19.0	25.0	19.9	0	37.1	44.9	82.0
School (2.9% of total area)	3.7	9.5	0.0	32.6	3.4	13.2	36.0	49.2
Office (8.1% of total area)	6.4	10.6	0.0	32.4	4.2	17.0	36.7	53.7
Retail shop (6.3% of total area)	39.1	34.2	1.0	1.2	19.0	73.3	21.2	94.4
Area weighted average	18.1	19.0	20.7	20.1	1.6	37.1	42.5	79.5

mostly by residential buildings and retail shops, while offices and school are closed. In the contrast, the needs for heating/cooling and DHW drop, respectively. The reasons are similar to previous scenarios. The average total system energy demand decreases by about 12.0%. The increased amount of electricity demand is nearly equal to the drops of the system energy demands.

11.4.4 Overall Comparison and Discussion

Figure 11.15 shows the variation of the average delivered electricity demand and the mean system energy demand in the base case and different confinement levels for one year duration. Comparing to level 1 (base case), the mean delivered electricity demand increases in range of 14.3–18.7%, while the average system energy demand decreases in a range of 7.1–12.0%. The stricter confinement measure leads to the lower increase percentage for the average delivered electricity demand, but higher decrease percentage in average system energy demand. In a 'soft' case as levels 2 and 3, both schools and office buildings have to stay open, resulting in a larger electricity need in total. While in a fully 'lock down' case as level 4, schools and offices are closed without any power consumption. So the overall electricity demand increases at a higher percentage when confinement measures are 'soft' in such a residential district. The system energy demands decrease with the tightness of confinement measures. When the lock down ratio rises, less and less heating/cooling and DHW are required in offices, schools, and retails. Meanwhile, although more DHW is needed in residential buildings, less heating is required due to more internal heat gains.

From Fig. 11.16, it is obvious that the electricity demand is higher in residential buildings and retail shops when confinement level increases, but school and offices need less electricity. These results are in line with the practice and assumptions. For instance, occupants need more electricity when they have to stay longer at home for living and working; retails require more power when they are assumed to open for longer time, in order to create opportunities for shopping peak shifting and social distance. Offices and school don't need so much electricity when they are close partially or fully.

In Fig. 11.17, the total system energy demands of each archetype at base case and different confinement levels are displayed. When the confinement becomes harder and harder, more and more internal heat gains from occupants, lighting and equipment can be expected in residential buildings, leading to a lower and lower heating need there. Although DHW and cooling demand increase, but their magnitude is much smaller than heating load. So the total system energy demand in residential buildings decreases when confinement level increases. School and office buildings vary oppositely as both school and offices need more heating due to less internal heat gains. Heating system has to stay open even when there is no occupant in offices and school, to ensure heathy indoor environment. So higher confinement level, more system energy demands are necessary in schools and offices. Retails shops need a bit more cooling in summer for longer opening, but they requires less heating meanwhile in winter when occupant density/ventilation is much smaller, causing a general drop in overall system energy load. However, this decrease percentage in retails seems very limited when the confinement level increases.

Figure 11.18 demonstrates the variation of energy demand with month, including the total aggregated delivered electricity and the system energy loads for the whole district. In this way, it explicitly exhibits the detailed varying patterns at the base case and at different confinement levels across a whole year. All the delivered electricity demands have a relatively smooth annual variation. In the base case, the monthly total delivered electricity demand has sharp drops in February and August. It is found that there are two main factors that influences the monthly total delivered electricity demand: (1) number of days in a month and (2) holidays. In February, the decrease in the total delivered electricity demand is mostly caused by the relatively smaller number

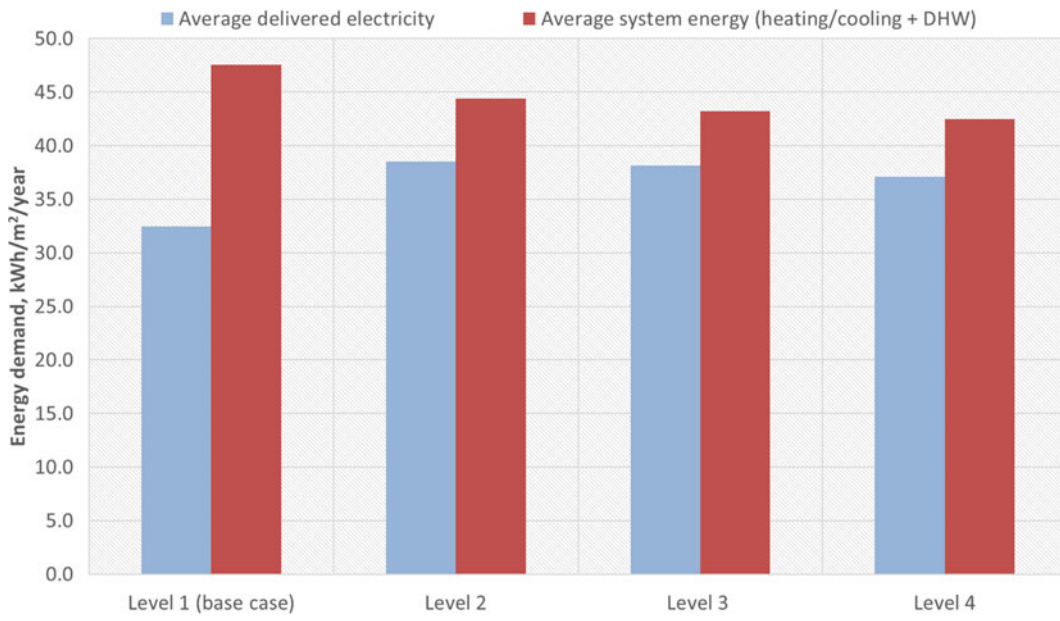


Fig. 11.15 Average delivered electricity and system energy demands of the whole district at base case and confinement levels

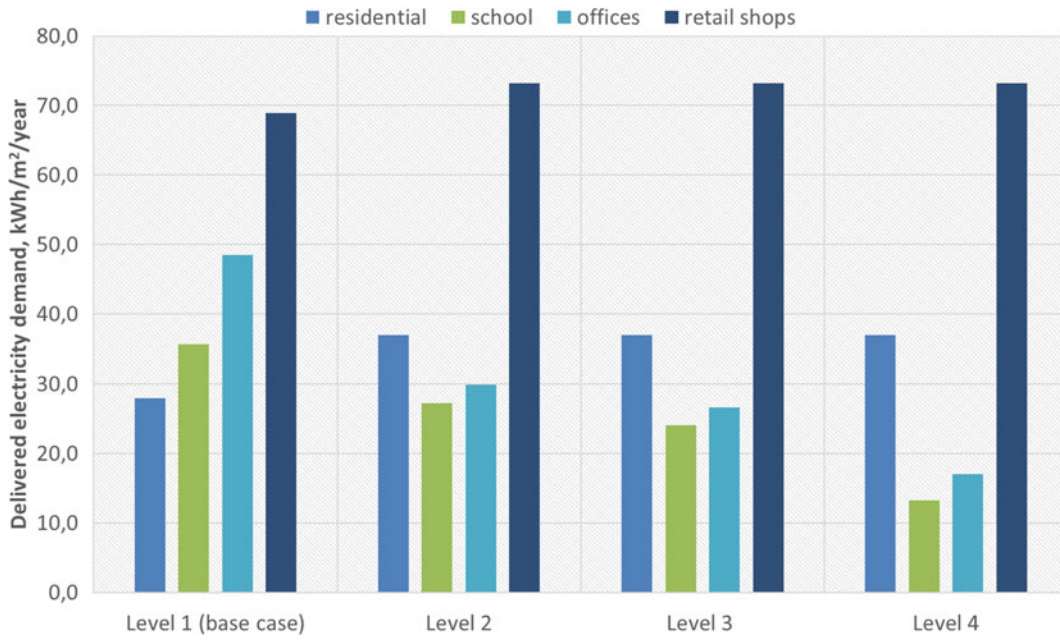


Fig. 11.16 Delivered electricity loads of each archetype at base case and confinement levels

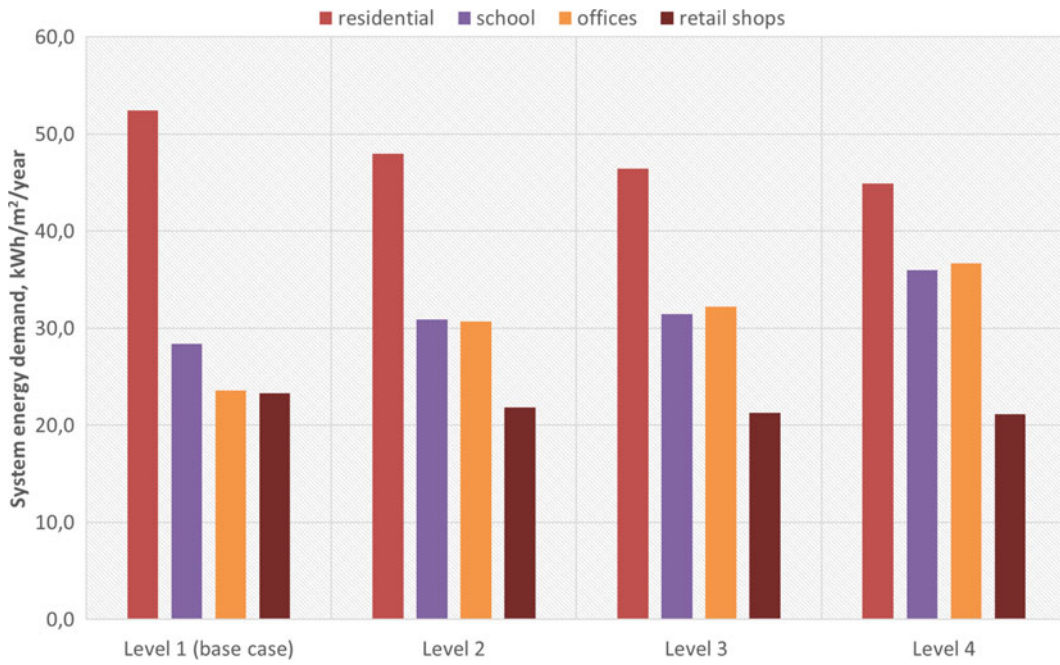


Fig. 11.17 Total system energy demands of each archetype at base case and confinement levels

of days in the months (i.e., 28 days as compared with 30 or 31 days in other months). In fact, compared with a month of 30 days, the decrease of days in February is about 6.7%, while the approximate electricity demand reduction is about 7.7%. It is also noticed that all the months with 30 days have lower delivered electricity demand than the others with 31 days, and February has only 28 days; so it is much lower. In addition, sport break usually takes place for one week in February and early November in Borlänge, and most family go out for sports holidays in the base case. This one-week sport break is another reason leading to decrease in the delivered electricity demand. Different from the summer holiday, only part of people with children usually take sport breaks. So the offices will keep open during the sport breaks, and only part residential buildings will have lower electricity demand in the base case. Therefore, the sport break has relatively smaller magnitude in electricity decrease than that of the day numbers. Thus, in the autumn, the impact of sport break on the demand decrease is not so significant, since the autumn sports break usually appears in early November with total 30 days in that month. So there is a different trend for the total aggregated electricity use in this two sport holiday periods. In the simulation, it assumes that most people take two weeks holidays in early August based on the local social custom in Borlänge city. Such long holiday will lead to significantly drops in electricity use at all types of buildings, due to closed offices/school and unoccupied residential buildings. In other confinement levels from level 2 to level 4, less outdoor holidays (such as sport breaks) are foreseen, comparing to that on the base case. In the simulation, during the holidays for the three confinement levels, it considers that most people will stay at home for longer time, without going out for sports activities. It is observed that the valleys of electricity demands are no longer in August, but electricity use in August is still relatively smaller, since offices and school will close for a short period during summer break. As a result, the aggregated electricity demand in August will be slightly lower than that in other months. While in February, as it has the least days in a month, the aggregated electricity demand is still much lower than that in the other

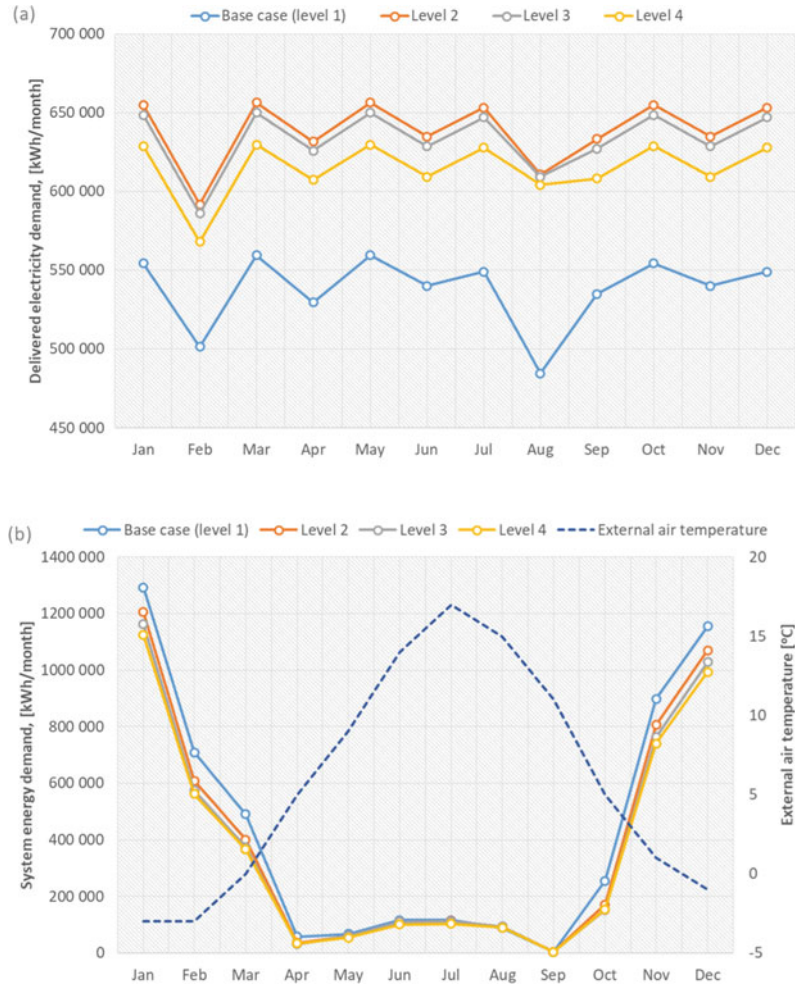


Fig. 11.18 Energy demands at base case and different confinement levels with months: **a** the delivered electricity variation; **b** the system energy variation

months, even with less outdoor holidays, resulting in the lowest monthly electricity demands in all these three levels. It is obvious that the confinement measures lead to higher delivered electricity loads than the base case all year around. This is because, in all the three confinement levels, retails are assumed to open longer; and more importantly, more occupants will stay longer at home for living and working in the simulated one year duration. These two changes result in larger needs in lighting and equipment in each month. In this district, residential buildings account for the largest proportion, so the total delivered electricity demand increases greatly in all the months, compared to the based case. However, by comparing level 2 to level 3 and 4, it is unsupervised that the higher level of confinement would not lead to higher delivered electricity. In level 2, school and office buildings have to be remained open, which have to consume electricity to maintain normal operation. In level 3, the open durations for school and offices are shorter than that in level 2, requiring less lighting and equipment power. In level 4, all school and offices are closed, and there is no electricity need any more. As the proportions of school and offices are not as large as that of residential buildings, such reduction in electricity need is not obvious in each month from level 2 to level 4.

In general, the system energy demands follow the principle trends for all 4 cases during a year,

which have peak demands in winter and much lower demands in transit seasons and summer. In the winter, heating demand dominates while in the summer cooling needs become prominent. During transit seasons, both heating and cooling demands are little, resulting in the lowest system energy demand in September. These variations are also consistent with the monthly mean external air temperature, which achieves lowest in January and peaks in July. It is clear that confinement causes lower system energy loads than the base case, which are in proportional to confinement level in each month.

This chapter, as a preliminary study, is the first step to investigate the impact of COVID-19 on energy patterns and demands in different buildings at a district. Although it doesn't address the impact of the detailed occupant behavior change on energy demand, the initial findings at the aggregated level could help policy makers to understand the corresponding overall energy patterns, demand and impact on different seasons in each building type. It is then possible to have benchmark for setup of new building standard for extreme crisis, such as lighting/equipment power density, ventilation rate, and building design guidance. It is also useful for policy makers to trigger the discussion about sharing of energy cost in the mode of working from home. Moreover, it can support city-level or even regional-level policymaking in careful design of confinement measures, planning of new district and its energy supply system/infrastructure, as well as emergent operation of energy system to guarantee sufficient energy supply to different buildings.

By comparing Tables 11.5 and 11.7, it is found that the increase or decrease in total energy demand of the whole district will be depending on the confinement level, determined by the net amount of the increased electricity need and the decreased system energy demands. Two 'soft' confinement scenarios lead to the increase in overall energy demand, while 'full lock down' scenario results in a slight lower total energy demand. So, it is not easy to say that a confinement measure will increase or decrease the overall energy consumptions, as it depends on the 'functions' of the buildings in the district. There might be a different result in an industrial district or a commercial district. In addition, the change trend in energy demand may be different in another context or another country, where the regulations are different. As a result, more investigation should be done in particular context and the approach proposed in this chapter can be replicated in different cases. Besides, discussions should be also carried out when considering energy demand on a system level, which should contain energy use in different sectors, such as building, transportation, and industry.

11.5 Limitations and Future Work

This chapter conducted a simulation study of a non-existing (virtual) neighborhood, which is still on the design and planning stage. As a result, there is no real data that can be used to validate the model. But we try to make sure the model as accurate as possible, by (1) using well-known UMI tool, (2) relaying on reasonable assumptions, building regulations, and statistics according to Swedish authority standards (for instance, SVEBY for residential building (Brukarindata Bostäder 2012), school (Brukarindata Undervisningsbyggnader 2015), and commercial building (Brukarindata Kontor Svebyprogrammet 2013), and with recommendation from the national building standards BBR 26 and BFS 2016:12 BEN in Sweden (OVK—obligatory ventilation control—boverket) (Boverkets föreskrifter och Allmänna Råd 2016:12) as well as industry, and (3) a uncertain analysis of key input parameters and comparison with energy demand requirement in different standards. This chapter's scope is limited to investigate the impact of different occupancy profiles (e.g., occupancy density and schedule), due to COVID-19, on energy demand. It relies on the existing building regulations, historical statistics, and the assumed confinement measurements, owing to lacking of real data. The detailed occupancy behavior changes (e.g., telecommuting scenarios) due to COVID-19 is not investigated as a result. For instance, it lacks data that describes how equipment

(i.e., computers, TV, etc.) and lighting using in building changes with occupancy in such a pandemic. Meanwhile, this chapter has restricted impacts on delivered electricity loads based on the rigid assumptions of overall operation schedules, which are lacking of explicit social background knowledge. In the future, it is important to have the real measurement data for such detailed behavior change to benchmark the parameter setup in the simulation tool. In terms of more accurate community energy model results, it is also significant to fully obtain compulsory social background knowledge, for instance, social customs, holiday mechanism, and various operating modes of different functional institutions. These together will not only further help to improve the simulation accuracy, but also make recommendations for solid update of building regulation, especially for emergencies as COVID-19 pandemic.

On another hand, compared to standard method used in this chapter, energy demand usually has a different correlation with occupancy profile in different types of buildings. For instance, based occupancy estimation by mobile phone, energy consumption differs by + 1% to − 15% for residential buildings and by − 4% to − 21% for commercial buildings, compared to standard methods (Barbour et al. 2019). In some buildings, even though the occupancy shows a significant correlation with the overall amount of electricity used, but it lacks correlation with the amount of cooling supplied by the HVAC system in campus buildings (Martani et al. 2012). This is because commercial and non-residential buildings (especially larger, older constructions) are often monitored and controlled through a facilities provider, which reduces the direct feedback of information to building occupants. With large-scale HVAC systems, the focus is often placed on maintenance rather than a rapidly responsive system that can adjust in real time to varying levels of occupancy (Martani et al. 2012). As a result, the standard approach in this chapter has its limitation, which might not be directly related to current benchmark classifications (i.e., office buildings). In the future, new models for occupancy and energy demand, such as (Barbour et al. 2019; Menezes et al. 2014), will be needed for designers to utilize at the early design stage for more accurate estimation.

11.6 Summary

This chapter evaluates the impact of the confined measures due to COVID-19 outbreak on the energy demand for a building mix in a 'virtual' district in Sweden, by proposing the dedicated occupancy schedules related to the different confinement scenarios.

In the level 1 (base case, normal life without confinement measure), the delivered electricity energy demand and the total system energy demand of the district are simulated at about 32.4 kWh/m^2/year and 47.6 kWh/m^2/year respectively in average, where retail shops have the highest energy demand due to the high needs in lighting and the equipment operation. Residential buildings rank the second since the heating demand and the DHW loads are high. Heating accounts for the largest proportion which therefore dominates the variation trend of total energy demand with the months over a year.

Comparing to level 1 (base case), in the scenarios with different levels of containment measures, the delivered electricity of the entire district increases in a range of 14.3–18.7%, contributed mainly by residential buildings and retail shops. However, the mean system energy demands, such as heating, cooling, and domestic hot water, decrease in a range of 7.1–12.0%. These two variation trends nearly cancel each other out, leaving the total energy demand almost unaffected in such a residential district within Swedish context. In addition, the delivered electricity demands in all the four cases have a relatively smooth variation across a year. The system energy demands follow the principle trends for all the four cases during a year, which have peak demands in winter and much lower demands in transit seasons and summer.

The increase or decrease in total energy demand of the whole district is depending on the confinement level, determined by the net amount of the increased electricity need and the decreased

other demands. Two 'soft' confinement scenarios lead to the increase in overall energy demand, while the 'full lock down' scenario results in a slight lower total energy demand. Therefore, it will be partial to say that a confinement measure will increase or decrease the overall energy consumptions, as it depends on the 'functions' of the buildings in the district. There might be a different result in a different building district, such as an industrial district or a commercial district. A single district cannot reflect the overall influence of the COVID-19 on the whole energy system. In addition, the energy demand may change differently in another context or another country, where the regulations are different. As a result, more investigation should be done in particular context and the approach proposed in this chapter can be replicated in different cases. Besides, a comprehensive discussion should be also conducted when considering energy demand on a system level, which include energy use in building, transportation and industry.

This chapter's scope is limited to investigate the impact of different occupancy profiles (e.g., occupancy density and schedule), due to COVID-19, on energy demand. It relies on the existing building regulations, historical statistics, and the assumed confinement measurements, owing to lacking of real data. The detailed occupancy behavior changes (e.g., telecommuting scenarios) due to COVID-19 is not investigated. In the future, it is important to have the real measurement data for such detailed behavior change to benchmark the parameter setup in the simulation tool for the improved accuracy. It is also significant to fully obtain compulsory social background knowledge, for instance, social customs, holiday mechanism, and various operating modes of different functional institutions.

There might be a few more waves of COVID-19 outbreak, which may lead to different levels of closures. The corresponding impacts on Swedish or global energy demands in different types of districts and sectors are necessary to be carried on. It is expected that this chapter opens up such initiative by conducting a preliminary study in a residential district through simulation. These initial findings will help mitigate the influence from COVID-19 with appropriate pre-preparations for new policy designs that can withstand future long-term shocks. The research results will help policy makers to understand the corresponding energy patterns and performance in different seasons. It is then possible to have benchmark for setup of new building standard for extreme crisis, such as lighting/equipment power density, ventilation rate and building design guidance. It is also useful for policy makers to trigger the discussion about sharing of energy cost in the mode of working from home. Moreover, it can support city-level or even regional-level policymaking in design of confinement measures, planning of new district, and its energy supply system/infrastructure, as well as emergent operation of energy system to guarantee sufficient energy supply to different buildings.

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12 Uncertainty-Based Near-Zero Energy Buildings Life-Cycle Performance Analysis

Pei Huang and Yongjun Sun

Abstract

Near-zero energy buildings (nZEBs) are considered as an effective solution to mitigating CO_2 emissions and reducing the energy usage in the building sector. A proper sizing of the nZEB systems (e.g. HVAC systems, energy supply systems, energy storage systems, etc.) is essential for achieving the desired annual energy balance, thermal comfort, and grid independence. Two significant factors affecting the sizing of nZEB systems are the uncertainties confronted by the building usage condition and weather condition, and the degradation effects in nZEB system components. The former factor has been studied by many researchers; however, the impact of degradation is still neglected in most studies. Degradation is prevalent in energy components of nZEB and inevitably leads to the deterioration of nZEB life-cycle performance. As a result, neglecting the degradation effects may lead to a system design which can only achieve the desired performance at the beginning several years. This chapter, therefore, proposes a life-cycle performance analysis (LCPA) method for investigating the impact of degradation on the longitudinal performance of the nZEBs. The method not only integrates the uncertainties in predicting building thermal load and weather condition, but also considers the degradation in the nZEB systems. Based on the proposed LCPA method, a two-stage method is proposed to improve the sizing of the nZEB systems. The study can improve the designers' understanding of the components' degradation impacts and the proposed method is effective in the life-cycle performance analysis and improvements of nZEBs. It is the first time that the impacts of degradation and uncertainties on nZEB LCP are analysed. Case studies show that an nZEB might not fulfil its definition at all after some years due to component degradation, while the proposed two-stage design method can effectively alleviate this problem.

Keywords

Uncertainty · Near-zero energy building · Degradation · Life-cycle performance · Design

Nomenclature

Abbreviations

DoD	Depth of discharge (*dimensionless*)

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EMS	Energy management system
ES	Energy storage
LCA	Life-cycle analysis
LCC	Life-cycle cost
LCP	Life-cycle performance
LCPA	Life-cycle performance analysis
PV	PV panel
RES	Renewable energy system
WT	Wind turbine

Symbols

a_t	Discount rate
A_R	Rotor area of a wind turbine
AC_t	Operational cost (HKD)
C_P	Power efficiency of a wind turbine
CAP_{AC}	Air-conditioning system capacity (kW)
CAP_{ES}	Energy storage system capacity (kWh)
CAP_{PV}	PV surface area (m^2)
CAP_{WT}	Capacity of wind turbine (kW)
CL	Cooling load (W)
CL_{sup}	Supply cooling capacity (W)
D_a	Average degradation rate
D_{CL}	Cumulative degradation rate of battery
E_{dem}	Annual electricity demand (kWh)
E_{sup}	Annual power production (kWh)
E_{store}	Energy stored in energy storage system (kWh)
I_{AM}	Combined incidence angle modifier for the PV cover material
I_T	Total amount of solar radiation incident on the PV collect surface (W/m^2)
IC	Initial cost (HKD)
P_{demand}	Power demand from nZEB (W)
P_{PV}	Power generated by a PV panel (W)
P_{WT}	Power generated by a wind turbine (W)
Pow_{charge}	Power charge of the energy storage system (W)
$Pow_{mismatch}$	Power mismatch of the nZEB (W)
Q_0	Amount when newly installed (it can be chiller capacity, pump efficiency, etc.)
Q_t	Amount in the tth year (it can be chiller capacity, pump efficiency, etc.)
r	Interest rate
R_k	Ranges of depth of discharge
t	Year of operation
T_{room}	Indoor air temperature (°C)
$T_{setpoint,cooling}$	Indoor temperature set-point in cooling condition (°C)
$T_{setpoint,heating}$	Indoor temperature set-point in cooling condition (°C)
$u_{ele,exp}$	Unit price of exported electricity (HKD/kWh)
$u_{ele,imp}$	Unit price of imported electricity (HKD/kWh)
u_{ac}	Unit price of air-conditioning system (HKD/kW)
u_{es}	Unit price of energy storage system (HKD/kWh)
u_{pv}	Unit price of PV panels (HKD/m^2)
u_{wt}	Unit price of wind turbine (HKD/kW)
U_0	Wind velocity (m/s)
$X(i)$	Data points in the DoD file

Greek symbols

η	Efficiency of the PV array
κ	Transmittance-absorptance product of the PV cover
τ_j	Value of cooling temperature set-point unmet hour in the jth hour (h)
ρ_{air}	Air density (kg/m^3)
$\Psi_{balance}$	Energy balance indicator (*dimensionless*)

$\Psi_{comfort}$	Cooling temperature set-point unmet hour indicator (h)
Ψ_{grid}	Grid independence indicator (*dimensionless*)
Γ	Time duration

12.1 Introduction

As reported by the many studies, buildings consume about 40% energy worldwide (ASHRAE 2011), and this percentage even increases to more than 60% in Hong Kong (EMSD 2017). Near-zero energy buildings (nZEBs) are considered as an effective solution for mitigating CO_2 emissions and reducing the energy usage in the building sector. Until now, a lot of efforts have been made on establishing regulations on nZEB for promoting its application. For instance, the European Directive on Energy Performance of Buildings introduces requirements that all the new buildings should be 'nearly net zero energy buildings' after 2020 (Buildings 2010, Lu et al. 2015). The U.S. government sets a target that 50% of commercial buildings achieve zero energy by 2040, and all commercial buildings achieve zero energy by 2050 (Pless and Paul Torcellini 2009). The International Energy Agency (IEA) Solar Heating and Cooling (SHC) programme sets up the task 40 'Towards net zero energy solar buildings' to develop a harmonized international definition framework including tools, innovative solutions, and industry guidelines (SHC 2008).

Proper sizing of nZEB systems is essential for ensuring that the nZEBs can perform as expected in terms of annual energy balance, thermal comfort, and grid independence (Lu et al. 2017; Sun et al. 2015; Zhang et al. 2016). A nZEB typically contains several types of systems, including energy consuming systems (e.g. HVAC systems, lighting systems, etc.), renewable energy systems (RES) (e.g. PV panels, wind turbine, etc.), energy storage systems, and thermal storage systems (Bekele and Tadesse 2012). The sizing of these nZEB systems is interconnected: the building peak cooling/heating load is the basis for sizing the HVAC system; the energy demand from energy consuming systems is the basis for sizing the RES systems; the energy demand and energy supply are the basis for sizing the energy storage system. Two significant factors affecting the sizing of nZEB systems are the uncertainties embodied in the building usage condition and weather condition, and the degradation effects in components of systems (Djunaedy et al. 2011; Gang et al. 2015; Huang et al. 2015; Zhou et al. 2016; Monforti and Gonzalez-Aparicio 2017). The former factor may easily lead to oversizing of nZEB components due to inaccurate evaluation of the building energy demand (Sun et al. 2015; Zhang et al. 2016; Lu et al. 2015, 2017). For instance, oversizing of HVAC systems is common, and some systems are oversized by as much as 100% (Djunaedy et al. 2011). The latter factor may lead to a design that cannot perform as expected during the whole service life (Huang et al. 2017).

Uncertainties are defined as the information gap between what the decision makers' present state of information and certainty (Aughenbaugh 2006, Nikolaidis 2005). For nZEB system design, uncertainties exist in the physical properties (e.g. thermal conductivity, density) of building envelope, in the internal heat gain (e.g. occupant and electrical facility density), in the weather condition, etc. (Hopfe 2009, Lu et al. 2017; Zhang et al. 2016). In recent years, the importance of uncertainties has been recognized by researchers, and many uncertainty-based design methods of nZEB systems have been developed (Sun et al. 2015; Kneifel and Webb 2016; Zhang et al. 2016; Lu et al. 2015, 2017; Cheng et al. 2017). For instance, Sun et al. (2015) and Zhang et al. (2016) proposed a multiple criteria design method for nZEB system under uncertainties, which selects the optimal design in the framework of multiple criteria decision making. Three performance indices, namely initial cost, thermal comfort, and power mismatch, were considered in that method. Lu et al. proposed a single-objective and multi-

objective optimization (Pareto optimization) approach of renewable energy systems of low/zero energy buildings and compared the performance of these two methods in system design (Lu et al. 2015). Notably, Yu et al. (2016) proposed a generic-algorithm-based system sizing method considering multiple criteria performance constrains. These methods are effective in handling the uncertainties related to the nZEB systems design. However, the degradation of the nZEB systems was neglected. A systematic way of investigating the degradation effects in nZEB systems and integrating these degradation effects into design is still lacking.

Degradation is prevalent in nZEB system components, from the chiller capacity, pump/fan efficiency in electricity demand side (Hunter 1941, Hu 2009, Huang et al. 2017), to the efficiency of PV panels and wind turbine in the electricity supply side (Jordan and Kurtz 2013; Cao et al. 2016; Tabrizi and Sanguinetti 2015), and to the capacity of energy storage system (Chawla et al. 2010; Alam and Saha 2016). Rosenthal investigated the performance of different PV systems and found that the degradation rates of 10 studied systems were larger than 1%/year (Rosenthal et al. 1993). The energy storage system capacity also degrades with its operation (Chawla et al. 2010). The degradation has a great influence on the system life-cycle performance. For instance, the degradation of chiller leads to decrease in the chiller maximum supply cooling capacity, which may result in inadequate thermal comfort during extreme hot weather conditions (Huang et al. 2017). The efficiency degradation of PV and wind turbine leads to decrease in the annual power production (Jordan and Kurtz 2013; Cao et al. 2016), which brings potential risk in achieving nZEB (Tabrizi and Sanguinetti 2015). The degradation of energy storage system capacity would increase the interaction of the nZEB with power grid and thus bringing great challenges to the stability and reliability of power grid (Chawla et al. 2010; Alam and Saha 2016). As a result, neglecting the degradation effects in the nZEB systems may produce a system design, which can only achieve the desired level of performance in the beginning several years of installation.

Life-cycle analysis (LCA) can help designers understand the life-cycle performance of the nZEB alternatives, thus enabling decision making to achieve the best design in terms of energy usage and carbon emission (Goggins et al. 2016). Some certifications, such as the LEED (LEED 2009) and Green globes and German Sustainable Building Certification (DGNB) (Schmidt 2012; GB 2014), even apply the LCA in their evaluation to grant a rating. In the field of nZEB, most of the existing LCA studies analyse from the building construction stage to destruction stage, divide the energy usage into embodied energy, operational energy and demolition energy, and calculate the energy usage in each phase by using some conversion factors (Hernandez and Kenny 2010; Berggren et al. 2013; Goggins et al. 2016). Only a limited number of studies focus on the nZEB operational phase and conduct detailed simulation to evaluate the nZEB life-cycle performance (Tripathy et al. 2017; Tabrizi and Sanguinetti 2015). However, the component degradation is rarely considered in the existing LCA studies. The impact of degradation on the nZEB life-cycle performance, such as thermal comfort, energy balance, and grid independence, is not studied, neither.

This chapter, therefore, proposes a life-cycle performance analysis (LCPA) method for investigating the longitudinal performance of the nZEBs. This analysis will not only integrate the uncertainties in predicting building thermal load and weather condition, but also take account of the degradation effects in the nZEB systems, including HVAC system, PV panels, wind turbine, and energy storage system. The objective is to improve the designers' understanding of the system performance under the impact of uncertainties and degradation effects. Based on the proposed LCPA method, a two-stage design method is proposed to improve the sizing of the renewable energy systems and energy storage system. Monte Carlo simulation will be

performed to produce statistical distributions of the nZEB thermal load and statistical life-cycle performance. Random deterioration method will be used for modelling the degradation effects in the nZEB systems. Case studies will be conducted to demonstrate the application of the LCPA method, and the performance of system obtained from the two-stage design method will be compared with the conventional sizing (which is designed without considering the degradation).

The contributions of the present study to the subject are briefly summarized as follows:

- An uncertainty-based life-cycle performance analysis method is developed, which takes both the uncertainties (existing in the prediction of nZEB energy demand and supply) and components' degradation into consideration. The LCPA method can help designers predict the nZEB long-term performance.
- To the authors' knowledge, it is the first time that the impacts of degradation and uncertainties on the nZEB life-cycle performance (e.g. thermal comfort, energy balance, grid independence, imported/ exported energy) are being analysed.
- This is one of the papers firstly indicating that an nZEB might not fulfil its definition at all after some years due to component degradation.
- To mitigate the impact of degradation, a novel two-stage design method is developed to improve the nZEB system life-cycle performance.
- The proposed two-stage design method is proven to be effective in reducing the nZEB life-cycle cost and meanwhile improving the nZEB life-cycle performance compared with the conventional design method (which does not consider component degradation).

12.2 Methodology

This section is organized as follows: Sect. 12.2.1 introduces the uncertainty-based LCPA method. Section 12.2.2 introduces a two-stage design method to improve the nZEB life-cycle performance based on the LCPA developed in Sect. 12.2.1.

12.2.1 Uncertainty-Based Life-Cycle Performance Analysis

Figure 12.1 shows process of uncertainty-based nZEB life-cycle performance analysis, which consists of four steps: quantification of uncertainties, nZEB systems sizing considering uncertainties, quantification of degradation rates, and life-cycle performance (LCP) analysis considering degradation.

12.2.1.1 Quantification of Uncertainties

In this step, the uncertain parameters for predicting building cooling load and in the weather condition are identified and are quantified using statistical distributions (e.g. normal distribution, uniform distribution, triangular distribution, etc.). The uncertainties confronted by the building thermal load prediction can be classified into three types, uncertainty in the physical parameters, uncertainty in the scenario parameters, and model form uncertainties, following (Hopfe 2009) and (Wang 2016).

The uncertainty in the physical parameters is related to the physical properties of materials. It is mostly identifiable as the standard input parameters in simulation and not influenced by designers. Examples of this category of parameters include thickness, density, thermal conductivity and heat capacity of walls, roofs and windows, etc. The uncertainty in the physical parameters is usually quantified to follow normal distributions (Domínguez-Muñoz et al. 2010; Handbook 2003; Hopfe 2009; Macdonald 2002; Zhang and Cheng 2017).

The uncertainty in scenario parameters refers to parameters that are relative to the real-time operation of the building during its life time. They are not measurable and hard to control. Examples of this category of parameters include weather condition (i.e. ambient temperature and relative humidity, solar radiation), casual heat

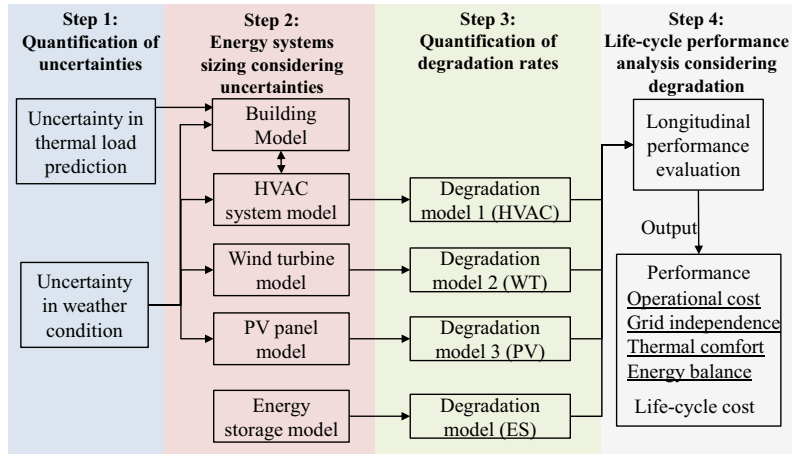


Fig. 12.1 Process of uncertainty-based ZEB life-cycle performance analysis

gain (i.e. the number of occupants, computers and light in use), internal and external shading coefficients, internal and external convection transfer rate, infiltration, etc. The uncertainty in scenario parameters is usually quantified using normal distributions or triangular distributions (Macdonald 2002; Hopfe 2009; Wang 2016; Zhang et al. 2017).

The model form uncertainty is caused by the discrepancy between the empirical data from physical experiments or outputs from a higher fidelity module and the module output which is based on a low fidelity description of the governing physics. Model form uncertainty exists since no model is a perfect representation of the physical reality (Wang 2016). Examples of this category of uncertainty include the mathematical models (e.g. the cooling load model, chiller plant model, cooling tower model, mechanical components, etc.) used in simulation software, such as EnergyPlus, and ESP-R. The model form uncertainty is usually quantified to follow normal distributions.

For simplicity, the model form uncertainty is not considered in this chapter. The results from EnergyPlus and Trnsys simulations are assumed to represent the physical reality well. The quantification of uncertainties in the physical and scenario parameters is summarized in Table 12.1.

12.2.1.2 nZEB System Sizing Considering the Quantified Uncertainties

First, the peak cooling/heating load of the building is predicted using the quantified distributions of the uncertain input parameters. A statistical distribution is produced for the peak cooling or heating load through Monte Carlo simulation. Second, the capacity of the chiller and boiler in the HVAC system is determined based on the distribution of the predicted peak cooling/heating load. The other components in the HVAC system are then determined based on the chiller and boiler size (Handbook 1996; Grondzik 2007; Huang et al. 2015). Third, the annual electricity demand (E_{dem} (kWh)) of the building is predicted based on the operational data of the HVAC system and other electricity facilities.

Then, the annual electricity demand is used for determining the size of PV panels and wind turbine (power generating equipment). The power generated by a PV panel (P_{PV} (W)) is calculated by Eq. (12.1) (Klein et al. 1998; Yu et al. 2016),

$$P_{\mathrm{PV}} = \kappa \times I_{\mathrm{AM}} \times I_{\mathrm{T}} \times \eta \times \mathrm{CAP}_{\mathrm{PV}} \qquad (12.1)$$

where κ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the

Table 12.1 Configuration of the case nZEB and HVAC system (Huang et al. 2017; Domínguez-Muñoz et al. 2010; Hopfe 2009, Macdonald 2002)

Category	Objective	Parameter	Distribution	Truncation
Scenario parameters: Internal heat gain	People gains	People per floor Area, person/m^2	$N(0.55,0.02^2)$	[0.5, 0.6]
		People activity level, W	$N(120,20^2)$	[70, 130]
		Fraction radiant	$T(0.25, 0.45, 0.65)$	
	Lighting gains	Watts per floor area, W/m^2	$N(19.4,1.29^2)$	[16, 22.8]
		Fraction radiant	$T(0.2, 0.3, 0.4)$	
	Equipment gains	Watts per person, W/person	$N(158, 30^2)$	[124, 229]
		Fraction radiant	$T(0.4, 0.5, 0.6)$	
Scenario parameters: Miscellaneous factors	Infiltration	Infiltration rate, ACH	$N(0.5, 0.005^2)$	
	Building air leakage	Effective air leakage area, cm^2	log-$N(640\ 64^2)$	
	Ductwork air leakage	Leakage rate %	$N(5, 0.5^2)$	
	Pipe and ducts	External convection coefficient, W/(m^2·K)	$N(1.9, 3.8^2)$	
	Ventilation	Ventilation per Person, m^3/(s·person)	$N(0.0024, 0.0001^2)$	
	Wall/Roof/Window	Internal convection coefficient, W/(m^2·K)	$T(1.59, 2.5, 4.08)$	
		External convection coefficient, W/(m^2·K)	$N(1.9, 3.8^2)$	
Design parameters: HVAC system	Condensate water pump	Rated power consumption, kW	$N(4.02, 0.20^2)$	
		Motor efficiency	$N(0.85, 0.05^2)$	[0.6, 0.85]
	Chilled water pump	Rated power consumption, kW	$N(2.55, 0.058^2)$	
		Motor efficiency	$N(0.85, 0.13^2)$	[0.6, 0.85]
	Cooling tower	Design fan power, kW	$N(3.87, 0.20^2)$	
		Design U-factor times area value, kW/K	$N(10.35, 1.035^2)$	
		Drift loss percent	$N(0.008, 0.0008^2)$	
	Zone supply fan	Fan total efficiency	$N(0.7, 0.02^2)$	[0.65, 0.75]
		Pressure rise, Pa	$N(75, 7.5^2)$	[52.5, 97.5]
		Motor efficiency	$N(0.85, 0.05^2)$	[0.6, 0.85]
	Chiller	Reference capacity, kW	$N(279, 14^2)$	
		Reference COP, kW/kW	$N(3.8, 0.19^2)$	

(continued)

Table 12.1 (continued)

Category	Objective	Parameter	Distribution	Truncation
Scenario parameters: Weather condition		Ambient temperature, °C	N(mean, 3.2^2)	[13, 35]
		Ambient relative humidity, %	N(mean, 11^2)	[40, 99]
		Solar radiation, W/m^2	N(mean, 56^2)	[0, 1111]
		Wind speed, m/s	N(mean, 2.3^2)	[0, 45]

combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (W/m^2) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; CAP_{PV} (m^2) is the PV surface area.

The power generated by a wind turbine (P_{WT} (W)) is calculated by Eq. (12.2) (Yu et al. 2016; Klein et al. 1998),

$$P_{WT} = \begin{cases} 0, & \text{if } U_0 \le 3 \\ C_P \times \rho_{air} \times A_R \times U_0^3, & \text{otherwise} \\ CAP_{WT}, & \text{if } U_0 \ge 14 \end{cases} \tag{12.2}$$

where C_P is power efficiency, which is a function of the axial induction factor; ρ_{air} (kg/m^3) is air density; A_R (m^2) is the rotor area; U_0 (m/s) is the wind velocity in the free stream; CAP_{WT} (W) is the capacity of wind turbine.

The annual power production by the RES (E_{sup} (kWh)) is calculated by Eq. (12.3),

$$E_{sup} = \left(\sum_{j=1}^{8760} P_{PV,j} + \sum_{j=1}^{8760} P_{WT,j} \right) / 1000 \tag{12.3}$$

where j indicates the operating hour. The PV size and wind turbine capacity are determined through satisfying Eq. (12.4).

$$E_{sup} = E_{dem} \tag{12.4}$$

Finally, the size of electrical battery system CAP_{ES} (kWh) is calculated by Eq. (12.5) and (12.6) (Sun 2015; Yu et al. 2016),

$$Pow_{mismatch,j} = \left(P_{PV,j} + P_{WT,j} - P_{demand,j} \right) / 1000 \tag{12.5}$$

$$CAP_{ES} = \max \left(\left| \sum_{j=1}^{n} Pow_{mismatch,j} \right|, \left| \sum_{j=2}^{n+1} Pow_{mismatch,j} \right|, \ldots, \left| \sum_{j=8761-n}^{8760} Pow_{mismatch,j} \right| \right) \tag{12.6}$$

where $Pow_{mismatch,i}$ (kW) is the power mismatch in the jth hour; $P_{demand,j}$ (W) is the power demand in the jth hour; n is the period (h) for sizing.

12.2.1.3 Quantification of Degradation Rates

Degradation Models

There are several models that have been developed for describing the degradation effects in mechanical components, including the failure rate function, random deteriorate rate, Markov processes, Brownian motion with drift, and gamma processes (De Wilde et al. 2011; Finkelstein 2008; Van Noortwijk 2009). An introduction to these methods is given in (Frangopol et al. 2004; Van Noortwijk 2009) and a comparison of them is shown in De Wilde et al.'s work (De Wilde et al. 2011).

In failure rate functions, the degradation condition is assumed to be either functioning state or failed state. Failure rate functions are effective for electrical and mechanical devices,

such as lamps. However, the failure rate cannot be measured for a particular component (Finkelstein 2008; De Wilde et al. 2011). Markov process is a stochastic process with the property that the future chain state only depends on the present state. Markov chains consist of an initial state distribution and a transition matrix. This method works well when the conditions/states of a system or component can be described by a limited number of categories (Frangopol et al. 2004; De Wilde et al. 2011). Brownian motion is a stochastic process with independent, real valued increments and decrements (also termed drift parameter) having a normal distribution. The Brownian motion is usually used for capturing the increasing or decreasing characteristic of alternative (Frangopol et al. 2004). A gamma process is a stochastic process with independent, non-negative increments having a gamma distribution. Gamma process models are more suitable to represent gradual monotonic degradation over time due to wear, fatigue, corrosion, erosion, etc. (Van Noortwijk 2009; De Wilde et al. 2011).

Another approach for modelling the degradation is random deterioration rate method. This method has a typical form shown in Eq. (12.7), in which the average rate of degradation per unit time is represented by a random quantity (Frangopol et al. 2004; Van Noortwijk 2009).

$$Q_t = Q_0 \times (1 - D_a \cdot t) \tag{12.7}$$

In this model, D_a indicates the average degradation rate in each year, t indicates the year of operation, Q_0 indicates the amount when newly installed (it can be chiller capacity, pump efficiency, PV efficiency, etc.). Due to its simple form and less requirement on the degradation process information, the random deteriorate rate method is widely used for modelling the degradation in the design stage (Van Noortwijk 2009; Frangopol et al. 2004; Huang et al. 2017). This chapter chooses this method for modelling the degradation.

Calculation of Degradation Rates

There are plenty study into the degradation of PV panels and wind turbines (Abete et al. 2000; Radue and Van Dyk 2010; Jordan and Kurtz 2013; Cao et al. 2016). The quantification of degradation rate of PV panels and wind turbines is quantified directly following the literature.

The energy storage system considered in this chapter is the electrical battery. The degradation rate of electrical battery is affected by factors such as material, temperature, and age. Most importantly, it is affected by the operation condition, namely the number and depth of cycles. This chapter mainly investigates the impact of operation condition on the energy storage system degradation. Figure 12.2 illustrates the relationship between the allowed maximum number of cycles and the Depth of Discharge (DoD). DoD is a dimensionless parameter. Typically, an electrical battery can operate more cycles at

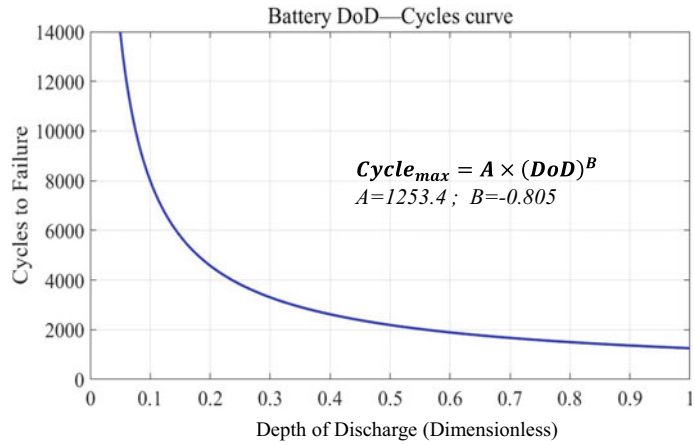
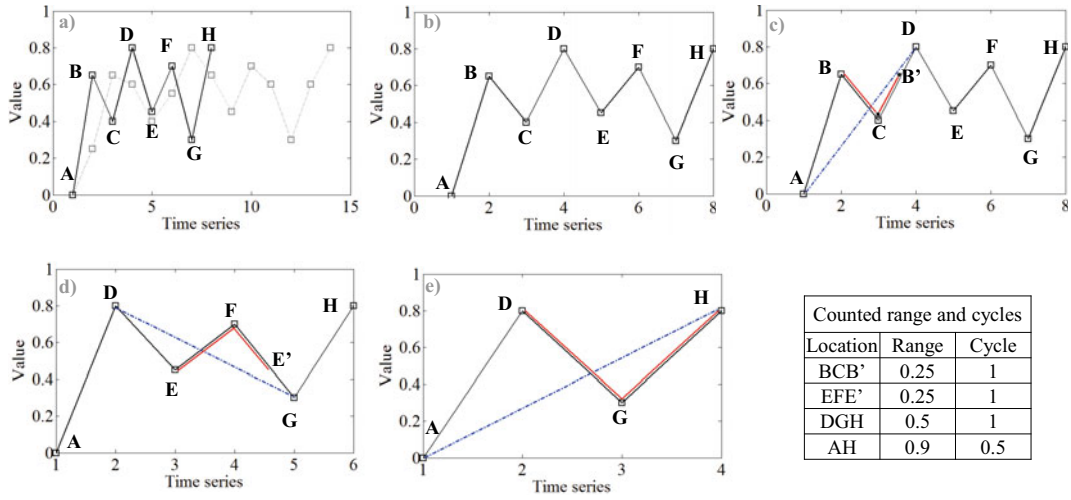


Fig. 12.2 Example of battery lifetime curve



Counted range and cycles		
Location	Range	Cycle
BCB'	0.25	1
EFE'	0.25	1
DGH	0.5	1
AH	0.9	0.5

Fig. 12.3 Illustration of Rainflow counting algorithm

lower DoDs. The handbook of batteries tabulated the impact of DoD on the cycle life of some typical battery systems (Linden and Reddy 2002). Based on the dynamic DoDs, the degradation of battery can be predicted using the Rainflow cycle counting algorithm (Chawla et al. 2010; Alam and Saha 2016).

The Rainflow cycle counting algorithm is a method for determine the number of fatigue cycles presented in a load-time history (Matsuishi and Endo 1968). It can be used to calculate the integral impact of varying load on the component degradation through decomposing the complex cycles into sub-cycles (Chawla et al. 2010; Alam and Saha 2016). Figure 12.3 gives an example of the application of Rainflow Counting algorithm. Details about how to implement the algorithm are introduced step by step.

Step 1: The time-series DoD file is reduced to contain only valleys and peaks. This can be done by identifying the reversal of the slop. For example, the original DoD file with 14 data points in Fig. 12.3a is reduced to a series with only 8 data points in Fig. 12.3b.

Step 2: Two ranges R_1 and R_2 are determined using three continuous points $(X(i), X(i+1), X(i+2))$ from the DoD files, as shown in Eq. (12.8)

$$\begin{aligned} R_1 &= |X(i) - X(i+1)|, \\ R_2 &= |X(i+1) - X(i+2)| \end{aligned} \tag{12.8}$$

Step 3: Using the rule shown in Eq. (12.9), a cycle is identified and calculated as a half (0.5) or full (1) cycle.

$$\text{Cycle} = \begin{cases} 0.5 & \text{if } R_1 \geq R_2, \text{ or } i = 1 \\ 1 & \text{if } R_1 < R_2 \end{cases} \tag{12.9}$$

Step 4: All the full cycles in the DoD file are identified using the rules introduced in Step 2 and 3, and R_1 is counted as the range corresponding to the full cycle. Then, the first point and second point selected in Step 2 are discarded. For instance in Fig. 12.3c, the range *CD* is larger than *BC*, therefore, a full cycle is identified as *BCB′* with range *BC*. Then point *B* and *C* are removed from the series. The same process is conducted to identify another full cycle *EFE′* (see Fig. 12.3d).

Step 5: Identify all three-point-series where $R_1 = R_2$, and count it as full cycles. Then, remove the first and second point after counting (see Fig. 12.3e). For instance in Fig. 12.3e, a full cycle *DGH* is identified.

Step 6: All the remaining data points are considered. Each of the remaining range is quantified as a half cycle. For instance, only *AH*

is left after Step 4 and Step 5, it is counted as half cycle and the range is *AH* (see Fig. 12.3e).

Figure 12.3 also summarizes the identified ranges (R_k) and cycles for the example.

Based on the counted ranges and cycles, the cumulative degradation rate can be calculated using Eq. (12.10) (Alam and Saha 2016).

$$D_{\mathrm{CL}} = \sum_{k=1}^{m} \frac{\text{Cycle of } R_k}{A \times (R_k)^B} \tag{12.10}$$

where A and B are coefficients determining the DoD cycle curve in Fig. 12.2.

12.2.1.4 Life-Cycle Performance Analysis with Degradation Effects Considered

The LCP of nZEB system is evaluated through integrating the uncertain inputs and degradation models into the building model and the energy system model. Figure 12.4 shows the process of uncertainty-based LCP analysis. In this chapter, the performance of nZEB system in the electricity demand side (i.e. nZEB and HVAC system) was simulated in EnergyPlus; the performance of nZEB system in the electricity supply side (i.e. PV system, WT system) was simulated in TRNSYS; and the electrical battery system was modelled in MATLAB.

The LCP is investigated based on an annual analysis. The operating time (in years) starts from 1, and it is imported into the degradation model to obtain the extent of degradation (represented by degradation terms) in that year. The degradation terms are used along with uncertain parameters and fixed parameters as inputs for predicting the annual performance. The predicted annual performance will be extracted and stored in a database. The process repeats until the operating time equals the service life (also in years).

There are many indices developed for evaluating the nZEB performance (Sun et al. 2015; Yu et al. 2016; Lu et al. 2017; Salom et al. 2014). In this chapter, four performance indices of nZEB

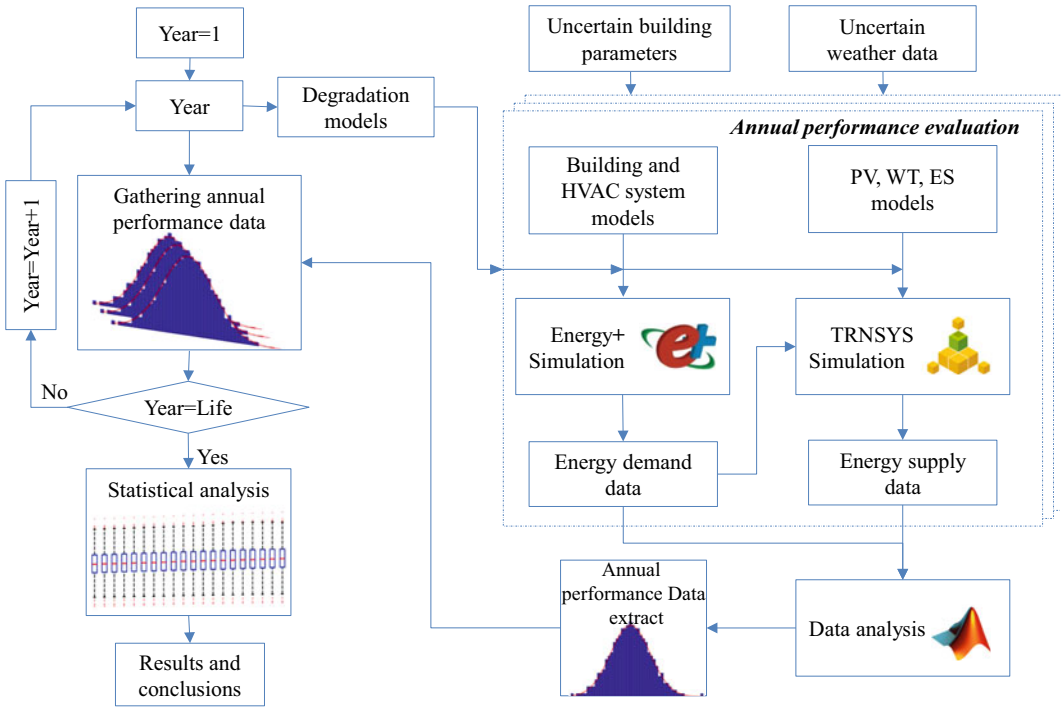


Fig. 12.4 Flow chart of uncertainty-based ZEB life-cycle performance analysis

are assessed, namely thermal comfort, energy balance, grid independence, and life-cycle cost.

Thermal comfort

Thermal comfort is an indicator expressing the users' satisfaction with the thermal environment. It is an integrative effect of the air temperature, humidity, air velocity, etc. For simplicity, in this chapter, the cooling/heating temperature set-point unmet hour is used for representing the extent of thermal comfort. The cooling/heating temperature set-point unmet hour indicates the number of hours that the indoor air temperature does not meet the cooling/heating set-point temperature (i.e. $T_{\text{room}} > T_{\text{setpoint,cooling}}$ or $T_{\text{room}} < T_{\text{setpoint,heating}}$). Equation (12.11) shows the calculation of cooling temperature set-point unmet hour Ψ_{comfort} (Yu et al. 2016),

$$\Psi_{\text{comfort}} = \sum_{j=1}^{8760} \tau_j \begin{cases} \tau_j = 1, & \text{if } \text{CL}_{\text{sup},j} \le \text{CL}_j \\ \tau_j = 0, & \text{if } \text{CL}_{\text{sup},j} > \text{CL}_j \end{cases} \tag{12.11}$$

where τ_j (h) represents the value of cooling temperature set-point unmet hour in the jth hour, $\text{CL}_{\text{sup},j}$ is the hourly supply cooling capacity, and CL_j is the hourly cooling load. A smaller Ψ_{comfort} value indicates a better thermal comfort performance. It should be noted that the cooling/heating temperature set-point unmet hour is not the only indicator of thermal comfort.

Energy balance

Energy balance is an indicator comparing the nZEB annual energy production and annual energy consumption. When the annual energy production is close to the annual energy demand, the building can be considered as nZEB. When the annual energy production is larger than the energy demand, the building becomes a plus energy building. The energy balance Ψ_{balance} is calculated by Eq. (12.12) (Yu et al. 2016)

$$\Psi_{\text{balance}} = \left(E_{\text{sup}} - E_{\text{dem}}\right)/E_{\text{dem}} \tag{12.12}$$

where E_{sup} (kWh) is the annual energy supply and E_{dem} (kWh) is the annual energy demand. A positive Ψ_{balance} value indicates the system can produce adequate energy to satisfy the building cooling load, while a negative value indicates the building cannot achieve net zero energy building.

Grid independence

Grid independence is an indicator representing how closely the nZEB depends on the power grid. The nZEB might need to import energy from or export energy to the power grid. The exchange of power presents a great challenge to the reliability and stability of the power grid. The grid independence Ψ_{grid} is calculated by Eq. (12.13) (Salom et al. 2011; Yu et al. 2016).

$$\Psi_{grid} = \frac{\Gamma_{power_{exchange}=0}}{\Gamma_{tot}} \tag{12.13}$$

where Γ represents the time duration, and $\text{power}_{\text{exchange}}$ (kW) denotes the hourly power exchange between a nZEB and the power grid. The subscript tot is the total number of hours for counting. The Ψ_{grid} value lies between 0 and 1. A large Ψ_{grid} value indicates better grid independence. The power exchange $\text{power}_{\text{exchange},j}$ (kW) is calculated by Eq. (12.14),

$$\text{power}_{\text{exchange},j} = \text{Pow}_{\text{mismatch},j} - \text{Pow}_{\text{charge},j} \tag{12.14}$$

where $\text{Pow}_{\text{charge},j}$ (kW) is the power charge of the energy storage system, which is calculated by Eqs. (12.15) and (12.16)

$$E_{\text{store},j} = \sum_{i=1}^{j-1} \text{Pow}_{\text{charge},i} \tag{12.15}$$

$$\text{Pow}_{\text{charge},j} = \begin{cases} \min\left(\left(\text{CAP}_{\text{ES}} - E_{\text{store},j}\right), \text{Pow}_{\text{mismatch},j}\right) & \text{if } \text{Pow}_{\text{mismatch},j} > 0 \\ \max\left(-1 \times E_{\text{store},j}, \text{Pow}_{\text{mismatch},j}\right) & \text{if } \text{Pow}_{\text{mismatch},j} < 0 \end{cases} \tag{12.16}$$

where $E_{\text{store},j}$ (kWh) is the energy stored in energy storage system in the jth hour. A positive

$\text{Pow}_{\text{charge},j}$ value indicates a charging process, while a negative value indicates a discharging process. The energy storage system works following two rules: (1) The excessive renewable energy is firstly stored in the energy storage system. Only when the energy storage system is fully charged, the excessive energy is exported into the power grid. (2) The insufficient renewable energy is firstly supplemented by the energy storage system. Only when the energy storage system is fully discharged, the rest part of required energy is imported from the grid.

Life-cycle cost

The life-cycle cost of the configured multiple-chiller plant is estimated using the Eq. (12.17) (Rosenquist et al. 2004; Hamdy et al. 2013).

$$\text{LCC} = \text{IC} + \sum_{t=1}^{N} a_t \cdot \text{AC}_t \tag{12.17}$$

where LCC is the life-cycle cost (HKD), IC is the initial cost (HKD), $\sum$ is the sum over lifetime from year 1 to year N, N is the lifetime of the appliance, t is the year that annual cost is calculated, a_t is the discount rate, which is calculated by Eq. (12.18) (Hamdy et al. 2013),

$$a_t = \frac{1-(1+r)^{-t}}{r} \tag{12.18}$$

where r is the interest rate. AC_t is the operational cost (HKD) in the tth year which is determined by Eq. (12.19)

$$\text{AC} = u_{\text{ele,imp}} \cdot E_{\text{imp}} - u_{\text{ele,exp}} \cdot E_{\text{exp}} \tag{12.19}$$

where $u_{\text{ele,imp}}$ and $u_{\text{ele,exp}}$ $(\text{HKD}/\text{kW}\cdot\text{h})$are the unit price of imported and exported electricity, and E_{imp} and E_{exp} $(\text{kW}\cdot\text{h})$ are the imported and exported electricity.

The initial cost is calculated by Eq. (12.20)

$$\text{IC} = u_{\text{ac}} \cdot \text{CAP}_{\text{AC}} + u_{\text{wt}} \cdot \text{CAP}_{\text{WT}} + u_{\text{pv}} \cdot \text{CAP}_{\text{PV}} + u_{\text{es}} \cdot \text{CAP}_{\text{ES}} \tag{12.20}$$

where u_{ac} (HKD/kW), u_{wt} (HKD/kW), u_{pv} (HKD/m^2), and u_{es} (HKD/kWh) are the unit price of each energy component, and C_{AC} (kW), C_{wt} (kW), C_{pv} (m^2), and C_{es} (kWh) are the number or size of each energy component.

12.2.2 A Two-Stage Design Method to Improve nZEB Sizing

Based on the nZEB life-cycle performance analysis method developed in Sect. 12.2.1, a two-stage design method is proposed to improve the sizing of nZEB so as to achieve a minimal life-cycle cost. Figure 12.5 shows the process of two-stage design method. At Stage 1, the RES (e.g. PV panel and wind turbine) sizes are improved. At Stage 2, the ES size is further improved based on the optimized RES sizes. The two-stage design method consists of six steps.

At Stage 1, a set of PV/WT size alternatives are firstly defined based on the PV/WT size determined from the standard sizing procedure (see Sect. 12.2.1.2). Second, the life-cycle cost of each PV/WT size alternative is evaluated using the uncertainty-based LCPA method introduced in Sect. 12.2.1. The ES size determined from the standard sizing procedure is used as fixed inputs in the evaluation. Third, the life-cycle cost of the specified alternatives is compared. The alternative with the lowest life-cycle cost is selected as the optimal design in stage 1.

Similarly, at Stage 2, a set of ES size alternatives are firstly defined based on the PV/WT size determined from the standard sizing procedure (see Sect. 12.2.1.2). Second, the life-cycle cost of each ES size alternatives is evaluated. The optimal PV/WT size selected from Stage 1 is used as fixed inputs in the evaluation. Third, the life-cycle cost of the defined alternatives is compared. The alternative with the lowest cost is selected as the ultimate optimal design.

The developed design method does not optimize all the design variables simultaneously. This is because improving all the design

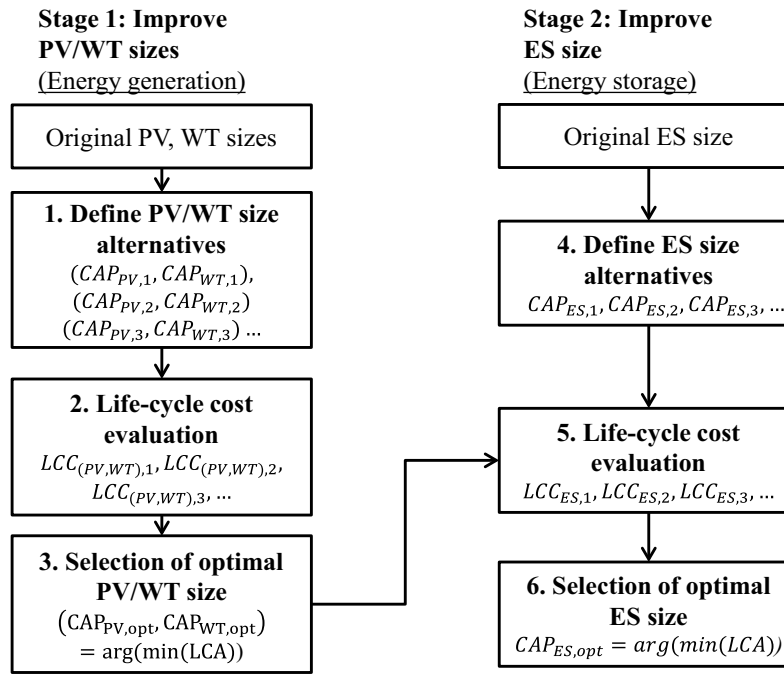


Fig. 12.5 Process of two-stage optimization of RES size

variables simultaneously would increase the computational burden unless more advanced searching algorithm is needed. As design is not the main focus of this chapter, this chapter chooses a simplified method and improves the size of each system separately. It should be noted this two-stage design method can only improve the design to achieve better performance. For the purpose of optimizing the nZEB system sizes, more advanced searching algorithms are needed.

12.3 Case Studies

12.3.1 Configuration of the Case nZEB and Systems

Case studies were performed to illustrate the procedures of nZEB life-cycle performance analysis, and application of the LCPA in improving system sizing. A case nZEB was constructed in simulation platforms (EnergyPlus and Trnsys). This nZEB was a simplified one-floor office building located in Hong Kong with a size of 25 m × 25 m × 3 m. The building is equipped with an HVAC system, power generation systems (PV panels and wind turbine), and an energy storage system (electrical battery). The schematic diagram of the case nZEB is shown in Fig. 12.6. The case building is connected to the power grid. The energy demand from the HVAC systems and nZEB electrical devices is firstly satisfied by the renewable energy system. The remaining energy demand is imported from the power grid. Figure 12.7 presents configuration of the HVAC system equipped in the case nZEB, which consists of a chiller, a cooling tower, a cooling coil, a constant-speed-driven chilled water pump, a constant-speed-driven condensate water pump, and a fan. The building was configured using the mean values of the parameters listed in Table 12.1. Table 12.1 also summarizes the statistical distributions of the uncertain parameters (Macdonald 2002; Hopfe 2009; Huang et al. 2017; Domínguez-Muñoz et al. 2010). Refer to Sect. 12.2.1.1 for the quantification of uncertain parameters.

Since heating is not needed during most time in Hong Kong, this chapter mainly considered the cooling condition of the HVAC system. The set-point of indoor air temperature was 25 °C and the relative humidity was set to 50%. Base

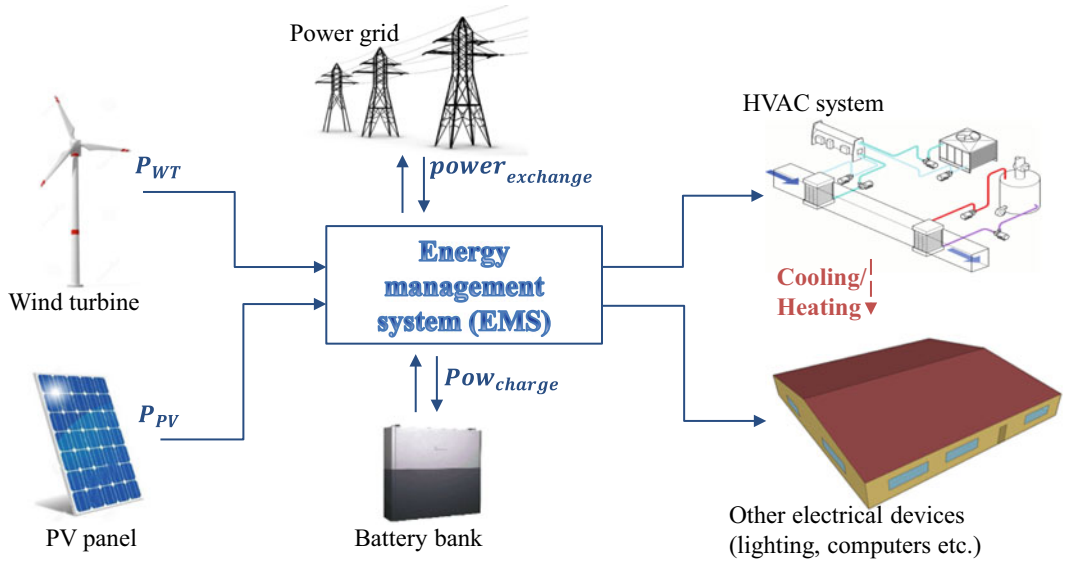


Fig. 12.6 Schematic diagram of the case nZEB

on the setting, the mean of building peak cooling load was calculated as 279 kW. The capacity of the HVAC system was sized as 279 kW. The configuration of other components in the HVAC system is also summarized in Table 12.1. The annual electricity demand of the nZEB was calculated to be 365,417 kWh (584.7 kWh/m^2). This chapter used a simple configuration that the PV panels and wind turbines were expected to produce the same amount of electricity. Using the sizing method introduced in Sect. 12.2.1.2, the area of PV panels was sized as 1298.6 m^2, and the required number of wind turbine was 16 (the rated power of wind turbine is 30 kW). The capacity of energy storage system was sized as 1571.8 kWh. Table 12.2 summarizes the price models used in this chapter.

12.3.2 Quantification of System Degradation Rates

The average degradation rates of HVAC system capacity, efficiency of PV panel, and wind turbine power production were quantified following existing literature to be 0.25%, 1.3%, and 1.6%, respectively (Jordan and Kurtz 2013; Cao et al. 2016; Huang et al. 2017). The degradation of chiller COP and efficiency of pumps and fans was also considered, and their degradation rates were quantified to be 0.25%, 0.2%, and 0.2% following literature (Huang et al. 2017).

The degradation rate in energy storage system was quantified based on one-year operational data. Figure 12.8 shows the DoD of the energy storage system over one-year period. It varies between 0 and 1 repeatedly. After peak-valley reduction of the DoD file, Rainflow counting algorithm was used to calculated the cumulative degradation rate in one year. The estimated average ranges and corresponding cycles are summarized in Table 12.3. Using Eq. (12.10), the annual degradation of energy storage system capacity in the case nZEB was calculated to be 5.73%.

The service life of the nZEB systems was considered to be 30 years (Berggren et al. 2013). For the simulation in each year, 100 samples were generated for each uncertain parameter using the Latin Hypercube Sampling (LHS) method (McKay et al. 2000). In total, the simulation repeated 3000 times.

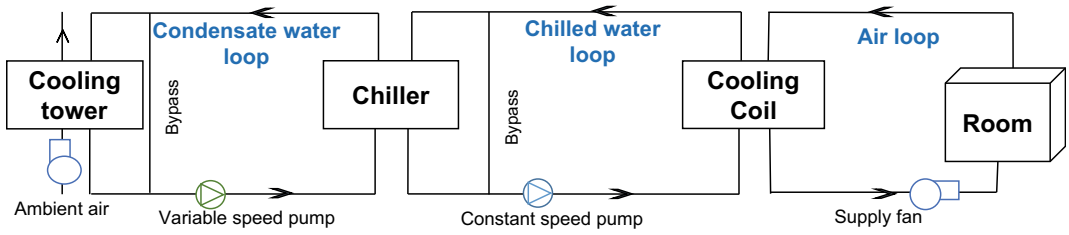


Fig. 12.7 Configuration of the HVAC system

Table 12.2 Summary of cost parameters

Pricing model			References
HVAC system	1400	HKD/kW	Yu et al. (2016)
PV panel	2000	HKD/m^2	Sun et al. (2015), Zhang et al. (2016)
Wind turbine	12,000	HKD/kW	
Energy storage system	600	HKD/(kWh)	
Electricity (buy on-peak 9:00–21:00)	2	HKD/(kWh)	Abdulla et al. (2016), Shen et al. (2016)
Electricity (buy off-peak other time)	0.5	HKD/(kWh)	
Electricity (sell)	0.22	HKD/(kWh)	Abdulla et al. (2016)

12.4 Results and Discussions

In the first part of this section, the simulation results of the case nZEB life-cycle performance are presented and analysed in detail. In the second part, the simulation results of the two-stage design method are presented: a set of alternatives are firstly defined and their LCCs are then calculated and compared.

12.4.1 Life-Cycle Performance Analysis Results

12.4.1.1 Energy Demand, Energy Supply, and Power Exchange

Figure 12.9 presents the varying of annual energy demand, energy supply, and power exchange with grid over the service life of the nZEB. For a given box, the central mark is the median, the edges of the box are the 25th and 75th percentiles, the whiskers represent the extreme data points, and the individual dots are outliers (McGill et al. 1978). As the operation year increases, the annual energy demand increases gradually (see Fig. 12.9a). This is caused by a combined effect of the decrease in chiller capacity, chiller COP, and efficiency of pumps and fans. After 30 years' operation, the annual energy demand increases 4.7%. The annual energy supply decreases 40% after 30 years (see Fig. 12.9b). This is caused by the decrease in power generation efficiency of PV panels and wind turbine. The evaluated annual energy demand has larger uncertainty (it has wider varying range in each year) than energy supply as more uncertain parameters are considered in the energy demand side (see Table 12.2). The annual power exchange with the grid changes from around 0 to -1.5357×10^5 kW · h (see Fig. 12.9c). This indicates that the nZEB can only achieve the goal near-zero energy in the beginning several years of usage. The varying of annual energy demand, energy supply, and power exchange all show a linear relationship with the operating year.

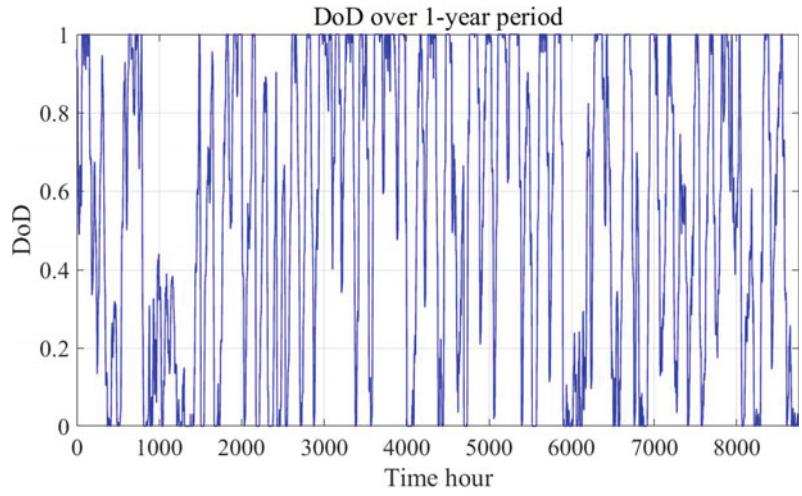


Fig. 12.8 DoD of the energy storage system during 1 year period

Table 12.3 Estimation of battery degradation based on sub-cycle identification

Range		Average range R_k	Cycle	Sub-degradation rate (%)
From	To			
0.9	1	0.95	23.5	1.80
0.8	0.9	0.85	6	0.42
0.7	0.8	0.75	3	0.19
0.6	0.7	0.65	7	0.39
0.5	0.6	0.55	2.5	0.12
0.4	0.5	0.45	6.5	0.27
0.3	0.4	0.35	7	0.24
0.2	0.3	0.25	11	0.29
0.15	0.2	0.175	6	0.12
0.1	0.15	0.125	18	0.27
0.05	0.1	0.075	53	0.53
0.025	0.05	0.0375	70	0.40
0	0.025	0.0125	297	0.70
Sum of degradation in 1 year				5.73

12.4.1.2 Thermal Comfort, Energy Balance, Operational Cost, and Grid Independence Indices

Figure 12.10 shows the varying of the four performance indices, namely cooling set-point unmet hour, operational cost, energy balance, and grid independence, over the service life. After 30 years, the average cooling set-point unmet hour increases 30%, and the increasing trend shows an exponential relationship (see Fig. 12.10a). The average operational cost increases from 60.7 to 456.5 kHKD (see Fig. 12.10b). The energy balance decreases from − 0.03 to − 0.63. The decreasing speed (see the decrease slope) becomes faster in the last few years than in the beginning several years (see Fig. 12.10c). The grid independence decreases from 0.69 to 0 (see Fig. 12.10d). The varying trend of grid independence changed dramatically

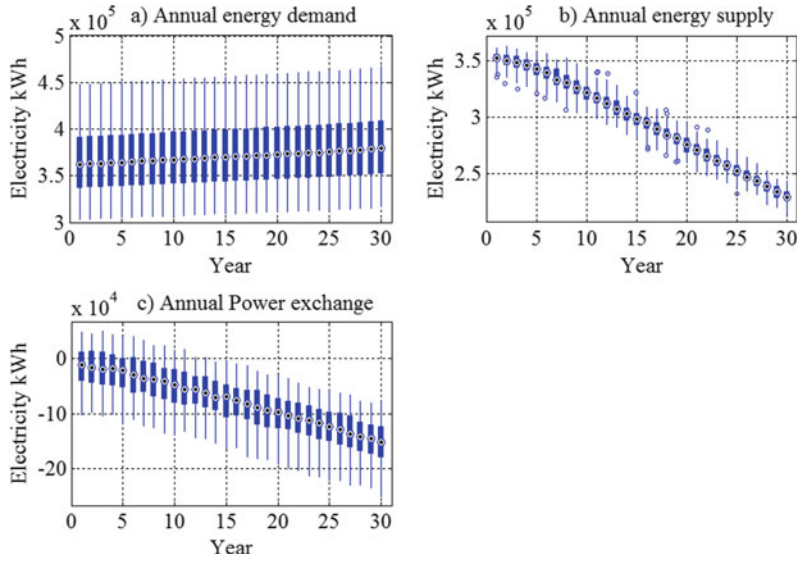


Fig. 12.9 Varying of annual energy demand, energy supply, and power exchange with operating year

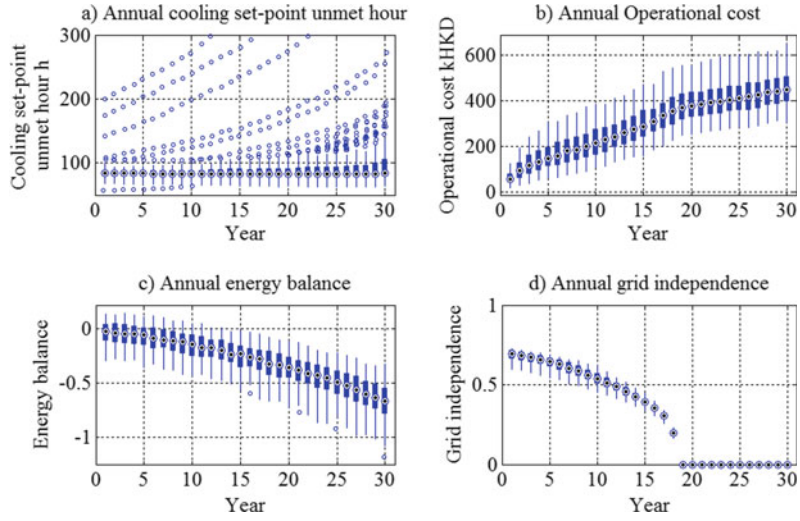


Fig. 12.10 Varying of four performance indices (cooling set-point unmet hour, operational cost, energy balance, and grid independence) with operating year

after 18 years. This is because after 18 years, the capacity of energy storage system already decreases to 0 (the average degradation rate was 5.73%). Under such circumstance, the nZEB fully depends on the power grid for regulating the unbalanced energy demand and supply. The varying trend of operational cost also changes after the energy storage system capacity decreases to 0. A nZEB can be considered performing well if its grid independence is larger than 0.5 (Yu et al. 2016). As can be seen, the case nZEB can achieve good grid independence only in the first 11 years.

12.4.1.3 Imported/Exported Energy from/to the Power Grid

Figure 12.11 shows the varying of annual imported and exported energy of the nZEB. The annual imported energy has an increasing trend. In the first 10 years, it increases slowly. Then after the 10th year, it increases faster. After 18 years, the increase speed reduces to a lower level. This is because as the operating year

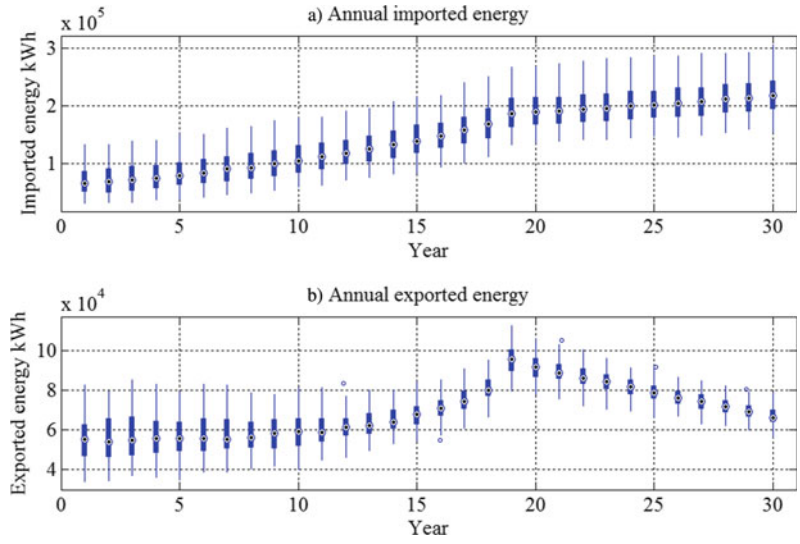


Fig. 12.11 Varying of annual imported and exported energy with operating year

increases, the annual energy supply will decrease, meanwhile the maximum stored energy in the energy storage system will also decrease. Both these two factors will lead to the increase in the annual imported energy. The varying of increasing speed after the 18th year is due to the fact that the energy storage system capacity already decreases to 0. The annual exported energy has a more complex varying trend. It increases slowly in the first 10 years. The increasing trend becomes faster after 10 years. The annual exported energy reaches peak in the 19th year. After the peak, it decreases gradually. The varying trend of exported energy is caused by two factors with the opposite impact: the decrease in energy supply and the decrease in energy storage system capacity. The decrease in energy supply would result in decreased exported energy, while the decrease in energy storage system capacity would lead to increase in exported energy (as less energy can be stored). Before the 19th year, the decrease of energy storage system capacity dominates the varying trend of exported energy. After the 19th year, the decrease of energy supply dominates the trend.

In summary, the system performance deteriorates dramatically after operating for years. Therefore, the sizing of nZEB systems without considering the degradation effects lacks long-term planning and would easily lead to a design that cannot perform well during the whole service life.

12.4.2 Results of Performance Improvements Using the Two-Stage Design Method

12.4.2.1 Life-Cycle Cost Results of the Two-Stage Design Method

The RES size and ES size of the case building were improved using the method proposed in Sect. 12.2.2. At Stage 1, 17 alternatives of wind turbine number were defined based on the original RES size (which is called Alt. 1). The wind turbine number increases from 16 to 32 at an interval of 1. The area of PV panels increases corresponding to the number of wind turbine to assure that they have the same annual energy production. The size of energy storage system was kept fixed. The life-cycle cost of these 17 alternatives were analysed and compared, as shown in Fig. 12.12a. As the number of wind turbine increases, the initial cost increases linearly (this is because a linear initial cost model was used). The operational cost decreases gradually, as the RES can supply more energy for local usage. The life-cycle cost firstly decreases

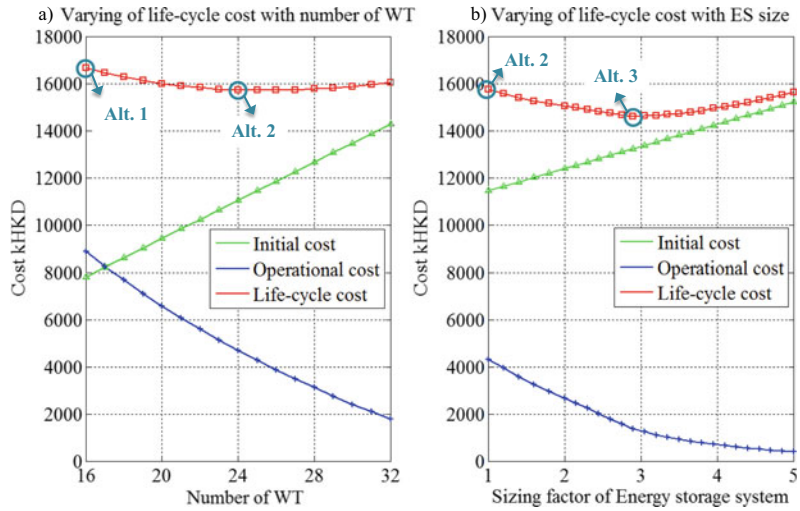


Fig. 12.12 Performance improvements using the two-stage design method

with the wind turbine number to a valley, and then it increases. The alternative of wind turbine number producing the minimal life-cycle cost was selected as Alt. 2. Table 12.4 summarizes the RES sizes of Alt. 1 and Alt. 2.

The number of wind turbine and area of PV panels of Alt. 2 was used as fixed inputs in Stage 2. A set of sizing factors, ranging from 1 to 5, were defined for the ES size. The real size of energy storage system is calculated by multiply the sizing factor with original size 1571.8 kW·h. Figure 12.12b shows the varying of life-cycle cost with the energy storage system size. The operational cost decreases gradually and finally it becomes stable around a constant value, which is determined by the size of PV panels and wind turbine. This is because the energy storage system only determines the maximum amount of stored energy. It does not produce extra energy to reduce the operational cost. Again, the life-cycle cost of each alternative is analysed and compared. The alternative of ES size producing the minimal life-cycle cost was selected as Alt. 3. Table 12.4 presents detailed configuration of Alt. 3.

Table 12.5 compares the life-cycle cost of three alternatives and calculates the percentage of cost saving from the two-stage method. Stage 1 improves the design from Alt. 1 to Alt. 2 and achieves a 5.77% cost saving. Stage 2 improves the design from Alt. 2 to Alt. 3 and further achieves a 6.84% saving. In total, the two-stage method cuts down the life-cycle cost 12.61%.

12.4.2.2 Comparison of Performance Indices Before and After Performance Improvements

The four performance indices of Alt. 3 were analysed, as shown in Fig. 12.13. Compared with the performance of Alt. 1 (see Fig. 12.10), the cooling set-point unmet hour does not change (see Fig. 12.13a). This is because the thermal comfort is mainly determined by the HVAC system size. In both cases, the HVAC system sizes were the same. The annual operational cost increases gradually in an exponential form (see Fig. 12.13b). The annual energy balance decreases gradually from 0.37 to − 0.09 (see Fig. 12.13c). Compared with Alt. 1, Alt. 3 can better achieve the goal near-zero energy.

The annual grid independence increases gradually until the 16th year, after then it decreases gradually. It performs better than Alt. 1 as during most time it has a grid independence larger than 0.5. The comparison shows the two-stage design method cannot only bring down the life-cycle cost, but also improve some performance indices (e.g. energy balance, and grid independence).

Table 12.4 Summary of sizes of each alternative

Component	Size of each Alter		
	1	2	3
WT (each)	16	25	25
PV panel (m^2)	1299	2029	2029
ES (kW·h)	1572	1572	4558

Table 12.5 Summary of LCC reduction from two-stage method

Stage	LCC (*k* HKD)			Reduced LCC	
	Alt. 1	Alt. 2	Alt. 3	Amount (*k* HKD)	Percentage (%)
Stage 1	16,705	15,741	–	963.8	5.77
Stage 2	–	15,741	14,599	1142.3	6.84
Sum				2106.1	12.61

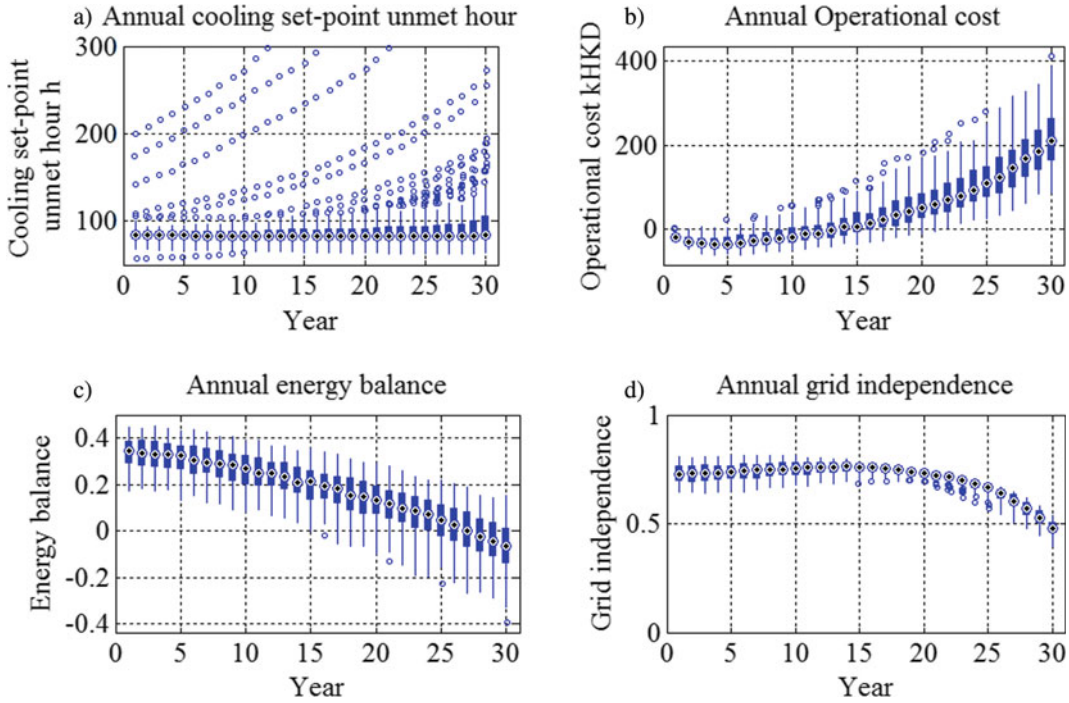


Fig. 12.13 Varying of four performance indices of the optimal design with operating year

12.5 Summary

Considering the systems' and components' degradation effects, this chapter has proposed a method to evaluate the life-cycle performance of nZEB under uncertainties, which consists of the following main steps: *quantification of uncertainties*, *nZEB systems design considering uncertainties*, *quantification of degradation rates*, and *life-cycle performance analysis considering degradation*. The uncertainties in the thermal load prediction and in the weather condition have been quantified using statistical

distributions. The degradation of nZEB systems has been modelled using the random deterioration rate method: the degradation rate of PV panel and wind turbine has been quantified using experimental data in literature, and the degradation rate of energy storage system has been investigated using the Rainflow counting algorithm. As the proposed LCPA method considers both the uncertainty and degradation effects, it can enhance the energy demand prediction accuracy and improve the designers' understanding of the components' degradation impacts on the nZEBs' performance. Based on the proposed LCPA process, a two-stage design method has been proposed to improve the sizing of nZEBs. The two-stage method first *improves the PV panel size and wind turbine size* and then *improves the energy storage system size*. The major findings are summarized as follows:

- Due to component degradation, the nZEB annual energy demand increased, the annual energy supply decreased. The annual power exchange also decreased (indicating more energy import from the grid).
- For the nZEBs designed from conventional method (which does not consider component degradation), the number of hours that AC system cannot provide sufficient cooling increased by 30% after 30 years in an exponential relationship with operation year.
- For the nZEBs designed from conventional method, they could not achieve the goal near-zero energy during most time of its service life.
- For the nZEBs designed from conventional method, their grid independence performance could even drop to 0 when electrical battery capacity decreased to 0. This would bring great challenges to the reliability and stability of the power grid.
- Through improving the renewable energy systems sizes, the life-cycle cost was reduced by 5.77%. Through improving the electrical battery sizes, the life-cycle cost was reduced by 6.84%.
- Compared with the conventional design method, the two-stage method could produce a design that had 12.61% lower life-cycle cost and could achieve desired grid independence performance during the whole service life.

As the proposed two-stage design method only explores a limited number of design alternatives of RES sizes, future work will investigate more alternatives and conduct a systematic study to optimize the sizing of RES considering the effects of uncertainties and degradation.

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13 Impact of the Demand Profile and the Normative Framework on a Residential Photovoltaic System

Marco Lovati and Xingxing Zhang

Abstract

In recent years, optimization-based design techniques are proposed for building-integrated photovoltaic systems (BIPV), or urban PV systems. Urban PV systems are subjected to an inter-play of shading and self-consumption issues such that it makes sense to determine the functional capacity and positioning for the system before to simulate it. This chapter, therefore, analyzes the effectiveness of one optimization approach and matches it against more traditional BIPV dimensioning methods. Three design methods are described and compared to a benchmark (i.e., the ideal optimum design): The minimum capacity required by the current Italian law, the PV capacity which has an annual cumulative production equal to the cumulative demand of the building and an optimization technique using a constant energy demand (e.g., in case the user has only energy bills or cumulative forecasts). The methods were all tested on a case study located in Firenze (Italy) consisting of a residential four stories building. The case study is currently undergoing energy improvement works for experimental purposes within the H2020 Energy Matching project. The results show that the optimization approach easily outperforms the other methods despite the simplified input data enabling a sensible improvement in Net Present Value (NPV). The optimization method, when fed constant data in absence of more realistic one, still leads to an improvement of NPV from + 24 to + 85% compared to the highest yielding traditional one (i.e., the Italian law) and can, after all, achieve from 93 to 98% of the actual optimum. In some countries, the net billing (or net metering) incentives are still in place: In such economic frameworks, because the self-consumption becomes less relevant, the optimization technique is not required.

Keywords

Residential photovoltaic · Urban photovoltaic · Economic analysis · Consumption profile · Prosumers · Self-sufficiency ratio

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13.1 Introduction

13.1.1 The Long-Term Trend in Photovoltaic (PV) Technology Development

The performance of photovoltaic (PV) electricity generation depends entirely on its technology as the quality of the resource cannot practically be modified by the owner or designer. Over the years, the scientific and industrial community pursued increases of efficiency, durability and scale of manufacturing while consistently decreasing the price of the technology in its every component and sub-process. This network of institutions and professionals has led to the continuous drop in prices of PV systems and it is expected to maintain the reduction going for the next 30 years, especially in scenarios where there is a strong installation of PV worldwide (Vartiainen et al. 2019). Of course, technology does not end at the production stage, system design also had a prominent role in pulling the yield up and levelized cost of electricity (LCOE) down.

13.1.2 Possible Developments of PV Technology Outside the City

The first key performance indicator (KPI) that comes to mind is the yield, i.e., amount of electricity produced during a period for each unit of power [kWh/kWp year], a yield-only mindset would tend to bring PV collectors out of cities to avoid shading by trees, buildings and other infrastructures. The avoidance of urban PV would eventually result in competitive pressure on other activities based on sunlight such as farming. There are approaches of higher yield farming such as the use of greenhouses, or complimentary use of PV such as the so-called agrivoltaics (Santra et al. 2017), but they might take decades to be economical in large scale production, and significant resources and precious metals would be required to build an infrastructure able to transport the electricity to the urbanized areas.

13.1.3 PV Design in the City: Cumulative KPIs

Assuming that plentiful integration of PV in the cities will at least complement the utility-scale production in the future, a wide and various literature can be found on the subject. When designing a PV system on a flat rooftop, one might want to organize the collectors in a series of tilted arrays so that they can have maximum yield [kWh/kWp year] (Santos and Rüther 2014; Martinez-Rubio et al. 2015), but that might not prove to be the best approach when space is an issue as it encourages to spread apart the arrays to avoid self-shading. Sometimes a similar approach, non-planar, was used to increase the yield over less irradiated surfaces such as facades (Valckenborg et al. 2016), or while also considering the effect of the non-planar array as a sun-shading device to selectively screen excessive solar radiation in apartment blocks (Lovati et al. 2016). The use of photovoltaic arrays as shade and in combination with electricity-based indoor temperature control has also been examined where the avoidance of over-heating is crucial, e.g., in a facility for the conservation of precious ancient biological material (Frasca et al. 2017). The non-planar approach can also be used without the aim to maximize the yield, but simply exploring the trade-off between yield and installed capacity over various facades (Hwang et al. 2012). The non-planar approach, despite being considered from an aesthetic perspective (Jayathissa et al. 2017), determines strongly the morphological appearance of any surface it is put on. A planar surface, aesthetic freedom aside, is simpler, thus cheaper, to manufacture while presenting fewer components prone to rupture. Given the importance of solar light for human vision and well-being, it has been proposed to mold the very shape of the urban fabric to maximize the solar access of buildings (Lobaccaro et al. 2012): This approach is viable albeit resource-intensive and implies a strong communal will to dedicate consistent engineering endeavors for the cause. In case of new buildings in an already urbanized area, the use of computer

simulation, particularly ray-tracing irradiance calculation, to efficiently harvest the solar resource while also minimizing the impact on the surroundings is a key item in the toolset of an ecologically wise society (Lobaccaro and Frontini 2014).

13.1.4 PV Design in the City: A Brief History of Self-Consumption

So far, all the studies examined focused solely on the quantity of energy that can be harvested over one year. The topic of production curve over-time is important because earth blocks sunlight with its own body and tinkers with its intensity due to an elliptical orbit and a visibly tilted rotation axe. For the sake of completeness, it should be reported that the solar resource is not variable in space, where it has been used for decades for powering unmanned probes for the exploration of the solar system and artificial satellites (Iles 2000). Nevertheless, the harvesting of sunlight in space currently presents serious technical and economic issues due to the transportation of PV systems in earth orbit or beyond and the transmission of the harvested energy back to earth. Amid dropping PV prices and increasing new capacity coming online every year, some studies started to consider the over-capacity issue, thus the matter concerning production and consumption curves: In the earlier days, it was not considered a design criteria in as much as a simple assessment of performance and was, more often than not, applied to stand-alone systems (Celik 2007). Over-time, the cumulative harvesting potential of a system became less prominent and the concept of value as determined by the need of energy in a particular point in time became increasingly crucial. The economic worth of a PV system was measured in a self-consumption-based scenario using KPI such as net present value (NPV) and payback time (PT) (Talavera et al. 2011). A complete literature review exists on the use of PV systems in urban environments and often without necessarily orienting them for maximum yield (Freitas and Brito 2019), an excerpt of notable studies is provided in the following paragraph to help better understand the research question soon to be investigated. Some studies don't go as far as to propose a design method for building-integrated PV (BIPV) systems but strive to provide tools to interpret their performance in a self-consumption driven business model (Salom et al. 2014). A clever graphical representation for the results in terms of self-consumption and self-sufficiency (Luthander et al. 2019) can aid a designer to evaluate among different solutions for the improvement of the contemporaneity of demand and production. When the aim is to match production and consumption with PV systems, the façades of buildings have to be considered alongside the roof, not only because that would increase the potential active surface (Fath et al. 2015), but also because it might be economically or environmentally beneficial for its superior matching with the demand in the local grid (Waibel et al. 2018). As a rule of thumb, adding a battery to the PV system would rule out the use of façade for energy harvesting, but the façade could also improve the matching on a seasonal time-frame (i.e., far beyond the storage potential of most realistic battery systems) (Lovati et al., 2019) when the electric consumption happens to be higher in winter. The preference for façade integration over roof integration is slightly more computationally expensive than the standard approach as it requires optimization instead of a simple sorting by annual cumulative irradiation (Freitas et al. 2018a). In some cases, e.g., in new constructions where the PV system uses but a fraction of the total resources for the construction, it is sufficient to consider the lifetime cumulative costs instead of performing a full financial NPV analysis (Talavera et al. 2019). In the context of scarce resources, such as extremely poor countries, the revenues for the salvage of system components as raw materials are considered in the economic analysis at the end of life of the system (Ndwali et al. 2019). In these strongly strained economic contexts, the PV electricity becomes vital not

only for the savings it can generate but also as a backup for temporary infrastructure failures that are fairly frequent. It is likely that in the future when the photovoltaic infrastructure will be manifold that of today, the scramble for outdated systems as raw material will be a sizeable industry. In some cases, when sufficient amounts of data have become available thanks to drones, Lidars and artificial intelligence for data acquisition (Brito et al. 2017), it is possible to perform city-wide analysis and build data repositories for effective integration of the PV systems over-time (Imenes and Kanters 2016). If in the early days of load-matching design principles the focus was often on a single building, it quickly became apparent the importance of the electric grid and the aggregation of the demand: The aggregation of different profiles (e.g., residential and offices or schools) allowed a better contemporaneity between consumption and production of PV overall (Vigna et al. u.d.). Even in cases of all residential district, it greatly improved all KPIs simultaneously thank to a lowering of the noise associated with the stochastic behavior of a single user (Lovati o.a.). It should also be noted that the grid infrastructure must be adapted in case there is the will to feed all the over-production of electricity from PV into the grid (Freitas et al. 2018b). In the future, as the technology becomes more mature, it is reasonable to believe that more and more sectors of energy production and use (e.g., transport, space heating and domestic hot water) will move from chemical, thermal or thermo-dynamic to electric sources: This will probably cause the installations of PV system to become on average larger or more ubiquitous (Huang et al. 2019). With the increasingly common use of the electric engine in cars, which are more efficient and simpler to manufacture than internal combustion ones and use a cleaner and more quiet technology, the use of PV electricity could be boosted thanks to a set of clever control strategies (Fachrizal and Munkhammar 2019). Also in the control strategy of charging cars, the excess of electricity sent into the grid could be seen as problematic, but it should be remembered that it is not mandatory to send all the excess electricity to the grid and the inverter could just curtail the excess electricity behind the meter (i.e., excess electricity should be sent to the grid only if it benefits it or if it is remunerated).

13.1.5 Novelty of This Chapter: What Happens When Hourly Input Data is not Available? Are Self-Consumption Optimization Techniques Still Valid?

The distinctive trait of the latest optimization-based techniques, which sets them aside from more traditional techniques of PV design, is that they do not require the geometry of a PV system as an input, but they provide it as an output instead. The capacity of a ground-mounted PV system is only constrained by either available space or budget. Urban PV systems are instead subject to an inter-play of shading and self-consumption issues such that it makes sense to determine the functional capacity and positioning for the system before to simulate it.

The studies carried out so far about the optimal features and performance of the PV system do not consider the effect of wrong or inaccurate input data for the optimization. The accurate forecast of complex systems such as the weather in the atmosphere or the consumption of electricity of a cluster of buildings are historically hard problems, and it is unlikely that they will progress much beyond the current state without major technological breakthrough (e.g., super-effectiveness of quantum computing technologies). Because of this, any data used in the design phase of a PV system based on self-consumption will be invariably flawed. In the following pages, a heavily inaccurate dataset, which is based only on the annual cumulative consumption and assumes the load curve as constant, is used for the optimization of a PV system. The results, both in the accuracy of the system capacity and its economic performance, are evaluated and matched against cumulative KPI-based design methods (Table 13.1).

Table 13.1 Benchmark of the studies that performed self-consumption driven capacity optimization of PV, a research gap is shown as none assessed the impact of using incomplete input data for the electric demand of the building

	Demand response study	Techno-economic optimization of PV	District scale	City scale	Assess impact of incomplete input
This chapter		✓			✓
Huang, et al. (2019)	✓	✓	✓		
Lovati et al.		✓	✓		
Vigna et al. (u.d.)		✓	✓		
Imenes and Kanters (2016)		✓		✓	
Brito et al. (2017)		✓	✓		
Talavera, Muñoz-Rodriguez et al. (2019)		✓			
Freitas et al. (2018a)		✓	✓		
Lovati, et al. (2019)		✓			

The results show the positive impact in economic performance due to the adoption of hourly based optimization even if the detailed consumption profile is not known. Even in the absence of hourly data, using the cumulative demand as a static value, the optimization technique shows an improvement of the *KPIs* with respect to traditional methods. Through the comparison with different sizing methods, it was shown why the use of optimization is a fundamental aspect to obtain a well-sized system which reduces the purchased energy maximizing the economic benefits of the investment. The *KPIs* analyzed in this chapter are the *NPV* and the *self-sufficiency* which have been used for comparing different design methods. Moreover, the effect of batteries was analyzed to demonstrate the impact of *self-consumption* and *self-sufficiency* indexes.

The existing research examined did not cover the case in which the electric demand of the building under exam is not available, and this suggests the following question: How sensible these techniques are to misshapen or downright constant demand profiles? This chapter thus analyzes the performance of the optimization approach and matches it against more traditional BIPV dimensioning methods. Section 13.2 describes the methodologies and input data. In Sect. 13.3, the results and discussion are presented. Summary are finally reported in Sect. 13.4.

13.2 Methodologies and Input Data

13.2.1 Methodologies

Different sizing methods that can be used by designers in the early design phase of a residential photovoltaic plant were compared using a case study (see 13.2.3 Case study description). Table 13.2 reports the early design methods studied and compared in this work, the PV system suggested by each method is then simulated and its KPIs are compared to establish its performance relative to other methods and to the benchmark (i.e., the theoretical optimum). In the context presented here, it is not possible to obtain a better result than the one obtainable with the benchmark method. In fact, being NPV the KPI measured and PV capacity the variable, benchmark method reliably sweep the capacity until it finds the maximum in NPV. Of course, this does not mean that this method is optimal in general terms, in fact, its optimality is only due to the fact that it is applied on a perfect forecast of the electric demand and of the techno-economic

Table 13.2 Early design methods compared in the study

	Method	Objective
1	D.Lgs. 28/2011	Install the minimum capacity suggested by the current Italian law
2	Annual cumulative	Install the PV capacity required to cover the annual cumulative demand
3	Average electric demand	Optimize the PV capacity assuming a constant electrical load
Benchmark	Hourly electric demand	Optimize the PV capacity using the real electrical demand of the building

parameters that affect the result. But this is the very point of the chapter: What happens if simplified data is fed into the algorithm? Does it still represent an advantage compared to traditional sizing methods?

Benchmark method employs the optimization algorithm shown in Sect. 13.2.2, it is unrealistically accurate because it uses the real hourly profile of the building (which in reality cannot be known but only guessed). This method exploits the full potential of the tool for the *PV* design. It was considered as the benchmark for the analysis of the other methods as it represents the optimum obtainable; indeed, the system designed with the different approaches is then simulated with hourly timestep using the real hourly demand to obtain its KPIs. The final goal is to evaluate the effect of using the Italian decree, the annual cumulative consumption or an annual average cumulative profile for the design of a *PV* plant compared with the results obtained with the benchmark.

Method 1 aims to respect the present Italian directive (DECRETO LEGISLATIVO n. 28 2011) for PV installation on new buildings. The directive sets the minimum installed nominal capacity according to the following formula:

$$P_{n,\min} = S/K = 7.08\,\text{kW} \tag{13.1}$$

where $P_{n,\min}$ expresses the minimum nominal capacity to be installed, S is the *GIA (Gross Internal Area)* of the building and K is a coefficient which varies according to the year of installation and has henceforth reached its present value of 50.

Method 2 (or annual cumulative) is an approach often used by designers who do not have a detailed hourly profile assumes a PV plant size large enough to match the annual cumulative demand of the building. The sizing process is based on changing the capacity of the plant until the annual cumulative production is equal to the annual cumulative consumption regardless of how much of the electricity produced is self-consumed on-site. The cumulative demand was obtained summing the whole year hourly demand profile [kW] of the building to have coherent values and be therefore able to compare the performance.

Method 3 This method is actually identical to the benchmark method, but it is flawed by the realistic assumption that the future hourly demand cannot be known, the designer usually only disposes of the annual cumulative demand of the building that he can infer from the previous energy bills (or from similar buildings in case of new construction). For this reason, method 3 assumes a constant hourly demand calculated averaging the annual cumulative consumption, this limitation has been chosen to assess the performance of such method when the actual hourly demand of the building is not available. Despite having a constant load curve, it is based on an hourly based optimization algorithm whose objective is to maximize the *NPV* at the end of a selected period (i.e., the planned lifetime of the system). The method performs a techno-economic optimization considering the avoided purchase of electricity from the grid due to self-consumption and, if available, the revenues from the grid for the sale of excess electricity. The algorithm is explained in more detail in Sect. 13.2.2 (Simulation and optimization environment), for now it is sufficient to remind that the result is profoundly affected by the power demand of the building.

13.2.2 Simulation and Optimization Environment

The benchmark method and method 3, with or without net billing, are based on an optimization procedure as described in (Lovati et al. 2019) and (Huang et al. 2019). The optimization is performed with fitness functions based on the genetic algorithm. The optimization algorithm works according to the workflow reported in Fig. 13.1. The input consists of a set of techno-economic parameters (see Table 13.3), a weather file containing information about local irradiation and temperature and a 3D representation of the building ad near shadings. The simulation of the PV system power [W_{el}], which is used to assume the hourly productivity of every system regardless of its design method, is linearly correlated (by a coefficient equal to the module efficiency) with the incoming irradiation [W], but corrected by a temperature coefficient of − 0.005 [°C] with respect to the temperature at Standard Test Condition (STC). The temperature of the module is calculated using an empirical Ross coefficient as described in Maturi et al. (2014). The incoming irradiation, which depends on the capacity and positioning of the PV system, is calculated for every hour of the year taking into account shadings and the reflectivity of the urban environment through a ray-tracing simulation using the open-source tool RADIANCE (Reinhart and Herkel 2000). In the present study, the battery capacity is not investigated, therefore is not a parameter of the optimization. In the present study, the reward function of the optimization algorithm, i.e., the value that the algorithm tries to maximize, is the Net Present Value (NPV) of the investment. The optimization algorithm varies the capacity of the system and runs a simulation of the NPV resulting from said capacity, the capacity that produces the highest NPV is therefore selected as the solution.

13.2.3 Case Study Description

The case study consists of a four-story building (see Fig. 13.2) located in Campi Bisenzio, Florence (Italy, 43°49′ N, 11°8′ E). The weather of the location (i.e., Mediterranean sub-oceanic) is characterized by dry summers and mild, wet winters. The cumulative irradiation for each month of the year is shown in Fig. 13.3: monthly cumulative horizontal irradiation.

The ground floor of the building is uninhabited and hosts common spaces and private

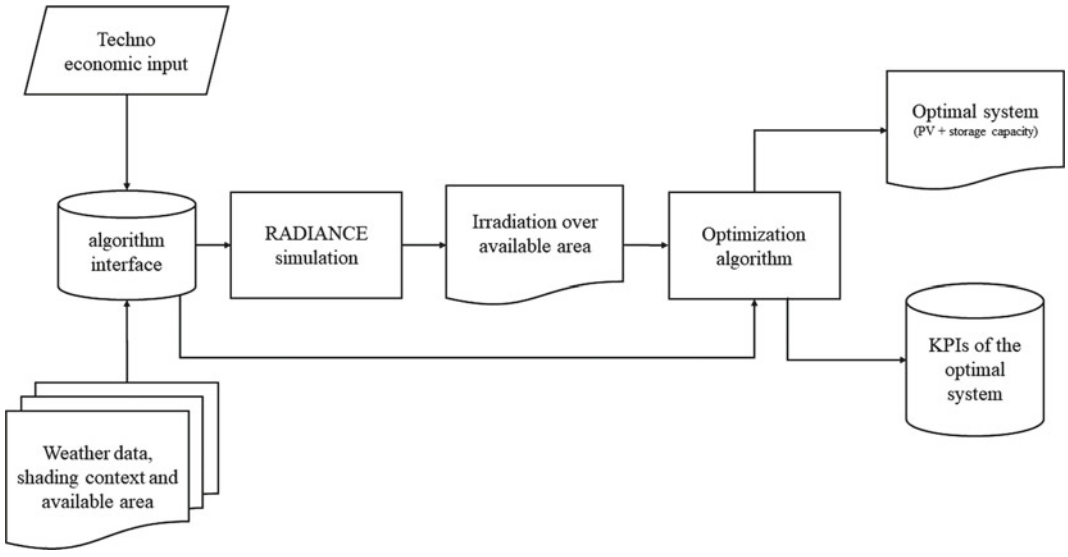
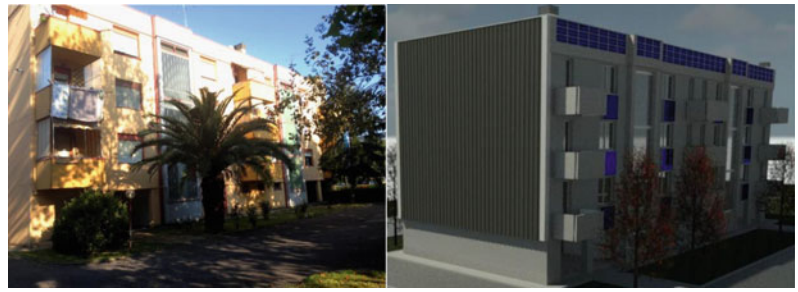


Fig. 13.1 Workflow of the optimization process in method 3 and in the Benchmark

Table 13.3 Input data

Parameter	Value
Efficiency [%]	16.5
Module dimensions [m]	1 × 1.6
Performance ratio [–]	0.75
Temperature coefficient [%/°C]	− 0.5
Price of electricity purchased [€/kWh]	0.18
Price of electricity sold [€/kWh]	0.05
Net billing premium [€/kWh]	0.1
Time horizon [years]	20
Cost of PV system [€/kWp]	1700
Annual maintenance costs [€/kWp year]	7.5
Linear annual growth of the electric load [%]	1
Linear annual efficiency losses [%]	0.75
Annual discount rate [%]	3.2
Linear annual growth of the cost for the electricity purchased [%]	0
Linear annual growth of the cost for the electricity sold [%]	0



(a) (b)

Fig. 13.2 Photo of the case study pre-renovation (**a**) and render of the same after renovation (**b**)

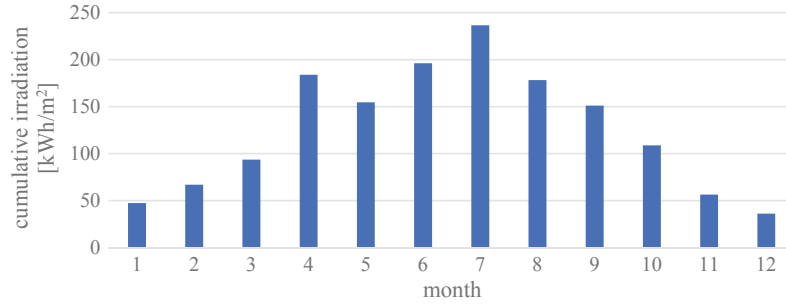


Fig. 13.3 Monthly cumulative horizontal irradiation

deposits for tools and light vehicles. The three out-of-ground stories comprise 14 inhabited households. The building is consistently aligned with most of the other buildings in town, thus has its main sides on South-East and North-West leaving a blind-short façade on South-West. The building is currently undergoing energy retrofit for research purposes within the framework of the H2020 project Energy Matching. The building will sport electrical and thermal energy

harvesting devices on its envelope and be equipped with state of the art HVAC, a centralized electric energy management system, a complete suite of sensors for energy performance and indoor air quality and off-grid autonomous ventilation machines integrated in the window frames.

13.2.4 Input Data

The tool requires a series of techno-economic inputs for the simulation and optimization of the plant. They are summarized in Table 13.3. Net billing and fixed tariff are assumed as the tariff schemes used for the analysis. The discount rate for the NPV calculation was set to 3.2% as the average global growth forecasted by International Monetary Fund (2019). The NPV analysis is extended over 20 years and no fiscal deductions are considered as a conservative set of assumptions.

The whole building is composed of 14 households whose electrical demands are assumed with LPG, Load Profile Generator (Pflugradt 2019), which is a behavioral-based tool for the modeling of realistic electric consumptions with one-minute timestep. It contains all kinds of activities due to occupant behavior in a household, as shown in Fig. 13.4a. The inner workings of LPG are well related in reference (Pflugradt 2017) and are illustrated in Fig. 13.4b. It performs a full behavior simulation of the people in a household and uses that to generate load curves. Every person is part of a household alongside other persons and appliances, with various numbers of needs. To satisfy its needs, each person might interact with different appliances according to its desire. The person then checks the expected satisfaction from each activity and chooses the one that promises to give the biggest boost. So if a person is tired, then he will sleep, if he is hungry, he will make some food (perhaps using an electric oven) and so on. To make this work, every desire grows a bit in every time step, although the amount of growth varies between different desires. During the simulation, this LPG generates load profiles in CSV files for the import into the simulation tool.

Two demand profiles from the LPG are used in the analysis: Profile 1 is obtained as the hourly resampled sum of the fourteen profiles, Profile 2 is the sum of Profile 1 and the hourly electric heating and cooling demands obtained with a dynamic simulation of the building using TRNSYS. The difference between Profile 1 and Profile 2 (which includes the heat pump) is shown in its hourly and monthly average in Figs. 13.5 and 13.6. The annual cumulative demands of Profile 1 and Profile 2 are equal to about 54.23 and 71.4 MWh, respectively. The average demands (obtained by applying a constant load such that the annual cumulative value remains unchanged) result equal to 6.19 and 8.15 kW, respectively.

The two profiles were chosen to simulate a situation both with and without electric heating and cooling (i.e., with gas or biomass for heating). The energy performance of the building and the electricity demand profiles was assessed by performing annual energy simulations with the use of a dynamic energy simulation software. To this end, a multi-zone TRNSYS model of the reference building and its energy system was developed (see schematic in Fig. 13.7).

Single dwellings are represented by single air-nodes characterized in terms of internal loads (artificial lighting, appliances, and human presence), infiltrations, and ventilation (decentralized heat recovery ventilation) as described in Table 13.4. The thermal demand was assessed for constant indoor air setpoint temperatures of 21 °C (heating) and 26 °C (cooling). The primary generator is a 16 kW air to water heat pump used for domestic hot water preparation and space heating/cooling delivery. The heat pump is equipped with an inverter and modulates the flow temperature according to a climatic curve. The primary generator is complemented by a gas boiler that is activated to cover the peak heating loads. The weather dataset used in the energy simulations is generated using the database Meteonorm 7 for the location of the

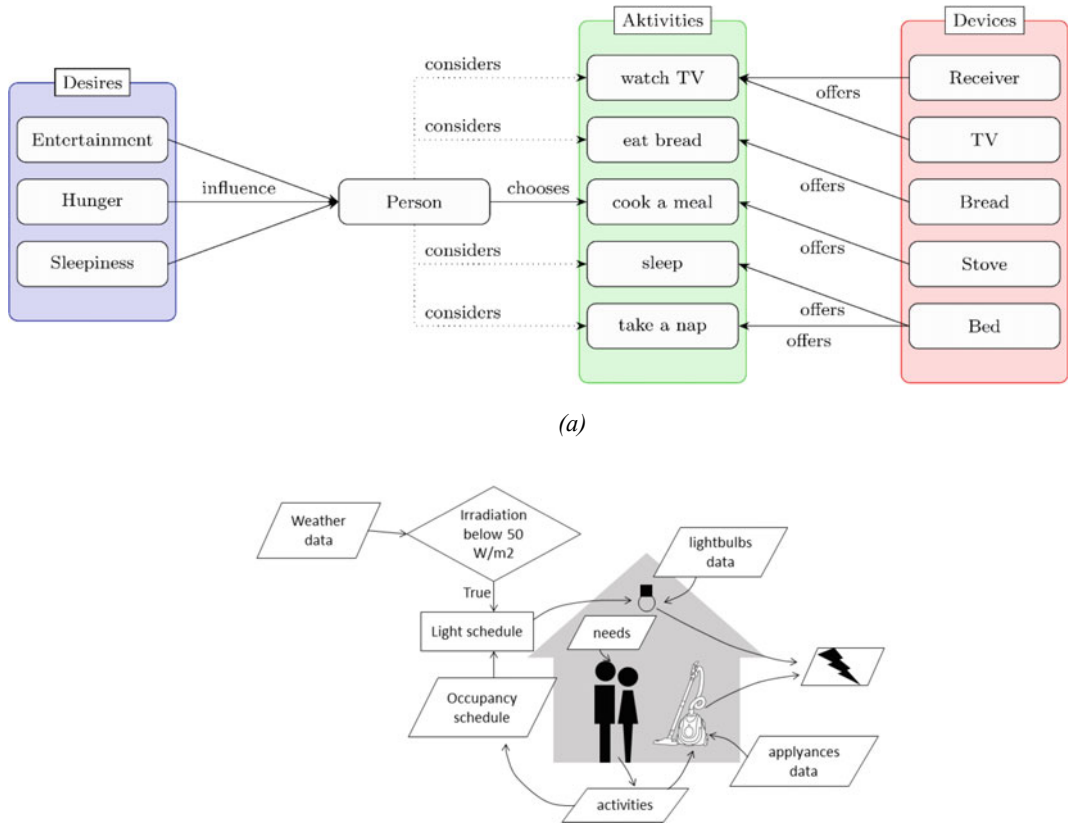


Fig. 13.4 **a** Behavior simulation (Pflugradt 2017); **b** workflow of the electric demand generation with Load Profile Generator (LPG) excluding the demand for heating and cooling. LPG has been used to forecast the electricity consumption of the 14 households in the building

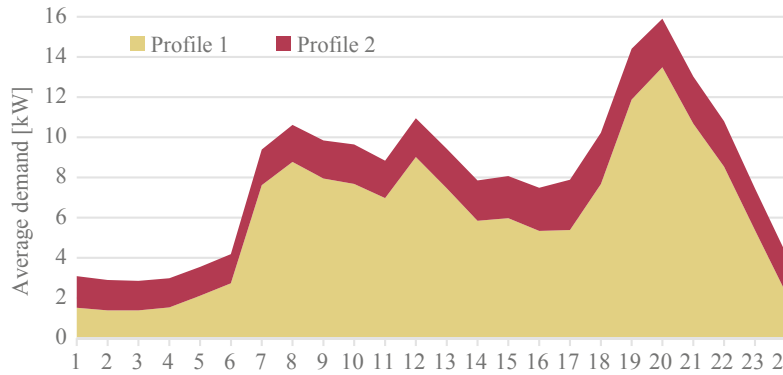


Fig. 13.5 Average electricity consumption used in the analysis for each hour of the day. Profile 1: 14 households, Profile 2: 14 households plus HP consumption, the HP does not show a strong variability during different hours of the day

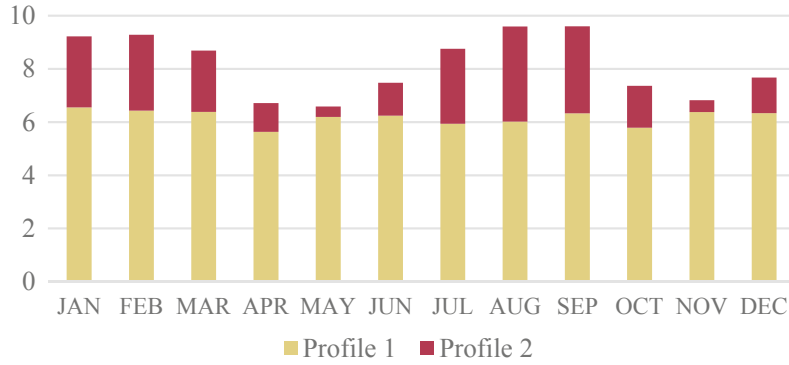


Fig. 13.6 Average electricity consumption used in the analysis for each month of the year. Profile 1: 14 households, Profile 2: 14 households plus HP consumption, the HP shows a strong variability during different months of the year

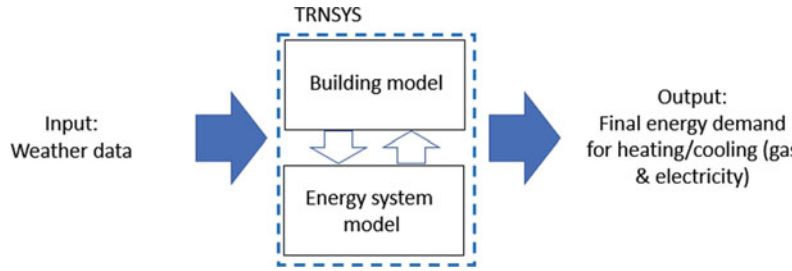


Fig. 13.7 Schematic of the TRNSYS model

reference building and contains hourly values of meteorological parameters such as ambient air temperature, humidity and solar radiation for a one-year period.

13.3 Results and Discussion

In the following session, one of several sizing methods of optimization based on self-consumption has been fed with simplified input data for the electric demand. The results show that, in absence of net billing incentive, the optimization method still produces better performing results compared to direct sizing methods.

In some countries, like for instance Italy, incentives such as net billing (also called net metering) are still active. For this reason, the comparison among the different design methods was performed both in the presence of the net billing incentive and in a regime of self-consumption.

13.3.1 Assuming Net Billing Incentive

In this subsection, results refer to the net billing tariff scheme, implemented in the tool according to ARERA (2012). In summary, feed-in electricity is valorized through the net billing premium, similar to a feed-in tariff (i.e., "Net billing premium [€/kWh]" in Table 13.3), up to a value equal to the annual demand. The surplus does not receive the premium and is therefore valorized only by the zonal electricity price (i.e., "Price of electricity sold [€/kWh]" in Table 13.3). PV plants with an electricity production higher than the annual consumption of the building are thus penalized.

(Method 1) D.Lgs. 28/2011: Italian legislative decree. As reported in Table 13.1, Method 1 for the sizing of the PV plant is based on the formula given by the Italian legislative decree D. Lgs. 28/2011. The resulting nominal power for the analyzed building is shown by Eq. 13.1.

Table 13.4 Assumptions for the modeling of heating and cooling demand using TRNSYS

Geometry and occupancy	Values	
Total heated floor area	825.5	m^2
Dwellings	12	–
U-value (external walls)	0.22	W/m^2K
U-value (glazing)	1.4	W/m^2K
g-value (glazing)	0.6	
Crowding index	0.04	p/m^2
Ventilation and infiltrations		
Design ventilation rate	40	$m^3/h/p$
Heat recovery efficiency (sensible)	75	%
Infiltration rate	0.16	1/h
Internal gains		
Human presence (active, 1.2 met)–Latent heat gain	0.08	kg/(h · p)
Human presence (active, 1.2 met)–Sensible heat gain	70	W/p
Appliances (on)	4.8	W/m^2
Appliances (stand-by)	0.4	W/m^2
Artificial lighting	2.7	W/m^2
Heating/cooling		
Space heating setpoint	21	°C
Space cooling setpoint	26	°C
DHW preparation demand (at 60 °C)	35	l/day/p

The size of the plant depends only on the GIA and it is not correlated to the demand profile as shown in Fig. 13.2. The annual production of the plant results equal to 7231.28 kWh. The plant is then simulated with the tool using the hourly demands. The NPV results equal to 4020.8€ for Profile 1 and 4103.5€ for Profile 2.

(Method 2) Cumulative is based on the annual cumulative demand of the building. In this approach, the goal is to size the system to produce the same amount of energy that is annually consumed by the building. Thus, the suggested size depends only on the cumulative demand and it results equal to 27.2 and 35.9 kWp for Profile 1 and Profile 2, respectively.

(Method 3) Average: Average electric load assumes that the designer knows the annual cumulative demand but does not know the hourly electrical profile. Instead of covering the annual cumulative demand, the system is optimized maximizing the NPV with the optimization shown in Fig. 13.1 but using a constant average load. The constant load is calculated as the annual cumulative demand divided by the number of hours. The suggested size depends on the average demand and it results equal to 28 and 34.9 kWp for Profile 1 and Profile 2, respectively.

(Method 4) Benchmark: Hourly electric demand. The hourly demand profile is used to optimize the capacity of the system. The results are used as a benchmark for the comparison with the other methods (higher level of detail of input used in the analysis). The optimal size of the plant results equal to 29.6 kWp (NPV 8746.8€) for Profile 1 and 38 kWp for Profile 2 (NPV 11,540€).

Figure 13.8 shows the comparison between the plant size suggested by the different methods considering net billing as the tariff scheme. It is possible to notice that all the methods, except Method 1, consistently suggest the PV capacity both for Profile 1 and Profile 2. The same result is reflected in the NPVs obtained for the four plants (Fig. 13.9). Method 1, since it depends

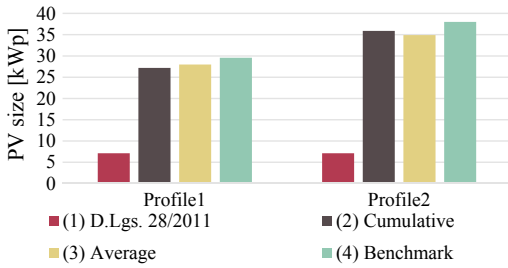


Fig. 13.8 PV size designed with different methods.Net billing used as the tariff scheme

only on the GIA of the building, in this case, suggests an undersized system that leads to NPV lower than the one obtained with the other methods for both Profile 1 and Profile 2. In the case of net billing, Method 2 and Method 3 lead to results similar to the ones obtained with the hourly timestep optimization. Net billing encourages the installation of PV systems that yearly produces the amount of energy that is consumed by the building. However, it is possible to notice a small difference between the suggested size. It is mainly caused by the lifetime growth input required by the optimization tool. The tool recognizes the benefit of installing a capacity slightly higher than the one suggested by the annual cumulative demand method because during the period of the analysis, the electric consumptions will gradually increase, and the performances of the PV system will decrease (see Table 13.3). As shown in Fig. 13.9, this does not have a relevant effect on the NPV, it can be considered optimal for all the three approaches.

Thus, from the economic point of view, both Method 2, Method 3 and the Benchmark method can be used for the correct design of the PV system if net billing is adopted. Net billing decreases the importance of direct self-consumption, while the electricity sold to the grid can be re-purchased at a low price. For this reason, a yearly cumulative-based approach can lead to well-designed plants. On the other hand, since Method 1 is independent of the demand profile, it often leads to the design of non-optimal systems and in general is insufficient for design strategies based on economic KPIs.

13.3.2 Results in a Self-Consumption Regime

The same case study is analyzed assuming a fixed tariff scheme instead of net billing. It was assumed a price for electricity sold equal to 0.05 €/kWh and a cost for electricity purchased equal to 0.18 €/kWh. As expected, hourly optimizations (Method 3 and Benchmark) assuming a fixed tariff scheme are different compared with the ones obtained assuming net billing. On the contrary, the system capacity designed with Method 1 and Method 2 does not depend on the normative framework, thus are equal to the one presented in the previous section. For brevity, results in terms of system size and NPV are summarized in Figs. 13.10, 13.11 and Table 13.5.

As shown in Fig. 13.11, the yearly cumulative method results in the poor design of the system. In general, it leads to oversized systems whose

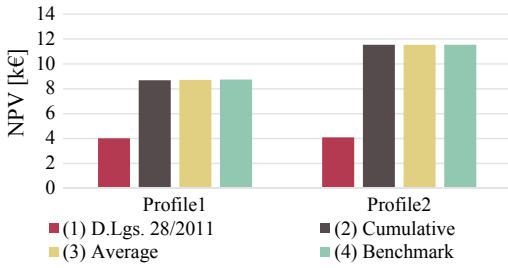


Fig. 13.9 NPV of systems designed with different approaches. Net billing used as the tariff scheme

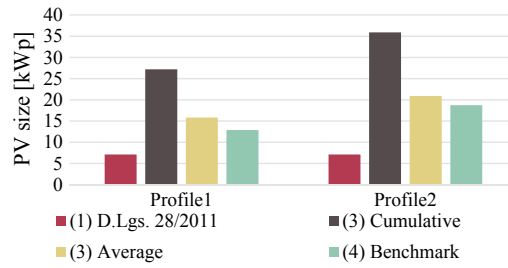


Fig. 13.10 PV size designed with different methods. Fixed tariff used as the tariff scheme

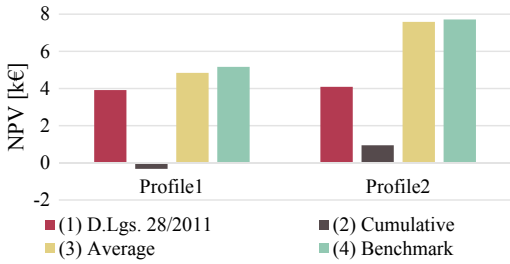


Fig. 13.11 NPV of systems designed with different approaches. Fixed tariff used as the tariff scheme

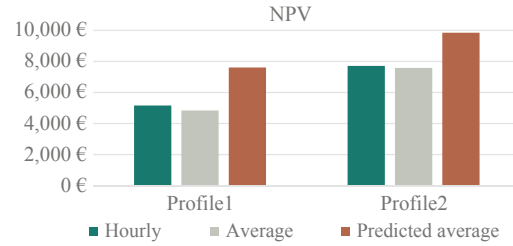


Fig. 13.12 Real and predicted NPVs obtained with different profiles as input

costs can't be repaid during the supposed lifetime of the plant (Profile 1) or results in a low value of the NPV (Profile 2). Considering the fixed tariff scheme, the average load method leads to acceptable results in terms of both PV size and NPV. Even if the PV system is slightly oversized compared to the Benchmark Method (+22.8% for Profile 1 and + 11.5% for Profile 2), the error in terms of NPV can be considered acceptable for the early design stage (-6.4% and -1.7%, respectively). Compared to net billing, a fixed tariff scheme penalizes oversized systems (bad load matching can lead to negative NPVs) and the optimal configuration can be found only with an hourly approach (evaluate at every timestep the energy flows).

However, it is important to stress the fact that even if the average electric demand method can be used to design the system capacity, it did not prove accurate predictions of the economic KPIs caused by a wrong prediction of the self-consumed energy. The average method during the optimization phase overestimates the predicted NPV (Fig. 13.12). Once the system is sized, to obtain the real NPV (yellow bars), the system is simulated using the real hourly profile as input. As already discussed, the real NPV of the system sized with Methods 3 and the Benchmark is similar for both Profile 1 and Profile 2 (Fig. 13.12). On a final note, results in this work are presented for two electrical demands (Profile 1 and Profile 2, Fig. 13.5). For brevity and clarity, the authors omitted to report the same conclusions obtained with three additional profiles.

13.3.3 Self-Sufficiency Versus NPV

The scope of a BIPV system is ultimately that of reducing the reliance of the building from the energy grid and by doing so reducing its specific CO_2 emissions. In the results from the previous section, it appears that Method 13.1 (i.e., the Italian law) has a decent performance when compared to the Benchmark (i.e., the ideal case). To fully appreciate the performance difference between an hourly optimization method and

Table 13.5 Size of the PV system designed with different approaches. Fixed tariff used as the tariff scheme

	Description	Profile	PV size [kWp]	NPV [€]
Method 1	D.Lgs. 28/2011	1	7.1	3916.0
Method 2	Annual cumulative	1	27.2	− 322.2
Method 3	Average electric demand	1	15.8	4836.5
Benchmark	Hourly electric demand	1	12.9	5166
Method 1	D.Lgs. 28/2011	2	7.1	4089.1
Method 2	Annual cumulative	2	35.9	941.2
Method 3	Average electric demand	2	20.9	7579.6
Benchmark	Hourly electric demand	2	18.7	7710.6

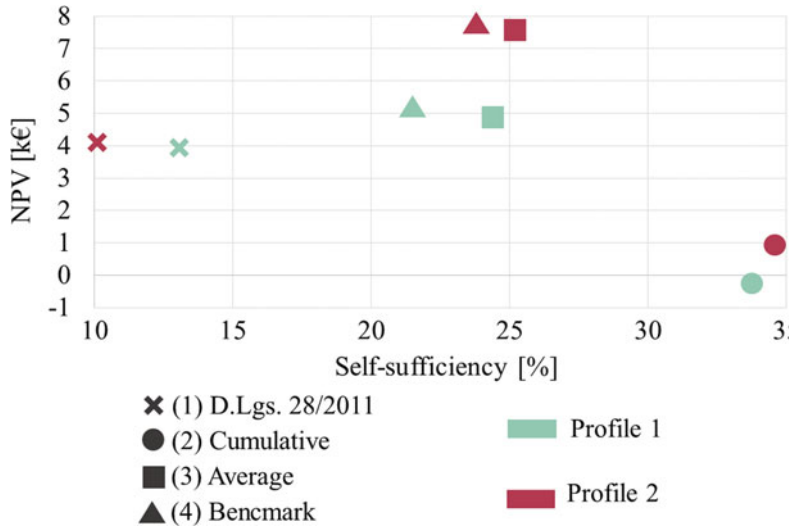


Fig. 13.13 Performance in terms of self-sufficiency and NPV of the different design strategies for both profiles

traditional ones, the NPV has been plotted against the self-sufficiency ratio (sometimes called solar fraction) as shown in Fig. 13.13.

13.4 Summary

The chapter shows the potential economic benefit associated with the adoption of an hourly based approach from the first phases of the early design process of a photovoltaic system. The hourly method was applied to a specific case study and compared with commonly used design approaches. The maximum potential NPV of the analyzed system is found through the optimization of the system capacity based on the hourly energy balances of the real consumption profile. Cumulative-based calculations, by lacking a correlation between the capacity of components and their performance, cannot aid the sizing of PV plants from the economical point of view (unless net billing is adopted). In the case of net billing, the yearly cumulative-based method gives acceptable results compared with hourly optimizations. The present Italian law is conservative and can lead to undersized systems characterized by NPVs lower than what is obtainable and poor self-sufficiency (sometimes referred to as solar fraction). In the case study shown, the Italian law leads to an NPV short of 50% of the maximum, suggesting the installation of a capacity (7.1 kWp) of about 24% the optimal one obtained considering net billing and Profile 1. On the contrary, oversized systems (such as those proposed by Method 2 "annual cumulative", see Table 13.5 or Fig. 13.10) can lead to economically unsustainable projects. The hourly approach takes into account the energy flows between grid, PV plant and load and can support the designers to obtain sustainable and profitable projects. Moreover, it was shown that the hourly approach can be applied with tolerably good results also when the hourly electrical profile is not available. Considering a constant average demand calculated from the yearly cumulative consumption led to systems of comparable capacity (ca. 122%) of the actual optimum.

In summary, it is possible to say that the tariff scheme strongly influences the early design phase of a PV system; cumulative-based methods lead to good results only if net billing is adopted. In the case of fixed tariff (or more complex schemes), it is possible to find the optimal configuration of the plant only using an hourly approach. In case of lacking the detailed demand profile, it is safe to use an average profile for the hourly optimization. Through the comparison with commonly used design methods, it is

possible to assert the importance of adopting hourly based approaches (with known or assumed electrical profile) to improve the quality and sustainability of PV projects.

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14 Generating Hourly Electricity Demand Data for Large-Scale Single-Family Buildings by a Decomposition–Recombination Method

Mengjie Han and Xingxing Zhang

Abstract

Household electricity demand has substantial impacts on local grid operation, energy storage, and the energy performance of buildings. Hourly demand data at district or urban level helps stakeholders understand the demand patterns from a granular time scale and provides robust evidence in energy management. However, such type of data is often expensive and time-consuming to collect, process, and integrate. Decisions built upon smart meter data have to deal with challenges of privacy and security in the whole process. Incomplete data due to confidentiality concerns or system failure can further increase the difficulty of modeling and optimization. In addition, methods using historical data to make predictions can largely vary depending on data quality, local building environment, and dynamic factors. Considering these challenges, this chapter proposes a statistical method to generate hourly electricity demand data for large-scale single-family buildings by decomposing time series data and recombining them into synthetics. The proposed method used public data to capture seasonality and the distribution of residuals that fulfill statistical characteristics. A reference building was used to provide empirical parameter settings and validations for the studied buildings. An illustrative case in a city of Sweden using only annual total demand was presented for deploying the proposed method. The results showed that the proposed method can mimic reality well and represent a high level of similarity to the real data. The average monthly error for the best month reached 15.9% and the best one was below 10% among 11 tested months. Less than 0.6% improper synthetic values were found in the studied region.

Keywords

Data generation · Time series decomposition · Hourly electricity demand · Large-scale buildings

14.1 Introduction

The United Nations issued a report recommending that the world adopt a set of Sustainable Development Goals (SDGs) to consider the details of the post-2015 development agenda (Sachs 2012). The final 2030 agenda aims to

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achieve a blueprint of 17 SDGs with 169 targets (Colglazier 2015), where Goal 7 ensures access to sustainable energy and Goal 11 aims to make cities resilient and sustainable. Energy-efficient buildings can make a significant contribution to meeting the goals because urban agglomerates consume roughly 80% of the global energy, of which buildings account for 40% (Di Foggia 2018). Understanding energy demand patterns of urban buildings will not only help develop efficient modeling approaches, but also enable optimal controls of power grid and district heating system operation (Huang et al. 2020; Johari et al. 2020). Among the different forms of energy, electricity is used directly for household appliances, heating and cooling in many buildings, which makes it worthwhile to set up a reliable framework for analyzing and managing electricity demand in urban buildings. In this regard, the acquisition of household electricity demand data on an hourly scale plays a crucial part in the modeling, which provides a more accurate basis for implementing management for each household and aggregated buildings.

The widespread popularity of smart meters in buildings enables a huge amount of fine-grained electricity consumption data to be collected during the past decade (Wang et al. 2019). However, acquiring large-scale such data is still not an easy task. There are three main barriers hindering the data acquisition. Firstly, due to the high-frequency data logging (i.e., the process of both collecting and storing data), the data can be overlooked and errors can still be made by humans or machines. Keeping a logger working correctly over a long period requires uninterrmitted monitoring and immediate measures to fix errors when they are detected. Deploying and maintaining such a framework in buildings at urban scale with the solver system standby is costly. In addition, although analytical algorithms can process huge quantities of data are available, many of these are not able to complete large amount multi-source data transformation and integration in a sufficiently short time period (Alahakoon and Yu 2016). Secondly, data privacy is considered a fundamental human right, and data security laws exist to protect personal information. Since electricity demand is highly related to family activities, data availability will largely rely on donation under the agreement between energy companies and customers. However, data donation has uncertainty and cannot be quickly generalized in a new context. Basic issues about smart meter data should be but have not been addressed: Who owns the smart meter data? Is it possible to disguise data to protect privacy and to not influence the decision-making of retailers? Thirdly, although many data-driven methods have been developed to avoid fully physical data collection (Andriopoulos et al. 2020; Li et al. 2021), the possibility of obtaining long-term hourly data with high accuracy is still unknown. For the method requiring a large number of covariates in the prediction, it will further increase the workload.

Thus, this chapter aims to propose a statistical method of data generation to simultaneously address the difficulties in the domain of data acquisition for urban building electricity demand on hourly scale. The method only uses a public dataset and a local reference dataset for determining parameters and providing validation. The contributions of this chapter are to

(1) Introduce a decomposition–recombination paradigm for extracting statistical features from public data and linking them to target buildings;
(2) Empirically investigate the most desirable context, e.g., time of a year and parameter settings, for the method to generate reliable electricity demand data in urban buildings that can be quickly and easily used for decision-making;
(3) Analyze how factors such as hour and day of a week can be utilized to further reduce the difference between the synthetic data and real data;
(4) Deploy the method in real buildings of different types and with different heating systems in Sweden as an illustrative example.

In the remaining sections of the paper, the importance of adopting hourly data and a data generation method will be briefly reviewed in

Sect. 14.2. Section 14.3 discusses the philosophy of time series decomposition and recombination. Both the public data and reference building will be presented in Sect. 14.4. Section 14.5 analyzes the results, and Sect. 14.6 presents a case study. Concluding remarks are given in the final section.

14.2 Modeling Hourly Electricity Demand

14.2.1 The Importance of Acquiring Hourly Data in Buildings

A temporal scale refers to the energy performance resolution of buildings and their related energy systems. The time varies from hours to days for buildings to respond to dynamic environment and achieve a steady-state condition. On the another hand, the time needed for an energy system to respond to a circumstance could be only hours, minutes, or even seconds (Srebric et al. 2015). Such different response speeds indicate that the time steps may differ in their orders of magnitude in order to manage the energy flows in a certain period. Currently, many studies choose the hourly energy data for building simulation, which is considered as the minimal temporal resolution to estimate the energy demand profiles (thermal demand Fonseca and Schlueter 2015; He et al. 2009) and electric demand (Richardson et al. 2010; Widén and Wäckelgård 2010)). Hourly energy data is basically capable of coping with transient building physics, energy flow, and dynamic weather/boundary scenarios in buildings. This shift to minute or second resolution data would increase the computational cost of simulations, which is not yet promising at the moment.

The acquisition of hourly energy demand data on a large scale plays a crucial part in understanding the energy patterns of households, which will provide a more accurate basis for implementing power management for each household. In fact, a lot of existing design (Zhang et al. 2016) and control optimization (Lu et al. 2015; Zhao et al. 2015) methods for energy systems for single households are using the hourly energy demand/supply as the inputs. Regarding the application for design optimization of energy systems in single-building level, a multiple criteria decision-making method was proposed using the annual hourly electricity demand and supply data in Zhang et al. (2016), which has made allowances for uncertainty. Regarding the application for control optimization of energy systems in single-buildings, a model predictive control was developed using a non-linear programming algorithm to schedule the operation of energy systems in grid-connected low energy buildings (Lu et al. 2015). The scheduling was based on the hourly electricity demand and supply data from a Zero Carbon Building in Hong Kong. Similarly, considering the different pricing strategies, an optimal scheduling method for the thermal energy storage system was developed taking into account the hourly electricity demand and supply (Zhao et al. 2015).

The hourly household energy demand/supply data are also important for urban planning (Huang and Sun 2019), and optimization of design (Huang et al. 2021; Sameti and Haghighat 2018) and control (Chaouachi et al. 2016; Fan et al. 2018; Huang et al. 2018, 2020) of the energy systems on an urban scale. Regarding the application for urban planning, a clustering-based grouping method was developed using the hourly energy demand/supply patterns for planning the building communities, which can maximize the compensation between large power demand and large renewable energy production (Huang and Sun 2019). Regarding the application for design optimization of urban energy systems, a mixed-integer linear programming optimization-based method was proposed for designing the distributed energy storage of a net-zero energy districts using the hourly household energy demand and renewable production (Sameti and Haghighat 2018). The proposed method can minimize both the total annual cost and equivalent CO_2 emission and meanwhile keeping a reliable system operation to cover the demand. Regarding the application for control optimization of urban energy systems, Fan et al. developed a coordinated control of distributed energy storage systems based on the individual

hourly household electricity demand and supply data using genetic algorithm (Fan et al. 2018). The developed control could coordinate the hourly charging/discharging rates of distributed energy storage systems to achieve the aggregated level optimum. Similarly, a coordinated control was developed to optimize the operation of distributed energy storage systems and EV charging loads and to improve the aggregated level performances based on the individual hourly household power demand and supply data (Huang et al. 2020).

In addition, the hourly household electricity demand data is also important for the implementation of peer-to-peer (P2P) energy trading, which is a new concept proposed in recent years (Lovati et al. 2020). Within in such P2P energy trading framework, the household prosumers (i.e., households with renewable energy systems installed which can thus also produce renewable power as well as consume) can trade their surplus power with the households that have a shortage of renewable production (due to the large power demands), and thus, the renewable energy local utilization and grid interaction performances at the aggregated level will be improved (Lovati et al. 2020). Despite a number of business models for P2P trading developed, most of these business models are based on hourly household electricity demand/supply data (An et al. 2021; Bandara et al. 2021; Lovati et al. 2021). For instance, Lovati et al. analyzed the P2P energy trading economic performances under different scenarios of PV ownerships and pricing strategies based on the 48 hourly household electricity demand and PV power production data (Lovati et al. 2021). A flocking-based decentralized double auction method was developed for P2P trading based on hourly electricity demand (Bandara et al. 2021).

14.2.2 Data Generation

14.2.2.1 GAN

Apart from supervised learning methods in generating electricity demand data (Pillai et al. 2014; Roth et al. 2020), a Generative Adversarial Network (GAN) has been comprehensively studied for generating synthetic data that resembles real data (Aggarwal et al. 2021; Goodfellow et al. 2014). Using samples from a simple distribution, a generator generates random objects from complex distribution. They are mixed with real objects for a discriminator to discriminate. The discriminator learns to give low scores to the generated data and high scores to the real data. The generator is updated to generate new objects that are likely to be assigned a high score the discriminator is fixed. A number of metrics are available for evaluating the stopping criteria of GAN (Borji 2019).

For electricity generation, smart meter data provides the basis for training a GAN (Kababji and Srikantha 2020). A recurrent GAN preceded by ARIMA modeling, Fourier transform and normalization was trained by one year of hourly building electrical meter data for non-residential buildings (Fekri et al. 2019). Stacked LSTMs were selected to store information for longer sequences. Later on, an ordinary GAN was applied to the same data set. Data was normalized and clustered to eliminate variations between each building before feeding into GAN with randomly sampled seed from the latent space of dimension 20 (Wang and Hong 2020). The generated mean and standard deviation of the key parameters of the load profiles are close to the real ones, and the KL-divergence was under 0.3. Using a small sample, GAN is also an efficient way of generating parallel predictions, where real data was mixed with the generated data to form a larger data set (Tian et al. 2019). After filtering irregular synthetic data, supervised learning algorithms were adopted to train a prediction model. It has been observed that GAN-generated data can reflect the running laws of the building energy demand data. In a similar way, the performance of using the conditional tabular GAN was enhanced by generating explanatory data that were further used to generate synthetic electricity load data via a regression model (Moon et al. 2020). The method can reduce the generation of irrational values. Another regression was trained for load forecasting using the mixed data. Taking auxiliary variables as input to the generator will

be more effective than using noise (Pang et al. 2019). Using a small data set, a conditional TimeGAN was recently proposed to capture temporal behavior in buildings for electricity load data generation (Baasch et al. 2021). In TimeGAN, the generator uses random noise vectors to generate samples that are encoded into latent codes, mixed with the codes from real data, for the discriminator to learn. Decoding transforms data into its original form. Similarly, an ensemble method based on conditional GANs was applied to simulate the variability in an occupant's electrical usage behavior as well as weather forecast errors. An elitist search strategy of the multi-population genetic algorithm was introduced to realize the communication among each sub-CGAN (Zhang and Guo 2020). An auxiliary classifier GAN-based method was proposed to generate load profiles under typical patterns (Gu et al. 2019). The generator takes both noise and labels of patterns as the input and the loss of the discriminator considers both the classification and the authenticity of sample points. The label input makes the generation controllable. By using convolutional layers to capture time dependence, the generated load pattern can represent various demand possibilities.

14.2.2.2 Statistical Methods

Statistical methods are another branch for modeling the time dependency (Magnano and Boland 2007) or the distribution of energy demand (Han et al. 2021; Paatero and Lund 2006). Markov property is viable to model the dependency of electricity load on a half-hour scale (McLoughlin et al. 2010). As an extension, a Hidden Markov Model (HMM) (Bishop 2006) is a generative mixture model where the sequential dependency of a state for the latent variable is only imposed on the value of previous state, which is regulated by a transition probability. On the other hand, an emission probability controls how the value of an observable variable is taken given a hidden state. Being capable of modeling high-resolution data, HMM adopted the range of percentile as observed states for generating electricity demand data, where hidden states are discrete values with one decimal place within the range (Jenkins et al. 2014; Patidar et al. 2014). A later piece of work combined a time series deseasonalization technique to HMM because ARIMA models can fully explore the correlations of a series. The combination was shown to be simpler to implement than pure HMM (Patidar et al. 2016). As for the distribution of the synthetic data, a double exponential distribution was adopted to characterize fatter tails compared with normal distribution (Magnano and Boland 2007). Random values were firstly generated by applying the inverse cumulative distribution function on a uniform sample. An ARMA modeling was then used to reflect temporal dependency. However, the generation process still requires original electricity demand data as well as other covariates observed on the studied site.

14.2.2.3 Summary

The GAN method uses both real and synthetic sample to train the networks. Assigning low scores to the synthetic ones will enable the generator to update quickly in the early stage of the training. The scoring effect becomes less significant as good synthetics can be generated. However, it is still challenging to examine whether it is the improvement of the generator or the limitation of the discriminator that assigns high score to synthetic data when stopping criteria are applied. Another limitation for GAN is the determination of latent variable to a new site of buildings. A trained generator can map latent variables to a complex distribution in which a sample of local energy demand is drawn. When the same generator is deployed to a new site for obtaining synthetic data, the mapping may fail to characterize the distributional features, which makes the selection of latent variable for the new site difficult.

Most of the above data generation methods either are problem-dependent or require covariates to be available. Some covariates, such as occupancy level and occupant habit of energy saving, are privacy variables and difficult to measure, thus making the prediction of electricity demand for a long period unrealistic. For GAN, large amount of training data will ensure the performance of the method. However, the data is

only available for limited number of countries and time periods. Implementation of GAN is not always feasible for most scenarios. A general scheme for the implementation without too strict constraints on data at the studied site is still missing. An alternative way is to use public data for information extraction. Time series decomposition will result in independent components that are individually treated, transferred or transformed to new series. A recombination process can be performed by adding them up, which is to be illustrated in this chapter.

14.3 Method

Hourly electricity demand data presents high time dependency on its preceding periods, which cannot be modeled by only using a probability distribution. Classical autoregressive or moving average models (Box et al. 2016) are much flexible when characterizing the dependency for arbitrary number of lag orders. However, simulation results from such models rely on a normally distributed error, which makes the value distribution poorly represented (Kegel et al. 2018). It is also inefficient to transfer data generation technique from other domain to energy demand because inherent correlation and seasonality patterns are unique for different domains. Thus, this chapter considers using public data in the same domain to analyze three components, which are trend, seasonality, and residual. These components are separately treated and then combined to form synthetic data that satisfies both statistical characteristics and the constraints of yearly demand at the site studied.

14.3.1 Time Series Decomposition and Recombination

Among the decomposition methods, seasonal and trend decomposition of time series using locally weighted regression (STL) (Cleveland et al. 1990; Cleveland 1979) is superior in dealing with hourly seasonality. The length of both trend and season smoothing window for STL can also be tailored to include a flexible number of neighboring observations. The robustness weights designated to extreme remainders in the outer loop ensure STL to be robust to outliers. Compared to other decomposition methods, STL-based algorithms have shown advantages in various areas due to the mechanism of local smoothing for long time period. For example, STL-based algorithm gives better forecasting results than empirical mode decomposition (EMD)-based ones in wave height forecast. STL algorithms have enormous advantage in terms of efficiency comparing with EMD (Yang et al. 2021). The advantages have also been highlighted in forecasting metro ridership (Chen et al. 2020). For electricity demand forecasting, the use of STL decomposition can achieve the optimal decomposition of the original sequence in the decomposition stage for the sequence with a clear seasonal trend where the variational mode decomposition (VMD) is a secondary step (Niu et al. 2021). Considering the advantages, we adopt STL to decompose hourly electricity demand data. For a data point y_t, $t = 1, 2, \ldots, N$, it can be decomposed into a trend component T_t, a seasonal component S_t, and a remainder component R_t. An additive representation is

$$y_t = T_t + S_t + R_t. \tag{14.1}$$

Since T_t can capture the base values, the mean values of S_t and R_t are equal to 0. In this work, S_t should reflect both daily and weekly periodicities.

14.3.2 Locally Weighted Regression

Locally weighted regression, also known as locally estimated scatterplot smoothing (Loess), performs well when solving the underfitting problem, which ensures an informative smoothing for long time series data y given independent variable x. The number of q data points that are the nearest to the fitting point x are identified in an interval, denoted as x_i, and are identified for regression. Each x_i is assigned a weight:

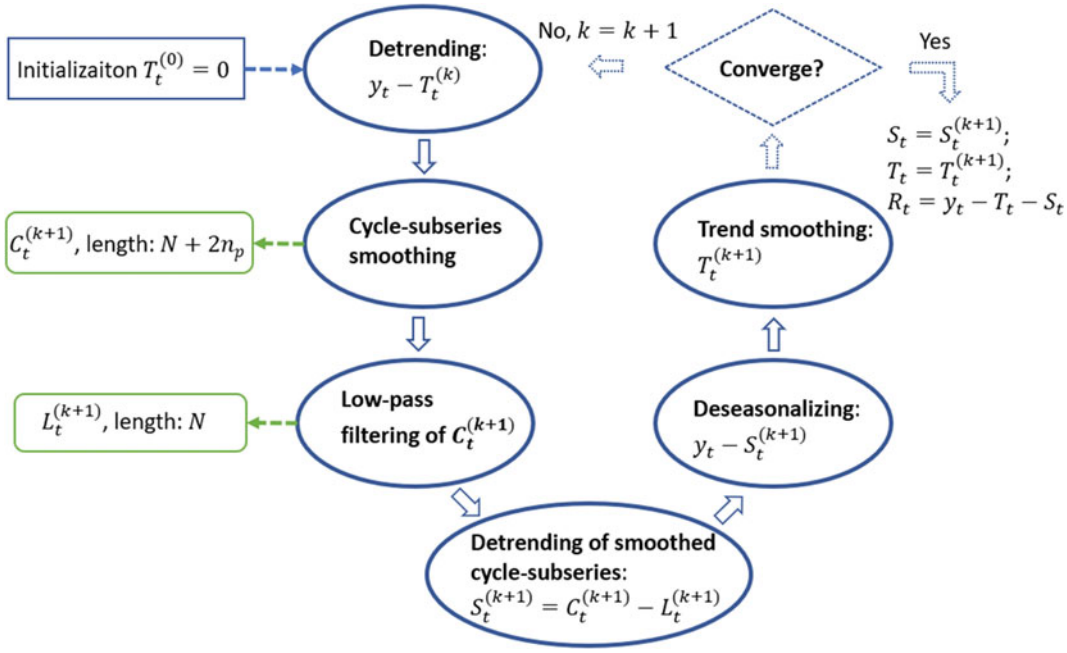

Fig. 14.1 Inner loop for STL

$$w_i = \omega\left(\frac{|x_i - x|}{\lambda_q(x)}\right), \tag{14.2}$$

where $\omega(u) = (1-u^3)^3$ for $0 \le u < 1$ and $\omega(u) = 0$ for $u \ge 1$. $\lambda_q(x)$ is the largest distance to x for all x_i s. A diagonal matrix W with elements w_i is then constructed to estimate the coefficients which is the same as the ordinary weighted least square: $\hat{\beta} = (X^T W X)^{-1} X^T W y$. The estimation is then replicated for all interesting points to get a smoothed curve.

14.3.3 Inner Loop and Outer Loop for STL

14.3.3.1 Inner Loop

A subseries is a set of observations that appear at the same time point in each seasonal fraction. In this work, the number of the sets, n_p, is set to be 168 representing each hour series in all weeks. Thus, each of the 168 sets is denoted as a cycle-subseries. The inner loop for STL updates the values of S_t and T_t until they converge. Given the estimates of them at the kth iteration: $S_t^{(k)}$ and $T_t^{(k)}$, the update at the $(k+1)$ th iteration is shown in Fig. 14.1, where six steps have been identified (Cleveland et al. 1990).

An initial value of $T_t^{(k)}$ is taken as 0. The *Detrending* step aims at ignoring the impact of trend component $y_t - T_t^{(k)}$ and preparing for extracting seasonal component for each of value in the whole sequence. The *Cycle-subseries smoothing* step uses the Loess to smooth each of the cycle-subseries. One more smoothed value is appended to both in front of the first week and after the last week, which means that the length of the smoothed sequence $C_t^{(k+1)}$, $t = 1, 2, \ldots, N$ is $N + 2n_p$. The *Low-pass filtering* step implements three moving average and one Loess filtering to $C_t^{(k+1)}$ to transform it to length N and to an even smoother curve $L_t^{(k+1)}$. Thus, the detrending to the smoothed cycle-subseries in the following step produces the seasonal component at the $(k+1)$ th iteration:

$$S_t^{(k+1)} = C_t^{(k+1)} - L_t^{(k+1)},\ t = 1, 2, \ldots, N, \tag{14.3}$$

which is subtracted from the original sequence thereby getting rid of the seasonal effect in the *Deseasonalizing* step. An additional Loess is adopted for smoothing and identifying the trend component $T_t^{(k+1)}$. A final convergence check will decide whether the update process should continue or stop.

14.3.3.2 Outer Loop

The outer loop sets robustness weight, ρ_t, to the remainders $R_t = y_t - T_t + S_t,\ t = 1, 2, \ldots, N$ to resist outliers. It will be a follow-up to the inner loop in this work to capture general patterns even though it is an optional loop. The larger the absolute value of R_t is, the small the ρ_t is assigned, which is computed as

$$\rho_t = \left(1 - \left(\frac{|R_t|}{6 \times \text{median}(|R_{1:N}|)}\right)^2\right)^2, \quad t = 1, 2, \ldots, N, \tag{14.4}$$

for $|R_t| < 6\ \text{median}(|R_{1:N}|)$. If $|R_t| \geq 6\ \text{median}(|R_{1:N}|)$, $\rho_t = 0$. As soon as ρ_t is obtained, it will then replace w_t in Loess for the cycle-subseries smoothing and trend smoothing steps in the inner loop.

14.3.4 Components

On the one hand, some Swedish buildings are heated by district heating in winter and transition seasons, which means that electricity is not primarily used for heating during these seasons. On the other hand, some buildings consume considerable electricity for running a heat pump, which makes it hard to reflect trend component by using seasons. Although temperature is the most dependent environmental factor across a year in Sweden, the trend component for a long period, for example, one year, seems to have a weak and insignificant impact on electricity demand. Herein, temperature is not the determinant of electricity demand (Vassileva et al. 2012). Moreover, the variations of electricity demand in winter are higher than in other seasons for multiple buildings and a large number of outliers are observed, which makes associating temperature to electricity demand vaguer (Quintana et al. 2021). For winter season in Sweden, specifically, the average electricity demand is 2.65 kWh/m^2 per month and for summer season, it is 1.52 kWh/m^2 per month in 2018. However, the respective standard deviations are 3.401 and 2.020 that are high. Since temperatures do not drastically change, using temperature to predict the trend of electricity demand is unreliable. Thus, the magnitude of the trend component is only modeled to share a same daily proportion to a reference building in the generated y_t under the constraint of yearly total demand.

As stated before, the seasonal component considers 168 h as a season or cycle, in which both weekly and daily periodicity can be captured. In this work, the length of the window for Loess smoothing of the cycle-subseries takes large value to obtain same seasonal patterns for all the weeks in a year. The extracted patterns will be used as the seasonal component in the generated data.

The remainder component reflects the stochastic information that incorporates any unknown randomness deviating from the model. Probability distribution and autocorrelation are two measures characterizing it (Kegel et al. 2018). Apart from the expectation, μ (equal to 0), and standard deviation, σ, skewness and kurtosis are also used as the feature of a probability distribution. If we denote μ_i as the i th order central moment, the skewness of R is the third standardized moment

$$\text{Skew}[R] = E\left[\left(\frac{R - \mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma^3}, \tag{14.5}$$

and the kurtosis measures is the fourth standardized moment

$$\text{Kurt}[R] = E\left[\left(\frac{R-\mu}{\sigma}\right)^4\right] = \frac{\mu_4}{\sigma^4}. \quad (14.6)$$

The skewness measures how asymmetric a distribution is and one of the sample moment estimations is (Joanes and Gill 1998)

$$\hat{g}_1 = \frac{\frac{1}{N}\sum_{i=1}^{N}(r_i - \overline{r})^3}{\left(\frac{1}{N}\sum_{i=1}^{N}(r_i - \overline{r})^2\right)^{3/2}}. \quad (14.7)$$

The kurtosis measures whether the data is heavy-tailed or light-tailed relative to normal distribution and one of the sample moment estimations is

$$\hat{g}_2 = \frac{\frac{1}{N}\sum_{i=1}^{N}(r_i - \overline{r})^4}{\left(\frac{1}{N}\sum_{i=1}^{N}(r_i - \overline{r})^2\right)^2}. \quad (14.8)$$

Autocorrelation is a correlation coefficient between values of the same variable at time t and $t+m$ instead of two different variables:

$$\gamma_m = \frac{\sum_{i=1}^{N-m}(r_i - \overline{r})(r_{i+m} - \overline{r})}{\sum_{i=1}^{N}(r_i - \overline{r})^2}. \quad (14.9)$$

In this work, we consider two autoregressive models, AR(1): $\tilde{R}_{t+1} = \varphi_1 \tilde{R}_t + \varepsilon_{t+1}$, and AR(2): $\tilde{R}_{t+1} = \phi_1 \tilde{R}_t + \phi_2 \tilde{R}_{t-1} + \varepsilon_{t+1}$, to model the remainders, where ε follows independently $N(0, \sigma_\varepsilon^2)$. The coefficients φ_1, ϕ_1 and ϕ_2 can be recursively obtained by solving the Yule-Walker equation (Box et al. 2016):

$$\hat{\varphi}_1 = \gamma_1, \quad (14.10)$$

and

$$\hat{\phi}_1 = \frac{\gamma_1(1-\gamma_2)}{1-\gamma_1^2} \quad (14.11)$$

$$\hat{\phi}_2 = \frac{\gamma_2 - \gamma_1^2}{1-\gamma_1^2}. \quad (14.12)$$

For AR(1), σ_ε^2 is estimated by $\text{Var}(r_t) - \hat{\varphi}_1 \cdot \text{Cov}(r_t, r_{t-1})$, where Var is the variance and Cov is the covariance. Similarly, the estimate of σ_ε^2 for AR(2) concerns the covariance of both lag order 1 and 2: $\text{Var}(r_t) - \hat{\phi}_1 \cdot \text{Cov}(r_t, r_{t-1}) - \hat{\phi}_2 \cdot \text{Cov}(r_t, r_{t-2})$.

Although the autoregressive relation can be modeled by AR(1) or AR(2), the normal assumption may prevent the remainders from taking diverse distributions that are more suitable for modeling the skewness and kurtosis. Therefore, after the remainders are generated from AR (1) or AR(2), they will be standardized and transformed to a uniform distribution U_t. The values are then fitted into an inverse distribution function, $F^{-1}(U_t)$, of a Pearson family. The Pearson families comprise eight different distribution groups that cover a wide range of combinations of moments (Pearson 1895, 1901, 1916). A 4-moment tuple, $<\hat{\mu} = 0,\ \hat{\sigma},\ g_1,\ g_2>$, computed from the public data will represent the distribution features (Nanopoulos et al. 2001) and be used to identify Pearson distribution. It will serve as the distribution of the generated remainder that is obtained by $F^{-1}(U_t)$.

14.3.5 Workflow

Based on the modeling framework and data described in Sect. 14.4, the workflow of the paper is summarized in Fig. 14.2. The public data with hourly electricity demand will be firstly decomposed into three components, where the remainders are used for extracting the autoregressive and distributive features. Data from a reference building in the study country, Sweden, is then decomposed only for calculating scale parameters for S_t and R_t. The median values of the ratio of reference data to public data are adopted to avoid the impacts from extreme values and to scale respective components. The recombination results will be validated at the reference building to determine the best scenario for modeling, which is also compared to the prediction from an ARIMA model. A day-and-hour adjustment will further improve the accuracy of recombination. The scheme is finally deployed to large-scale buildings with only yearly total demand available in the same region

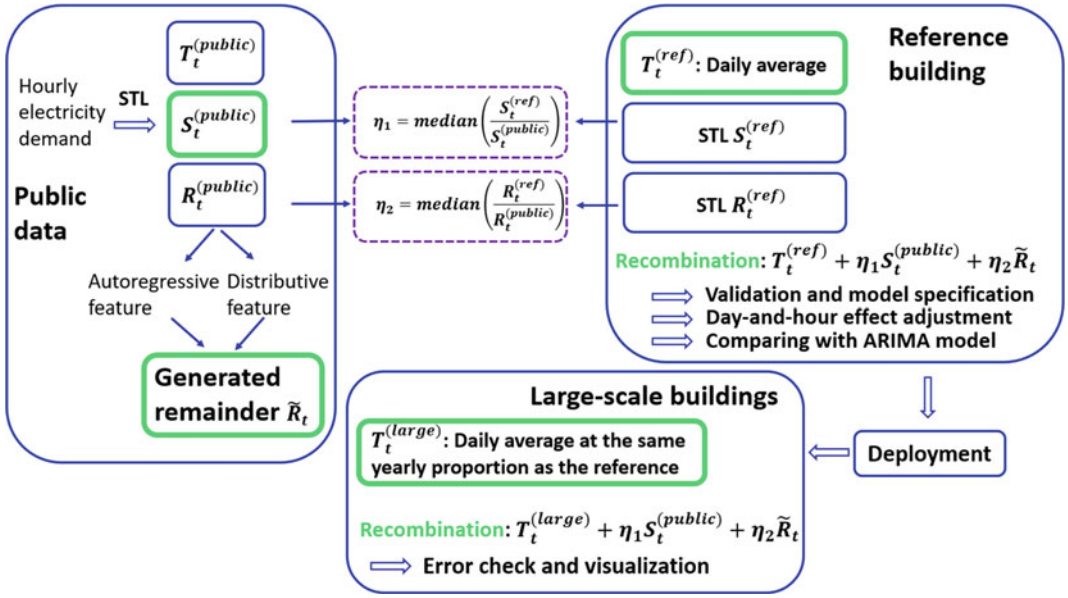

Fig. 14.2 Modeling framework and workflow

as the reference building. A check of irrational synthetics provides an assessment of the proposed method, and visualization illustrates the spatial distribution of household electricity demand.

14.4 Data

14.4.1 Public Data

A lot of efforts have been made for creating public energy measurements at hourly rate for providing test datasets or benchmarking modeling algorithms (Miller et al. 2020; Miller and Meggers 2017). However, there is still a lack of public data across all countries to reflect local electricity demand patterns. Sweden, for example, is a country in the Nordic region with more than a twelve-degree span in latitude and almost a twelve-degree in longitude, which indicates various climate types, urban environments, and building types. Even though climate is not a direct determinant of electricity demand, this chapter adopts a public dataset for capturing seasonality and statistical characteristics of residuals collected in British Columbia in Canada, which has the same latitude as Sweden.

The data was donated from residential customers of a provincial power utility (Makonin 2019). Most of them have three years' observations of hourly electricity demand with the rate of missing values lower than 1%. Twenty-eight buildings are available of nine different house types with detailed information including the direction a house is facing, region of weather station, HVAC system, etc. Both the house type and HVAC system are crucial for linking the data from the Canadian houses to the Swedish ones. One example of the public buildings (numbered as 1 in the list) with the type of bungalow and forced air gas furnace, gas fireplace, and heat pump as the type was selected in 2014 to illustrate the method. As seen in Fig. 14.3 for an aggregated plot, higher daily demands are observed in winter, which is also the case in most Swedish buildings. We can also presume from the figure that the frequency of using air conditioning in summer is very low.

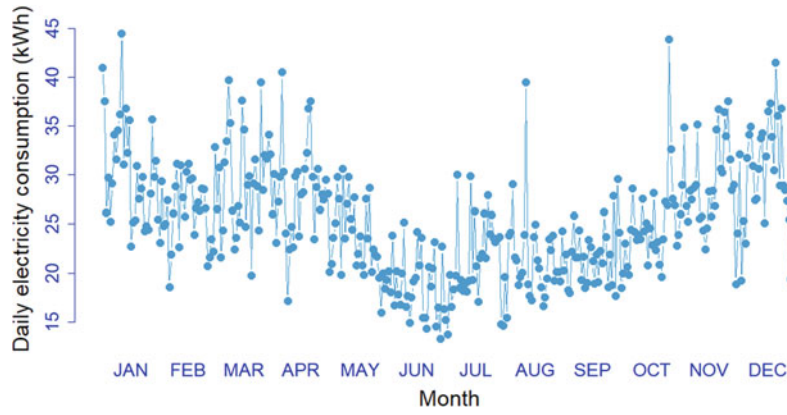

Fig. 14.3 Public data of aggregated electricity demand for Canada in 2014

14.4.2 Data for the Reference Building

Synthetic data are fictitious, making it important validate it against real data. To test the accuracy of the synthetic data obtained from the generative model, in this work, a dataset containing sub-hour measurements for household electricity demand in typical Swedish buildings was adopted as a reference. This dataset, which is initially collected by the Swedish Energy Agency to understand the structures and patterns of electricity demand in the Swedish residential buildings, includes end-use metering data for 400 households in Sweden (Zimmermann 2009). The choice of households in this dataset was made not only based on the type of buildings and their heating systems but also on different socioeconomic factors such as age and size of the household. This means that this dataset is a good representative of the variations in household electricity demand in Sweden. Almost half of the data presented in this dataset belong to single-family buildings with electric or non-electric heating systems. Among them, 22 were monitored for one year and the rest for one month.

Considering the wide distribution of households presented in this dataset, it has been decided to choose a household with a total electricity demand close to the average and with similar properties, i.e., building type and system, as the buildings found in the Canadian public data (Sect. 14.4.1). The chosen data, therefore, belong to a single-family building equipped with an electrical heating system, i.e., mainly a heat pump, and with total electricity demand of 18,223 kWh/year. This is to mention that the HVAC system and its operation in the Swedish reference building are not completely identical to the Canadian one. Yet, with accepting some degrees of uncertainty, it is assumed that during most hours of the year both buildings run on electricity and electrical heating system.[1]

14.5 Results

In this section, we present the decomposition results of the public data and use the reference building to assist the calculation of scale parameters for seasonality and remainder. The combination results are validated to select the best scenario for modeling, which is followed by day-and-hour adjustment.

14.5.1 Remainder Component of the Public Data

The remainders are the most stochastic component compared to the other two since they can reflect both system uncertainty and unpredictable interventions, which may be location or building dependent. Thus, autoregressive models aim to

[1] In the Canadian building, if the temperature falls below 2 °C, the forced air gas furnace instead of the heat pump supplies the heat demand of the building.

capture general dependencies between neighboring units rather than focusing on a specific day and hour. Similarly, the distributional transformation ensures that the overall values of the remainder resemble those taken from the domain of electricity demand in buildings. Another reason to generate a new sequence is that the remainders can be reproduced in batches depending on the time span and the number of individuals.

The decomposed remainders are shown in Figs. 14.4 and 14.5, where the hourly values and histograms, as well as the autocorrelation functions (ACFs), are plotted for 11 months by using and not using a robust outer loop. Both figures present daily and weekly seasonality where the ones without the loop have weak correlations. The ACF in Fig. 14.4 has more negative values due to the larger variance than that in Fig. 14.5. On the other hand, the remainders with robust outer loops tend to have densely concentrated remainders around 0. However, neither of them presents a normally distributed picture, which can also be seen in Table 14.1. In Table 14.1, empirical moments of the remainders are calculated for the two scenarios. For normal distribution, the skewness equals 0 and the kurtosis equals 3. The estimated $\hat{g}_1$ and $\hat{g}_2$ are considered to be much larger than those values, respectively. The empirical moments will be further used to identify Pearson distributions in Sect. 14.5.1.2.

14.5.1.1 Autoregressive Generation

Time dependency captured by AR(1) and AR(2) was treated before the distributional characteristics in the generation process. The autoregressive models can also be estimated from the remainders of the public data using Eqs. (10)–(12) and the estimate of σ_ε^2. Using the estimates of φ_1, ϕ_1, ϕ_2 and σ_ε^2 in Table 14.1, the generated remainders present stationarity for both autoregressive models after 1000 iterations. A fraction between the 1001 and 2000 iteration is shown for the process without a robust outer loop in Fig. 14.6, where neither trend nor heteroscedasticity is observed. The generation process continues until a sufficiently large number of $\tilde{R}_t$s is obtained (100,000 in this work). A similar pattern can also

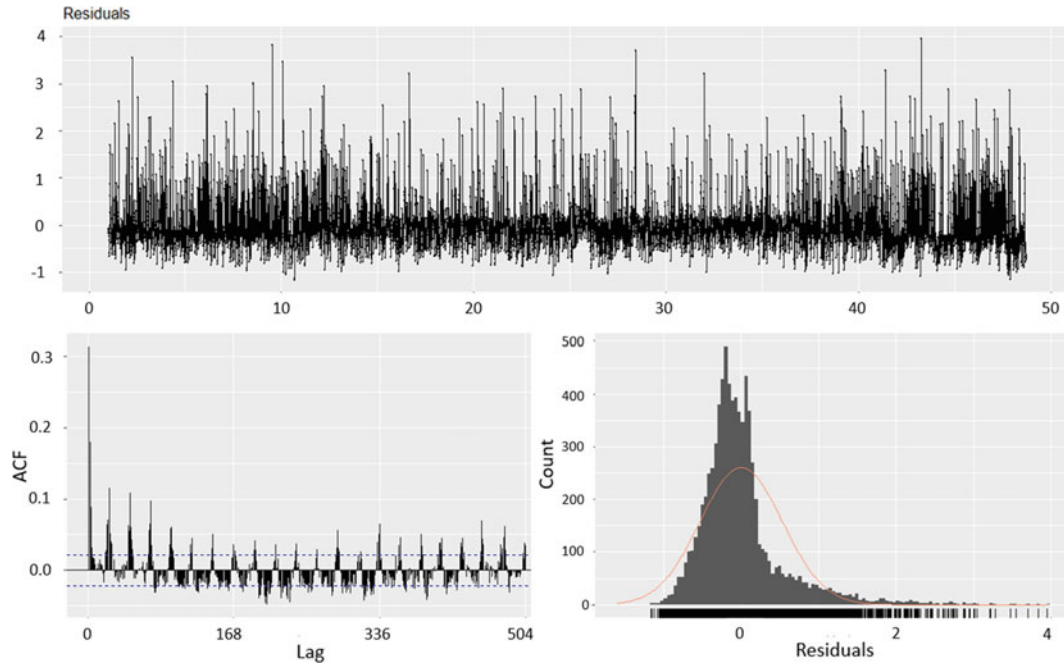

Fig. 14.4 Distribution of the remainder and ACF without robust outer loop

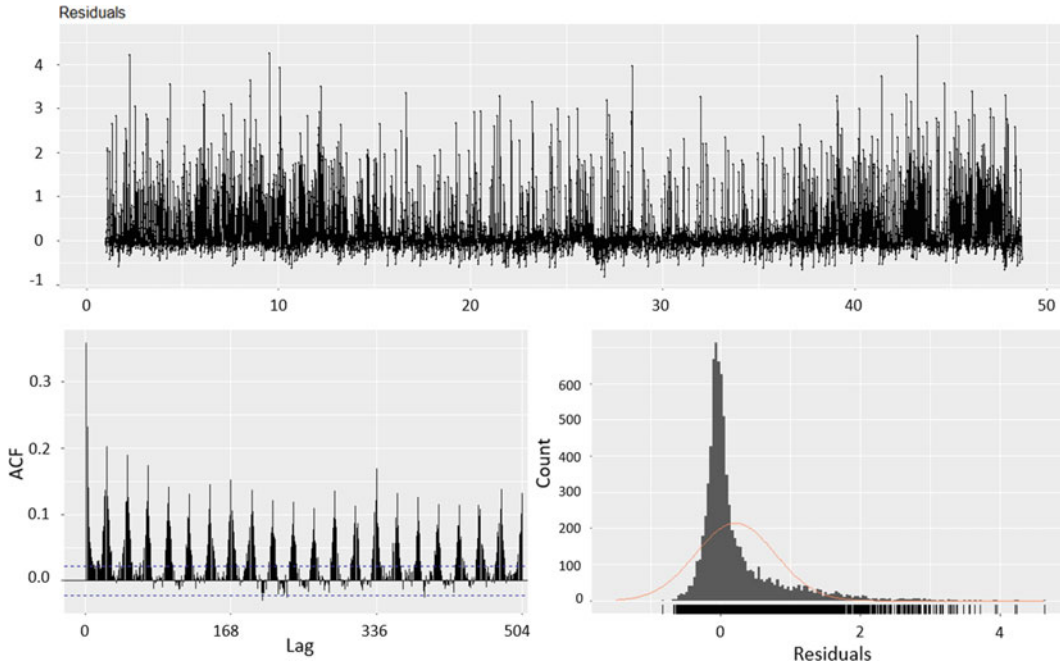

Fig. 14.5 Distribution of the remainder and ACF with robust outer loop

Table 14.1 Empirical moments autoregressive model for remainder

Robustness	Empirical moments				AR(1)		AR(2)		
	$\hat{\mu}$	$\hat{\sigma}$	$\hat{g}_1$	$\hat{g}_2$	$\hat{\varphi}_1$	$\hat{\sigma}_\varepsilon^2$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\sigma}_\varepsilon^2$
No	0.00	0.53	1.95	9.32	0.31	0.26	0.28	0.09	0.25
Yes	0.21	0.57	2.38	10.15	0.36	0.28	0.31	0.12	0.28

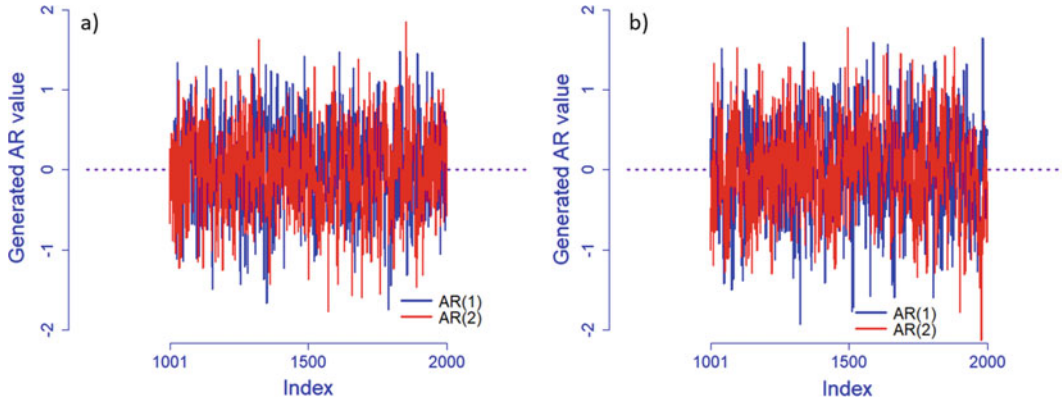

Fig. 14.6 Generated values from AR(1) and AR(2), **a** without robust outer loop; **b** with robust outer loop

be observed for the robust decomposition with outer loop.

14.5.1.2 Transformed Distribution

The generated $\tilde{R}_t$s, which are normally distributed, in Sect. 14.5.1.1 were first standardized. They were then transformed into a uniform distribution by applying $u_t = \Phi(\varepsilon_t)$ for $u_t \in [0, 1]$, where Φ is the cumulative distribution function (CDF) of standard normal distribution. Last, remainders $R_t s$ that belong to the Pearson family were obtained by using the inverse CDF, $F^{-1}(u_t)$, to map u_t onto the quantiles of specific distribution.

The method of moments estimator (Johnson et al. 1994) determines a Pearson distribution with moments matching the empirical remainders. Two types were found from the public dataset based on the empirical moments: type I (also known as aka Beta) and type VI (also known as aka Beta Prime). The Pearson type I distribution

$$f_{\mathrm{I}}(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \left(\frac{x-\lambda}{s}\right)^{\alpha-1} \left(1-\frac{x-\lambda}{s}\right)^{\beta-1} \tag{14.13}$$

ffor $\alpha > 0,\ \beta > 0,\ s \neq 0$ and $0 < \frac{x-\lambda}{s} < 1$ is location-scale transformation of Beta distribution, where s is the scale parameter and λ is the location parameter. The Pearson type VI distribution

$$f_{\mathrm{VI}}(x) = \frac{\Gamma(\alpha+\beta)}{|s|\Gamma(\alpha)\Gamma(\beta)} \left(\frac{x-\lambda}{s}\right)^{\alpha-1} \left(1+\frac{x-\lambda}{s}\right)^{-\alpha-\beta} \tag{14.14}$$

for $\alpha > 0,\ \beta > 0,\ s \neq 0$ and $\frac{x-\lambda}{s} > 0,$ is location-scale transformations of Beta prime distribution. Detailed estimation is shown in Table 14.2.

It can be seen from Fig. 14.7 that, from the distribution perspective, no obvious differences are found between the AR(1) and AR(2) models for both scenarios. While asymmetric and leptokurtic distributions are observed for both scenarios, type I Pearson family seems to produce smaller variance and sharper top. It can also be seen that the generated remainders, $\tilde{R}_t$, mimic those from the real data well as presented in Figs. 14.4 and 14.5. Thus, the identified autocorrelations and probability distributions are adequate to characterize the inherent residuals in the domain of electricity demand in buildings that are similar to the Canadian one.

14.5.2 Seasonal Component of the Public Data

In this work, we set the Loess window for seasonal extraction to be periodic, which implies that the values of each hour appearing in each of the weeks will be the same. In another words, the seasonal component is extracted as homogeneously as possible to be able to reflect the periodical effect for the entire sequence. It can be seen from Fig. 14.8 that both scenarios present a clear daily cycle comprising two peaks and two valleys, except for Sunday. The morning peak for workdays takes place around 8:00 and the evening one is around 18:00. Weekends have relatively low morning peaks and high evening peaks (this also applies for Friday evening). It is not surprising that the first valley appears when people are asleep and the second appears when there are fewer people in the house in the afternoons. We consider that this is a typical pattern for the seasonal component. In addition, the robust outer loop process tends to shrink the effect for all the days in a week.

Table 14.2 Moments estimator of Pearson distribution

Robustness	Type	Parameters
No	VI	$\alpha = 1.46,\ \beta = 23.03,\ \lambda = -0.61,\ s = 9.17$
Yes	I	$\alpha = 0.42,\ \beta = 7.26,\ \lambda = -0.40,\ s = 7.35$

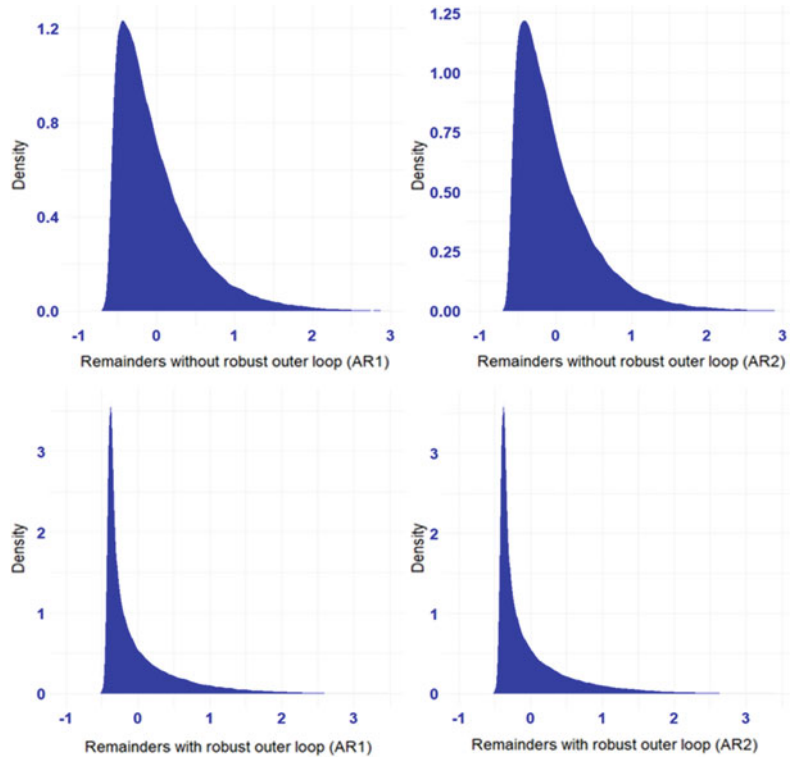

Fig. 14.7 Generated remainders

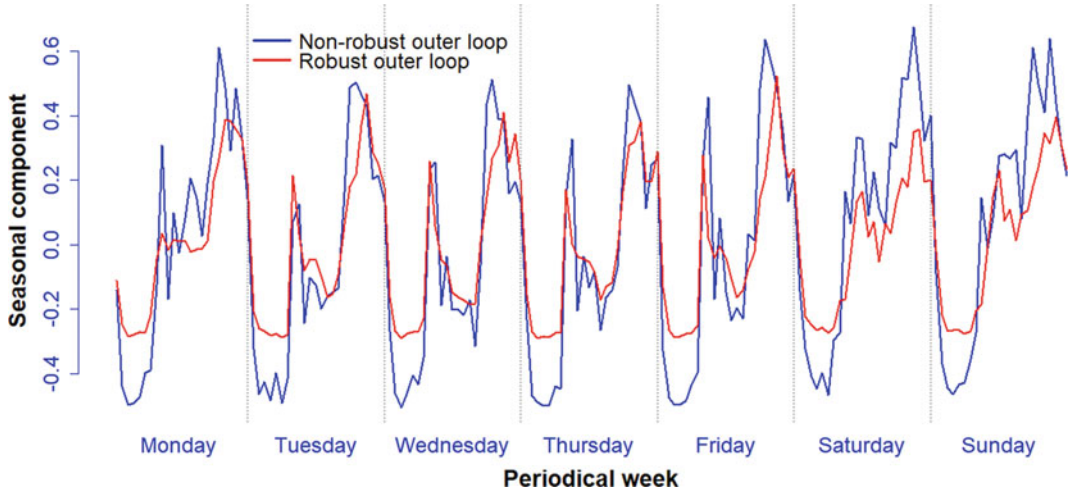

Fig. 14.8 Seasonal component for each day from 00:00 to 24:00 in a periodical week

14.5.3 Trend Component

As stated in Sect. 14.3.4, the trend component is more complex to extract than the other two. On the one hand, there is an unclear correlation between the electricity demand and other factors, which makes the estimation even inaccurate on the hourly scale. On the other hand, it is

dependent on the span of the Loess window or the data points used for extracting trend. While too large span can eliminate local varieties, too small span can lose the generality to capture long-term effect. Considering the contradictive issue, daily averages are adopted to represent the trend component. A linear model will be further fitted to adjust the deviations between the generated data and real data when the method is deployed to buildings with unknown daily or monthly values.

14.5.4 Recombination

We denote $T_t^{(\text{public})}$, $S_t^{(\text{public})}$ and $R_t^{(\text{public})}$ as the components of the public data set and $T_t^{(\text{ref})}$, $S_t^{(\text{ref})}$ and $R_t^{(ref)}$ as components for the reference building, i.e., the Swedish building. Since $T_t^{(\text{ref})}$ is represented by the daily average, two scalars, η_1 and η_2, are introduced such that $S_t^{(\text{ref})} = \eta_1 S_t^{(\text{public})}$ and $R_t^{(\text{ref})} = \eta_2 R_t^{(\text{public})}$, which complies the expectation property: $E\left[S_t^{(\text{ref})}\right] = E\left[S_t^{(\text{public})}\right] = 0$ and $E\left[R_t^{(\text{ref})}\right] = E\left[R_t^{(\text{public})}\right] = 0$. We used the validation data set to extract $S_t^{(\text{ref})}$ and $R_t^{(\text{ref})}$ for specifying both scalars. We also tested different values of $R_t^{(\text{public})}$ to start the first hour of a day in the recombination and the experiments showed that the value at around 50% quantile (Fig. 14.7) gave the best outcome. Monthly spans were considered for estimation to retain the generality, meaning that both scalars were calculated for each month separately. The median values of $\frac{S_t^{(\text{ref})}}{S_t^{(\text{public})}}$ and $\frac{R_t^{(\text{ref})}}{R_t^{(\text{public})}}$ were taken as the estimates to resist extreme values. Doing so, we were able to investigate the performance of the proposed method for both scenarios of outer loops and both AR models. An illustrative case was presented in Fig. 14.9, where daily relative errors were plotted for all eleven months. The errors were computed as the mean hourly difference in absolute values divided by the mean daily electricity demand, where the weekends and workdays were further distinguished.

Among all the months in the year 2007 except January, most of the errors of February are below 20%. The lowest one can achieve 10%, and meanwhile, the highest one is no more than 30%, which is considered to be the best month across the year. On the whole, summer months have higher errors than winter months because the heat pumps used in winter can provide more

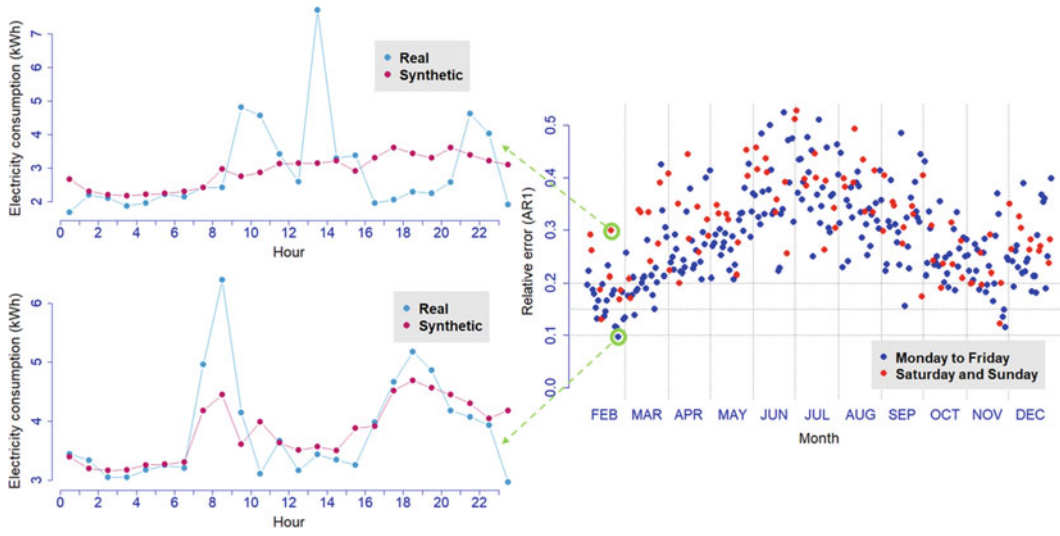

Fig. 14.9 Average daily error (absolute value in percentage) of generated value and two example days without robust outer loop for AR(1)

regular patterns compared with other types of use. Comparing to December and January, on the one hand, February has much fewer holidays in Sweden thus making the electricity demand more consistent. The temperature starts to increase in March where the electricity usage from heat pumps begins decreasing.

A comprehensive monthly comparison of mean errors and standard deviations (s.d.) shows that the proposed method works best for February without the robust outer loop (Table 14.3). However, the difference between two AR models is negligible. Thus, for simplicity, we adopted the non-robust decomposition with AR(1) modeling the remainders to link Canadian data to the Swedish context. The synthetic sequences that best imitated and poorly imitated real demand were singled out in Fig. 14.9. For the best day, the only mismatching occurs around the morning peak, while for the worst one, irregular weekend peaks hinder the recombination from imitating the reality. It can also be seen that the synthetics tend to have smaller variances, which again implies that some extreme peaks or valleys cannot be well reached. Thus, a linear model was fitted to correct the effects of specific hours and days of a week causing the difference.

The linear model considers two factors, the day of a week and the hour of a day, as the independent variables, and the deviations for the real values against synthetic values as the dependent variable. With these settings, the model output can be directly used to further correct individual predictions. A distribution graph (Fig. 14.10) shows that the synthetics underestimate the demands 8:00–10:00 and 16:00–20:00. This has also been observed 11:00–16:00 for some of the weekends. On the other hand, some overestimations are found 23:00–1:00 for most of the workdays. The results of the model fitting are presented in Table 14.4. Although some coefficients are not statistically significant at the 5% level, they are adopted in the adjustment due to that even a minor change in the adjustment can have an effect on multiple days. An example which helps explain Table 14.4 is to consider a Wednesday at 12 o'clock. The adjustment to the synthetics is to multiply a coefficient $1 - 0.038 + 0.167 - 0.076 = 1.053$ on the value. This will be applied to all hours on all days in February. By doing so, the average error has decreased to 0.159 without changing the standard error, and both the day with the best results and the day with the worst results gain lower errors (0.090 and 0.288, respectively).

The synthetic data obtained from the decomposition–recombination method was further compared to ARIMA predictions with seasonal effects. Since ARIMA does not

Table 14.3 Average absolute error per month and standard deviation

Month	Non-robust AR (1)		Non-robust AR (2)		Robust AR(1)		Robust AR(2)	
	Mean	s.d	Mean	s.d	Mean	s.d	Mean	s.d
Feb	**0.178**	**0.050**	**0.178**	**0.050**	**0.183**	**0.047**	**0.183**	**0.046**
Mar	0.247	0.078	0.246	0.078	0.248	0.081	0.248	0.083
Apr	0.281	0.065	0.281	0.064	0.284	0.064	0.284	0.063
May	0.303	0.061	0.303	0.059	0.304	0.063	0.305	0.062
June	0.387	0.088	0.387	0.089	0.395	0.088	0.394	0.087
July	0.386	0.068	0.385	0.070	0.395	0.077	0.394	0.080
Aug	0.351	0.061	0.350	0.062	0.375	0.055	0.376	0.051
Sep	0.315	0.075	0.315	0.075	0.326	0.078	0.326	0.075
Oct	0.261	0.055	0.261	0.054	0.274	0.056	0.274	0.056
Nov	0.223	0.058	0.224	0.058	0.233	0.069	0.233	0.068
Dec	0.269	0.061	0.269	0.062	0.283	0.060	0.285	0.063

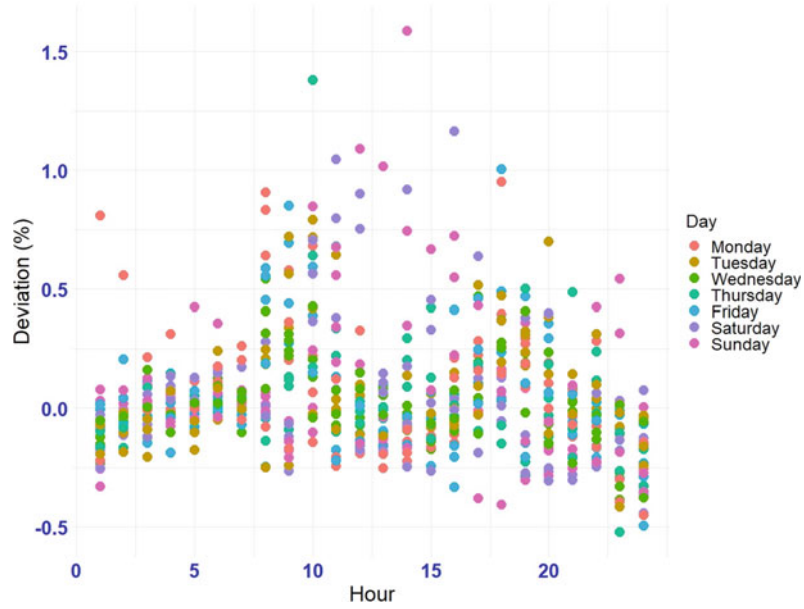

Fig. 14.10 Deviations in percentage for the real values against synthetic values for each hour and day of the week

Table 14.4 Estimates of hour effect with *p*-values less than 0.05

Hour	Estimate	Hour	Estimate	Hour	Estimate	Day	Estimate
2	0.067	10	0.473***	18	0.284***	Tue	0.008
3	0.089	11	0.261***	19	0.176**	Wed	− 0.038
4	0.115	12	0.167**	20	0.118	Thu	− 0.028
5	0.094	13	0.066	21	0.034	Fri	− 0.012
6	0.162*	14	0.175**	22	0.048	Sat	− 0.003
7	0.119	15	0.090	23	− 0.059	Sun	0.018
8	0.287***	16	0.171**	24	− 0.154*	Intercept	− 0.076
9	0.252***	17	0.222***				

Significant level: *0.05; **0.01; ***0.001

independently model the seasonality, we directly compared the errors of month average, the best day, and the worst day for February. To make the comparison fair, ARIMA used real data collected from last 11 days in January for estimating parameters. The predictions were also adjusted by using the difference of daily means between the real and predicted values, where both additive (ADD) and multiplicative (MUL) adjustments were considered. The comparisons of the errors are given in Table 14.5, where the proposed method outperforms the compared ones and the adjustment methods perform equally. Although the adjusted predictions can provide smaller errors against ARIMA, they are still inferior to the synthetic data generated by our proposed method.

1. A case of generating hourly data for buildings at the district level

Table 14.5 Comparisons of the decomposition–recombination method and ARIMA

	Methods			
	Decomposition–Recombination	ARIMA	ARIMA (ADD adjustment)	ARIMA (MUL adjustment)
Month average	0.159	0.262	0.228	0.228
The best day	0.090	0.142	0.123	0.123
The worst day	0.288	0.460	0.366	0.366

The proposed method was generalized and deployed to 72 residential buildings in a district in the municipality Uppsala, Sweden. These buildings are single-family buildings that share the most similarities with the given single-buildings, character, bungalow and special, in the Canadian dataset. These single-family buildings were constructed before the year 1989 and have a combination of heating systems including district heating, gas, oil, or wood boiler, heat pump and direct electric heating systems. If it is assumed that the non-electric heating systems such as district heating and boiler and similarly, forced air gas furnace, have no considerable effect on the electricity demand of a building, then, it would be possible to classify the Swedish buildings based on the Canadian building archetypes. An illustration of the district is presented in Fig. 14.11.

All the buildings only have aggregated electricity demand for a year. We assume that these buildings have similar daily demand patterns as the reference building and calculate the trend component at the same proportion across a year obtained from the reference building. Given the previous results, we generated data for February. In addition, both η_1 and η_2. were re-scaled by multiplying a coefficient of the ratio of monthly total. Since we are not interested in any specific date and both hour of a day and day of a week have an impact, we constructed the synthetics for a typical week from Monday to Sunday. Thus, 168 data points were generated for each building. Another issue is to check the number of negative values generated for mimicking the values that are close to 0. It turned out to be 0.58%, that is trivial, and all the negative ones were set to be 0. Figure 14.12 presents illustrative maps for the

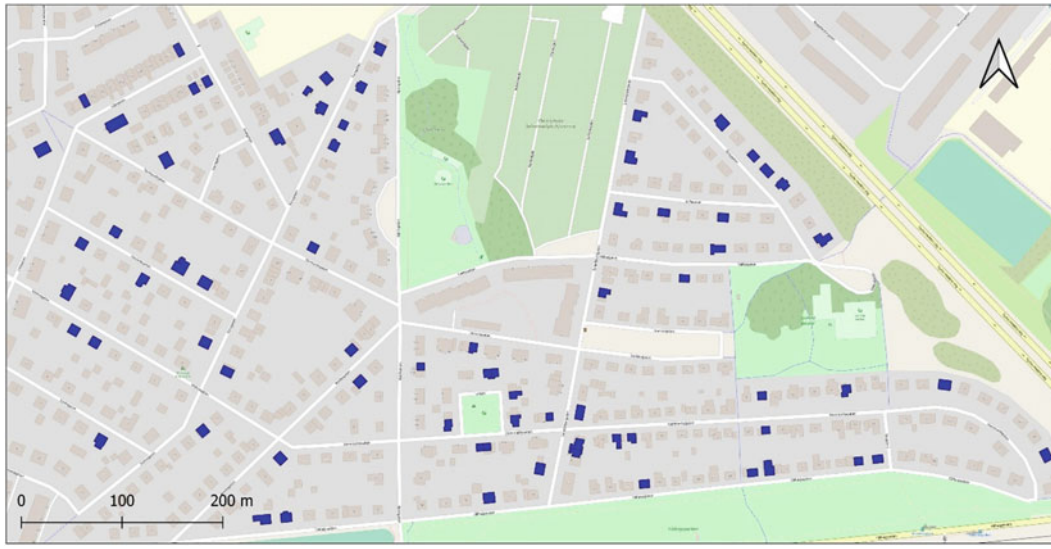

Fig. 14.11 Overview of the illustrative district

Fig. 14.12 Spatial distribution of household electricity demand for a typical Wednesday: **a** 18:00–19:00, **b** 23:00–24:00

spatial distribution of household electricity demand over the district for a Wednesday evening between 18:00 and 19:00 and a Wednesday night between 23:00 and 24:00. Clearly, during the evening hours when the electricity demand is highly dependent on the occupants' presence at home, the electricity demand over the district is higher. However, during night, the intensity of household electricity demand is reduced.

14.6 Summary

Statistical methods are usually used as an alternative to generate data that is either too costly to collect or impossible to obtain. Hourly electricity demand in large-scale buildings is one such domain where privacy concerns can even impose restrictions to data acquisition and use.

Compared to machine learning methods, the statistical methods introduce parameters that have actual meanings to express the modeling outcomes.

This chapter thus proposed to use a seasonal and trend decomposition of time series using a locally weighted regression method as well as a recombination process to synthesize hourly electricity demand data for large-scale buildings in Sweden, in which only the annual demands and HVAC systems are available. A public dataset was adopted to extract key components in the domain of building electricity demand. A local reference building was employed to provide model validation and empirical parameter settings to associate the public data with target ones. It has shown that the method works best for the month February and the average error can be controlled under 16%. The smallest standard deviation was also found for February, which implies the highest reliability. The method was further deployed to a district with 72 buildings in the fourth largest municipality, Uppsala, of Sweden to show how a typical week looks like. The overall modeling indicated a negligible rate of irrational data points that were generated by the method.

There are still many remaining challenges in energy data generation for a large area of buildings, which forms future research topics. First, methods for handling buildings without electrical heating systems and their specific operational schedule need to be developed since the method proposed in this work relies on regular electricity used from the electrical heating system, in particular, heat pumps. This is usually the case in summer. However, it is important to be aware of the use of air conditioners in developing the method. Second, the decomposition method is dependent on the choice of different values of parameters, e.g., the Loess window of the low-pass filter and trend component. A uniform and systematic method of determining such values still lacks. Last, more studies using statistical method should focus on relevant domains, such as energy demand for heating, building CO_2 emission, and occupancy level.

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15 Design Optimization of Distributed Energy Storage Systems by Considering Photovoltaic Power Sharing

Pei Huang and Xingxing Zhang

Abstract

Proper energy storage system design is important for performance improvements in solar power shared building communities. Existing studies have developed various design methods for sizing the distributed batteries and shared batteries. For sizing the distributed batteries, most of the design methods are based on single building energy mismatch, but they neglect the potentials of energy sharing in reducing battery capacity, thereby easily causing battery oversizing problem. For sizing the shared batteries, the existing design methods are based on a community aggregated energy mismatch, which may avoid battery oversizing but cause another severe problem, i.e. excessive electricity losses in the sharing process caused by the long-distance power transmissions. Therefore, this chapter proposes a hierarchical design method of distributed batteries in solar power shared building communities, with the purpose of reducing the battery capacity and minimizing the energy loss in the sharing process. The developed design method first considers all the distributed batteries as a virtual 'shared' battery and searches its optimal capacity using genetic algorithm. Taking the optimized capacity as a constraint, the developed method then optimizes the capacities of the distributed batteries for minimizing the energy loss using nonlinear programming. Case studies on a building community show that compared with an existing design method, the proposed design can significantly reduce the battery capacity and electricity loss in the sharing process, i.e. 36.6% capacity reduction and 55% electricity loss reduction. This chapter integrates the considerations of aggregated energy needs, local PV power sharing, advanced community control, and battery storage sharing, which will be useful to optimize three functions (energy efficiency, energy production and flexibility) in a positive energy district towards energy surplus and climate neutrality.

Keywords

PV · Distributed energy storage · Design optimization · Energy sharing · Building community

Nomenclature

CAP	Capacity
$Cost^{c,0}$	Building cluster annual electricity costs after installing battery (€)

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$\text{Cost}^{c,1}$	Building cluster annual electricity costs before installing battery (€)
E_d	Hourly energy demand (kWh)
E_s	Hourly energy supply (kWh)
E_m	Hourly energy mismatch (kWh)
E_m^1	Hourly energy mismatch after surplus sharing (kWh)
E_m^2	Hourly energy mismatch after surplus sharing and own battery regulating (kWh)
E_{bat}	Hourly energy taken from or stored in battery (kWh)
$E_{bat,own}$	Hourly energy taken from or stored in o. building' battery (kWh)
$E_{bat,other}$	Hourly energy taken from or stored in other buildings' batteries (kWh)
E_{grid}	Hourly energy exchange with the grid (kWh)
$E_{grid,-}$	Hourly energy exported to the grid (kWh)
$E_{grid,+}$	Hourly energy imported from the grid (kWh)
I_{AM}	Combined incidence angle modifier for the PV cover material
I_T	total amount of solar radiation incident on the PV collect surface (W/m^2)
n	Number of buildings

Greek Symbols

ρ	Unit cost of battery (€/(kWh))
ϕ	Amount of electricity stored in the battery (kWh)
η	Efficiency
τ	Transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle
χ_{buy}	Price of purchasing electricity from the grid (€/(kWh))
χ_{sell}	Price of selling electricity to the grid (€/(kWh))

Subscript

i	The ith hour
j	The jth building
Charge	Battery charging power
Discharge	Battery discharging power
rem	Battery remaining storage
Store	Battery power storage
trans	Power transmission among different buildings
PV	PV panels
th	threshold
limit	Maximum or minimum limit of battery charging

Superscript

c	Building community
*	Optimal scenario

Abbreviation

CES	Community energy storage (i.e. Centralized system design)
GA	Genetic algorithm
HES	Household energy storage (i.e. Distributed system design)
NLP	Non-linearly programming
PB	Payback period (Year)
PED	Positive energy district
SC	Self-consumption

15.1 Introduction

Buildings are large energy end-users worldwide (Zhang et al. 2020). In both E.U. and U.S., above 40% of total primary energy is consumed in the building sector (Cao et al. 2016). To mitigate the large carbon emissions in the building sector, increasing solar photovoltaic (PV) are installed in buildings due to its easy scalability, installation and relatively low maintenance. The Swedish Energy Agency set a target that building-installed PV systems contribute 5–10% of the total electricity generation by 2040 (Agency 2017). Worldwide, the capacity of PV systems installed in the residential sector is predicted to rise to 1.8 TW by 2040 from 104 GW in 2014 (G, 2015). The installation of PV systems transforms the building's role in the urban energy systems from electricity 'consumers' to electricity 'prosumers', i.e. electricity consumers which also produce on-

site electricity from renewable energy sources (Fichera et al. 2020). A detailed comparison between the electricity consumers and prosumers is given in Parag and Sovacool (2016). When building prosumers produce more energy than their energy demand, they are called 'positive energy buildings'. And when multiple building prosumers are grouped in a building community, they can form a positive energy district (PED), which is defined by IEA as energy-efficient and energy-flexible building areas with surplus renewable energy production and net zero greenhouse gas emissions (Agency). Solar power shared building community is the type of positive energy district in which buildings can share their surplus PV power with other buildings (Coughlin et al. 2011). Such energy sharing can help increase the community aggregated-level PV power self-consumption (i.e. the amount of PV power used on-site instead of exporting to the power grid) and thus reduce the grid power usage. A study conducted in Luthander et al. (2016) shows that a basic energy sharing among 21 residential buildings in Sweden, i.e. aggregate the electricity demand and supply of all the buildings, can easily improve the PV power self-consumption by over 15%.

Many studies have been conducted to facilitate the energy sharing techniques in solar PV power shared building communities from perspectives of *microgrid technology* (Ferroamp 2018; Strunz et al. 2014; Huang et al. 2019), *electricity trading business models* (Parag and Sovacool 2016; Zhang et al. 2018b), and *community designs* (Jafari-Marandi et al. 2016) etc. Regarding the ***microgrid technology***, some studies have recommended using DC (direct current) microgrid for PV power sharing considering the large converting losses between AC (alternating current) power and DC power, since the PV panels, battery storage, and many modern loads (e.g. pumps, compressors and servers) are often operating with DC power (Huang et al. 2019). In the EU2020 Energy Matching project, the Ferroamp developed an EnergyHub DC microgrid system for power sharing within a building community and bidirectional power flow between the DC grid and utility AC grid (Ferroamp 2018). The operating voltage of the DC microgrid is 760 V, and the communication of EnergyHub is based on TCP/IP protocol. Another example is the DC microgrid developed by Chen et al. (2013) with an operating voltage of 380 V and communication with the EMS based on RS-485 or ZigBee protocol. Regarding the ***business models***, Parag and Sovacool (2016) identified and proposed three promising potential prosumer markets related to *prosumer grid integration* (i.e. prosumers provide services to a microgrid), *peer-to-peer models* (i.e. prosumers interconnect directly with each other, buying and selling energy services) and *prosumer community groups* (i.e. a group of prosumers pools resources or forms a virtual power plant). Similarly, Zhang et al. (2018b) developed a four-layer hierarchical system architecture model (i.e. power grid layer, ICT layer, control layer and business layer) to identify and categorize the key elements and technologies involved in P2P energy trading. In the developed model, they simulated the bidding process in business layer using game theory. Regarding the ***community design***, Jafari-Marandi et al. (2016) proposed a homogeneity index to quantify the diversity of load/supply profiles and evaluated the energy sharing potentials under different homogeneity index with different levels of diversity. Notably, Huang et al. (Huang and Sun 2019) developed a clustering-based grouping method to design building communities from a large number of buildings, which can help maximize the energy sharing potentials in each sub-community. Both these two studies conclude that a large diversity in the load/supply profiles inside a community would create more potentials in energy sharing. These energy-sharing-related studies can effectively improve the local balance between electricity load and supply and thus the PV power self-utilization at the community-level. However, due to the large intermittent characteristics of PV power and limited energy sharing potentials, in most cases, energy sharing alone cannot completely balance the electricity load and supply in buildings. Energy storage systems, which

conducts direct regulation on the electricity demand profile, are another effective tool for balancing the local electricity load and supply.

Existing studies have developed many design methods for the distributed energy storage systems (named 'individual design' in this chapter). For instance, Baniasadi et al. (2020) developed a particle swarm optimization (PSO) algorithm-based design method to size the electrical energy storage and thermal energy storage system in a building with the purpose of reducing life-cycle cost of the PV-battery system. Considering the demand prediction uncertainty, battery degradation and maintenance, in (Huang et al. 2018) a genetic algorithm-based design optimization method was developed, which uses the energy system life-cycle costs as the fitness function and the users' performance requirements as the constraints. Considering the uncertainty, system degradation and climate change, Lu et al. (2020) developed a robust design optimization method for selection of energy systems in zero energy buildings. They evaluated three scenarios: (1) Deterministic design; (2) Markov chain-based robust design without considering the aging effect; (3) Markov chain-based robust design considering the aging effect. Similarly, Bozorgavari et al. (2020) developed a robust planning method of the distributed battery energy storage system from the viewpoint of distribution system operation with the goal of enhancing the power grid flexibility. They consider a set of factors including the degradation and operation costs of energy storages systems, the revenues, the technical limits of the network indexes and uncertain renewable energy sources. They model the planning problem using non-linear programming method. The abovementioned design methods size each individual battery separately based on single building's energy mismatch, while the possible energy sharing among buildings in reducing the required capacity of energy storage is neglected. Considering the possible energy sharing among different buildings, Sameti and Haghighat (2018) developed a mixed-integer linear programming (MILP) optimization-based method to design the distributed energy storages of a net-zero energy district in Switzerland. Their developed method takes into account factors including the location of host buildings, type of energy storage technologies and associated size, the energy distribution network layout and the optimal operating strategy. Pareto analysis was used to identify the best integrated district energy system which minimizes both the total annualized cost and equivalent CO_2 emission while ensuring the reliable system operation to cover the demand. This chapter considers the surplus sharing (i.e. use one building's surplus power to meet other buildings' electricity demand), but the storage sharing (i.e. store one building's surplus power in other buildings' batteries) is not considered.

Some studies have investigated the community shared energy storage system design (named 'group design' in this chapter) and its performances. For instance, Parra et al. (2016) designed a method to calculate the optimal community energy storage (CES) systems for end-user applications based on the levelized cost, which considers round-trip efficiency and durability. Their simulation-based case studies showed that the application of a community energy storage to 100 houses could reduce the levelized cost by 56% by shifting demand compared to a single house energy storage installation. Based on the results, they concluded that the application of a community shared energy storage could result in a good solution to facilitate the usage of distributed renewable energy generation and manage the loads. Sardi et al. (2017) developed a framework for designing CES in an existing residential community system with rooftop solar PV units. Their proposed method first determines the optimal CES location, which minimizes the annual energy loss based on Centre of Gravity (COG) theory. Then, it optimizes the CES capacity that meets a user-defined annual load factor. Last, it optimizes the operational characteristics of CES to flatten the daily demand profile and improve the voltage profile. Their study results indicated that 22% of community energy storage could reduce the annual purchased energy cost from the grid by 11.1% and the annual energy loss cost by 36.9%. Dong et al. (2020b) developed an agent-based model

for simulating the operation of household energy storage (HES) systems and CES both for PV installed residential building community. Using the developed model and operation strategy, they analysed the performances of different types of systems from technical, economical and environment aspects. Their study results indicated that both HES and CES could significantly reduce the peak grid power imports and exports, improve the community self-consumption rate and self-sufficiency rate, and contribute to high energy saving. However, this chapter only considers the surplus sharing but neglects the storage sharing in the HES. As a result, the performance of the HES is not as good as the CES under the same aggregated capacity. In another study, Dong et al. (2020a) investigated the impacts of the heterogeneous building electricity demand on the battery performances (i.e. self-consumption rate, self-sufficiency rate and battery degradation) using the same agent-based modelling as (Dong et al. 2020b). They concluded that the change in community demand was insignificant to the overall self-consumption rate and self-sufficiency rate of the community and was unlikely to cause significant CES capacity degradation.

In the CES, there are actually two forms of energy sharing: surplus sharing (i.e. use the surplus PV power to meet the electricity needs in other buildings) and storage sharing (i.e. store or take electricity from other buildings' batteries) (Huang et al. 2020a). The buildings first share their surplus PV power with other buildings with insufficient PV power production (i.e. surplus sharing). Then, the remaining surplus PV power will be stored in the shared CES (i.e. storage sharing) if the aggregated surplus power is larger than the aggregated deficiency, or the remaining power shortage will be taken from the shared CES if the aggregated deficiency is larger than the aggregated surplus power. Contributed by such energy sharing, the CES typically performs better than the conventional HES, which does not enable energy sharing or only enable very limited energy sharing (e.g. only surplus sharing enabled in Dong et al. (2020b), Huang et al. (2020a)) In recent years, with the development of advanced energy storage controls for energy sharing, such as the *simultaneous approaches* (which optimize all the energy storage's operation simultaneously with the goal of achieving aggregated-level optimum) (Gao and Sun, 2016; Zhang et al. 2018a), *bottom-up approaches* (which optimize single energy storage's operation one by one in sequence until all the energy storages are optimized to achieve the aggregated-level optimum) (Fan et al. 2018; Prasad and Dusparic 2019), and *top-down approaches* (which first identify the community aggregated-level optimum and then coordinate single energy storage's operation to achieve the aggregated-level optimum) (Huang et al. 2020b), the HES can achieve nearly the same level of energy sharing and thus the similar performances as the CES system. In fact, due to the frequent low-voltage energy exchanges with the CES system which can be located in a long distance from the buildings, there can be significant amount of electricity losses due to such long-distance power charging/discharging. The HES, on the other hand, can store most of the electricity near the buildings and thus reduce the energy losses due to long-distance power transmission.

To sum up, the existing individual design methods (e.g. Baniasadi et al. (2020, Huang et al. (2018)) size the distributed batteries according to single building's energy mismatch, but the potentials of energy sharing in reducing battery capacity is mostly neglected, thus easily leading to oversized systems with high initial investment and high battery storage losses. While the existing group design methods (e.g. Parra et al. (2016), Sardi et al. (2017), Dong et al. (2020b)) size the centralized battery according to the building community's aggregated energy mismatch, which can effectively reduce the required capacity by energy sharing. However, there are large electricity losses due to the long-distance low-voltage power transmission. Moreover, these design approaches of CES are not applicable to the scenarios that buildings need to install their own batteries or there is no space for installing a CES with large size. Therefore, this chapter proposes a hierarchical design method for the distributed batteries in solar PV power shared building community, with the purpose of

reducing the required battery capacity by applying energy sharing and minimizing the electricity loss in the energy sharing process. The developed design method first considers all the distributed batteries as a virtual 'shared' battery and searches the optimal capacity of the virtual 'shared' battery using genetic algorithm. Such optimized capacity of the virtual 'shared' battery is considered as the minimized capacity required by the whole building community to achieve the required community-level performance (e.g. meet a required self-consumption rate). Based on the optimized aggregated capacity at community-level, the developed method then optimizes the capacity of the distributed batteries installed in each building using non-linear programming with the objective of minimizing the storage sharing (and thus the associated power loss due to long-distance power transmission). For validation purpose, the developed design method is compared with two existing design methods (i.e. individual design for distributed batteries and group design for centralized battery) based on a virtual building community located in Sweden. The major contributions of this chapter to the subject are summarized as follows.

- A hierarchical design optimization method is developed to improve the cost-effectiveness of distributed battery system in solar PV power shared building community.
- The developed design can make use of the surplus sharing and storage sharing in the building community to keep more PV power inside the building community, thus eliminating the requirements of large-sized battery and power exports to the grid.
- The performances of the developed design are compared with a conventional individual design for distributed batteries (i.e. the battery is sized based on single building's power mismatch, and energy sharing is conducted after battery regulation) and a group design for centralized battery (i.e. the battery is sized based on the aggregated buildings' power mismatch, and energy sharing is conducted before and in the battery regulation) in aspects of economy and energy.
- The detailed energy sharing processes, including surplus sharing and storage sharing, are analyzed in a Swedish community, which can be easily replicated in other contexts.

This chapter is organized as follows: Sect. 15.2 describes the overall hierarchical design of distributed battery system for the solar PV power shared building community. Section 15.2.3 presents the detailed building model and energy system models. In Sect. 15.3, the developed design is applied on a case building community located in Sweden, and its performance is compared with two existing designs/scenarios. The summary is provided in Sect. 15.4.

15.2 Methodology

This section first introduces the energy sharing and battery sharing scenarios. The proposed hierarchical design of the distributed batteries in PV power shared building communities is then introduced. After that, the building and system modelling for validating the proposed design method are presented.

15.2.1 Basic Idea of Energy Sharing and Typical Design Scenarios

Existing studies have proven that energy sharing is effective in enhancing the renewable energy self-consumption rates (i.e. the percentage of renewable energy that is used onsite) at the building community level. Such enhanced self-consumption can further contribute to the reduced economic costs, improved energy efficiency, and reduced grid interactions (Huang et al. 2020a). The energy sharing can be implemented by installing an energy sharing microgrid to connect the buildings within the solar PV

power shared building community (Huang et al. 2019). In this chapter, the energy sharing within a solar powered building community is further classified into two types: surplus sharing (i.e. use the surplus PV power to meet the electricity needs in other buildings) and storage sharing (i.e. store or take electricity from other buildings' batteries). Table 15.1 summarizes the three scenarios of battery sizing and the energy sharing configurations, and Fig. 15.1 presents the operation strategies.

Scenario 1 (***Individual sizing*** for distributed batteries): Each building has its own individual battery. The surplus PV power production (as compared with electricity demand) of the building will first be stored in its own battery. After the battery is fully charged, the remaining surplus power will be sent to the building community to meet the electricity demands of other buildings with insufficient power production (i.e. surplus sharing). If there is still surplus power remaining after meeting the building community's power needs, the remaining surplus power will be sold to the power grid. On the other hand, when there is PV power deficiency, the buildings will take electricity following the sequence from *(i)* their own batteries, *(ii)* building community and *(iii)* the power grid. In this scenario, only surplus sharing is enabled. The buildings cannot exchange electricity with other buildings' batteries.

Scenario 2 (***Group sizing*** for centralized battery): The buildings in the community have one centralized battery. For such centralized battery scenario, the surplus power from one building will first be used to meet the power needs from buildings with insufficient PV power production (i.e. surplus sharing). After compensating the large power supply and large power needs inside the building community, the remaining surplus/insufficient power will be stored-in/taken-from the centralized battery (i.e. storage sharing). If there is still surplus/insufficient power remaining, such remaining surplus power/insufficient power will be exported-to/imported-from the power grid.

Scenario 3 (***Proposed hierarchical sizing*** for distributed batteries): Inspired by the centralized battery design and energy sharing operation logic, this chapter proposes the following operation scenario for distributed batteries design with both surplus sharing and storage sharing enabled. The surplus power from one building will first be used to meet the power needs from buildings with insufficient PV power production (i.e. surplus sharing). Then, the remaining surplus/insufficient electricity will be stored-in/taken-from its own battery. Next, after its own battery being fully charged/discharged, the remaining surplus/insufficient electricity will be stored-in/taken-from other buildings' batteries in the community (i.e. storage sharing). If there is any remaining surplus/insufficient power, such remaining surplus/insufficient power will be exported-to/imported-from the power grid.

In Scenario 1, the capacity of the distributed battery is determined by the power mismatch (i.e. deviation between power demand and power supply) of each individual building, i.e. individual sizing. Since the battery is usually installed inside the building, the energy loss due to power transmission in battery charging/discharging is low. In Scenario 2, the capacity of the centralized battery is determined

Table 15.1 Three scenarios with different sizing and energy sharing configurations

Scenarios	Explanation	System type	Priority of power exchanges	Sharing type
Scenario 1	Individual sizing	Distributed battery system	Building → Own Battery → Community → Grid	Surplus sharing
Scenario 2	Group sizing	Centralized battery system	Building → Community → Centralized battery → Grid	Surplus sharing & storage sharing
Scenario 3	Hierarchical sizing	Distributed battery system	Building → Community → Own battery → Centralized battery → Grid	Surplus sharing & storage sharing

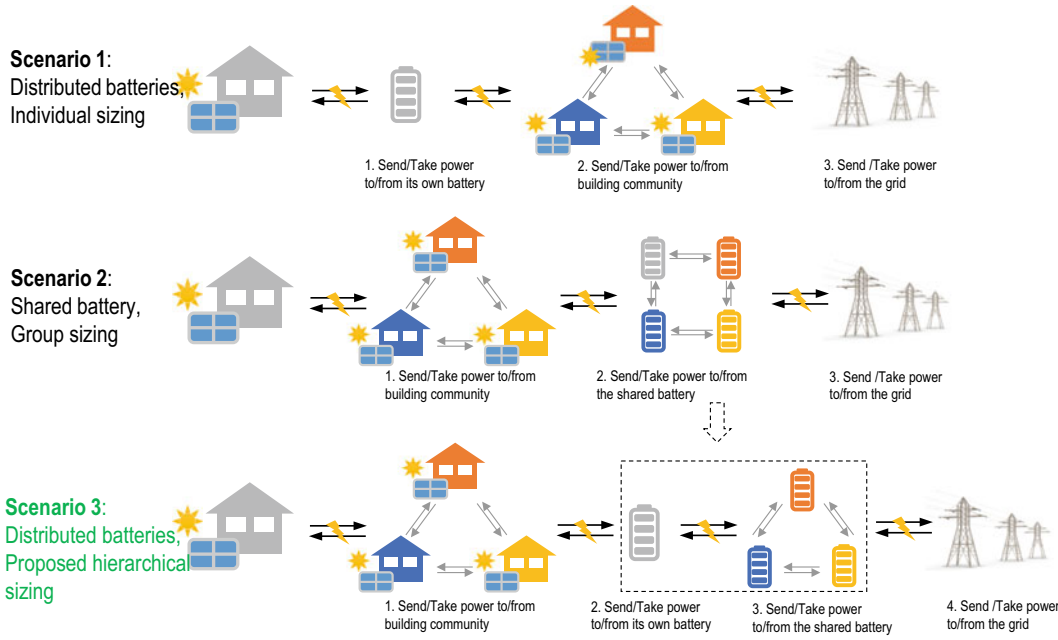

Fig. 15.1 Different operation strategies and design approaches of battery system in solar power shared building community

by the aggregated power mismatch of the whole building community. Since the positive-power-mismatch buildings can compensate with the negative-power-mismatch buildings (i.e. surplus sharing), the required capacity of the centralized battery can be significantly reduced at the aggregated level. However, on the other hand, due to the increased distance between buildings and centralized battery, and the frequent low-voltage power exchanges with the centralized battery large energy loss exists. The proposed hierarchical sizing (Scenario 3) combines the merits of both the individual sizing and group sizing with reduced battery capacity and reduced transmission loss.

15.2.2 A Hierarchical Design of Distributed Batteries for a Solar Power Shared Building Community

This sub-section presents the developed hierarchical design method for distributed batteries in solar power shared building community. The proposed design will combine the merits of both individual design (i.e. low energy loss due to power transmission from/to battery) and group design (i.e. reduced battery capacity due to energy sharing). The hierarchical design consists of four steps, as depicted by Fig. 15.2. In Step 1, the PV power production and electricity demand of each individual building and is evaluated and then aggregated to obtain the power supply/demand of the whole building community. In Step 2, using the aggregated-level power supply/demand as inputs, the capacity of a virtual 'shared' battery is optimized using genetic algorithm (*GA*) according to the user-required energy performance (e.g. to reach a specific self-consumption). The optimized capacity of the virtual 'shared' battery is considered as the minimized capacity required by the whole building community to achieve the required energy performance. In Step 3, the capacity of the distributed batteries installed in each building is optimized using non-linear programming (*NLP*) to minimize the storage sharing (i.e. power exchanges with other batteries) and thus the associated energy loss. The aggregated capacity

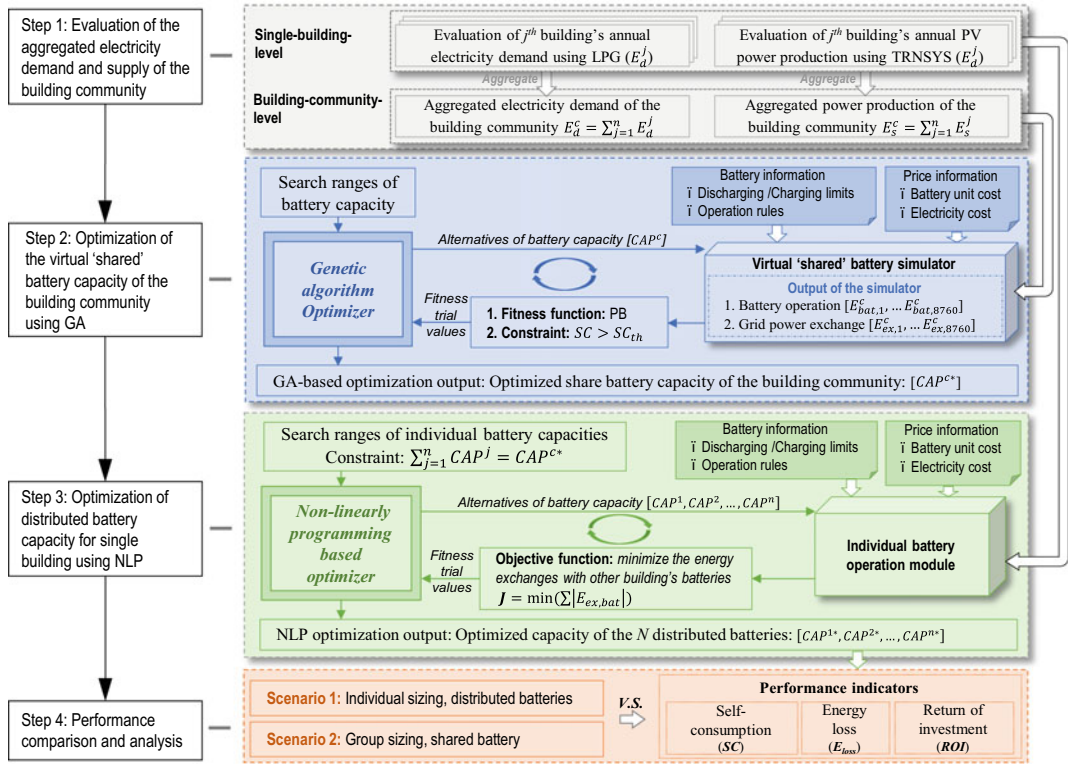

Fig. 15.2 Basic idea of the hierarchical design of distributed batteries for solar power shared building communities

of all the distributed batteries should equal the capacity of the virtual 'shared' battery obtained in Step 2. In Step 4, the performances of the proposed hierarchical sizing are compared with the two common designs, namely individual sizing and group sizing (i.e. Scenarios 1 and 2 in Table 15.1). The details of each step are introduced below.

15.2.2.1 Step 1: Evaluation of the Aggregated Electricity Demand and Supply of the Building Community

In this step, the aggregated electricity demand and supply of the building community are evaluated. The hourly electricity demand ($[E^c_{\mathrm{d},1}, E^c_{\mathrm{d},2}, \ldots, E^c_{\mathrm{d},8760}]$ (kWh)) of the building community equals the aggregated hourly electricity demand of each single building ($[E^j_{\mathrm{d},1}, E^j_{\mathrm{d},2}, \ldots, E^j_{\mathrm{d},8760}]$ (kWh)) (j indicates the jth building), and its hourly PV power production ($[E^c_{\mathrm{s},1}, E^c_{\mathrm{s},2}, \ldots, E^c_{\mathrm{s},8760}]$ (kWh)) equals the aggregated hourly PV power production of each single building ($[E^j_{\mathrm{s},1}, E^j_{\mathrm{s},2}, \ldots, E^j_{\mathrm{s},8760}]$ (kWh)), as depicted by Eqs. (15.1) and (15.2). The electricity demand and PV power generation of each individual building are calculated using the models presented in Sect. 15.2.3.

$$E^c_{\mathrm{d},i} = \sum_{j=1}^{n} E^j_{\mathrm{d},i} \quad (i = 1, 2\ldots, 8790\,\mathrm{hr}) \quad (15.1)$$

$$E^c_{\mathrm{s},i} = \sum_{j=1}^{n} E^j_{\mathrm{s},i} \quad (i = 1, 2\ldots, 8760\,\mathrm{hr}) \quad (15.2)$$

Based on the aggregated power generation and power demand, the hourly power mismatch ($E^c_{\mathrm{m},i}$ (kWh)) at the building-community-level is calculated using Eq. (15.3).

$$E^{c}_{\mathrm{m},i} = E^{c}_{\mathrm{d},i} - E^{c}_{\mathrm{s},i} \quad (i = 1, 2\ldots, 8760\,\mathrm{hr}) \tag{15.3}$$

15.2.2.2 Step 2: Optimization of the Virtual 'Shared' Battery Capacity of the Building Community Using GA

In this step, the GA is used to search the optimal battery capacity (CAP^{c*} (kWh)) that minimizes the payback period (*PB*) of battery while meeting the user-required PV power self-consumption rates ($\mathrm{SC}_{\mathrm{th}}$). In each generation (i.e. iteration), a set of design alternatives of battery capacity (CAP^{c}) are first generated by the GA optimizer. Then, the performances (i.e. fitness value in the GA) of each design alternative are evaluated and compared. Several optimal alternatives will be selected and used as parent generations to produce design alternatives of the child generations. The process will continue until the optimal solution does not change after several generations. In this chapter, minimizing the payback period is set as the fitness function as an example, see Eq. (15.4). Note that the fitness functions can be flexibly changed according to the users' needs.

$$J_{\mathrm{fitness}} = \min(\mathrm{PB}) \quad \text{s.t. } \mathrm{SC} \geq \mathrm{SC}_{\mathrm{th}} \tag{15.4}$$

The PB is calculated as the by Eq. (15.5), which is calculated as the ratio of the investment of the battery and electricity cost savings contributed by battery installation.

$$\mathrm{PB} = \frac{\mathrm{CAP}^{c} \cdot \rho}{\mathrm{Cost}^{c,0} - \mathrm{Cost}^{c,1}} \tag{15.5}$$

CAP^{c} (kWh) is the aggregated battery capacity; ρ (€/(kWh)) is the unit cost of the battery. $\mathrm{Cost}^{c,0}$ (€) and $\mathrm{Cost}^{c,1}$ (€) are annual electricity costs before and after installing battery, which are calculated by Eq. (15.6).

$$\mathrm{Cost}^{c,0/1} = \sum_{i=1}^{8760} E^{c,0/1}_{\mathrm{grid},i} \times \chi_i, \begin{cases} \chi_i = \chi_{\mathrm{buy}}, & \text{if } E^{c}_{\mathrm{grid},i} > 0 \\ \chi_i = \chi_{\mathrm{sell}}, & \text{if } E^{c}_{\mathrm{grid},i} \leq 0 \end{cases} \tag{15.6}$$

$E^{c}_{\mathrm{grid},i}$ (kWh) is the building community's energy exchange with the power grid in the *i*th hour, which is calculated as the deviation between the energy mismatch ($E^{c}_{\mathrm{m},i}$, as calculated by Eq. (15.3) and the battery charging/discharging rates ($E^{c}_{\mathrm{bat},i}$), see Eq. (15.7). χ_i (€/kWh)) is the hourly electricity price. χ_{buy} (€/(kWh)) is the price of purchasing electricity from the power grid, and χ_{sell} (€/(kWh)) is the feed-in-tariff.

$$E^{c}_{\mathrm{grid},i} = E^{c}_{\mathrm{m},i} - E^{c}_{\mathrm{bat},i} \tag{15.7}$$

In this chapter, the battery is considered to be continuously operating. The calculation of the charging/discharging states of the virtual 'shared' battery is described as follows.

***Discharging state*:** When the community-level power mismatch ($E^{c}_{\mathrm{m},i}$) is larger than zero, the building community has insufficient PV power generation, and thus the battery is in discharging state. The battery discharging rates $E^{c}_{\mathrm{bat},i}$ (kWh) should be smaller than both the amount of electricity stored in the battery and the battery charging limits, as shown by Eq. (15.8).

$$E^{c}_{\mathrm{bat},i} = \begin{cases} \min\left(\phi_i, E^{c}_{\mathrm{charge,limit}}\right), & \text{if } E^{c}_{\mathrm{m},i}/\eta_{\mathrm{discharge}} > \min\left(\phi_i, E^{c}_{\mathrm{charge,limit}}\right) \\ E^{c}_{\mathrm{m},i}/\eta_{\mathrm{discharge}}, & \text{if } E^{c}_{\mathrm{m},i}/\eta_{\mathrm{discharge}} \leq \min\left(\phi_i, E^{c}_{\mathrm{charge,limit}}\right) \end{cases} \tag{15.8}$$

$E^{c}_{\mathrm{charge,limit}}$ (kWh) is the maximum charging/discharging rates of the battery in each hour, and $\eta_{\mathrm{discharge}}$ is the battery discharge efficiency. ϕ_i (kWh) is the amount of electricity stored in the battery in the *i*th hour, which is calculated by Eq. (15.9).

$$\phi_i = \eta_{\text{discharge}} \cdot \phi_{i-1} + E^c_{\text{bat},i} \tag{15.9}$$

***Charging state*:** When the community-level power mismatch ($E^c_{\text{m},i}$) is smaller than zero, the building community has surplus PV power generation, and thus the battery is in charging state. The battery charging rates $E^c_{\text{bat},i}$ (kWh) should be smaller than both the remaining storage capacity of the battery and the battery charging limits, as shown by Eq. (15.10).

$$E^c_{\text{bat},i} = \begin{cases} -\min\left(\text{CAP}^c - \phi_i, E^c_{\text{charge,limit}}\right), & \text{if } -E^c_{\text{m},i} \cdot \eta_{\text{charge}} \geq \min\left(\text{CAP}^c - \phi_i, E^c_{\text{charge,limit}}\right) \\ E^c_{\text{m},i} \cdot \eta_{\text{charge}}, & \text{if } -E^c_{\text{m},i} \cdot \eta_{\text{charge}} < \min\left(\text{CAP}^c - \phi_i, E^c_{\text{charge,limit}}\right) \end{cases} \tag{15.10}$$

η_{charge} is the battery discharge efficiency. The self-consumption rate (SC), i.e. the percentage of PV power that is consumed on-site, is calculated by Eq. (15.11), in which $\left|\sum E^c_{\text{grid},i,-}\right|$ is the aggregated amount of electricity exported to the grid ($E^c_{\text{grid},i}$ with negative values).

$$\text{SC} = \frac{\sum_{i=1}^{8760} E^c_{\text{s},i} - \left|\sum E^c_{\text{grid},i,-}\right|}{\sum_{i=1}^{8760} E^c_{\text{s},i}} \tag{15.11}$$

The output of the GA search is the optimal capacity of the virtual 'shared' battery (CAP^{c*}) which has the minimal *PB* while meeting a user-required *SC*. The aggregation of the distributed battery capacities, to be optimized in Step 3, should be equal to CAP^{c*}.

15.2.2.3 Step 3: Optimization of Distributed Battery Capacity for Single Building Using NLP

In this step, the capacity of distributed batteries ($[\text{CAP}^1, \text{CAP}^2, \ldots, \text{CAP}^n]$ (kWh)) installed in individual buildings is optimized using NLP based on the virtual 'shared' battery capacity. The objective function of the NLP is expressed by Eq. (15.12), which aims at minimizing the amount of storage sharing (i.e. the required energy exchanges with other buildings' batteries). The lower limit and upper limit of the capacity are set as [0, 0, …, 0] and [CAP^{c*}, CAP^{c*}, …, CAP^{c*}], representing that the capacity of individual battery should be in the range of 0 and the aggregated capacity. By minimizing the required energy exchanges with other buildings' batteries, the energy loss due to long-distance low-voltage power transmission can be significantly reduced.

$$\begin{aligned} J_{\text{NLP}} &= \min\left(\sum_{i=1}^{8760}\sum_{j=1}^{50}\left|E^j_{\text{bat,other},i}\right| / \eta_{\text{trans}}\right) \\ \text{s.t.} &\sum_{j=1}^{n} \text{CAP}^j = \text{CAP}^{c*} \end{aligned} \tag{15.12}$$

$E^j_{\text{bat,other},i}$ (kWh) is amount of energy stored-in/taken-from other buildings' batteries, as depicted in Fig. 15.3. Figure 15.3 displays the operation logic of battery operation and energy sharing in the proposed hierarchical design in the ith hour for each individual building. η_{trans} is the power transmission efficiency. For more detailed operation logic and calculation of $E^j_{\text{bat,other},i}$, please refer to Fig. 15.15 in the Appendix. The key parameters are explained below. To make the main text concise, the detailed calculation of these parameters is presented in Fig. 15.16 in the Appendix.

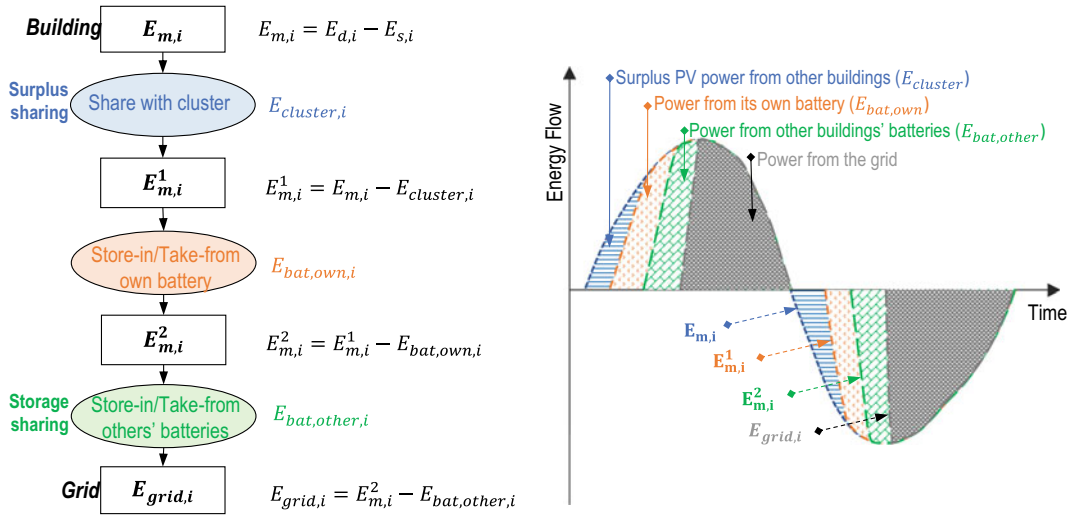

Fig. 15.3 Operation logic of battery and energy sharing in the proposed hierarchical design

- $E_{\mathrm{m},i}$ (kWh): the hourly energy mismatch (calculated as the deviation between the hourly power demand and power production).
- $E_{\mathrm{cluster},i}$ (kWh): the amount of surplus power sharing with other buildings in the power grid. This value is determined by the energy mismatch of each individual building
- $E^1_{\mathrm{m},i}$ (kWh): the hourly energy mismatch after surplus sharing.
- $E_{\mathrm{bat,own},i}$ (kWh): the amount of energy stored-in/taken-from its own battery, i.e. the battery charging/discharging rates, as calculated by Eqs. (15.8), (15.9) and (15.10).
- $E^2_{\mathrm{m},i}$ (kWh): the hourly energy mismatch after surplus sharing and own battery regulating.
- $E_{\mathrm{bat,other},i}$ (kWh): the amount of storage sharing, i.e. energy stored-in/taken-from other buildings' batteries. Its calculation is presented in Appendix 15.16.
- $E_{\mathrm{grid},i}$ (kWh): the hourly energy exchanges with the power grid after surplus sharing, own battery regulating, and storage sharing.

The hierarchical design will minimize the aggregated storage sharing (i.e. aggregated $E_{\mathrm{bat,other},i}$) and thus maximize the usage of the buildings' own batteries (i.e. aggregated $E_{\mathrm{bat,own},i}$). The output of the NLP is the optimized capacity of the distributed batteries $[\mathrm{CAP}^{1*}, \mathrm{CAP}^{2*}, \ldots, \mathrm{CAP}^{n*}]$. In this chapter, the NLP algorithm was implemented in Matlab. Considering the convergence and computational efficiency, the '*fmincon Interior Point Algorithm*' in Matlab was used as the solver. For more details of the principles, please refer to MathWorks (2020). The computational time was around 390 s on an Intel® Core™ i5 computer.

15.2.2.4 Step 4: Performance Comparison and Analysis

After obtaining the optimized design of the distributed batteries, the building-community-level performances are analysed and compared with the two existing design approaches: individual design (distributed battery) and group design (centralized battery), i.e. Scenarios 1 and 2 in Sect. 15.2.1. In the individual design, the capacity of each building's battery is designed based on its own energy mismatch. GA is used to search the optimal capacity that minimize the *PB* of the individual battery while achieving a user-defined *SC* for the individual building. In the group design, the capacity of the centralized battery is designed based on the building-community-level energy mismatch. GA is used to search the optimal capacity that minimize the

PB of the centralized battery while achieving a user-defined *SC* for the whole building community.

15.2.3 Buildings and System Modelling

This sub-section introduces the electricity demand modelling, PV and electrical storage system modelling. This chapter will consider two different battery scenarios, i.e. distributed batteries and centralized battery.

15.2.3.1 Electricity Demand Modelling

This chapter considered a virtual building cluster located in Ludvika, Dalarna region, Sweden. The district is made up of 50 independent households. The residential district is equipped with a direct current (DC) microgrid as described in Huang et al. (2019) for energy sharing. The number of occupants was set to range between 1 and 5 to represent the various scenarios in practise. Meanwhile, some special but practical scenarios, such as working at home (which may lead to high electricity demand), have also been considered in configuring these buildings. The electricity demand of these families is generated using the LPG (LoadProfileGenerator) (Pflugradt 2017). LPG is a behaviour-based tool used to simulate the energy consumption in time intervals down to one minute. It contains various activities caused by the behaviour of different family members in each household, as shown in Fig. 15.4. The internal functioning of LPG is described in Pflugradt and Muntwyler (2017). The tool performs a complete behavioural simulation of occupants in the family and uses it to generate load curves. Every occupant is part of a household along with other occupants and appliances, and every person has various needs at any time point. To meet the occupant's needs, a person can interact with different devices with the possibility of generating electric loads. To decide what to do, each person checks the expected satisfaction associated with each activity and chooses the activity that provides the best improvement. For example, if a person is tired, then he/she will sleep; and if he/she is hungry, he will prepare some food (perhaps using an electric oven), and so on. To accomplish this task, each desire has a small increase at each time step of the simulation. The amount of increase varies between different desires (i.e. different desires have different recharge times). After the simulation, LPG generates a log file in CSV format to be imported into the energy system simulation. Note that the buildings are assumed to the district heating system, and thus the heating equipment is not considered in the electricity demand calculation.

15.2.3.2 PV System Modelling

The power generation from the PV panel P_{PV} (kW) is calculated by Eq. (15.13) (Klein et al. 2004),

$$P_{PV} = \tau \times I_{AM} \times I_T \times \eta_{PV} \times CAP_{PV} \quad (15.13)$$

where τ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (W/m^2) is the total amount of solar radiation incident on the PV collect surface; η_{PV} is the overall efficiency of the PV array; CAP_{PV} (m^2) is the PV surface area.

15.2.3.3 Electrical Battery Modelling

This chapter used simplified electrical battery models. The electricity stored in the battery is calculated using a simplified model, as expressed by Eq. (15.9). It is estimated as the aggregated hourly charging rates (as calculated by Eq. (15.8)) and discharging rates (as calculated by Eq. (15.10)) (Sun et al. 2018).

15.3 Case Studies and Results Analysis

In the case studies, the 50 buildings with demand data modelled in Sect. 15.3.1 were used to test the performances of the proposed hierarchical design method. The weather data of Ludvika was used to model the local PV power productions.

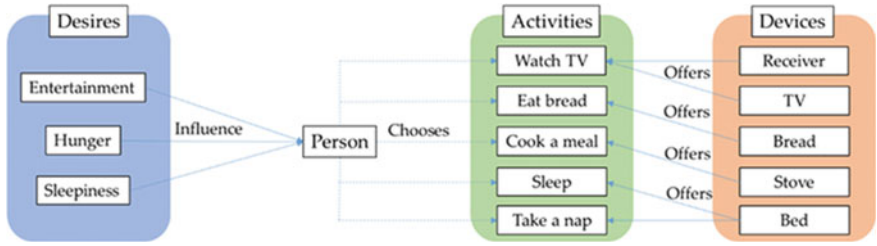

Fig. 15.4 Workflow for the generation of electricity loads using LPG (every person has a set of desires which influence the choice of activities, which in turn use devices which might be associated with an electric load)

The PV system capacity was sized to achieve the zero-energy goal that its annual aggregated PV production equals the annual aggregated electricity demand. This section first presents the electricity demand/supply information of the buildings. Then, the performances of the proposed design are compared with the two existing designs.

15.3.1 Building Electricity Demand, Renewable Power Generation and Electricity Mismatch

Figure 15.5 displays the annual aggregated electricity demand/supply of the 50 buildings in the building community. Note the demand bars overlap with the supply bars since the PV systems are sized to produce the same amount of power as the annual power demand of each building. For most of these single-family houses, the annual aggregated electricity demand/supply is in the range of 1640–7000 kWh. Among these buildings, building number 10 has the largest electricity demand, which is about 11,660 kWh (because of more occupants). The electricity demand/supply data will be used as the inputs to test the battery system performance under different designs and operation scenarios.

Figure 15.6 presents the hourly electricity demand, PV power production and energy mismatch of the 50 buildings of the community in a selected summer week. In the selected summer week, most of the hourly electricity demand of the 50 buildings is in the range of 0–4 kWh, and most of the hourly PV power production is in the range of 0–10 kWh. The hourly electricity mismatch mostly lies in the range of –8–5 kWh. In the time slot with simultaneous positive energy

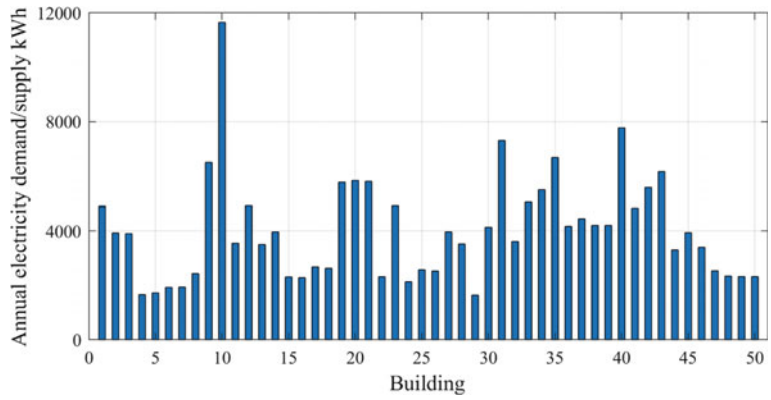

Fig. 15.5 Annual aggregated electricity demand/supply of the 50 buildings in the community (note the demand bars overlap with the supply bars since the PV systems are sized to produce the same amount of power as the annual power demand of each building)

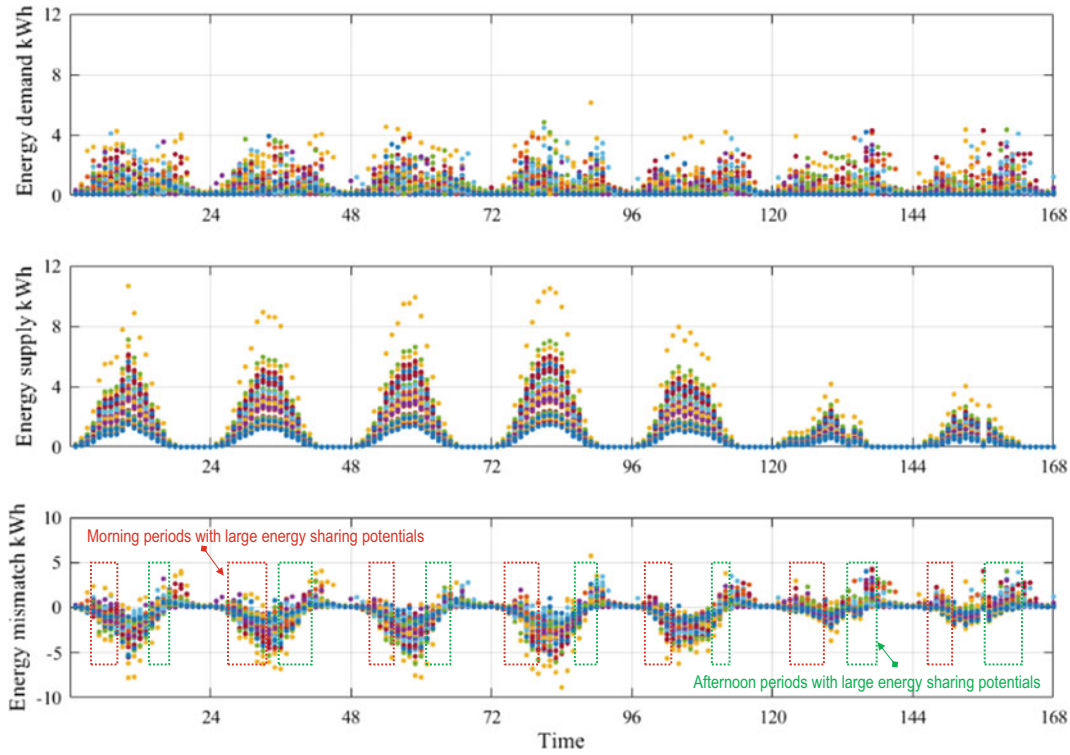

Fig. 15.6 Hourly electricity demand, PV power production and power mismatch of the 50 buildings in the community in a selected summer week

mismatch (i.e. the building has surplus PV power production) and negative energy mismatch (i.e. the building has insufficient PV power production), the positive energy mismatch can compensate with the negative energy mismatch and thus creates potentials for energy sharing. There are two periods in each day with large energy sharing potentials, either in the morning or in the afternoon. At night, the PV power production is close to zero for all buildings, and thus all the buildings are in short of PV power (i.e. no building with surplus PV power). While at noon, the PV power production reaches maximum for all the buildings simultaneously, and thus most of the buildings have surplus power production (i.e. no building with PV power deficiency).

Table 15.2 summarizes the techno-economic parameters used in this chapter for performance evaluation of each design. The price of buying electricity from the grid was set to be 0.16 €/(kWh). Considering the negative impacts on the grid stability and safety, the feed-in-tariff was set lower than the purchasing price (i.e. 0.05 €/(kWh)) (Huang et al. 2019). Since this chapter does not consider the economics of individual batteries, the prices of power trading within the building community are not considered. Please note in practise, the setting of prices for power trading within the building community should provide economic incentives to the stakeholders so that they would prefer trading within the community to trading with the grid. With reference to Cole and Frazier (2019), the unit price of battery was set to be 250 €/(kWh) including the cost of installation. The maximum charging/discharging rates of each battery were assumed to be 30% of its capacity. The user-required PV power self-consumption rate was set as 60%. This value will be used as constraint threshold in the battery design. Table 15.2 also summarizes the parameters related to energy efficiency in each process, e.g. battery charging/discharging

Table 15.2 Techno-economic parameters (Parra et al. 2016; Huang et al. 2019; Cole and Frazier 2019)

Category	Input name	Value
Economic parameters	Price of electricity bought from the grid (€)	0.16
	Price of electricity sold to the grid (€)	0.05
	Price of electricity bought from the building community (€)	0.1
	Price of electricity sold to the building community (€)	0.1
	Cost of the storage system including installation (€/kWh)	250
Technical parameter	Maximum charging/discharging rates of battery (% Capacity)	30%
	User-required PV power self-consumption rate	60%
	Battery charge/discharge efficiency	92%
	Battery storing efficiency	92%
	Surplus sharing efficiency (due to power transmission loss)	92%
	Storage sharing efficiency (due to power transmission loss)	92%

(i.e. roundtrip), battery storing, and power transmission loss.

Due to the physical constraints, the buildings in an energy sharing community are usually located in the same location. Considering the relatively small differences in the distances between different buildings, this chapter used the same transmission efficiency for energy sharing among different buildings for simplicity.

15.3.2 Performance Comparison at Community(Cluster)-Level

Using the electricity demand and PV power production data as inputs, the three different design methods (see Table 15.1) have been used to design the battery system in the building community (i.e. building-cluster). In Scenario 1, the capacity of distributed battery is sized to achieve a 60% self-consumption rate (as specified in Table 15.2) for each individual building with the minimized payback period (the individual design cannot set the community-level performance as the design goal). In Scenario 2, the capacity of the centralized battery is sized to achieve a 60% self-consumption rate for the whole building community with the minimized payback period. In Scenario 3, the aggregated capacity of the distributed battery is sized to achieve a 60% self-consumption rate for the whole building community with the minimized payback period.

Table 15.3 compares the design results and economic performances of the three methods. Under the individual design and operation scenario, the aggregated capacity of the distributed batteries was 322.1 kWh. While the capacity of the centralized battery in Scenario 2 and the aggregated capacity of the distributed batteries in Scenario 3 were both 204 kWh. The aggregated battery capacities were the same in Scenarios 2 and 3, since the proposed design used the virtual 'shared' battery capacity (obtained from group design) as benchmark to instruct the sizing of distributed batteries. Meanwhile, since in Scenarios 2 and 3 the buildings can share their surplus PV power production with other buildings, the need of battery for storing the excessive PV power is reduced. The aggregated battery capacity was significantly reduced (i.e. 36.6% decrease) compared with Scenario 1. Correspondingly, the initial investment of battery was significantly reduced in Scenarios 2 and 3 (i.e. 36.6% decrease). The community-level annual cost saving of Scenario 1 was about 5.3% higher than the cost savings in Scenarios 2 & 3. This is because in Scenario 1, the aggregated battery capacity was much larger, which could help keep more surplus power inside the community and thus reduce the grid power imports. The payback

Table 15.3 Comparison of the design results under different scenarios

	Scenario 1: individual design	Scenario 2: group design	Scenario 3: proposed design
Aggregated battery capacity (kWh)	322	204	204
Battery investments (€)	80,525	51,000	51,000
Cost saving per year (€/year)	4129	3912	3917
Payback period (year)	19.5	13.0	13.0

period of the individual design was 19.5 years, while payback periods in the group design and proposed design were both 13 years (i.e. 33.3% decrease).

Figure 15.7 presents the annual PV power self-consumption rates at the building-community-level under the three different designs without considering the energy loss (i.e. Fig. 15.7a, the valued used in the constraint check) and with energy loss considered (i.e. Fig. 15.7b). As shown in Fig. 15.7a, all the three designs meet the user-required threshold for self-consumption rate (i.e. 60%). The community-level self-consumption rate is 61% in Scenario 1, slightly higher than the threshold. This is because the individual design takes the single building's self-consumption rate as the design constraint. After meeting the individual building's self-consumption rate target, by enabling surplus sharing (as explained in Sect. 15.2.1), the self-consumption rate at the building-community-level could be further increased slightly. However, 36.6% increase in the aggregated battery capacity only leads to about 1% increase in the community-level self-consumption, which is not so economical. Figure 15.7b displays the self-consumption rate considering the energy losses in the battery storage, battery charging/discharging process, and power transmission loss in the energy sharing process (using the efficiency parameters listed in Table 15.2). When the energy loss is considered, the self-consumption rate in the Scenario 1 decreases to be close to Scenarios 2 and 3, due to the relatively larger energy loss.

Figure 15.8 compares the amount of energy losses in different processes under the three scenarios. In Scenario 1, the energy losses occur in the battery charging/discharging, battery storing, and the surplus sharing process. While in Scenarios 2 and 3, the energy losses also occur in the storage sharing process. In all the three scenarios, the energy loss in battery storage accounts for the largest percentage (i.e. 89.9%, 64.2% and 73.2%, respectively). In Scenario 1, the energy loss in battery storing is much larger than Scenarios 2 and 3 (about 50.1% increase). This is because the aggregated battery capacity is much larger, and thus more electricity can be stored in the battery. The energy loss due to surplus sharing in Scenario 1 is much smaller than Scenarios 2 & 3. This is because after the battery

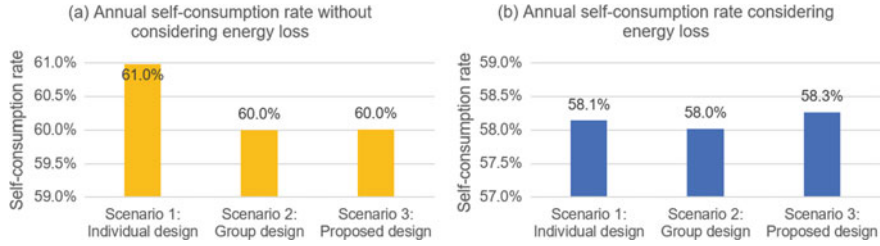

Fig. 15.7 Self-consumption of the whole building community under different scenarios **a** Ideal case without considering energy loss, **b** With energy loss considered

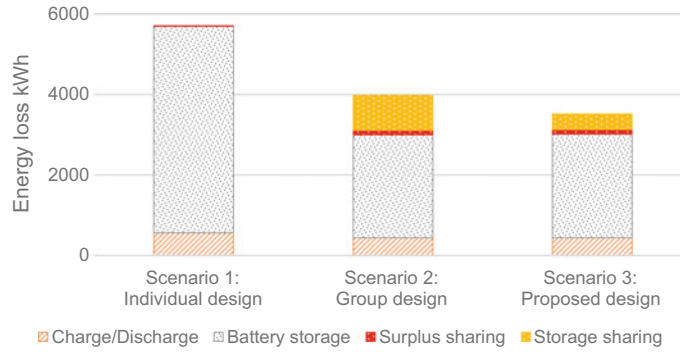

Fig. 15.8 Energy losses of the whole building community in different scenarios

regulation of each single building's energy mismatch, the remaining energy mismatch of most buildings will approach zero, and thus reducing the potentials of surplus sharing. While in Scenarios 2 & 3, surplus sharing is implemented before the battery regulating, when there is large diversity between different buildings' energy mismatch, and thus there are more potentials of surplus sharing (and more losses due to surplus sharing as well). The energy loss due to storage sharing in Scenario 3 is smaller than the loss in Scenario 2 (about 2412 kWh decrease). This is because in the distributed battery configuration, the buildings can use their own batteries as part of the electricity storage and thus reduce the need of storage sharing. Such reduced energy loss in storage sharing contributed to a slight increase in the community-level self-consumption rates (i.e. about 0.3% as shown in Fig. 15.7b).

Table 15.4 summarizes the design results of individual batteries under the individual design (i.e. Scenario 1) and the proposed design (i.e. Scenario 3). The sizing results are very different under the two design strategies. For some buildings, the optimal battery capacity is smaller in the proposed design, while for some buildings, the optimal battery capacity is larger. This is because in the individual design, minimizing the payback period was used as the fitness function to optimize the individual battery capacity. While in the proposed design, minimizing the power loss caused by energy sharing (see Step 3 in the proposed method) was set as the fitness function to optimize the individual battery capacity. At the building community level, the aggregated battery capacity is much smaller compared to the individual design (see Table 15.3).

Figure 15.9 presents the aggregated electricity storage in the batteries of the building community under the three scenarios. The aggregated electricity storage is the same in Scenarios 2 and 3 (i.e. both with a storage capacity of 204 kWh). Due to a larger aggregated battery capacity in Scenario 1 (i.e. 322 kWh), on average more electricity is stored compared with Scenarios 2 and 3. The batteries typically become fully charged in the period 12:00–18:00 from May to August (i.e. Day-120 to Day-240). While in the evenings of these months, the electricity stored in the battery cannot be fully consumed by the buildings, see the storage in the period 0:00–6:00 from May to August. As a result, with the cumulation of surplus PV power, there is a high state of charge during the summer period in Scenario 1. Scenarios 2 and 3 are slightly better than Scenario 1 in aspects of night discharging, as most of the stored electricity can be consumed. Since the storage loss is proportional to the amount of electricity storing in the battery, there is more storage loss in Scenario 1 compared with Scenarios 2 and 3 (see the comparison in Fig. 15.8).

Figure 15.10 displays the hourly surplus sharing in each hour under the three scenarios. The surplus sharing is the same in Scenarios 2 and 3 (see the principals and explanations in Sect. 15.2.1). The surplus sharing only occurs during daytime when PV can produce power. In Scenario 1, the surplus sharing process is after the individual battery's regulating (when most of

Table 15.4 Summary of the design results of individual batteries using different methods: individual design (Scenario 1) and proposed design (Scenario 3)

Building	1	2	3	4	5	6	7	8	9	10
Individual design	8.7	7.3	7.2	3.1	3.4	3.1	3.1	5.0	10.1	9.8
Proposed design	1.8	4.3	2.5	0.6	3.3	0.4	7.6	4.8	2.0	9.5
Building	11	12	13	14	15	16	17	18	19	20
Individual design	5.8	8.3	5.4	5.4	4.1	4.1	6.3	5.8	9.9	10.5
Proposed design	1.1	7.0	0.1	6.7	0.0	4.0	9.0	0.1	0.1	7.6
Building	21	22	23	24	25	26	27	28	29	30
Individual design	10.7	4.5	8.9	2.7	3.5	3.3	8.0	4.4	2.0	8.1
Proposed design	1.3	2.5	0.1	0.0	2.5	5.4	0.1	3.9	9.5	6.6
Building	31	32	33	34	35	36	37	38	39	40
Individual design	16.0	7.0	6.1	10.6	7.9	5.6	4.5	5.7	8.5	11.5
Proposed design	1.6	2.5	2.6	6.2	3.7	2.6	11.1	1.8	1.8	5.2
Building	41	42	43	44	45	46	47	48	49	50
Individual design	6.5	9.6	9.2	3.1	3.3	7.2	3.1	5.0	4.6	4.6
Proposed design	4.7	17.4	0.8	1.1	0.1	9.6	0.0	12.4	7.9	6.4

The unit of capacity is kWh

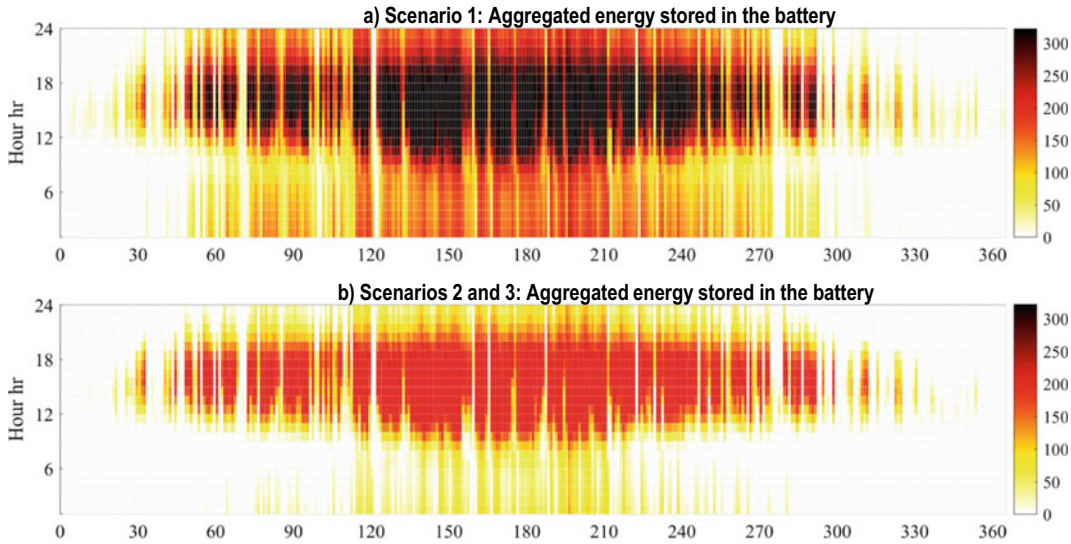

Fig. 15.9 Aggregated amount of electricity stored in the building community batteries under the three scenarios (note the aggregated energy storage is the same in Scenarios 2 & 3). The unit for the colour bar is kWh

the buildings' remaining energy mismatch will approach zero). As a result, the amount of surplus sharing is very limited in Scenario 1. In Scenarios 2 and 3, there is large surplus sharing in the early morning (especially at 8:00 in the summer period from May to August, Day-120 to Day-240) and late afternoon. This is because in these two periods there is large possibility of simultaneous positive-energy-mismatch and negative-energy-mismatch in the buildings

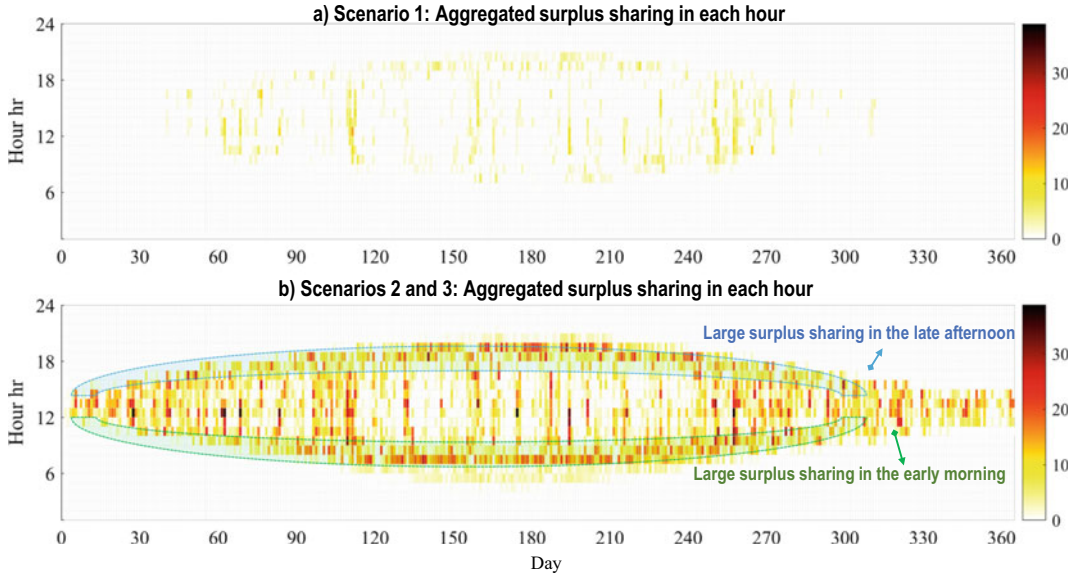

Fig. 15.10 Aggregated amount of surplus sharing in the building community under the three scenarios (note the surplus sharing is the same in Scenarios 2 & 3). The unit for the color bar is kWh

(as explained in Fig. 15.6). At night, all the buildings have positive energy mismatch (due to zero PV power production), while at noon, most of the buildings have negative energy mismatch (i.e. maximized PV power production). In Scenarios 2 and 3, the surplus sharing reduces the battery needs of buildings for keeping the surplus power inside the community. Due to a lack of solar irradiation in winter, the surplus sharing is very limited.

Figure 15.11 displays the amount of storage sharing in Scenario 2 (i.e. centralized battery design) and Scenario 3 (i.e. distributed battery design). Note that in Scenario 1, there is no storage sharing. The storage sharing in Scenario 2 is much larger than Scenario 3, since any charging/discharging of the centralized battery is considered as the storage sharing. In Scenario 3, only the part of electricity stored in other buildings' batteries is considered as storage sharing. Due to the long-distance power transmission for storage sharing, the electricity loss due to storage sharing in Scenario 2 is nearly two times the loss in Scenario 3, as highlighted in Fig. 15.8. The storage sharing process occurs mostly in the morning (9:00–11:00) and in the evening (18:00–24:00) in the summer period from May to August (i.e. Day-120 to Day-240). In the summer mornings, the distributed battery of each building will gradually become fully charged due to the large solar irradiance but at different speeds (because of different PV size, power demand, and storage capacity). Some batteries are fully charged more quickly than others, and thus the buildings installing these batteries can store their surplus PV power in other buildings' batteries which are not fully charged yet. While in the summer evenings, the buildings, which used up all the stored electricity in its own batteries, can take electricity from other buildings' batteries. At summer noon, as most of the batteries are already fully charged, there is not any storage sharing. Due to a lack of solar irradiation in winter, the storage sharing is also very limited.

Figure 15.12 presents the building-community-level hourly electricity exchanges with the power grid under the three scenarios. The community-level hourly electricity exchanges are nearly the same in Scenario 2 and Scenario 3 (with the same aggregated battery capacity and both enable full energy sharing). For all these three scenarios, there are large

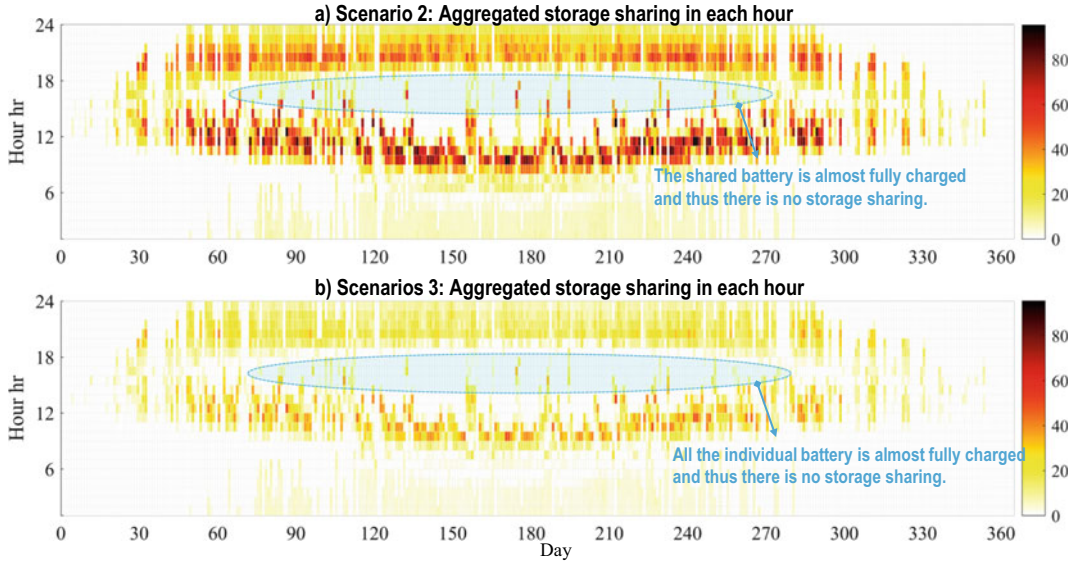

Fig. 15.11 Aggregated amount of storage sharing in the building community batteries under the two scenarios (note the storage sharing is not enabled in Scenario 1). The unit for the colour bar is kWh

electricity exports to the grid in the afternoon (i.e. 12:00–18:00) from May to August (i.e. Day-120 to Day-240). There are large electricity imports from the grid in the morning (i.e. 7:00–10:00) and evening (i.e. 19:00–23:00) in the winter period (i.e. Day-300 to Day-30 next year). In Scenario 1, due to the installation of larger sized batteries at aggregated level, the amount of electricity exported to the grid is relatively smaller than Scenarios 2 and 3 (e.g. see Region A in Fig. 15.12). The amount of electricity imported from the grid is also smaller in Scenario 1 (e.g. see Region B in Fig. 15.12). Such reduced grid power exchanges contributed to 5.3% more electricity cost savings in Scenario 1 compared with Scenarios 2 and 3 (as calculated in Table 15.3).

15.3.3 Performance Comparison of a Single Building

Section 15.3.2 presents the overall performances at the building community level. This section selects one of the buildings from the community as an example and analyses its detailed operation and electricity exchanges in a typical summer week. Figure 15.13 illustrates the hourly energy exchanges of the building in a selected summer week under the three different scenarios. A negative value of energy flow indicates surplus PV power production, while a positive value represents larger electricity demands.

In Scenario 1, during daytime the battery is first charged until it becomes fully charged (see the blue regions). Then, part of the remaining surplus is shared with other buildings in the building community. However, the amount of surplus sharing is very limited since most of the buildings in the community do not lack PV power. The remaining part is exported to the power grid (see the grey regions). During nighttime, since there is no surplus PV power production (as no solar irradiation), the building takes electricity from its own battery.

In Scenario 2, during daytime the building first shares part of its surplus power with the buildings in shortage of PV power (i.e. surplus sharing, see the green regions). Then, it stores part of the surplus power in the centralized battery (i.e. storage sharing, see the yellow regions). Since some of other buildings do not have surplus power and thus do not need the battery storage, this building can take use more storage

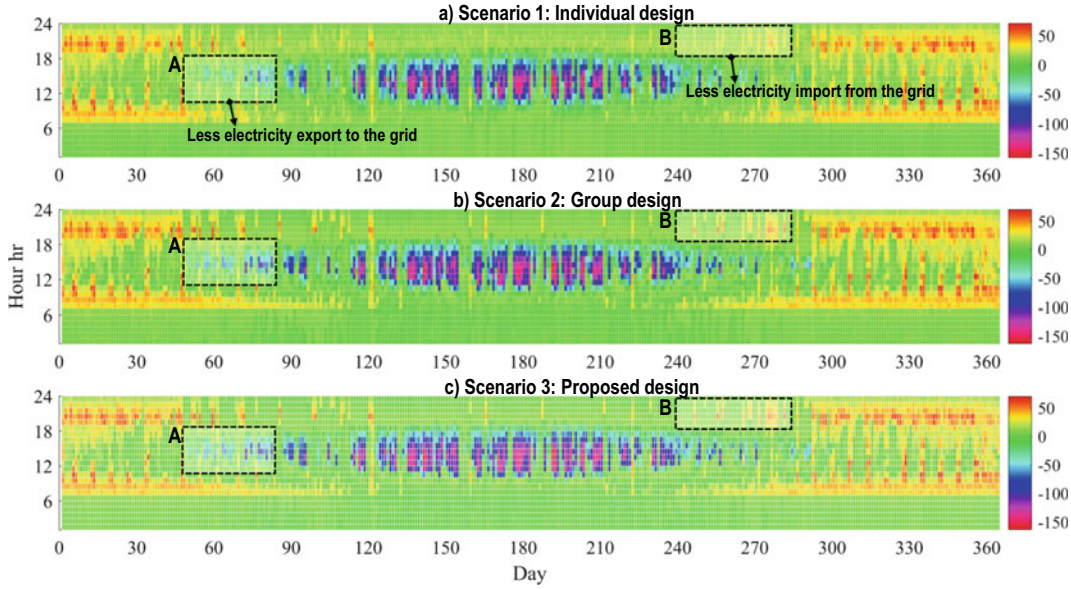

Fig. 15.12 Hour energy exchanges with the power grid of the building community under the three scenarios. The unit for the color bar is kWh

capacity from the centralized battery to store its surplus power compared with Scenario 1. After the centralized battery is fully charged, the remaining surplus power is exported to the grid (see the grey regions). During nighttime, the building takes electricity from either the centralized battery or the power grid.

In Scenario 3, the battery storage of surplus PV power is further classified into storing in the building's own battery (see the blue regions) and storing in other buildings' batteries (i.e. storage sharing, see the yellow regions). The sum of the own battery storing and shared storing in Scenario 3 equals the storage in the centralized battery in Scenario 2. Under the same aggregated level battery usage, by dividing the battery into self-usage and shared-usage, the amount of storage sharing can be significantly reduced, which will help reduce the electricity losses due to the long-distance low-voltage power transmission.

Figure 15.14 shows the daily exported electricity to the grid of the building in the selected week. In some days, the amount of electricity exports to the grid is smaller in Scenarios 2 & 3. This is because in these days (e.g. Day-1, Day-2, Day-3, Day-6 and Day-7) the building shared most of surplus electricity with other buildings. In some days, the amount of electricity exports to the grid is smaller in Scenario 1. This is because in these days the energy sharing potentials are very limited and as a result the battery regulating will dominate the performance: the battery capacity in Scenario 1 is 10.6 kWh while 6.2 kWh in Scenario 3, and thus more electricity can be stored inside the community in Scenario 1.

15.4 Summary

This chapter has proposed a hierarchical design optimization of distributed batteries in solar power shared building community. The developed design method first considers all the distributed batteries as a virtual 'shared' battery and searches the optimal capacity of the virtual 'shared' battery using genetic algorithm.

Then, the developed method optimizes the capacity of the distributed batteries installed in each building using non-linear programming with the objective of minimizing the storage sharing (and thus the associated power loss due

Fig. 15.13 Hourly energy exchanges of a building in a selected summer week under the three different scenarios: **a** Individual design, **b** Group design and **c** Proposed design

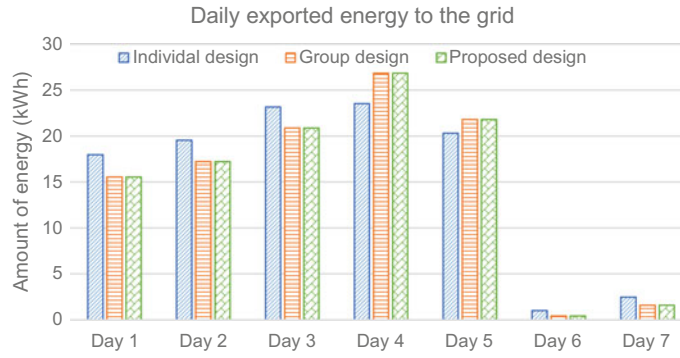

Fig. 15.14 Comparison of daily energy exported to the power grid of a building in the selected summer week for the three different scenarios

to long-distance power transmission). For validation, the developed design method has been compared with two existing design methods (i.e. *individual design* for distributed batteries and *group design* for centralized battery) based on a virtual building community located in Sweden. Case studies have shown that the developed design can achieve the user-required PV power self-consumption rate at building-community-level with a much smaller aggregated capacity compared with the *individual design*. Meanwhile, the energy loss due to storage sharing can be decreased compared with the *group design*. This chapter has also investigated the surplus sharing and storage sharing inside the building community under the Sweden context. The major findings are summarized as follows:

- *Overview of design results*: At a required community-level self-consumption rate of 60%, the proposed design reduced the aggregated capacity of the distributed batteries by 36.6% compared with individual design (i.e. Scenario 1). The payback period was reduced by 33.3%. By taking advantage of energy sharing, the proposed design can improve the cost-effectiveness of distributed battery system in solar powered building community.
- *Impacts of capacity on performances*: With battery capacity increases, the electricity cost savings will increase as more PV power can be kept on-site. But the initial investments and electricity loss in battery storing will increase as well. Such side-effects do not guarantee that a larger battery capacity is always good.
- *Time when surplus sharing occurs*: In the individual design, the surplus sharing occurs mostly in the summer afternoon when the majority batteries are fully charged. While in the group design and propose design, the surplus sharing occurs mostly in the early summer morning (at 8:00) and late summer afternoon. This is because in these two periods there is large possibility of simultaneous positive energy mismatch and negative energy mismatch in the buildings. In total, the amount of surplus sharing increased by 3.8 times in Scenarios 2 and3 compared with Scenario 1 by simply changing the sequence of sharing.
- *Time when storage sharing occurs*: The storage sharing occurs mostly in summer mornings (9:00–11:00) and evening (18:00–24:00). In summer mornings, the distributed battery of each buildings will gradually become fully charged due to the large solar irradiance but at different speeds. Some batteries become fully charged more quickly than others, and thus the buildings installing these batteries can store their surplus PV power in other buildings' batteries which are not fully charged yet. While in the summer evenings, the buildings, which used up all the stored electricity in its own batteries, can take electricity from other buildings' batteries.
- *Reduction of storage sharing power loss*: Compared with the group design of

centralized battery, the proposed design effectively reduced the amount of storage sharing, and thus the power loss due to the relatively long-distance low-voltage power transmission. The reduction in power loss reached over 55%.

This chapter does not consider the impacts of PV investment, battery degradation and the power trading business models on the design of individual batteries. In addition, this chapter used fixed grid electricity price and feed-in-tariff. Different regions/countries can have different electricity price scheme and PV power feed-in policy, which will have large impacts on the electricity costs savings and thus the investment payback. Thus, future work will consider various electricity prices strategies and business models for the power trading within a PED (considering the PV investment and battery degradation) and investigate their impacts on the distributed electrical energy storage system design performance. This chapter proposed a deterministic design (i.e. assuming all inputs perfectly known), while the uncertainty in the building electricity demand and PV power production is not considered. Neglecting the uncertainty might lead to the designed systems not performing as expected. Future work will also investigate these uncertainties and develop more robust design methods of energy storage systems for PEDs.

Appendix

Figure 15.15 presents the operation strategy of the proposed design for distributed battery system, which can maximize the energy sharing inside the building community. Figure 15.15a shows the operation strategy for buildings with surplus PV power. When a building has surplus power, it first sends the surplus power to meet the large electricity demand in other buildings. If there is any remaining surplus power, the building will use such power to charge its own battery. After its own battery is fully charged, the remaining surplus power will be stored in other buildings' batteries. After all the batteries in the building community are fully charged, the remaining surplus power will be sent to the power grid. Figure 15.15b shows the operation strategy for buildings with insufficient PV power production. The building will first take electricity from the buildings with surplus power production. If there is still power deficiency, the building will take power from its own battery. Until its battery is fully discharged, the building will take power from other buildings' batteries. After all the batteries in the community are fully discharged, the building will purchase power from the power grid to meet its remaining power deficiency.

Figure 15.16 illustrates the calculation of battery operation and energy sharing in each hour in the proposed hierarchical design. It consists of the following seven steps.

Step 1: Calculate hourly energy mismatch (i.e. $E_{\mathrm{mis},j}$, deviation between demand and supply) of all the buildings in the community.

Step 2: Calculate hourly aggregated energy mismatch of the building cluster ($E_{\mathrm{mis,tot}}$) and the amount of surplus sharing (E_{share}). The amount of surplus sharing is calculated as the smaller one of the aggregated surplus PV power and the aggregated power deficiency. If $E_{\mathrm{mis,tot}}$ is larger than zero, it indicates all the surplus power can be kept inside the building community. The surplus power will be shared by the buildings in shortage of power according to the ratio of insufficiency. While if $E_{\mathrm{mis,tot}}$ is smaller than zero, it indicates all the PV power shortage can be met inside the community. The remaining surplus power exported to the power grid will be determined according to the ratio of sufficiency.

Step 3: Determine the *Remaining Energy Mismatch* 1 (REM 1) of each building ($E^1_{\mathrm{mis},j}$) after sharing energy with other buildings. $E^1_{\mathrm{mis},j}$ is calculated as the

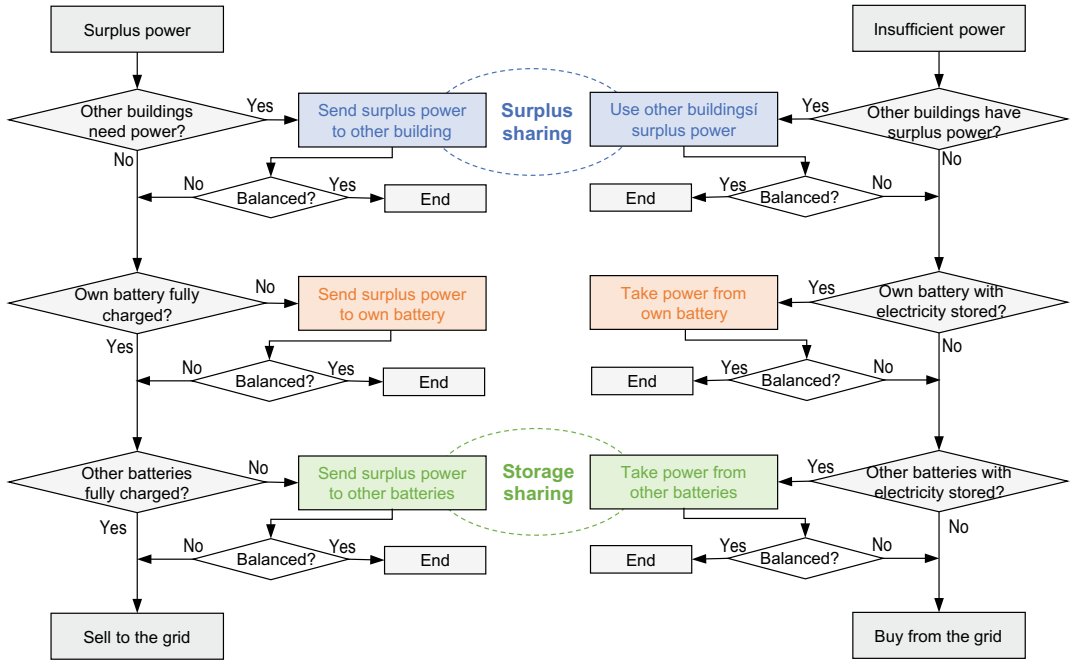

Fig. 15.15 Operation strategy for the distributed battery system with maximized energy sharing **a** Operation strategy for buildings with surplus PV power; **b** Operation strategy for buildings with insufficient power

deviation of $E_{\mathrm{mis},j}$ and the amount of surplus power sharing.

Step 4: Determine operation of each building's own battery (i.e. charging/discharging rates $E_{\mathrm{bat},j}$, stored energy $E_{\mathrm{store},j}$, and remaining capacity $E_{\mathrm{rem},j}$) based on the REM1. The battery operation should follow the rules described in Sect. 15.2.2, Eq. (15.8)–(15.10).

Step 5: Determine the *Remaining Energy Mismatch* 2 (REM2) of each building ($E^2_{\mathrm{mis},j}$) after battery regulating. $E^2_{\mathrm{mis},j}$ is calculated as the deviation of $E^1_{\mathrm{mis},j}$ and the battery charging/discharging rate $E_{\mathrm{bat},j}$.

Step 6: Determine the amount of energy sharing with other buildings' batteries. If there is remaining surplus power (i.e. $\sum E^2_{\mathrm{mis},j} < 0$), the remaining aggregated surplus power will be stored in the batteries not fully charged according to the ratio of sufficiency. While if there is remaining power deficiency (i.e. $\sum E^2_{\mathrm{mis},j} > 0$), the remaining aggregated power deficiency will be taken from the batteries not fully discharged according to the ratio of deficiency.

Step 7: Update the battery operation and calculate the energy exchange with the grid of each building ($E^3_{\mathrm{mis},j}$) after storage sharing. $E^3_{\mathrm{mis},j}$ is calculated as the deviation of $E^2_{\mathrm{mis},j}$ and the battery charging/discharging rate after storage sharing $E^2_{\mathrm{bat},j}$.

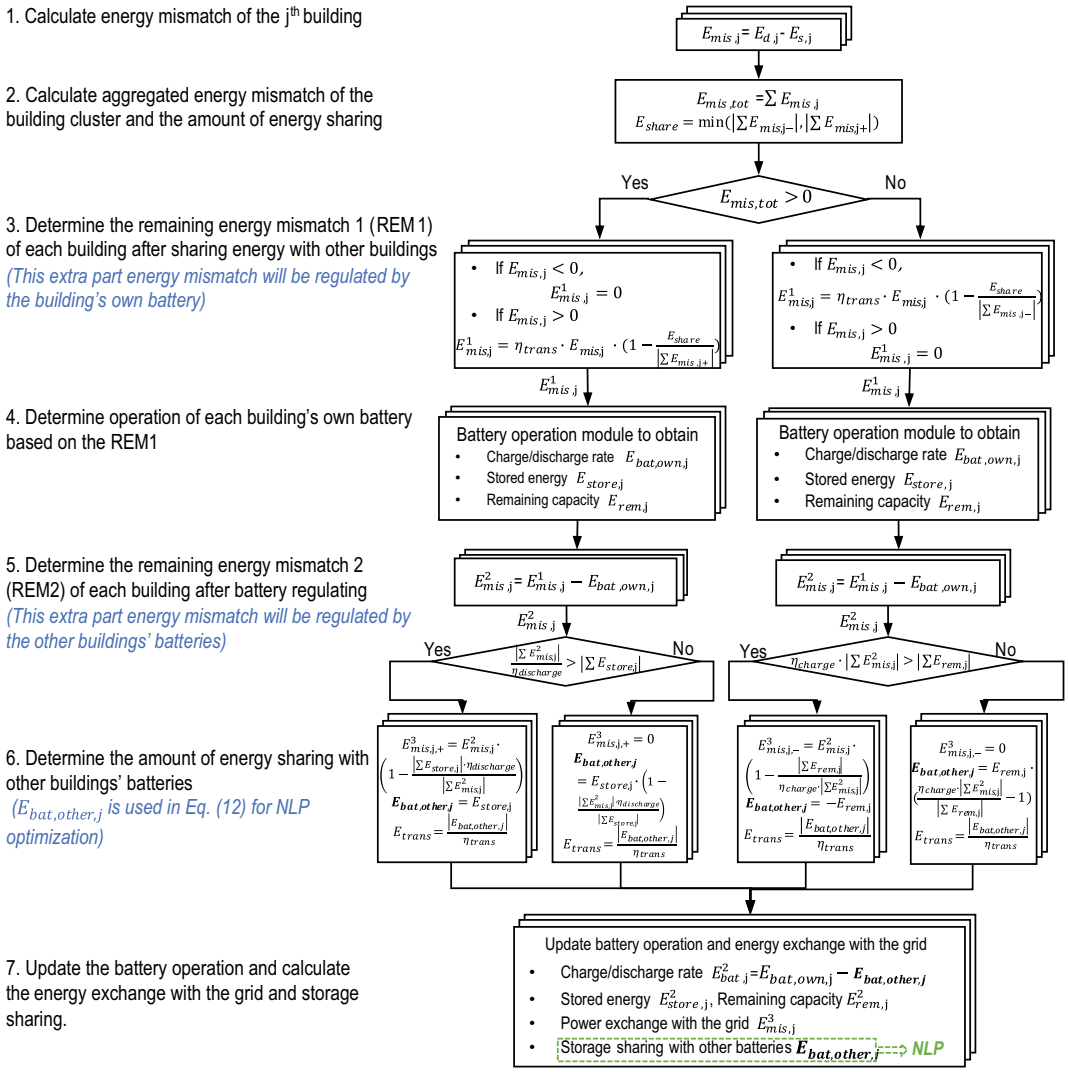

Fig. 15.16 Calculation of the battery operation and energy sharing in the proposed hierarchical design

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16 Geographic Information System-Assisted Optimal Design of Renewable-Powered Electric Vehicle Charging Stations in High-Density Cities

Pei Huang and Yongjun Sun

Abstract

The crowded urban environment and busy traffic lead to heavy roadside pollutions in high-density cities, thereby causing health damages to city pedestrians. Electric vehicle (EV) is considered as a promising solution to such street-level air pollutions. Currently, in high-density cities, the number of public charging stations is limited, and they are far from enough to form a complete charging network with a high coverage ratio that can provide easy and convenient charging services for EV users. Concerns and worries on being unable to find a charging port when needed become a major hurdle to EV practical applications. Meanwhile, greener and cheaper renewable energy is recommended to replace fossil fuel-based grid energy that is commonly used in existing charging stations. Thus, this study proposes a novel geographic information system (GIS)-assisted optimal design method for renewable-powered EV charging stations in high-density cities. By selecting the optimal locations and optimal number of the renewable-powered charging stations with the considerations of the existing charging stations and renewable potentials, the proposed method is able to minimize the life cycle cost of the charging stations while satisfying a user-defined area coverage ratio. Using Hong Kong as an example, case studies have been conducted to verify the proposed design method. The design method can be used in practice to help high-density cities build their public charging networks with cost-effectiveness, which will promote EV practical applications and thus alleviate the roadside air pollutions in high-density cities.

Keywords

Renewable energy · Charging station design · Electric vehicle · Geographic information system

16.1 Introduction

In high-density cities, due to the crowded urban environment and busy road traffic, the vehicle emissions cannot be easily dissipated, and they are trapped in the narrow space enclosed by the crowded buildings to form heavy and enduring air pollutions at street level (Committee 2014). Many governments and organizations have established strict regulations to control the street-

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level air pollutions. For instance, the Environmental Protection Department of Hong Kong has implemented a wide range of measures to reduce the emissions from motor vehicles, e.g., phasing out pre-Euro IV diesel commercial vehicles, using roadside remote sensing equipment to strengthen emission control, setting up low emission zones, and tightening the emission standards (Department 2018a). The National Environment Agency of Singapore has set up high standards of exhaust emission and fuel quality for all vehicles (Government 2019). Unfortunately, even with tremendous efforts from local governments, the roadside air pollutions in high-density cities still cannot be controlled at acceptable levels. In Hong Kong, the street-level NO_2 concentration exceeded the required level by up to 134 ug/m^3, and the respirable suspended particulates surpassed the allowed threshold by up to 10 ug/m^3 (Group 2016). The roadside air pollutions are proved to have diverse health damages to people. For instance, short-term exposures may lead to nose and throat irritations, shortness of breath, coughing, and chest tightness (Department 2018b), while medium/long-term exposures could result in more severe health impacts including risk increase of cancer (Protection 2017). Thus, the roadside air pollutions in high-density cities must be resolved.

Electric vehicle (EV) is considered as a promising solution to the roadside air pollution and associated health damages since it directly cut off the pollutions from the source (Seddig et al. 2019). Many governments have made efforts in promoting its practical applications, such as tax reductions, one-for-one replacement scheme, and subsidies for EV purchase and EV licenses (Department 2019). Unlike the conventional fuel-based vehicles, EV takes much longer time in its charging process. For instance, the regular charging takes 6–8 h, and even the fastest supercharging still takes about 30 min (Scientists 2018). Due to the slow charging processes, significantly more charging stations will be needed for EVs if the equivalently convenient services as conventional gas stations are trying to achieve. Existing studies have developed some design methods of charging stations for improving the EV deployments. For instance, Wang et al. investigated the siting and sizing problem of fast charging stations in a highway network, where the budget constraint and the service capacities of charging spots are considered (Wang et al. 2018). However, renewable energy integration was not considered. Luo et al. developed a comprehensive optimization model concerning the joint planning of distributed generators and electric vehicle charging stations, which involves spatially dispatchable characteristic at the planning stage of distribution systems and deploys power devices in a cost-effective way (Luo et al. 2019). However, their method is not applicable to high-density cities due to the limited spaces for installing the PV panels.

There are three typical chargers for EVs, i.e., private charging piles (Tesla, Accessed in 2019), public charging stations (Luo et al. 2018), and mobilized chargers (Council 2018). The EV owners can build a charging pile in their own garages. The EVs can be charged in the garages at night when the EV owners are off work at home. When the EV owners are not able to build their own charging piles, they can charge their vehicles in the public charging stations. To date, many public charging stations have been constructed worldwide, and the number keeps growing (Tesla, Accessed in 2019). Considering that the traffic jams and the limited existing charging stations may jointly cause EV users unable to charge conveniently, the Hong Kong Productivity Council developed a flexible mobilized EV charger, which offers multi-standard charging services and can fully recharge five to six EVs at the same time (Council 2018).

To promote the wide deployment of EVs, it is essential to provide EV users easy and convenient charging services. In high-density cities, developments of the private charging piles and the mobilized EV chargers may not be able to achieve such an objective due to the following reasons. In a high-density city, the limited garage space may not allow the wide installations of the private charging piles, and thus the EV users become more dependent of the public charging stations. A mobilized EV charger, in nature a

vehicle with a slightly larger battery, may worsen the busy traffic that is already notorious in a high-density city. In addition, the limited battery capacity of the mobilized charger also prevents its wide applications in practice. For this reason, the development of the public charging stations becomes critical for EV deployment in the high-density cities. Currently, in most high-density cities, existing charging stations are limited, and they are far from sufficient to form a complete charging network with a high coverage ratio (Sadeghi-Barzani et al. 2014). For instance, the total number of public charging stations in Hong Kong is merely 23, which can hardly cover the whole city and thus cannot provide EV users with convenient charging services. As a result, concerns and fears on being unable to find a charging port when needed have become a major hurdle to the growth of EV applications in high-density cities (Council 2018).

Meanwhile, existing charging stations still utilize high-polluting fossil fuel-based grid energy rather than cheaper and greener renewable energy. To further improve associated environmental performance, renewable energy (e.g. solar and wind) has been highly recommended to replace conventional grid energy for powering the charging stations (Abbasi et al. 2019). The Department of Energy (DOE) has offered typical templates in utilizing diverse renewable sources to power charging stations (Energy, Accessed on March 2019). Note that when solar energy is selected, the available area for photovoltaic (PV) panels is a major factor that influences the total renewable energy generation and consequently the system configuration in practice (Quddus et al. 2018). Currently, there are two typical types of system configurations of renewable powered charging stations, i.e., distributed family-based one and centralized city-based one. In the United States, house roof area is generally large enough to install PV panels to power distributed family-based EV charging port (Energy 2013). With an integrated battery, energy collected during the day is stored and made available for other periods, turning a house into a personal utility (Energy 2013). In the city of Yantai of China, taking advantage of its large open land in the suburban area, a centralized PV system, occupying 5000 m^2, has been constructed to power the EV charging stations for all the electric buses in the city (Hongwei 2018). The charging stations are also connected with the grid in case of insufficient solar energy generation in cloudy or rainy days.

The above-mentioned two solar-powered charging system configurations are proved to be effective, but they are not suitable for high-density cities due to the extremely limited land resource. In other words, in a high-density city, there is neither enough roof area for each family to install PV panels to power its own charging pile nor enough open land for the city to install a centralized PV system to power the charging stations. This chapter, therefore, proposes a novel geographic information system (GIS)-assisted optimal design method of solar powered charging stations for promoting EV applications in high-density cities. By selecting the optimal locations and optimal number of the renewable powered charging stations with the considerations of the existing charging stations and renewable potentials, the proposed method is able to minimize the life cycle cost of the charging stations while satisfying a user-defined area coverage ratio. To verify the proposed method, case studies were conducted in a real district of Hong Kong. The results were analyzed and compared with other non-optimal options in terms of life cycle cost and district coverage ratio. Meanwhile, this chapter also investigated the impacts of user-defined coverage ratio on the optimal design results.

The proposed design method can be used in practice to help high-density cities complete its public charging network with cost-effectiveness. Providing easy and convenient charging services for EV users, the public charging network will promote EV practical applications and thus can alleviate the roadside air pollutions in high-density cities. The contributions of the present study to the subject are briefly summarized as follows.

- The study adopts the geographic information system techniques for convenient and quick

collection of geographic information of massive buildings, and the information is then used to estimate the building roof solar energy potentials and help select proper locations of the renewable powered charging stations.

- The study proposes a GIS-assisted design method to optimize the charging station locations and number for minimized life cycle cost while satisfying a user-defined area coverage ratio.
- The study reveals the impacts of user-defined area coverage ratio on the optimal design results and the total life cycle costs.

The rest of this chapter is organized as follows. Section 16.2 presents the details of the GIS-assisted optimal design method of renewable powered EV charging stations in high-density cities. Section 16.3 presents a case study by applying the proposed design method in the Kowloon district of Hong Kong. Section 16.3.3 investigates the impacts of user-defined coverage ratio on the performances of the designs. Conclusive remarks are given in Sect. 16.4.

16.2 Geographic Information System-Assisted Optimal Design of Renewable Powered Electric Vehicle Charging Stations

This section introduces the developed optimal design method of renewable-powered EV charging stations in high-density cities. Satisfying a user-defined coverage ratio, the proposed method aims to optimize the locations and number of renewable-powered charging stations to minimize the life cycle cost.

Figure 16.1 presents the basic idea of the proposed design method. It consists of five steps. In Step 1, geographic information system (GIS) techniques are used for quick and convenient collection of the geographic information of

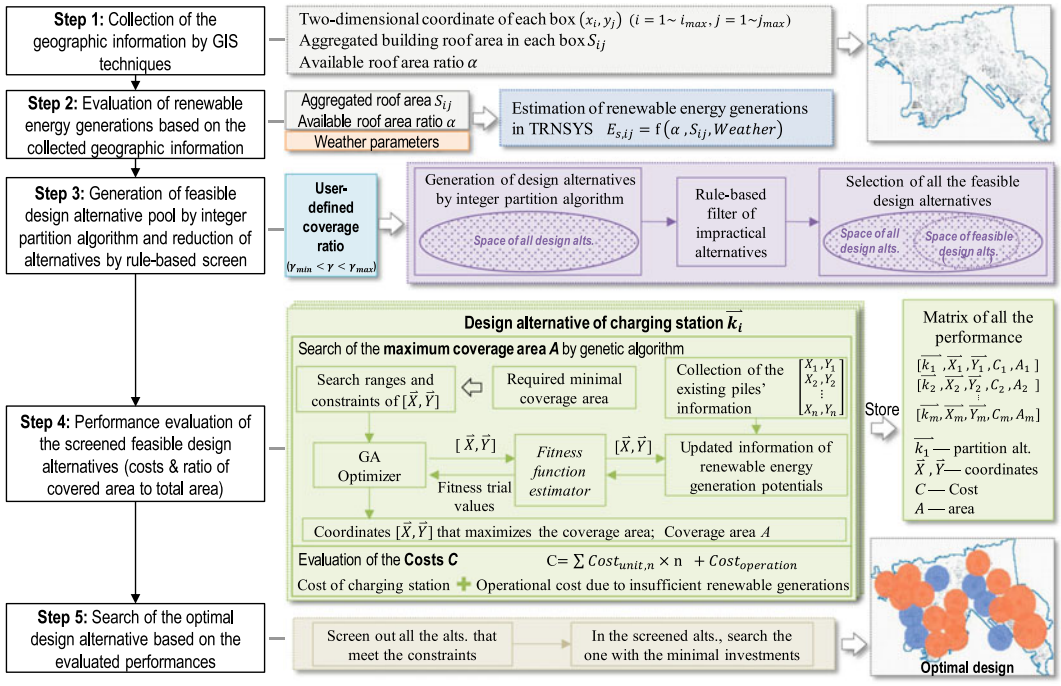

Fig. 16.1 Flowchart of the GIS assisted design method for renewable powered EV charging stations in high-density cities

massive buildings in a district or a whole city. Note that conventional techniques (e.g., survey and field measurement) are time-consuming or even impossible to collect the geographic information (e.g., roof area) of a large number of buildings. The studied district is first discretized by partitioning it into a number of equally sized grids. Then, the geographic information of each grid, including the location, aggregated roof area, and usable roof area ratio, is collected. In step 2, the roof-based solar energy potential in each grid is estimated using the usable roof area collected from step 1 together with the local solar radiation. The solar energy potentials will be used to estimate the operating cost of the charging stations. In step 3, a pool of possible design alternatives is generated by using an integer partition algorithm, and a simple rule-based filter is developed to rule out the unreasonable alternatives, thereby reducing the pool size. In step 4, the performances (i.e., maximum coverage ratio and associated life cycle cost) of each design alternative are evaluated. The charging stations could locate anywhere on the map, leading to different coverage ratios and the associated life cycle costs. Genetic algorithm (GA) is applied for searching the optimal charging station locations that can maximize the area coverage. The associated life cycle cost is also estimated. Note that a maximized area coverage indicates a minimized coverage overlaps among individual charging stations, which is beneficial to achieving a defined area coverage with reduced cost. In step 5, the optimal design alternative is identified by comparing the performance (i.e., maximum coverage ratio and life cycle cost) of the feasible design alternatives obtained from step 4. The details of each step are introduced as follows.

16.2.1 Building Geographical Locations and Roof Areas Obtained Using Geographic Information System Technique

In this step, the geographic information of the studied districts is collected. Due to massive buildings considered, it is impractical to collect the data by conventional techniques (e.g., survey and field measurement). Thus, GIS techniques are used for the quick and convenient collection of the geographic information. The GIS is also prominent at capturing, storing and managing spatially referenced vector and raster data (Miles and Ho 1999).

The processes of geographic information collection and compiling are briefly introduced as follows. First, the original map data of the studied district (e.g., boundaries, latitudes and longitudes of buildings, building area) are collected from the OpenStreetMap (OSM) database (OpenStreetMap, Accessed December 18, 2018). The OSM database is a comprehensive and complete map database with detailed building geographical information of many cities/districts. To ensure the collected information can be easily and mathematically analyzed, the studied district is discretized into small grids (Wang et al. 2019). Then, the information, including the coordinates of each grid and the aggregated building roof area in each grid, is calculated and stored in a two-dimensional matrix. The discretization of the studied districts into small grids and collection of the aggregated geographic information (i.e., locations and aggregated roof area) are automatically implemented in ArcGIS, which is a platform for compiling geographic data and analyzing mapping information (OpenStreetMap, Accessed December 18, 2018). After a series of automatic processing, including creating fishnet, intersecting, calculating geographic and spatial joining, the locations and aggregated roof area data of each grid can be obtained. More details of the associated GIS techniques can be found in ESRI (2019).

Due to factors such as equipping roof-mounted cooling towers and green plants, in most cases not all the roof area can be used for installing PV panels. The area measurement tool in Google Earth will be used to assist the evaluation of the usable roof area ratios. It should be mentioned that the coordinates (indicating building geographical locations) and the aggregated usable building roof areas (indicating renewable potentials) will be used in the

subsequent operating cost evaluation for search of the optimal design alternative.

Note that the owner of roof area can be a second party, different from the charging network developer. To encourage the usage of roof area, the charging station owner/developer can provide economic incentives/subsidies to a second party for renting the space. Meanwhile, the local government can play a lead role in promoting roof-top renewable applications for a greener city. In a high-density city like Hong Kong, there are many public housing buildings directly owned and managed by the government. It will provide chances to install rooftop PV panels for the development of renewable based charging network.

16.2.2 Estimation of Renewable Generation Potentials Based on the Collected Geographic Information

Based on the collected geographic information by GIS techniques, the renewable energy generation in each grid is evaluated in this step. The renewable energy system considered in this chapter is the roof-mounted PV panels. The annual renewable energy generations of the PV panel $E_{s,ij}$ (kW h) is calculated by Eq. (16.1) (Huang et al. 2018b). The estimation of renewable energy generation will be conducted in the commonly used software, i.e., TRNSYS.

$$E_{\mathrm{pv},ij} = \sum_{t=1}^{8760} \left(\tau \times I_{\mathrm{AM},t} \times I_T \times \eta \times \left(\alpha \times S_{ij}\right)\right) \tag{16.1}$$

where τ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1 (Huang and Sun 2019a); I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (kW/m^2) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of PV array; α is the available roof area ratio; S_{ij} (m^2) is the PV surface area, which equals to the aggregated roof area in the specified grid.

The renewable energy system is connected with the power grid (Huang and Sun 2019c). In the operation period, the surplus renewable energy of a charging station will be exported to the power grid, while when the renewable energy is insufficient, the charging station will use the grid energy to charge the EVs. Such process has similar mechanisms to the *energy certificate systems* (Ltd 2014) and *power purchase agreements* (Bruck et al. 2018), which are both used for renewable energy generation and distribution on the power grid.

16.2.3 Generation of a Feasible Design Alternative Pool and Reduction of the Alternatives by a Rule-Based Screen

In this step, a feasible design alternative pool of charging stations is generated by using an integer partition algorithm. Since not all the alternatives are able to produce the optimal performance, a simple rule-based screen is then conducted to filter the obvious unreasonable design alternatives.

16.2.3.1 Initialization of Charging Station Number

Based on a user-defined coverage ratio of the charge station network, the total number of designed charging stations will be initialized in this step. The minimum number of charging stations required is calculated by Eq. (16.2)

$$N_{\min} = \frac{\gamma \times S_{\mathrm{tot}} - S_{\mathrm{cover_exist}}}{S_{\mathrm{cs}}} \tag{16.2}$$

where γ is the user-defined coverage ratio that the charging network should meet, S_{tot} (m^2) is the total area of the districts, $S_{\mathrm{cover_exist}}$ (m^2) is the aggregated area of the district that is already covered by the existing charging stations, and S_{cs} (m^2) is the covered area of a charging station. The required number of charging stations will increase when the user-defined coverage ratio

increases. Meanwhile, the total costs of the charging stations (including initial costs and operational costs) will also increase dramatically. Due to the concern of budget, a maximum required number of charging stations can be defined based on the available budgets.

16.2.3.2 Generation of Possible Design Alternatives by Using an Integer Partition Algorithm

In this step, all possible design alternatives are first generated by the integer partition algorithm. The number of charging stations in one location is in fact influenced by available renewable resource. When there is more renewable energy available in one location, multiple charging stations can be constructed. Considering that there can be multiple charging stations in one location, the integer partition algorithm will be used to identify all the possible design alternatives of charging stations.

Integer partition examines how to represent a given integer n with the sum of one or more positive integers x_i, i.e. $n = x_1 + x_2 + \cdots + x_m$ (Stojmenović and Zoghbi 1998). In such representation, the order of x_i is of no consequence, and thus two partitions of an integer n are distinct only if they differ with respect to the x_i they contain. Extensive studies have been conducted to investigate the integer partition problem. The theory of partition problem was introduced by Andrews (1976). Based on such theory, many integer partition algorithms have been developed, such as parallel algorithms (Akl 1993), Tree traversal related algorithms (Fenner and Loizou 1983), and loop-free algorithms (Fenner and George 1981). Due to its simplicity and convenience, this chapter selects the algorithm developed in Stojmenović and Zoghbi (1998) for partitioning integers.

For a given n, one way of integer partition is considered as 1 design alternative. For instance, there are 7 ways of partition when n equals 5, and thus there are 7 design alternatives (see Table 16.1). The partitioned integer n varies from $N_{\min}$ to $N_{\max}$, as specified in Sect. 16.2.3.1. These design alternatives (when n varies from $N_{\min},\ldots,N_{\max}$) together form the space of all possible design alternatives. When the number of a charging station is 1, its coverage area is S_{cs} (m^2). Similarly, for a charging station of number n, its coverage area is $n \times S_{cs}$ (m^2).

16.2.3.3 Rule-Based Filter of Impractical Alternatives

Many design alternatives, generated by the integer partition algorithm from Sect. 16.2.3.2, are obviously not economic and thus they should be ruled out for the final optimal search. To improve the optimization efficiency through reducing the number of design alternatives, a rule-based screen is introduced to rule out the non-economic alternatives. In this chapter, one rules

Table 16.1 An example of design alternative pool using the integer partition algorithm

Alternatives	Number of charging stations in each location				
	Location 1	Location 2	Location 3	Location 4	Location 5
1	5	–	–	–	–
2	4	1	–	–	–
3	3	2	–	–	–
4	3	1	1	–	–
5	2	2	1	–	–
6	2	1	1	1	–
7	1	1	1	1	1

will be adopted as below to rule out the non-economic alternatives.

Rule: The design alternatives $\vec{k}_k$, with the number of charging stations larger than a threshold value $k_{\max}$, should be removed. This is because in comparison with distributed charging stations, placing too many charging stations in a same location could reduce the possibility of utilizing all renewable energy, thereby increasing operating costs as well as life cycle costs.

$$\text{Remove } \vec{k}_k, \text{if } \max\left(\vec{k}_k\right) \geq k_{\max} \qquad (16.3)$$

The threshold value $k_{\max}$ is determined by the radius of the largest circle that can be put inside the studied district. For any design alternative, its maximum number of charging stations at a same location should not be larger than $k_{\max}$. Otherwise, the coverage area of the charging stations will exceed the district, leading to useless area coverage and increased operating cost.

After applying the rule-based filter, the feasible design alternatives are selected out. Their performance will then be evaluated in the next step.

16.2.4 Performance Evaluation of the Feasible Design Alternatives

This step evaluates the performances of the screened feasible design alternatives in terms of the maximum coverage area and total costs. For a design alternative, its life cycle cost will approach the minimum when it has the largest area coverage ratio. This is because once the number of charging stations is determined, the initial cost is nearly fixed, and thus the life cycle costs are mainly determined by the operating cost. With the increased coverage ratio, the coverage overlaps among individual charging stations will decrease. As a result, the operational cost, mainly caused by the grid energy imports, will decrease when there is insufficient renewable generation. Thus, in the genetic algorithm, maximizing the coverage area is directly used as the fitness function. The locations of charging stations are the variables to be optimized. Based on the searched locations, the overlaps of charging station coverages are determined, and then the operating cost is estimated with the available renewable energy. Last, the life cycle cost is calculated.

16.2.4.1 Search of the Maximum Coverage Area by Genetic Algorithm

When charging stations are placed in different locations, their effective coverage area can be different. Since genetic algorithm (GA) is effective in searching the global or near optimal solutions (Huang et al. 2018c), it was used for identifying the maximum coverage area. For each alternative (e.g., alternative $\vec{k}_k$ with N charging stations placed in m locations), the GA algorithm searches the optimal X–Y coordinates (e.g. $[x_1, x_2,\ldots,x_m; y_1, y_2,\ldots,y_m]$) of the designed charging stations that maximize the coverage ratio. In each generation of GA, the trials of the X–Y coordinates (e.g. $[x_1, x_2,\ldots,x_m; y_1,y_2,\ldots,y_m]$) are generated by the GA optimizer to evaluate the effective coverage ratio. Figure 16.2 presents the chromosomes in the GA optimization. There are $2 \times m$ variables to be optimized in each GA optimization. x_i is within the range [0,233], and y_i is within the range [0,163]. Both x_i and y_i can be represented by an 8-bit binary number. In total, the chromosome has $16 \times m$ bits. More details about the GA can be found in Chipperfield and Fleming 1995).

The fitness function of the GA search is expressed by Eq. (16.4), which aims at maximizing the coverage ratio of the designed charging stations by searching the optimal X–Y coordinates.

$$J_{\text{area}} = \max(\gamma) \qquad (16.4)$$

γ is the coverage ratio of the designed charging stations under the X–Y coordinate trials, which is calculated by comparing the area covered by charging stations with the total district area considered, see Eq. (16.5).

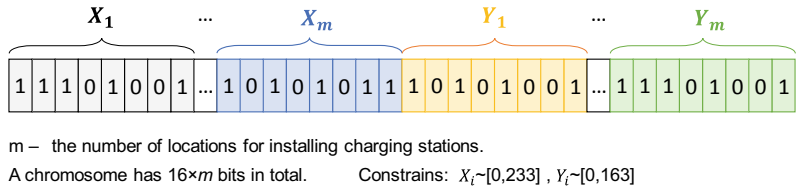

Fig. 16.2 Illustration of the chromosomes used in the GA optimization

$$\gamma = \frac{S_{\text{cover_exist}} + S_{\text{cover_new}}}{S_{\text{tot}}} \tag{16.5}$$

$S_{\text{cover_exist}}$ (m^2) and $S_{\text{cover_new}}$ (m^2) are the coverage area of the existing charging stations and the newly designed charging stations, respectively. S_{tot} (m^2) is the area of the whole district. The coverage area (S_{cover} (m^2)) of the charging stations in one location is determined by the number of charging stations n.

$$S_{\text{cover}} = n \times S_{\text{cs}} \tag{16.6}$$

$$S_{\text{cs}} = \pi R^2 \tag{16.7}$$

where R (m) is the coverage radius of one charging station. This chapter assumed that the EVs are evenly distributed in the considered district. Under such an assumption, the coverage area is simply determined through dividing the number of EVs that a charging station can accommodate by the statistical EV distribution density. Please note that the proposed method can also consider more complex EV distributions and then search for the corresponding optimal number and locations of the charging stations.

The X–Y coordinates should meet the following constraints. The x_1, x_2,…,x_m should be within the range [$x_{\min}$, $x_{\max}$], and y_1, y_2,…,y_m should be within the range [$y_{\min}$, $y_{\max}$].

$$\begin{aligned} x_{\min} \le x_1, x_2, \ldots, x_m \le x_{\max}, \\ y_{\min} \le y_1, y_2, \ldots, y_m \le y_{\max} \end{aligned} \tag{16.8}$$

Note that the overlapped area should not be repeatedly considered when there are overlaps between the coverage of different charging stations. This is conducted by the following process: (1) Based on the original renewable production matrix (i.e., $\left[E^0_{\text{pv},ij}\right]_{I\times J}$ obtained in Sect. 16.2.2), the renewable production in the coverage of the 1st charging station is calculated by aggregating the production in each grid (i.e., represented by index i and j) within its coverage. (2) After calculation for the first charging station, the renewable production in the calculated grids are reset to 0. The original renewable production matrix is updated to be $\left[E^1_{\text{pv},ij}\right]_{I\times J}$. (3) Based on the updated renewable production matrix (i.e., $\left[E^1_{\text{pv},ij}\right]_{I\times J}$), the renewable production in the coverage of the 2nd charging station is calculated by aggregating the production in each grid within its coverage. The process is repeated until all the charging stations are calculated. By resetting the renewable production of the calculated grids to 0 and updating the renewable production matrix, the repeated calculations of the overlapped areas among different charging stations are avoided.

16.2.4.2 Analysis of Life Cycle Cost

The considered costs of charging stations consist of two parts, including initial cost ($\text{Cost}_{\text{initial}}$ (USD)) for installing charging stations and renewable energy systems (i.e., PV panels), and operational cost ($\text{Cost}_{\text{operation}}$ (USD)) due to insufficient renewable energy supply during the considered life cycles (T (years)), as described by Eq. (16.9).

$$C = \text{Cost}_{\text{initial}} + \text{Cost}_{\text{operation}} \times T \tag{16.9}$$

The initial cost ($\text{Cost}_{\text{initial}}$ (USD)) includes the charging station cost ($\text{Cost}_{\text{station}}$ (USD)), and renewable energy systems cost (Cost_{pv} (USD)), which are calculated by Eqs. (16.11) and (16.12), respectively.

$$\text{Cost}_{\text{initial}} = \sum_{k=1}^{m} \text{Cost}_{\text{station},k} + \text{Cost}_{\text{pv}} \quad (16.10)$$

$$\text{Cost}_{\text{station},k} = \alpha^{n_k} \times \text{Cost}_{\text{charge,unit}} \times n_k \quad (16.11)$$

Usually, the initial cost of charging stations will gradually increase with the increased number of charging stations, but the increase is not in a simple linear relationship, since a large scale charging station (e.g., two centralized charging stations) can save some costs in electricity cables, installation costs, land area, etc. as compared with several small-scale charging stations which have the same aggregated number (e.g., two distributed charging stations). With reference to Commission (2013) and Shah et al. (2018), this chapter considered a discount coefficient α in the evaluation of charging stations costs. In Eq. (16.11), $\text{Cost}_{\text{charge,unit}}$ (USD) indicates the unit initial cost of a charging station, and n_k indicates the number of charging stations in the kth ($k = 1,\ldots,m$) location.

$$\text{Cost}_{\text{pv}} = \text{Cost}_{\text{pv,unit}} \times S_{\text{pv}} \quad (16.12)$$

The initial cost of PV panels (Cost_{pv} (USD)) is considered to be proportional to the area of the PV panels (Huang et al. 2018a). $\text{Cost}_{\text{pv,unit}}$ (USD/m^2) is the unit price for installing PV panels, and S_{pv} (m^2) indicates the area of the installed PV panels.

The operation cost ($\text{Cost}_{\text{operation}}$ (USD)) of the solar-powered EV charging stations is considered since the renewable generations in some areas may not be enough to meet the EV electricity demand, and thus the power grid electricity is needed. The operational cost is determined based on the renewable energy shortage, as calculated by Eq. (16.13) (Huang et al. 2019).

$$\text{Cost}_{\text{operation}} = \rho \times \sum_{k=1}^{m} (-\Delta E_k) \quad (16.13)$$

where ρ (USD/(kW h)) is the electricity price in the local market, and ΔE_k (kW h) is the renewable energy shortage in the kth location, which is calculated by Eq. (16.14).

$$\Delta E_k = \begin{cases} E_{\text{pv},k} - E_{\text{th,unit}} \times n_k, & \text{if } E_{\text{pv},k} < E_{\text{th,unit}} \times n_k \\ 0, & \text{if } E_{\text{pv},k} > E_{\text{th,unit}} \times n_k \end{cases} \quad (16.14)$$

where $E_{\text{pv},k}$ (kW h) is the annual renewable energy production by the PV panels in the kth location, which is calculated by Eq. (16.1). n_k represents the number of charging station in the kth location. $E_{\text{th,unit}}$ (kW h) indicates the threshold of annual energy demand required by the EVs, which should be set based on the statistical analysis of the local EV density. Note that when there is renewable shortage, the insufficient electricity will be imported from the power grid.

Note that the proposed method can calculate the operating cost with smaller time intervals (e.g., minute interval and second interval) if the provided inputs can achieve such high time resolutions. A smaller time interval can help improve the accuracy of the operating cost estimation. In practical applications, high-time-resolution data (e.g., solar radiation) could be difficult to be attained and thus cause the differences between the estimated operating cost and the actual one. To reduce such differences, a proper sized storage unit can be installed.

16.2.5 Search for the Optimal Design Alternative by Comparing the Obtained Performance

After performance evaluation of each feasible design alternative, the optimal design that meets the user-defined coverage ratio is identified in this step. The search consists of two sub-steps. First, the feasible design alternatives that meet the user-defined coverage ratio are selected. Then, these selected design alternatives are sorted by their life cycle costs. The one with the minimal costs is selected as the optimal design and will be finally used for the design of EV charging stations.

The outputs of the proposed design methods consist of the number (m) of locations with

charging stations constructed, the coordinates of each location ($[x_1, x_2,\ldots,x_m; y_1, y_2,\ldots,y_m]$), the number of charging stations ($[n_1, n_2,\ldots,n_m]$) in each location, the coverage ratio (γ) and the total costs (C).

16.3 Application of the Proposed Design Method in Hong Kong

To illustrate the proposed method, case studies were conducted in the Kowloon District of Hong Kong. The radius of the coverage area was set to be 750 m as an example. Note that in practice, this radius of coverage should be determined and adjusted based on the statistical analysis of EV density. The threshold of renewable insufficiency $E_{\text{th,unit}}$ was set to 5×10^7 kW h with reference to the density of EVs in Hong Kong (Department 2018c). Table 16.2 summarizes the parameters related to the cost evaluation of solar-powered EV charging stations.

The user-defined coverage ratio of EV charging stations was set as 40% as an example in this section. Different user-defined coverage ratios have been further discussed in Sect. 16.2.2.

16.3.1 Geographic Information Collected by Geographic Information System

The geographic information of the Kowloon district was collected by GIS techniques (OpenStreetMap, Accessed December 18, 2018). Figure 16.3 presents the collected geographic information. The red curve indicates the boundary of the studied district, and the area with dark gray color indicates the building roofs. The whole district was discretized by partitioning it into 50 × 50 m grids, i.e., the length and width of each grid equals 50 m. In total, there were 37,979 (X-coordinate 233 × Y-coordinate 163) grids. Using the GIS technique, the building area in each grid was also obtained and stored in a matrix with dimensions of 233 × 163.

Due to factors such as equipping roof-mounted cooling towers (e.g., see Fig. 16.4a) and green plants, in most cases not all the roof area can be used for installing PV panels. To investigate the available roof area ratio, 500 buildings (consisting of commercial buildings, residential buildings and factories) were selected in different locations of Kowloon, and their available roof areas were estimated by Google Earth. An example is presented in Fig. 16.4a. Figure 16.4b presents the statistical analysis results of the available roof area ratio based on the estimated ratio of these 500 buildings. The mean of available roof area ratio was 70%. This ratio was used to evaluate the renewable energy generations (see Eq. (16.1)) in the following steps.

16.3.2 Renewable Energy Generation Evaluation Results

After obtaining the geographic information by GIS techniques, the renewable energy generations in each grid was evaluated. Using the

Table 16.2 Parameter values used for cost calculation

ID	Parameter	Definition	Value	Reference
1	α	Discount rate due to increase of charging station scales	0.98	Shah et al. (2018)
2	ρ	Electricity prices (USD/(kW h))	0.2	Huang and Sun (2019b)
3	$\text{Cost}_{\text{pv,unit}}$	Unit price of PV panels (USD/(m^2))	150	Huang et al. (2018a)
4	$\text{Cost}_{\text{charge,unit}}$	Unit price of charging stations (USD)	2×10^4	Xylia et al. (2017)
5	T	Considered life cycles (year)	30	Huang et al. (2018a)

Fig. 16.3 Collected building roof area information from GIS (the dark color indicates buildings)

(a) Illustration of the available and unavailable are in a case building

(b) Statistical analysis of the available roof area ratio from 100 typical buildings

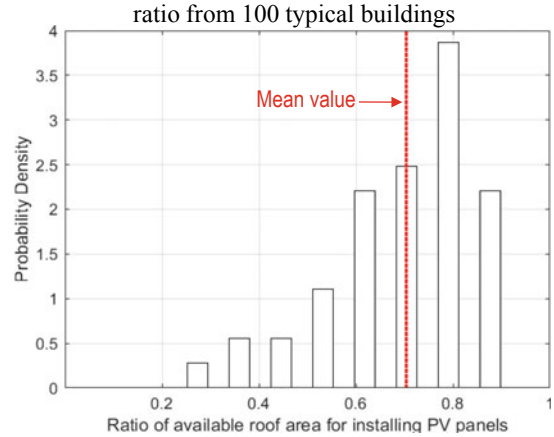

Fig. 16.4 Estimation of the available roof area ratio for installing PV panels

weather data in Hong Kong, the annual renewable energy generations was calculated in TRNSYS. Figure 16.5 presents the renewable energy generations in each grid. The blue dotted curve is the boundary of Kowloon district. In some grids, due to the high density of buildings,

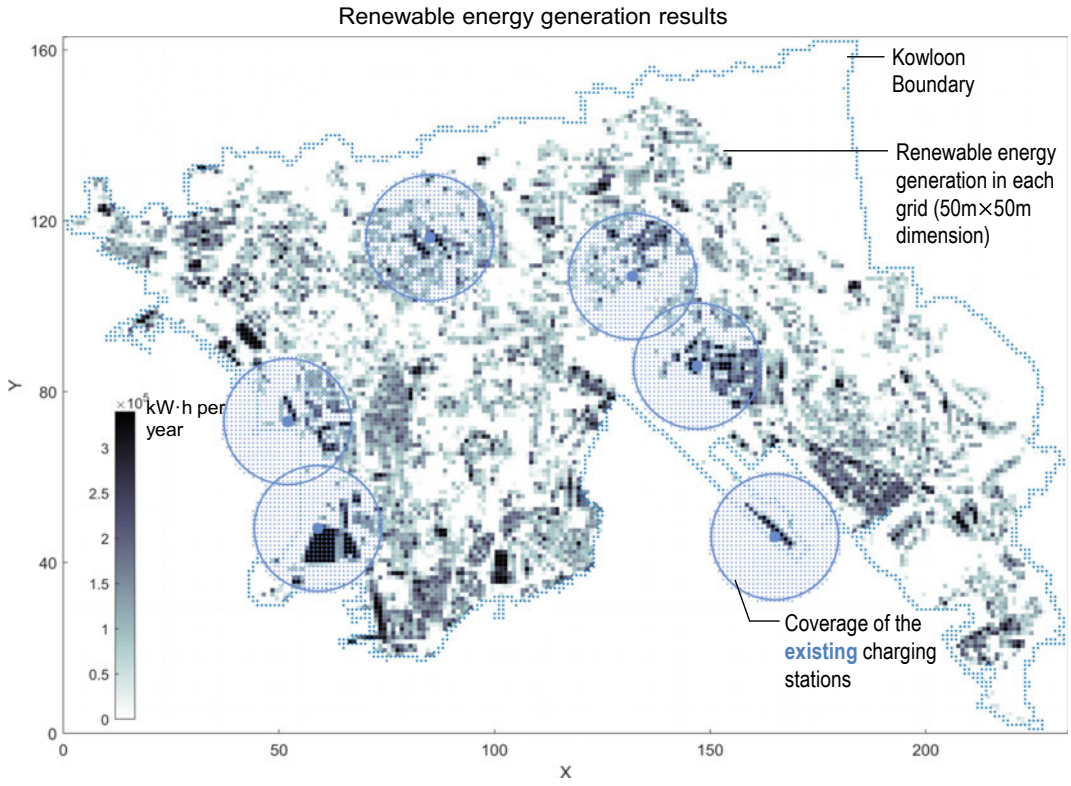

Fig. 16.5 Evaluation results of renewable energy generations in each microgrid in Kowloon

the annual renewable generations can be as high as 3.4×10^5 kW h. While in some grids, due to the low density of buildings, the annual renewable generation can be very small. In some mountain districts, there are no buildings constructed, and thus the renewable energy generation is equal to 0.

According to the Tesla Hong Kong website (Tesla, Accessed in 2019), currently there are 6 public charging stations located in the Kowloon District. The geographic information (i.e. latitude and longitude) of these six existing charging stations are summarized in Table 16.3. Then, these latitude and longitude coordinates were converted

Table 16.3 Survey of existing public charging stations

ID	Name	Original coordinates		Converted coordinates	
		Latitude	Longitude	X	Y
1	Festival Walk Supercharger	22.33757	114.1742	85	116
2	Mikiki Supercharger	22.33359	114.1969	132	107
3	E-Max Supercharger	22.32396	114.2042	147	86
4	Olympian City Supercharger	22.31802	114.1584	52	73
5	Elements Supercharger	22.3068	114.1615	59	48
6	Kai Tak Supercharger	22.30611	114.2132	165	46

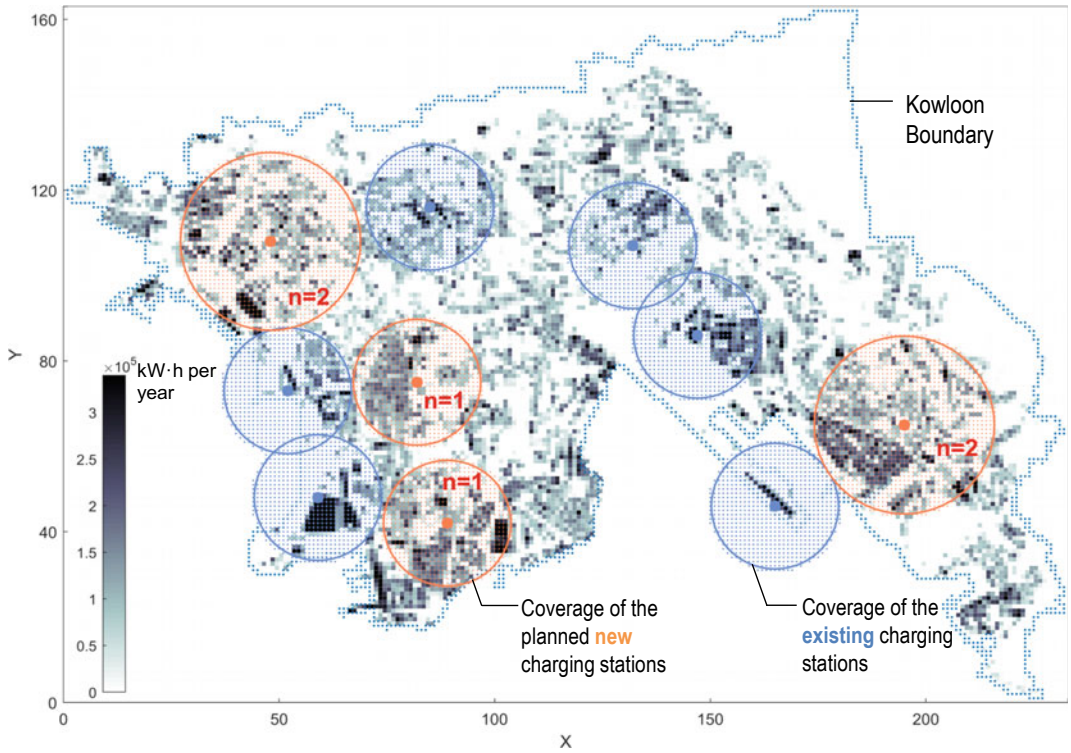

Fig. 16.6 Optimized location and number of charging stations when the required coverage ratio was 40%

to the coordinates in the grid matrix obtained in Sect. 16.3.1 for further mathematical analysis. The scale of these charging stations was considered to be 1. Figure 16.6 highlights the locations of the six existing charging stations (i.e., the blue filled dots) and their coverage area (i.e., the blue circles) in the Kowloon districts. The locations of the existing charging stations as well as the renewable energy generation information were used as the inputs in the following design optimization.

16.3.3 Initialization of Design Alternatives

After obtaining the renewable energy generation and the locations of the existing charging stations, the possible design alternatives were generated by the integer partition algorithm. Calculated by Eq. (16.2), the minimum number of required charging stations was 5 for a user-defined coverage ratio of 40%. Considering budget limit, the maximum number of designed charging stations was set as 20. When N varied from 5 to 20, the total number of possible partition ways was 2702 (see Fig. 16.8). The maximum number of charging stations in one location was set to 8, with reference to the size of the Kowloon district on the map. After applying the rules-based filter, the number ofs feasible design alternatives was reduced to 1826 (see Fig. 16.8). Table 16.4 summarizes the counting results of the removed and remained design alternatives under different number of charging stations.

16.3.4 Performance Evaluation and Search Results of the Optimal Design

After obtaining the feasible design alternatives, the cost performance and the maximum coverage ratio of each alternative were evaluated. To guarantee that the GA algorithm terminated by

Table 16.4 Counting of design alternatives under different number of charging stations

Number of charging stations	5	6	7	8	9	10	11	12
Number of partitions	7	11	15	22	30	42	56	77
Number of removed alternatives	1	1	1	1	2	4	7	12
Number of remained alternatives	6	10	14	21	28	38	49	65
Number of charging stations	13	14	15	16	17	18	19	20
Number of partitions	101	135	176	231	297	385	490	627
Number of removed alternatives	19	30	45	67	96	137	190	263
Number of remained alternatives	82	105	131	164	201	248	300	364
Total number of alternatives						2702		
Total number of remained alternatives						1826		

reaching the fitness limit, the GA parameters were tuned using the trial and error method. The number of generations was set as 100, the population size was set as 500, and the termination tolerance was set as 1e-6. Based on the user-defined coverage ratio (i.e., 40%), 1814 design alternatives, which had a coverage ratio over 40%, were selected and sorted. The one with the minimal cost was selected as the optimal design. The coverage ratio of the optimal design was 43.2%, and the total cost was about 584 kUSD.

Figure 16.6 displays the locations and number of charging stations in the optimal design alternative on the map. The blue dots indicate the locations of the existing charging stations, and the blue circles represent their coverage areas. The orange dots indicate the locations of the newly designed charging stations, and the orange circles represent their coverage ranges. n indicates the number of charging stations in one location. As depicted by Fig. 16.6, adding six more charging stations could meet the user-defined coverage ratio. The optimal design has two features. First, there were very limited overlaps between the designed charging stations. Reducing the overlaps between the coverage ranges of different charging stations can help increase the utilization of renewable energy, and thereby lowering down the operational cost. Another feature is that most of the charging stations were installed in the locations where there were sufficient renewable energy generations. This is because increased renewable generations can help reduce the operational cost.

To better explain the performances, the optimal design was compared with four near-optimal designs, in which the total number of charging stations was the same (i.e. 6). In Fig. 16.7, ΔE (10^6 kW h) indicates the amount of renewables shortage in each location (calculated by Eq. (16.9)), C_{in}, C_{op}, C_{tot} (kUSD) represents the initial costs, operational costs due to renewables shortage, and total costs of the design alternatives, respectively, and γ is the coverage ratio of the design alternative. Three different regions were selected for detailed analysis.

In Region *a*, the renewable energy generations can sufficiently supply up to two centralized charging stations (i.e., Fig. 16.7a, d, e). When more than two charging stations are installed (i.e., Fig. 16.7c) or distributed charging stations are installed (i.e., Fig. 16.7b), there will be renewable energy shortages. This is because the layout of two centralized charging stations fits the distribution of renewable energy generations better in this region. For instance, as shown in Fig. 16.7b, there was a 3×10^6 kW h renewable energy shortage when installing two distributed charging stations. The renewable energy shortage will increase to 20×10^6 kW h when installing 3 charging stations, as shown in Fig. 16.7c. Consequently, the increased renewable energy shortage will lead to increased operational costs.

In Region *b*, the renewable energy generations can sufficiently supply up to three distributed

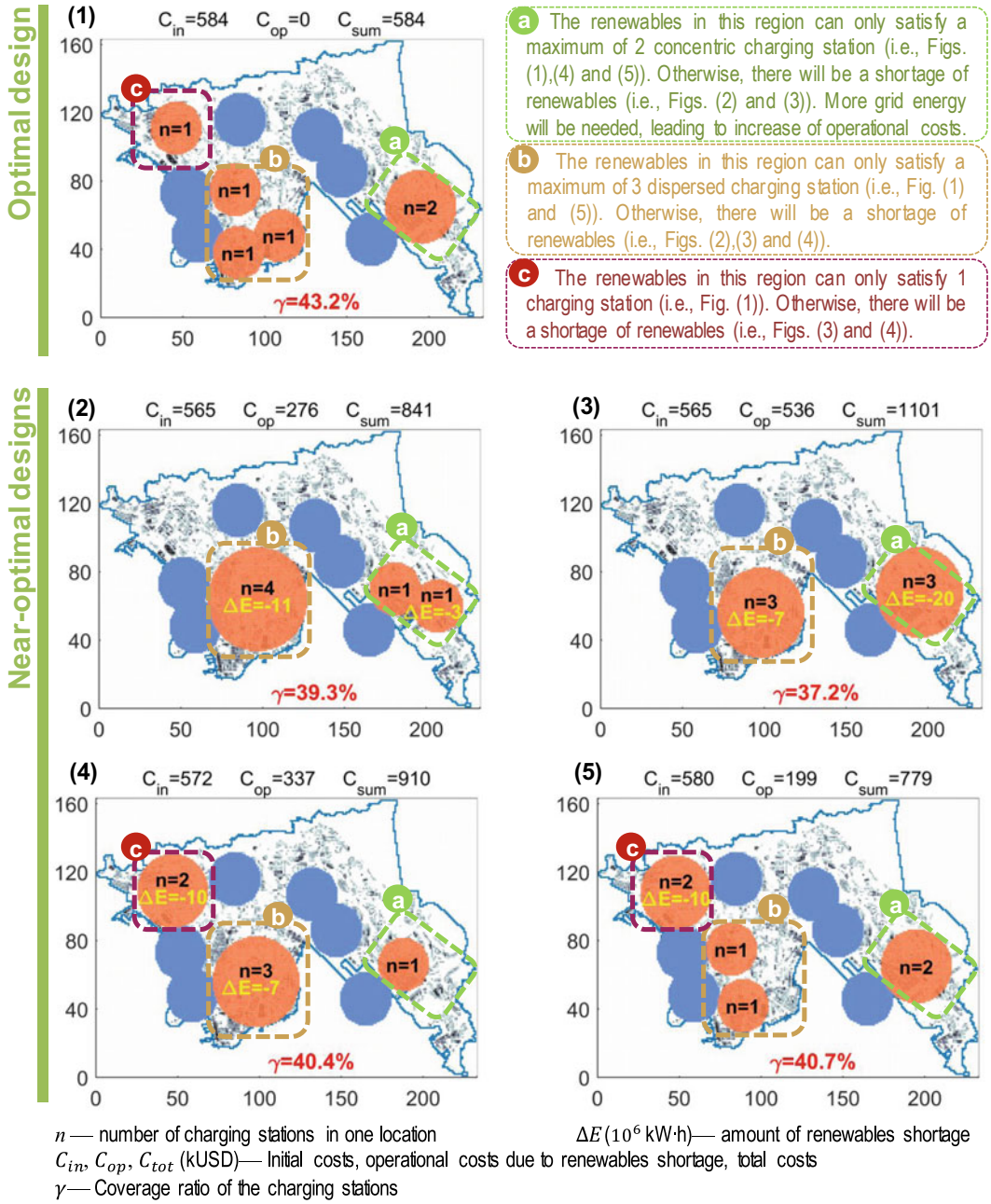

Fig. 16.7 Performances comparison of the proposed design with four near-optimal designs

charging stations (i.e., Fig. 16.7a, e). When more than three charging stations are installed (i.e., Fig. 16.7b) or centralized charging stations are installed (i.e., Figs. 16.7c, d), there will be renewable energy shortages. This is because the layout of three distributed charging stations can better fit the distribution of renewable energy generations in this region. For instance, as shown in Fig. 16.7c, d, there was a 7×10^6 kW h renewable energy shortage when installing three

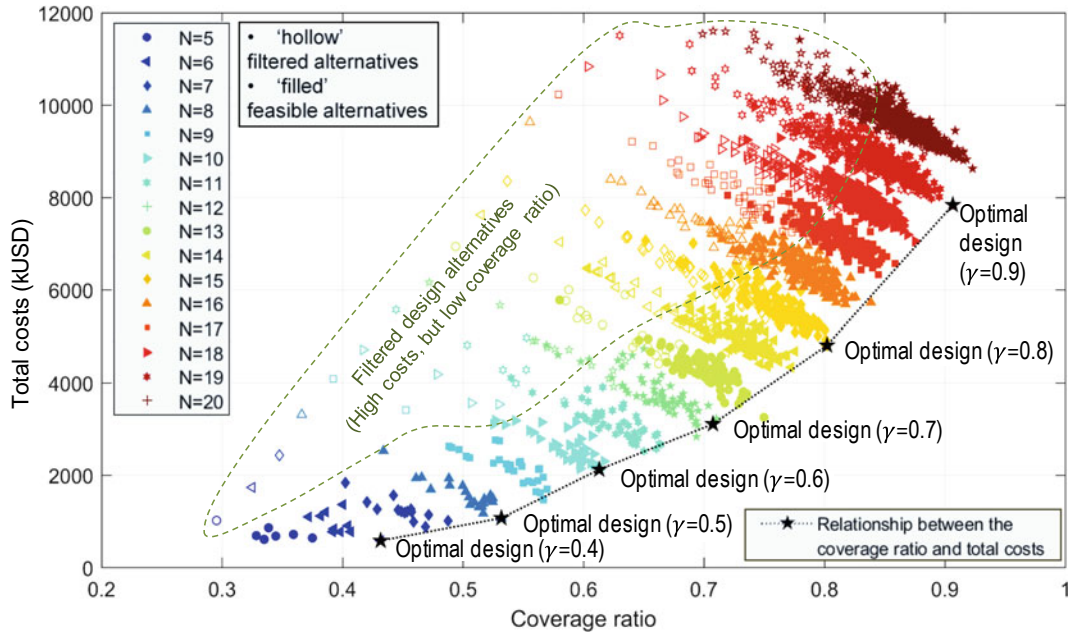

Fig. 16.8 Performances of all the design alternatives and relationship between the minimum required investments and the required coverage ratio

centralized charging stations. The renewable energy shortage will increase to as high as 11×10^6 kW h when installing four charging stations, as shown in Fig. 16.7b.

In Region *c*, the renewable energy generations can sufficiently supply up to one charging station (i.e., Fig. 16.7a). The renewable energy shortages will be resulted when more than one charging stations were installed (i.e., Fig. 16.7d, e). This is due to the limited renewable energy available in this region. For instance, as shown in Figs. 16.7d, e, there was a 10×10^6 kW h renewable energy shortage when installing two centralized charging stations.

With the increase of renewable energy shortage, the operational costs of design alternatives will gradually increase. With the centralization of charging stations, the initial costs of design alternatives will gradually decrease. Compared with the four near-optimal designs, the initial cost of the optimal design (i.e., Fig. 16.7a) was the largest due to the dispersion of charging stations. However, because of the proper design of charging station locations and numbers, the renewable energy shortage of the optimal design was well controlled at 0, and thus its operational cost was the smallest. Overall, the optimal design showed the best economic performance.

The optimal design also had the largest coverage ratio (i.e. 43.2%) compared with the four near-optimal designs which had the same total number of charging stations. Even though the aggregated area of the circles was similar in different alternatives, their effective coverages of occupancy area were different, since the buildings were not evenly distributed in the Kowloon district.

16.4 Discussions

The user-defined coverage ratio has significant impacts on the optimal design of renewable-powered EV charging stations. This section analyzed the performance of all the design alternatives and investigated the relationship between the required coverage ratios and the costs.

Figure 16.8 presents the coverage ratio and cost performance of all the design alternatives, including the removed design alternatives by the pre-set two rules in Sect. 16.2.3.3 (i.e., the 'hollow' marks) and the remaining feasible design alternatives (i.e., the 'filled' marks). As can be seen, the rule-based filter was effective in selecting most of the non-optimal design alternatives, which had low coverage ratios but high total costs (see the region within the green dashed curve). N indicates the total number of charging stations, and the different values were highlighted by the different marks and colors. Both the average total costs and average coverage ratio gradually increased with the increase of charging station total number. The optimal design under a set of user-defined coverage ratios, ranging from 0.4 to 0.9 with an interval of 0.1, was searched and marked by black pentagrams in Fig. 16.8. The total costs of the optimal design increased in an exponential relationship when increasing the required coverage ratio.

The number of possible design alternatives will increase with $N_{\min}$, leading to increased computational load. There are two means to improve the computational efficiency. First, a large district can be portioned into several small districts, which break downs a complex optimization problem into several simpler ones, thereby reducing the complexity and computation load. Second, since searching of the optimal locations for each alternative is independent, parallel computing techniques can be employed to reduce search duration. By district partitioning and parallel computing, the computing time can be significantly reduced.

The layouts of the six selected optimal designs, which were determined under the required coverage ratios ranging from 0.4 to 0.9, are presented in Fig. 16.9. It can be seen that the overlaps between the area coverages of different charging stations increased with the increase of the required coverage ratios. When the required coverage ratio γ was larger than 0.7, there was a dramatic increase in the operational costs. This was because the overlaps between the area coverages of different charging stations would increase inevitably and significantly for high required coverage ratios. Due to the increased overlaps of coverage area and increased number of charging stations, the total costs will increase dramatically when the required coverage ratio is large. The developed design method is effective in optimizing the number and locations of charging stations in high-density cities under different user-required coverage ratios.

16.5 Summary

This chapter has proposed a novel geographic information system (GIS)-assisted optimal design method of renewable powered electric vehicle charging stations for high-density cities. By selecting the optimal locations and number of renewable powered charging stations considering the existing charging stations and renewable potentials, the proposed method is able to minimize the life cycle cost of the charging stations while satisfying a user-defined area coverage ratio. The developed method consists of the five steps. First, the geographic information of massive buildings in the studied district is collected by GIS techniques. The studied district is discretized into equally sized grids, and the geographic information of each grid (i.e., locations, aggregated roof area, and usable roof area ratio) is collected. Then, using the collected usable roof area together with the local solar information, the roof based solar energy potential in each grid is estimated. Next, a pool of feasible design alternatives is generated by using an integer partition algorithm, and a rule-based screen is applied for reducing the number of alternatives. After that, the performances (i.e., maximum coverage ratio and minimal life cycle cost) of all the selected feasible design alternatives are evaluated. Finally, the optimal design is identified by comparing the evaluated performance. Case studies have been conducted in the Kowloon district of Hong Kong. For validation purpose, the coverage ratio and cost performance of the optimal design have been compared with several near-optimal designs, and the performance improvements have been analyzed. The impacts of the required coverage ratio on the optimal

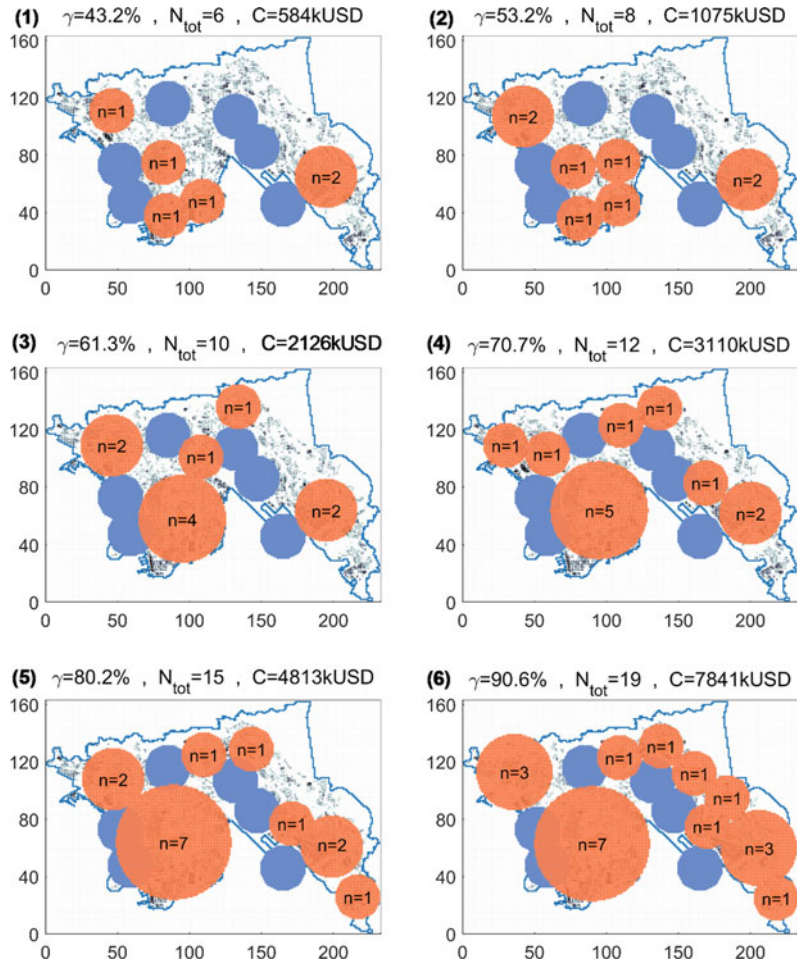

Fig. 16.9 Optimized location and number of charging stations under different required coverage ratio

design have also been investigated. The major findings are summarized as follows.

- With the assistance of GIS techniques, the proposed design method is able to conveniently and quickly collect the geographic information of massive buildings, which is then used to estimate the building roof solar energy potentials and help select proper locations of the renewable powered charging stations.
- By locating the charging stations in sites with large solar power potential and minimizing the overlaps between different charging stations' coverage, the proposed method is effective in minimizing the life cycle cost of charging stations while satisfying a user-defined area coverage ratio. In the case studies, the coverage ratio of the optimal design was 43.2%, which was much higher than the ratios of the four near-optimal designs (i.e., 37.2–40.7%). The life cycle cost of the optimal design was 584 kUSD, which was much lower than the costs of the near-optimal designs (i.e., 779–1101 kUSD).
- The rule-based filter can effectively reduce the considered number of design alternatives, thereby largely improving the computational efficiency.
- The life cycle cost of the renewable powered charging stations increases in an exponential relationship with the increase of the required coverage ratio. There was a dramatic increase in total cost when the required coverage ratio is large, since the operational cost will

increase inevitably and dramatically due to the increased overlaps between the coverage range of different charging stations.

It should be mentioned that one limitation of this chapter is that the 3D information (e.g., height and shape) of the buildings and their impacts on the solar radiation capture of the adjacent buildings were not considered due to lack of proper computation-efficient modeling tools. This may cause differences between the predicted renewable potentials and actual renewable generations. In the future, with more advanced district-/city-level solar radiation modeling tools developed, the PV power generations can be more accurately simulated, thus improving the performance of the developed method. The density of vehicles is an important factor that may affect the design of charging network. Future work will take account of such factor with reference to the practical data. The renewable energy generations may be affected by uncertain factors such as weather conditions. Future work will also take such uncertain factors into consideration.

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Clustering Nearly Zero Energy Buildings for Improved Performance

17

Pei Huang and Yongjun Sun

Abstract

Collaborations among nZEBs (e.g., renewable energy sharing) can improve nZEBs' performance at the community level. To enable such collaborations, the nZEBs need to be properly grouped. Grouping nZEBs with similar energy characteristics merely brings limited benefits due to limited collaboration existed, while grouping nZEBs with diverse energy characteristics can bring more benefits. In the planning of nZEB communities, due to the large diversity of energy characteristics and computation complexity, proper grouping that maximizes the collaboration benefits is difficult, and such a grouping method is still lacking. Therefore, this chapter proposes a clustering-based grouping method to improve nZEB performance. Using the field data, the grouping method first identifies the representative energy characteristics by advanced clustering algorithms. Then, it searches the optimal grouping alternative of these representative profiles that has the optimal performance. For validation, the proposed grouping method is compared with two cases (the nZEBs are either not grouped or randomly grouped) in aspects of economic costs and grid interaction. The study results show that the developed method is effective in improving nZEBs' performance at the community level. The proposed method will provide the decision makers a means to group nZEBs, which maximizes the collaboration benefits and thus assists the planning of nZEB communities.

Keywords

Nearly zero energy building · Community · Clustering · Grouping · Collaborations

17.1 Introduction

Buildings consume over 40% of the overall energy worldwide (Kolokotsa et al. 2011). Nearly zero energy buildings (nZEBs) are widely considered as a promising solution to the increasing energy problem in the building sectors (Crawley et al. 2009; Hernandez and Kenny 2010; Marszal et al. 2011). The nZEBs are such buildings that are equipped with renewable energy systems and have annual renewable energy generations no smaller than their energy demands (Marszal et al. 2011; Hernandez and Kenny 2010). Many countries and organizations

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have established regulations on nZEBs for enhancing their applications, such as the ZEBRA 2020 (nearly zero energy building strategy 2020) established by 17 European Member States, and the 'nearly/net zero energy buildings target' launched by the European Directive on Energy Performance of Buildings. In recent years, the increasing application of nZEBs has boosted interest in the next-generation buildings called nearly zero energy buildings community (sometimes called nZEB cluster) (Odonkor and Lewis 2015, 2016; Deng et al. 2014). The nZEB community is based on a collaboration concept that the nZEBs can freely share their renewable energy generations, storage, and information with each other inside the community (Odonkor and Lewis 2016; Li and Wen 2017; Jafari-Marandi et al. 2016). The collaborations among the nZEBs can bring many benefits at the community level, such as reducing operational costs and reducing grid interaction (Sun et al. 2018). Some efforts have been made to explore the implementation of nZEB community, such as the Clinton Climate Initiative C40 program (Group 2011), IEA ECBCS Annex 51 (Jank 2012), the EnEff Stadt project (Jank 2010), and the U.S. Army net-zero installation program (IE&E 2012). The scale of a nZEB community usually should not be very large due to factors such as stability issues, low inertia, control complexity (Olivares et al. 2014).

To improve the performance of nZEBs community, researchers have developed some demand response controls that enables energy sharing and battery sharing inside a community (Odonkor and Lewis 2015, 2016; Hu et al. 2014; Jafari-Marandi et al. 2016; Muhssin et al. 2018; Razmara et al. 2017; Lu et al. 2018). For example, Odonkor and Lewis (2015) developed a dynamic optimization method for nZEB community based on a bi-level operation framework. In their method, the distributed system operational parameters are optimized in the first level and the centralized system operational parameters are optimized in the second level. Based on the bi-level decision method, Odonkor and Lewis also developed a robust control method of the nZEB community using a Pareto band considering the uncertainty in the energy demand prediction (Odonkor and Lewis 2016). Similarly, Hu et al. (2014) proposed a bi-level operation method to optimize nZEB community operation using an augmented multi-objective particle swarm optimization (AMOPSO). These methods try to improve the nZEB community performance by optimizing single nZEBs' operation. They can be classified to the bottom-up methods (from single nZEB level to community level). Notably, Huang et al. developed a top-down control method (from community level to single nZEB level) of nZEBs, which first considers the whole nZEB community as 'one' nZEB, optimizes its operation, and then coordinates the operation of single nZEB. The abovementioned studies focused on solving the problem how to enable collaborations among nZEBs inside a community but neglected the problem how much collaborations benefits can be obtained from collaborations.

For a nZEB community, even though a very good control method is used, the performance improvement potentials will still be quite limited if the collaboration benefits among nZEBs are small (Jafari-Marandi et al. 2016). For example, grouping two similar office buildings together would bring very limited collaboration benefits, since they can hardly share renewable energy with each other due to the similarity in their energy characteristics. In contrast, grouping an office building and a residential building together would bring more benefits due to the diversity of their energy characteristics. The office building and residential building might have surplus renewable generations in different periods so that they can share them with each other (Jafari-Marandi et al. 2016; Huang et al. 2018b). The diversity in the energy characteristics can help enhance the performance improvement potentials in different aspects, such as reducing the energy bills and relieving the grid stress (Sun et al. 2018).

Many methods have been developed to identify the diverse energy characteristics of buildings based on the clustering algorithms (Lara-Santillán et al. 2014; Panapakidis et al. 2015; Chicco et al. 2006; Hsu 2015; Tardioli et al. 2018). For instance, Panapakidis et al. developed a two-stage clustering method to obtain the

representative load profiles. They used different characteristic parameters to simply the load profiles and applied three clustering algorithms (*K*-means and *K*-means-based algorithms) in the pattern recognition (Panapakidis et al. 2015). Chicco et al. compared three data reduction techniques and six clustering methods for classifying electricity customers (Chicco et al. 2006). Hsu compared the accuracy and stability of different clustering methods in the prediction of building energy demand (Hsu 2015). Notably, Jafari-Marandi et al. applied a self-organizing map-based clustering algorithm to identify representative buildings, proposed a homogeneity index to evaluate the diversity of building groups, and used a genetic algorithm-based control method to evaluate the building group performance improvements of different diversities. They found a large diversity in the building energy characteristics would produce more performance improvement potentials than the buildings with small diversity (Jafari-Marandi et al. 2016). These methods are effective in identifying the diversities of building energy characteristics; however, how to increase these diversities to maximize the collaboration benefits still remains unsolved. A systematic grouping method of nZEBs is still lacking.

To sum up, the collaboration benefits of nZEBs are determined by the diversities of nZEB energy characteristics. Grouping the nZEBs with large diversities of energy characteristics can significantly increase the collaboration benefits. Until now, such a grouping method is still lacking. To fill this gap, this chapter proposes a grouping method of nZEBs to maximize the performance improvement potentials by collaborations. Given a large number of nZEBs, the grouping method first divides them into different clusters based on their energy characteristics and then conducts an exhaustive search of the optimal grouping alternative of the representative energy profiles (obtained from clustering) from all the possible alternatives. Using the field data, the performance of the grouping method is analyzed in aspects of economic cost and grid interaction. The objective is to provide the decision maker with a systematic grouping method of nZEBs into nZEB communities that can achieve the optimal performance at the community level. The contributions of the present study to the subject are briefly summarized as follows:

- A grouping method of nZEBs has been developed. The grouping method can maximize the diversities of the nZEBs' energy characteristics inside the community, thus increasing the collaboration benefits and the performance improvement potentials.
- The performance improvement potentials of the developed grouping method have been investigated by comparing it with the case that the nZEBs are not grouped and the case that the nZEBs are randomly grouped.
- The performance improvement potentials of grouping method have been investigated under two control strategies (minimizing economic cost and minimizing grid interaction) using advanced demand response control method.

17.2 A Grouping Method of nZEBs for Performance Improvements

This section first introduces the similarity and diversity of power mismatch curves (PMCs). A large diversity of nZEBs' energy characteristics enables more energy sharing than nZEBs with a large similarity. Then, a grouping method of nZEBs is presented, which can increase the diversity between nZEBs in each group, thereby increasing benefits from collaboration at the nZEB community level.

17.2.1 Illustration of Similarity and Diversity of Power Mismatch Curves

Figure 17.1 illustrates the similarity and diversity of power mismatch curves (PMCs). A negative power mismatch value indicates producing

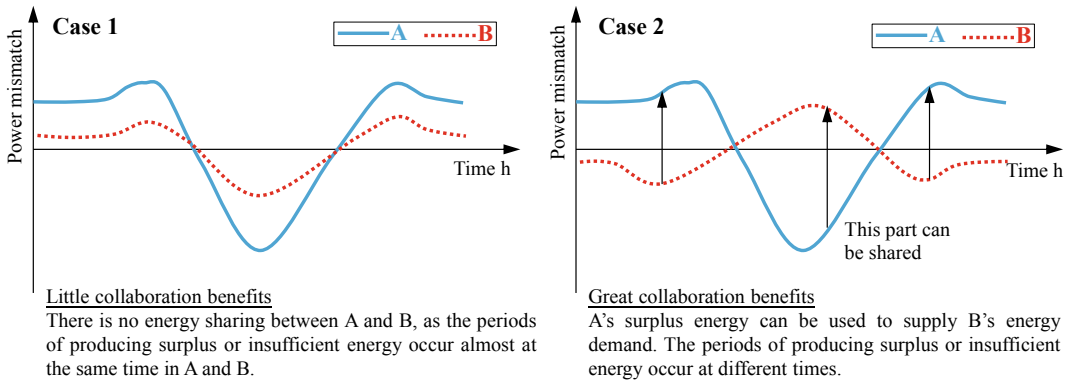

Fig. 17.1 Comparison of similarity and diversity in power mismatch curves

surplus renewable energy, while a positive value indicates there are insufficient energy productions. For two nZEBs, when their periods of producing surplus or insufficient renewable energy occur almost at the same time, their PMCs are considered to be similar (e.g., Case 1). Grouping such two nZEBs would bring little benefits as there is no energy sharing between the two nZEBs. In contrast, for two nZEBs, when their periods of producing surplus or insufficient renewable energy are different, their PMCs are considered to be diverse (e.g., Case 2). Grouping such two nZEBs can bring great benefits as the two nZEBs can share energy with each other.

Grouping nZEBs as a community enables collaborations (i.e., energy sharing and battery sharing) among nZEBs inside the community. In order to maximize the collaboration benefits, nZEBs with diverse power mismatch curves should be put into one group (e.g., Case 2); while nZEBs with similar power mismatch curves should be separated (e.g., Case 1). In other words, a good grouping should increase the diversity of the nZEBs PMCs in the group as much as possible.

17.2.2 A Grouping Method of nZEBs for Performance Improvements

Figure 17.2 presents the flowchart of the developed grouping method of nZEB communities for performance improvements. In Step 1, the PMCs are clustered based on their characteristics using clustering algorithm. Then a representative power mismatch curve (RPMC) is produced for each cluster of PMCs. The RPMCs will be used for the following grouping analysis. In Step 2, alternatives of communities of RPMCs are generated using combination algorithm, and their performance is evaluated. The one with the optimal performance is selected as the ultimate design. In Step 3, optimal grouping alternative is compared with the alternative that there is no grouping and the alternative obtained using random grouping. Details of each step are introduced as follows. The objective of grouping is to increase the diversities between the PMCs of nZEBs inside the communities, so as to increase potentials of performance improvements by collaborations.

17.2.2.1 Clustering of Power Mismatch Curves to Identify the Representative Energy Characteristics

In nZEBs grouping, when the number of nZEBs is very large, it would be insufficient in the computation to explore all the possible combinations of nZEBs. An alternative way is to cluster the nZEBs based on their PMC characteristics and calculate their RPMCs and then using the RPMCs to conduct grouping analysis. This section introduces the process of PMC clustering.

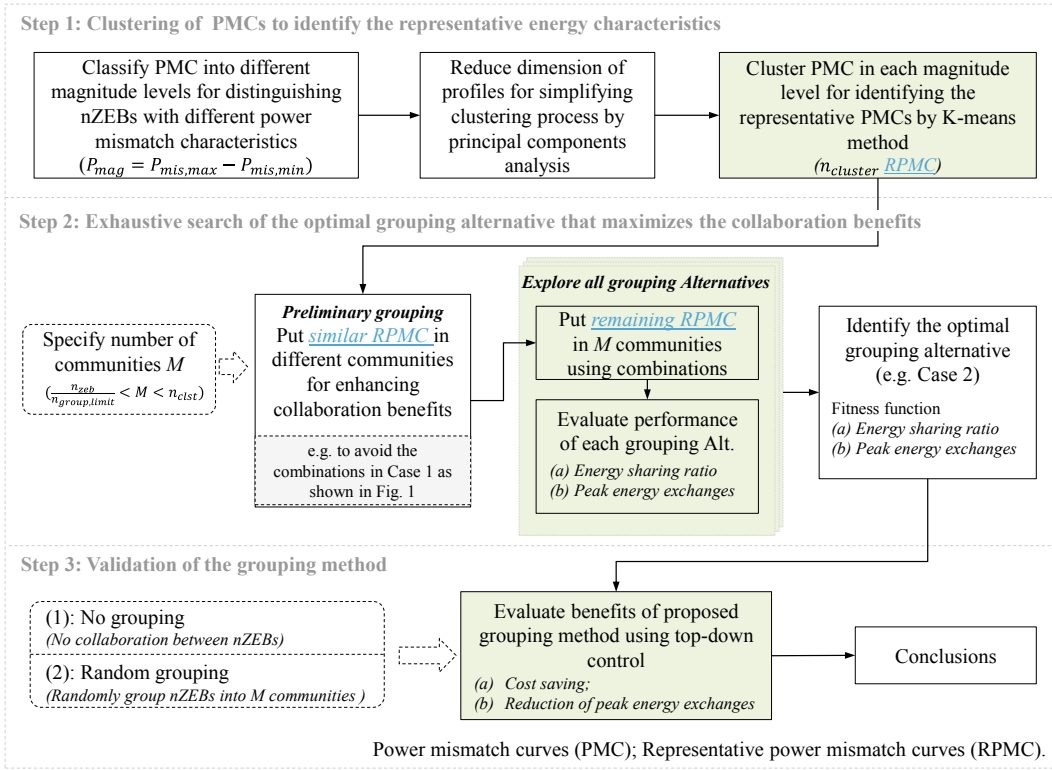

Fig. 17.2 Flowchart of the optimal grouping method

Classifying of PMCs into different magnitude levels for distinguishing between them.

The grouping method first classifies the nZEBs into different magnitude levels based on their power mismatch curves (Panapakidis et al. 2015). For each nZEB, a magnitude parameter (P_i kW) is obtained using Eq. (17.1)

$$P_i = \max\left(P_{\text{mis},i}\right) - \min\left(P_{\text{mis},i}\right) \tag{17.1}$$

where $P_{\text{mis},i}$ (kW) is the power mismatch of the ith nZEB. $P_{\text{mis},i}$ (kW) is calculated by Eq. (17.2).

$$P_{\text{mis},i} = P_{\text{demand},i} - P_{\text{supply},i} \tag{17.2}$$

where P_{demand} (kW) is the power demand of the ith nZEB during the 24 h period, and $P_{\text{supply},i}$ (kW) is the power supply. Based on the magnitude parameter calculated for each nZEB, several magnitude levels can be defined.

Dimensionality reduction of profiles for simplifying clustering process by principal components analysis.

Before clustering, characteristic parameters are usually identified to reduce dimensions of the original power mismatch profiles, so as to simply the clustering process. Several commonly used characteristic parameters including the time-domain indices (e.g., daily load pattern), load shape indices (e.g., ratio of minimum to average), frequency-domain indices (e.g., phase-related values and harmonic amplitudes), and indices obtained from dimensionality reduction techniques (e.g., principle components analysis) (Panapakidis et al. 2015; Tardioli et al. 2018). Due to its wide adaption, this chapter chooses principle components analysis (PCA) to produce characteristic parameters. The PCA aims to extract the important components from the original dataset which can effectively represent the

potential structure of the data (Panapakidis et al. 2015). This is conducted by applying an orthogonal transformation to transfer a dataset of possibly correlated parameters ($[X]_{n\times m}$) into a set of linearly uncorrelated parameters named principal components ($[C]_{n\times m}$), see Eq. (17.3) (Tibshirani et al. 2013).

$$[C]_{n\times m} = [X]_{n\times m} \times [W]_{m\times m} \tag{17.3}$$

$[W]_{m\times m}$ is the covariance matrix of the dataset $[X]_{n\times m}$. The first r principal components are calculated by Eq. (17.4).

$$[C]_{n\times r} = [X]_{n\times m} \times [W]_{m\times r} \tag{17.4}$$

Clustering of PMCs for identifying the representative PMCs by K-means method.

There are many clustering methods developed, such as the Euclidean distance framework, Hierarchical clustering, *K*-means clustering, Fuzzy K-means, Modified Follow-the-Leader, and SOMs (Panapakidis et al. 2015; Tardioli et al. 2018; Hsu 2015). A systematic introduction of these methods can be found in Tardioli et al. (2018). The *K*-means clustering is applied in this chapter since it can produce good performance and meanwhile it is simple to conduct (Panapakidis et al. 2015). The *K*-means clustering aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean (Hartigan and Wong 1979). This algorithm is based on an optimization procedure where data vectors are relocated between clusters with the minimization of an objective function until an optimal partition is obtained (Tibshirani et al. 2013):

$$\mathrm{f} = \min_{C_1,\ldots C_k} \left\{ \sum_{k=1}^{K} W(C_k) \right\} \tag{17.5}$$

where $C_1, \ldots, C_k$ are datasets containing indices of the observations in each cluster and $W(C_k)$ is a dissimilarity measure based on the squared Euclidean distance, which is calculated by Eq. (17.6).

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i'=\in C_k} \sum_{j=1}^{p} \left(x_{ij} - x_{i'j}\right)^2 \tag{17.6}$$

In the *K*-means algorithm, each point of the dataset is associated with a given centroid. The position of the centroids is recalculated and updated as the barycenter of the clusters resulting from the previous step. Points are then re-associated to each cluster, and cluster memberships are evaluated again. The recursive loop is repeated until the centroid position converges.

For each cluster, a representative power mismatch curve (corresponding to the centroid in K-means algorithm) will be generated. The RPMCs represent the mean of PMC in each cluster. n_{clst} RPMCs will be produced corresponding to the n_{clst} clusters. These RPMC will be used in the grouping analysis.

17.2.2.2 Exhaustive Search of the Optimal Grouping Way that Maximizes the Collaboration Benefits

Generation of grouping alternatives

In this step, the alternatives of grouping will be generated using combinations and evaluated. The one with the best performance will be selected as the optimal grouping way. First, the number of nZEB communities M is specified by designers. M should satisfy Eq. (17.7)

$$\frac{n_{\mathrm{zeb}}}{n_{\mathrm{group,limit}}} < M < n_{\mathrm{clst}} \tag{17.7}$$

where n_{zeb} is the total number of nZEBs, $n_{\mathrm{group,limit}}$ is the upper limit of the number of nZEBs inside a nZEB community, which is determined by the control capacity and the connection complexity. n_{clst} is the number of clusters obtained in Step 2.

After specifying the number of nZEB communities, a preliminary grouping is conducted by putting similar RPMCs into different communities. For instance, $n_{\mathrm{clst,smlr}}$ ($n_{\mathrm{clst,smlr}} < M$) RPMCs

with similar characteristics will be separately put into $n_{\text{clst,smlr}}$ communities. The preliminary grouping aims to reduce the combination of similar RPMCs like Case 1 in Fig. 17.1.

Then, an exhaustive search is conducted to explore all the alternatives of grouping. The total number of the grouping alternatives is determined by Eq. (17.8) by combinations.

$$N_{\text{alt}} = M^{\wedge a} - (M-1)^{\wedge a} \times \binom{b}{1} - (M-2)^{\wedge a} \times \binom{b}{2} - \cdots - \left(n_{\text{clst,smlr}}\right)^{\wedge a} \times \binom{b}{b} \tag{17.8}$$

where a is the number of the remaining RPMCs not used in the preliminary grouping, and b is the number of communities that contains no nZEB in the preliminary grouping. a and b are calculated by Eqs. (17.9) and (17.10), respectively.

$$a = n_{\text{clst}} - n_{\text{clst,smlr}} \tag{17.9}$$

$$b = M - n_{\text{clst,smlr}} \tag{17.10}$$

$\binom{b}{k}$ is calculated by Eq. (17.11), indicating all the combinations of selecting k elements from b elements.

$$\binom{b}{k} = \frac{b!}{k!(b-k)!} \tag{17.11}$$

After obtaining all the alternatives of grouping, the performance of each alternative is evaluated.

Performance indicators for identifying the optimal grouping alternative.

This chapter considers the economic performance and grid interaction performance. The operational cost is used as the economic performance indicator. A lower economic cost indicates more energy sharing among the nZEBs in each group. The peak energy exchange with the grid is used as the grid interaction performance indicator. Reducing the peak energy exchanges can help mitigate the grid stress for maintaining power balance and thus improving nZEB grid friendliness.

The economic cost of each grouping alternative (containing M nZEB communities) is calculated by Eq. (17.12),

$$J_{\text{Cost}} = \sum_{j=1}^{M} \sum_{i=1}^{24} P^{j}_{\text{ex},i} \times \tau \times \chi_i \tag{17.12}$$

where M indicates the number of communities, $P^{j}_{\text{ex},i}$ (kW) is the sum of power exchanges with the grid of the jth community in the ith hour, τ is the time during (1 h), χ_i (HKD/kW h) is the real-time electricity cost in the ith hour. $P^{j}_{\text{ex},i}$ (kW) is calculated by Eq. (17.13),

$$P^{j}_{\text{ex},i} = \sum_{k=1}^{k'} P^{j,k}_{\text{ex},i} \tag{17.13}$$

where k' is the number of nZEBs inside the jth community. $P^{j,k}_{\text{ex},i}$ (kW) is the power exchanges of the kth nZEB inside the jth community in the ith hour. It is calculated by Eq. (17.14),

$$P^{j,k}_{\text{ex},i} = P^{j,k}_{\text{mis},i} + P^{j,k}_{\text{charge},i} \tag{17.14}$$

where $P^{j,k}_{\text{mis},i}$ (kW) is the power mismatch of the kth nZEB inside the jth community in the ith hour (refer to Eq. (17.2), and $P^{j,k}_{\text{charge},i}$ (kW) is the battery charging rates. For simplicity, in the exhaustive search, the battery operation is not considered, thus $P^{j,k}_{\text{charge},i}$ is set to be 0.

The peak energy exchanges (kW h) of each grouping alternative (containing M nZEB communities) is calculated by Eq. (17.15)

$$J_{\text{Grid}} = \sum_{j=1}^{M} \max\left(\left|P^{j}_{\text{ex},1} \times \tau\right|, \left|P^{j}_{\text{ex},2} \times \tau\right|, \ldots \left|P^{j}_{\text{ex},24} \times \tau\right|\right) \tag{17.15}$$

where $P^{j}_{\text{ex},1}$ (kW) is the power exchanges with the grid of the jth community, which is calculated by Eq. (17.14). The two performance indicators are considered separately. Each of them will produce an optimal grouping alternative.

For validating the proposed grouping method, the performance of the optimal grouping

alternative is compared with two cases. In the first case, the nZEBs are not grouped, i.e., there is no collaboration among the nZEBs. In the second case, the nZEBs are randomly grouped. The collaborations are enabled inside each community similar to the proposed grouping method.

17.3 Building and Systems Modeling

17.3.1 Building Modeling

In this chapter, the energy demand data of buildings are obtained either from statistical analysis of existing buildings (Institute 2016) or DOE building templates (DOE 2018). In total, there are 33 sets of energy demand data obtained. 8 of these buildings belong to residential buildings, and 25 of them are commercial buildings. Table 17.1 summarizes the types of buildings and the sources of data. Details of these buildings can be found in Institute (2016), DOE (2018).

17.3.1.1 Renewable Energy System Modeling and Battery Modeling

The power generation of PV panel P_{PV} (kW) is calculated by Eq. (17.16) (Huang et al. 2018a; Klein et al. 2004),

$$P_{PV} = \tau \times I_{AM} \times I_T \times \eta \times CAP_{PV} \quad (17.16)$$

where τ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (W/m^2) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; CAP_{PV} (m^2) is the PV surface area.

The power generated by a wind turbine P_{WT} (kW) is calculated by Eq. (17.17) (Klein et al. 2004; Huang et al. 2018a)

$$P_{WT} = \begin{cases} 0, & \text{if } U_0 \le 3 \\ C_P \times \rho_{air} \times A_R \times U_0^3, & \text{otherwise} \\ CAP_{WT}, & \text{if } U_0 \ge 14 \end{cases} \quad (17.17)$$

where C_P is power efficiency, which is a function of the axial induction factor; ρ_{air} (kg/m^3) is air density; A_R (m^2) is the rotor area; U_0 (m/s) is the wind speed in the free stream; CAP_{WT} (kW) is the capacity of the wind turbine.

The electricity stored in the battery is calculated using a simplified model, as expressed by Eq. (17.18). It is estimated by the charging rates (Sun et al. 2018),

$$\emptyset_{i-1} = \emptyset_0 + \left(P_{charge,1} + P_{charge,2} + \cdots + P_{charge,i-1}\right) \times \tau \quad (17.18)$$

where $\emptyset_{i-1}$ (kW h) is the energy stored in the battery in the $(i-1)$th hour, $P_{charge,1}$, $P_{charge,2}$, … (kW) are the battery charging rates in each hour, τ is the time duration (1 h). The battery operation should satisfy the following constraints.

$$0 \le \emptyset_0 + \left(P_{charge,1} + P_{charge,2} + \cdots + P_{charge,i-1}\right) \times \tau \le CAP_{bat} \quad (17.19)$$

CAP_{bat} (kW h) is the battery capacity, which is designed by Eq. (17.20).

$$CAP_{bat} = \max\left(\left|\sum_{i=1}^{n} P_{mis,i}\right|, \left|\sum_{i=2}^{n+1} P_{mis,i}\right|, \ldots, \left|\sum_{i=8761-n}^{8760} P_{mis,i}\right|\right) \quad (17.20)$$

where n (h) is a pre-specified period.

17.4 Case Studies and Results Analysis

The field data of the 33 buildings summarized in Table 17.1 were used in the case studies. Their energy demand data was directly used. Their renewable energy generations were simulated in TRNSYS under the weather condition of Hong

Table 17.1 Summary of the building energy demand data and data sources

Type	ID	Name		Type	ID	Name		Type	ID	Name	
Residential buildings	1	Manufactured Home Mixed AC	Institute (2016)	Commercial buildings	12	Hotel	Institute (2016)	Commercial buildings	23	Full service restaurant	DOE (2018)
	2	Multiple Family Mixed Cooling	Institute (2016)		13	Large office	Institute (2016)		24	Quick service restaurant	DOE (2018)
	3	Single Family Central AC	Institute (2016)		14	Medium office	Institute (2016)		25	Large office	DOE (2018)
	4	Single Family Heat Pump AC	Institute (2016)		15	Small office	Institute (2016)		26	Medium office	DOE (2018)
	5	Single Family Large	Institute (2016)		16	Bank	Institute (2016)		27	Small office	DOE (2018)
	6	Residential: low	DOE (2018)		17	Large retail	Institute (2016)		28	Primary school	DOE (2018)
	7	Residential: medium	DOE (2018)		18	Midrise apartment	DOE (2018)		29	Secondary school	DOE (2018)
	8	Residential: high	DOE (2018)		19	Large hotel	DOE (2018)		30	Standalone retail	DOE (2018)
Commercial buildings	9	Small retail	Institute (2016)		20	Small hotel	DOE (2018)		31	Strip mall	DOE (2018)
	10	Church	Institute (2016)		21	Hospital	DOE (2018)		32	Supermarket	DOE (2018)
	11	Education, K-12	Institute (2016)		22	Out patient	DOE (2018)		33	Warehouse	DOE (2018)

Kong. The renewable energy systems included a wind turbine and PV panels. The renewable energy system of each building was sized to produce the amount of renewable energy equal to its energy demand. This chapter used 5 sets of renewable energy systems with different energy generation ratios of the wind turbine to PV panels (0/1, 0.25/0.75, 0.5/0.5, 0.75/0.25, and 1/0) for each building. Each combination of building and renewable energy systems was considered as one unique nZEB. The selection of different renewable system configuration ratios was in corresponding to the configuration diversity in practice. By using different configuration ratios of renewable energy systems, the number of nZEBs was extended to 165 from 33. This extension was reasonable as in practice there might be multiple buildings that have similar energy usage patterns but different renewable energy systems. Each nZEB was equipped with a battery for reducing its dependence on the power grid. The battery capacities were calculated by Eq. (17.20).

Table 17.2 summarizes the price models used in this chapter. With reference to the electricity prices of a real power company (Power 2018), the prices for purchasing electricity from the grid were set as 2 HKD/kW h and 1 HKD/kW h in the on-peak hours and off-peak hours, respectively. Meanwhile, the feed-in-tariff was set as 0.2 HKD/kW h considering the grid stress caused by the renewable energy exports. In practice, the feed-in-tariff may be much cheaper or even forbidden as the grid concerns associated negative impacts on its power balance and supply quality (Zhao et al. 2015). To encourage renewable energy sharing among nZEBs, the prices of electricity purchased from and sold to the community were set to be 0.5 HKD/kW h, which is lower than the price of electricity purchased from the grid but higher than that of electricity exported to the grid. Note that if the price of electricity purchased from the community was higher than 1 HKD/kW h or the price of electricity sold to the community was lower than 0.2 HKD/kW h, there would be no economic incentive for the nZEBs to share renewable energy in a cluster.

The 24 h energy demand base shape profile (Institute 2016) and the corresponding energy supply base shape profile of each nZEB were used for analysis following existing studies (Panapakidis et al. 2015; Tardioli et al. 2018). In the base shape profile, the value of energy demand/supply in each hour is the average value in this particular hour of 365 days.

17.4.1 Clustering of Power Mismatch Curves

The 165 PMCs were classified into 5 levels based on their magnitudes, as shown in Fig. 17.3. Level 1 had a magnitude between 0–10 kW and contained 41 nZEBs. Level 2 had a magnitude between 10–100 kW and contained 59 nZEBs. Level 3 had a magnitude between 100–400 kW and contained 40 nZEBs. Level 4 had a magnitude between 400–1000 kW and contained 16 nZEBs. Level 5 had a magnitude between 1000–3000 kW and contained 9 nZEBs.

Then PCA was conducted to reduce the dimensionality of the PMC profiles. 9 principal components were selected as they can explain over 95% of the variance. Then K-means method was used to cluster the PMCs in each group. In

Table 17.2 Electricity price (Sun et al. 2018; Lu et al. 2015; Huang et al. 2018b)

Electricity price (HKD/kW h)			
Grid	Buy-from-grid	On-peak (8–21)	2
		Off-peak (other)	1
	Feed-in-tariff		0.2
Cluster	Buy-from-cluster		0.5
	Sell-to-cluster		0.5

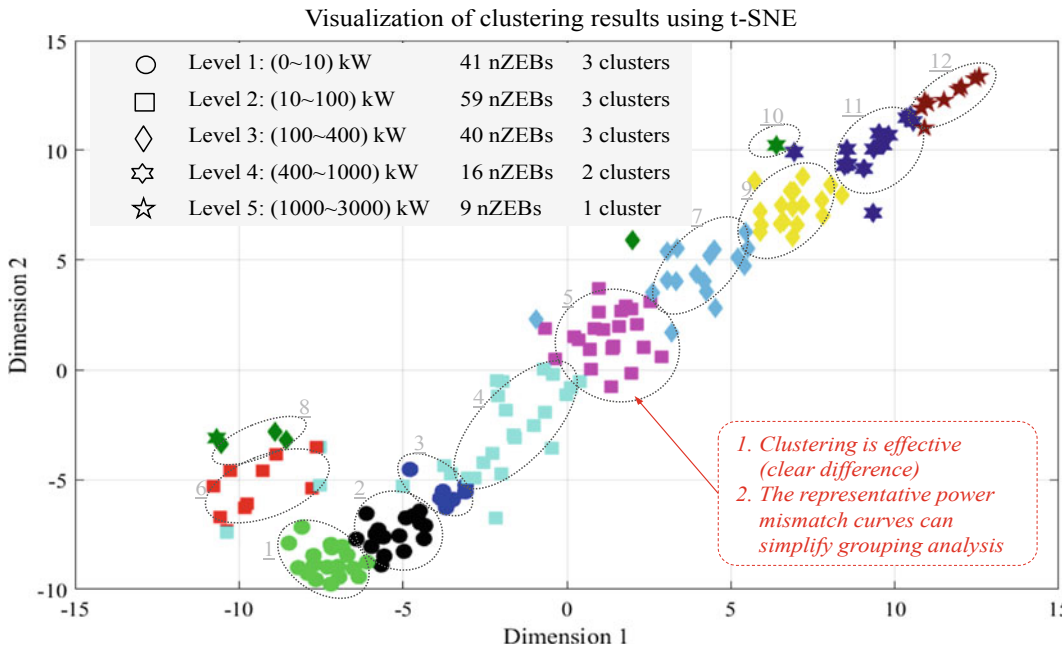

Fig. 17.3 Visualization of clustering results using t-SNE

each of the 1st, 2nd, and 3rd magnitude level, the nZEBs were partitioned into 3 clusters. The nZEBs in the 4th level was partitioned into 2 clusters. As there are only 9 nZEBs in the 5th level, all these 9 nZEBs were classified into 1 cluster. In total, all the nZEBs were classified into 12 clusters.

Figure 17.3 visualizes the clustering results using t-SNE (Maaten and Hinton 2008). Each marker shape represents a magnitude level. Each color represents a cluster. As can be seen, the clusters were clearly distinguished from each other. The clustering is effective to identify the nZEBs with diverse energy characteristics and obtain the corresponding representative power mismatch curves. Figure 17.4 presents the PMCs and RPMCs of the 12 clusters corresponding to Fig. 17.3. The RPMCs of the 12 clusters were used for grouping analysis.

The 165 nZEBs were grouped into 4 communities. As the RPMCs of cluster 9, 11, and 12 had similar characteristics, they were put into three separate communities. In total, there are 4^9–3^9 = 242,461 alternatives of grouping, as calculated by Eq. (17.8).

17.4.2 Operational Cost Evaluation Results

In this section, the operational cost was used as the indicator for selecting the optimal grouping alternative. The operational costs of all the 242,461 grouping alternatives were evaluated and compared. The grouping alternative with the lowest operational cost was selected as the ultimate design.

17.4.2.1 *Grouping Results for Minimizing the Operational Cost*

Figure 17.5 presents the optimal alternative of grouping the 12 RPMCs into 4 communities. As can be seen, community 1, 2, and 4 can achieve significant energy sharing as there is large diversity between the RPMCs. In each community, the periods of producing surplus renewable energy and insufficient energy were very different between each nZEBs. Through collaboration, the nZEBs can share their surplus renewable energy with other nZEBs whose renewable generations cannot meet their energy demands.

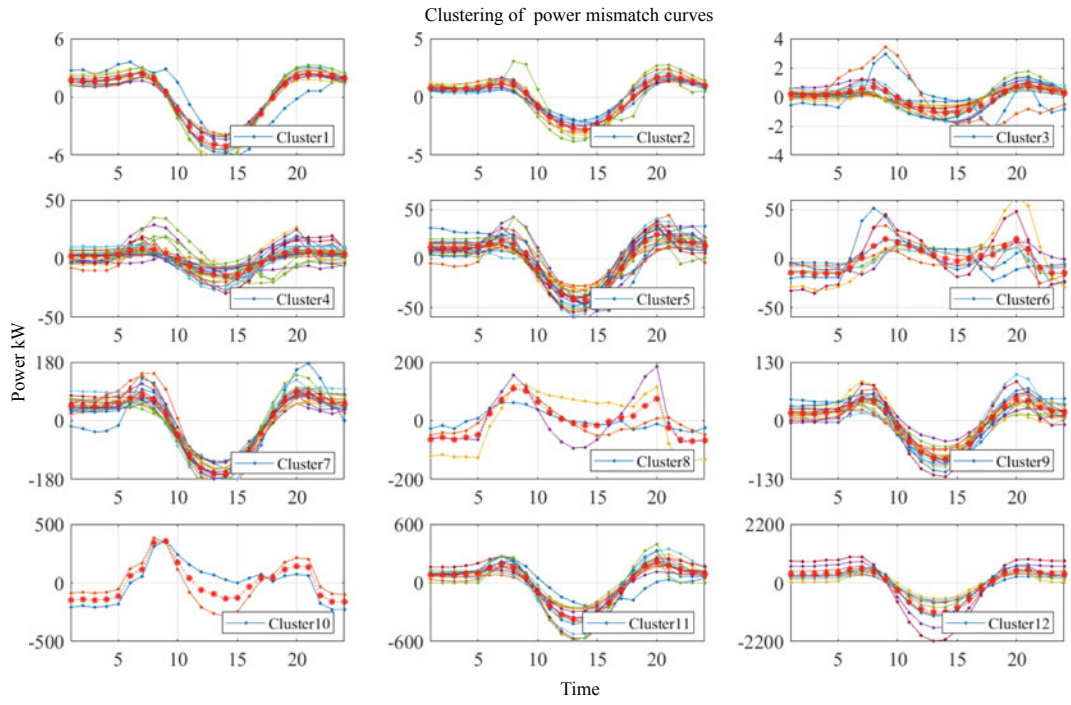

Fig. 17.4 Power mismatch curves and representative power mismatch curves of the 12 clusters

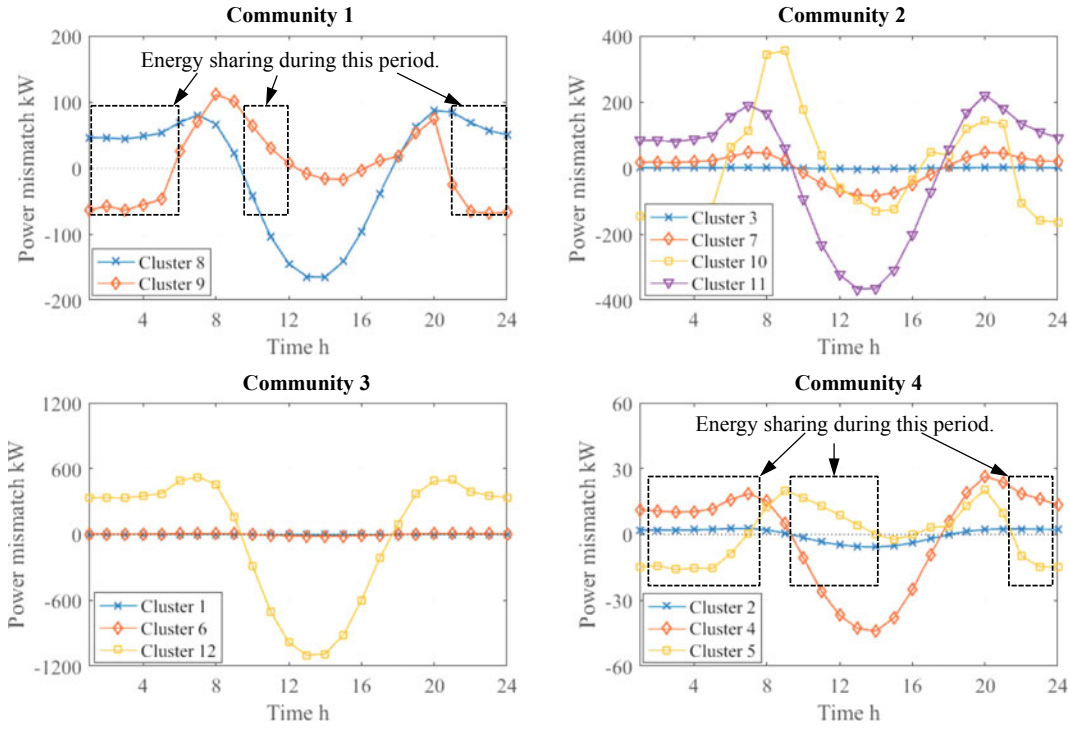

Fig. 17.5 Representative power mismatch curves in each nZEB community

In this way, the nZEBs with insufficient renewable energy generations can significantly reduce unnecessary high-priced energy imports from the grid, thus reducing operational costs. In community 3, the amount of energy sharing is very small. This is because the magnitude of RPMC 12 is larger than the magnitude of RPMC 1 and 6. The periods of energy sharing in community 1 and 4 are also highlighted in Fig. 17.5.

Then, the PMCs in each cluster were put into the community corresponding to the grouping alternatives of the 12 RPMCs. In total, there were 23 nZEBs in community 1, 49 nZEBs in community 2, 56 nZEBs in community 3, and 37 nZEBs in community 4.

17.4.2.2 Operational Costs Comparison with no Grouping

The operational costs of the grouping alternative were compared with the case that nZEBs were not grouped. In the case without grouping, each individual nZEB was connected to the power grid individually, and there was no collaboration among nZEBs. For comparison purpose, in the case nZEBs are not grouped, the performance of nZEBs was simulated separately and accumulated following the grouping alternative as shown in Fig. 17.5.

Figure 17.6 compares the hourly power exchanges with grid and operational costs of the four nZEB communities. The top-down control developed by Huang et al. (2018b) was used to optimize the operation of the batteries of nZEBs in each community. A brief introduction of the top-down control was given in the Appendix. For comparing the costs, at the end of the day, the electricity stored in the battery was sold to the grid. Using the price setting in Table 17.2, the hourly operational costs of each community was calculated. As can be seen, grouping the nZEBs can effectively reduce the high-priced energy imports (1 HKD/kW h during the off-peak period) from the grid and meanwhile reduce energy exports to the grid at a low price (0.2 HKD/kW

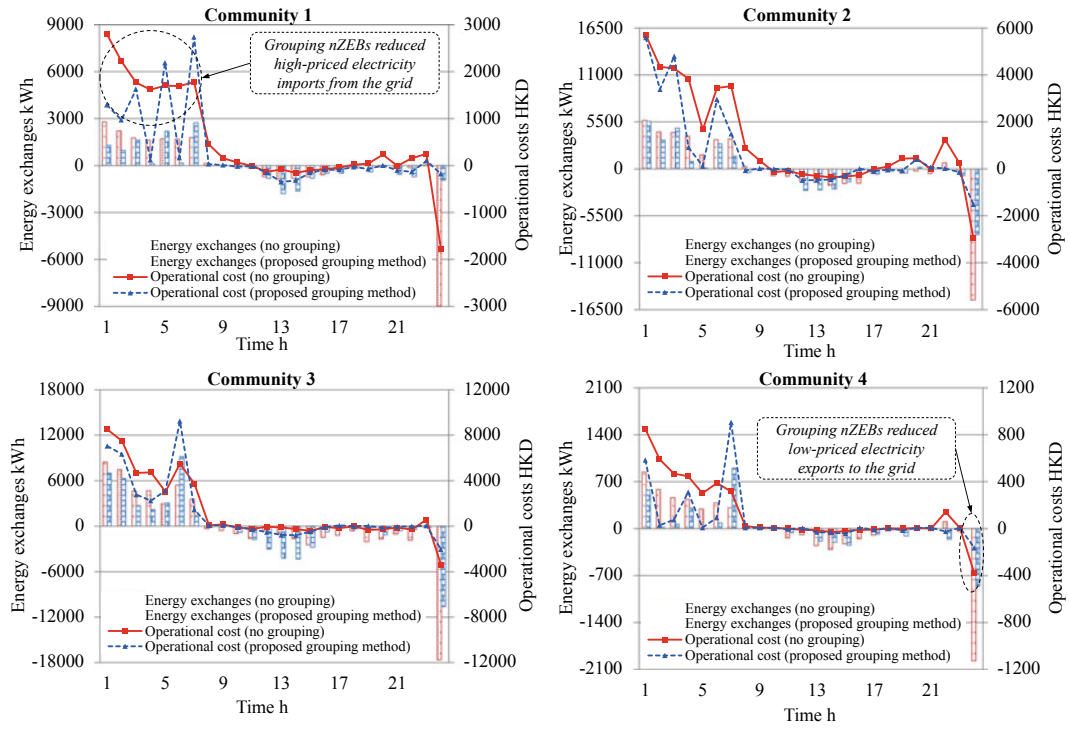

Fig. 17.6 Comparison of the hourly power exchanges with grid and operational costs of the four nZEB communities

h). This is because grouping enables collaborations among the nZEBs, which allows the nZEBs to share their surplus renewable energy with other nZEBs which has insufficient renewable generations. The daily operational costs of each nZEB community were summarized in Table 17.3. There is a significant cost saving from 1st to 5th hour in both community 1 and 4 (see Fig. 17.6). This is contributed by the large diversity in the PMCs during this period (see community 1 and 4 in Fig. 17.5).

Figure 17.7 compares the sum of hourly power exchanges with the grid and the hourly operational costs of the case using proposed grouping method (enabling collaborations) and the case without grouping (no collaboration). The total cost saving was summarized in Table 17.3. As a whole, grouping nZEBs as communities can significantly reduce energy imports from the grid at a high price and reduce energy exports at a low price. This is contributed by the collaborations among nZEBs in each community. By grouping the nZEBs and enabling collaborations among the nZEBs in each community, the daily operational cost can be reduced by 30% in total.

17.4.2.3 Operational Costs Comparison with Random Grouping

The operational costs of the proposed grouping method were compared with random grouping for validation. Due to the high computational costs of the top-down control, a limited number of 100 grouping alternatives were produced using random grouping. In each of the 100 alternatives, the 12 RPMCs were randomly put into the four communities. Then, the nZEBs in each of the 12 clusters were put into the corresponding communities.

Figure 17.8 compares the operational costs of the 100 random grouping alternatives with the grouping using the proposed method (49,970 HKD) and the case without grouping (71,460 HKD). The operational costs of alternatives obtained from random grouping were between the alternative using the proposed grouping method and the alternative without grouping. Both the proposed grouping method and random grouping can produce better cost performance than the case without grouping. This is within expectation as grouping the nZEBs enables collaborations among them. Thus the nZEBs can

Table 17.3 Comparison of operational costs

Community	1	2	3	4	Sum
Case without grouping (HKD)	12,357	25,437	30,763	2902	71,460
Case using proposed grouping method (HKD)	7329	15,716	25,360	1565	49,970
Cost saving (HKD)	5029	9720	5403	1337	21,489
Cost saving percentage (%)	41	38	18	46	30

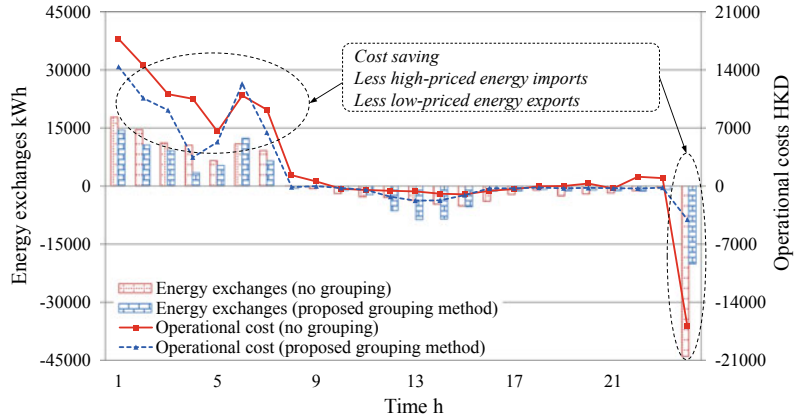

Fig. 17.7 Comparison of the hourly power exchanges with grid and operational costs of all the 165 nZEBs

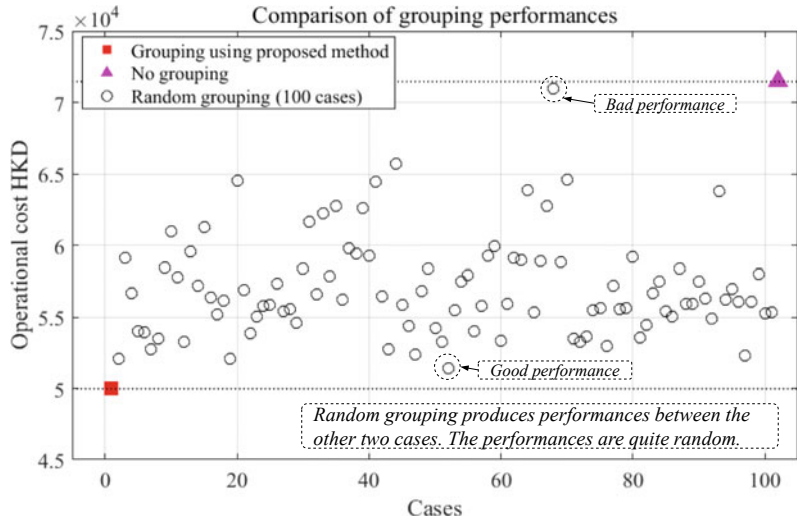

Fig. 17.8 Comparison of the operational costs of all the 165 nZEBs using different methods

share their surplus renewable energy and reduce high-priced energy imports from the grid. The proposed grouping method can produce better performance than random grouping, as it conducts an exhaustive search of all the possible alternatives and identifies the one that has the maximum energy sharing. Random grouping is simple to conduct, but the results were not robust.

17.4.3 Peak Energy Exchange Evaluation Results

In this section, the peak energy exchanges were used as the indicator for selecting the optimal grouping alternative. The peak energy exchanges of all the 242,461 alternatives of grouping were evaluated and compared. The grouping alternative with the minimum peak energy exchange was selected as the ultimate design.

17.4.3.1 *Grouping Results for Minimizing the Peak Energy Exchanges*

Figure 17.9 presents the optimal alternative of putting the 12 RPMCs into 4 communities. As can be seen, community 2 and 4 can achieve significant energy sharing as there is large diversity between the RPMCs. In each community, the periods of producing surplus renewable energy and insufficient energy were very different between the nZEBs. Through collaboration, the nZEBs can share their surplus renewable energy with other nZEBs whose renewable generations cannot meet their energy demands. In this way, the nZEBs can reduce unnecessary large energy imports/exports with the grid, thus reducing peak energy exchanges at the community level. The periods of energy sharing in community 2 and 4 are also highlighted in Fig. 17.9.

Then, the PMCs in each cluster were put into the communities corresponding to the grouping alternatives of the 12 RPMCs. In total, there were 108 nZEBs in community 1, 16 nZEBs in community 2, 18 nZEBs in community 3, and 23 nZEBs in community 4.

17.4.3.2 Peak Energy Exchanges Comparison with No Grouping

The peak energy exchanges of the grouping alternative were compared with the case that nZEBs were not grouped. In the case without grouping, each individual nZEB was connected to the power grid individually, and there was no collaboration among nZEBs. For comparison purpose, in the case that nZEBs are not grouped, the performance of nZEBs was simulated separately and accumulated following the grouping alternative as shown in Fig. 17.9.

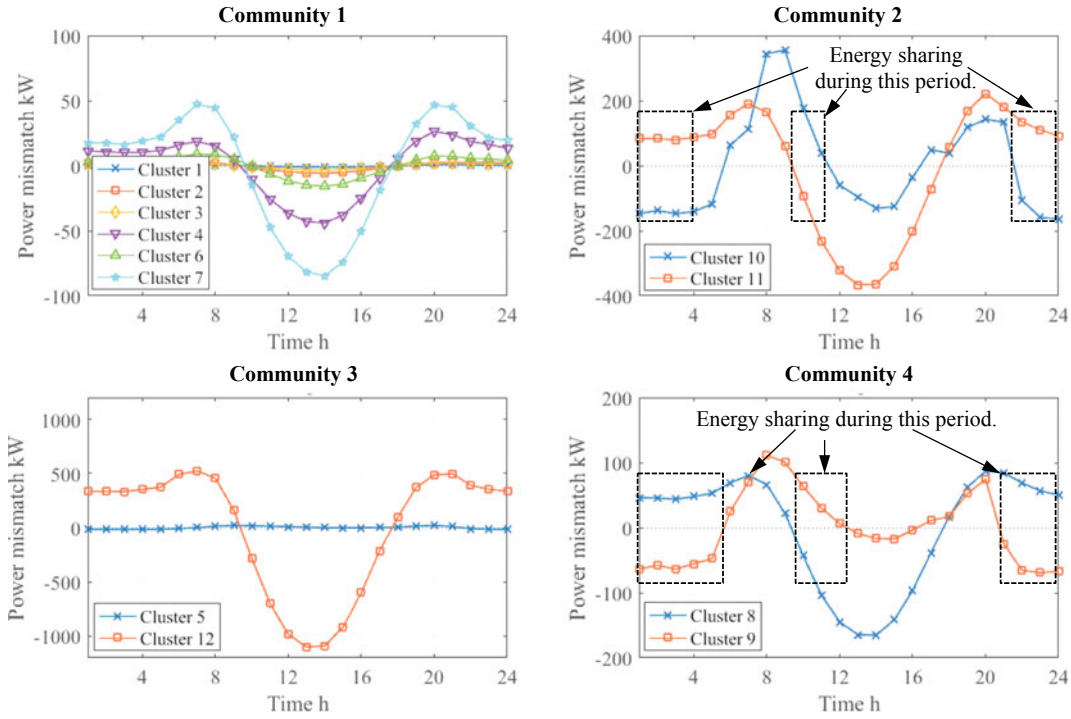

Fig. 17.9 Representative power mismatch curves in each nZEB community

Figure 17.10 compares the hourly power exchanges with the grid of the four nZEB communities. Similarly, the top-down control developed by Huang et al. (2018b) was used to optimize the operation of the batteries of nZEBs in each community. Grouping the nZEBs can effectively reduce the peak energy exchanges with the grid. This is because grouping enables collaborations among nZEBs, which allows the nZEBs to share their surplus renewable energy and thereby reducing unnecessary large energy imports/exports with the grid. The daily peak energy exchanges of each nZEB community were summarized in Table 17.4.

Figure 17.11 compares the sum of hourly power exchanges with the grid of the case using proposed grouping method (enabling collaborations) and the case without grouping (no collaboration). The total reduction in peak energy exchanges was summarized in Table 17.4. As a whole, grouping nZEBs can reduce the peak energy exchanges with the grid. This is contributed by the collaborations among nZEBs in each community. By grouping the nZEBs and enabling collaborations among the nZEBs, the daily peak energy exchanges can be reduced by 16% in total.

17.4.3.3 Peak Energy Exchanges Comparison with Random Grouping

The peak energy exchanges of the proposed grouping method were compared with random grouping for validation. Again, 100 grouping alternatives were produced using random grouping.

Figure 17.12 compares the peak energy exchanges of the 100 random grouping alternatives with the grouping using the proposed method (8910 kW h) and the case without grouping (10,637 kW h). The peak energy exchanges of alternatives obtained from random grouping were between the alternative using the proposed grouping method and the alternative without grouping. Both the proposed grouping method and random grouping can produce better

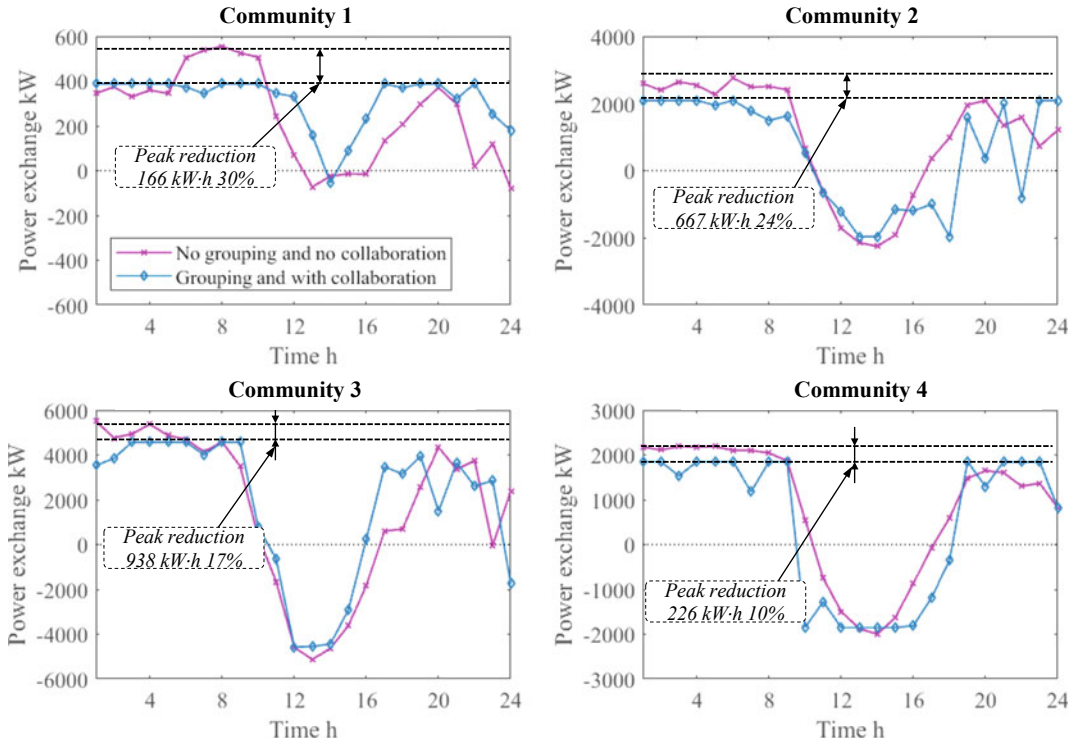

Fig. 17.10 Comparison of hourly power exchanges with the grid of the four nZEB communities

Table 17.4 Comparison of peak energy exchanges

Community	1	2	3	4	Sum
Case without grouping (kW h)	555	2755	5512	2187	10,637
Case using proposed grouping method (kW h)	389	2088	4574	1961	8910
Reduction in peak energy exchanges (kW h)	166	667	938	226	1727
Reduction percentage (%)	30	24	17	10	16

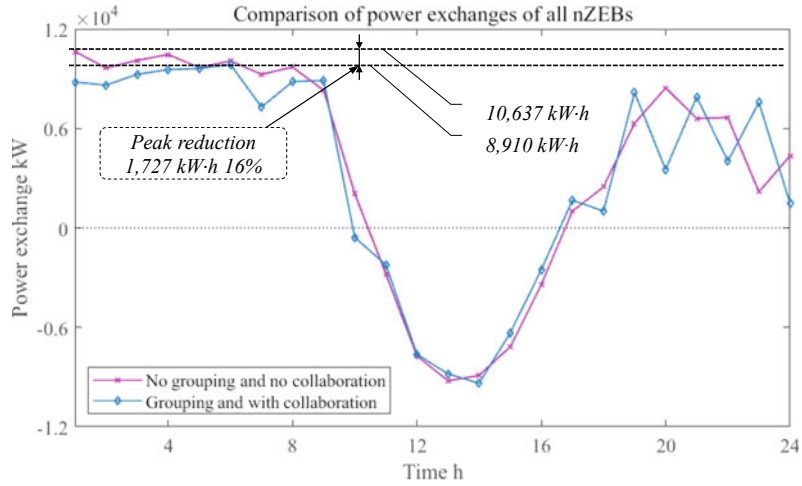

Fig. 17.11 Comparison of the hourly power exchanges with grid and operational costs of all the 165 nZEBs

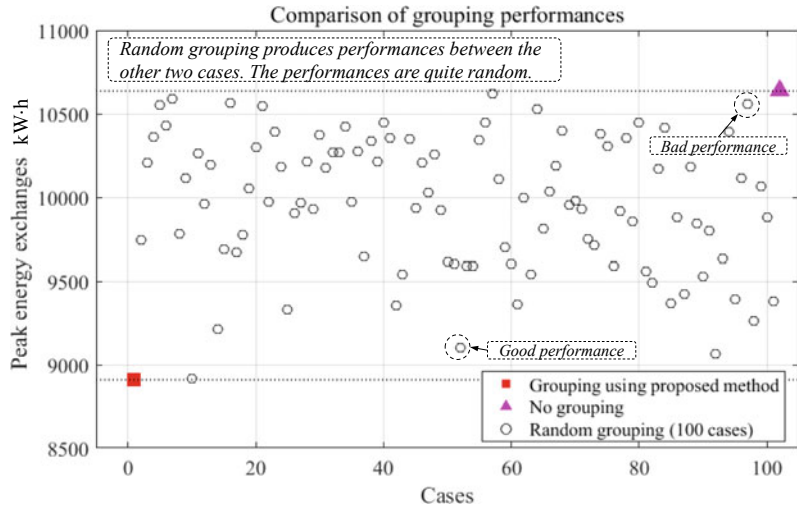

Fig. 17.12 Comparison of the daily peak energy exchanges with the grid of all the 165 nZEBs using different methods

grid interaction performance than the case without grouping. This is because grouping the nZEBs enables collaborations among them. By sharing the surplus renewable energy with the nZEBs with insufficient renewable generations, the nZEBs can reduce unnecessary large energy imports/exports from/to the grid. The proposed grouping method can produce better performance than random grouping, as it conducts an exhaustive search of all the possible alternatives and identifies the one that has the maximum energy sharing.

17.5 Summary

This chapter has proposed a grouping method of nZEBs for performance improvements at the community level. Given a large number of nZEBs, the grouping method first classifies the nZEBs into different clusters based on their PMC characteristics and then conducts an exhaustive search of the optimal grouping alternative of the RPMCs from all the possible alternatives. The developed grouping method has been compared with the case without grouping and the case nZEBs are randomly grouped in aspects of economic performance and grid interaction performance. The major findings are stated as below.

- The proposed grouping method can maximize the diversities of the nZEB PMCs inside the communities. Inside a community, a large diversity of the nZEBs PMCs can increase the collaboration benefits (e.g., more energy sharing), thus improving the performance at the community level.
- In aspects of economic cost, by sharing the nZEBs' surplus renewable energy with the nZEBs with insufficient renewable generations, the proposed grouping method significantly reduce the high-priced electricity energy imports from the grid. Thus, the cost savings can be improved by 30% compared with the case that nZEBs are not grouped.
- In aspects of grid interaction, by sharing the nZEBs' surplus renewable energy with the nZEBs with insufficient renewable generations, the proposed grouping method can effectively eliminate the unnecessary large energy exchanges with the grid. Thus, the peak energy exchanges can be reduced by 16% compared with the case that nZEBs are not grouped.
- As the developed grouping method maximizes the diversities in the PMCs of nZEBs inside a group, it can achieve more robust performance (more cost savings and more reduction in peak energy exchanges) than the random grouping method.

The computational cost of the proposed grouping method is high. Future work will try to explore more advanced searching algorithms to enhance the grouping method.

Appendix: The Top-Down Control Method

Figure 17.13 presents a flowchart of the top-down control developed by Huang et al. (2018b). The aim of the top-down control is to adjust the operation of individual nZEB in the next 24 h to achieve the optimal performance at the community level. This top-down control method consists of two steps. In Step 1, all the nZEBs inside a community is considered as 'one' nZEB (its power supply/demand/ battery storage is the sum of the individual nZEB power supply/ demand/battery storage), and its operation in the coming 24 h is optimized using a genetic algorithm. In Step 2, the operation of individual nZEB inside the community is coordinated using the non-linear programming based on the nZEB community operation obtained in Step 1. The community operational parameters are used as constraints of individual nZEB operation, e.g., the sum of the power exchange of individual nZEB (obtained in Step 2) equals the power exchange of nZEB cluster (obtained in Step 1). Please refer to Huang et al. (2018b) for details of the top-down control method.

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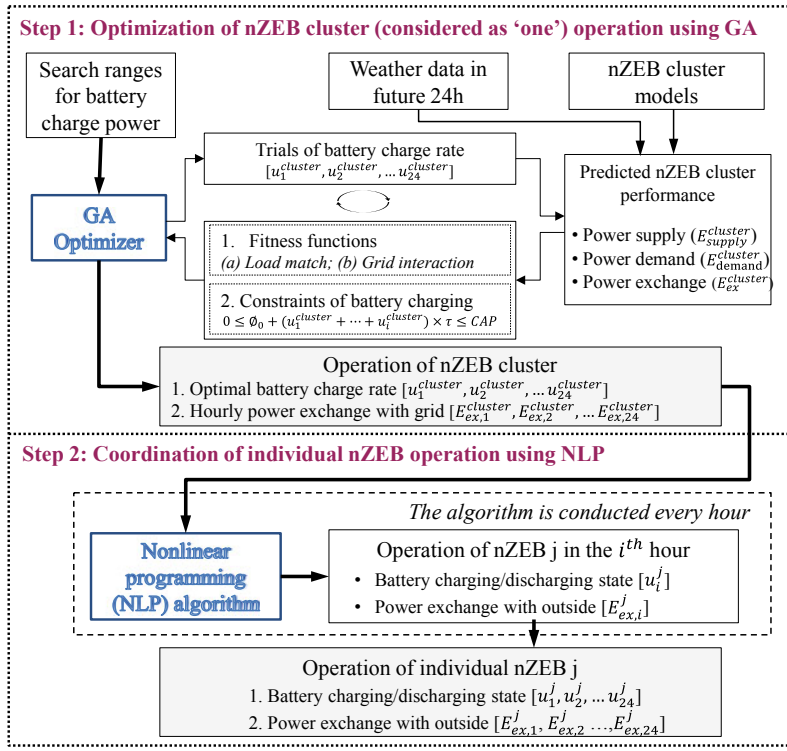

Fig. 17.13 Flowchart of the top-down control method (Huang et al. 2018b)

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Dynamic Pricing for Improving Bi-Directional Interactions with Reduced Power Imbalance

18

Yongjun Sun and Pei Huang

Abstract

Active supply–demand interactions in a smart grid are essential for reducing grid power imbalance which is important for the security and efficiency of power supply. A key element to the success of such interactions is the proper pricing strategy. The latest game-theory-based dynamic pricing methods require *information exchanges* not only between the supply and demand sides, but also among individual buildings, since they make decisions for one building's demand response based on/influenced by operations of the others. However, in practical applications in which a number of buildings are considered, the latter information exchanges are extremely difficult due to the concerns of privacy, communication complexity and high computation load. Therefore, this chapter proposed a genetic algorithm-based dynamic pricing method for improving bi-directional interactions with reduced power imbalance, which does not require information exchanges among individual buildings. In this chapter, at the demand side, targeting at minimizing daily electricity cost, a non-linear programming-based demand response control is performed in individual buildings at a dynamic price given by the grid operator genetic algorithm optimizer. Targeting at reducing grid power imbalance, the genetic algorithm optimizer is used by the grid operator to search for a better dynamic price based on the aggregated demand response results. Such interaction will continue until the grid power imbalance cannot be further reduced. The impacts of demand elasticity are also investigated on performance improvements. The proposed pricing method can be used in practical applications to improve dynamic pricing of a smart grid for reduced grid power imbalance and thus increased operation efficiency.

Keywords

Demand response · Power imbalance · Bi-Directional interactions · Dynamic pricing · Smart grid

18.1 Introduction

The global electricity demand is increasing exponentially, and this amount is expected to double by 2030 (Boßmann and Staffell 2015).

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Smart grid is widely considered as a promising solution to the increasing energy and environmental problems. Smart grid refers to modern electricity networks that can intelligently integrate the behavior and actions of all users (power suppliers, consumers and those that do both) connected to it in order to efficiently deliver sustainable, economic and secure electricity supplies (Fang et al. 2012). It is beneficial to the power suppliers, consumers, the grid operator (i.e., transmission system operator (Wang et al. 2015; Hassan and Radman 2010). For the power suppliers, the smart grid provides accurate data on the overall demand profiles and the status of the electricity systems. With such information, the power suppliers can optimize their production management in line with more clearly quantified demand for improved security and efficiency (Hledik 2009). For the consumers, the smart grid provides more reliable and secure services (e.g., reduced outages) and offers greater control for economic savings (Hledik 2009). For the grid operator, the smart grid provides more information and resources for power dispatch, so as to minimize the downtimes and failures of power system (Momoh 2009). Furthermore, the energy loss in the grid can be reduced as an easier balance between supply and demand can be made.

To realize the benefits of the smart grid, the active interactions between the supply side and the demand side are essential (Kakran and Chanana 2018). A key element in determining the interactions between the supply and demand sides is the pricing strategy (Dutta and Mitra 2017). The supply side collects the necessary real-time demand data and determines proper electricity prices based on the collected demand information. Such pricing motivates the demand side to alter their load profiles in the manner expected by the supply side. The stress of the power grid can therefore be reduced, and the utilization efficiency of the existing generators and available renewable energy resources is improved. For demand side, optimal demand response control for load profile management will be implemented to achieve economic savings under dynamic pricing (Cui et al. 2015; Xue et al. 2015). Unfortunately, in most cases, the consumers are charged a fixed electricity price no matter how much electricity is consumed (i.e., *fixed pricing*). In such pricing strategy, the fluctuation of prices is borne by the power supplier. Since the consumers are not affected by the prices, the demand in the grid can show large fluctuations with low valleys at night and large peaks during daytime (Simshauser and Downer 2014). These large fluctuations will dramatically reduce the profits of the power supplier (e.g., increased initial costs due to more capacity required, increased operational costs for purchasing more fuels), and reduce the supply reliability and system efficiency (Srinivasan et al. 2017). An effective pricing strategy to mitigate these negative effects is *dynamic pricing*, in which the electricity price is varying all the time and the consumers are charged by their instant energy demands (Faruqui and Palmer 2012). In such pricing strategy, the electricity price is usually set high in peak hours and low in off-peak periods. For reducing the electricity bills, the consumers will tend to shift some flexible demands from peak to off-peak periods, and as a result the large fluctuations of the demand can be effectively reduced.

Buildings are the major electricity consumers in the grid. Demand response (DR) control of buildings plays important roles in the interaction between the supply side and demand side. Demand response indicates altering the consumers' electricity usage from their normal consumption by incentive-based dynamic pricing techniques (Gao et al. 2015; Xue et al. 2015; Cui et al. 2015). Until now, many DR control methods have been developed (Nojavan et al. 2017; Cui et al. 2015; Xue et al. 2015, 2014; Huang et al. 2018c; Sun et al. 2013). For instance, Cui et al. developed a fast power demand response strategy for smart grid applications (Cui et al. 2015), which provides immediate and stepped power demand reduction through shutting down chillers without sacrificing the indoor thermal comfort. They also investigated the potentials of both active and passive cold storages in energy demand alternation. Xue et al. proposed an interactive building power demand management

strategy (Xue et al. 2014). They quantified the demand and effective demand alteration characteristics of commercial buildings. Their study also proposed proper building thermal storage model to predict power demand alteration potentials. Huang et al. developed a top-down demand response control method of renewable energy buildings, which enables renewable energy sharing among buildings inside the cluster (Huang et al. 2018c). In their method, the charging/discharging rates of batteries are adjusted in response to the dynamic electricity prices and the on-site renewable energy generations. Notably, Sun et al. conducted a systematic review of the DR controls using different types of facilities, including building thermal mass, thermal energy storage systems and phase change materials (Sun et al. 2013).

In recent years, many dynamic pricing methods considering building demand response have been developed by researchers for enhancing the application of smart grid (Lu et al. 2018; Srinivasan et al. 2017; Ma et al. 2016; Meng and Zeng 2013; Saghezchi et al. 2014; Dehnavi and Abdi 2016). For instance, Lu et al. developed a dynamic pricing demand response algorithm for a hierarchical electricity market considering both service provider's profit and customers' costs (Lu et al. 2018). They formulated the dynamic pricing problem as a discrete finite Markov decision process, and then adopted reinforcement learning algorithm to identify the optimal prices that maximize the total benefits of service provider and customers. Srinivasan et al. treated the pricing problem as a game process between the utility company and the end-users. They developed a game-theory-based dynamic pricing strategy for the Singapore electricity market, and tested the proposed pricing method in both the residential and commercial sectors (Srinivasan et al. 2017). Meng and Zeng modeled the interactions between the power supplier and consumers as a *1-leader, N-follower* Stackelberg game (Meng and Zeng 2013). They adopt genetic algorithms to maximize the profits of the power supplier and linear programming to minimize the payment bills of the consumers. Similarly, Ma et al. proposed a multi-party energy management method for building cluster based on non-cooperative game-theory (Ma et al. 2016).

The existing game-theory-based dynamic pricing methods require *information exchanges* not only between the supply and demand sides, but also among the demand side buildings, since the decision of one building's demand response is made based on operation of the others. However, in real application, the latter information exchanges are difficult due to the concern of privacy, communication complexity and high computation. Therefore, this chapter proposed a genetic algorithm-based dynamic pricing method for improving bi-directional interactions with reduced power imbalance, which does not require information exchanges among individual buildings. In the proposed method, at the demand side, targeting at minimizing daily electricity cost, a non-linear programming-based demand response control is performed in individual buildings at a dynamic price given by the genetic algorithm optimizer from the grid operator. Targeting at reducing grid power imbalance, the grid operator then uses a genetic algorithm optimizer to search for a better dynamic price based on the aggregated demand response results. Such interaction will continue until the grid power imbalance cannot be further reduced. The performances of the developed dynamic pricing method are analyzed under two different types of supply (by thermal power plants/renewable energy systems). The impacts of demand elasticity on the grid power imbalance are also investigated. The objective is to reduce the grid power imbalance, so as to achieve the maximum benefits of the smart grid. The contributions of the present study to the subject are briefly summarized as follows:

- Unlike most of the game-theory-based pricing method, the proposed method does not require information exchanges among individual buildings and thus can be easily implemented in practice for reducing grid power imbalance. The method is especially useful as a number of buildings are considered.
- Utilizing the bi-directional supply–demand reactions in the smart grid, the proposed

method, based on an optimal search method (i.e., GA), is able to select proper dynamic pricing which can effectively reduce the grid power imbalance. Such pricing method is not only applicable to the conventional energy supply from thermal power plants but also the emerging renewable energy supply.
- The demand elasticity impacts on the grid power balance have been revealed.

18.2 Main Challenge of Bi-Directional Interactions

Figure 18.1 depicts the bi-directional interactions between the grid operator and the end-users from the demand side. In such interactions, after receiving the day-ahead electricity generation from the supply side, the grid operator will determine and announce the preliminary dynamic pricing to consumers. In response to the received dynamic electricity prices, the consumers will plan their own DR controls, and identify their hourly energy demands at the given prices. Then, the buildings will send back the demand information to the grid operator. Based on the obtained energy demand, the grid operator will make proper adjustments in the dynamic prices to reduce the grid power imbalance. The main challenge in such bi-directional interactions is that how to ensure each price adjustment will lead to a further reduced grid power imbalance, and more importantly, after limited times of interactions, the grid power imbalance can be minimized. Aiming at overcoming such a challenge, this chapter proposes a dynamic pricing method, from the perspectives of the grid operator (i.e., power distribution) and the demand side, to improve bi-directional interactions with reduced power imbalance.

In fact, the reduced power imbalance, resulting from the joint efforts of the grid operator and the end-users, will also benefit the power suppliers in term of more economic power generations (e.g., reductions of costly power reserves). In return, the supply side can share the reduced cost with the end-users lowering the average electricity prices. In other words, a win–win situation (i.e., reductions of both utilities' operation costs and end-users' electricity bills) can be achieved. Note that the wholesale electricity market comprises three market places according to different time scales: day-ahead market, intraday market, and real-time market (Wang et al. 2015). The day-ahead scheduling conducts dispatch on a day-ahead basis. The intraday market is for the settlement hour-ahead before operation. The real-time market further compensates the real-time imbalance in a very short-term forward dispatch (Ellison et al. 2012). This chapter mainly focuses on reducing the power imbalance in the day-ahead scheduling market. The grid operator receives the expected day-ahead electricity productions provided by the supply side, then it searches the proper day-ahead electricity prices to regulate the demand side electricity usage to reduce imbalance with the expected electricity supply.

18.3 A Genetic Algorithm-Based Dynamic Pricing for Improving Bi-Directional Supply–Demand Interactions with Reduced Power Imbalance

Figure 18.2 presents flowchart of the genetic algorithm-based dynamic pricing method. It does not require information exchanges among individual buildings, since the buildings interact directly with the grid operator. Each building makes its own decision of demand response without the necessity to take account of other buildings' actions. For the grid operator, targeting at reducing grid power imbalance, the genetic algorithm is applied for generating trials of dynamic prices based on the expected energy supply and aggregated demand response results. While at the demand side, targeting at minimizing daily electricity cost, a non-linear programming-based demand response control is performed in individual buildings at a dynamic price given by the genetic algorithm optimizer from the grid operator. In each iteration, the power supplier produces several trails of

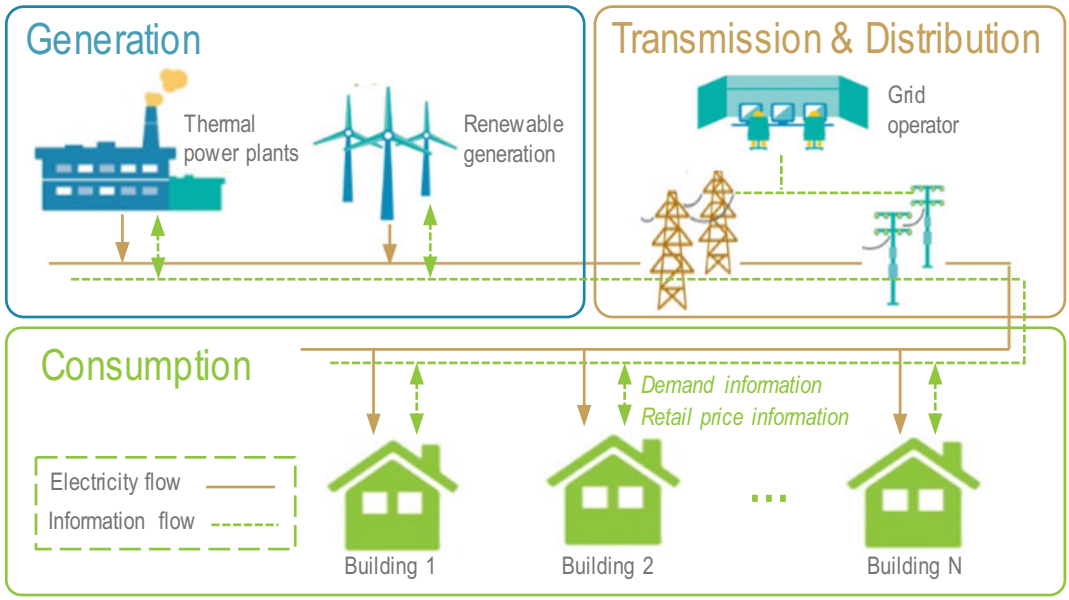


Fig. 18.1 Schematics of the power distribution and bi-directional interactions

dynamic prices and sends the price information to the buildings. The buildings will then conduct demand response controls to maximize their cost benefits (by shifting the flexible energy demands). Next, the buildings will send the altered demand information to the grid operator. The grid operator will evaluate the match between expected energy supply and the energy demand after demand response control. The iterative process will repeat until there is no further reduction in the grid power imbalance. Details of the processes of the developed dynamic pricing method are introduced as follows.

The electricity prices are set the same for all buildings in Fig. 18.2. Note that it is also reasonable to set different prices for different types of buildings, e.g., different prices for residential buildings and industrial buildings, as they could consume different quantities of electrical power and require different levels of power quality (Lu et al. 2018). To demonstrate the developed method, this chapter used a simple scenario that all buildings are charged with the same electricity prices. In case that the grid operator needs to charge different buildings at different electricity prices, the developed method is also applicable by simply adding several more sets of electricity price trials in the GA optimizer.

18.3.1 Grid Operator Action: Search of the Optimal Dynamic Prices by Genetic Algorithm

For the grid operator, genetic algorithm is applied for searching the optimal dynamic electricity prices that can minimize the grid power imbalance between the expected energy supply and the grid energy demand. Genetic algorithm is selected due to its powerful search capability and easy application (Huang et al. 2018a). In each iteration, for the given energy supply (expected by the supply side) $\left[E_{s,1}, E_{s,2}, \ldots, E_{s,24}\right]$ $(kW{\cdot}h)$, the genetic algorithm first produces a set of trials of electricity prices $[p_1^s, p_2^s, \ldots p_{24}^s]$ $(HKD/(kW{\cdot}h))$. The price information is sent to the buildings in the grid, and the buildings adjust their energy demands based on the electricity

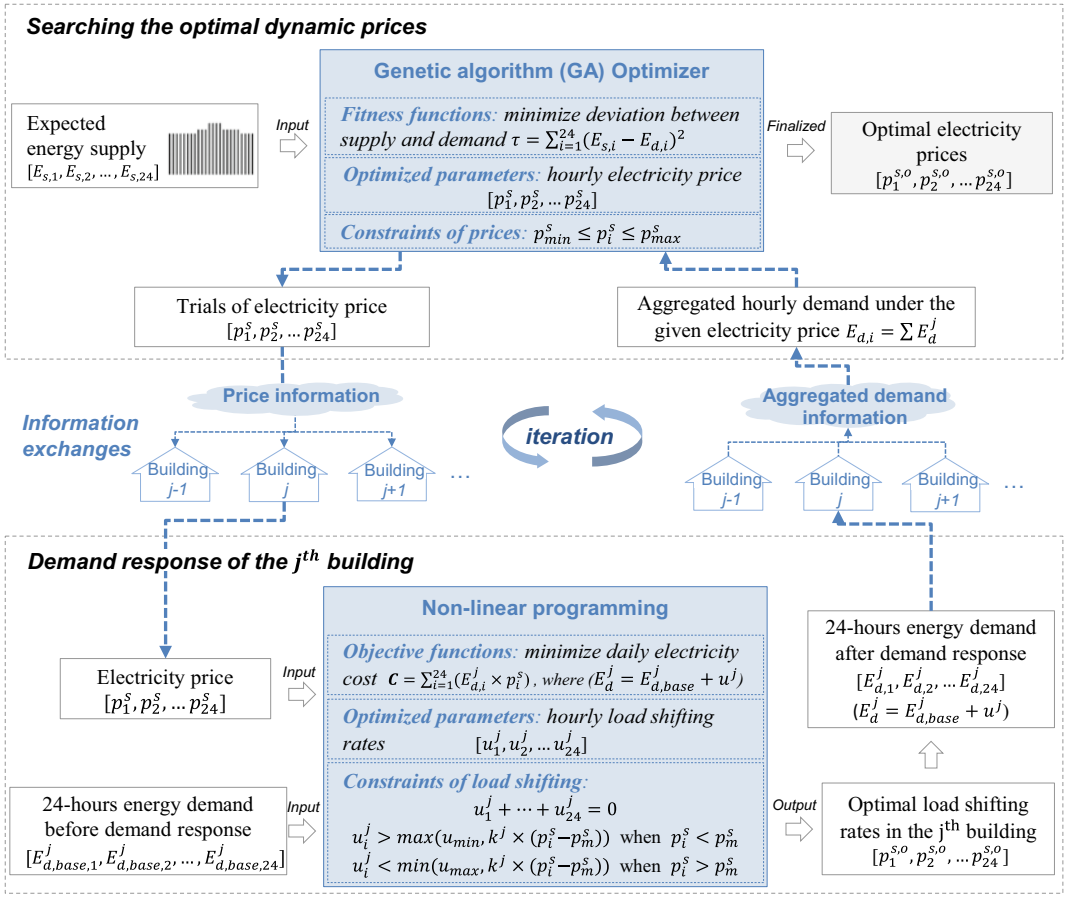


Fig. 18.2 Flowchart of the genetic algorithm-based dynamic pricing method for reducing grid power imbalance

prices. Then, the building demand information $[E_{d,1}, E_{d,2}, \ldots, E_{d,24}]$ $(kW{\cdot}h)$ are aggregated and sent to the grid operator. The aggregated grid demand $E_{d,i}$ $(kW{\cdot}h)$ in the i^{th} hour is calculated by Eq. (18.1)

$$E_{d,i} = \sum_{j=1}^{N} E_{d,i}^{j} \tag{18.1}$$

where $E_{d,i}^{j}$ $kW{\cdot}h$ is the energy demand of the j^{th} building in the i^{th} hour, and N represents the number of buildings in the grid.

The fitness function τ of the genetic algorithm is set as the average deviation between the energy supply and energy demand, as presented in Eq. (18.2),

$$\tau = \min\left(\sqrt{\sum_{i=1}^{24} \left(E_{s,i} - E_{d,i}\right)^2 / 24}\right) \tag{18.2}$$

where $E_{s,i}$ $(kW{\cdot}h)$ and $E_{d,i}$ $(kW{\cdot}h)$ are the expected energy supply and actual energy demand in the i^{th} hour, respectively. The objective is to minimize the deviation between the energy supply and demand. The hourly electricity price should be limited in a certain range. The constraints of the hourly electricity price are shown in Eq. (18.3)

$$p_{\min}^{s} \le p_i^s \le p_{\max}^{s}, \quad (\boldsymbol{i} = 1, 2, \ldots, 24) \tag{18.3}$$

where p_{min}^{s} $(HKD/(kW{\cdot}h))$ is the lower limit of the electricity price, and p_{max}^{s} $(HKD/(kW{\cdot}h))$ is the upper limit of the electricity price. The

iteration in genetic algorithm will repeat until the fitness function coverages, i.e., there is not further reduction in the grid power imbalance.

18.3.2 Demand Side Action: Demand Response of an ndividual Building at a Given Dynamic Price

After receiving the price information ($[p_1^s, p_2^s, \ldots p_{24}^s]$ $(HKD/(kW{\cdot}h))$ from the grid operator, each building at the demand side conducts demand response control to adjust its own energy demand for reducing the electricity bills (Cui et al. 2017). Non-linear programming is used to search the optimal load shifting rates ($[u_1^j, u_2^j, \ldots u_{24}^j]$ $(kW{\cdot}h)$ for each building. The objective function $\boldsymbol{C_j}$ of the demand response control is to minimize the electricity costs of the j^{th} building, as expressed by Eq. (18.4),

$$\boldsymbol{C_j} = \min\left(\sum_{i=1}^{24}\left(E_{d,i}^j \times p_i^s\right)\right) \tag{18.4}$$

where $E_{d,i}^j$ $(kW{\cdot}h)$ is the energy demand of the j^{th} building in the i^{th} hour, and p_i^s $(HKD/(kW{\cdot}h))$ is the electricity price in the i^{th} hour. The building's energy demand is calculated by Eq. (18.5) (Gao and Sun 2016),

$$E_{d,i}^j = E_{d,base,i}^j + u_i^j \tag{18.5}$$

where $E_{d,base,i}^j$ $(kW{\cdot}h)$ is the base energy demand of the j^{th} building in the i^{th} hour before demand response, u_i^j $(kW{\cdot}h)$ is the demand shifting rates of the j^{th} building in the i^{th} hour. A positive u_i^j indicates load shift-in; while a negative u_i^j indicates load shift-out.

The load shifting rates should meet the following constraints.

i. The sum of load shifting rates during one-day period equals to 0. This constraint guarantees the demand response does not reduce the consumer's needs on electrical devices usage.

ii. $$u_1^j + u_2^j + \ldots + u_{24}^j = 0 \tag{18.6}$$

iii. The load shifting rates in each hour should be limited in a certain range, since part of the hourly demand cannot be shifted to other time slot, such as the refrigerator and lighting. Otherwise, the users' comfort and satisfaction will be reduced dramatically. The percentage of maximum amount of demand that can be shifted is termed elasticity (Kirschen et al. 2000; CONSULTING 2010)

$$u_{\min}^j \le u_i^j \le u_{\max}^j, \quad (\boldsymbol{i} = 1, 2, \ldots, 24) \tag{18.7}$$

$u_{\min}^j$ (kW·h) and $u_{\max}^j$ (kW·h) are the lower and upper limits of the load shifting rates of the j^{th} building, which are calculated by an elasticity coefficient φ of the demand, as expressed by Eqs. (18.8) and (18.9) (Kirschen et al. 2000; CONSULTING 2010)

$$u_{min}^j = -1 \times \varphi \cdot max(E_{d,base}^j) \tag{18.8}$$

$$u_{max}^j = \varphi \cdot max(E_{d,base}^j) \tag{18.9}$$

The elasticity coefficient will determine the ranges of load shifting rates (i.e., maximum/minimum values). These ranges will be used in the non-linear programming algorithm as constraints for searching the optimal solutions of hourly load shifting rates which can achieve the most economic savings.

iv. Following (CONSULTING 2010; Dehnavi and Abdi 2016; Srinivasan et al. 2017; Kirschen et al. 2000), in this chapter, the maximum load shift-in and shift away rates are assumed to have an approximate linear relationship with the electricity price. In response to the electricity price increase, the customers

will tend to shift away more demand to the time slots when the electricity price is low. Considering the upper and lower limits of load shifting rates, the load shifting rates should satisfy Eqs. (18.10) and (18.11),

$$u_i^j > \max(u_{min}, k^j \times (p_i^s - p_m^s)), \text{when } p_i^s > p_m^s \quad (18.10)$$

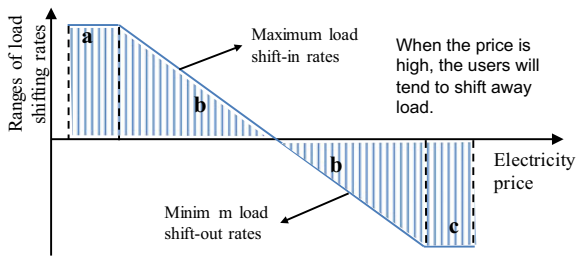
$$u_i^j < \min(u_{max}, k^j \times (p_i^s - p_m^s)), \text{when } p_i^s < p_m^s \quad (18.11)$$

where (p_m^s *HKD/(kW·h)*) is the mean electricity prices, k^j (*kW·h/HKD*) is a coefficient (negative value) representing the load shifting rates in response to the price changes. Note that this chapter adopts a simple load shifting model. Other load shifting models can also be used, as in practice different buildings may have different load shifting rates due to different preferences and usage characteristic (Kirschen et al. 2000).

Figure 18.3 presents the ranges of load shifting rates in response to electricity prices considering constraints (ii.) and (iii.). In Section a, the building reaches its maximum load shifting rates. In Section c, the building reaches its minimum load shifting rates. In Section b, the load shifting rates changes with the electricity price in a linear relationship.

The developed dynamic pricing method can improve the supply–demand interactions with reduced grid power imbalance. Genetic algorithm is applied by the grid operator to search the optimal prices, and non-linear programming is applied at the demand side for demand response control. By such bi-directional supply–demand interactions, the grid power imbalance can be reduced effectively. For the supply side, the reduced power imbalance can bring benefits, such as increased economic profits due to reduced operational costs and increased supply reliability and stability. For the demand side, the reduced power imbalance can bring benefits, such as reduced electricity bills. Note that the expected energy supply should be determined with reference to the historical energy demand and supply profiles in the power grid, and it is merely the amount of energy that the supply side expects the grid demand to approach, so as to maximize its benefits. It does not represent the real power supply during operation. This is because electricity is a non-storable commodity, in practice, the amount of electricity supply should be equal to the demand in the grid.

The 'fmincon' solver and the sequential quadratic programming (SQP) algorithm in Matlab were used to search for the optimal solutions of the non-linear programming problem. In the SQP algorithm, the function solves a quadratic programming (QP) subproblem at each iteration. The 'fmincon' solver updates an estimate of the Hessian of the Lagrangian at each iteration using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) formula (Powell 1978). The SQP algorithm has the advantages of strict feasibility with respect to bounds, robustness to non-double results, refactored linear algebra routines, and reformulated feasibility routines (Mathworks 2019). More details of the algorithm can be found in (Mathworks 2019).



Section **a**: when prices are low, demand is relatively inelastic – the price is too low for the customer to care.

Section **b**: when prices are moderately high, prices are relatively elastic and customers will curtail or shift consumption in response to changes in price.

Section **c**: when prices are very high, demand is once again inelastic – the only consumption left that has not been curtailed is nondiscretionary.

Fig. 18.3 Illustration of relationship between electricity price and load shifting rates

18.3.3 Building Modeling

In this chapter, 33 buildings are considered to be connected in a grid. The energy demand data of these buildings are obtained either from statistical analysis of existing buildings (Institute 2016, Huang and Sun 2019a) or from the DOE building templates ((DOE) 2018, Huang and Sun 2019a). Eight of the 33 buildings are residential buildings, and twenty-five of them are commercial buildings. The information of these buildings is summarized in Table 18.1. The building demands are investigated under the weather condition of Hong Kong.

18.3.4 Case Studies and Results Analysis

Two sets of expected energy supply data were used for validating the proposed dynamic pricing method. One set of energy supply data is flat in both the peak-hour and off-peak hours (see Fig. 18.6), which represents the generations of conventional thermal power plants using coals and gases. The other set has a peak (see Fig. 18.11), which represents the generations of renewable energy systems using PV panels and wind turbines. The shapes of these two types of supply were both determined with reference to existing literature and database (Xue et al. 2014; Nojavan et al. 2017; Huang and Sun 2019b; Zhang et al. 2016, 2018; Cui et al. 2019). In these two sets of data, the daily energy supply was approximately equal to the daily energy demand. The purpose of applying these two different sets of data was to demonstrate the effectiveness of the proposed pricing method under different types of energy supply in reducing the grid power imbalance. These two sets of energy supply data were used as inputs in the genetic algorithm for searching the optimal electricity prices. Compared with the conventional thermal power plant supply, the renewable energy supply is more fluctuating and not stable due to the uncertainty in weather conditions. Since

Table 18.1 Summary of the building energy demand data and data sources (Institute 2016; Huang and Sun 2019a; [DOE] 2018)

	ID	Name	Type	ID	Name	Type	ID	Name
Residential buildings	1	Manufactured Home Mixed AC	Commercial buildings	12	Hotel	Commercial buildings	23	Full service restaurant
	2	Multiple Family Mixed Cooling		13	Large office		24	Quick service restaurant
	3	Single Family Central AC		14	Medium office		25	Large office
	4	Single Family Heat Pump AC		15	Small office		26	Medium office
	5	Single Family Large		16	Bank		27	Small office
	6	Residential: low		17	Large retail		28	Primary school
	7	Residential: medium		18	Midrise apartment		29	Secondary school
	8	Residential: high		19	Large hotel		30	Standalone retail
Commercial buildings	9	Small retail		20	Small hotel		31	Strip mall
	10	Church		21	Hospital		32	Supermarket
	11	Education, K-12		22	Out patient		33	Warehouse

extensive studies have investigated the renewable generations with the impacts of weather uncertainty (Huang et al. 2018a, b, Al Irsyad et al. 2019), This chapter will not analyze the renewable generation uncertainty. The power production results (i.e., shape of renewable supply) from the existing studies are directly used as inputs for investigating the performance of the proposed method in reducing the grid power imbalance.

The field data of the 33 buildings summarized in Table 18.1 were used in the case studies. A typical day in summer was selected for analysis. In Hong Kong, due to the hot weather in summer, the electricity demands of buildings typically have large fluctuations with large peaks in the daytime and small valleys at midnight, which is challenging for the grid operator to balance the grid demand and supply. In other periods of the year, the electricity demands have relatively smaller fluctuations, and thus the grid regulating becomes relatively easier. A typical summer day approximately represents the worst scenario. If the proposed method can work well in the summer period, it will work well in other periods with less-challenging grid imbalance.

Figure 18.4 presents the hourly aggregated energy demand of the 33 buildings in the selected day. The energy demand profiles of individual buildings are illustrated by the blue curves in Figs. 18.7 and 18.12. Each of the buildings has a distinct load profile. The peak energy demand approached 5,800 kW·*h* and the peak hours were from 12:00 to 17:00. The energy demand information was used as inputs in the demand response control.

The demand elasticity coefficient was set as 25% in the case studies (Cui et al. 2018). For each building, the minimum/maximum load shifting rates were set to be −25%/25% of its peak energy demand. A positive value indicates load shifting-in and a negative value indicates load shifting-out. Note that the minimum/maximum load shifting rates of each building are affected by its own demand elasticity and may be different in each building. The case studies adopted 25% as an example. The impacts of demand elasticity were analyzed in Sect. 18.5.1.3 and Sect. 18.5.2.3. With reference to the local electricity market, the upper limit of the electricity price was set to be 2.5 *HKD/(kW·h)* and the lower limit was set as 0.5 *HKD/(kW·h)*.

For the genetic algorithm, in each generation the population size was set to 1000, and the number of parents was 400. The mutation rate was set as 0.01, and the maximum generation was 100. The total run time for conducting this number of simulation is approximately 30 min in a 64-bit computer with Intel®Core™i7-3770U CPU and 16 GB RAM.

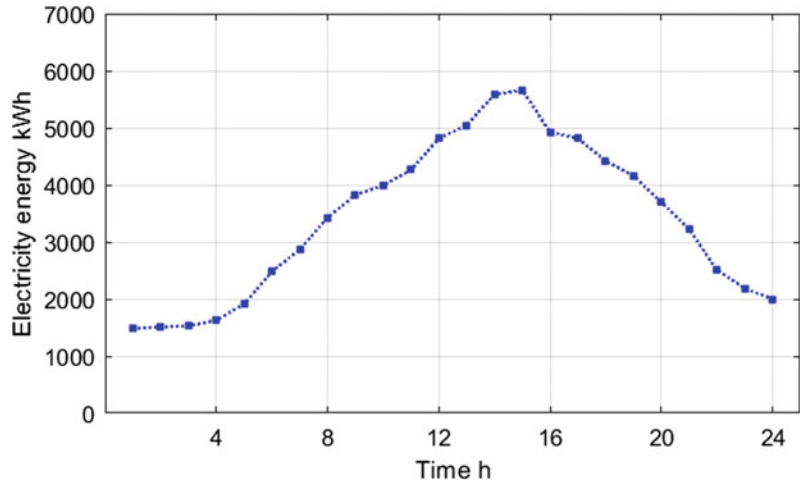


Fig. 18.4 The aggregated energy demand of the 33 buildings

18.3.5 Grid Power Imbalance Reduction as Energy Supply from Thermal Power Plants

18.3.5.1 Optimal Dynamic Prices and Grid Power Imbalance

This section presents the grid power imbalance reduction as energy supply from conventional power plants (using coal). For such electricity generations, when the supply is flat and does not fluctuate dramatically, the supply side can reach the maximum profits (Xue et al. 2014; Nojavan et al. 2017). Considering the high energy demands during peak hours, several more generators were assumed to be turned on in the peak period, and thus the supply in the period 11:00 ~ 20:00 was larger than the off-peak period, as presented in Fig. 18.6.

Figure 18.5 presents the optimized hourly dynamic electricity price in this case and compared its performance in reducing the power imbalance with the fixed price (which equals the average of the dynamic price). The hourly power mismatch was reduced significantly when the dynamic electricity pricing scheme was implemented. The peak power mismatch was reduced by as much as 615 kW (i.e., 43%), and the average power mismatch was reduced by 455 kW (i.e., 58%). The proposed dynamic pricing method is effective in reducing the grid power imbalance as energy supply from thermal power plants. Figure 18.6 compares the expected energy supply, the total grid energy demand, and the total energy demand after applying demand response control under the optimized hourly electricity prices (see Fig. 18.5). Referring to Fig. 18.6, the base energy demand in the grid (see the blue dashed curve with square marks) was larger than the expected energy supply in the periods 9:00 ~ 10:00 and 13:00 ~ 17:00. In order to reduce the electricity demand during this period, the grid operator set the electricity price relatively higher in these two periods, as shown in Fig. 18.5. In response to the high electricity price in these two periods, the consumers would shift part of the energy demand to other time slots with lower electricity prices. Note that since the deviation between supply and demand was relatively smaller in 11:00 and 12:00, correspondingly the electricity prices in these two hours were lower than other peak hours. From 1:00 to 6:00, the base electricity demand in the grid was smaller than the expected energy supply. To encourage more energy usage during this period, the grid operator set the electricity price relatively lower. In response to the low electricity price, the consumers would shift part of the energy demand to these time slots. By such price setting, the grid operator can lead the grid demand to approach the expected energy supply.

As can be seen from Fig. 18.6, by applying the proposed dynamic pricing method, the grid

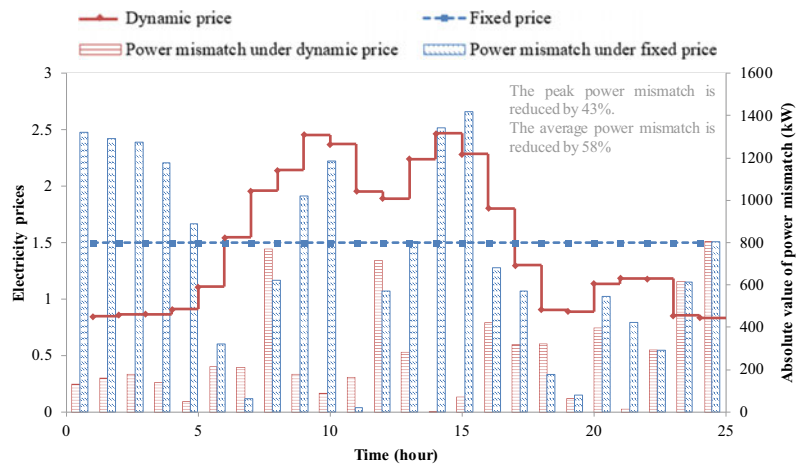


Fig. 18.5 Optimized dynamic pricing results as energy supply from thermal power plants

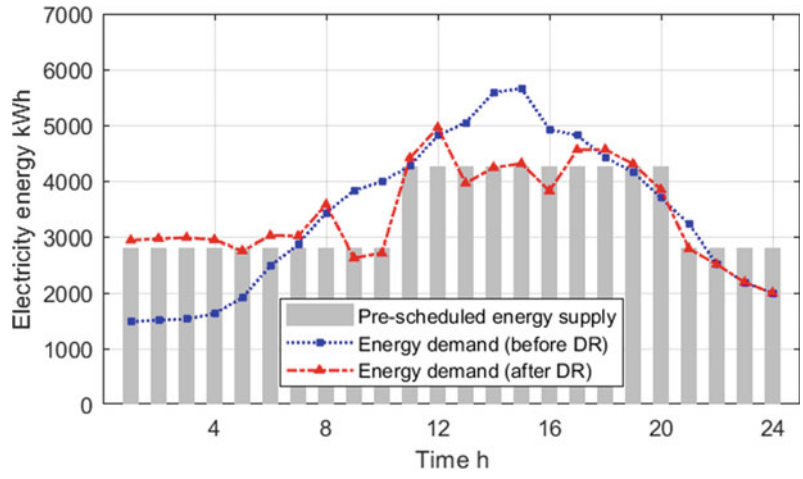


Fig. 18.6 Comparison of the expected energy supply (by thermal power plants), total energy demand, and total energy demand in response to the dynamic electricity prices

power imbalance was reduced significantly. The average deviation between the energy supply and demand (calculated by Eq. (18.2)) was reduced from 851 kW·*h* to 359 kW·*h* by applying the dynamic pricing (about 57.8% reduction in the power imbalance). Such reduced grid power imbalance can effectively improve the grid operational reliability and stability and meanwhile reduce energy wastes.

18.3.5.2 Single Building's Demand Response and Associated Cost Savings

Figure 18.7 presents the energy demand of each single building before and after demand response control under the optimal dynamic electricity prices. As can be seen, during the periods of 9:00 ~ 10:00 and 13:00 ~ 15:00, there was significant decrease in the energy usage in each building. This was because the electricity prices were relatively high in this period: the buildings would try to shift away the flexible energy demands to other periods with lower electricity prices, so as to reduce their own electricity bills. Since the electricity prices during the period of 1:00 ~ 7:00 were relatively low, the flexible energy demands were shifted in to this period. Figure 18.7 also presents the electricity cost saving of each building via the load shifting under the given dynamic price. The buildings can achieve a minimum cost saving of 5.7% (i.e., Building 29) and a maximum cost saving of 10.2% (i.e., Building 33). Such results proved that as energy supply from the thermal power plants, the developed dynamic pricing method not only benefits the grid side by reducing the grid power imbalance (see Fig. 18.6), but also benefits the involved buildings by reducing their electricity costs.

18.3.5.3 Elasticity Impacts on the Grid Power Balance

This section analyzes the impacts of elasticity on the grid power imbalance reduction when the electricity is supplied by thermal power plants. Elasticity indicates the percentage of maximum load shifting-in/out rates of the building (refer to Eqns. 18.8 and 18.9). The grid power balance performances in eight scenarios with different elasticity values (0, 5%, 10%, 15%, 20%, 25%, 30% and 35%) were analyzed. When elasticity was 0, the buildings cannot shift any demand. It has the same values as the demand presented in Fig. 18.4, and it is used as benchmark in evaluating the reduction in grid power imbalance. Note that the elasticity coefficients were set the same for all the 33 buildings. The maximum/minimum load shifting rates of all the 33 buildings will change simultaneously when the elasticity value increases from 5 to 35%. Correspondingly, due to the changed constraints, the simulated load shifting rates in each hour will also change.

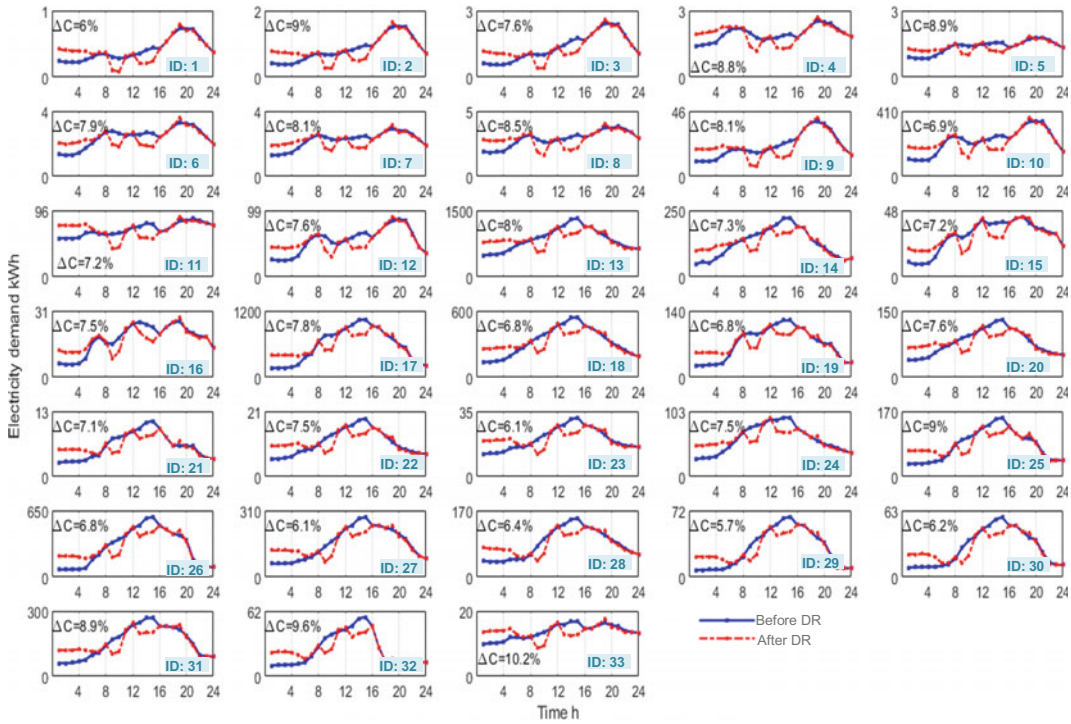


Fig. 18.7 Energy demand change and associated cost saving of each single building after load shifting under the dynamic price as energy supply from thermal power plants

Figure 18.8 compares the energy supply by thermal power plant with the grid energy demand (after applying dynamic pricing and demand response). As can be seen, when the elasticity increased from 0 to 20%, the grid power balance became better; when the elasticity further increased from 20 to 35%, the grid power balance became worse. This is because when the demand elasticity is relatively small, the total flexible demand is smaller than the deviation between supply and demand. As a result, due to the limited demand shifting ability, large deviations between supply and demand still exists. In other words, the energy demand is 'under-tuned'. On the other hand, when the demand elasticity is relatively high, the total flexible demand is larger than the deviation between supply and demand. As the consumers will always shift as much flexible demand to the low-price periods as possible to maximize their own economic benefits, large deviations between supply and demand may occur again, due to the 'over-tune' of consumers' demands.

Figure 18.9 compares the average deviations (calculated by Eq. 18.2) between the energy demand and expected energy supply (see the red bars). The reduction in the grid power imbalance are also highlighted in the blue bars by comparing with the benchmark case (elasticity = 0). As can be seen, with the increase of demand elasticity, the average deviation first decreased and then gradually increased. When the demand elasticity is 20%, the deviation reaches the minimum. The grid power imbalance was reduced by as much as 59% compared with the benchmark case (see the red bar). When the elasticity was smaller than 20%, the energy demand was 'under-tuned' due to the limited demand shifting ability. While when the elasticity was larger than 20%, the energy demand was 'over-tuned' as

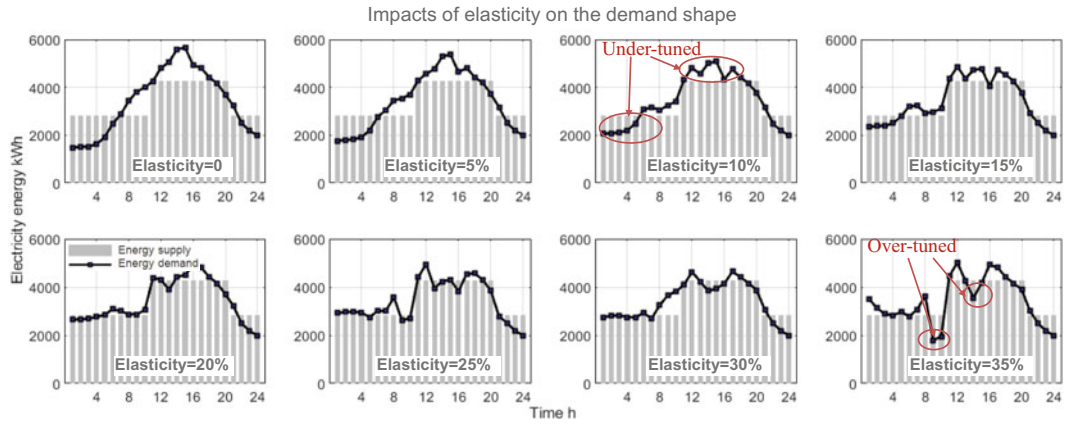


Fig. 18.8 Impacts of demand elasticity on the grid power balance as energy supply from thermal power plants

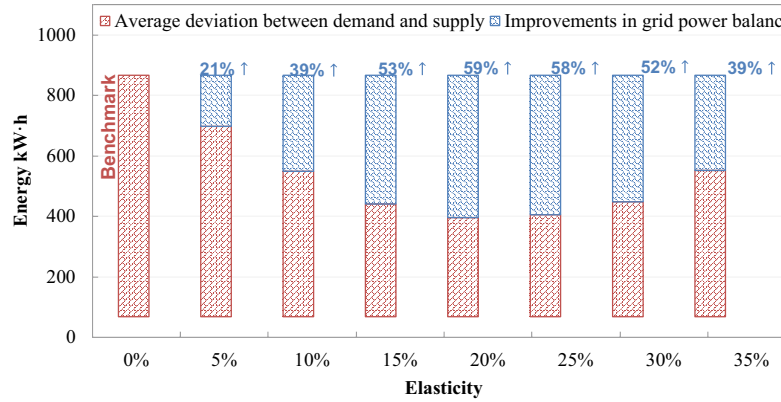


Fig. 18.9 Comparison of the demand elasticity impacts on the grid power balance as energy supply from thermal power plants

consumers shifted more flexible demand than the energy deviation. Note that for different supply and demand, the optimal demand elasticity leading to the minimal grid power imbalance can be different.

18.3.6 Grid Power Imbalance Reduction as Energy Supply from Renewables

18.3.6.1 Optimal Dynamic Prices and Grid Power Balance

This section presents the grid power imbalance reduction as energy supply from renewable energy systems (i.e., PV panels and wind turbines). Figure 18.11 presents the energy supply data. The energy supply shape was determined referring to the data in existing literature (Huang and Sun 2019b; Zhang et al. 2016, 2018).

Figure 18.10 presents the optimized hourly dynamic electricity prices in this case and compared its performance in reducing the power imbalance with the fixed prices (which equals the average of the dynamic price). The hourly power mismatch was reduced significantly when the dynamic electricity pricing scheme was implemented. The maximal power mismatch was reduced by as much as 1,171 kW (i.e., 69%), and the average power mismatch was reduced by 881 kW (i.e., 81%). Such results show that the proposed dynamic pricing method is effective in reducing the grid power imbalance as energy supply from renewable energy systems. Figure 18.11 compares the expected energy supply, the total grid energy demand, and the total

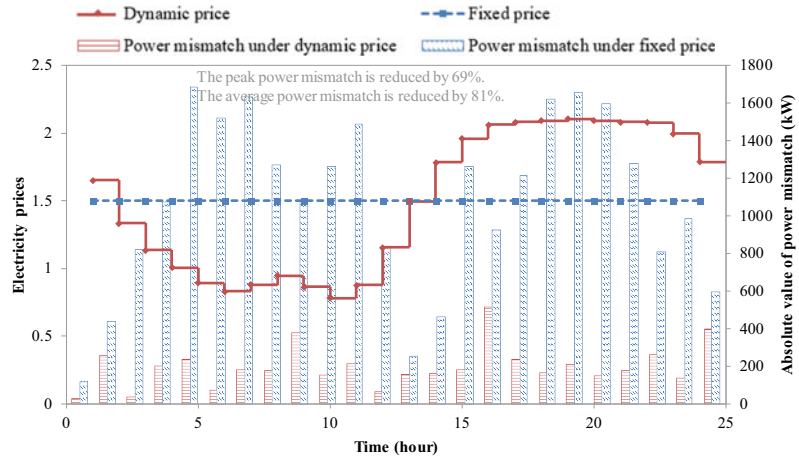


Fig. 18.10 Optimized dynamic pricing results as energy supply from renewable energy systems

energy demand after applying demand response control based on the optimized hourly electricity prices (see Fig. 18.10). Referring to Fig. 18.11, the base energy demand in the grid (see the blue dashed curve with square marks) was larger than the expected energy supply from 14:00 to 24:00. In order to reduce the electricity demand during this period, the power supplier set the electricity price relatively higher in this period, as shown in Fig. 18.10. In response to the high electricity price, the consumers would shift part of the energy demand to other time slots with lower electricity prices. From 2:00 to 12:00, the base electricity demand in the grid was smaller than the expected energy supply. To encourage more energy usage during this period, the grid operator set the electricity price relatively lower. In response to the low electricity price, the consumers would shift part of the energy demand to these time slots. By such price setting, the grid operator can lead the grid demand to approach the expected energy supply.

As can be seen from Fig. 18.11, by applying the proposed dynamic pricing, the grid power imbalance was reduced dramatically. The average deviation between the energy supply and demand (calculated by Eq. 18.2) was reduced from 1,129 kW·*h* to 279 kW·*h* by applying the dynamic pricing (about 76.9% reduction in the grid power imbalance). Such reduced grid power

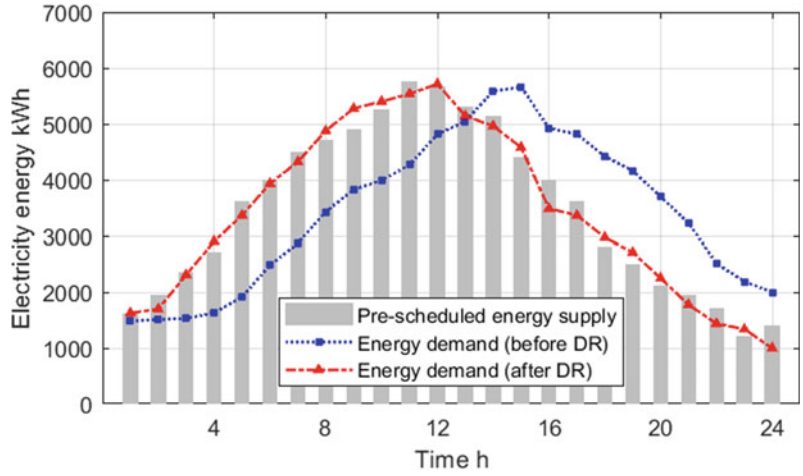


Fig. 18.11 Comparison of the expected energy supply (by renewable energy systems), total energy demand, and total energy demand in response to the dynamic electricity prices

imbalance can effectively improve the grid operational reliability and stability and meanwhile reduce energy wastes.

18.3.6.2 Single Building's Demand Response and Associated Cost Savings

Figure 18.12 presents the energy demand of each single building before and after demand response control based on the optimal dynamic electricity prices. As can be seen, during the period of 14:00 ~ 24:00, there was significant decrease in energy usage in each building. This was because the electricity prices were relatively high in this period: the buildings would try to shift away the flexible energy demands to other periods with lower electricity prices, so as to reduce the electricity bills. Since the electricity prices during the period of 2:00 ~ 12:00 were relatively low, the flexible energy demands were shifted to this period. Figure 18.12 also presents the electricity cost saving of each building via load shifting under the given dynamic price. The buildings can achieve a minimum cost saving of 8.1% (i.e., Building 29) and a maximum cost saving of 16.3% (i.e., Building 2). The cost savings were higher than the first case that energy is supplied from thermal power plants (i.e., 5.7% ~ 10.2%). This was because of the increased amount of load shifted: in total 13,321 kW·h energy demand was shifted in the second case (i.e., energy supply from renewable energy systems), while only 8,058 kW·h energy demand was shifted in the first case (i.e., energy supply from thermal power plants). Such results also proved that as energy supply from the renewable systems, the developed dynamic pricing method can benefit both the grid side (by reducing the aggregated power imbalance, see Fig. 18.11) and the associated individual buildings (by reducing the electricity costs).

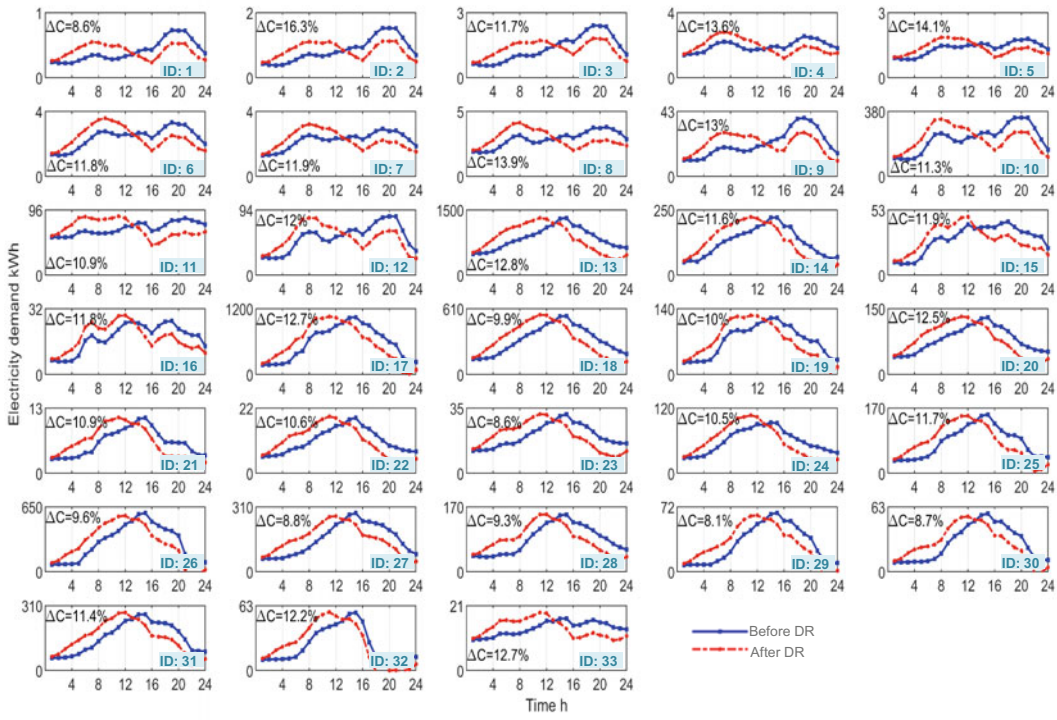


Fig. 18.12 Energy demand change and associated cost saving of each single building after load shifting under the dynamic price as energy supply from renewable energy systems

18.3.6.3 Elasticity Impacts on the Grid Power Balance

This section analyzes the impacts of elasticity on the grid power balance performances when the electricity is supplied by renewable energy systems. Again, the grid power balance performances in eight scenarios with different elasticity values (0, 5%, 10%, 15%, 20%, 25%, 30% and 35%) were analyzed. The first scenario (Elasticity = 0) was used as benchmark in evaluating the reduction in grid power imbalance.

Figure 18.13 compares the energy supply (by renewable energy systems) with the grid energy demand (after dynamic pricing and demand response). As can be seen, when the elasticity increased from 0 to 25%, the grid power balance became better; when the elasticity further increased from 25 to 35%, the grid power balance became worse. This is because when the demand elasticity is relatively small, the total flexible demand is smaller than the deviation between supply and demand. As a result, due to the limited demand shifting ability, large deviation between supply and demand still exists. The energy demand is 'under-tuned'. On the other hand, when the demand elasticity is relatively high, the total flexible demand is larger than the deviation between supply and demand. As the consumers will always shift as much flexible demand to the low-price periods as possible to maximize their own economic benefits, large deviation between supply and demand may occur due to the 'over-tune' of consumers.

Figure 18.14 compares the average deviations (calculated by Eq. 18.2) between the energy demand and the expected energy supply (see the red bars). The reduction in the grid power imbalance are also highlighted in the blue bars by comparing with the benchmark case (elasticity = 0). As can be seen, with the increase of demand elasticity, the average deviation first decreased and then gradually increased. When the demand elasticity was 25%, the deviation reaches the minimum. The grid power imbalance was reduced by as much as 81% compared with the benchmark case. When the elasticity was smaller than 25%, the energy demand was 'under-tuned' due to the limited demand shifting ability. While when the elasticity was larger than 25%, the energy demand was 'over-tuned' as consumers shifted more flexible demand than the energy deviation. Note that for different supply and demand, the optimal demand elasticity leading to the minimal grid power imbalance can be different.

18.4 Summary

This chapter has proposed a genetic algorithm-based dynamic pricing for improving bi-directional supply–demand interactions, which

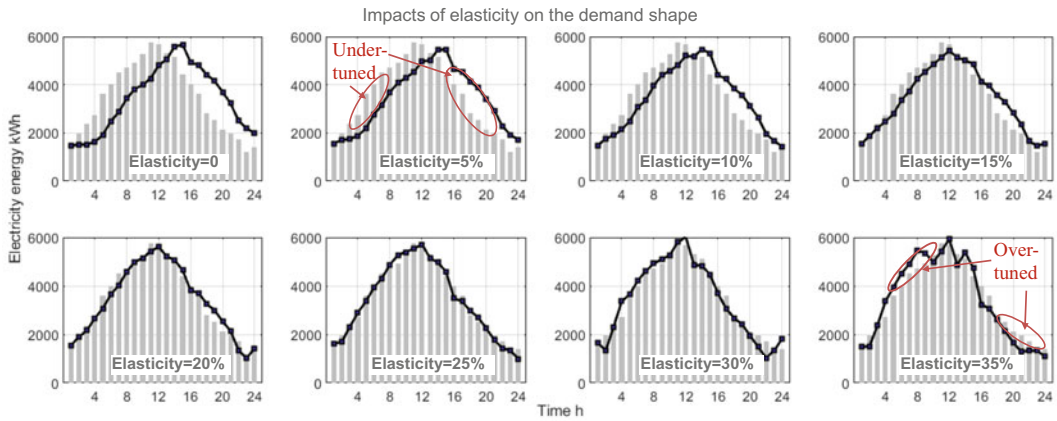


Fig. 18.13 Impacts of demand elasticity on the grid power balance as energy supply from renewable energy systems

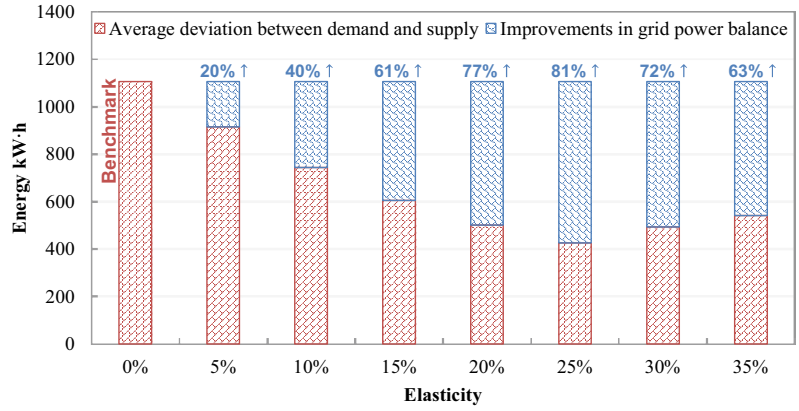


Fig. 18.14 Comparison of the demand elasticity impacts on the grid power balance as energy supply from renewable energy systems

can effectively reduce the grid power imbalance, so as to achieve the maximum benefits of the smart grid. At the demand side, a non-linear programming-based demand response control has been performed in individual buildings at a dynamic price given by the genetic algorithm optimizer from the grid operator, targeting at minimizing daily electricity cost. For the grid operator, the genetic algorithm optimizer has been used to search for a better dynamic price based on the aggregated demand response results, targeting at reducing grid power imbalance. The developed dynamic pricing method has been validated using two different types of energy supply: by thermal power plants and by renewable energy systems. The impacts of demand elasticity on the grid power balance have also been analyzed. The major findings are stated as follows:

- The developed pricing method can effectively improve bi-directional supply–demand interactions with reduced grid power imbalance, so as to achieve the maximum benefits of the smart grid. The proposed method is applicable to both the conventional thermal power generations and renewable energy generations.
- For the energy supply by conventional thermal power plants, the developed dynamic pricing method can reduce the grid power imbalance by as much as 59% in the case studies. For the energy supply by renewable energy systems, the developed method can reduce the grid power imbalance by as much as 81% in the case studies.
- An optimal demand elasticity coefficient exists in each microgrid for achieving the best grid power balance. When the demand elasticity is small, due to the limited demand shifting ability, large deviation between supply and demand may still exists. While when the demand elasticity is high, as the consumers will always shift as much flexible demand to the low-price periods as possible to maximize their economic savings, large deviation between supply and demand may occur again.

It should be noted that this chapter used a simplified demand elasticity model and did not consider the diverse energy usage characteristics in different types of buildings. Future work will develop more sophisticate elasticity models to take account of these diverse energy characteristics. Future work will also take account of the demand/supply prediction uncertainty to improve the robustness of the proposed dynamic pricing method.

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Hierarchical Coordinated Demand Response Control for Building Cluster

19

Yongjun Sun and Pei Huang

Abstract

Demand response (DR) control is one of the common means used for building peak demand limiting. Most of the existing DR controls focused on performance optimization of single building, and thus may cause new undesirable peak demands at building group, imposing stress on the grid power balance and limiting the economic savings. A few latest studies have demonstrated the potential benefits of DR coordination, but the proposed methods cannot be applied in large scales. The main reason is that, for DR coordination of multiple buildings, associated computational load and coordination complexity, increasing exponentially with building number, are challenges to be solved. This chapter, therefore, proposes a hierarchical DR control to optimize the building operations in large scales for building group peak demand reduction. The hierarchical control first considers the building group as a 'virtual' building and searches the optimal performance that can be achieved at building group using genetic algorithm (GA). To realize the GA-based optimal performance, it then coordinates each single building's operation using non-linear programming (NLP). For validations, the proposed method has been applied on a case building group, and the study results show that the hierarchical DR control can overcome the challenges of excessive computational load and complexity. Moreover, in comparison with conventional independent DR control, it can achieve better performances in aspects of peak demand reduction and economic savings. This chapter provides a coordinated DR control method for application in large scales, which can improve the effectiveness and efficiency in relieving the grid stress, and reduce the electricity bills of the end-users.

Keywords

Peak demand limiting · Demand response · Building group coordination · Economic cost · Grid interaction

19.1 Introduction

Buildings consume over 40% of the overall energy worldwide. With increasing population, continuing urbanization and increasing economic, the energy consumption in the building sector continues to increases. The rapid increase

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of electrical demand has imposed great stress on the power grid (IEA 2014). To eliminate such large stress, peak demand limiting in the building sector is necessary (Albadi and El-Saadany 2007). Peak demand limiting can bring significant benefits to both the consumers (e.g., economic saving) and the power suppliers (e.g., reduced capital costs, improved reliability and efficiency). Under a peak demand pricing strategy (Albadi and El-Saadany 2007), the building is usually charged for two parts: the amount of consumed electricity and the peak demand in a billing period. With effective peak demand management in buildings, the International Energy Agency (IEA) reported a possible annual saving of $10–$15 billion for U.S. market (Sadineni and Boehm 2012). Load shifting is one of the main means commonly used for peak demand management in buildings. Load shifting indicates taking advantage of different electricity prices via shifting the peak-hour demand to the off-peak hour (Agency 2010). The implementation of a load shifting control strategy contains three parts: load prediction, cooing charging and discharging controls (Sun et al. 2013). There are typically four facilities widely used for load shifting, namely building thermal mass (Yin et al. 2010; Sun et al. 2010; Gao and Sun 2016), thermal energy storage (Gao and Sun 2016; Roth et al. 2006; Kuznik et al. 2011), combined use of building thermal mass and thermal energy storage (Gao and Sun 2016; Henze et al. 2004; Hajiah and Krarti 2012), and phase change materials (Gao and Sun 2016; Kuznik et al. 2008; Scalat et al. 1996).

Demand response (DR) control is an essential aspect in load shifting for eliminating the undesirable peak demands in the power grid and reducing the electricity bills of buildings. Demand response enables the end-users to participate in the power grid operation by decreasing or altering their energy usage from their normal consumption in response to time-based electricity prices or other forms of financial incentives. Many DR control methods have been developed in recent years. For instance, Cui et al. proposed a fast power DR strategy for smart grid applications (Cui et al. 2015). Without sacrificing the thermal comfort in conditioning zones, their developed method enabled immediate and stepped power demand reduction by means of shutting down chillers. The potentials of both active and passive cold storages in energy demand alternation were also investigated. Xue et al. developed an interactive building DR strategy for commercial buildings (Xue et al. 2014). They first quantified the demand and effective demand alteration potentials of commercial buildings and then proposed an interactive DR control method to optimize the thermal storage's operation under dynamic prices. Considering the building energy demand prediction uncertainty, Gao et al. proposed a generic algorithm (GA)-based robust DR control for commercial buildings (Gao et al. 2015). Compared with those deterministic DR controls, Gao et al.'s method can significantly improve the building performances under uncertainty. Notably, Sun et al. conducted a systematic review of the DR controls in the building sector. Different load shifting facilities, namely building thermal mass, thermal energy storage systems and phase change materials, were reviewed in their study (Sun et al. 2013).

The above conventional independent DR controls (i.e., Cui et al. 2015; Xue et al. 2014; Gao et al. 2015) focus on single building's performance optimization, while they ignore the more important grid-concerned building group performance (Mohsenian-Rad et al. 2010; Muratori et al. 2014). Unlike them, the coordinated controls pay more attention to the improvements of building group performance by coordinating individual buildings' operation (Jafari-Marandi et al. 2016; Odonkor and Lewis 2015, 2016; Hu et al. 2014; Huang et al. 2018). Shen et al. compared the building group performances of a conventional independent building DR control and a coordinated DR control (Shen et al. 2016). They found that the conventional independent DR could cause new undesirable peak demands of building group which could be much larger than the original ones and impose stress on grid, while the coordinated controls can solve these problems. Sun et al. developed a collaborative DR control of zero energy buildings for building group performance

improvements, in which the control of each building was conducted in sequence, and the optimization of one building's operation was based on the previously optimized building' operation (Fan et al. 2018). With the building number increase, the collaborative DR control has increased chance to converge at near-optimal results due to the fact that the final performance of a sequential optimization will suffer from the imperfection of any previous steps. Notably, Gao et al. developed a genetic algorithm (GA)-based coordinated DR control for peak demand reduction of the building group (Gao and Sun 2016). In the method, the thermal charging/discharging rates of all the buildings inside the group were optimized simultaneously, and minimization of the group-level peak demand was used as the fitness function in GA. This method could effectively reduce the peak demand at the group level, but it cannot be applied in large scales when a large number of buildings are concerned. This is due to the excessive computational loads and coordination complexity, as there are too many variables to be optimized at the same time.

To sum up, most of the existing DR controls focused on single building's peak demands reduction, which could cause new undesirable peak demands at building group larger than the original one and impose stress on grid. Only a few studies considered coordination of DR controls at building group. However, these existing coordinated controls cannot be applied in a large scale, due to their excessive computational loads. This chapter, therefore, proposes a computation-efficient hierarchical demand response control method, which is applicable in large scales for building group peak demand reduction. The hierarchical control first considers the building group as a 'virtual' building and searches the optimal performance that can be achieved at building group using genetic algorithm (GA). To realize the GA-based optimal performance, it then coordinates each single building's operation using non-linear programming (NLP). The objective is to provide a practical demand response control method for application in large scales, which can effectively improve building group performances. For validation purpose, the proposed method has been applied on a case group of eight buildings, and its performances have been compared with a conventional independent demand response control. The developed method can not only effectively and efficiently relieve the grid stress and reduce the operation costs and net emission of the power supplier, but also effectively reduce the consumers' electricity bills. The contributions of the present study to the subject are briefly summarized as follows:

- A hierarchical demand response control method of building group has been developed for large scale application, with significant computation reduction.
- The developed method can coordinate single building's operation to jointly minimize the building group peak demands.
- The performances of the proposed method have been compared with a conventional independent demand response control and an existing coordinated demand response control.
- The performance improvements have been analyzed in aspect of peak demand reduction and consumers' economic saving.

This chapter is organized as follows. Section 19.2 presents the proposed hierarchical demand response control. Section 19.3 describes the building, HVAC system, and PCM storage tank models used in this chapter. Section 19.4 presents the case studies conducted and analyzes the results. A summary is given in Sect. 19.5.

19.2 A Hierarchical Demand Response Control for Improved Building Group Performances

This section first compares the structures of the conventional independent demand response control and the coordinated demand response control. Then, the developed hierarchical coordinated demand response control is introduced.

19.2.1 Basic Ideas of the Independent DR Control and Coordinated DR Control

Figure 19.1 compares the control structures of the independent DR control and coordinated DR control. In the independent DR control, each building is controlled by an independent local controller. The local controller optimizes the thermal storage of each building to minimize their own peak energy demands. These conventional independent DR controls can effectively reduce single building's peak electrical demands. However, they could cause new undesirable peak demands at building group, especially when many buildings shifts demand to the same low-price period (Gao and Sun 2016). In the coordinated DR controls, the buildings in the building group are controlled by a central controller. The central controller coordinates each building's thermal storage operation to shift their peak demands to different periods, so as to minimize the building group peak electrical demands.

19.2.2 A Hierarchical Coordinated DR Control for Improved Building Group Performances

This sub-section presents the developed hierarchical demand response control method. Since HVAC systems always represent the largest energy end-users in buildings, this chapter considers HVAC systems as the main study object. Meanwhile, for demand shifting purpose, this chapter also considered an active thermal energy storage system, which is integrated with the HVAC system.

Figure 19.2 presents the flowchart of the developed method. The aim of the hierarchical control is to coordinate the operation of each single building inside a group, so as to achieve the optimal performances at the building group. The hierarchical demand response control includes four steps. In Step 1, all the buildings in the building group are considered as a 'virtual' building, and the electrical demand and load shifting capacity of the 'virtual' building are predicted: its electrical demand is the aggregated demand of all buildings inside the group, and its demand shifting ability is the aggregated demand shifting ability of all the buildings. In Step 2, the operation of the 'virtual' building is optimized using genetic algorithm (GA). The optimized performance of the 'virtual' building is considered to be the best performances that the building group can achieve (Shen et al. 2016). In Step 3, the operation of each single building inside the building group is coordinated using non-linear programming (NLP) based on the 'virtual' building's operation obtained from Step 2. In Step 4, the performances of the proposed hierarchical DR control are compared with a conventional independent DR control and an

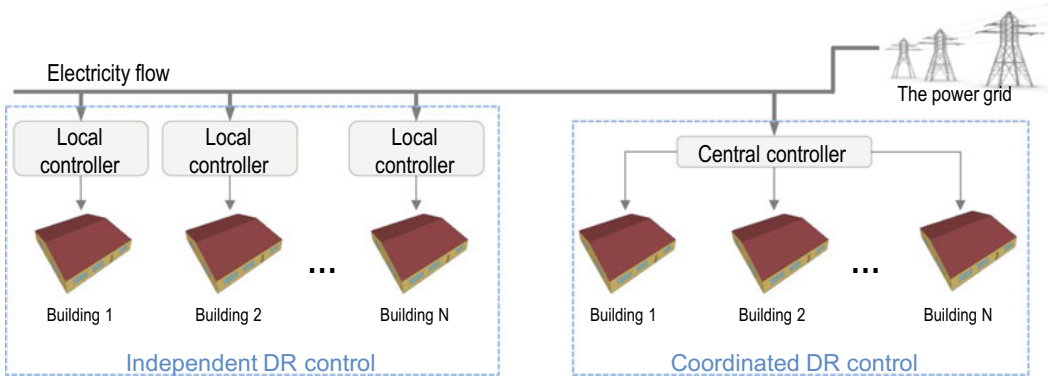

Fig. 19.1 Comparison of the independent DR control and coordinated DR control

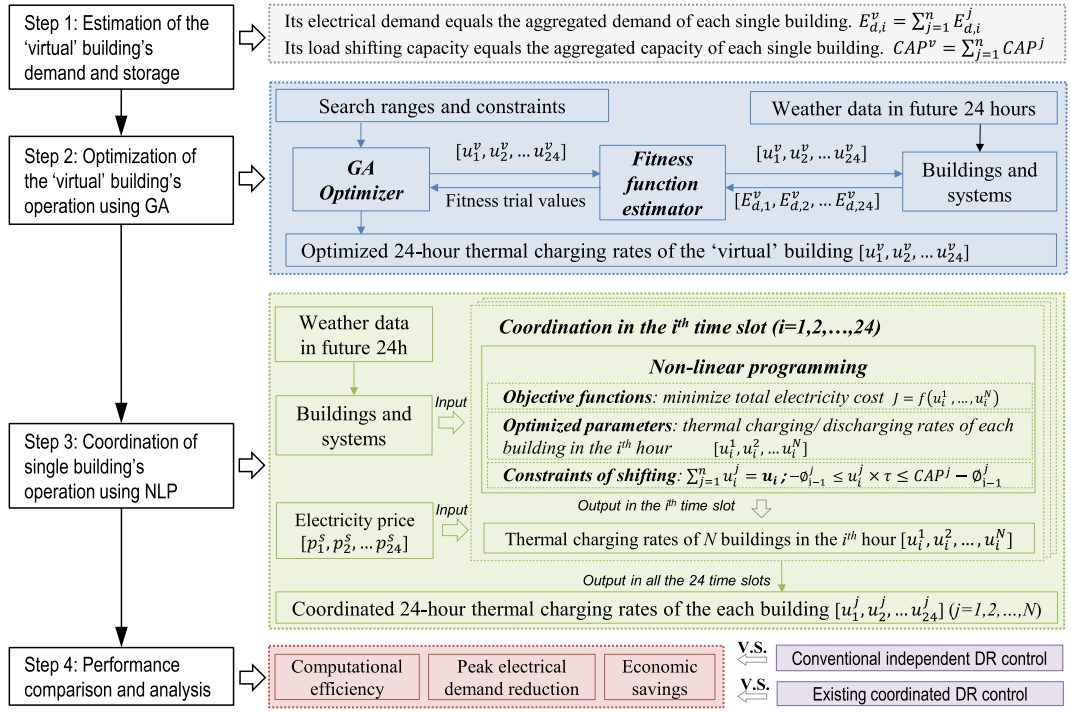

Fig. 19.2 Flowchart of the hierarchical coordinated demand response control

existing coordinated DR control. The details of each step are introduced below.

Step 1: Estimation of the 'virtual' building's demand and storage

In this step, all the buildings inside the group are considered as a 'virtual' building. Its hourly energy demand ($E^v_{d,i}$ (kW·h)) is the aggregated hourly energy demand of each single building ($E^j_{d,i}$ (kW·h)) (i indicates time with a unit of hour), and its load shifting capacity (CAP^v (kW·h)) is the aggregated load shifting capacity of each single building (CAP^j (kW·h)).

$$E^v_{d,i} = \sum_{j=1}^{n} E^j_{d,i} \tag{19.1}$$

$$CAP^v = \sum_{j=1}^{n} CAP^j \tag{19.2}$$

Step 2: Optimization of the 'virtual' building's operation using GA

The energy demand of each single building is calculated using the models presented in Sect. 19.3. The GA algorithm searches the optimal charging/discharging rates of the thermal storage tank (see Sect. 19.3.3 for the detailed models) that can minimize the peak energy demands at the building group. For example, the storage tank is charged in advance during off-peak periods; then it discharges the cooling during on-peak hours, so that the electricity usage during the on-peak hours can be reduced. In the GA simulation, the inputs mainly include the thermal energy charging/discharging rates to be optimized, the future 24-h weather data (e.g., temperature, solar radiation, etc.), building parameters, HVAC systems parameters, and thermal storage parameters. In each generation of GA, trials of 24-h thermal storage hourly charging/discharging rates (i.e., $\left[u^v_1, u^v_2, \ldots u^v_{24}\right]$ kW) are generated by the GA optimizer. The virtual building's hourly energy demand ($E^v_{d,i}$ (kW·h)) in the future 24 h is then predicted using

the building and system models (see models given in Sect. 19.3).

The operation of thermal storage tanks should meet the following two constraints: (1) The thermal storage charging amount could not exceed the remaining thermal storage capacity. (2) The thermal storage discharging amount could not exceed the stored thermal energy in the tank. These two constraints are expressed by Eq. (19.3) (Lu et al. 2015; Sun et al. 2018),

$$0 \le \varnothing_0^v + \left(u_1^v + u_2^v + \ldots + u_i^v\right) \times \tau \le CAP^v \quad \text{where i} = 1, 2, \ldots, 24 \tag{19.3}$$

where $\varnothing_0^v$ (kW·h) is the amount of thermal energy initially stored in the tank, τ is the duration of thermal charging/discharging (i.e., 1 h in this chapter).

This chapter considers the strategy to minimize the peak electrical demand of the building group. Following this control goal, a fitness function is determined, as expressed by Eq. (19.4) (Salom et al. 2011).

$$J_{grid} = \min(E_{d,peak}^v) \tag{19.4}$$

where $E_{d,peak}^v$ (kW·h) is the peak electrical demand, which is calculated by Eq. (19.5).

$$E_{d,peak}^v = max\left(\left|E_{d,1}^v\right|, \left|E_{d,2}^v\right|, \ldots \left|E_{d,24}^v\right|\right) \tag{19.5}$$

The outputs of the GA search are the 'virtual' building's thermal charging/discharging rates ($[u_1^v, u_2^v, \ldots u_{24}^v]$ (kW)) in the next 24 h. Then the hourly electrical demand ($[E_{d,1}^v, E_{d,2}^v, \ldots E_{d,24}^v]$ (kW·h)) is calculated in the simulation platform. The optimized thermal charging/discharging rates of the 'virtual' building are used in Step 2.

Step 3: Coordination of single building's operation using NLP

In this step, the single building's thermal charging/discharging rates (i.e., u_i^j of the j^{th} building in the i$^{\text{th}}$ hour) are coordinated using NLP based on the 'virtual' building's operation (Zhao et al. 2015). The NLP is conducted in each hour and will be repeated 24 times for obtaining the building's daily operation. The fitness function of the NLP is expressed by Eq. (19.6) and (19.7), which aims at minimizing the electricity costs of the building group.

$$J_{NLP} = \min\left(Cost_{all,i}\right) \tag{19.6}$$

$$Cost_{all,i} = \sum_{j=1}^{n} \left(E_{d,i}^j \times \chi_i\right)^2 \tag{19.7}$$

In order to reduce the uneven allocation of the thermal charging/discharging rates (otherwise only a few buildings take benefits from the demand response), the square of each building's operational cost are used in the fitness function. $E_{d,i}^j$ (kW·h) is the energy demand of the j^{th} building in the i^{th} hour after applying the u_i^j (kW) amount of thermal charging/discharging, which is calculated by the models presented in Sect. 19.3. χ_i (HKD/(kW·h)) is the electricity price in the i$^{\text{th}}$ hour, which is usually different between off-peak hours and on-peak hours.

In the i$^{\text{th}}$ hour, the optimized parameters in the NLP are the hourly thermal charging/discharging rates of all the buildings inside the building group, i.e., $[u_i^1, u_i^2, \ldots, u_i^N]$ (kW), where N indicates the number of buildings in the building group. The thermal charging/discharging rates in each hour should follow the constraints below.

i. The sum of thermal charging/discharging rates of each building (u_i^j (kW)) should equal the thermal charging/discharging of the 'virtual' building (u_i^v (kW)) (obtained from Step 2).

$$\sum_{j=1}^{N} u_i^j = u_i^v \tag{19.8}$$

ii. For each single building, the thermal charging amount must be smaller than the remaining storage of thermal storage tank, and the thermal discharging amount must be smaller than the amount of thermal energy

stored in the tank. There are 2N inequality constraints for N buildings.

$$-\varnothing^{j}_{(i-1)} \le u^{j}_{i} \times \tau \le CAP^{j} - \varnothing^{j}_{(i-1\circ)} \quad (\boldsymbol{j} = 1, 2, \ldots \mathrm{N}, \text{ respectively}) \tag{19.9}$$

where τ is the charging duration (i.e., 1 h), CAP^{j} (kWh) is the thermal storage capacity of the j^{th} building, $\varnothing^{j}_{\mathrm{i}-1}$ (kW·h) is the thermal energy stored in the j^{th} building's thermal storage tank. $\varnothing^{j}_{\mathrm{i}-1}$ (kW·h) is calculated by Eq. (19.10).

$$\varnothing^{j}_{\mathrm{i}-1} = (u^{j}_{1} + u^{j}_{2} + \ldots + u^{j}_{i-1}) \times \tau \tag{19.10}$$

In total, for a building group with N buildings, there are N unknown parameters to be solved, there is 1 equality constraints, and there are $2N$ inequality constraints.

Step 4: Performance comparison and analysis
After obtaining the optimized operation of each single building, the performances of the proposed hierarchical control are compared with two existing controls in aspects of computational efficiency, peak demand reduction and consumers' economic saving. The two existing controls include a conventional independent DR control without any coordination (Shen et al. 2016) and an existing coordinated DR control developed in (Gao and Sun 2016). In the independent DR control (Shen et al. 2016), GA was used for searching the optimal thermal charging/discharging rates of each building, which is similar to the control optimization of the 'virtual' building (as presented in Step 2 in Fig. 19.2). After obtaining the individual buildings' optimal operation, their electrical demands were aggregated for evaluating the building group peak energy demand. In the coordinated DR control (Gao and Sun 2016), the thermal charging/discharging rates of all the eight buildings were optimized simultaneously using GA, and the minimization of the building group peak electrical demand was used as the fitness function.

19.3 Building and Systems Modeling

This section presents the modeling of buildings, HVAC systems, and thermal storage tanks. The building and building system models were modeled in Trnsys. The studied systems mainly include an air-conditioning system with a thermal storage tank integrated.

19.3.1 Building Modeling

This chapter considered eight buildings with different functions, namely a hotel, an office building, a shopping center, a restaurant, strip malls, a retail store, a primary school, and an apartment. These buildings were built in Trnsys. Following the Building Energy Code in Hong Kong (EMSD 2007), the detailed load schedules (e.g., occupancy schedules, electrical device operation schedules, and lighting schedules) for each building were determined. For all these eight buildings, the indoor air temperature set-point was 25 °C and the relative humidity was set as 50%. The ventilation rate and infiltration rate were set as 15 l/s/person and 0.2ACH, respectively, following the standards in Hong Kong (EMSD 2007). Since all the eight buildings were considered to locate in the same city, the differences in electrical demands were mainly caused by the various building usage schedules.

19.3.2 HVAC System Modeling

The considered air-conditioning system contains a cooling coil, a chiller, a cooling tower, three pumps, and a fan. Figure 19.3 presents the schematic diagram of the air-conditioning system. The details of each model are introduced below.

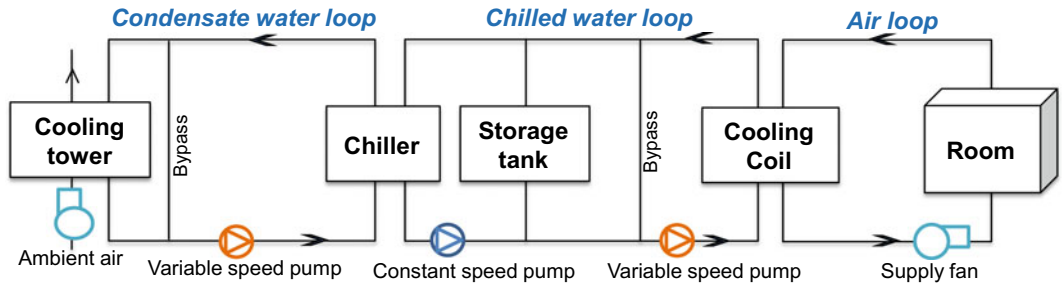

Fig. 19.3 Schematics of the case HVAC system with thermal storage tank integrated

19.3.2.1 Chiller Model

The chiller model developed by (Wang 1998) is used to estimate the chiller power consumption based on the given condenser inlet water temperature, evaporator cooling load, and evaporator outlet water temperature. The overall heat transfer coefficients of the evaporator (UA_{ev}) and condenser (UA_{cd}) are used directly to calculate refrigerant vapor compression power (P_{com} (kW)). These coefficients are calculated by Eqs. (19.11 and 19.12) (Ma and Wang 2009).

$$c_1 M_{w,ev}^{-0.8} + c_2 Q_{ev}^{-0.745} + c_3 = \frac{1}{UA_{ev}} \tag{19.11}$$

$$c_4 M_{w,ev}^{-0.8} + c_5 (Q_{ev} + P_{com})^{1/3} + c_6 = \frac{1}{UA_{cd}} \tag{19.12}$$

$M_{w,ev}$ (kg/s) is the chilled water flow rate, and Q_{ev} (kW) is the supplied cooling of individual chiller. The parameters (c_1-c_6) are coefficients that can be identified using a regression method based on real chiller operational data (Ma and Wang 2009). The chiller power consumption is calculated based on the compressor power consumption, as expressed by Eq. (19.13),

$$P_{chiller} = c_7 + c_8 P_{com} + c_9 P_{com}^2 \tag{19.13}$$

where (c_7–c_9) are coefficients obtained by regression based on a real chiller operational data.

19.3.2.2 Cooling Tower Model

This chapter adopts a semi-physical cooling tower model developed by (Power 2018) to evaluate the air flow rate that is required for keeping the cooling water supply temperature at its set-point. The calculation is mainly based on three parameters, namely, the condenser water outlet temperature, outdoor air temperature, and cooling water flow rate.

Using the heat transfer coefficient of the cooling tower (UA) calculated by the classical ε-NTU method, the required air flow rate ($M_{a,tower}$ (kg/s)) is calculated by Eq. (19.14).

$$\mathrm{UA} = D_0 \left(\frac{M_w}{M_{w,des}}\right)^m \left(\frac{M_{a,tow}}{M_{a,des,tow}}\right)^n \frac{c_{p,af}}{c_{p,a}} \tag{19.14}$$

D_0, m and n are determined based on operational data. $c_{p,af}$ (J/(kg·K)) is the fictitious air specific heat which is calculated by Eq. (19.15).

$$c_{p,af} = \frac{h_{a,out} - h_{a,in}}{T_{wb,out} - T_{wb,in}} \tag{19.15}$$

M is mass flow rate; T is temperature; h represents enthalpy; subscripts w, a, des, in, out, wb, *and* tow represent water, air, designed value, in, out, wet bulb temperature, and cooling tower, respectively. After $M_{a,tower}$ (kg/s) is determined, the power consumption of the fans in the cooling towers can be estimated using the affinity law, as described by Eq. (19.18).

19.3.2.3 Air Handling Unit Model

This chapter uses a simplified air handling unit (AHU) model following (Gao and Sun 2016) to simulate the required chilled water flow rate and the outlet water temperature from the AHU coil

based on the given water inlet temperature and the total cooling load. The adopted AHU model approximately considers the temperature difference ($\Delta T_{w,AHU}$ ($^{\circ}C$)) between the inlet and outlet water to be constant, as expressed by Eq. (19.16). The water flow rate ($\dot{m}_{w,AHU}$ (kg/s)) is calculated by Eq. (19.17).

$$T_{w,out,AHU} = T_{w,in,AHU} + \Delta T_{w,AHU} \tag{19.16}$$

$$\dot{m}_{w,AHU} = \frac{\mathrm{CL}}{c_{p,w}\left(T_{w,out,AHU} - T_{w,in,AHU}\right)} \tag{19.17}$$

$c_{p,w}$ (J/(kg·K)) is the water specific heat, and CL (kW) is the cooling load.

19.3.2.4 Pump Model

Considering the small power ratio of the pumps, the affinity law is used for simple calculation of the pump power consumption. The power consumption of variable speed pumps P_{pump} (kW) is calculated by Eq. (19.18) (Gao and Sun 2016), which has a linear relationship with the third power of water flow rate $\dot{m}_w$ (kg/s).

$$P_{pump} = \beta \dot{m}_w{}^3 \tag{19.18}$$

β is a coefficient determined based on operational data.

19.3.3 PCM Storage Tank Model

This chapter adopts the PCM storage tank model developed by (Cui et al. 2014) to evaluate the amount of charged thermal energy (i.e., cooling amount) and the discharged thermal energy. The maximum charged/discharged thermal energy in each hour is calculated by Eq. (19.19).

$$\left|u^j_{max}\right| = c_{p,w}\dot{m}^j_{w,max}\left(T_{w,out,tank} - T_{w,in,tank}\right) \tag{19.19}$$

$T_{w,out,tank}$ ($^{\circ}C$) is the chilled water temperature at the outlet of the tank, and $T_{w,in,tank}$ ($^{\circ}C$) is the chilled water temperature at the inlet of the tank. $\dot{m}^j_{w,max}$ (kg/h) is the maximum charging/discharging flow rate of the j^{th} building's thermal storage tank. The thermal energy stored in thermal storage tank is calculated by aggregating the hourly charging/discharging energy, as presented by Eq. (19.10).

19.4 Case Studies and Results Analysis

A group of eight buildings, namely a hotel, an office building, a shopping center, a restaurant, strip malls, a retail store, a primary school, and an apartment (introduced in Sect. 19.3.1), was considered in the case studies. These eight buildings were assumed to be located in Hong Kong. Each of the buildings was equipped with a set of air-conditioning system as well as a thermal storage tank (see Fig. 19.3) for thermal conditioning and demand shifting. A typical summer week in July was selected for analysis. This chapter adopted the time-of-use electricity pricing. The detailed electricity prices are presented in Table 19.1.

Table 19.2 presents the configurations of the HVAC system parameters (i.e., chiller capacity, cooling tower capacity, and thermal storage capacity) in each building. In the AHU model, the temperature difference $\Delta T_{w,AHU}$ was set to be 5 $^{\circ}C$. The parameter β in the pump was set as

Table 19.1 Electricity price (Power 2018)

No	Type	Period	Price (HKD/kW·h)
1	Peak	13:00 ~ 18:00	2
2	Normal	7:00 ~ 12:00 19:00 ~ 21:00	1.5
3	Valley	1:00 ~ 6:00 19:00 ~ 21:00	1

Table 19.2 Configuration of the HVAC system parameters in each building

No	Building	Chiller capacity (kW)	Cooling tower capacity (kW)	Thermal storage capacity ($kW{\cdot}h$)
1	Hotel	238	288	300
2	Office	322	372	400
3	Shopping center	232	282	290
4	Restaurant	266	316	330
5	Strip malls	170	220	210
6	Retail store	185	235	230
7	Primary school	411	461	530
8	Apartment	121	171	150

0.6. For the thermal storage system, the PCM material E8 was selected due to its proper thermal dynamic properties (Cui et al. 2014). Its density was 1469 kg/m^3 and its latent heat was 140 kJ/kg (Cui et al. 2014).

To guarantee the GA algorithm terminates by reaching the fitness limit, the GA parameters have been tuned using the trial and error method. The number of generations was set as 100, the population size was set as 2,000, and the termination tolerance was set as 1e-6.

19.4.1 Building Energy Demand Profiles

Adopting the models introduced in Sect. 19.3, the hourly electrical demands of the eight buildings in the selected summer week were predicted. As shown in Fig. 19.4, the peak electrical demands of hotel occurred at around 01:00 ~ 07:00 and 22:00 ~ 24:00, the peak demand of the shop occurred at around 17:00 ~ 18:00, the peak demand of the office, strip malls, and retail store occurred at around 15:00 ~ 17:00, the peak demand of the restaurant occurred at around 12:00 ~ 14:00 and 19:00 ~ 20:00 (mealtime), the peak demand of the primary school occurred at around 15:00 ~ 16:00, and the peak demand of the apartment occurred at around 18:00 ~ 20:00. These different electrical demand characteristics provided possibilities of coordination for the peak demand reduction at building group.

Figure 19.5 presents the aggregated energy demands of all the eight buildings. The aggregated peak energy demand occurred at around 13:00 ~ 17:00 and could reach as high as 838 kW·h. The off-peak period occurred at 00:00 ~ 06:00 and the electrical demands were between 210 ~ 250 kW·h. The peak-to-average ratio of the aggregated demand was as high as 1.64. Such large fluctuation in the electrical demand would not only jeopardize the stable and reliable operation of the power grid, but also increase the initial costs of the power supplier due to the large generation capacity required.

19.4.2 Computational Efficiency Comparison

This section compares the performances of the conventional DR control, the existing coordinated DR control, and the proposed DR control in aspect of computational efficiency. The simulation is conducted in a 64-bit computer with Intel®Core™i7-3770U CPU and 16 GB RAM. Table 19.3 summarizes the principles, and computational performances of each control. Both the conventional independent DR control and the proposed hierarchical DR control can search the optimal solutions well under the given number of buildings. In these two methods, the number of variables to be simultaneously optimized in each GA simulation is 24 (i.e., the number of hours in a single day as the optimization is conducted in a daily interval). The proposed coordinated DR

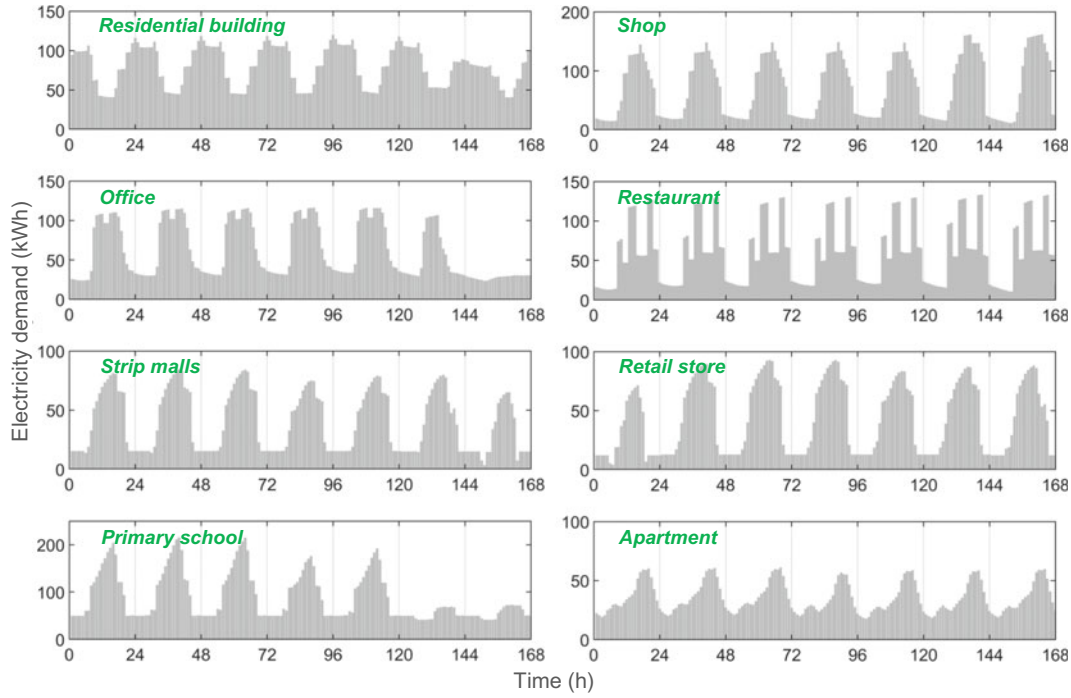

Fig. 19.4 Hourly electrical demands of the eight studied buildings

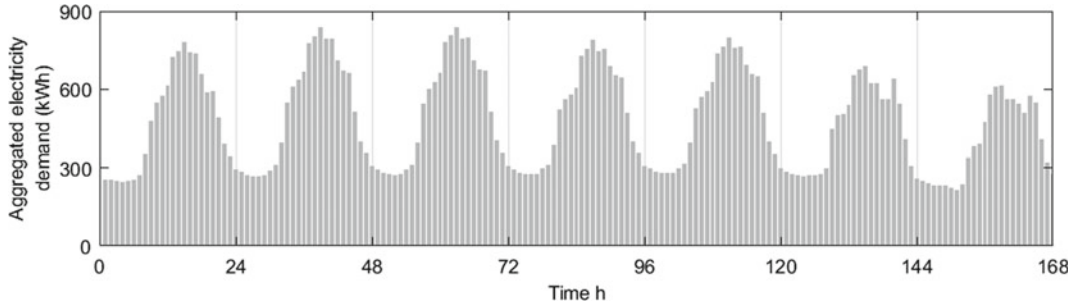

Fig. 19.5 Aggregate hourly electrical demands of the eight buildings

control is the most computational efficient, as the GA search, which has high computational load, is conducted only one time for the 'virtual' building (the NLP is faster and more computation-efficient compared with GA). While in the conventional independent DR control, the GA search is conducted eight times corresponding to the eight buildings, and thus it is not as computation-efficient as the proposed coordinated DR control. The existing coordinated DR control (Gao and Sun 2016) could not find the optimal solutions, due to the large computational loads. In fact, there are $24 \times 8 = 192$ (i.e., 24 is the number of hours in a day, and 8 is the number of buildings) variables to be optimized simultaneously, which imposes great load and complexity on the searching algorithm.

Since the existing coordinated DR control still could not identify the optimal solutions after hours' search, it was considered not applicable in practice especially when a large number of buildings considered. Thus, in the following two

Table 19.3 Comparison of the computational performances

Controls	Principles	Number of variables to be simultaneously optimized in GA	Call times of GA engine	Computational load* (hour)	Convergence or not?
Conventional independent DR control (Shen et al. 2016)	GA was used for searching the optimal thermal charging/discharging rates of each building independently. The minimization of the single building's peak electrical demands was used as the fitness function	24	8	2.2	√
Existing coordinated DR control (Gao and Sun 2016)	The thermal charging/discharging rates of all the eight buildings were optimized simultaneously using GA. The minimization of the building group peak electrical demand was used as the fitness function	192	1	4.1	×**
Proposed hierarchical DR control	All the eight buildings were considered as a 'virtual' building, and the 'virtual' building's operation is optimized using GA. Then, to realize the GA-based optimal performance, each single building's operations were coordinated using NLP	24	1	0.3	√

* The simulation is conducted in a 64-bit computer with Intel®Core™i7-3770U CPU and 16 GB RAM
**The existing coordinated DR control cannot identify the optimal solutions after hours' search, it was considered not applicable in practice especially when a large number of buildings considered. Thus, in the following two sub-sections, only the performances of the conventional independent DR control and proposed hierarchical DR control were presented

sub-sections, only the performances of the conventional independent DR control and proposed hierarchical DR control were presented.

19.4.3 Single Building's Performance Comparison

In this section, the single building's hourly electrical demands after applying the different controls were presented and compared. Figure 19.6 compares the hourly thermal charging/discharging rates of each building under the hierarchical DR control (see the orange curves) and conventional DR control (see the green dashed curves). Figure 19.7 compares the hourly energy demand without DR control (see the gray bars), demand after applying the hierarchical DR control (see the orange curves), and the demand after applying the conventional DR control (see the green dashed curves). As can be seen, at the single building level, the conventional DR control can more effectively reduce the peak electrical demand of each building than the hierarchical DR control.

Under the conventional DR control, the electrical energy demands of the office building, shopping center, restaurant, strip malls, retail store, primary school, and an apartment during

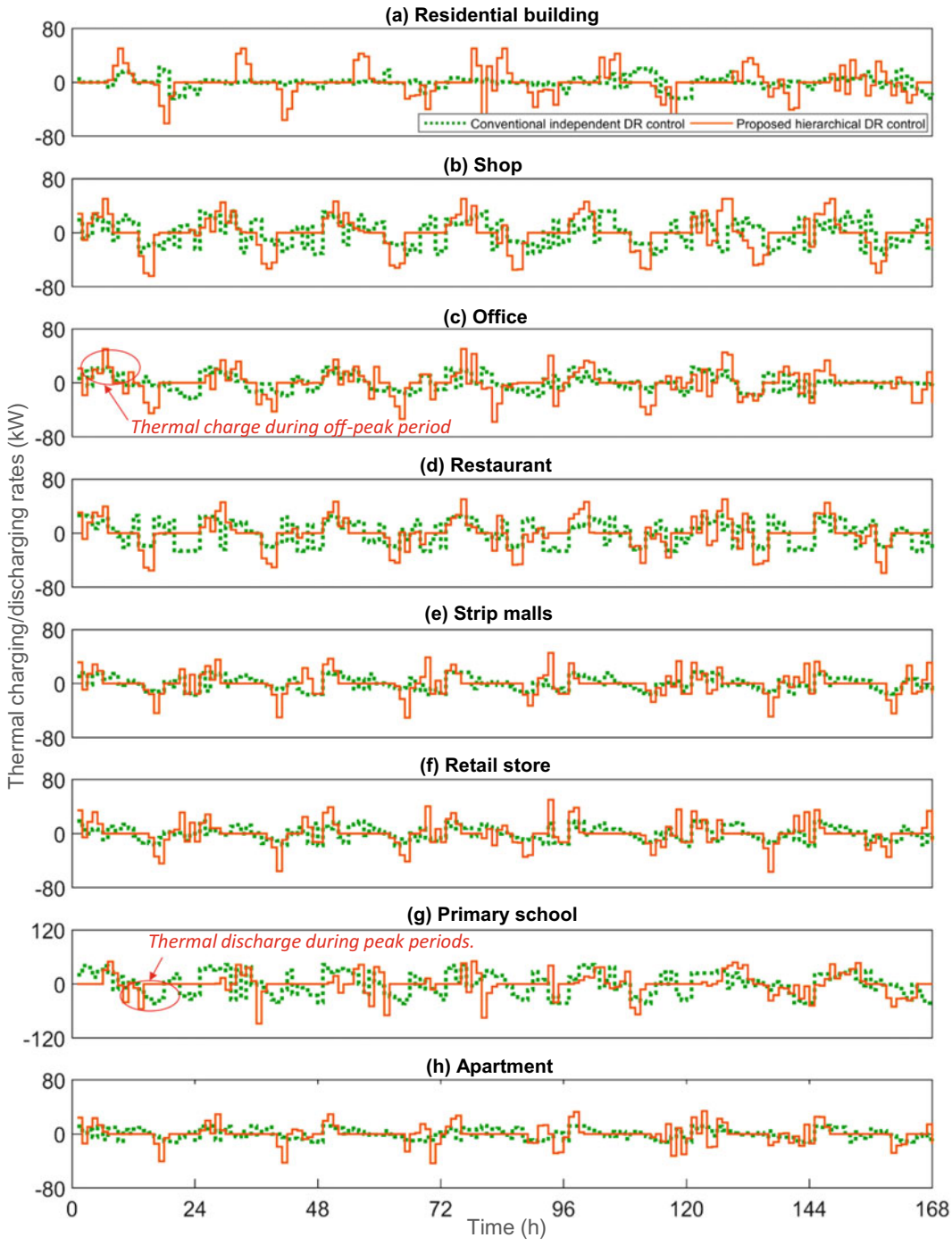

Fig. 19.6 Comparison of the thermal charging/discharging rates under different controls

peak periods were effectively shifted to the off-peak periods. The load shifting was contributed by the thermal charging during the off-peak periods (nighttime) and thermal discharging during the peak periods (daytime). For the hotel, its peak demand occurs during both 01:00 ~ 07:00 and

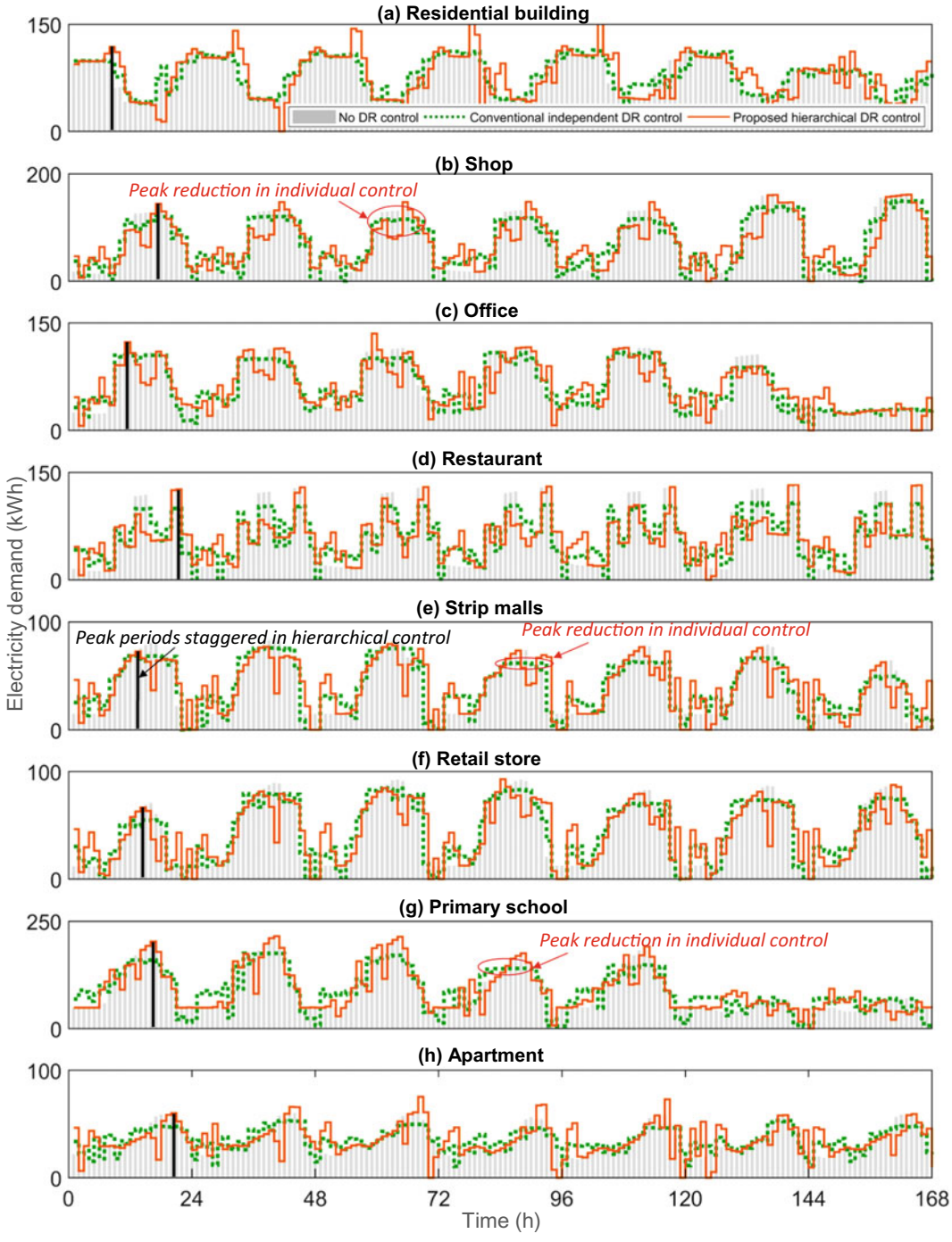

Fig. 19.7 Comparison of the energy demand under different controls

22:00 ~ 24:00. Since the thermal storage should be charged first before discharging thermal energy, there is no thermal discharging in the hotel during the period of 01:00 ~ 07:00, and as a result the peak demand in this period was not reduced. During the period of 10:00 ~ 14:00, the

thermal tank was charged with thermal energy. This amount of cooling was discharged during the period of 22:00 ~ 24:00, and thus there was a reduction in the electrical demand during this period. Under the developed hierarchical DR control, the peak demand of single building was not effectively reduced. In fact, the peak demand of some buildings increased after the hierarchical control. This is because the hierarchical DR control focuses on coordinating each building's peak demands to occur in different periods, instead of optimization at single building level.

19.4.4 Building Group Performance Comparison

This section compares the performances of the two controls at the building group. The building group electrical demand is the aggregated demand of each single building. As shown in Fig. 19.8, both the two controls significantly reduced the building group peak energy demands. Compared with the conventional DR control, the hierarchical DR control was more effective in peak reduction at the building group. This was because the hierarchical control focused on coordination of single building's operation at the building group. As shown in Fig. 19.7, the hierarchical DR control coordinated each building's peak demands to occur in different periods. For instance, the peak period of the hotel was shifted to around 08:00, the peak period of the office building was shifted to around 17:00, the peak period of the shopping center was shifted to around 11:00, the peak period of the restaurant was shifted to around 21:00, the peak period of the strip malls was shifted to around 13:00, the peak period of the retail store was shifted to around 14:00, the peak period of the primary school was shifted to around 16:00, and the peak period of the apartment was shifted to around 20:00. By shifting the peak demand of each building to different periods, the peak demand was reduced significantly at the building group.

Figure 19.9 compares the daily peak electrical demands without any DR controls and the demands after applying the two controls. Compared with the base case, both the two controls significantly reduced the peak energy demand. The daily peak reduction in the conventional DR control was due to the demand shifting from the peak periods to the off-peak periods. And the peak reduction in the hierarchical DR control was due to the coordination to stagger the peak periods of each building. Figure 19.10 further compared the daily peak demand reduction amount under the two controls. The conventional DR control reduced the daily peak demand by 49 ~ 102 kW·h (8.1 ~ 11.7% reduction compared with the base case), while the hierarchical DR control reduced the daily peak demand by 112 ~ 174 kW·h (14.3 ~ 22.3% reduction compared with the base case).

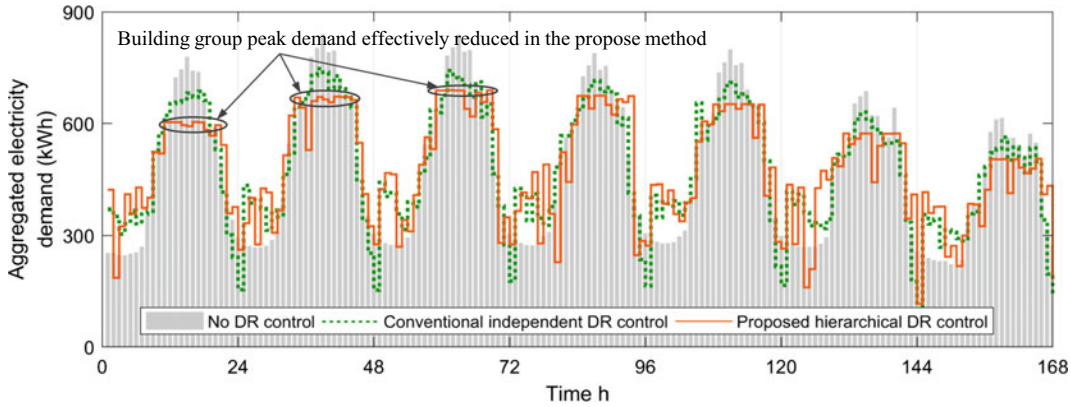

Fig. 19.8 Comparison of the aggregated energy demand of all the building under different controls

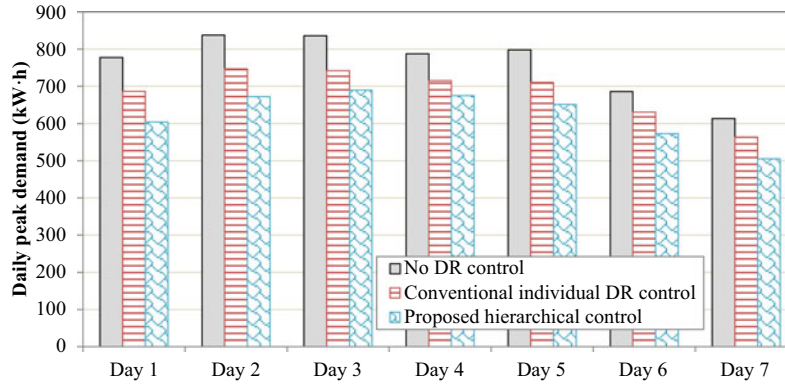

Fig. 19.9 Comparison of the daily peak electrical demand under different controls

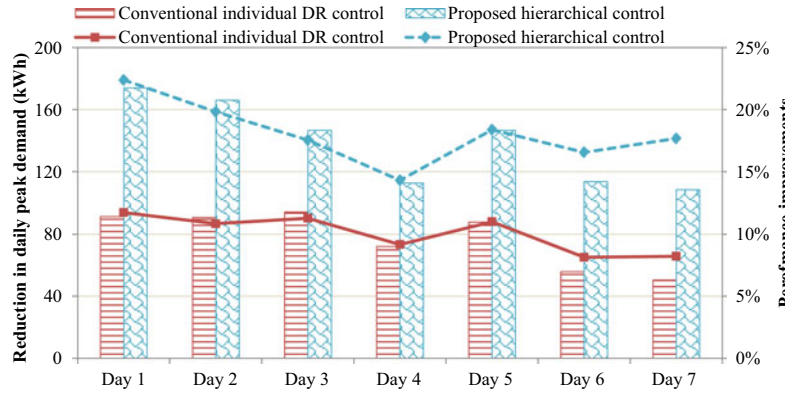

Fig. 19.10 Comparison of the reduction in daily peak electrical demand under different controls

Table 19.4 compares the peak-to-average ratio of the aggregated electrical demands under different controls. By limiting the peak demand, both the two controls could effectively and significantly reduce the peak-to-average ratio. Since there was more peak reduction in the hierarchical control, it had better peak-to-average ratio performances compared with the conventional control. The flattened electrical demand shape can help improve the stability, reliability, and efficiency of the power grid operation (Gao and Sun 2016; Sun et al. 2013; Xue et al. 2014).

Figure 19.11 compares the daily electricity costs without any DR controls and the costs after applying the two controls. Compared with the base case, both the two controls significantly reduced the daily electricity costs. This was due to the shifting of electrical demand from the high-price period (i.e., 13:00 ~ 18:00, see Table 19.1) to the low-price period (i.e., 1:00 ~ 6:00 and 19:00 ~ 21:00, see Table 19.1). The hierarchical DR control produced more electricity shifting, and thus it led to more cost reduction. Figure 19.12 further compared the daily electricity costs reduction amount under the two controls. The conventional DR control reduced the daily electricity costs by 462 ~ 862 *HKD* (2.6 ~ 4.2% reduction compared with the base case), while the hierarchical DR control reduced the peak demand by 646 ~ 1350 *HKD* (3.7 ~ 8.5% reduction compared with the base case).

19.5 Summary

This chapter has proposed a hierarchical demand response control method of buildings for peak demand reduction at the building group. The hierarchical control first considers the building group as a 'virtual' building and searches the optimal performance that can be achieved at building group using genetic algorithm. To

Table 19.4 Comparison of the peak-to-average ratio of the aggregated energy demands under different controls

Controls		Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
No DR control	Peak-average-ratio	1.64	1.62	1.61	1.58	1.59	1.54	1.55
Conventional independent DR control	Peak-average-ratio	1.45	1.44	1.43	1.43	1.41	1.41	1.42
	Improvements	*11.8%*	*10.9%*	*11.3%*	*9.6%*	*11.0%*	*8.1%*	*8.2%*
Proposed hierarchical DR control	Peak-average-ratio	1.27	1.30	1.33	1.35	1.29	1.27	1.28
	Improvements	*22.6%*	*19.6%*	*17.8%*	*14.4%*	*18.4%*	*17.1%*	*17.2%*

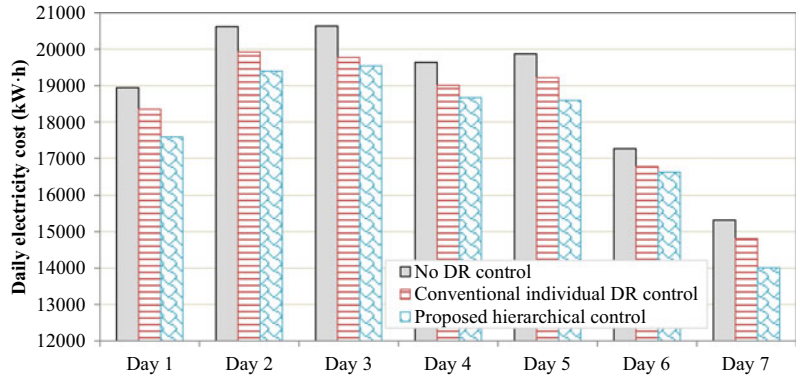

Fig. 19.11 Comparison of the daily electricity costs under different controls

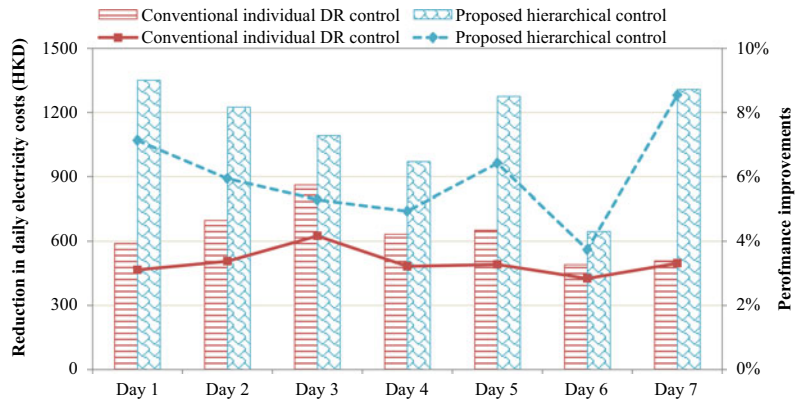

Fig. 19.12 Comparison of the reduction in daily electricity costs under different controls

realize the GA-based optimal performance, it then coordinates each single building's operation using non-linear programming. For validation, the developed hierarchical DR control has been applied on a group of eight buildings, and its computational efficiency, peak electrical demand reduction, and cost reduction performances have been compared with a conventional independent DR control and an existing coordinated DR control. The developed hierarchical control can improve the effectiveness and efficiency in relieving the grid stress, reduce operation costs and net emission for the supply side, reduce the electricity bills of the end-users, and reduce the computational load. The major findings are summarized as follows:

- The hierarchical control provides a mechanism to coordinate each single building's operation for improved building group performances.
- In aspect of computational efficiency, the hierarchical control can dramatically reduce the computational load, as the computation-consuming GA engine is called only one time for the 'virtual' building; while the computational load in the conventional independent demand response control is around eight times of the proposed method, as the GA engine is called eight times. The existing coordinated demand response control could not well identify the optimal solutions under the eight building case due to the large computational load.
- In aspect of electrical demands, the hierarchical control achieved more daily peak demand reduction (5% $\sim$ 11% approximately) than the conventional control in the case studies. This was because coordination of the individual building's operation was more effectively than reducing single building's peak demand in building group peak reduction.
- In aspect of economic costs, the hierarchical control achieved more daily electricity cost savings (1% $\sim$ 5% approximately) than the conventional control in the case studies. This was because more electrical demands were shifted from the high-price periods to off-peak periods.

Demand uncertainty is a significant fact affecting the performance of demand response control. Future work will take account of the demand prediction and develop more robust controls at the building group.

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Optimization of Near-Zero Energy Buildings Cluster with Top-Down Control

20

Pei Huang and Yongjun Sun

Abstract

Nearly zero energy buildings (NZEBs) are considered as a promising solution to the mitigation of the energy problems. A proper control of the energy system operation of the nZEB cluster is essential for improving load matching, reducing grid interaction and reducing energy bills. Existing studies have developed many demand response control methods to adjust the operation of energy systems to improve performances. Most of these studies focus on optimizing performances at individual-nZEB-level while neglecting collaborations (e.g., energy sharing and battery sharing) between nZEBs. Only a few studies consider the collaborations and optimize the system operation at nZEB-cluster-level, yet they cannot take full advantage of nZEB collaborations as optimization is conducted in a bottom-up manner lacking global coordination. This chapter, therefore, proposes a top-down control method of nZEBs for optimizing performances at the cluster level. The top-down control method first considers the nZEB cluster as 'one' and optimizes its energy system operation using the genetic algorithm (GA), and then it coordinates the operation of every single nZEB inside the cluster using non-linear programming (NLP). The top-down control enables collaborations among nZEBs by coordinating single nZEB's operations. Such collaborations can bring significant performance improvements in different aspects. For instance, in aspect of economic cost, the collaborations can reduce the high-priced energy imports from the grid by sharing the surplus renewable energy with nZEBs which have insufficient energy generations. The proposed top-down control has been compared with a traditional non-collaborative control. The study results show that the top-down control is effective in improving performances at cluster level.

Keywords

Near-Zero energy building cluster · Top-Down control · Economic cost · Load matching · Grid interaction

20.1 Introduction

With the increasing population, continuing urbanization and increasing economy, the global energy usage continues to increase. It is reported

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that buildings consume over 40% of overall energy worldwide (Kolokotsa et al. 2011). Nearly zero energy buildings (NZEBs) are considered as a promising solution to the mitigation of the energy problems. There are different definitions for nZEBs (Crawley et al. 2009; Hernandez and Kenny 2010; Marszal et al. 2011; Sartori et al. 2012; Wells et al. 2018; Deng et al. 2014). The basic idea is that the overall primary energy usage does not exceed the energy generation by the on-site renewable energy systems during a specific period (usually one year) (Hernandez and Kenny 2010; Cao et al. 2016; Ascione et al. 2016; Wells et al. 2018; Zhang et al. 2016; Lu et al. 2015b; Deng et al. 2014). Until now, many countries have established related policies and regulations on nZEBs for promoting its application (Recast 2010; Crawley et al. 2009; Wells et al. 2018). For instance, the European Directive on Energy Performance of Buildings launched the goal of 'nearly net zero energy buildings' for all the new buildings from 2020 (Recast 2010). 17 European Member States established the ZEBRA 2020 (nearly zero energy building strategy 2020) for the purpose of monitoring market uptake of nZEBs across Europe and generating data and evidence for policy evaluation and optimization (2020, 2014). The California Public Utilities Commission of the USA sets a nearly zero energy target that all new residential buildings should achieve by 2020 and all new commercial buildings should achieve by 2030 (Crawley et al. 2009).

The wide application of nZEBs has led to an interest in next-generation buildings known as nearly zero energy building clusters (Odonkor and Lewis 2015; Deng et al. 2014). Inside the cluster, the distributed systems are capable of exploiting inter-dependencies such as shared energy generation, shared storage and collaborative load management (Odonkor and Lewis 2016; Li and Wen 2017; Hu et al. 2014; Jafari-Marandi et al. 2016). By enabling free energy and information exchanges between individual buildings, the entire building cluster can achieve maximum energy efficiency (Li and Wen 2017). Some organizations have made first efforts to explore the implementation of nZEB clusters, such as the IEA ECBCS Annex 51 (Jank 2012; Broers 2014), the EnEff Stadt project funded by German (Jank 2010), the World Bank Energy Sector Management Assistance Program (ESMAP 2012), Energy efficient cities initiative C40 program (ESMAP 2012), the Clinton Climate Initiative C40 program (Group 2011) and the U.S. Army net zero installation program ((IE&E) 2012). The NZEB cluster is also called smart community is some studies (Anandalaskhmi et al. 2013; Ahmadi et al. 2015).

Demand response control is widely recognized as an effective method to improve grid power balance and reduce nZEB's energy bills (Gao et al. 2015; Zhao et al. 2015; Barbato and Capone 2014; Hussain et al. 2015; Cao et al. 2016; Aghajani et al. 2017; Yang et al. 2014). The demand response control indicates changes in electricity usage by end-users from their normal consumption patterns in response to the electricity price variation over time, or to incentive payments designed for encouraging lower electricity usage during high market prices or when grid reliability is jeopardized (Qdr 2006; Gao et al. 2015; Aghajani et al. 2017). It is usually achieved by adjusting the building energy systems operation, such as the operation/control of HVAC system, thermal storage, battery and on-site generation systems (Hussain et al. 2015; Barbato and Capone 2014). For instance, Zhao et al. developed a model predictive control based optimal scheduling strategy using nonlinear programming algorithm to schedule the operation of electricity and cold generation/use and the cold storage 24 h ahead. Two cases (i.e., selling electricity to the grid was allowed or not allowed) were considered in their study (Zhao et al. 2015). Lu et al. developed an optimal scheduling approach based on mixed-integer nonlinear programming. They used the developed method to control energy system operation in buildings integrated with energy generation and thermal energy storage and compared it with rule-based control (Lu et al. 2015a). Gao et al. developed a robust demand response control of commercial buildings based on genetic algorithm (GA) considering the uncertainty in the prediction of building energy

demand (Gao et al. 2015). Their method could achieve robust control performance under the impacts of uncertainty compared with other deterministic control. Most of these control methods focus on optimizing the nZEB performance at individual-nZEB-level. The collaborations between nZEBs were neglected and the performances at nZEB-cluster-level were not optimized.

The demand side response control at cluster level is different from the control at the individual level (which optimizes single building operation individually). The nZEB cluster control should pay more attention to coordination of individual nZEB's operation to take advantages of collaborations, so as to improve the nZEB cluster performances (Jafari-Marandi et al. 2016; Odonkor and Lewis 2015, 2016; Hu et al. 2014). For instance, Jafari-Marandi et al. developed a genetic algorithm-based bi-level distributed decision framework. Using the developed method, they evaluated the performance improvements of the nZEB cluster with centralized renewable energy systems and batteries (Jafari-Marandi et al. 2016). Odonkor and Lewis also developed an adaptive bi-level decision method to optimize the operation of the nZEB cluster. Their case cluster includes distributed renewable energy system and battery for individual nZEB and a centralized cooling system. In their method, the individual nZEB operational parameters (e.g., temperature set-point, battery state and ice storage state) are optimized in the first level, and the operational parameters of the centralized cooling system are optimized at the second level (Odonkor and Lewis 2015). Based on this method, Odonkor and Lewis further developed a robust nZEB cluster operation control method using Pareto band considering the energy demand side uncertainties (Odonkor and Lewis 2016). Similarly, Hu et al. developed a bi-level operation decision framework to optimize nZEB cluster operation based on an augmented multi-objective particle swarm optimization (AMOPSO) (Hu et al. 2014). These methods are effective in improving the nZEB cluster performance, but they only take limited advantage of collaborations between nZEBs as they optimize in a bottom-up manner (from single nZEB to nZEB cluster) lacking global coordination of single nZEBs. As studied by Sun. et al., in the ideal case, the performance improvements through cluster collaborative control could reach as high as 118%, 40% and 56% in terms of economic cost, load matching and grid interaction, respectively (Sun et al. 2018). The existing nZEB cluster control method could hardly achieve such great performance improvements.

To sum up, most of existing demand response control methods focus on single building level optimization and neglects collaborations among nZEBs. The existing cluster level demand response control methods merely perform simple and limited collaborations among nZEBs and thus they may not be able to achieve the optimal results at cluster level. Therefore, this chapter proposes a top-down control method to optimize nZEB cluster performances at nZEB-cluster-level in the coming 24 h. In the first step, the top-down control method considers the nZEB cluster as 'one' nZEB and optimizes its operation using the genetic algorithm (GA). In the second step, the top-down control coordinates the operation of every single nZEB inside the cluster using nonlinear programming (NLP). The operation parameters of nZEB cluster obtained from the first stage are used as constraints in the second stage. The performances of top-down control are analyzed in aspects of load matching and grid interaction. Load matching refers to how the local energy generation compares with the building load by measuring the degree of overlap between generation and load profiles. Grid interaction refers to the energy exchange between the building and a power grid, which takes aspects of the unmatched part of generation or load profiles (Salom et al. 2011). The objective is to provide decision makers a control method of the nZEB cluster, which can improve the nZEB cluster performances as high as possible. The contributions of the present study to the subject are briefly summarized as follows:

- A top-down control method of the nZEB cluster has been developed. The top-down control method enables collaborations between nZEBs by coordinating single nZEB's operation.
- The collaborations (energy sharing and battery sharing) of nZEBs inside the cluster have been investigated.
- The collaboration benefits of top-down control have been investigated under two control strategies: maximizing load matching and minimizing grid interaction.

20.2 Control for Performance Optimization at nZEB-Cluster-Level

In this section, the individual nZEB control and nZEB cluster control are first introduced. Then the proposed top-down control (which belongs to nZEB cluster control) is presented.

20.2.1 Basic Idea of Individual nZEB Control and nZEB Cluster Control

Figure 20.1(a) presents the schematics of conventional individual nZEB control (Sun et al. 2018). Each nZEB is controlled by an independent local controller to optimize its performance. Corresponding to nZEB-cluster-level control, this can be called individual-nZEB-level control. As can be seen, communication and collaboration are not allowed between individual nZEB in this type of control. When a nZEB has surplus energy production and its battery is already fully charged, it has to export the excessive amount of energy to the power grid. On the other hand, when it has insufficient energy production and its battery is already fully discharged, it has to import the insufficient amount of energy from the grid. Since there is no collaboration between nZEBs, even though the individual nZEB performances (such as economic cost, load matching and grid interaction) are optimized, the nZEB performance at the nZEB-cluster-level might not be optimal.

Figure 20.1(b) presents the schematics of nZEB cluster control (Sun et al. 2018; Li and Wen 2017; Broers 2014). All the nZEBs in the cluster are controlled by a central controller. The central controller allows power exchange inside the nZEB cluster. This type of connection and control enables nZEB collaboration from two aspects: energy sharing and battery sharing (Sun et al. 2018). Energy sharing indicates using renewable energy from other nZEBs, and battery sharing indicates storing/using energy to/from other nZEB's battery. For instance, the surplus

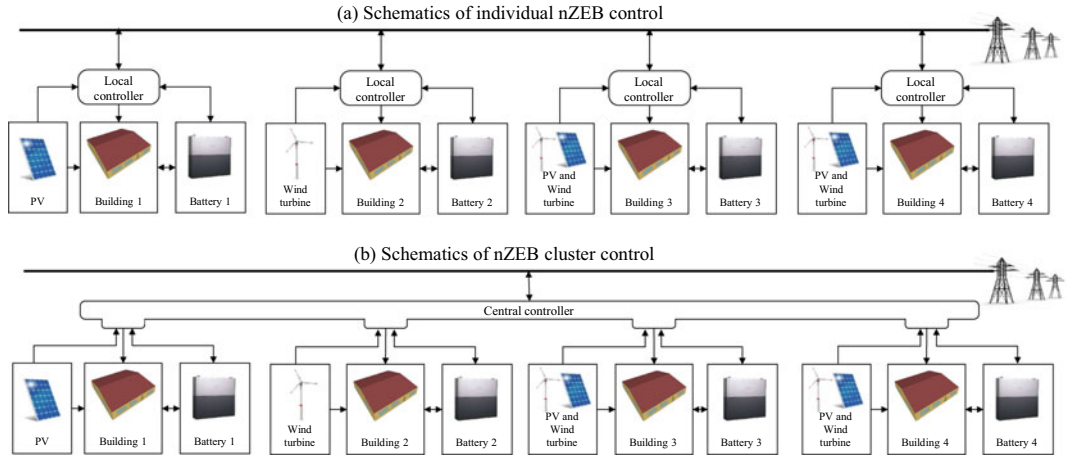

Fig. 20.1 Comparison of individual nZEB control and nZEB cluster control (a), schematics of individual nZEB control (b), schematics of nZEB cluster control

energy of nZEB 1 can be used to supply energy demand of nZEB 3 (energy sharing). The electricity stored in electrical battery 4 can be exported to nZEB 2 (battery sharing). Energy sharing and battery sharing bring benefits to both the nZEB cluster and the power grid. From the aspect of the nZEB cluster, the economic cost could be reduced, as less electricity might be required from the power grid. From the aspect of the power grid, the stability and reliability could be improved, since the power exchange with the grid might be reduced due to the power compensation inside the nZEB cluster.

20.2.2 Top-Down Control Method for Performance Optimization at nZEB-Cluster-Level

Figure 20.2 presents the flowchart of the developed top-down control (corresponding to the schematic diagram shown in Fig. 20.1(b)). The aim of top-down control is to adjust the operation of individual nZEB in the next 24 h to achieve the optimal performance at nZEB-cluster-level. This top-down control method consists of two steps. In Step 1, the nZEB cluster operation in the coming 24 h is optimized at nZEB-cluster-level using GA. The nZEB cluster is considered as 'one' nZEB: its power supply is the sum of the individual nZEB power supply, its power demand is the sum of the individual nZEB power demand, and its battery storage capacity is the sum of the individual nZEB battery capacity. In ideal cases, the nZEB cluster performance obtained from GA is considered to be optimal at nZEB-cluster-level (Sun et al. 2018). In Step 2, the operation of individual nZEB inside the cluster is coordinated using the NLP based on the nZEB cluster operation obtained in Step 1. Several operational constraints should be satisfied, e.g., the sum of the power exchange of individual nZEB (obtained in Step 2) equals the power exchange of nZEB cluster (obtained in Step 1). Similar to other nZEB control methods (Lu et al. 2015a; Gao et al. 2015; Zhao et al. 2015; Sun et al. 2018; Hu et al. 2012; Jafari-Marandi et al. 2016), it is assumed the predicted energy demand and supply data are accurate in the top-down control.

Step 1: Optimization of nZEB cluster (considered as 'one') operation using GA

In Step 1, the nZEB cluster is considered as 'one' nZEB: in the i^{th} hour, its power supply ($E^{cluster}_{supply,i}(kW)$) equals the sum of the power supply of each individual nZEB ($E^{j}_{supply,i}(kW)$), its power demand ($E^{cluster}_{demand,i}(kW)$) is the sum of power demand of each individual nZEB ($E^{j}_{demand,i}(kW)$), and the nZEB cluster electricity battery capacity ($CAP^{cluster}(kWh)$) is the sum of the battery capacity of each individual nZEB ($CAP^{j}(kWh)$).

$$E^{cluster}_{supply,i} = \sum_{j=1}^{n} E^{j}_{supply,i} \tag{20.1}$$

$$E^{cluster}_{demand,i} = \sum_{j=1}^{n} E^{j}_{demand,i} \tag{20.2}$$

$$CAP^{cluster} = \sum_{j=1}^{n} CAP^{j} \tag{20.3}$$

The GA-based optimization method mainly adjusts the charging/discharging rates of the electrical battery. For instance, charge the electrical battery during the off-peak period so as to avoid importing power from the grid during on-peak hours. The inputs mainly include the search ranges for the battery charging rates, weather data in the next 24 h, nZEB and nZEB systems models. In each GA simulation, a set of trials of battery hourly charging rates over the next 24 h (i.e., $\left[u^{cluster}_{1}, u^{cluster}_{2}, \ldots, u^{cluster}_{24}\right]$ kW) is generated by the GA optimizer. The nZEB cluster hourly power demand ($E^{cluster}_{supply,i}(kW)$) and hour power supply ($E^{cluster}_{demand,i}(kW)$) in the next 24 h are predicted using the virtual nZEB and nZEB systems models (see Eqs. 20.1, 20.2 and 20.3). Using the trials of battery charging rates and the hourly power demand and supply data, the nZEB cluster hourly power exchange with the grid

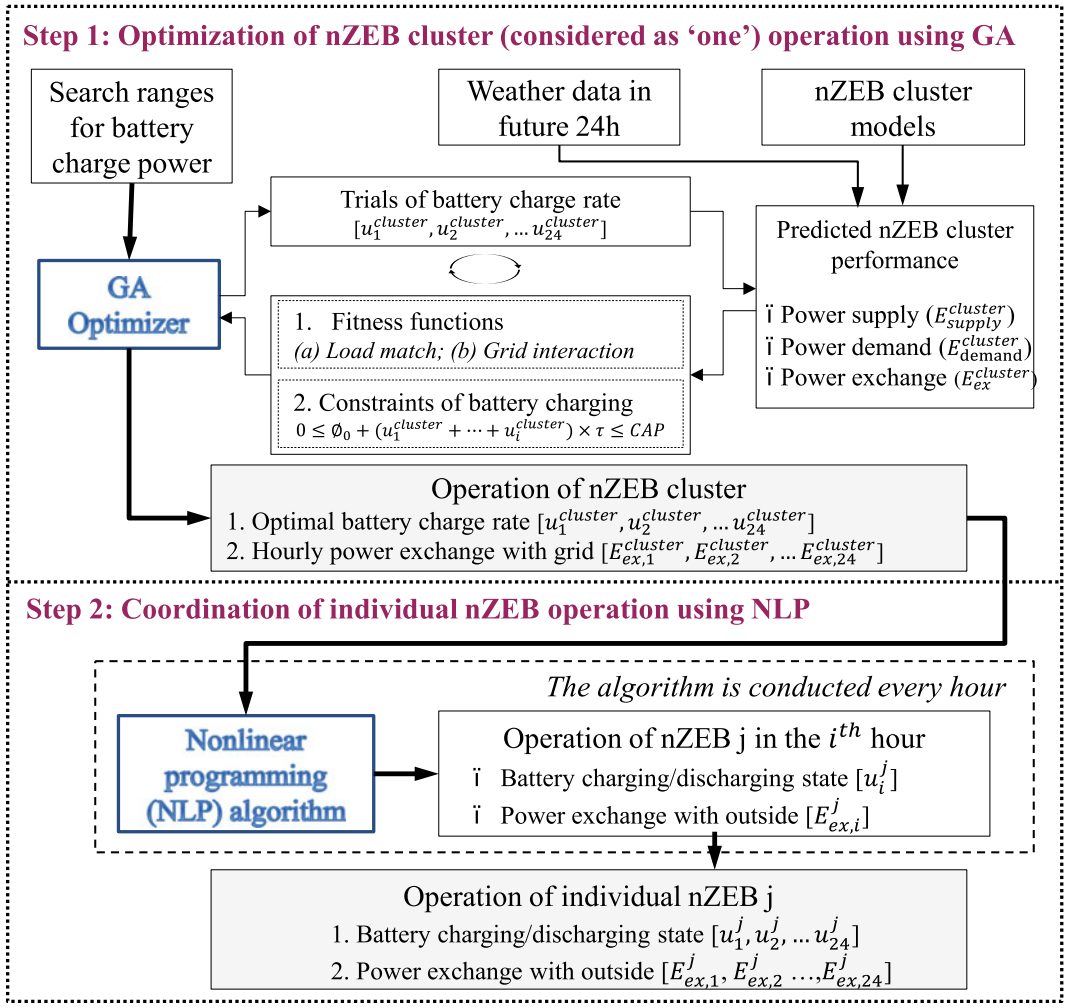

Fig. 20.2 Flowchart of the top-down control

$(E_{ex,i}^{cluster}(kW))$ can be predicted by Eq. (20.4) (Sun et al. 2018).

$$E_{ex,i}^{cluster} = E_{demand,i}^{cluster} - E_{supply,i}^{cluster} + u_i^{cluster} \quad (20.4)$$

There are two constraints associated with the electrical battery operation: (20.1) The battery charging amount could not exceed the remaining storage capacity of the battery (20.2) The battery discharging amount could not exceed the stored electricity in the battery. These two constraints are expressed by Eq. (20.5) (Lu et al. 2015a; Sun et al. 2018),

$$0 \le \varnothing_0 + (u_1^{cluster} + u_2^{cluster} + \ldots + u_i^{cluster}) \times \tau \le CAP^{cluster} \quad (20.5)$$

where $\varnothing_0^{cluster}(kWh)$ is the initial amount of electricity stored in the battery, τ is the charging duration which is set to be 1 h. Please note that the battery operates continuously: at the beginning of each day, the battery initial stored energy should be equal to the amount of energy stored at the end of last day.

Depending on the goal of control, different fitness functions can be selected. This chapter considers two different control strategies: (1) to maximize the nZEB cluster load matching, (2) to

minimize the grid interaction. Corresponding to the two control goals, two fitness functions are established. These two fitness functions are separately considered. The fitness function for maximizing the nZEB cluster load matching is expressed in Eq. (20.6). It aims to maximize the fraction of load covered by the power generated on-site (Sun et al. 2018).

$$J_{match} = \max\left(\sum_{i=1}^{24} E_{onsite,i}^{cluster} / \sum_{i=1}^{24} E_{demand,i}^{cluster}\right) \tag{20.6}$$

where $E_{onsite,i}^{cluster}(kW)$ is the amount of load covered by the power generated on-site in the i^{th} hour. It is calculated by Eq. (20.7)

$$E_{onsite,i}^{cluster} = \min(E_{supply,i}^{cluster} - u_i^{cluster}, E_{demand,i}^{cluster}) \tag{20.7}$$

The fitness function for minimizing the nZEB cluster grid interaction is expressed in Eq. (20.8). It aims to minimize the daily peak power exchange with the power grid, which can help relieve the grid stress for maintaining power balance and thus improving the nZEB cluster friendliness (Salom et al. 2011).

$$J_{grid} = \min(E_{peak}^{cluster}) \tag{20.8}$$

where $E_{peak}(kW)$ is the peak power exchange, which is calculated by Eq. (20.9).

$$E_{peak}^{cluster} = max\left(\left|E_{ex,1}^{cluster}\right|, \left|E_{ex,2}^{cluster}\right|, \ldots \left|E_{ex,24}^{cluster}\right|\right) \tag{20.9}$$

The output from the GA optimization is the nZEB cluster battery charging rates ($[u_1^{cluster}, u_2^{cluster}, \ldots, u_{24}^{cluster}]$) in the next 24 h. Then the hourly power exchange with grid ($[E_{ex,1}^{cluster}, E_{ex,2}^{cluster}, \ldots E_{ex,24}^{cluster}]$) is calculated by Eq. (20.4). The nZEB battery charging rates and power exchange are used as constraints in Step 2.

Step 2: Coordination of individual nZEB operation using NLP

After obtaining the optimal operation at the nZEB-cluster-level, the individual nZEB operation is coordinated using NLP. Figure 20.3 presents the flowchart of coordinating individual nZEB operation using NLP (Zhao et al. 2015). The NLP is conducted in each hour to obtain the individual nZEB operation (i.e., $u_i^j, E_{ex,i}^j$) in each hour. In total, the NLP is repeated 24 times for obtaining individual nZEB operation in one day.

In the i^{th} hour, the inputs of the NLP include the individual nZEB operational data [$E_{supply,i}^j$, $E_{demand,i}^j, u_i^j, E_{ex,i}^j$, (j = 1,…,n, where n is the number of nZEBs inside cluster)], and the nZEB cluster operational data [$u_i^{cluster}$, $E_{ex,i}^{cluster}$]. The known parameters includes the $E_{supply,i}^j, E_{demand,i}^j$ of the j^{th} nZEB (which are predicted using the future weather data, nZEB and nZEB system models) and $u_i^{cluster}$, $E_{ex,i}^{cluster}$ of nZEB cluster (obtained from GA simulation in Step 1). The unknown parameters includes $u_i^j, E_{ex,i}^j$ of the j^{th} nZEB. It should be noted that the nZEB cluster operation parameters $(u_i, E_{(ex,i)})$ are used in the constraints. In the objective function, only the individual nZEB operation parameters $(E_{supply,i}^j, E_{demand,i}^j, u_i^j, E_{ex,i}^j)$ are used.

These inputs of individual nZEB operation $(E_{supply,i}^j, E_{demand,i}^j, u_i^j, E_{ex,i}^j)$ are then imported into the objective function in the NLP solver, as shown in Eq. (10). This objective function is different from the fitness function in the GA optimization.

$$\begin{aligned} J_{NLP} = f\Big(&E_{supply,i}^1, E_{demand,i}^1, u_i^1, E_{ex,i}^1, \\ &E_{supply,i}^2, E_{demand,i}^2, u_i^2, E_{ex,i}^2, \ldots, \\ &E_{supply,i}^n, E_{demand,i}^n, u_i^n, E_{ex,i}^n\Big) \end{aligned} \tag{20.10}$$

A typical objective function is the sum of the operational cost of each individual nZEB, as expressed in Eq. (20.11) (Sun et al. 2018)

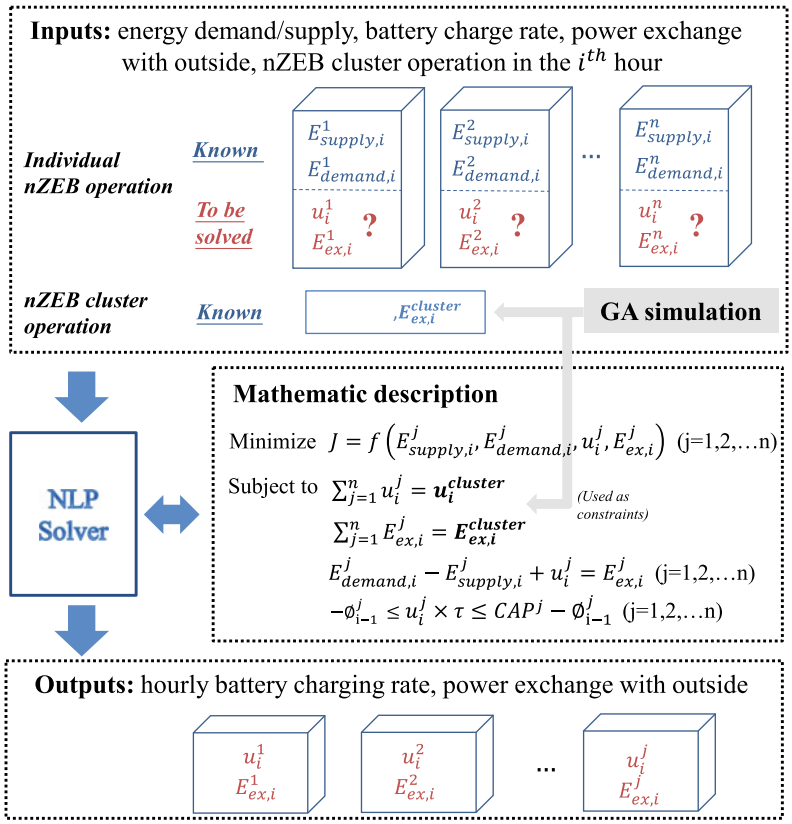

Fig. 20.3 Flowchart of coordinating individual nZEB operation using NLP

$$J_{NLP,i} = \sum_{j=1}^{n} E^j_{ex,i} \times \tau \times \chi_i \qquad (20.11)$$

where χ_i(HKD/kWh) is the real-time electricity cost in the i^{th} hour. It is usually different between off-peak hours and on-peak hours. A positive $E^j_{ex,i}$ value indicates importing electricity, while a negative $E^j_{ex,i}$ value represents exporting electricity. The feed-in-tariff is usually lower than the importing electricity price, as exporting power to the grid might worsen the power grid stability and reliability (Gao et al. 2015; Lu et al. 2015a; Zhao et al. 2015; Sun et al. 2018).

The objective function should be subject to the following constraints.

i. The sum of charging rates of individual nZEB (u^j_i (kW)) equals the charging rate of the nZEB cluster ($u^{cluster}_i$ (kW)) (obtained from Step 1).

$$\sum_{j=1}^{n} u^j_i = u^{cluster}_i \qquad (20.12)$$

ii. The sum of power exchange of individual nZEB ($E^j_{ex,i}$ (kW)) equals the power exchange of the nZEB cluster ($E^{cluster}_{ex,i}$ (kW)) (obtained from Step 1).

$$\sum_{j=1}^{n} E^j_{ex,i} = E^{cluster}_{ex,i} \qquad (20.13)$$

iii. The energy balance of individual nZEB. There are n equality constraints for n nZEBs.

$$E^j_{(demand,i)} - E^j_{(supply,i)} + u^j_i = E^j_{(ex,i)} \quad (j = 1, 2, \ldots n, \text{respectively}) \qquad (20.14)$$

iv. The charging amount should not exceed the remaining storage of battery, and the discharging amount should not exceed the amount of electricity stored in the battery.

There are n inequality constraints for n nZEBs.

$$-\varnothing_{i-1}^{j} \le u_i^j \times \tau \le CAP^j - \varnothing_{i-1}^{j} \quad (j = 1, 2, \ldots n, \text{respectively}) \tag{20.15}$$

where τ is the charging duration (which is set to 1 h), CAP^j (kWh) is the battery capacity of the j^{th} nZEB, $\varnothing_{i-1}^{j}$ (kWh) is the energy stored in the battery of the j^{th} nZEB. $\varnothing_{i-1}^{j}$ (kWh) is calculated by Eq. (20.16).

$$\varnothing_{i-1}^{j} = \varnothing^{j}{}_0 + u_1^j + u_2^j + \ldots + u_{i-1}^j \tag{20.16}$$

where $\varnothing^{j}{}_0$ is the initial stored energy in the battery of the j^{th} nZEB. In total, for an nZEB cluster with n nZEBs, there are $2n$ unknown parameters to be solved, there are $(n+2)$ equality constraints and there are n inequality constraints. The outputs of the NLP are $u_i^j, E_{ex,i}^j$ of the j^{th} nZEB.

The top-down control method uses GA at cluster level and NLP at individual level. GA is effective in searching the global or near optimal solution but is inefficient in computation [6]. In contrast, NLP is efficient in computation but might produce local optimal solutions [6]. The goal at cluster level is to optimize cluster's operational parameters, while at individual level is to coordinate single nZEB's operations. Based on the goals at different levels and characteristics of the two algorithms, GA and NLP are selected for cluster-level-optimization and individual-level-coordination.

In the case studies, the performance of the top-down control method is compared with an conventional individual-GA control (which optimize the operation of each individual nZEB using GA without considering the collaboration between nZEBs (Lu et al. 2015a; Sun et al. 2018), see Fig. 20.1(a)). The performance improvement percentages of the load matching (IP_{match}) and grid interaction (IP_{grid}) are calculated by Eqs. (20.17 and 20.18), respectively.

$$IP_{match} = \left(J_{match,top_down} - J_{match,ind_GA}\right)/J_{match,ind_GA} \tag{20.17}$$

$$IP_{grid} = \left(J_{grid,ind_GA} - J_{grid,top_down}\right)/J_{grid,ind_GA} \tag{20.18}$$

where J_{match} and J_{grid} were determined by Eqs. (20.6 and 20.8) under the top-down control and individual-GA control, respectively. It should be noted that for load matching, a larger value of indicates better performance; while for grid interaction, a smaller value of indicates a better performance.

To sum up, this section first introduces individual nZEB control and nZEB cluster control, and then compares their overall performance. Second, a new top-down control method, enabling full collaborations, is proposed. The top-down control optimizes the cluster level operations in the first step and then coordinates individual nZEB's operations in the second step. The detailed implementation of the top-down control is also described.

20.3 Building and Systems Modeling

This section introduces the modeling of the nZEB and nZEB systems. Trnsys was used to simulate the nZEB and nZEB systems. The studied systems include an air-conditioning system, two renewable energy systems (i.e., PV panel and wind turbine) and an electricity storage system (i.e., battery).

20.3.1 Building Modeling and HVAC System Modeling

A three-floor building was built in Trnsys. Each floor has a dimension of 20m × 10m. In each floor, there are four windows (4m × 1.5m) embedded in each wall. The indoor air

temperature was set at 25 °C and the relative humidity was set as 50%. The ventilation rate was 15 l/s/person and the infiltration rate was set to 0.2ACH. The building was used for four different functions: hotel, shopping center, office and restaurant. Corresponding to different functions, the internal occupancy schedules, electrical equipment operation schedules and lighting schedules were determined following the Hong Kong Building Energy Code (EMSD 2007). As the four buildings had the same geometry and were used under the same weather condition, the differences in energy demand were mainly caused by usage schedules of the building.

The studied air-conditioning system included a cooling coil, a chiller, a cooling tower, two pumps and a fan. The schematic diagram of the air-conditioning system is shown in Fig. 20.4.

The power consumption of the chiller $Pow_{chiller}$ (kW) is calculated by Eq. (20.19), (Klein et al. 2004; Handbook 2009; Zhang and Cheng 2017)

$$Pow_{chiller} = FFLP \times Q/COP_{nom} \quad (20.19)$$

where COP_{nom} is the nominal chiller COP, Q (kW) is the actual supplied cooling capacity, and $FFLP$ is the fraction of full load power. The power consumption of pumps Pow_{pump} (kW) are calculated by Eq. (20.20)

$$Pow_{pump} = \dot{m}_w \times g \times h/\eta_p/1000 \quad (20.20)$$

where $\dot{m}_w$ (kg/s) is the water mass flow rate, g is gravity (9.81 m/s^2), h (m) is the water head and η_p is the overall efficiency of pump. The power consumption of fan Pow_{fan} (kW) is calculated by Eq. (20.21)

$$Pow_{fan} = \dot{m}_a \times \Delta P/\rho/\eta_f/1000 \quad (20.21)$$

where $\dot{m}_a$ (kg/s) is the air mass flow rate, ΔP (Pa) is the pressure difference between fan inlet and outlet, ρ (kg/m^3) is the air density, and η_f is the overall efficiency of fan.

20.3.2 Renewable Energy System Modeling and Battery Modeling

The power generation of PV panel P_{PV} (kW) is calculated by Eq. (20.22) (Huang et al. 2018; Klein et al. 2004),

$$P_{PV} = \tau \times I_{AM} \times I_T \times \eta \times CAP_{PV} \quad (20.22)$$

where τ is the transmittance-absorptance product of the PV cover for solar radiation at a normal incidence angle, ranging from 0 to 1; I_{AM} is the combined incidence angle modifier for the PV cover material, ranging from 0 to 1; I_T (W/m^2) is the total amount of solar radiation incident on the PV collect surface; η is the overall efficiency of the PV array; CAP_{PV} (m^2) is the PV surface area.

The power generated by a wind turbine P_{WT} (kW) is calculated by Eq. (20.23) (Klein et al. 2004; Huang et al. 2018)

$$P_{WT} = \begin{cases} 0, \ if \ U_0 \le 3 \\ C_P \times \rho_{air} \times A_R \times U_0^3, otherwise \\ CAP_{WT}, \ if \ U_0 \ge 14 \end{cases} \quad (20.23)$$

where C_P is power efficiency, which is a function of the axial induction factor; ρ_{air} (kg/m^3) is air

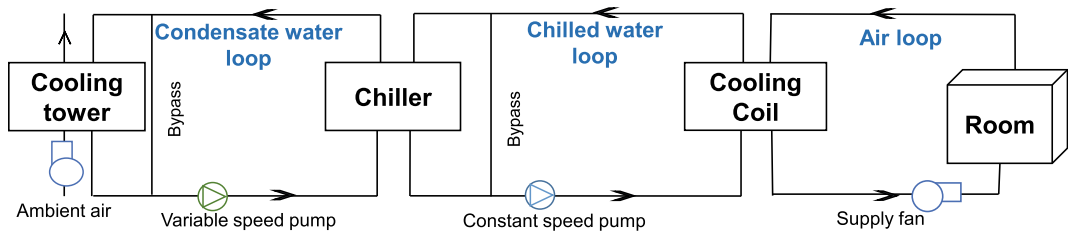

Fig. 20.4 Schematics of the air-conditioning system

density; A_R (m^2) is the rotor area; U_0 (m/s) is the wind speed in the free stream; CAP_{WT} (kW) is the capacity of wind turbine.

The electricity stored in the battery is calculated using a simplified model, as expressed by Eq. (20.16). It is estimated by the charging rates (Sun et al. 2018).

20.4 Case Studies and Results Analysis

Four nZEBs of different types, a hotel, an office, a shopping center and a restaurant, were considered in this chapter. These four buildings were studied under the weather condition in Hong Kong. Each of the nZEB was installed with its own air-conditioning system, renewable energy system and electrical battery system (see Fig. 20.1(b)). The case studies presented and analyzed the results of one week (in July). Table 20.1 summarizes the price models used in this chapter. With reference to the electricity prices of a real power company (Power 2018), the prices for purchasing electricity from the grid were set as 2 HKD/kWh and 1 HKD/kWh in the on-peak hours and off-peak hours, respectively. Meanwhile, the feed-in-tariff was set as 0.2 HKD/kWh considering the grid stress caused by the renewable energy exports. Note that in practice, the feed-in-tariff could be much cheaper or even forbidden as the grid concerns associated negative impacts on its power balance and supply quality (Zhao et al. 2015). To encourage renewable energy sharing among nZEBs, the prices of electricity purchased from and sold to the cluster were set to be 0.5 HKD/kWh, which is lower than the price of electricity purchased from the grid but higher than that of electricity exported to the grid. Note that if the price of electricity purchased from cluster was higher than 1 HKD/kWh or the price of electricity sold to cluster was lower than 0.2 HKD/kWh, there would be no economic incentive for the nZEBs to share renewable energy in a cluster.

The proposed top-down control was compared with the conventional individual-GA control. Both the two controls were applied to optimize the operation of the nZEBs. In the GA simulation, the population size was set to 100, the termination tolerance was set to 1e-6 and the number of generations was set as 100. These GA parameters have been tuned using the trial and error method to ensure the GA simulation terminates by reaching the fitness limit. The comparison was conducted under two different control strategies: maximizing cluster load matching and minimizing cluster grid interaction. These two strategies were implemented by using different fitness functions (i.e., Eqs. 20.6 and 20.8) in the GA algorithm. Please note that cost minimization (Eq. 20.11) only applies for the NLP-based coordination in the top-down control. With the cost consideration (Eq. (11)), the NLP will identify the most economic coordination solution to realize the cluster level optimal results generated from the GA. A 5% energy loss was assumed for energy transmission inside the nZEB cluster (Peng and Tao 2017). As the calculation of load coverage by renewable energy and peak energy exchanges are not affected by energy stored in battery every day, these two indicators were compared on a daily basis. In contrast, the operational cost was compared on a weekly basis.

Table 20.1 Electricity price (Sun et al. 2018; Lu et al. 2015a)

Electricity price (HKD)			
Grid	Buy-from-grid	on-peak (8–21)	2
		off-peak (other)	1
	Feed-in-tariff		0.2
Cluster	Buy-from-cluster		0.5
	Sell-to-cluster		0.5

20.4.1 Building Energy Demand and Renewable Energy Supply

Using the models presented in Sect. 20.3 and the TMY data in Hong Kong, the annual energy demand of each individual nZEB is predicted. The hourly energy demand results of the selected week are presented in Fig. 20.5(a). As can be seen, the peak energy demand of hotel occurred during the night, the peak energy demand of the other three buildings occurred during the daytime. There were three energy demand peaks for the restaurant corresponding to the three mealtimes. These different load characteristics provided possibilities for nZEB collaboration.

The capacities of renewable energy systems were sized to produce the same amount of energy as the annual energy demand, thus achieving the goal of net zero. Considering renewable system configuration diversity in practice, the renewable energy systems were sized using different energy generation ratio of wind turbine to PV panels (0/1, 0.3/0.7, 0.6/0.4 and 1/0 for hotel, shop, restaurant and office, respectively). The capacities of renewable energy systems and electrical batteries of the four case nZEBs are listed in Table 20.2. The battery capacity was sized to guarantee a nZEB to operate normally for 12 h without interacting with the grid (Huang et al. 2018). As the calculated battery capacities of the four nZEBs were close to 1000 kWh, their capacities were selected as 1000 kWh. Figure 20.5(b) presents the hourly renewable energy supply of the four nZEBs. In the 6th and 7th day, during the most time, the energy generation of the office approached zero. This is because the wind speed in these two days was very low (the office was not equipped with PV panels).

20.4.2 Load Matching Evaluation Results

In this chapter, the load matching mainly considered maximizing the load coverage by the onsite renewable energy generation. The purpose was to reduce the dependence on the grid. This

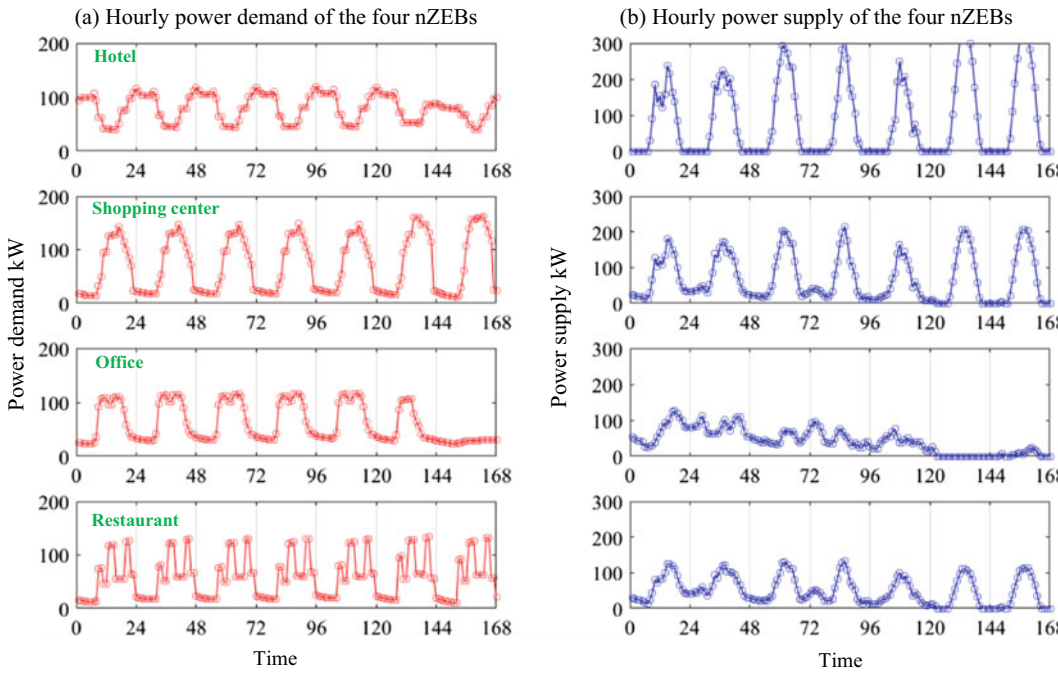

Fig. 20.5 Demand and supply profiles (a), Hourly energy demand of the four nZEBs (b), Hourly energy supply of the four nZEBs

Table 20.2 Capacities of renewable energy systems and electrical batteries of four case nZEBs

No	Building type	Wind turbine capacity (kW)	PV panel area (m^2)	Electrical battery Capacity (kWh)
1	Hotel	——	347	1000
2	Shopping center	80	207	1000
3	Office	186	——	1000
4	Restaurant	100	110	1000

section first presented the operational details of individual nZEB and then presented the load matching and operational cost results of the nZEB cluster.

20.4.2.1 Individual nZEB Battery Charging, Battery Energy and Power Exchange Results

Figure 20.6 presents the battery charging controls and the associated battery stored energy results of the four nZEBs using the two controls. The dotted red curves with circles were results from top-down control, while the dotted blue curves with diamonds were results from individual-GA control.

Figure 20.7 presents power exchanges results of each individual nZEB using the two controls. The individual-GA control could limit nZEB's exported energy well at 0 during the most time (see curves of shop, office and restaurant in Fig. 20.7(b)); while top-down control could hardly achieve this goal. This is because individual-GA control aims at optimizing at individual-nZEB-level. However, a good control at individual-nZEB-level could not lead to the optimal performance at cluster level, as collaborations between nZEBs were neglected. As shown in Fig. 20.9(a), the top-down control could limit the cluster's exported energy at 0 much better than individual-GA control. This is because the top-down control coordinates

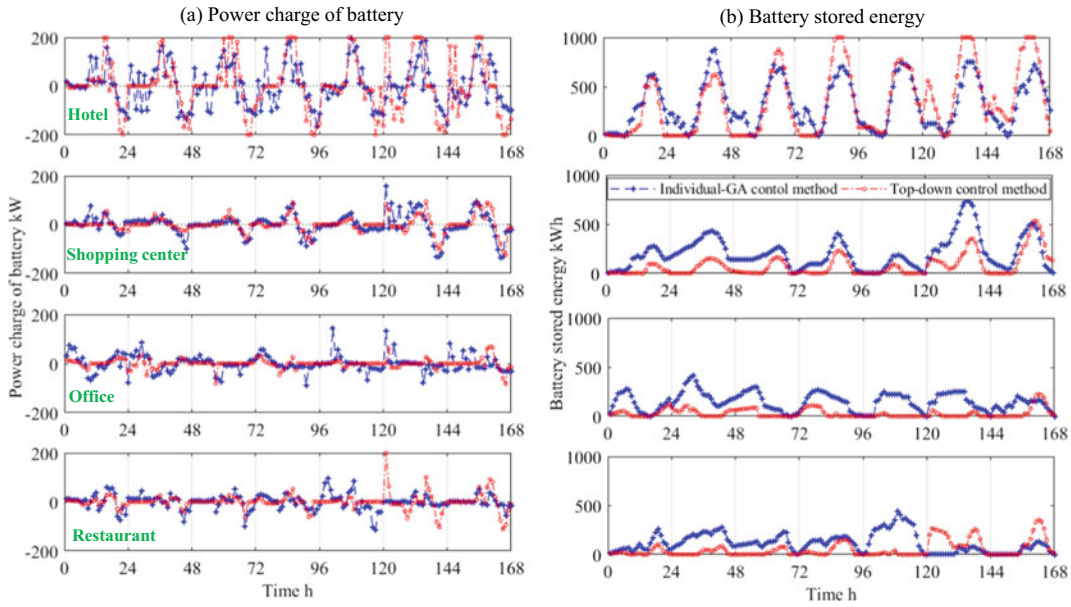

Fig. 20.6 Comparison of top-down control and individual-GA control (a), power charge of battery (b), Battery stored energy battery

collaborations between nZEBs, thus the cluster would reduce exporting energy to the grid to increase load matching. The power export of the hotel could hardly be controlled to 0 as during daytime it had large power mismatch (see Fig. 20.5) and its battery was already fully charged (see Fig. 20.6(b)).

20.4.2.2 nZEB Cluster Load Matching and Operational Cost Results

Figure 20.8 compares the daily load coverage by renewable energy of the two controls. The top-down control could effectively increase the nZEB cluster load coverage by renewable energy than the individual-GA control. The performance improvements percentage of load matching (which is calculated by Eq. 20.17) could be as high as 45%.

The top-down control produces better load matching performances at cluster level as it enables collaborations among nZEBs. Such collaborations allow nZEBs to share the surplus energy with other nZEBs which have insufficient energy generations or store the surplus energy in other nZEBs' batteries, thereby reducing unnecessary energy exports to the grid. With the reduced energy exports to the grid, the daily load coverage by the on-site renewable energy generation of the cluster can be improved significantly.

In order to demonstrate the collaboration capability of top-down control, the energy flow during a period of 4 h (137th to 140th hour) was analyzed and presented. Table 20.3 summarizes the energy mismatch (sum of power mismatch), energy charge (sum of power charge) and energy exchange (sum of power exchange) of each individual nZEB during this period.

Figure 20.10 presents the energy flow inside the nZEB cluster during this period. Please note this way of energy flow is merely presented as an example. Other ways of energy flow are also possible. The transmission loss was not displayed for simplicity.

- The hotel had surplus energy supply as the sum of power mismatch was negative (power mismatch is calculated by power demand minus power supply).
- The shop, office and restaurant had insufficient energy supply.
- As Battery 1 was already fully charged (its initial stored energy was 1000 kWh), the surplus energy of hotel (280 kWh) was exported to the other nZEBs, part (186 kWh)

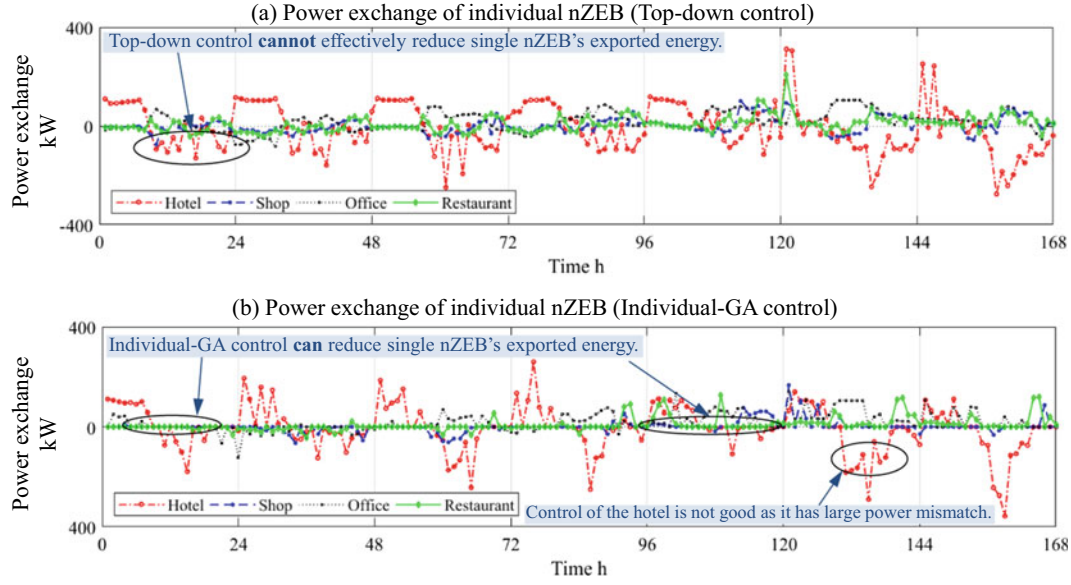

Fig. 20.7 Power exchange of individual nZEB (a), Top-down control (b), Individual-GA control

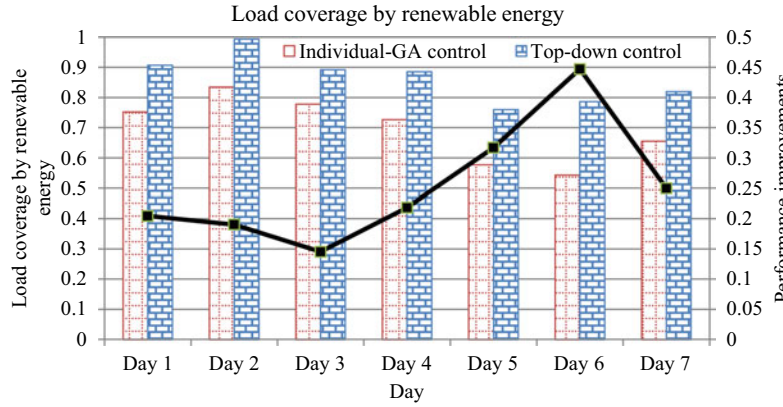

Fig. 20.8 Comparison of daily load coverage by renewable energy of individual-GA control and top-down control

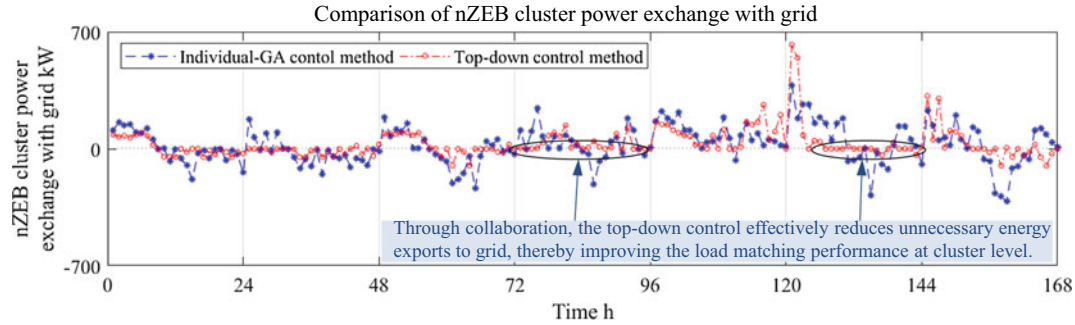

Fig. 20.9 Comparison of nZEB cluster power exchange with the grid using the two controls

Table 20.3 Summary of energy flow during 137 h ∼ 140 h (4 h period as an example)

nZEB	Energy mismatch (kWh)	Energy charge (kWh)	Energy exchange (kWh)	Battery initial stored energy (kWh)
Hotel	−280	−170	−450	1000
Shop	274	−88	186	309
Office	166	−26	140	26
Restaurant	150	−26	124	184
Energy from Grid (*a positive value indicates importing from the grid*)				0

was used to supply the shop, part (94 kWh) was used to supply the office.

- The remaining insufficient energy demand of shop (88kWh) was supplied from its own battery.
- Battery 3 only had 26 kWh initial stored energy. All its 26 kWh energy was supplied to the office. The remaining insufficient energy demand (46 kWh) was supplied from Battery 1.
- For the restaurant, part (124 kW) of its insufficient energy was supplied by Battery 1, the remaining part (26 kW) was supplied by its own battery.

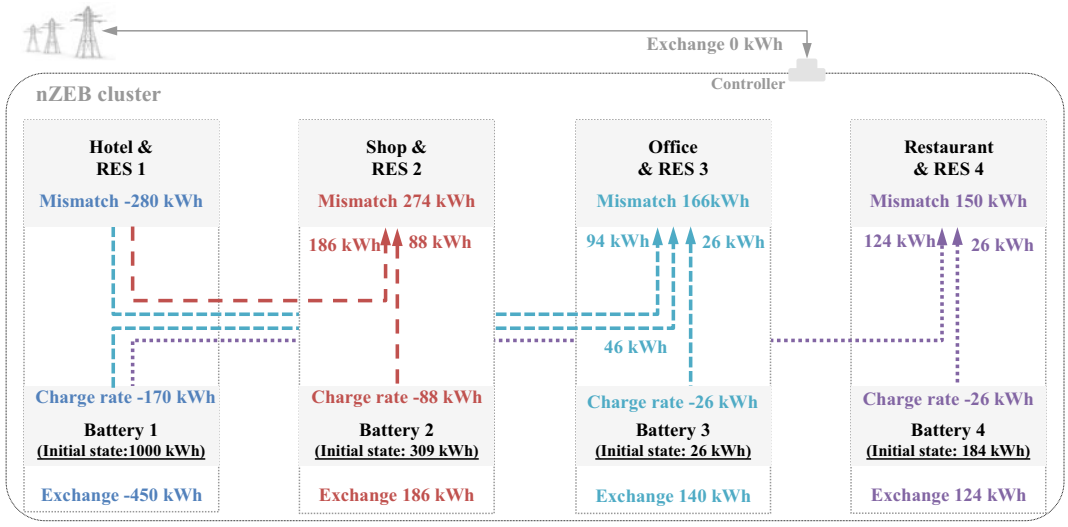

Fig. 20.10 Illustration of energy flow inside the nZEB cluster using the top-down control—an example

Using Battery 1 to supply energy demand of office and restaurant is battery sharing, and using the surplus energy of hotel to supply the demand of shop and office is energy sharing. Due to the energy sharing and battery sharing capability of top-down control, the nZEB cluster load matching performances could be significantly improved.

Using the electricity price in Table 20.4, the weekly operational cost of two control methods was calculated by Eq. (20.11) and summarized in Table 20.4. At the end of the week, the energy stored in the battery was sold to the grid with the given feed-in-tariff. Under the strategy of maximizing load matching, the top-down control could also achieve an operational cost reduction of 5,037 HKD (39%). The top-down control can achieve cost savings mainly because it enables collaborations among nZEBs, i.e., energy sharing and battery sharing insider a cluster (see Fig. 20.10). Through such collaborations, the nZEBs can share surplus renewable energy with other nZEBs whose renewable generations cannot meet their energy demands. In this way, the nZEBs with insufficient renewables can significantly reduce unnecessary high-priced energy imports from the grid, thereby reducing operational cost. Figure 20.11 presented the hourly energy sharing inside the nZEB cluster using the top-down control. A larger energy sharing indicates more benefits from nZEB collaborations. In individual-GA control, there is no energy sharing.

20.4.3 Grid Interaction Evaluation Results

In this chapter, the grid interaction mainly considered minimizing the peak energy exchanges between the nZEB cluster and the grid. As the energy exchanges include both energy import and energy export, the minimization of the peak energy exchanges should consider both.

Table 20.4 Comparison of weekly operational cost of two control method

	Weekly operational cost (HKD)
Individual-GA control	12,928
Top-down control	7,890
Cost saving by top-down control	5,037
Percentage	39%

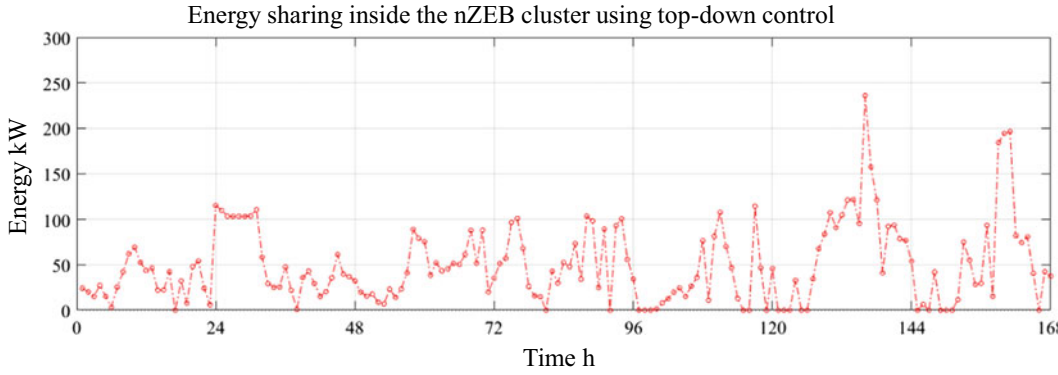

Fig. 20.11 Energy sharing inside the nZEB cluster using top-down control

20.4.3.1 Individual nZEB Battery Charging, Battery Energy and Power Exchange Results

Figure 20.12 presents the battery charging controls and the associated battery stored energy results of the four nZEBs using the two controls. The battery operation of individual nZEB is similar in different controls. Figure 20.13 presents power exchanges results of each individual nZEB using the two controls. Similar to load matching results, the individual-GA control has a better control performance at individual-nZEB-level as it aims to optimize the operation of individual nZEB. This can be seen from Fig. 20.13(b): the peak hourly energy exchange of each nZEB was well controlled in limited ranges. For instance, in the first day, the peak imported energy of hotel was limited at 99 kWh from the 1st to the 9th hours, and the peak exported energy was limited at −99 kWh in the 16th and 17th hour. The top-down control did not limit the hourly energy exchanges of individual nZEB (see Fig. 20.13(a)).

20.4.3.2 nZEB Cluster Grid Interaction and Operational Cost Results

Figure 20.14 compared the daily peak energy exchanges of the two controls. The top-down control could effectively reduce the nZEB cluster peak energy exchanges than the individual-GA control. The performance improvements percentage of grid interaction (which is calculated by Eq. (18)) could be as high as 82% on a daily basis.

The top-down control results in better nZEB cluster's grid interaction performance than individual-GA control. This is because the top-down control enables collaborations among nZEBs. Through such collaboration, the nZEBs can eliminate the unnecessary large energy imports/exports by sharing the surplus renewable energy with other nZEBs. Meanwhile, avoiding the useless peak limiting at individual-nZEB-level, the top-down control can coordinate single nZEB's operations to more effectively limit the daily peak energy exchanges at nZEB-cluster-level. The collaboration capability of top-down control has been demonstrated in Fig. 20.10. This can be seen from Fig. 20.15: the peak hourly energy exchange of the nZEB cluster was well controlled in limited ranges. For instance, in the first day, the peak imported energy was limited at 88 kWh from 1st to 9th hour, and the peak exported energy was limited at -88 kWh in the 10th, 12th and 14th hour.

Under the strategy of minimizing grid interaction, the weekly operational cost of two control methods was calculated and summarized in Table 20.5. The top-down control could achieve an operational cost reduction of 13,665 HKD (55%). The top-down control can achieve cost saving because it reduced high-priced electricity energy imports from the grid by collaborations. Figure 20.16 presented the hourly energy sharing inside the nZEB cluster using the top-down control.

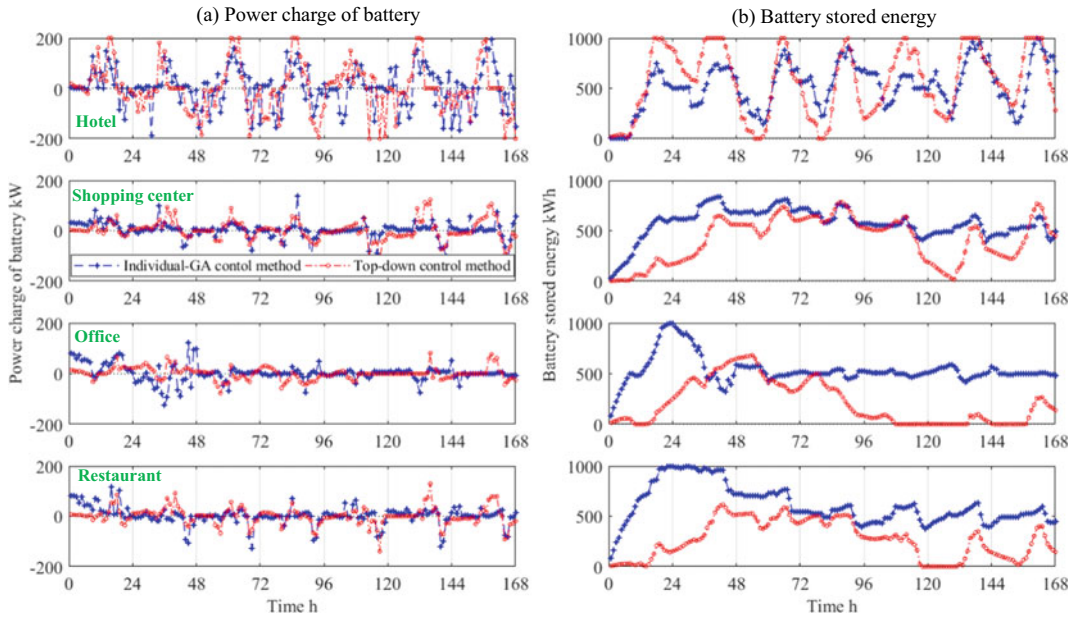

Fig. 20.12 Comparison of top-down control and individual-GA control (a), power charge of battery (b), Battery stored energy battery

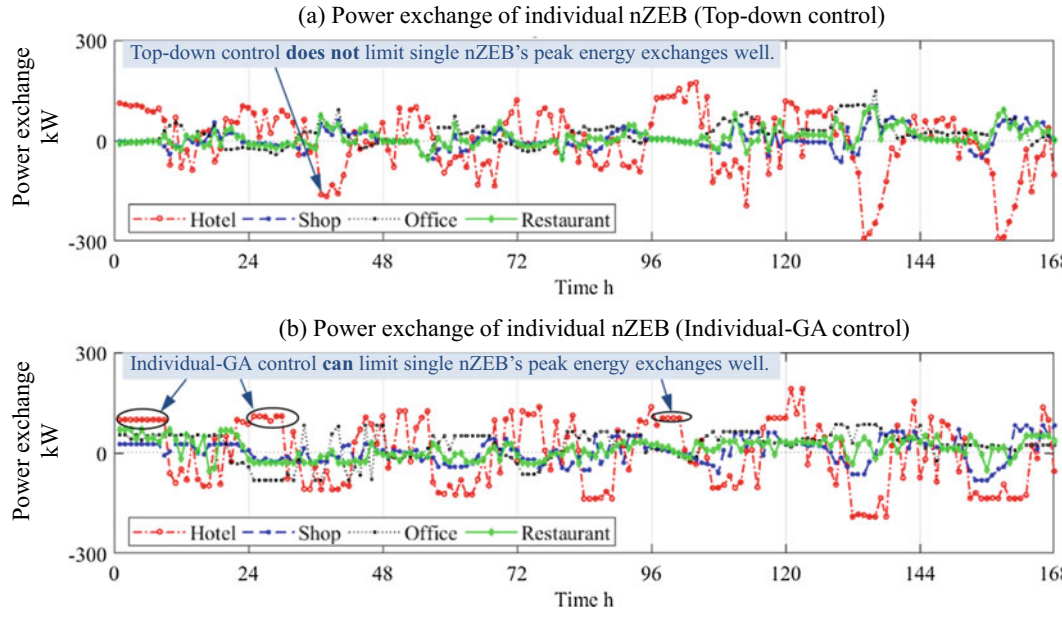

Fig. 20.13 Power exchange of individual nZEB (a), Top-down control (b), Individual-GA control

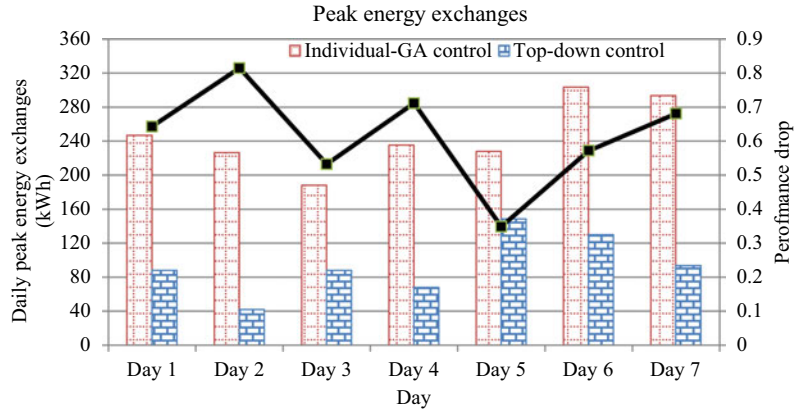

Fig. 20.14 Comparison of daily peak energy exchanges of individual-GA control and top-down control

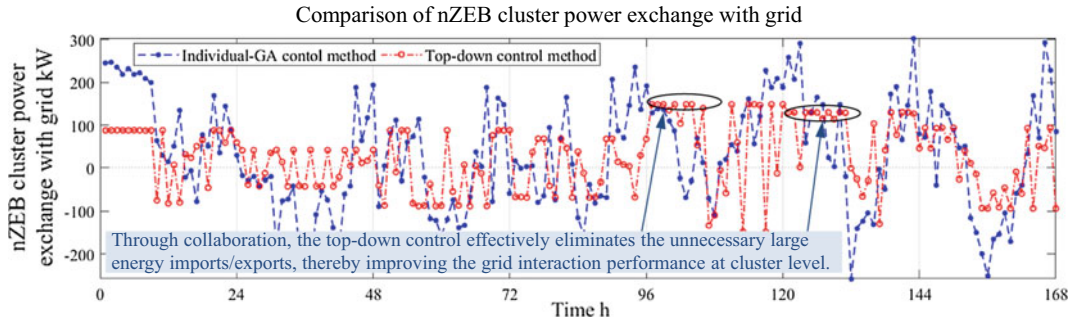

Fig. 20.15 Comparison of nZEB cluster power exchange with the grid using the two controls

Table 20.5 Comparison of weekly operational cost of two control method

	Weekly operational cost (HKD)
Individual-GA control	24,719
Top-down control	11,054
Cost saving by top-down control	13,665
Percentage	55%

20.5 Summary

This chapter has proposed a top-down control method to optimize the performances of nZEBs at cluster level. The top-down control first optimizes the nZEB cluster (considered as 'one') operation using GA and then coordinates the operation of individual nZEB using NLP. The collaboration benefits of top-down control have been investigated under two control strategies: maximizing load matching and minimizing grid interaction. The top-down control has been compared with a conventional individual-GA control. The major findings are stated as below.

- The top-down control provides a mechanism to collaborate the nZEBs inside a cluster for performance improvement in different aspects.
- In aspects of load matching, the collaborations enable nZEBs to share surplus renewable energy with other nZEBs or to store the

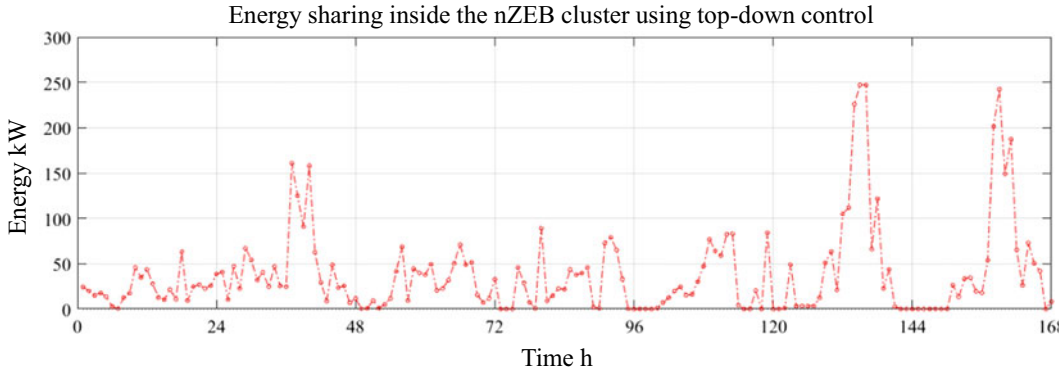

Fig. 20.16 Energy sharing inside the nZEB cluster using top-down control

surplus energy in other nZEBs' batteries, thereby largely reducing the total energy exports to the grid. With the reduced energy exports to the grid, the daily load coverage by the on-site renewable energy generations of the cluster can be improved by 45%.

- In aspects of grid interaction, the collaborations enable nZEBs to share the surplus renewable energy with other nZEBs, thereby eliminating the unnecessary energy exchanges with the grid. Meanwhile, avoiding useless peak limiting at individual level, the top-down control can coordinate single nZEB's operation to more effectively limit the daily peak energy exchanges at cluster level. Thus, the grid interaction performance can be improved by 82%.
- In aspects of economic cost, the collaborations can significantly reduce the high-priced energy imports from the grid by sharing the surplus energy with nZEBs whose renewable generations cannot meet their energy demands. Thus, the cost savings can be improved by 55%.

The top-down control closely relies on the predicted nZEB energy demand and supply. Thus the prediction accuracy of the energy demand and supply is crucial to the performances of the top-down control. In the future work, uncertainty in the prediction of nZEB energy demand and supply will be integrated, and a more robust nZEB cluster control method will be developed. Adopting a simplified battery model, this chapter has not considered the energy losses in the battery charging/discharging and energy storage processes. More sophisticated battery models need to be developed to consider the energy losses in the future cluster control developments.

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